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GOOD MORNING TEST (GMT)

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1. Exercise RIMPAC 2026.

Q. Which of the following best describes the Rim of the Pacific (RIMPAC) exercise?

(a) A biennial multinational maritime exercise hosted by the U.S. Navy to enhance interoperability among partner navies in the Indo-Pacific region.
(b) A regional security dialogue among Pacific island nations aimed at disaster management and climate resilience.
(c) A naval exercise conducted exclusively by QUAD countries to ensure freedom of navigation in the South China Sea.
(d) A joint military exercise organised by ASEAN to strengthen maritime law enforcement capabilities.

Answer: (a) A biennial multinational maritime exercise hosted by the U.S. Navy to enhance interoperability among partner navies in the Indo-Pacific region.

Explanation

Option (a): Rim of the Pacific (RIMPAC) is the world's largest international maritime exercise. It is hosted biennially by the U.S. Navy's Indo-Pacific Command through the U.S. Pacific Fleet in and around the Hawaiian Islands. The exercise aims to strengthen interoperability, operational readiness, and collective security among participating nations while promoting a free and open Indo-Pacific. **Hence, option (a) is correct.**

Option (b):RIMPAC is not a regional dialogue mechanism focused on climate resilience or disaster governance. It is a military maritime exercise involving naval, air, and amphibious operations. **Hence, option (b) is incorrect.**

Option (c): RIMPAC is not limited to QUAD countries. It includes participation from a large number of nations across different regions and is not exclusively focused on freedom of navigation operations in the South China Sea**. Hence, option (c) is incorrect.**

Option (d): RIMPAC is neither organised by ASEAN nor limited to maritime law enforcement. It is hosted by the U.S. Navy and covers a broad spectrum of naval warfare and security operations. **Hence, option (d) is incorrect.**

Value Addition

Exercise RIMPAC

Feature	Details
Full Form	Rim of the Pacific Exercise
Nature	Multinational Maritime Exercise
Host	U.S. Navy Pacific Fleet
First Conducted	1971
Frequency	Biennial (Every 2 Years)
Venue	Hawaiian Islands and Pacific Ocean
Edition (2026)	30th Edition
Theme (2026)	"Partners: Integrated and Prepared"
Indian Participation	Since 2014

Major Activities

- Anti-Submarine Warfare (ASW)
- Air Defence Operations
- Missile Proficiency Exercises
- Mine Countermeasure Operations
- Amphibious Assault Exercises
- Maritime Interdiction Operations
- Humanitarian Assistance and Disaster Relief (HADR)
- Gunnery and Live-Fire Training

Significance

- Enhances interoperability among partner navies.
- Strengthens collective maritime security.
- Promotes a free, open, and rules-based Indo-Pacific.
- Improves joint operational readiness in complex maritime environments.
- Usually conducted around **Hawaii and Southern California**.
- Includes naval, air, amphibious, and submarine operations.

In NEWS: World's Biggest Maritime Exercise, RIMPAC 2026, Sees Participation From 30 Nations & 30 Plus Warships In Hawaii

2. PM Family Care Tracker (PM-FCT)

Q. With reference to the PM Family Care Tracker (PM-FCT), consider the following statements:

1. It is a family-centric digital platform designed to monitor health, nutrition, education, and welfare indicators from pregnancy up to 18 years of age.
2. It maintains separate databases for each government welfare scheme and does not integrate information from health, education, or civil registration systems.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (a) 1 only

Explanation

Statement 1: PM Family Care Tracker (PM-FCT) is a digital platform aimed at strengthening maternal and child health, nutrition, and family welfare. It digitally tracks key developmental milestones from pregnancy until a child attains 18 years of age, ensuring timely delivery of eligible government benefits and services. **Hence, Statement 1 is correct.**

Statement 2: PM-FCT is specifically designed as a family-centric integrated platform, rather than maintaining separate records for different schemes. It seeks convergence across departments by integrating data from birth and death registration systems, as well as health, nutrition, and education databases. **Hence, Statement 2 is incorrect.**

Value Addition

Feature	Details
Full Form	PM Family Care Tracker (PM-FCT)
Nature	Family-centric digital monitoring platform
Launched As	Pilot Project in Gujarat
Coverage	Pregnancy to 18 years of age
Objective	Strengthen maternal and child health, nutrition and family welfare
Key Monitoring Areas	Antenatal care, postnatal care, immunisation, nutrition, growth, education and adolescent health
Special Feature	Digital Health Passport for families
Alerts	Missed vaccinations, school dropouts and other service gaps
Data Integration	Health, nutrition, education, birth and death registration databases

Key Benefits

- Timely delivery of welfare benefits.
- Reduction in maternal and infant mortality.
- Better monitoring of malnutrition.
- Improved inter-departmental convergence.
- Early intervention through automated alerts and notifications.

IN NEWS: PM Family Care Tracker launched in Gujarat to digitally monitor maternal and child health

3. Tipping Points

Q.The term "Climate Tipping Point", frequently seen in the context of climate change, best describes:

(a) A temporary fluctuation in global temperature that reverses naturally over time.
(b) A critical threshold in the climate system beyond which large-scale and often irreversible changes may occur.
(c) A target fixed under the Paris Agreement to limit global warming below 2°C.
(d) A mechanism through which greenhouse gases are removed from the atmosphere.

Answer: (b) A critical threshold in the climate system beyond which large-scale and often irreversible changes may occur.

Explanation

Option (a): Climate tipping points do not refer to short-term or naturally reversible fluctuations in temperature. They involve major systemic changes that can become self-reinforcing and difficult to reverse.**Hence, option (a) is incorrect.**

Option (b): According to the Intergovernmental Panel on Climate Change (IPCC), tipping points are critical thresholds in a system that, when exceeded, can lead to significant changes in the state of the

system, often irreversible in nature. Once crossed, these changes may continue even if warming is later reduced.**Hence, option (b) is correct.**

Option (c): The Paris Agreement sets global temperature goals, but a climate tipping point is not a policy target. It is a scientific concept related to abrupt changes in Earth's climate system.**Hence, option (c) is incorrect.**

Option (d): Removal of greenhouse gases refers to carbon sequestration or carbon dioxide removal techniques, not tipping points.**Hence, option (d) is incorrect.**

Value Addition

Climate Tipping Points

Feature	**Details**
Defined By	IPCC
Meaning	Critical threshold causing major climate system change
Nature	Often abrupt, self-reinforcing and irreversible
Earlier Scientific Estimate	Around 4°C warming
Current Concern	Several tipping points may be triggered beyond 1.5°C warming
Impact	Accelerated climate change and cascading global effects

Major Climate Tipping Points

1. Greenland Ice Sheet

- World's second-largest ice sheet.
- Complete melting could raise global sea levels by about 7.2 metres.
- Greenland is warming 3–4 times faster than the global average.
- Currently the largest contributor to global sea-level rise.

2. Atlantic Meridional Overturning Circulation (AMOC)

- Major ocean current system transporting heat across the Atlantic.
- Functions like a global ocean conveyor belt.
- Weakening due to freshwater input from ice melt and increased rainfall.
- A collapse could drastically alter weather patterns in Europe, Africa and the Americas.

3. Amazon Rainforest

- Largest tropical rainforest on Earth.
- Produces nearly half of its own rainfall through moisture recycling.
- Deforestation and repeated droughts are pushing parts of the Amazon from a carbon sink to a carbon source.

4. Permafrost Thaw

- Permafrost is ground that remains frozen for at least two consecutive years.
- Stores massive quantities of carbon accumulated over thousands of years.
- Thawing releases carbon dioxide (CO_2) and methane (CH_4), intensifying global warming.

5. Antarctic Ice Sheets

- Rapid warming and glacier retreat are being observed.
- Continued melting may contribute up to 1 metre of sea-level rise by 2100.

6. Coral Reef Die-Off

- Coral reefs support immense marine biodiversity.
- Provide coastal protection and livelihood support.
- Scientists warn that many reefs may face near-total collapse by 2050 if warming continues.

In NEWS:India seeks clarity as 'tipping points' rock Bonn climate talks

4. Vizhinjam Port

Q.With reference to the Vizhinjam International Seaport, consider the following statements:

1. It is a strategic deep-water transshipment port located close to major international shipping routes.
2. It is India's first semi-automated port.
3. The breakwater constructed at the port is the deepest in India.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation

Statement 1:

Vizhinjam Port is located at Vizhinjam near Thiruvananthapuram in Kerala. It is situated only about 10 nautical miles from one of the world's busiest East-West international shipping corridors, through which a significant share of global container

traffic passes.Its naturally deep waters and proximity to major maritime trade routes make it highly suitable for handling large mother vessels without requiring significant diversion from established shipping lanes.**Hence, Statement 1 is correct.**

Statement 2:

Vizhinjam is India's first semi-automated port. The port incorporates advanced technologies such as:

- Fully automated yard cranes
- Remotely operated ship-to-shore cranes
- Digital cargo management systems
- AI-enabled Vessel Traffic Management System (VTMS)

These technologies improve efficiency, reduce turnaround time and enhance operational safety.**Hence, Statement 2 is correct.**

Statement 3:

The breakwater at Vizhinjam extends for nearly 3 kilometres and is recognized as the deepest breakwater in India.The port also enjoys a natural draft of around 20 metres, enabling it to handle ultra-large container vessels without extensive dredging.**Hence, Statement 3 is correct.**

Value Addition

Vizhinjam International Seaport: At a Glance

Feature	**Details**
Location	Vizhinjam, Thiruvananthapuram, Kerala
Type	Deep-water International Container Transshipment Port
Operational Model	Public-Private Partnership (PPP)
Operator	Adani Ports and SEZ Ltd.
State Partner	Government of Kerala
Cost	Around ₹8,900 crore
Natural Draft	~20 metres
Port Category	India's First Dedicated Transshipment Port
Automation	India's First Semi-Automated Port

Trans shipment Port

A port where cargo is transferred from one ship to another before reaching the final destination.

Draft

The vertical distance between the waterline and the bottom of a ship. A deeper draft allows larger vessels to dock.

Breakwater

A structure built offshore to protect a harbour from waves and currents.

Littoral Drift

Movement of sand and sediments along the coastline due to wave action.

In NEWS: MSC arm to buy 49% stake in Vizhinjam port for $1.4 bn

5. National Company Law Appellate Tribunal (NCLAT)

Q.With reference to the National Company Law Appellate Tribunal (NCLAT), consider the following statements:

1. It can hear appeals against orders of the Competition Commission of India (CCI).
2. Its orders are enforceable in the same manner as decrees of a civil court.
3. It was constituted under the Companies Act, 2013.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation

Statement 1:NCLAT hears appeals against orders of the Competition Commission of India (CCI).Apart from hearing appeals against orders of the National Company Law Tribunal (NCLT), NCLAT also acts as the appellate authority for certain orders passed by the Competition Commission of India (CCI). This role strengthens the institutional framework for competition regulation and corporate governance in India. **Hence, Statement 1 is correct.**

Statement 2: Orders passed by NCLAT are enforceable like decrees of a civil court. NCLAT has certain powers of a civil court while regulating its own procedure and is not bound by the CPC. Its orders are legally binding and enforceable in the same manner as a decree issued by a civil court. It can:

- Summon and examine witnesses.

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- Receive evidence through affidavits.
- Require production of documents.
- Issue commissions for examination of witnesses.

Hence, Statement 2 is correct.

Statement 3:NCLAT was constituted under the Companies Act, 2013. The National Company Law Appellate Tribunal (NCLAT) is a quasi-judicial body constituted under Section 410 of the Companies Act, 2013. It became operational on 1 June 2016 and serves as the appellate tribunal for company law and insolvency-related matters. **Hence, Statement 3 is correct.**

Value Addition

National Company Law Appellate Tribunal (NCLAT)

Feature	Details
Established Under	Section 410, Companies Act, 2013
Operational Since	1 June 2016
Nature	Quasi-Judicial Appellate Tribunal
Headquarters	New Delhi
Appointed By	Central Government
Further Appeal Lies To	Supreme Court of India
Objective	Speedy resolution of corporate and insolvency disputes

Appeals Heard by NCLAT

Authority	Appeal Lies to NCLAT
National Company Law Tribunal (NCLT)	Yes
Insolvency and Bankruptcy Board of India (IBBI)	Yes
Competition Commission of India (CCI)	Yes
National Financial Reporting Authority (NFRA)	Yes

Powers of NCLAT

- Regulates its own procedure.
- Possesses powers of a civil court.
- Summons witnesses and records evidence.
- Enforces production of documents.
- Orders are executable as civil court decrees.
- Civil courts have no jurisdiction over matters assigned to NCLAT.
- Injunctions cannot be granted against actions taken by NCLAT under its lawful authority.

Comparison: NCLT vs NCLAT

Aspect	NCLT	NCLAT
Role	Original Adjudicating Authority	Appellate Authority
Established Under	Companies Act, 2013	Companies Act, 2013
Deals With	Company and insolvency cases at first instance	Appeals against NCLT and other regulators
Further Appeal	NCLAT	Supreme Court

In NEWS:NCLAT upholds order to pay PF dues to Jet Airways staff

6. Unified Health Interface (UHI)

Q.With reference to the Unified Health Interface (UHI), consider the following statements:

1. It is an interoperable digital network under the Ayushman Bharat Digital Mission (ABDM).
2. It enables consent-based sharing of health data through the Health Information Exchange framework.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Explanation

Statement 1: It is an interoperable digital network under the Ayushman Bharat Digital Mission (ABDM).The Unified Health Interface (UHI) is an interoperable digital network developed under the

Ayushman Bharat Digital Mission (ABDM). It acts as the service layer of ABDM and enables seamless interaction between patients and healthcare providers across multiple digital health platforms.UHI eliminates platform dependency and allows citizens to discover, book, and access healthcare services using any UHI-enabled application.UHI is not a mobile application **Hence, Statement 1 is correct.**

Statement 2: It enables consent-based sharing of health data through the Health Information Exchange framework.UHI facilitates secure and consent-based exchange of health information through the Health Information Exchange (HIE) framework of ABDM.

The network uses:

- ABHA (Ayushman Bharat Health Account) as the patient identifier.
- Healthcare Professionals Registry (HPR) for doctor verification.
- Health Facility Registry (HFR) for facility verification.
- Health Information Exchange (HIE) for secure and consent-driven sharing of health records.

This ensures that patients retain control over their health data and can share it only after providing consent.**Hence, Statement 2 is correct.**

Value Addition

Unified Health Interface (UHI)

Feature	**Details**
Mission	Ayushman Bharat Digital Mission (ABDM)
Nodal Agency	National Health Authority (NHA)
Nature	Interoperable Digital Health Network
Role	Service Layer of ABDM
Launch Objective	Improve accessibility, discoverability and interoperability in healthcare
Patient Identifier	ABHA (Ayushman Bharat Health Account)

IN NEWS: Union Health Minister Shri Jagat Prakash Nadda Launches Unified Health Interface – The Interoperable Network for Digital Health Services

7. Pradhan Mantri Krishi Sinchayee Yojana(PMKSY)

Q.With reference to the Pradhan Mantri Krishi Sinchayee Yojana (PMKSY), consider the following statements:

1. PMKSY is a Centrally Sponsored Scheme aimed at expanding cultivable area under assured irrigation.
2. The Long-Term Irrigation Fund (LTIF) and Micro Irrigation Fund (MIF) are maintained with NABARD.
3. Water budgeting under PMKSY is limited only to the agriculture sector**.**

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation

Statement 1: PMKSY is a Centrally Sponsored Scheme launched to improve physical access to water on farms, enhance irrigation coverage, and expand cultivable area under assured irrigation**. Hence, Statement 1 is correct.**

Statement 2: PMKSY has two dedicated irrigation funds maintained through NABARD:

- Long-Term Irrigation Fund (LTIF)
- Micro Irrigation Fund (MIF) **Hence, Statement 2 is correct.**

Statement 3: Water budgeting under PMKSY is not confined to agriculture alone. It covers household, agriculture, and industrial sectors for efficient water management. **Hence, Statement 3 is incorrect.**

Value Addition

Fund	**Purpose**
LTIF	Funding and fast-tracking incomplete major and medium irrigation projects
MIF	Financial assistance to States for micro-irrigation at concessional rates

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Value Addition

Components of PMKSY

Component	Objective
Har Khet Ko Pani	Irrigation access to every farm
IWMP	Watershed development and conservation
M-CADWM	Modern irrigation infrastructure
OFWM	Efficient on-farm water management

Monitoring Mechanism

Body	Chairperson
National Steering Committee (NSC)	Prime Minister
National Executive Committee (NEC)	Vice Chairman, NITI Aayog

IN NEWS:PMKSY completes over 10 years of irrigation-led agricultural transformation

8. Financial Action Task Force(FATF)

Q.With reference to the Financial Action Task Force (FATF), consider the following statements:

1. It was established by the G7 countries to address concerns related to money laundering.
2. Its Secretariat is hosted by the Organisation for Economic Co-operation and Development (OECD) in Paris.
3. It is a specialized agency of the United Nations.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation

Statement 1: The Financial Action Task Force (FATF) was established in Paris Summit in 1989 by the G7 countries to combat the growing threat of money laundering and later expanded its mandate to include terrorist financing and proliferation financing.**Hence, Statement 1 is correct.**

Statement 2: The FATF Secretariat is hosted by the Organisation for Economic Co-operation and Development (OECD) at its headquarters in Paris, France.**Hence, Statement 2 is correct.**

Statement 3:FATF is not a specialized agency of the United Nations. It is an independent intergovernmental body that works closely with the UN, IMF, World Bank and other international organizations.**Hence, Statement 3 is incorrect.**

Value Addition

Feature	Details
Established	1989
Established By	G7 Countries
Headquarters	Paris, France
Secretariat Hosted By	OECD
Nature	Intergovernmental Body and NOT a UN agency.
Members	FATF has 40 members (38 jurisdictions and 2 regional organizations: European Commission and Gulf Cooperation Council).
Objective	Combat Money Laundering, Terrorist Financing and Proliferation Financing

Value Addition

FATF Lists

List	Meaning
Grey List	Jurisdictions under Increased Monitoring
Black List	High-Risk Jurisdictions Subject to Countermeasures

In NEWS: India says FATF remains indispensable in fight against terror financing

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9. United Nations Economic and Social Commission for Asia and Pacific(UNESCAP)

Q.With reference to the United Nations Economic and Social Commission for Asia and the Pacific (UNESCAP), consider the following statements:

1. Its mandate is to promote economic and social development in the Asia-Pacific region through regional cooperation.
2. Disaster Risk Reduction is one of the areas in which UNESCAP carries out its work.
3. Its headquarters is located in Bangkok, Thailand.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation

Statement 1: The United Nations Economic and Social Commission for Asia and the Pacific (UNESCAP) is the principal economic and social development arm of the United Nations for the Asia-Pacific region. Its mandate is to promote economic and social development by fostering cooperation among member States and associate members and by addressing common regional development challenges.**Hence, Statement 1 is correct.**

Statement 2:UNESCAP undertakes activities in a wide range of sectors including:

- Macroeconomic Policy
- Poverty Reduction
- Financing for Development
- Trade and Investment
- Transport Connectivity
- Environment and Development
- Information and Communication Technology (ICT)
- Disaster Risk Reduction
- Social Development

The Commission actively supports member countries in building resilience against natural disasters and climate-related risks.**Hence, Statement 2 is correct.**

Statement 3:UNESCAP is headquartered in Bangkok, Thailand, which serves as the regional hub for its economic, social and sustainable development initiatives across the Asia-Pacific region.**Hence, Statement 3 is correct.**

Value Addition

United Nations Economic and Social Commission for Asia and the Pacific (UNESCAP)

Feature	Details
Established	Established in 1947 as ECAFE (Economic Commission for Asia and the Far East); renamed ESCAP in 1974.
Type	UN Regional Commission
Headquarters	Bangkok, Thailand
Parent Organisation	United Nations
Members	53 Member States
Associate Members	9
Region Covered	Asia-Pacific

Mandate of UNESCAP

- Promote economic growth and social development.
- Foster regional and sub-regional cooperation.
- Support implementation of Sustainable Development Goals (SDGs).
- Address common developmental and environmental challenges.
- Facilitate policy dialogue and capacity building among member countries.

In NEWS: UN report flags $800 billion climate fund gap in Asia-Pacific, calls for synergistic action

10.White - Rumped Vulture

Q.With reference to the White-rumped Vulture (*Gyps bengalensis*), consider the following statements:

1. It is also known as the Indian White-backed Vulture.
2. It is native to South and Southeast Asia and feeds mainly on carcasses.
3. It is listed as Endangered under the IUCN Red List and placed in Schedule II of the Wildlife Protection Act, 1972.

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Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation

Statement 1: The White-rumped Vulture (*Gyps bengalensis*) is also known as the Indian White-backed Vulture or Oriental White-backed Vulture. It is one of the most well-known vulture species of the Indian subcontinent. **Hence, Statement 1 is correct.**

Statement 2: The species is native to South and Southeast Asia, occurring in countries such as India, Pakistan, Bangladesh, Nepal, Bhutan, Myanmar, Thailand, Laos, Cambodia and Vietnam.Like most vultures, it is a scavenger and feeds primarily on animal carcasses, playing a crucial ecological role in waste disposal and disease control. **Hence, Statement 2 is correct.**

Statement 3: The White-rumped Vulture is classified as Critically Endangered under the IUCN Red List due to its catastrophic population decline caused mainly by the veterinary drug Diclofenac.Further, it receives the highest level of legal protection under Schedule I of the Wildlife (Protection) Act, 1972.**Hence, Statement 3 is incorrect.**

Value Addition

White-rumped Vulture (*Gyps bengalensis*)

Feature	**Details**
Scientific Name	*Gyps bengalensis*
Other Names	Indian White-backed Vulture, Oriental White-backed Vulture
Family	Accipitridae
Type	Old World Vulture
Distribution	South and Southeast Asia
Habitat	Plains, agricultural landscapes, villages and urban fringes
Diet	Carrion (animal carcasses)
Ecological Role	Natural scavenger and disease regulator
IUCN Status	Critically Endangered
WPA, 1972	Schedule I



Key Characteristics

- Smallest among the resident *Gyps* vultures of India.
- Possesses an unfeathered head and neck adapted for scavenging.
- Adults have:
  - Dark blackish plumage
  - White neck ruff
  - Distinct white patch on lower back (rump)
- Broad wings and short tail.
- Males and females are nearly identical in appearance.

Population Collapse

Major Cause:

Diclofenac Poisoning

- Diclofenac is a veterinary anti-inflammatory drug used in cattle.
- Vultures feeding on carcasses of treated livestock suffer kidney failure.
- The species experienced a decline of over 99% since the 1990s.

Other Threats

- Electrocution from power lines.
- Habitat degradation.
- Food scarcity.
- Poisoning and contamination.

Conservation Measures

- **B**an on veterinary Diclofenac.
- Captive breeding programmes.
- Vulture Conservation Breeding Centres.
- Vulture Safe Zones.
- Monitoring through radio-tagging and satellite tracking

In NEWS: Radio-tagged white-rumped vulture dies of electrocution in Nilgiris

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11.Credit Rating Agency(CRA)

Q.Consider the following statements regarding Credit Rating Agencies (CRAs) in India:

1. All Credit Rating Agencies in India are exclusively regulated by the Securities and Exchange Board of India (SEBI).
2. Credit Rating Agencies are government-owned statutory bodies established under the SEBI Act.
3. A credit rating is a forward-looking opinion about a borrower's ability to repay their debts in the future.

Which of the statements given above is/are correct?

(a) 3 only
(b) 1 and 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 3 only

Explanation

Statement 1:.While Credit Rating Agencies (CRAs) that rate securities and debt instruments are regulated by SEBI under the SEBI (Credit Rating Agencies) Regulations, 1999, they are not exclusively regulated by SEBI. Depending on the sector, other regulators such as the RBI, IRDAI, and PFRDA may prescribe norms regarding the use of credit ratings. **Hence, Statement 1 is incorrect.**

Statement 2: The statement is incorrect. Credit Rating Agencies are not government-owned statutory bodies. They are private entities registered with SEBI that assess the creditworthiness of borrowers and debt instruments. **Hence, Statement 2 is incorrect.**

Statement 3: The statement is correct. A credit rating is a forward-looking opinion regarding the ability and willingness of a borrower to meet its debt obligations on time. It indicates the probability of default and helps investors assess credit risk. **Hence, Statement 3 is correct.**

Value Addition:

Credit Rating

- An assessment of the creditworthiness of a borrower, company, government, or debt instrument.
- Expresses the likelihood of timely repayment of debt obligations.
- It is an opinion, not a guarantee of repayment.

Regulatory Framework

- Regulated by Securities and Exchange Board of India under the SEBI (Credit Rating Agencies) Regulations, 1999.

Major Credit Rating Agencies in India

- CRISIL (India's first CRA, established in 1987)
- ICRA
- CARE Ratings
- India Ratings and Research
- Acuite Ratings

Rating Scale

Rating	Meaning
AAA	Highest degree of safety
AA	High degree of safety
A	Adequate safety
BBB	Moderate safety (Investment Grade)
BB and below	Speculative/Non-Investment Grade

- SEBI regulates CRAs, but ratings are widely used by banks, insurers, mutual funds, and other financial institutions.
- Sovereign ratings for countries are issued by global agencies such as Moody's, S&P Global Ratings, and Fitch Ratings.

In NEWS:What are India's problems with most credit rating agencies?

12.73rd Constitutional Amendment Act ,1992

Q.Consider the following statements regarding the 73rd Constitutional Amendment Act, 1992:

1. The establishment of a three-tier Panchayati Raj structure—at the village, intermediate, and district levels—is strictly mandatory for all States across India without exception.
2. A person who has attained the age of 22 years cannot be disqualified from being a member of a Panchayat solely on the ground that they have not reached the age of 25 years.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2

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(d) Neither 1 nor 2

Answer: (b) 2 only

Explanation

Statement 1: The establishment of a three-tier Panchayati Raj structure is not strictly mandatory for all States without exception. While Article 243B mandates a three-tier system (village, intermediate, and district levels), it provides a specific exemption: States with a total population not exceeding 20 lakhs have the option to skip the creation of the Panchayat at the intermediate level. **Hence, Statement 1 is incorrect.**

Statement 2: According to Article 243F of the Constitution, a person cannot be disqualified from being chosen as, or for being, a member of a Panchayat on the ground that they have not attained the age of 25 years, provided they are at least 21 years of age. Therefore, a 22-year-old individual meets the minimum age requirement and cannot be disqualified on these specific grounds**. Hence, Statement 2 is correct.**

Value Addition:

Key Institutional Framework

- Part and Schedule: The Act added a new Part IX to the Constitution titled "The Panchayats" and introduced the Eleventh Schedule, which contains 29 functional items placed under the purview of Panchayats.
- Gram Sabha (Article 243A): Envisaged as the foundation of the entire system, it is a village assembly consisting of all registered voters within the area of the village panchayat.
- Exempted Regions (Article 243M): The provisions of this Act do not automatically apply to the states of Nagaland, Meghalaya, and Mizoram, nor do they apply to the Scheduled Areas and Tribal Areas referred to in Article 244.

Compulsory vs. Voluntary Provisions

Compulsory Provisions (Mandatory)	**Voluntary Provisions (Discretionary)**
Organization of a Gram Sabha in a village or group of villages.	Granting representation to MPs and MLAs in the Panchayats at different levels.
Direct elections of members at the village, intermediate, and district levels.	Providing reservations of seats or offices for backward classes (OBCs).
Mandatory 1/3rd reservation of seats for women (both in membership and chairpersons).	Devolution of powers to prepare plans for economic development and social justice.
Fixing a tenure of 5 years and ensuring fresh elections within 6 months of dissolution.	Granting financial powers to Panchayats to levy, collect, and appropriate local taxes.
Setting up a State Finance Commission and a State Election Commission.	Giving Panchayats autonomy over the 29 matters listed in the 11th Schedule.

1. Types of Panchayat Funds (Article 243H)

The financial system of Panchayats relies on four main streams to prevent them from becoming completely dependent on state handouts:

- Internal Resource Generation: Power to levy, collect, and appropriate local duties, taxes, tolls, and fees (e.g., house tax, market fees).
- Consolidated Fund of the State: Grants-in-aid assigned directly out of the state's revenues.
- Devolution via State Finance Commission: A fixed percentage share of the state government's net tax collection.
- Central Infrastructure Schemes: Direct fund transfers from the Union Government for centrally sponsored schemes (like MGNREGA).

2. The Strategic Role of the State Election Commission (Article 243K)

- Complete Independence: The State Election Commissioner is appointed by the Governor. To safeguard their autonomy, they cannot be removed from office except in the same manner and on the exact same grounds as a Judge of a High Court.
- Conditions of Service: The conditions of service and tenure cannot be varied to their disadvantage after appointment.

The PESA Act, 1996 (Extension to Scheduled Areas)

Because the 73rd Amendment did not automatically apply to regular tribal zones under Article 243M, Parliament enacted the Provisions of the Panchayati Raj (Extension to the Scheduled Areas) Act, 1996 to extend Part IX to Fifth Schedule areas across 10 states.

Key Structural Highlights of PESA:

- Legal Alignment: State legislations on Panchayats must be customized to stay in harmony with

traditional tribal customs, practices, and community resource management models.

- Gram Sabha Supremacy: Unlike regular rural areas where the Gram Panchayat holds executive power, PESA makes the Gram Sabha (the entire village assembly) the absolute powerhouse.
- Mandatory Approvals: The Gram Sabha's prior recommendation is legally mandatory for:
  - Granting prospecting licenses or mining leases for minor minerals.
  - Land acquisition for development projects and rehabilitating displaced tribal communities.
  - Managing minor water bodies and local water security designs.

Absolute Enforcement Powers Given to PESA Gram Sabhas:

1. The absolute power to enforce prohibition or to regulate and restrict the sale and consumption of any intoxicant.
2. The ownership rights over Minor Forest Produce (MFP).
3. The power to prevent land alienation in the Scheduled Areas and to take restoration actions if tribal land is illegally taken.
4. The authority to control local plans, resources, and budgets for tribal sub-plans, including controlling local money lending practices.

IN NEWS:Yes and no - Governments must respect the decision-making of gram sabhas

13.Petroglyphs

Q.The term "Petroglyphs", recently seen in the news, refers to:

(a) Ancient paintings made on cave walls using natural pigments
(b) Symbolic images engraved or carved on rock surfaces by removing part of the rock
(c) Megalithic stone monuments used for burial purposes
(d) Prehistoric inscriptions written on clay tablets

Answer: (b) Symbolic images engraved or carved on rock surfaces by removing part of the rock

Explanation

Option (a): Paintings made on cave walls are generally known as rock paintings or petrographs, not petroglyphs. **Hence, option (a) is incorrect.**

Option (b): Petroglyphs are symbolic images carved, pecked, incised, scratched, polished, or engraved on rock surfaces by removing part of the rock. They represent one of the earliest forms of human artistic expression and communication**. Hence, option (b) is correct.**

Option (c): Megalithic monuments are large stone structures associated with burial and commemorative practices. They are different from petroglyphs. **Hence, option (c) is incorrect.**

Option (d): Petroglyphs are not inscriptions on clay tablets. Such records are associated with ancient writing systems like Mesopotamian cuneiform. **Hence, option (d) is incorrect.**

Value Addition

Petroglyphs

Feature	Details
Meaning	Rock engravings or carvings
Origin of Term	Greek words *Petros* (stone) + *Glyphein* (to carve)
Method	Pecking, incising, scratching, abrading, drilling, polishing
Period	Mainly prehistoric
Found In	All continents except Antarctica
Significance	Early art, communication, territorial markers, astronomical symbols

Petroglyphs vs Petrographs

Petroglyphs	Petrographs
Carved or engraved on rock	Painted or drawn on rock
Material removed from rock surface	Pigments applied on rock surface
More durable	More susceptible to weathering

Examples in India

- Edakkal Caves
- Konkan Petroglyphs of Maharashtra
- Leh Petroglyph Sites, Ladakh

IN NEWS: What are petroglyphs and why are they in the news?

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14.Strawberry Moon

Q.With reference to astronomical phenomena associated with the Moon, consider the following statements:

1. A Blue Moon refers to the full moon that occurs closest to the Earth in its orbit.
2. The Strawberry Moon derives its name from the reddish colour it displays during June.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (d) Neither 1 nor 2

Explanation

Statement 1:A Blue Moon does not refer to the full moon closest to Earth. The commonly accepted definition is the second full moon occurring within a single calendar month. A full moon closest to Earth is known as a Supermoon, whereas a full moon occurring near apogee is called a Micromoon. **Hence, Statement 1 is incorrect.**

Statement 2: The Strawberry Moon does not derive its name from its colour. The term originated from the Algonquian Native American tribes, who associated June's full moon with the brief season of wild strawberry harvesting. Although the Moon may sometimes appear orange or golden near the horizon due to atmospheric scattering, this is unrelated to the name. **Hence, Statement 2 is incorrect.**

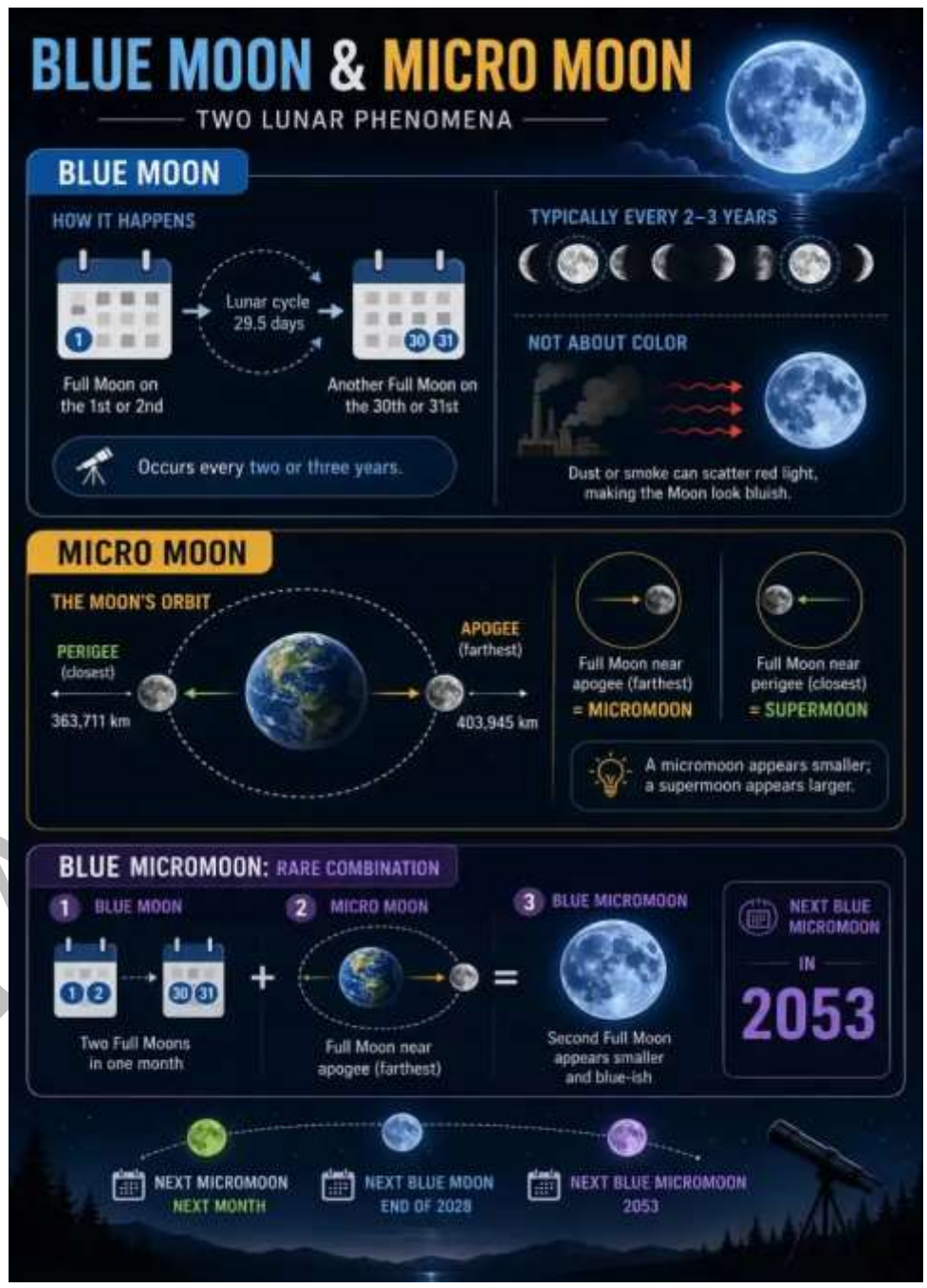
Value Addition

Moon's Orbit

Position	Distance from Earth (Approx.)
Perigee	363,711 km
Apogee	403,945 km

Full Moon Terminology

Term	Meaning
Strawberry Moon	June full moon named after strawberry harvesting season
Blue Moon	Second full moon in a calendar month
Supermoon	Full moon occurring near perigee (closest point to Earth)
Micromoon	Full moon occurring near apogee (farthest point from Earth)



Why does the Strawberry Moon appear larger?

- Due to the Moon Illusion, an optical illusion occurring when the Moon is near the horizon.
- The Moon is not actually larger; it only appears so when viewed alongside trees, buildings, and other terrestrial objects.

Related Facts

- Lunar eclipses occur only during a Full Moon.
- Solar eclipses occur only during a New Moon.
- The next Blue Micromoon (Blue Moon + Micromoon together) is expected in 2053.

Types of Lunar Eclipses

(i) Total lunar eclipses: The Moon moves into the inner part of Earth's shadow, or the umbra. Some of the sunlight passing through Earth's atmosphere reaches the Moon's surface, lighting it dimly. Colors with shorter wavelengths — the blues and violets —

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scatter more easily than colors with longer wavelengths, like red and orange.

(ii) Partial lunar eclipse: An imperfect alignment of Sun, Earth and Moon results in the Moon passing through only part of Earth's umbra. The shadow grows and then recedes without ever entirely covering the Moon.

(iii) Penumbral eclipse: The Moon travels through Earth's penumbra, or the faint outer part of its shadow. The Moon dims so slightly that it can be difficult to notice.

In NEWS: What makes Strawberry Moon different from Blue Moon and Micromoon?

15.World Bank and Climate Change Action Plan

Q.Consider the following statements:

1. The World Bank officially began operations in 1946 and initially focused on post-World War II reconstruction in Europe.
2. The International Development Association (IDA) raises capital from global financial markets to finance development projects.
3. The Climate Change Action Plan (CCAP) was launched in 2020

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (c) 1 and 3 only

Explanation

Statement 1: The World Bank originated from the Bretton Woods Conference (1944) and officially commenced operations in 1946. Its initial mandate was to assist in the reconstruction of Europe after World War II before expanding its focus to development and poverty reduction.**Hence, Statement 1 is correct.**

Statement 2: The International Development Association (IDA) does not raise capital from international financial markets. It provides concessional loans and grants to the poorest countries and is largely funded through contributions from donor countries.

The institution that raises capital from global financial markets is the International Bank for Reconstruction and Development (IBRD).**Hence, Statement 2 is incorrect.**

Statement 3: The Climate Change Action Plan (CCAP) was launched by the World Bank Group in 2020 for a five-year period ending in 2026 to integrate climate action into development financing 2021–2025 implementation period, launched in 2020..**Hence, Statement 3 is correct.**

Value Addition:

Headquarters

World Bank Headquarters – Washington, D.C.

Established

- Founded during the Bretton Woods Conference (1944)
- Began operations in 1946

World Bank Group Institutions

1. International Bank for Reconstruction and Development(IBRD)-Loans to middle-income and creditworthy low-income countries.
2. International Development Association(IDA)-Concessional loans and grants to the poorest countries.
3. International Finance Corporation(IFC)-Private sector financing.
4. Multilateral Investment Guarantee Agency(MIGA)-Political risk insurance.
5. International Centre for Settlement of Investment Disputes(ICSID)-Settlement of investment disputes

Institution	Main Function
IBRD	Raises money from capital markets
IDA	Funded by donor contributions
IFC	Supports private sector
MIGA	Investment guarantees
ICSID	Arbitration of investment disputes

Recent Development (2026):

- The World Bank announced that it would retire the 45% climate-finance target.
- However, it stated that it would continue supporting countries in meeting their Nationally Determined Contributions (NDCs) under the Paris Agreement.
- The Bank will continue tracking:
  - Net greenhouse gas emissions
  - Beneficiaries with enhanced climate resilience
  - Climate co-benefits reporting

In NEWS:World Bank may scrap funding target for climate projects

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16.Carbon Capture, Utilization and Storage(CCUS)

Q.Consider the following statements regarding Carbon Capture, Utilisation and Storage (CCUS):

1. Carbon Capture, Utilisation and Storage (CCUS) aims to capture carbon dioxide emissions and prevent them from re-entering the atmosphere through storage or utilisation.
2. Direct Air Capture (DAC) is one of the techniques used under CCUS to capture carbon dioxide directly from the atmosphere.
3. Under CCUS, unutilized carbon dioxide can be permanently stored in geological formations such as saline aquifers and basalt formations.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation

Statement 1: Carbon Capture, Utilisation and Storage (CCUS) refers to a group of technologies that capture carbon dioxide (CO_2) emissions from large stationary sources such as power plants, steel plants, cement industries, and refineries. The captured CO_2 is then either utilised in industrial processes or stored permanently to prevent its release into the atmosphere.**Hence, Statement 1 is correct.**

Statement 2: Direct Air Capture (DAC) is an advanced carbon-capture technology that removes carbon dioxide directly from ambient air rather than from industrial exhaust streams. It is recognized as one of the capture methods under the CCUS framework, though it is currently expensive and at an early stage of deployment.**Hence, Statement 2 is correct.**

Statement 3: Captured CO_2 that is not utilised can be stored permanently in geological formations. Common storage options include saline aquifers, depleted oil and gas reservoirs, and basalt rock formations. Such storage is intended to ensure long-term sequestration of carbon dioxide. **Hence, Statement 3 is correct.**

Value Addition

India's CCUS Target

- As per NITI Aayog's report on CCUS, India aims to achieve 750 Million Tonnes Per Annum (MTPA) of CO_2 capture, utilisation, and storage.

Significance of CCUS

- Decarbonisation of hard-to-abate sectors such as steel, cement, fertilisers, petrochemicals, and oil & gas.
- Supports production of Blue Hydrogen.
- Helps achieve India's Net Zero target by 2070.
- Promotes a Circular Carbon Economy through conversion of captured CO_2 into useful products.

Budget 2026–27

- The Union Budget 2026–27 announced an allocation of ₹20,000 crore over five years to support the development and deployment of CCUS technologies.

International Example

- The CarbFix Project (Iceland) injects captured CO_2 into basalt formations, where it mineralises into solid rock, providing a permanent storage solution.

IN NEWS: A unified policy architecture for India's energy future

17.Interplanetary Coronal Mass Ejections(ICME)

Q.Consider the following statements regarding Interplanetary Coronal Mass Ejections (ICMEs):

1. Interplanetary Coronal Mass Ejections (ICMEs) are massive eruptions of magnetized plasma released from the Sun's outer atmosphere.
2. When Earth-directed ICMEs interact with Earth's magnetic field, they can trigger geomagnetic storms.
3. ICMEs are most frequently generated during the minimum phase of the Sun's 11-year activity cycle.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation

Statement 1: Interplanetary Coronal Mass Ejections (ICMEs) are the interplanetary counterparts of Coronal Mass Ejections (CMEs). They are massive clouds of magnetized plasma expelled from the Sun's corona (outer atmosphere) and travel through interplanetary space.**Hence, Statement 1 is correct.**

Statement 2: When Earth-directed ICMEs encounter Earth's magnetosphere, they can disturb the geomagnetic field and generate geomagnetic storms. These storms may affect satellites, navigation

systems, communication networks, and power infrastructure.**Hence, Statement 2 is correct.**

Statement 3: The Sun follows an approximately 11-year solar activity cycle consisting of solar minimum and solar maximum phases. The occurrence of CMEs and ICMEs increases significantly during the solar maximum, when solar activity is at its peak. Therefore, ICMEs are not most frequently generated during solar minimum.**Hence, Statement 3 is incorrect.**

Value Addition

ICMEs

- ICMEs are the interplanetary manifestations of Coronal Mass Ejections (CMEs).
- They consist of:
  - Magnetized plasma
  - Charged particles
  - Embedded magnetic fields
- They propagate through the heliosphere and may take 1–5 days to reach Earth.

ICMEs Importance

- They are among the most important drivers of Space Weather.
- They can produce:
  - Severe geomagnetic storms
  - Radiation storms
  - Auroral displays (Northern and Southern Lights)

Impacts on Earth

ICMEs can adversely affect:

- Satellite operations
- GPS and navigation systems
- Radio communications
- Aviation routes, especially polar flights
- Power transmission grids
- Spacecraft electronics
- Astronaut safety

Geomagnetic Storm

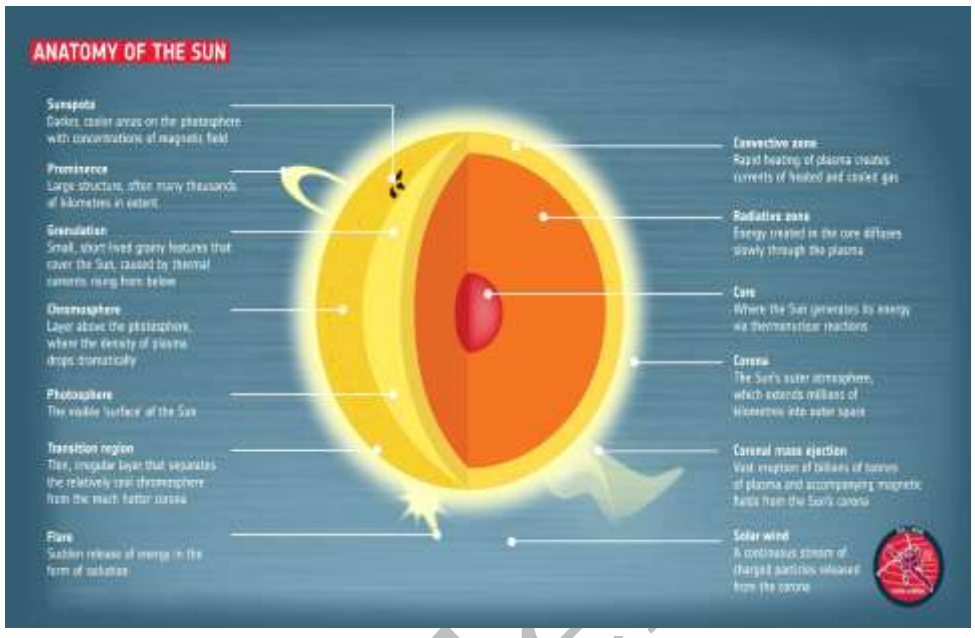
A geomagnetic storm occurs when a strong ICME interacts with Earth's magnetosphere, causing rapid variations in Earth's magnetic field.

Astronomical Unit (AU)

- 1 AU = Average Earth–Sun distance
- Approximately 149.6 million km
- Most ICME observations are studied at 1 AU, where Earth is located.

Solar Cycles

- Solar activity varies in an approximately 11-year cycle.
- Solar Maximum: High sunspot activity, more CMEs and ICMEs.
- Solar Minimum: Reduced solar activity and fewer eruptions.



Recent Indian Study

- Conducted by researchers from the Indian Institute of Astrophysics.
- Analysed ICME behaviour from 1995–2024.
- Covered:
  - Solar Cycle 23
  - Solar Cycle 24
  - Rising phase of Solar Cycle 25
- Found that nearly 45% of magnetic ejecta exhibited heating signatures at 1 AU, challenging the earlier assumption that ICMEs only cool as they expand.

Facts

Term	Description
CME	Coronal Mass Ejection originating from the solar corona
ICME	CME observed in interplanetary space
Magnetosphere	Region dominated by Earth's magnetic field
Aurora	Visible light display caused by charged particles interacting with Earth's atmosphere
Space Weather	Conditions in space influenced by solar activity affecting Earth and technological systems
Solar Wind	Continuous stream of charged particles from the Sun

IN NEWS:Thermal signatures of three decades of solar storms at Earth reveal new clues for space weather forecasting

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18.SHE-LEAPS

Q.Consider the following statements regarding the SHE-LEAPS platform:

1. It is a digital platform designed to promote and track both farm and non-farm rural enterprises led by Self-Help Group (SHG) women.
2. The platform is developed by the National Bank for Agriculture and Rural Development (NABARD) to support the Lakhpati Didi initiative.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (a) 1 only

Explanation

Statement 1:SHE-LEAPS (Self Help Group Enterprise Livelihoods Acceleration Platform) is a digital platform designed to identify, promote, monitor, and scale both farm and non-farm enterprises run by women members of Self-Help Groups (SHGs). It aims to facilitate enterprise growth, improve market access, and support sustainable livelihoods for rural women. **Hence, Statement 1 is correct.**

Statement 2: The SHE-LEAPS platform has been developed by the Ministry of Rural Development (MoRD) under the Deendayal Antyodaya Yojana–National Rural Livelihoods Mission (DAY-NRLM). It is not developed by NABARD. Developed by Digital India Corporation (DIC) under the LokOS framework, supporting the Ministry of Rural Development's Lakhpati Didi targets. While it supports the broader objective of creating more Lakhpati Didis, it is not a NABARD initiative. **Hence, Statement 2 is incorrect.**

Value Addition

SHE-LEAPS

Full Form: Self Help Group Enterprise Livelihoods Acceleration Platform for SHGs

Nodal Ministry

- Ministry of Rural Development (MoRD)

Implemented Under

- Deendayal Antyodaya Yojana – National Rural Livelihoods Mission

Objectives

- Promote women-led rural entrepreneurship.
- Track growth of SHG enterprises.
- Facilitate market linkages.
- Improve access to credit and financial services.
- Support transition from subsistence livelihoods to sustainable enterprises.

Key Features

- Digital profiling of SHG enterprises.
- Enterprise performance tracking.
- Data-driven policy support.
- Monitoring of farm and non-farm livelihood activities.
- Identification of high-growth enterprises for targeted support.

Lakhpati Didi Initiative

- Aims to enable rural SHG women to earn at least ₹1 lakh per year through sustainable livelihood activities.
- Implemented under DAY-NRLM.
- Focuses on skill development, entrepreneurship, financial inclusion, and market access.

DAY-NRLM

Launch: 2011

Ministry: Ministry of Rural Development

Objective: Reduce rural poverty through promotion of self-employment and organization of poor households into SHGs.

In NEWS: SHE-LEAPS: Digital Empowerment for Lakhpati Didis

19.GAGAN

Q. Consider the following statements regarding the GPS Aided GEO Augmented Navigation (GAGAN) system:

1. It is an independent regional navigation satellite system developed by ISRO to completely replace the American Global Positioning System (GPS) over Indian airspace.
2. It is the first Satellite-Based Augmentation System (SBAS) globally to be certified for operations within the equatorial region.
3. The space-based payload segment of GAGAN is hosted on India's geostationary communication satellites, such as GSAT-8 and GSAT-10.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Correct Answer: (b) 2 and 3 only

Explanation

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Statement 1:GAGAN is not an independent regional navigation satellite system, nor does it replace GPS. Instead, it is a Satellite-Based Augmentation System (SBAS) that operates on top of the existing GPS network. It functions by monitoring GPS signals, calculating real-time atmospheric/clock errors, and broadcasting correction messages to improve accuracy and integrity. India's completely independent positioning system is NavIC (Navigation with Indian Constellation)**. Hence, Statement 1 is incorrect.**

Statement 2:GAGAN holds a significant global distinction as the first SBAS certified to handle the highly volatile ionospheric disturbances characteristic of the equatorial region. This makes India one of a select few nations alongside the US (WAAS), Europe (EGNOS), and Japan (MSAS) to operationalize specialized satellite-driven aviation landing assistance. **Hence, Statement 2 is correct.**

Statement 3: The space segment of GAGAN relies on specialized transponder payloads carried aboard India's indigenous geostationary communication satellites, specifically GSAT-8, GSAT-10, and GSAT-15, which broadcast the corrected navigation coordinates back to the aircraft receivers. **Hence, Statement 3 is correct.**

Value Addition:

1. The Three Structural Segments of GAGAN

The system continuously corrects GPS data using a strictly coordinated pipeline:

- The Ground Segment (INRES): Consists of 15 precisely surveyed Indian Reference Stations spread across the country that pick up standard GPS signals and detect any subtle signal delays or positional drift.
- The Master Control (INMCC): Comprises 2 processing centers that take data from INRES to compute real-time ionospheric corrections and build safety integrity updates.
- The Uplink Segment (INLUS): Utilizes 3 Indian Land Uplink Stations to beam these final correction packages directly up to the space segment.

2. GAGAN vs. NavIC (Crucial UPSC Distinction)

Parameter	GAGAN	NavIC
System Classification	Satellite-Based Augmentation System (SBAS).	Independent Satellite Navigation System.
Core Function	Corrects, refines, and enhances existing GPS signals.	Provides direct positioning, timing, and standalone navigation.
Primary Beneficiary	Civil Aviation (enables precise satellite-guided runway approaches).	Multi-sectoral commercial, civilian, and strategic military defense usage.
Coverage Envelope	Indian Flight Information Region (FIR) and neighboring regional airspace.	Sovereign territory of India plus an extended region of up to 1,500 km beyond its borders.

In NEWS:GAGAN: Navigating India's Skies with Precision

20.CITES Project

Q.Consider the following statements regarding the Centralised IT-Enabled System (CITES) Project:

1. It provides a single centralized national database architecture for the Employees' Provident Fund Organisation (EPFO).
2. It has been designed by the Centre for Development of Advanced Computing (C-DAC).

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Explanation

Statement 1: The Centralised IT-Enabled System (CITES) Project is an EPFO initiative aimed at modernising service delivery through automation and rule-based processing. A key feature of the project is the creation of a single centralized national

database, replacing the earlier fragmented database architecture.**Hence, Statement 1 is correct.**

Statement 2: The CITES platform has been designed by the Centre for Development of Advanced Computing (C-DAC), India's premier R&D organization in advanced computing and information technology.**Hence, Statement 2 is correct.**

Value Addition

Employees' Provident Fund Organisation (EPFO)

- Statutory body under the Ministry of Labour and Employment.
- Administers:
  - Employees' Provident Fund (EPF)
  - Employees' Pension Scheme (EPS), 1995
  - Employees' Deposit Linked Insurance Scheme (EDLI)

Universal Account Number (UAN)

- A unique 12-digit number allotted to every EPF member.
- Remains unchanged throughout an employee's career.
- Links multiple Member IDs under a single account.

Centre for Development of Advanced Computing (C-DAC)

- Scientific society under the Ministry of Electronics and Information Technology.
- Established in 1988.
- Known for development of India's PARAM supercomputers.

Major Benefits of CITES

- Nationwide portability of EPFO services.
- Faster claim settlement.
- Simplified PF transfers.
- Improved service history maintenance.
- Reduced paperwork and manual intervention.
- Online KYC updation.

Future Roadmap

- Foundation for direct UPI-based EPF withdrawals.
- Integration of pension processing and claim settlements through a centralized platform.

In NEWS: What is Centre's new EPFO upgrade

21.National Rainfed Area Authority

Q.Consider the following statements regarding the National Rainfed Area Authority (NRAA):

1. It was established as an attached advisory body under the Ministry of Rural Development to address structural challenges in rainfed agriculture.
2. The Governing Board of the Authority is chaired by the Union Minister of Rural Development and co-chaired by the Union Agriculture Minister.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (d) Neither 1 nor 2

Explanation

Statement 1: The National Rainfed Area Authority (NRAA) was established in 2006 as an advisory body under the Government of India to address challenges in rainfed agriculture and presently functions under the Department of Agriculture and Farmers Welfare **.** **Hence, Statement 1 is incorrect.**

Statement 2: The Governing Board of NRAA is chaired by the Union Agriculture Minister and co-chaired by the Union Minister of Rural Development, not vice versa. **Hence, Statement 2 is incorrect.**

Value Addition

National Rainfed Area Authority (NRAA)

Feature	**Details**
Establishment	2006
Ministry	Ministry of Agriculture and Farmers' Welfare
Headquarters	New Delhi
Nature	Advisory body
Focus Area	Sustainable development of rainfed regions

Mandate of NRAA

- Promote sustainable agriculture in rainfed areas.
- Improve productivity and livelihood security of farmers.
- Encourage integrated watershed development.
- Promote climate-resilient and resource-efficient farming systems.
- Support holistic development through farming and livelihood-based approaches.

Governance Structure

Governing Board

- Provides policy direction and coordination.

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- Chaired by the Union Agriculture Minister.
- Co-chaired by the Union Minister of Rural Development.

Executive Committee

- Headed by a full-time Chief Executive Officer (CEO).
- Consists of technical experts and representatives of stakeholder Ministries.
- Supported by five full-time technical experts.

Recent Development

Recently, the NRAA achieved a significant milestone in the conservation of wild rice (Oryza rufipogon) genetic resources in Sonitpur district of Assam, highlighting its role in promoting climate-resilient agriculture and biodiversity conservation.

In NEWS:NRAA-Funded Wild Rice Conservation Project Secures Major Milestone in Assam

22.Academic Bank of Credit

Q.Consider the following statements regarding the Academic Bank of Credit (ABC) and Automated Permanent Academic Account Registry (APAAR):

1. The Academic Bank of Credit (ABC) is a digital platform regulated by the University Grants Commission (UGC) for storing, transferring, and redeeming academic credits earned by students.
2. Students registered under the ABC system are provided with a unique APAAR ID linked to Aadhaar and DigiLocker for maintaining lifelong academic records.
3. Academic credits deposited in the Academic Bank of Credit remain valid indefinitely and can be redeemed multiple times across different educational programmes.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation

Statement 1: The statement is correct. The Academic Bank of Credit (ABC) is a digital platform established by the Ministry of Education and regulated by the University Grants Commission (UGC) for storing, managing, transferring, and redeeming academic credits earned by students from recognised institutions. **Hence, Statement 1 is correct.**

Statement 2: The statement is correct. Students registering on the ABC portal receive a unique ABC ID or APAAR ID linked to Aadhaar and DigiLocker, enabling secure maintenance and sharing of academic records. **Hence, Statement 2 is correct.**

Statement 3: The statement is incorrect. Credits deposited in the Academic Bank of Credit are generally valid for a maximum period of seven years or as prescribed by the relevant discipline. Further, once credits are redeemed, they cannot be reused**.** **Hence, Statement 3 is incorrect.**

Value Addition

Academic Bank of Credit (ABC)

Feature	Details
Launch Framework	National Education Policy (NEP) 2020
Regulator	University Grants Commission (UGC)
Purpose	Digital storage, transfer and redemption of academic credits
Eligibility	Students enrolled in recognised Higher Education Institutions (HEIs)
Credit Validity	Credits are not permanently valid and may expire according to applicable academic regulations; redeemed credits cannot ordinarily be reused.

APAAR

Full Form: Automated Permanent Academic Account Registry

- Provides a unique lifelong academic identity to students.
- Linked with Aadhaar and DigiLocker.
- Facilitates seamless access to academic records across institutions.
- Supports digital verification and academic mobility.

Key Features of ABC

- Academic Mobility: Transfer credits across institutions and programmes.
- Multiple Entry and Exit (MEE): Supports flexible learning pathways under NEP 2020.
- SWAYAM Integration: Up to 40% credits may be earned through SWAYAM courses.
- National Credit Framework (NCrF): Covers academic, vocational, and experiential learning.

- National Academic Depository (NAD): Acts as the secure repository for academic awards and records.

IN NEWS:Academic Bank of Credits and APAAR

23.National Green Hydrogen Mission(NGHM)

Q.Consider the following statements regarding the National Green Hydrogen Mission (NGHM):

1. Green hydrogen is produced through electrolysis of water using renewable energy sources.
2. The mission is being implemented by the Ministry of Power.
3. It seeks to promote indigenous manufacturing capabilities in the hydrogen value chain.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (b) 1 and 3 only

Explanation

Statement 1: Green hydrogen is produced through the process of electrolysis, where water (H_2O) is split into hydrogen and oxygen using electricity generated from renewable energy sources such as solar, wind, or hydropower. Since the electricity used comes from renewable sources, the process results in negligible greenhouse gas emissions.**Hence, Statement 1 is correct.**

Statement 2: The National Green Hydrogen Mission is being implemented by the Ministry of New and Renewable Energy (MNRE), not by the Ministry of Power.The Union Cabinet approved the Mission in January 2023 with a financial outlay of ₹19,744 crore.**Hence, Statement 2 is incorrect.**

Statement 3: One of the major objectives of the Mission is to develop indigenous manufacturing capacity across the green hydrogen value chain, particularly in electrolyser manufacturing, storage systems, transportation infrastructure, and related technologies. This is intended to reduce import dependence and support the vision of Aatmanirbhar Bharat.**Hence, Statement 3 is correct.**

Value Addition

Green Hydrogen

Hydrogen produced through the electrolysis of water using electricity generated entirely from renewable energy sources is known as Green Hydrogen.

Electrolysis Process

Water + Renewable Electricity → Hydrogen + Oxygen

Hydrogen generated can be used as:

- Industrial feedstock
- Clean fuel
- Energy storage medium
- Transportation fuel

Types of Hydrogen

Type	Production Method	Carbon Emissions
Grey Hydrogen	Natural gas/coal without carbon capture	High
Blue Hydrogen	Fossil fuels with Carbon Capture Utilisation and Storage (CCUS)	Moderate
Green Hydrogen	Electrolysis using renewable energy	Near Zero
Brown/Black Hydrogen	Coal gasification	Very High

National Green Hydrogen Mission (NGHM)

Launch

- Approved by Union Cabinet: January 2023
- Nodal Ministry: Ministry of New and Renewable Energy (MNRE)
- Total Outlay: ₹19,744 crore

Vision

To make India a global hub for:

- Production
- Utilisation
- Export of Green Hydrogen and its derivatives.

Major Objectives

Energy Security

- Reduce dependence on imported fossil fuels.
- Improve energy self-reliance.

Decarbonisation

Support India's:

- Net Zero target by 2070
- Nationally Determined Contributions (NDCs)

Industrial Competitiveness

Decarbonise hard-to-abate sectors such as:

- Fertilizers
- Refineries
- Steel
- Shipping
- Mobility

Indigenous Manufacturing

Develop domestic capacity for:

- Electrolysers
- Fuel cells
- Storage systems
- Hydrogen infrastructure

Strategic Interventions for Green Hydrogen Transition (SIGHT)

The flagship component of NGHM.

SIGHT provides incentives for:

1. Domestic manufacturing of electrolysers.
2. Production of Green Hydrogen.

Purpose:

- Lower production costs.
- Encourage private investments.
- Create global-scale manufacturing capabilities.

Green Hydrogen Hubs

The Mission envisages development of dedicated Green Hydrogen Hubs where:

- Renewable energy availability is high.
- Production infrastructure can be concentrated.
- Industrial demand exists. These hubs will facilitate large-scale production, storage, and utilization.

Expected Outcomes by 2030

Target	Expected Outcome
Green Hydrogen Production	At least 5 Million Metric Tonnes (MMT) annually
Renewable Energy Addition	About 125 GW
Investments	More than ₹8 lakh crore
Employment	More than 6 lakh jobs
Fossil Fuel Import Reduction	Over ₹1 lakh crore
Emission Reduction	Nearly 50 MMT annually

Implementation Phases

Phase I (2022–23 to 2025–26)

Focus on:

- Creating demand and supply.
- Electrolyser manufacturing.
- Deployment in:
  - Refineries
  - Fertilizer sector
  - City gas distribution

Pilot projects in:

- Steel
- Mobility
- Shipping

Phase II (2026–27 to 2029–30)

Focus on:

- Commercial-scale deployment.
- Expansion into:
  - Steel
  - Railways
  - Shipping
  - Aviation
  - Heavy transport

Green hydrogen is expected to become cost-competitive with fossil fuels.

Green Hydrogen Importance for India

- India currently consumes around 6 million tonnes of hydrogen annually.
- Demand may rise to 28 million tonnes by 2050.
- Most hydrogen today is produced from fossil fuels (grey hydrogen).
- Green hydrogen can significantly reduce:
  - Carbon emissions
  - Import dependence
  - Energy insecurity

IN NEWS:India Advances Global Green Hydrogen Leadership under National Green Hydrogen Mission

24.Gulf of Aden

Q.Which of the following countries share a coastline with the Gulf of Aden?

1. Yemen
2. Oman
3. Djibouti
4. Somalia

Select the correct answer using the code given below:

(a) 1, 2, and 4 only
(b) 1, 3, and 4 only
(c) 2 and 3 only
(d) 1, 2, 3, and 4

Answer: (b) 1, 3, and 4 only

Explanation

Statement 1: Yemen

Yemen forms the northern coastline of the Gulf of Aden. The gulf lies between Yemen on the Arabian Peninsula and the Horn of Africa. **Hence, Statement 1 is correct.**

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Statement 2: Oman

Oman does not border the Gulf of Aden. Oman lies east of Yemen and has coastlines along the Arabian Sea, the Gulf of Oman, and the Strait of Hormuz. **Hence, Statement 2 is incorrect.**

Statement 3: Djibouti

Djibouti is located at the western entrance of the Gulf of Aden near the strategic Bab-el-Mandeb Strait, which connects the Gulf of Aden to the Red Sea. **Hence, Statement 3 is correct.**

Statement 4: Somalia

Somalia forms a major part of the southern coastline of the Gulf of Aden and occupies a strategic position along important international shipping routes. **Hence, Statement 4 is correct.**

Value Addition

Gulf of Aden

Feature	Details
Type	Deep-water gulf in the northwestern Indian Ocean
Location	Between Yemen (north) and Somalia & Djibouti (south)
Connects	Arabian Sea with the Red Sea
Strategic Strait	Bab-el-Mandeb Strait
Importance	Major international maritime trade route

Countries Bordering the Gulf of Aden

- Yemen
- Djibouti
- Somalia



Bab-el-Mandeb Strait

- Connects the Gulf of Aden with the Red Sea.
- Separates Yemen from Djibouti and Eritrea.
- One of the world's most important maritime chokepoints.
- Vital for trade between Europe and Asia through the Suez Canal.

In NEWS:Navy foils piracy attempt in Gulf of Aden; three vessels still in captivity, IMO warns

25.AIR SUVIDHA 2.0

Q.With reference to AIR SUVIDHA 2.0, which one of the following statements best describes its primary objective?

(a) To facilitate immigration clearance through biometric verification.
(b) To strengthen border health surveillance by digitally screening incoming international travelers.
(c) To provide travel insurance coverage to international passengers.
(d) To integrate airline ticketing systems with airport security networks.

Answer: (b)To strengthen border health surveillance by digitally screening incoming international travelers.

Explanation

Option (a): AIR SUVIDHA 2.0 is not designed for immigration clearance through biometric verification. Immigration procedures are handled separately by the Bureau of Immigration. The portal focuses specifically on health screening and disease surveillance of incoming international passengers**. Hence, option (a) is incorrect.**

Option (b): AIR SUVIDHA 2.0 is a fully contactless and paperless health self-declaration portal that strengthens border health surveillance by digitally screening international arriving passengers. It helps identify travelers from high-risk disease transmission areas and enables timely public health interventions**. Hence, option (b) is correct.**

Option (c):AIR SUVIDHA 2.0 does not provide travel insurance coverage or any financial protection to passengers. Its primary function is public health monitoring and disease surveillance. **Hence, option (c) is incorrect.**

Option (d): The portal is not intended to integrate airline ticketing systems with airport security networks. Its purpose is health data collection, risk assessment, and coordination among health and surveillance agencies**. Hence, option (d) is incorrect.**

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Value Addition

AIR SUVIDHA 2.0 Portal

Feature	Details
Launched By	Ministry of Civil Aviation and Delhi International Airport Limited (DIAL)
Nature	Digital health self-declaration and screening portal
Target Group	International arriving passengers
Objective	Strengthen border health surveillance and prevent cross-border transmission of infectious diseases
Mode	Fully contactless and paperless
Submission Window	Up to 24 hours before arrival in India
Verification	QR Code/PDF-based digital validation

Key Features

- Mandatory online Self-Declaration Form (SDF).
- Captures travel history, exposure risks, and symptom details.
- Eliminates paper-based health screening.
- Facilitates seamless airport processing.

Integrated Agencies

The portal shares data in real time with:

- Airport Health Officers (APHOs)
- Bureau of Immigration
- State Surveillance Officers
- Integrated Disease Surveillance Programme (IDSP)

Significance

- Enables rapid contact tracing and quarantine measures.
- Enhances preparedness against global health emergencies.
- Strengthens surveillance at Points of Entry (PoEs).
- Aligns India's border health measures with the World Health Organization International Health Regulations (IHR 2005).

In NEWS: Ministry of Civil Aviation launches AIR SUVIDHA 2.0 portal for Ebola health screening

26. Marine Products Export Development Authority(MPEDA)

Q.With reference to the Marine Products Export Development Authority (MPEDA), consider the following statements:

1. MPEDA was created by merging the Marine Products Export Promotion Council into it.
2. MPEDA is responsible for promoting exports of both marine and agricultural products.
3. It functions under the Ministry of Commerce and Industry.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (c) 1 and 3 only

Explanation

Statement 1:MPEDA was established in 1972 through an Act of Parliament. The erstwhile Marine Products Export Promotion Council, established in 1961, was merged into MPEDA on 24 August 1972**.** **Hence, Statement 1 is correct.**

Statement 2:MPEDA is responsible for the promotion and regulation of exports of marine products only. Promotion of agricultural and processed food exports is undertaken by the Agricultural and Processed Food Products Export Development Authority. **Hence, Statement 2 is incorrect.**

Statement 3:MPEDA functions under the Ministry of Commerce and Industry and serves as the nodal agency for the promotion of marine product exports from India**.** **Hence, Statement 3 is correct.**

Value Addition

Marine Products Export Development Authority (MPEDA)

Feature	Details
Establishment	1972
Nature	Statutory Body
Parent Ministry	Ministry of Commerce and Industry
Headquarters	Kochi, Kerala

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Origin	Formed by merging the Marine Products Export Promotion Council into MPEDA
Objective	Promotion and regulation of marine product exports

Major Functions

- Promotion of marine product exports.
- Registration of exporters of marine products.
- Registration of fishing vessels, processing plants and storage facilities.
- Fixing standards and specifications for marine products meant for export.
- Regulation of marine product exports.
- Development and management of offshore and deep-sea fisheries.

Infrastructure

- Five Quality Control Laboratories at Kochi, Nellore & Bhimavaram, Bhubaneswar and Porbandar.
- Fifteen ELISA Screening Laboratories across maritime states.
- 18 Regional/Sub-Regional/Desk Offices across India.
- Trade Promotion Offices in New Delhi, Tokyo and New York.

IN NEWS: MPEDA Concludes Second National Skill Olympiad on Seafood Value Addition at Seafood Expo Bharat 2026

27.Mini Ratna

Q.With reference to the Miniratna status granted to Central Public Sector Enterprises (CPSEs), consider the following statements:

1. A CPSE must have earned profits continuously for the last three years and possess a positive net worth to be eligible for Miniratna Category-II status.
2. Miniratna CPSEs can be granted status even if they depend upon budgetary support from the Government.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (a) 1 only

Explanation

Statement 1: The statement is correct. To qualify for Miniratna Category-II status, a CPSE must have made profits continuously for the last three years and maintain a positive net worth. **Hence, Statement 1 is correct.**

Statement 2: The statement is incorrect. Miniratna CPSEs should not depend on budgetary support or Government guarantees. They are expected to be financially self-sustaining and should not have defaulted on repayment of Government loans or interest. **Hence, Statement 2 is incorrect.**

Value Addition

Miniratna Categories

Category	**Eligibility**
Miniratna Category-I	Profit in last 3 years, pre-tax profit of ₹30 crore or more in at least one of the 3 years, positive net worth
Miniratna Category-II	Profit in last 3 years and positive net worth

Investment Autonomy

Category	**Investment Limit**
Miniratna-I	Up to ₹500 crore or net worth, whichever is lower
Miniratna-II	Up to ₹300 crore or 50% of net worth, whichever is lower

Other Ratna Categories at a Glance

Category	**Key Requirement**	**Autonomy Level**
Maharatna	Navratna status + high turnover, net worth and profit + global presence	Highest
Navratna	Miniratna-I + Schedule A CPSE + composite score ≥ 60/100	Moderate
Miniratna	Profitability and positive net worth criteria	Limited

IN NEWS:Ministry of Steel grants Miniratna Category-I status to MECON Limited

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28.Vikram -1 Rocket

Q.With reference to the Vikram-1 Rocket, consider the following statements:

1. Vikram-1 is India's first privately developed orbital-class rocket developed by Skyroot Aerospace.
2. All four stages of Vikram-1 use solid-fuel propulsion systems.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (a) 1 only
Explanation

Statement 1: Vikram-1 is India's first privately developed orbital-class rocket and has been developed by Skyroot Aerospace. It marks a significant milestone in India's private space sector. **Hence, Statement 1 is correct.**
Statement 2: Vikram-1 is a four-stage launch vehicle in which the first three stages use solid-fuel propulsion, while the fourth stage uses hypergolic liquid propulsion powered by Raman engines. **Hence, Statement 2 is incorrect.**

Value Addition
Vikram-1 Rocket

Feature	Details
Developer	Skyroot Aerospace
Type	Orbital-class launch vehicle
Stages	Four
Propulsion	First three stages: Solid fuel; Fourth stage: Hypergolic liquid fuel
Payload to LEO	Up to 350 kg
Payload to SSO	Up to 260 kg
Structure	All-carbon composite
Special Feature	3D-printed engines and rapid launch capability

Raman Engines

- Hypergolic liquid engines.
- Use MMH (Monomethylhydrazine) as fuel and NTO (Nitrogen Tetroxide) as oxidizer.
- Ignite automatically upon contact, eliminating the need for an ignition system.

Hypergolic	Cryogenic
Ignites on contact	Requires ignition system
MMH + NTO	Liquid Hydrogen + Liquid Oxygen
Used in Vikram-1 upper stage	Used in heavy-lift launch vehicles like LVM3

In NEWS: Vikram-1, India's first-ever privately developed orbital rocket, gears up for launch

29.e-SARAS Platform

Q. The term 'eSARAS Platform' is sometimes mentioned in the news with reference to:

(a) Providing a unified digital architecture for real-time tracking of rural-to-urban seasonal migration.
(b) Creating an exclusive online digital storefront for products manufactured by women Self-Help Groups.
(c) Integrating micro-cold storage units along national highways to reduce post-harvest agricultural losses.
(d) Automating the transfer of direct benefit endowments to landless agricultural laborers.

Answer: (b) Creating an exclusive online digital storefront for products manufactured by women Self-Help Groups.
Explanation
Option (a):eSARAS is not a platform for tracking rural-to-urban migration or maintaining migration databases. Its primary objective is promoting products made by women Self-Help Groups (SHGs). **Hence, option (a) is incorrect.**
Option (b): eSARAS is the official e-commerce platform developed by the Ministry of Rural Development under DAY-NRLM to provide an exclusive digital marketplace for products manufactured by women Self-Help Groups and their federations. **Hence, option (b) is correct.**
Option (c): eSARAS is not related to cold-chain infrastructure, warehousing, or reduction of post-harvest losses. **Hence, option (c) is incorrect.**

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Option (d): eSARAS does not facilitate direct benefit transfers or welfare payments to agricultural labourers. It is a digital marketing platform for SHG products. **Hence, option (d) is incorrect.**

Value Addition
eSARAS Platform

Feature	Details
Developed By	Ministry of Rural Development
Under	Deendayal Antyodaya Yojana – National Rural Livelihoods Mission (DAY-NRLM)
Nature	Official e-commerce marketplace
Beneficiaries	Women Self-Help Groups (SHGs) and their federations
Objective	Market access and digital empowerment of rural women entrepreneurs
Reach	More than 8.62 crore women SHG members

Key Features

- Direct-to-consumer sales model eliminating multiple intermediaries.
- Integrated with Open Network for Digital Commerce.
- Accessible through UMANG.
- Hosts products across apparel, food, personal care, handicrafts, toys, and home décor.

Special Initiative
SARAS Shakti Collection

- Premium curated product range.
- Designed for institutional and corporate buyers.
- Promotes high-quality SHG products in larger markets.

In NEWS:eSARAS: Expanding Markets, Empowering Rural Women

30.Default Bail

Q.With reference to the concept of Default Bail, consider the following statements:

1. It is a statutory right available to an accused when the investigation is not completed within the prescribed period.
2. The right can be claimed only after the expiry of the prescribed period and before a valid charge sheet is filed.
3. It is available only in cases involving offences punishable with imprisonment of less than ten years.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only
Explanation
Statement 1: The statement is correct. Default bail, also known as statutory bail, is a right available to an accused person when the investigating agency fails to complete the investigation and file the charge sheet within the period prescribed by law. This right is provided under Section 187 of the Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023 (corresponding to Section 167(2) of the Code of Criminal Procedure, 1973). **Hence, Statement 1 is correct.**
Statement 2: The statement is correct. The right to default bail becomes enforceable only after the expiry of the prescribed investigation period. However, the accused must exercise this right before a valid charge sheet is filed. If the charge sheet is filed before the accused applies for default bail and is prepared to furnish bail, the right ceases to exist. **Hence, Statement 2 is correct.**
Statement 3: The statement is incorrect. Default bail is not restricted to offences punishable with imprisonment of less than ten years. It is available for all categories of offences where the investigating agency fails to complete the investigation within the statutory period prescribed for that offence. The applicable period may vary depending on the nature of the offence and the governing law. **Hence, Statement 3 is incorrect.**

Value Addition
Default Bail (Statutory Bail)

Feature	Details
Provision	Section 187, BNSS, 2023 (earlier Section 167(2), CrPC, 1973)
Nature	Statutory Right

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Purpose	Protect personal liberty against prolonged detention without completion of investigation
Constitutional Basis	Article 21 – Right to Life and Personal Liberty
Trigger	Failure to complete investigation and file charge sheet within prescribed period

Investigation Time Limits

Nature of Offence	**Maximum Period for Investigation**
Offence punishable with imprisonment up to 10 years	60 days
Offence punishable with death, life imprisonment, or imprisonment of 10 years or more	90 days

Key Principles

- Default bail is an indefeasible right once the statutory period expires and the accused applies for bail.
- It is not automatic; the accused must apply for bail and be prepared to furnish sureties if required.
- The right can be exercised only before a valid charge sheet is filed.
- Filing of a charge sheet after grant of default bail does not automatically cancel the bail.
- Cancellation of default bail requires grounds similar to those applicable for cancellation of regular bail.

Supreme Court Position

The Supreme Court has consistently held that default bail is a safeguard against arbitrary and prolonged incarceration and is closely linked to the constitutional guarantee of personal liberty under Article 21. The State cannot continue to detain an accused indefinitely merely because the investigation remains incomplete.

BNSS Linkage

Under the Bharatiya Nagarik Suraksha Sanhita (BNSS), the principle of default bail continues substantially in the same manner as under the Code of Criminal Procedure, ensuring time-bound investigation and protection of individual liberty.

In NEWS: Default bail only if charge sheet is not filed on time: SC

31.Foreign Direct Investment

Q.With reference to India's Foreign Direct Investment (FDI) policy, consider the following statements:

1. Investments under the Automatic Route do not require prior approval of the Government.
2. The Government may shift specified categories of investments from the Government Route to the Automatic Route.
3. The Automatic Route applies uniformly to all sectors and all source countries.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3
(e)

Answer: (a) 1 and 2 only

Explanation:

Under the **Automatic Route**, the non-resident investor or the Indian company does not require any prior approval from the Reserve Bank of India (RBI) or the Government of India for the investment. They only need to notify the RBI within a specified timeframe after the investment is made.**Hence Statement 1 is correct.**

The Government of India reviews the FDI policy continuously and frequently shifts sectors or specific investment categories from the **Government Route** (which requires prior approval from the respective ministry) to the **Automatic Route** to promote the ease of doing business.**Hence Statement 2 is correct.**

The Automatic Route does *not* apply uniformly.

- **Sector restrictions:** Certain sensitive sectors are completely prohibited from receiving FDI (like lottery, gambling, and chit funds), while others have specific caps or require the Government Route exclusively.
- **Country restrictions:** Under Press Note 3 (2020), any investment from a country that shares a land border with India (such as China, Pakistan, Bangladesh, etc.) strictly requires prior government approval, meaning the Automatic Route does not apply to them regardless of the sector.**Hence Statement 3 is incorrect.**

In NEWS: Centre opens power bids to four China-linked firms

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32.Article 371

Q. With reference to the special provisions under Article 371, consider the following States:

1. Maharashtra
2. Gujarat
3. Nagaland
4. Mizoram
5. Andhra Pradesh
6. Karnataka

How many of the above have special provisions under Article 371 or its sub-clauses?

(a) Only three
(b) Only four
(c) Only five
(d) All six

Answer: (d)All six
Explanation:
Part XXI of the Constitution of India contains **"Temporary, Transitional and Special Provisions"**. While Article 370 (which dealt with Jammu and Kashmir) was effectively abrogated, **Article 371 and its sub-clauses (A to J)** continue to provide special provisions for several states to protect their tribal cultures, maintain law and order, or address regional economic imbalances.
All six states listed in the question are covered under these articles:

Article	Covered State(s)	Key Objective / Focus
Article 371	**Maharashtra** & **Gujarat**	Establishment of separate development boards for regions like Vidarbha, Marathwada, Saurashtra, and Kutch.
Article 371A	**Nagaland**	Protects Naga customary laws, religious practices, and land ownership from parliamentary interference unless approved by the State Assembly.
Article 371D & 371E	**Andhra Pradesh**	Deals with equitable opportunities in public employment/education and the establishment of a Central University in the state.
Article 371G	**Mizoram**	Protects Mizo religious and social practices, customary laws, and land ownership ownership/transfer.
Article 371J	**Karnataka**	Special provisions for the Hyderabad-Karnataka region (now Kalyana-Karnataka) to establish a development board and ensure local quotas in jobs/education.

Quick Mnemonics to Remember Articles 371A to 371J
A popular way to remember the sequence of states from **A to J** is the phrase:
"Nagaland **A**ssam **M**anipur **A**ndhra **S**ikkim **T**ripura (Skip) **E**... **M**izoram **A**runachal **G**oa **K**arnataka**"** *(Note: Skip 'T' and 'E' in Namaste to transition smoothly into the remaining letters)*

- **371A: N**agaland
- **371B: A**ssam
- **371C: M**anipur
- **371D & E: A**ndhra Pradesh / Telangana
- **371F: S**ikkim
- **371G: M**izoram
- **371H: A**runachal Pradesh
- **371I: G**oa
- **371J: K**arnataka

In NEWS: Ladakh to have customised self-governance framework

33.Teesta River

Q. Consider the following statements with reference to the Teesta River:

1. It originates in the eastern Himalayas.
2. It is a tributary of the Brahmaputra River.
3. It flows through both India and Bangladesh.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

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Answer:(d)1, 2 and 3

Explanation:

- The Teesta River originates in the eastern Himalayas from the **Tso Lhamo lake / Pahunri glacier** area in North Sikkim, at an altitude of over 5,000 meters.**Hence Statement 1 is correct.**
- It is a major tributary of the **Brahmaputra River** (known as the *Jamuna* in Bangladesh). It meets the Brahmaputra at Fulchhari in Bangladesh.**Hence Statement 2 is correct.**
- The river flows down through the Indian states of **Sikkim** and **West Bengal** before entering **Bangladesh**, making it a vital transboundary river system shared by both countries.**Hence Statement 3 is correct.**



Visualizing the Basin Flow

The map below illustrates how the Teesta acts as a critical drainage artery carved directly through the steep eastern Himalayan terrain before feeding into the vast plains of the Brahmaputra system.

In NEWS: Will factor all related developments in approach to Teesta issue: India

34.Unmanned Aerial Vehicles

Q. With reference to Unmanned Aerial Vehicles (UAVs), consider the following statements:

1. Kamikaze drones are intended to destroy themselves while engaging the target.
2. Anti-Unmanned Aerial Vehicle (Anti-UAV) systems may employ both kinetic and electronic warfare measures.
3. Electronic warfare systems can neutralise hostile drones without physically destroying them.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d)1,2 and 3

Explanation:

Kamikaze drones—also known as **loitering munitions**—are a mix of a cruise missile and a drone. They fly over an area looking for a target and, once located, intentionally crash into it to detonate their onboard explosive payload, destroying themselves in the process.**Hence Statement 1 is correct.**

Modern Anti-UAV (Counter-UAV) defensive frameworks use a layered approach. They rely on **kinetic measures** (physically striking the drone down using machine guns, nets, programmable ammunition, missiles, or interceptor drones) as well as **electronic warfare measures** (interfering with signals) to disable the threat.**Hence Statement 2 is correct.**

Electronic warfare (EW) counter-drone systems neutralize hostile UAVs by targeting their control links or guidance networks without causing physical fragmentation. They do this by **jamming** communication frequencies (severing the link between the pilot and the drone) or **spoofing** GPS signals (forcing the drone's navigation system to believe it is elsewhere, triggering a forced landing or return-to-home protocol).**Hence Statement 3 is correct.**

In NEWS: DAC gives approval for military purchase worth ₹52,000 crore

35.Indus Water Treaty

Q. With reference to the Indus Waters Treaty, consider the following statements:

1. India enjoys unrestricted use of the waters of the Eastern Rivers.
2. India is permitted to specify non-consumptive uses of the Western Rivers in accordance with the Treaty.
3. The Treaty prohibits India from constructing any hydroelectric projects on the Western Rivers.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

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Answer: (a) 1 and 2 only

Explanation:

- Under the Indus Waters Treaty (IWT) signed in 1960, the waters of the three **Eastern Rivers** (Sutlej, Beas, and Ravi) are allocated to India for **unrestricted use**. India has full rights to exploit, store, and divert these waters.**Hence Statement 1 is correct.**
- The waters of the three **Western Rivers** (Indus, Jhelum, and Chenab) are allocated primarily to Pakistan. However, India is explicitly permitted to use these waters for **non-consumptive uses** as well as specified domestic, agricultural, and power generation purposes according to the criteria laid down in the Treaty Annexures.**Hence Statement 2 is correct.**
- The Treaty **does not prohibit** India from constructing hydroelectric projects on the Western Rivers. Instead, it explicitly allows India to build **Run-of-the-River (RoR) hydroelectric plants**. These plants generate electricity by utilizing the natural flow of the water without creating massive storage reservoirs that would permanently block or drastically reduce the flow into Pakistan.**Hence Statement 3 is incorrect.**

Key Allocation of the Indus System

River Group	Rivers Included	Primary Allocation & India's Rights
Eastern Rivers	Ravi, Beas, Sutlej	**India:** Unrestricted use (~20% of the total Indus system water).
Western Rivers	Indus, Jhelum, Chenab	**Pakistan:** Primary allocation. **India's Rights:** Limited to domestic, non-consumptive, agricultural, and Run-of-the-River power generation.

IN NEWS: 'No revival of Indus treaty if Pak. does not end terror'

36.Specialised Investment Funds

Q.Consider the following statements with reference to Specialised Investment Funds (SIFs):

1. They are regulated by the Securities and Exchange Board of India (SEBI).
2. They are intended to bridge the gap between conventional mutual funds and Portfolio Management Services (PMS).
3. They permit investment strategies involving derivatives and hedging, subject to regulatory conditions.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d)1,2 and 3

Explanation:

- Specialised Investment Funds (SIFs) are a relatively new investment vehicle introduced under the regulatory umbrella of the **Securities and Exchange Board of India (SEBI)**. They function via specific amendments mapped to the *SEBI (Mutual Funds) Regulations, 1996*.**Hence Statement 1 is correct.**
- SIFs were explicitly launched by SEBI to act as a **"missing middle" bridging asset class**. They sit comfortably between traditional mutual funds (accessible to retail investors with very strict diversification caps) and high-ticket Portfolio Management Services (PMS) or Alternative Investment Funds (AIFs) which mandate a steep minimum ticket size of ₹50 lakh to ₹1 crore. SIFs lower the point of entry for advanced strategies with a threshold of **₹10 lakh**.**Hence Statement 2 is correct.**
- Unlike traditional long-only mutual funds that are restricted to utilizing derivatives strictly for plain portfolio rebalancing or hedging, SIFs have far more flexibility. They are permitted to actively trade in **derivatives and unhedged long-short positions** (up to 25% of net assets as a naked or unhedged short exposure) to generate alpha and compound returns even in flat or falling market regimes, subject to hard SEBI boundaries.**Hence Statement 3 is correct.**

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Comparison of the Wealth Architecture

Parameter	Traditional Mutual Funds	Specialised Investment Funds (SIFs)	Portfolio Management Services (PMS)
Minimum Entry Ticket	No legal lower limit (SIPs from ₹100)	**₹10 Lakh**	**₹50 Lakh**
Structure	Standard Pooled Vehicle	Standard Pooled Vehicle	Segregated/Individual Demat Account
Derivative Liberty	Strictly for basic hedging/rebalancing	Advanced long-short (up to 25% unhedged)	Flexible mandates per client agreement

In NEWS: SIFs see rapid adoption; asset base grows to ₹13,500 cr. by May-end

37.Enforcement Directorate

Q. Consider the following statements:

1. The Enforcement Directorate is responsible for the enforcement of the Foreign Exchange Management Act, 1999.
2. The Enforcement Directorate is also the designated agency for enforcing the Prevention of Money Laundering Act, 2002.
3. The Enforcement Directorate functions under the Ministry of Home Affairs.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a)1 and 2 only
Explanation:
The Enforcement Directorate (ED) is a specialized financial investigation agency mandated to investigate and enforce the provisions of the **Foreign Exchange Management Act, 1999 (FEMA)**, which deals with foreign exchange violations and cross-border trade/investment irregularities.**Hence Statement 1 is correct.**
The ED is also the premier designated agency responsible for tracing, attaching, and prosecuting assets derived from scheduled offenses under the **Prevention of Money Laundering Act, 2002 (PMLA)**. In addition, it enforces the Fugitive Economic Offenders Act, 2018 (FEOA).**Hence Statement 2 is correct.**
The Enforcement Directorate does not function under the Ministry of Home Affairs. Instead, it is an administrative wing of the **Department of Revenue**, which falls under the **Ministry of Finance**, Government of India.**Hence Statement 3 is incorrect:**

In NEWS: ED to share foreign funds probe findings with the CBI

38.National Company Law Tribunal

Q. Consider the following statements:

1. The National Company Law Tribunal (NCLT) is a statutory body constituted under the Companies Act, 2013.
2. Appeals against the orders of the National Company Law Tribunal lie before the National Company Law Appellate Tribunal (NCLAT).
3. Appeals against the orders of the National Company Law Appellate Tribunal lie directly to the High Court concerned.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only
Explanation:
The National Company Law Tribunal (NCLT) is a quasi-judicial statutory body set up under **Section 408 of the Companies Act, 2013**. It was formally constituted on June 1, 2016, based on the recommendations of the Justice Eradi Committee to handle corporate disputes, winding-up of companies, and insolvency matters.**Hence Statement 1 is correct.**

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Under **Section 410 of the Companies Act, 2013**, any person or entity aggrieved by an order passed by the NCLT can file an appeal before the designated appellate authority: the **National Company Law Appellate Tribunal (NCLAT)**.**Hence Statement 2 is correct.**

Appeals against the decisions of the NCLAT **do not** go to the High Court. Instead, under **Section 423 of the Companies Act, 2013** (and similarly under Section 62 of the Insolvency and Bankruptcy Code, 2016), any person aggrieved by an order of the NCLAT may file an appeal **directly to the Supreme Court of India** on any substantial question of law.**Hence Statement 3 is incorrect.**

In NEWS: Manufacturing justice

39.Notifiable Disease

Q. Consider the following statements:

1. A notifiable disease is one that must be reported to the designated public health authority upon its diagnosis.
2. The list of notifiable diseases is uniformly prescribed by a Central law applicable throughout India.
3. Notification of diseases facilitates disease surveillance and public health planning.

Which of the statements given above are correct?

(a) 1 and 3 only
(b) 2 only
(c) 1 and 2 only
(d) 1, 2 and 3

Answer: (a) 1 and 3 only

Explanation:

A notifiable disease is legally defined as any medical condition that clinicians, diagnostic laboratories, and hospitals are mandated by law to report to government public health authorities immediately upon clinical or laboratory diagnosis. **Hence Statement 1 is correct.**

There is no uniform, single list of notifiable diseases prescribed across all of India by a Central law. Public health and sanitation fall primarily under the **State List** (Entry 6) of the Seventh Schedule of the Constitution, while the prevention of infectious diseases shifting across borders falls under the **Concurrent List** (Entry 29). Consequently, **individual state governments are responsible** for fixing and updating their own distinct list of notifiable diseases based on localized geographic risks and regional endemic outbreaks. For instance, the Delhi government recently moved to declare rabies notifiable, while other states have added conditions like snakebites or specific local fevers under the general framework of the *Epidemic Diseases Act, 1897*.**Hence Statement 2 is incorrect.**

The primary policy intent behind making a disease notifiable is to act as an early-warning signal. It triggers tracking and surveillance networks (like the *Integrated Disease Surveillance Programme - IDSP*), helps epidemiologists isolate transmission chains, map outbreak patterns, allocate medical countermeasures, and outline long-term public health planning.**Hence Statement 3 is correct.**

In NEWS: Counting cancer

40.Financial Instruments

Q. Consider the following statements:

1. A derivative is a financial instrument whose value is derived from an underlying asset.
2. Hedging is a strategy used to reduce financial risk.
3. Every derivative transaction is undertaken solely for speculative purposes.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 only
(d) 1, 2 and 3

Answer:(a)1 and 2 only

EXPLANATION:

- By definition, a derivative is a security or financial contract whose price and value are not intrinsic but are completely dependent on or derived from one or more **underlying assets** (such as stocks, bonds, commodities, currencies, interest rates, or market indexes).**Hence Statement 1 is correct.**
- Hedging is a risk management strategy used to offset or reduce the risk of adverse price movements in an asset. For example, an exporter might buy a currency forward contract to "lock in" an exchange rate, protecting themselves against a sudden drop in the value of that foreign currency.**Hence Statement 2 is correct.**

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● Derivative transactions are not undertaken *solely* for speculative purposes. While speculators do use derivatives to bet on future price movements to make quick profits, derivatives are equally used for **hedging** (reducing existing risk) and **arbitrage** (making risk-free profits by exploiting price differences for the same asset across different markets). The word "solely" makes this statement factually wrong.**Hence Statement 3 is incorrect.**

The Three Participants in a Derivative Market

Participant	Core Motivation / Strategy
Hedger	Wants to **reduce or eliminate risk** by locking in future prices to protect an existing business operation or asset portfolio.
Speculator	Intentionally **takes on risk** to bet on future market directions in hopes of achieving high returns from price fluctuations.
Arbitrageur	Look for **price mismatches** of the exact same asset in different markets to buy low in one and simultaneously sell high in the other for risk-free profit.

In News:SIFs see rapid adoption; asset base grows to ₹13,500 cr. by May-end

41.Trade Deficit

Q.Consider the following:

1. Reduction of non-tariff barriers
2. Market access for pharmaceuticals
3. Promotion of foreign direct investment
4. Diversification of export basket

Which of the above measures can help reduce India's bilateral trade deficit?

(a) 1, 2 and 4 only
(b) 1 and 3 only
(c) 2 and 4 only
(d) 1, 2, 3 and 4

Answer: (d)1,2,3 and 4
Explanation:
Reduction of non-tariff barriers (1): Non-tariff barriers include strict regulations, quotas, or complex licensing requirements imposed by trading partners that make it hard for Indian goods to enter their markets. Reducing these barriers directly opens up foreign markets for Indian exports.**Hence option 1 is correct.**
Market access for pharmaceuticals (2): India is a global leader in generic medicines ("the pharmacy of the world"). Gaining better market access for pharmaceuticals in countries where India runs a trade deficit (like China) allows India to leverage its competitive advantage and boost high-value exports.**Hence option 2 is correct.**
Promotion of foreign direct investment (3): Attracting FDI can boost domestic manufacturing capacity (e.g., through initiatives like "Make in India"). This helps in two ways: it creates products that can substitute for expensive imports, and it scales up production to increase exports globally.**Hence option 3 is correct.**
Diversification of export basket (4): Relying on just a few export products makes a country vulnerable to market shifts. Expanding the variety of goods and services India exports ensures more stable, consistent, and widespread revenue streams from trading partners.**Hence option 4 is correct.**

In NEWS: Chinese investment influx good for wider ties, says India's envoy

42.Biological Products

Q. Consider the following:

1. Monoclonal antibodies
2. Generic medicines
3. Vaccines
4. Biosimilars

Which of the above are biological products?

(a) 1 and 4 only
(b) 2 and 3 only
(c) 1, 3 and 4 only
(d) 1, 2, 3 and 4

Answer: (c)1, 3 and 4 only
Explanation:
Biological Products (Biologics)
Biological products are medicines manufactured in, extracted from, or synthesized using **living organisms** (such as bacteria, yeast, or mammalian cells). They are highly complex, large-molecule

mixtures that are difficult to characterize or replicate perfectly.

- **Monoclonal antibodies (1):** These are laboratory-made proteins designed to mimic the immune system's ability to fight off harmful pathogens or target cancer cells. They are grown in living cell cultures.
- **Vaccines (3):** Made from weakened or killed forms of a microbe, its toxins, or its genetic material (like mRNA) to stimulate an immune response, vaccines are inherently biological.
- **Biosimilars (4):** A biosimilar is a biologic medical product that is highly similar to another already-approved biological medicine. Think of it as the "generic" version of a biologic, but because it is grown in living cells, it cannot be a perfect identical copy.

Chemical Drugs (Small Molecules)

- **Generic medicines (2):** Traditional generic medicines are **chemically synthesized** in a lab using predictable chemical reactions. They have simple, small-molecule structures. Because they are purely chemical, a generic medicine is an *exact, identical copy* of the original brand-name drug's active ingredient (unlike a biosimilar, which is just "highly similar").**Hence option(2) is correct**

In NEWS: India's essential medicine list lags behind global benchmark; unrevised in 4 years

43.Pharmaceutical classifications

Q. Consider the following statements with reference to pharmaceutical classifications:

I. Biologics and biosimilars are manufactured using living cell substrates, making them inherently more structurally complex than chemically synthesized generic small-molecule drugs.
II. Because biosimilars are identical copies of their reference biological products, they do not require independent clinical trials to establish therapeutic equivalence.
III. Monoclonal antibodies and mRNA vaccines represent advanced classes of biological therapeutics that leverage cellular machinery rather than direct chemical reaction routes.

Which of the relationships among the above statements is/are correct?

1. Statement II contradicts the regulatory premise implicit in Statement I.
2. Statement III provides technological instances that validate the manufacturing complexity described in Statement I.
3. Statements I, II, and III together establish that biosimilars and small-molecule generics follow the same bio-equivalence protocols.

Code:
(a) 1 only
(b) 1 and 2
(c) 2 and 3
(d) 3 only

Answer: (b) 1 and 2

Explanation:

Statement II is factually incorrect because biosimilars are *highly similar*, not identical, to their reference products due to the micro-variability inherent in living cells. Thus, they require rigorous clinical evaluation to prove no clinically meaningful differences exist. This micro-variability and complexity means Statement II contradicts the premise of Statement I (making relationship 1 correct). Statement III correctly identifies advanced biologics that utilize biological systems, exemplifying the complexity stated in Statement I (making relationship 2 correct).

In NEWS: India's essential medicine list lags behind global benchmark; unrevised in 4 years

44.UAPA

Q. With reference to the Unlawful Activities (Prevention) Act (UAPA), consider the following statements:

1. The Central Government can designate both organisations and individuals as terrorists under the Act.
2. An individual designated as a terrorist is automatically convicted of terrorist offences.
3. The designation can be challenged before a Review Committee.

Which of the statements given above is/are correct?

(a) 1 only
(b) 1 and 3 only
(c) 2 and 3 only

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(d) 1, 2 and 3

Answer: (b)1 and 3 only
Explanation:

- Under Section 35 of the Unlawful Activities (Prevention) Act, 1967 (UAPA), the Central Government has the power to notify and declare an organisation as a terrorist organisation (First Schedule). Following the **2019 amendment** to the Act, the Central Government was explicitly empowered to also designate **individuals** as terrorists (Fourth Schedule) under the same criteria.**Hence Statement 1 is correct.**
- Designation as a terrorist by the executive branch does **not** automatically translate into a criminal conviction. A conviction requires a formal trial, judicial scrutiny, and a verdict by a competent court of law following standard legal procedures. The executive designation serves functions such as freezing assets, restricting travel, and aiding investigative agencies like the NIA.**Hence Statement 2 is incorrect.**
- Under Section 36 of the UAPA, an individual or organization aggrieved by the designation can first apply to the Central Government for de-notification. If that application is rejected, the applicant has the legal right to challenge the decision before a high-level **Review Committee** appointed by the government, which is headed by a sitting or retired High Court Judge.**Hence Statement 3 is correct.**

In NEWS: Centre lists six Indians,17 Pakistan nationals as terrorists under UAPA

45.Large Language Models

Q. With reference to Large Language Models (LLMs), consider the following statements:

1. They are trained on large volumes of text data to generate and understand natural language.
2. They can perform tasks such as coding, summarisation, translation and question answering.
3. All LLMs are open-source and freely downloadable.

Which of the statements given above is/are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (b)1 and 2 only
Explanation:

LLMs are deep learning algorithms trained on massive datasets containing billions of words to recognize, summarize, translate, predict, and generate natural language text.**Hence Statement 1 is correct.**
Modern LLMs are highly versatile and widely used for text-based tasks including writing code, summarizing documents, language translation, and answering complex questions.**Hence Statement 2 is correct.**
Not all LLMs are open-source. While open-source models exist (such as Llama or Mistral), many prominent models (such as OpenAI's GPT-4 or Anthropic's Claude) are closed-source, proprietary, and only accessible via paid APIs or web interfaces.**Hence Statement 3 is incorrect.**

In NEWS: A race to catch up

46.Standard Model

Q. Which of the following are elementary particles according to the Standard Model?

1. Quarks
2. Leptons
3. Protons
4. Electrons

Select the correct answer using the code below.

(a) 1 and 2 only
(b) 1, 2 and 4 only
(c) 1, 3 and 4 only
(d) 1, 2, 3 and 4

Answer:(b) 1, 2 and 4 only
Explanation:
According to the **Standard Model of Particle Physics**, elementary particles are the fundamental building blocks of matter that cannot be broken down into anything smaller.

- **Quarks (1):** These are elementary particles. There are six types (flavors) of quarks: up, down, charm, strange, top, and bottom.
- **Leptons (2):** These are also elementary particles. The lepton family includes six particles: the electron, muon, tau, and their three corresponding neutrinos.

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- **Electrons (4):** The electron is a specific type of **lepton**, making it a fundamental, elementary particle.
- **Protons (3) are NOT elementary particles:** A proton is a composite particle (specifically a hadron/baryon). It is made up of smaller constituent elementary particles: **two up quarks and one down quark** bound together by gluons. Because it has an internal structure, it is not elementary.**Hence option(b) is correct.**

In NEWS: On the cusp of big upgrade, CERN collider faces hard question

47.Phonons

Q. With reference to phonons, consider the following statements:

1. Phonons are quantized units of lattice vibrations in a solid.
2. Phonons play an important role in the thermal conductivity of materials.
3. Phonons are elementary particles listed in the Standard Model of Particle Physics.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a)1 and 2 only
Explanation:
Phonons are **quasiparticles** that represent quantized collective excitations or vibrations in a rigid crystalline structure (lattice) of a solid. Just as a photon is a quantized packet of light/electromagnetic energy, a phonon is a quantized packet of vibrational/vocal energy within a crystal lattice.**Hence Statement 1 is correct.**
In non-metallic solids (like insulators and semiconductors), electrical conductors rely on electrons to transfer heat, but heat conduction is primarily governed by the movement and scattering of phonons. They carry thermal energy through the lattice structure, directly determining the material's thermal conductivity.**Hence Statement 2 is correct.**
Phonons are **not** elementary particles, nor are they listed in the Standard Model of Particle Physics. They are **quasiparticles** (or collective excitations)—they do not exist as independent particles in empty space (vacuum). Instead, they are emergent phenomena that only exist inside a condensed matter medium (like a solid crystal).**Hence Statement 3 is incorrect.**

In NEWS: Scientists use budget tech to probe long-standing physics mystery

48.Raman spectroscopy

Q. Assertion (A): Raman spectroscopy is widely used to investigate the structure and properties of materials.
Reason (R): Raman spectroscopy measures changes in the energy of light scattered by molecules or crystal lattices.

Codes:
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer: (a)Both A and R are true, and R is the correct explanation of A.
Explanation
- **Assertion (A) is true:** Raman spectroscopy is a powerful, non-destructive chemical analysis technique. It is widely utilized across materials science, chemistry, nanotechnology, and pharmaceuticals to identify molecular structures, analyze chemical compositions, map phases, and investigate physical properties (like stress or crystallinity) of materials.
- **Reason (R) is true:** The fundamental principle of Raman spectroscopy relies on **Raman scattering** (inelastic scattering of light). When a monochromatic light source (usually a laser) illuminates a sample, it interacts with molecular vibrations or phonons in a crystal lattice. Most photons scatter elastically (Rayleigh scattering, no energy change), but a tiny fraction scatters **inelastically**. This means they lose or gain kinetic energy during the collision.
- **R is the correct explanation of A:** Because the technique precisely measures these shifting energy levels (the "Raman shift"), it acts as a unique structural fingerprint. Since every material has its own characteristic vibrational modes, measuring the scattered light's energy change is exactly what allows scientists to determine the material's specific molecular structure and properties.

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Key Concepts of Raman Scattering

When the incident light interacts with a molecule or lattice, the inelastic scattering results in two types of energy shifts:

1. **Stokes Scattering:** The scattered photon has *lower* energy (longer wavelength) than the incident photon because the molecule absorbed some energy to jump to a higher vibrational state.
2. **Anti-Stokes Scattering:** The scattered photon has *higher* energy (shorter wavelength) because the molecule was already in an excited vibrational state and dropped to a lower ground state, imparting extra energy to the photon.

In NEWS: Scientists use budget tech to probe long-standing physics mystery

49.White Dwarf

Q. Consider the following statements regarding a white dwarf:

1. It represents the final evolutionary stage of low- and intermediate-mass stars such as the Sun.
2. It no longer undergoes nuclear fusion in its core.
3. It may still possess a planetary system orbiting it.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a)1 and 2 only

Explanation

- A white dwarf is the stellar remnant left behind after a low- to intermediate-mass star (stars with masses up to about 8 times the mass of our Sun) has exhausted its nuclear fuel, shed its outer layers as a planetary nebula, and collapsed. Our Sun will eventually evolve into a white dwarf.**Hence Statement 1 is correct.**
- A white dwarf is a "dead" star in terms of nuclear reactions. It no longer undergoes nuclear fusion in its core because it lacks the mass, gravitational pressure, and temperature required to fuse carbon and oxygen. Instead, it shines simply by radiating away its leftover, trapped thermal energy over billions of years. It is supported against gravitational collapse by **electron degeneracy pressure**.**Hence Statement 2 is correct.**
- Observational astronomy has confirmed that white dwarfs can possess planetary systems. While the red giant phase of a star's evolution will destroy close-in planets, outer planets, asteroids, and comets can survive. Astronomers have discovered intact giant planets orbiting white dwarfs, as well as debris disks formed from pulverized rocky bodies.**Hence Statement 3 is correct.**

In NEWS: Planet shows star's death does not have to doom its worlds

50.CAR T-cell therapy

Q. With reference to CAR T-cell therapy, consider the following statements:

1. It involves genetically modifying a patient's own T lymphocytes.
2. The modified T cells are designed to recognize and attack specific cancer cells.
3. CAR T-cell therapy is a form of passive immunization.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a)1 and 2 only

EXPLANATION:

CAR T-cell therapy (Chimeric Antigen Receptor T-cell therapy) is an autologous treatment, meaning it uses a patient's own cells. A patient's T lymphocytes (a type of white blood cell essential for the immune system) are extracted from their blood via a process called leukapheresis and then genetically modified in a laboratory using a disarmed virus or gene-editing tool.**Hence Statement 1 is correct.**

During the laboratory modification, a specific gene is inserted into the T cells that instructs them to produce **Chimeric Antigen Receptors (CARs)** on their surface. These synthetic receptors act like a GPS, specifically engineered to lock onto and destroy particular proteins (antigens) found on the surface of target cancer cells (such as the CD19 protein found in certain leukemias and lymphomas).**Hence Statement 2 is correct.**

CAR T-cell therapy is classified as **adoptive cell transfer (ACT)** or active cellular immunotherapy, not passive immunization. Traditional *passive*

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immunization involves transferring pre-formed, short-lived immune proteins (like antibodies or antivenoms) from one individual or animal to another, providing immediate but temporary protection without creating cellular memory. In contrast, CAR T-cells are living, genetically engineered cells that dynamically expand, multiply, and can persist in the patient's body for years, providing active, long-term surveillance.**Hence Statement 3 is incorrect.**

In NEWS:CAR T-cell therapy 'sees' solid tumours thanks to protein

51.Torrential rain

Q. Consider the following statements regarding Torrential Rain:

1. Torrential rain refers to extremely heavy rainfall occurring over a short duration, often resulting in flooding and waterlogging.
2. The occurrence of torrential rain in India is exclusively associated with tropical cyclones originating in the Bay of Bengal.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (a) 1 only
Explanation
Statement 1:
Torrential rain refers to exceptionally heavy rainfall occurring within a short period of time. Such rainfall often exceeds the capacity of natural drainage systems and urban infrastructure, resulting in waterlogging, flash floods, landslides, and disruption of transport and economic activities. In India, torrential rainfall is commonly experienced during the southwest monsoon season and during intense weather events. **Hence, Statement 1 is correct.**
Statement 2:
Torrential rainfall in India is not exclusively associated with tropical cyclones originating in the Bay of Bengal. It can result from a variety of meteorological conditions such as monsoon depressions, low-pressure systems, western disturbances, convective thunderstorms, cloudbursts, and orographic uplift along mountain ranges such as the Western Ghats and the Himalayas**.**
Hence, Statement 2 is incorrect.
Value Addition
Torrential Rain

Feature	Details
Meaning	Extremely heavy rainfall occurring over a short duration
Associated Hazards	Flash floods, urban flooding, landslides, waterlogging
Common Seasons in India	Southwest Monsoon (June–September) and Northeast Monsoon (October–December)
Major Triggering Systems	Monsoons, cyclones, thunderstorms, low-pressure systems

Major Causes of Torrential Rain
Monsoon Dynamics
- Moisture-laden southwest monsoon winds bring large quantities of water vapour from the Arabian Sea and Bay of Bengal.
- Strong monsoon currents can lead to intense rainfall episodes.

Cyclonic Systems
- Tropical cyclones and depressions generate large-scale convergence of moisture.
- Often result in prolonged heavy rainfall over coastal and inland regions.

Orographic Effect
- Moist air rises when it encounters mountain barriers.
- Cooling and condensation lead to intense rainfall on windward slopes.

Convective Activity
- High surface heating causes rapid uplift of moist air.
- It leads to thunderstorms and short-duration heavy rainfall.

Climate Change
- Rising temperatures increase atmospheric moisture content.
- Intensifies the frequency and magnitude of extreme rainfall events.

Major Impacts
- Urban flooding and waterlogging.
- Damage to crops and agricultural land.
- Loss of life and property.

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- Landslides in hilly regions.
- Spread of water-borne diseases.
- Disruption of transport and communication networks.

In NEWS:Torrential rain kills 3, injures 4 in Maharashtra

52.United Nations Conference on Trade and Development(UNCTAD)

Q.Consider the following statements regarding the United Nations Conference on Trade and Development (UNCTAD):

1. It was established by the United Nations General Assembly in 1964.
2. It functions as a specialized agency of the United Nations.
3. It serves as a forum for intergovernmental consensus-building on trade and development issues.
4. India is a member of UNCTAD.

Which of the statements given above are correct?

(a) 1, 3 and 4 only
(b) 1 and 2 only
(c) 2, 3 and 4 only
(d) 1, 2, 3 and 4

Answer: (a) 1, 3 and 4 only

Explanation

Statement 1:

UNCTAD was established in 1964 by the United Nations General Assembly (UNGA) as a permanent intergovernmental body to address issues relating to trade, investment, finance, and development, particularly from the perspective of developing countries.**Hence, Statement 1 is correct.**

Statement 2:

UNCTAD is not a specialized agency of the United Nations like World Health Organization or United Nations Educational, Scientific and Cultural Organization.Instead, it is a permanent intergovernmental body that forms part of the UN Secretariat and reports to the UN General Assembly and ECOSOC.**Hence, Statement 2 is incorrect.**

Statement 3:

One of UNCTAD's core functions is to provide a platform where member states can discuss and build consensus on issues relating to:

- International trade
- Investment flows
- Sustainable development
- Debt and finance
- Technology transfer
- Globalization and development **Hence, Statement 3 is correct.**

Statement 4:

India is one of the 195 member states of UNCTAD and actively participates in its discussions relating to trade, development financing, technology transfer, and South-South cooperation.**Hence, Statement 4 is correct.**

Value Addition

United Nations Conference on Trade and Development (UNCTAD)

Feature	Details
Established	1964
Established By	United Nations General Assembly (UNGA)
Headquarters	Geneva
Nature	Permanent Intergovernmental Body
Parent Structure	Part of the UN Secretariat
Membership	195 Countries
India's Status	Member
Reports To	UNGA and ECOSOC

Core Functions of UNCTAD

1. Research and Analysis

Provides policy-oriented research on:

- International trade
- Global investment trends
- Development finance
- Debt sustainability
- Digital economy

2. Consensus Building

Acts as a forum where governments negotiate and discuss development-related economic issues.

3. Technical Assistance

Provides capacity-building support to developing and least-developed countries.

Major Reports Published by UNCTAD

Report	Focus Area
Trade and Development Report	Global economy and development trends
World Investment Report	Foreign Direct Investment (FDI)
Least Developed Countries Report	Issues facing LDCs
Digital Economy Report	Digital trade and technology
Economic Development in Africa Report	African development challenges

UNCTAD helps countries:

- Integrate effectively into the global trading system.
- Diversify economies away from commodity dependence.
- Address debt vulnerabilities.
- Improve investment climates.
- Harness technology for sustainable development.
- Promote inclusive and equitable globalization.

UNCTAD	WTO
UN body established in 1964	Independent international organization established in 1995
Discussion, research, policy support	Rule-making and dispute settlement
Focus on development concerns	Focus on global trade rules
No dispute settlement mechanism	Has a formal dispute settlement system

IN NEWS:India to chair UNCTAD's 9th session of Intergovernmental Group of Experts on Consumer Protection in Switzerland

53.BIMSTEC

Q.Consider the following statements regarding BIMSTEC:

1. BIMSTEC serves as a bridge between South Asia and Southeast Asia.
2. BIMSTEC includes five SAARC countries and two ASEAN countries.
3. India proposed the BODHI Programme for capacity building during the 6th BIMSTEC Summit.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation

Statement 1:BIMSTEC (Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation) connects the countries of South Asia and Southeast Asia through a common regional framework centered around the Bay of Bengal. It is therefore often described as a bridge between South Asia and Southeast Asia. **Hence, Statement 1 is correct.**

Statement 2:BIMSTEC consists of seven member countries:

- Bangladesh
- Bhutan
- India
- Nepal
- Sri Lanka
- Myanmar
- Thailand

Among them:

SAARC members: Bangladesh, Bhutan, India, Nepal, Sri Lanka (5)

ASEAN members: Myanmar and Thailand (2)

Thus, BIMSTEC includes five SAARC countries and two ASEAN countries.**Hence, Statement 2 is correct.**

Statement 3: At the 6th BIMSTEC Summit (2025) hosted by Thailand, India proposed the BODHI Programme (BIMSTEC for Organized Development of Human Resource Infrastructure). The initiative aims to strengthen regional cooperation through:

- Skill development
- Training programmes
- Scholarships

- Capacity building of professionals and officials across BIMSTEC countries**Hence, Statement 3 is correct.**

Value Addition

BIMSTEC at a Glance

Feature	**Details**
Full Form	Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation
Established	1997
Founding Declaration	Bangkok Declaration
Secretariat	Dhaka
Member Countries	7
Population	About 1.7 billion people
Combined GDP	About USD 5.2 trillion

BIMSTEC Members

South Asia	**Southeast Asia**
Bangladesh	Myanmar
Bhutan	Thailand
India	
Nepal	
Sri Lanka	

Key Highlights of the 6th BIMSTEC Summit (2025)

- Adoption of Bangkok Vision 2030
- Proposal for BODHI Programme
- BIMSTEC Centres of Excellence in India
- Promotion of Digital Public Infrastructure (DPI)
- Proposal to link UPI with BIMSTEC payment systems
- BIMSTEC Athletics Meet (2025)
- First BIMSTEC Games (2027)

BIMSTEC - India

- Supports Act East Policy
- Complements Neighbourhood First Policy
- Serves as an alternative platform to the stalled South Asian Association for Regional Cooperation
- Strengthens India's role in the Bay of Bengal and Indo-Pacific region
- Enhances connectivity, trade, energy, and security cooperation

IN NEWS:India and Japan did well to ring-fence their ties from multilateral accords

54.Dr.Shyama Prasad Mukherjee

Q.The following statements describe a prominent personality in modern Indian history:

1. He became the youngest Vice-Chancellor of Calcutta University.
2. He was a member of the Constituent Assembly of India.
3. He served as President of the Mahabodhi Society.
4. He founded the Bharatiya Jana Sangh in 1951.

Who among the following is being described above?

(a) Shyama Prasad Mukherjee
(b) Deendayal Upadhyaya
(c) B. R. Ambedkar
(d) Madan Mohan Malaviya

Answer: (a) Shyama Prasad Mukherjee

Explanation

Clue 1: Youngest Vice-Chancellor of Calcutta University

Dr. Shyama Prasad Mukherjee became the Vice-Chancellor of Calcutta University in 1934 at the age of 33, making him one of the youngest Vice-Chancellors in India's history.

Clue 2: Member of the Constituent Assembly

He entered the Constituent Assembly from Bengal and participated in constitutional discussions before joining the Union Cabinet.

Clue 3: President of the Mahabodhi Society

He served as President of the Mahabodhi Society, which worked for the preservation and promotion of Buddhist heritage and sacred relics.

Clue 4: Founder of Bharatiya Jana Sangh

In 1951, he founded the Bharatiya Jana Sangh (BJS), which later evolved into the Bharatiya Janata Party (BJP). **Hence, Option (a) is correct.**

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Value Addition: Dr. Shyama Prasad Mukherjee

Feature	Details
Born	6 July 1901
Father	Sir Ashutosh Mukherjee
Profession	Educationist, Lawyer, Politician
Vice-Chancellor	Calcutta University (1934–38)
Union Minister	Industry and Supply Minister (1947–50)
Founded	Bharatiya Jana Sangh (1951)
Opposed	Article 370 and Permit System in J&K
Died	23 June 1953, Srinagar

Option (b): Deendayal Upadhyaya

- He was an important ideologue of the Bharatiya Jana Sangh.
- Propounded the philosophy of Integral Humanism (Ekatma Manav Darshan).
- Served as President of the Jana Sangh in 1967.
- However, he did not found the Jana Sangh.
- He was neither Vice-Chancellor of Calcutta University nor a member of the Constituent Assembly. **Therefore, Option (b) is incorrect.**

Option (c): B. R. Ambedkar

- Chairman of the Drafting Committee of the Constitution.
- First Law Minister of independent India.
- Architect of the Indian Constitution.
- Founder of the Independent Labour Party and later the Scheduled Castes Federation.
- Converted to Buddhism in 1956. **Therefore, Option (c) is incorrect.**

Value Addition

Feature	Details
Chairman	Drafting Committee
First Law Minister	Independent India
Major Work	Annihilation of Caste
Conversion	Buddhism (1956)
Bharat Ratna	1990 (Posthumous)

Option (d): Madan Mohan Malaviya

- Founder of Banaras Hindu University (1916).
- Four-time President of the Indian National Congress.
- Prominent freedom fighter and educationist.
- Played a major role in promoting modern education in India. **Therefore, Option (d) is incorrect.**

Value Addition

Feature	Details
Founder	BHU
Congress President	Four times
Popular Title	Mahamana
Bharat Ratna	2014 (Posthumous)
Death	1946

In NEWS:Modi hails Syama Prasad Mookerjee as 'champion of national unity'

55.Asiatic lion

Q.Consider the following statements regarding the Asiatic Lion:

1. The Asiatic Lion is found exclusively in India and its natural population is confined to Gujarat.
2. The Asiatic Lion is listed in Appendix I of CITES and Schedule I of the Wildlife (Protection) Act, 1972.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Explanation

Statement 1:

The Asiatic Lion (Panthera leo persica) survives in the wild only in India. Its population is concentrated in the Saurashtra region of Gujarat, including:

- Gir National Park and Wildlife Sanctuary
- Girnar Wildlife Sanctuary
- Coastal and revenue landscapes of Saurashtra

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Although lions are expanding their range beyond Gir, they remain naturally confined to Gujarat. **Hence, Statement 1 is correct.**

Statement 2:

The Asiatic Lion receives the highest level of legal protection:

- Wildlife (Protection) Act, 1972 – Schedule I
- CITES – Appendix I

Appendix I includes species threatened with extinction and permits international trade only under exceptional circumstances. **Hence, Statement 2 is correct.**

Value Addition



Asiatic Lion

Feature	Details
Scientific Name	Panthera leo persica
Common Name	Indian Lion / Persian Lion
Endemic To	India
Major Habitat	Gir Landscape, Gujarat
IUCN Status	Vulnerable (2025)
CITES	Appendix I
WPA, 1972	Schedule I

Lion Census 2020

- Population: 674 lions
- Spread across:
  - 9 districts
  - 53 talukas
  - Nearly 30,000 sq km of Saurashtra landscape

56. Urban cooperatives bank

Q.Consider the following statements regarding Urban Cooperative Banks (UCBs):

1. The primary objective of Urban Cooperative Banks is to provide banking and credit facilities to small borrowers, traders, self-employed persons, and micro and small enterprises in urban and semi-urban areas.
2. Urban Cooperative Banks can accept deposits from the public and provide loans to eligible customers similar to commercial banks, subject to regulatory norms.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Explanation

Statement 1:

Urban Cooperative Banks (UCBs) were established to meet the banking and credit needs of urban and semi-urban populations, particularly:

- Small borrowers
- Traders
- Artisans
- Self-employed persons
- Micro, Small and Medium Enterprises (MSMEs)

Their objective is to promote financial inclusion and provide affordable credit to sections that may have limited access to formal banking channels. **Hence, Statement 1 is correct.**

Statement 2:

Like commercial banks, UCBs perform core banking functions such as:

- Accepting savings, current and fixed deposits
- Providing housing loans
- Personal loans
- MSME loans
- Working capital finance

However, these activities are subject to prudential norms and regulations prescribed by the Reserve Bank of India under the Banking Regulation Act, 1949. **Hence, Statement 2 is correct.**

Value Addition

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Urban Cooperative Banks (UCBs)

Feature	Details
Nature	Primary Cooperative Banks in urban and semi-urban areas
Ownership	Cooperative institutions owned by members
First Urban Cooperative Credit Society	Anyonya Sahakari Mandali (1889), Baroda
Banking Regulator	Reserve Bank of India
Registration	State Cooperative Societies Act or Multi-State Cooperative Societies Act, 2002
Main Objective	Mobilize savings and provide credit to local communities

Banking Functions of UCBs

Deposits

- Savings Deposits
- Current Deposits
- Fixed Deposits
- Recurring Deposits

Lending Activities

- Housing Loans
- Personal Loans
- Gold Loans
- MSME Credit
- Working Capital Loans
- Education Loans

Regulatory Framework

RBI Regulates

- Banking licence
- Branch expansion
- Prudential norms
- Interest rates
- Capital adequacy
- Inspection and supervision

State Governments/Registrar of Cooperative Societies Regulate

- Registration
- Elections
- Audit
- Management
- Liquidation

IN NEWS: Shah to launch several projects as Cooperation Ministry completes 5 years

57. Gaganyaan mission

Q.Which of the following statements regarding SOLVE are correct?

1. SOLVE is a sub-orbital test platform developed for Gaganyaan missions.
2. Its solid stage is derived from the PSLV strap-on motor.
3. It has been developed primarily to test the human-rated LVM3 launch vehicle.
4. It provides flexibility to simulate different crew module recovery conditions.

Select the correct answer using the code given below:

(a) 1, 2 and 4 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2, 3 and 4

Answer: (a) 1, 2 and 4 only

Explanation

Statement 1:ISRO has developed the Sub-Orbital Launch Vehicle for Experiments (SOLVE) as a dedicated test platform for conducting various Gaganyaan-related experiments, particularly those associated with the crew module's recovery and deceleration systems.

Unlike orbital launch vehicles, SOLVE is designed for sub-orbital missions, where the vehicle reaches a specified altitude and returns to Earth without completing an orbit. **Hence, Statement 1 is correct.**

Statement 2: The solid stage of SOLVE has been derived from the Polar Satellite Launch Vehicle (PSLV) strap-on motor, with modifications made to meet Gaganyaan-specific requirements.

These modifications include:

- Development of a slow burn-rate propellant.
- Straight nozzle configuration.
- Secondary Injection Thrust Vector Control (SITVC). **Hence, Statement 2 is correct.**

Statement 3: The primary objective of SOLVE is not to test the human-rated LVM3 launch vehicle.

Instead, SOLVE is intended to:

- Validate the crew module's deceleration system.
- Test parachute deployment sequences.
- Simulate recovery conditions.
- Verify safe splashdown mechanisms.

The Human-Rated LVM3 (HLVM3) is being independently developed and tested as the launch

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vehicle for the actual Gaganyaan mission. **Hence, Statement 3 is incorrect.**

Statement 4: According to ISRO, SOLVE enables testing under a variety of mission conditions by carrying the crew module to different altitudes and validating recovery procedures.

This flexibility allows engineers to assess:

- Parachute deployment performance.
- Descent characteristics.
- Splashdown behaviour.
- Recovery system reliability. **Hence, Statement 4 is correct.**

Value Addition

SOLVE (Sub-Orbital Launch Vehicle for Experiments)

Feature	Details
Full Form	Sub-Orbital Launch Vehicle for Experiments
Developed By	Indian Space Research Organisation
Purpose	Gaganyaan test missions
Type	Solid motor-based Sub-Orbital Launch Vehicle
Test Location	Satish Dhawan Space Centre
First Ground Test	July 2026
Derived From	PSLV Strap-on Motor

SOLVE - Importance

SOLVE is intended to validate:

Deceleration System

The crew module must slow down safely before landing.

Parachute Deployment Sequence

A series of approximately **10 parachutes** are deployed in stages.

Crew Module Recovery

The module is recovered after splashdown in the sea.

Mission Safety

Crew safety is the most critical requirement in human spaceflight.

Gaganyaan Mission

Feature	Details
Mission Type	Human Spaceflight Programme
Implementing Agency	ISRO
Approved By	Government of India (2018)
Objective	Send Indian astronauts to Low Earth Orbit and safely return them
Orbit	Low Earth Orbit (~400 km)
Mission Duration	Up to 3 days (planned)
Launch Vehicle	Human-Rated LVM3 (HLVM3)
Crew Size	Up to 3 astronauts

Will make India the **fourth country** to independently demonstrate human spaceflight capability after:

1. Soviet Union / Russia
2. United States
3. China

Major Components of Gaganyaan

1. Crew Module (CM)

- Habitable capsule carrying astronauts.
- Equipped with life-support systems.
- Returns safely to Earth.

2. Service Module (SM)

- Provides propulsion.
- Supplies power and support systems.

3. Human-Rated LVM3

- Modified version of LVM3.
- Enhanced reliability and safety standards.

4. Crew Escape System (CES)

- Rapidly separates astronauts from the rocket during emergencies.

Key Gaganyaan Tests Conducted So Far

Test	Purpose
TV-D1 (Test Vehicle Demonstration-1)	Crew Escape System validation
IADT-01	Integrated Air Drop Test
IADT-02	Crew module recovery validation
SOLVE Ground Test	Validation of parachute and deceleration systems

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Mission MITRA	Behavioural and physiological studies of astronauts

Human Spaceflight Challenges

Launch Phase

- High acceleration loads.
- Extreme vibration.

Orbit Phase

- Microgravity adaptation.
- Life-support reliability.

Re-entry Phase

- Temperatures exceeding 1,500°C.
- Communication blackout.

Recovery Phase

- Precise parachute deployment.
- Safe splashdown and retrieval.

Previous Human Spaceflight Milestones of India

Rakesh Sharma Mission (1984)

- First Indian in space.
- Flew aboard Soviet spacecraft Soyuz T-11.
- Conducted under the Soviet Intercosmos Programme.

IN NEWS:ISRO conducts test of key component for Gaganyaan mission

58.Insolvency bankruptcy code

Q.Which of the following statements regarding the Insolvency and Bankruptcy Code (IBC), 2016 are correct?

1. Insolvency Professionals manage the affairs of the corporate debtor during the resolution process.
2. Information Utilities maintain and authenticate financial information relating to debt and default.
3. National Company Law Tribunal (NCLT) is the adjudicating authority for insolvency proceedings involving companies and LLPs.
4. Debt Recovery Tribunal (DRT) adjudicates insolvency proceedings relating to companies and LLPs.

Select the correct answer using the code given below:

(a) 1, 2 and 3 only
(b) 1 and 4 only
(c) 2 and 3 only
(d) 1, 2, 3 and 4

Answer: (a) 1, 2 and 3 only

Explanation

Statement 1: Under the Insolvency and Bankruptcy Code (IBC), when a Corporate Insolvency Resolution Process (CIRP) is initiated, the management of the Corporate Debtor (CD) is suspended and transferred to an Insolvency Professional (IP), who acts initially as an Interim Resolution Professional (IRP) and subsequently as a Resolution Professional (RP).

The IP:

- Takes control of the corporate debtor's operations.
- Preserves and protects the assets of the company.
- Collects and verifies claims of creditors.
- Constitutes the Committee of Creditors (CoC).
- Invites and evaluates resolution plans.

Thus, Insolvency Professionals play a central role in managing the affairs of the corporate debtor during insolvency proceedings.**Hence, Statement 1 is correct.**

Statement 2: Information Utilities (IUs) are specialized electronic repositories established under the IBC to collect, store, verify, and authenticate financial information submitted by creditors and debtors.

Their functions include:

- Maintaining records of loans and liabilities.
- Recording instances of default.
- Providing authenticated evidence during insolvency proceedings.
- Reducing disputes regarding debt claims.

The information maintained by IUs serves as reliable evidence before the adjudicating authorities.**Hence, Statement 2 is correct.**

Statement 3: Under the IBC:

- Insolvency cases relating to Companies and Limited Liability Partnerships (LLPs) are heard by the National Company Law Tribunal (NCLT).
- Appeals against NCLT orders lie before the National Company Law Appellate Tribunal (NCLAT).

The NCLT admits insolvency applications, supervises CIRP, approves resolution plans, and orders liquidation when necessary.**Hence, Statement 3 is correct.**

Statement 4:

The Debt Recovery Tribunal (DRT) does not deal with insolvency proceedings of companies and LLPs.

Under the IBC:

- **NCLT** → Companies and LLPs
- **DRT** → Individuals and Partnership Firms

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Therefore, the statement incorrectly assigns jurisdiction of companies and LLPs to DRT.**Hence, Statement 4 is incorrect.**

Value Addition

Four Pillars of the Insolvency and Bankruptcy Code (IBC)

Pillar	Function
Insolvency and Bankruptcy Board of India (IBBI)	Regulates insolvency ecosystem
Insolvency Professionals (IPs)	Manage insolvency resolution process
Information Utilities (IUs)	Store and authenticate financial information
Adjudicating Authorities	NCLT and DRT

Adjudicating Authorities under IBC

Category	Adjudicating Authority
Companies	NCLT
LLPs	NCLT
Individuals	DRT
Partnership Firms	DRT

. Under IBC, corporate insolvency matters are exclusively handled by NCLT, while DRT deals with non-corporate insolvency.

Insolvency Professionals (IPs)

Key Responsibilities

- Take over management of Corporate Debtor.
- Verify claims of creditors.
- Constitute Committee of Creditors (CoC).
- Run the company as a going concern.
- Invite and examine resolution plans.
- Assist Adjudicating Authority.

Regulator

- Registered and regulated by the Insolvency and Bankruptcy Board of India (IBBI).

Information Utilities (IUs)

Purpose

To ensure:

- Transparency
- Authenticity of financial information
- Faster insolvency resolution

Example

A loan default recorded and authenticated by an IU can serve as strong evidence before NCLT without lengthy disputes over documentation.

Corporate Insolvency Resolution Process (CIRP)

Feature	Provision
Minimum Default Threshold	₹1 crore
Initial Timeline	180 days
Extension	90 days
Maximum Limit (including litigation)	330 days
Approval of Resolution Plan	66% voting share of CoC

Committee of Creditors (CoC)

Composition

- Primarily Financial Creditors.
- Operational Creditors generally do not have voting rights.

Functions

- Appoint Resolution Professional.
- Evaluate Resolution Plans.
- Decide future of Corporate Debtor.
- Approve plan by at least 66% voting share.

Pre-Packaged Insolvency Resolution Process (PPIRP)

Feature	Details
Introduced For	MSMEs
Nature	Hybrid insolvency framework
Timeline	120 days
Objective	Faster and less disruptive resolution

In NEWS:Explained: can the ED attach a firm's assets after it enters insolvency?

59.Kaziranga National Park

Q. Which of the following statements regarding Kaziranga National Park are correct?

1. It was declared a UNESCO World Heritage Site in 1985.

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2. It contains the world's largest population of one-horned rhinoceroses.
3. The Gangetic River Dolphin is found in its ecosystem.
4. The River Manas flows through the core area of Kaziranga National Park.

Select the correct answer using the code given below:

(a) 1, 2 and 3 only
(b) 1 and 4 only
(c) 2 and 3 only
(d) 1, 2, 3 and 4

Answer: (a) 1, 2 and 3 only

Explanation

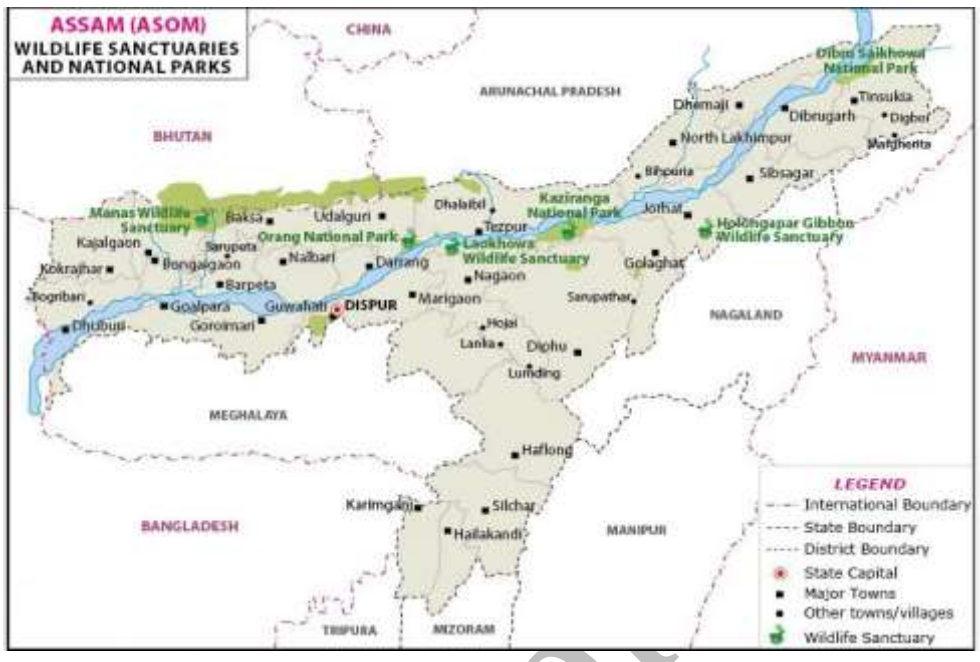
Statement 1: Kaziranga National Park was inscribed as a UNESCO World Heritage Site in 1985 due to its outstanding biodiversity and its importance as the primary habitat of the Greater One-Horned Rhinoceros. UNESCO recognized Kaziranga as one of the finest examples of ongoing ecological and biological processes in the Brahmaputra floodplain ecosystem.**Hence, Statement 1 is correct.**

Statement 2: Kaziranga is globally renowned for hosting the largest wild population of the Greater One-Horned Rhinoceros (Rhinoceros unicornis). Conservation efforts undertaken by the Government of Assam and wildlife authorities have transformed Kaziranga into the most important stronghold of this species. The park supports nearly two-thirds of the global population of the Greater One-Horned Rhinoceros.**Hence, Statement 2 is correct.**

Statement 3: The Gangetic River Dolphin inhabits the river systems associated with Kaziranga, particularly the Brahmaputra and its tributaries flowing through the park landscape. The floodplain wetlands, channels, and riverine habitats provide suitable ecological conditions for the species. The Gangetic River Dolphin is India's National Aquatic Animal.**Hence, Statement 3 is correct.**

Statement 4: The principal river associated with Kaziranga is the Diffalu River, a tributary of the Brahmaputra. Another tributary, Moradifalu, flows along the southern boundary of the park. The Manas River is associated with Manas National Park and not Kaziranga National Park.**Hence, Statement 4 is incorrect.**

Value Addition



Kaziranga National Park: At a Glance

Feature	Details
Location	Assam
UNESCO World Heritage Site	1985
National Park Status	1974
Landscape	Brahmaputra Floodplain
Major River	Diffalu
River Boundary	Brahmaputra River (North)
Southern Boundary	Karbi (Mikir) Hills
Tiger Reserve	Yes
Elephant Reserve	Yes

Geographical Significance

Kaziranga is situated:

- Between the Brahmaputra River in the north and the Karbi Hills in the south.
- In the floodplain ecosystem of the Brahmaputra Valley.
- It represents one of the largest undisturbed natural areas in Northeast India.

Important Rivers

River	Association
Diffalu	Flows through Kaziranga
Moradifalu	Southern boundary of Kaziranga
Brahmaputra	Northern boundary
Manas	Associated with Manas National Park

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Major Fauna of Kaziranga

Species	**Importance**
Greater One-Horned Rhinoceros	World's largest population
Royal Bengal Tiger	High tiger density
Asian Elephant	Major population
Wild Water Buffalo	Significant population
Eastern Swamp Deer	Important habitat
Hoolock Gibbon	India's only ape
Capped Langur	Found in park
Gangetic River Dolphin	Riverine ecosystem

Major Flora

Kaziranga is dominated by:

- Tall Elephant Grass
- Marshlands
- Tropical Moist Mixed Forests
- Semi-evergreen Forests
- Water Hyacinths
- Lotus
- Water Lilies

The grassland ecosystem supports large herbivore populations.

Conservation Status of Key Species

Species	**IUCN Status**
GreaterOne-Horned Rhinoceros	Vulnerable
Bengal Tiger	Endangered
Asian Elephant	Endangered
Gangetic River Dolphin	Endangered
Hoolock Gibbon	Endangered

Comparison: Kaziranga vs Manas

Feature	**Kaziranga**	**Manas**
State	Assam	Assam
UNESCO World Heritage Site	Yes	Yes
Famous For	One-Horned Rhino	Tiger, Golden Langur, Pygmy Hog
Major River	Diffalu	Manas River
Landscape	Floodplain Grassland	Himalayan Foothills

In NEWS:Kaziranga bids adieu to iconic patrol elephant Joymala

60.Pandavani

Q.The term "Pandavani", sometimes seen in the news, refers to:

(a) A traditional martial dance form of Rajasthan depicting episodes from the Ramayana.
(b) A classical Sanskrit theatre tradition performed exclusively in temples of Odisha.
(c) A folk storytelling tradition of Chhattisgarh that narrates episodes from the Mahabharata through music and dramatic performance.
(d) A tribal harvest festival celebrated in the Bastar region of Chhattisgarh.

Answer: (c) A folk storytelling tradition of Chhattisgarh that narrates episodes from the Mahabharata through music and dramatic performance.

Explanation

Option (a): Pandavani is not a martial dance form and is not associated with Rajasthan. It is primarily a narrative folk tradition centered on storytelling through music and performance.**Hence, Option (a) is incorrect.**

Option (b): Pandavani is neither a classical Sanskrit theatre tradition nor confined to temple performances. It belongs to the oral folk tradition of Chhattisgarh.**Hence, Option (b) is incorrect.**

Option (c): Pandavani is one of India's most distinctive oral storytelling traditions originating from Chhattisgarh. The term literally means "the story of the Pandavas." It involves the musical narration of episodes from the Mahabharata, often by a single lead performer who combines singing, storytelling, dialogue delivery, and dramatic enactment.**Hence, Option (c) is correct.**

Option (d): Pandavani is a performing art form and not a tribal harvest festival.**Hence, Option (d) is incorrect.**

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Value Addition
Pandavani

Feature	Details
Origin	Chhattisgarh
Meaning	"Story of the Pandavas"
Theme	Episodes from the Mahabharata
Nature	Oral storytelling and musical performance tradition
Performance Style	Singing, narration, acting and improvisation
Lead Performer	Usually a single storyteller supported by musicians

Key Features

- Based primarily on the Mahabharata, especially the exploits of the Pandavas.
- Relies heavily on oral tradition.
- The lead artist narrates, sings, and enacts multiple characters.
- Accompanied by instruments such as:
  - Harmonium
  - Tabla
  - Dholak
  - Manjira

The Tambura/Ektara as a Prop

A unique feature of Pandavani is the use of a tambura (or ektara).

During narration, it symbolically transforms into:

- Bhima's mace
- Arjuna's bow
- A sword
- A chariot
- Other objects required by the story

This creates dramatic effect without elaborate stage arrangements.

Styles of Pandavani

1. Vedamati Style

- Narration-oriented style.
- Performer generally remains seated.
- Greater emphasis on singing and storytelling.
- Minimal dramatic enactment.

2. Kapalik Style

- More theatrical and expressive.
- Performer moves across the stage.
- Uses gestures, voice modulation, and enactment.
- Battle scenes and character portrayals are dramatized.

Renowned Pandavani Artists

Teejan Bai

- Brought Pandavani to national and international audiences.
- Known for popularising the Kapalik style.
- Recipient of:
  - Padma Shri
  - Padma Bhushan
  - Padma Vibhushan

Ritu Verma

In NEWS: Chhattisgarh's Pandavani legend Teejan Bai dies at 70

61. Direct Seeded Rice

Q. Consider the following statements regarding Direct Seeded Rice (DSR):

1. In Direct Seeded Rice, rice seeds are sown directly in the field without raising and transplanting seedlings from a nursery.
2. Direct Seeded Rice requires continuous flooding of fields throughout the crop-growing period, similar to the conventional transplanted rice system.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (a) 1 only

Explanation

Statement 1:

Direct Seeded Rice (DSR) is a cultivation technique in which rice seeds are sown directly into the field, eliminating the need for raising seedlings in a nursery and subsequently transplanting them.

The method can be adopted through:

- Dry seeding
- Wet seeding
- Water seeding

By skipping nursery preparation and transplantation, DSR reduces labour requirements, saves time, and lowers cultivation costs.**Hence, Statement 1 is correct.**

Statement 2:

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A major advantage of DSR is that it does not require continuous flooding of fields, unlike the conventional puddled transplanted rice system.

DSR can reduce water consumption by nearly 30–50%, making it particularly suitable in areas facing:

- Water scarcity
- Delayed monsoons
- Erratic rainfall
- Groundwater depletion

Continuous flooding is generally associated with traditional transplanted rice cultivation and not with DSR. **Hence, Statement 2 is incorrect.**

Value Addition

Direct Seeded Rice (DSR):

Feature	Details
Method	Direct sowing of rice seeds in the field
Nursery	Not required
Transplantation	Not required
Water Requirement	Lower than conventional rice
Labour Requirement	Lower
Suitable Soils	Medium to heavy textured soils (Loam to Clay)
Unsuitable Soils	Light textured and poorly drained soils
Objective	Water-saving and cost-effective rice cultivation

Advantages of DSR

Water Conservation

- Saves up to 30–50% water compared to transplanted rice.
- Reduces pressure on groundwater resources.

Labour Saving

- Eliminates labour-intensive nursery raising and transplanting.
- Particularly useful where agricultural labour is scarce.

Reduced Cost of Cultivation

- Lower labour and irrigation costs.
- Increases farmers' profitability.

Climate Benefits

- Reduces methane emissions due to reduced flooding.
- Promotes climate-resilient agriculture.

Better Cropping Intensity

- Early sowing and harvesting facilitate timely planting of succeeding crops.

Women and Youth Friendly

- Mechanized DSR reduces drudgery and creates opportunities for custom-hiring services.

Challenges in DSR

Weed Infestation

- Most serious challenge.
- Absence of standing water allows weeds to grow aggressively.

Higher Seed Requirement

- Requires relatively higher seed rates.

Bird and Pest Damage

- Exposed seeds are vulnerable to birds, rodents, and insects.

Poor Crop Establishment

- Uneven germination may occur under adverse weather conditions.

Lodging Risk

- Plants may be more susceptible to lodging under certain conditions.

DSR vs Conventional Transplanted Rice

Parameter	Direct Seeded Rice	Transplanted Rice
Nursery Raising	Not Required	Required
Transplantation	Not Required	Required
Water Requirement	Lower	Higher
Labour Requirement	Lower	Higher
Cost of Cultivation	Lower	Higher
Methane Emissions	Lower	Higher
Weed Problem	Higher	Lower
Groundwater Use	Lower	Higher

Climate-Smart Agriculture Practices

- Direct Seeded Rice (DSR)
- System of Rice Intensification (SRI)
- Zero Tillage
- Precision Farming
- Micro-Irrigation (Drip & Sprinkler)

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Related Government Initiatives

- Indian Council of Agricultural Research promotes water-efficient rice cultivation techniques.
- The National Mission for Sustainable Agriculture encourages climate-resilient agricultural practices.

In NEWS:How direct seeded rice could be the future of rice farming

62.Indira Point

Q.Consider the following statements regarding Indira Point:

1. Indira Point is located on Great Nicobar Island.
2. It is located near the strategically important Six Degree Channel.
3. It is situated in the Gulf of Mannar Biosphere Reserve.
4. It lies south of Galathea Bay.

Which of the statements given above are correct?

(a) 1, 2 and 4 only
(b) 1 and 3 only
(c) 2, 3 and 4 only
(d) 1, 2, 3 and 4

Answer: (a) 1, 2 and 4 only

Explanation

Statement 1: Indira Point is situated at the southern tip of Great Nicobar Island, the largest island of the Nicobar group. It is the southernmost point of India and falls under the Nicobar district of the Andaman and Nicobar Islands. **Hence, Statement 1 is correct.**

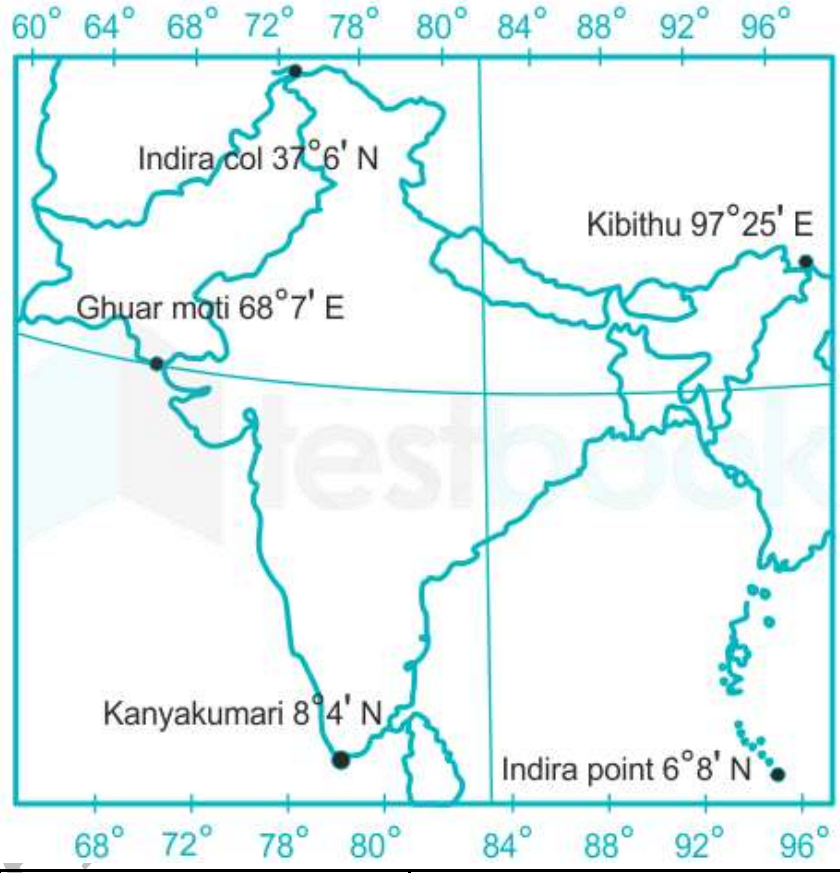
Statement 2: The Six Degree Channel lies between Great Nicobar Island and the Indonesian island of Sumatra. It is one of the busiest maritime routes in the eastern Indian Ocean through which a significant volume of global trade and energy shipments passes. Due to its proximity to this channel, Indira Point possesses immense strategic and geopolitical significance for India's maritime security and Indo-Pacific interests. **Hence, Statement 2 is correct.**

Statement 3: The Gulf of Mannar Biosphere Reserve is located along the southeastern coast of Tamil Nadu between India and Sri Lanka..Indira Point is situated on Great Nicobar Island, nearly 1,500 km away from the Gulf of Mannar region. Therefore, it has no geographical association with the Gulf of Mannar Biosphere Reserve. **Hence, Statement 3 is incorrect.**

Statement 4: Galathea Bay is located on the southeastern coast of Great Nicobar Island. Indira Point lies further south and represents the southernmost extremity of the island. Therefore, Indira Point is situated south of Galathea Bay. **Hence, Statement 4 is correct.**

Value Addition

Indira Point:



Feature	Details
Location	Great Nicobar Island
Union Territory	Andaman & Nicobar Islands
District	Nicobar District
Southernmost Point of India	Yes
Earlier Name	Pygmalion Point / Parsons Point
Renamed	1985
Named After	Indira Gandhi
Lighthouse Commissioned	1972
Tsunami Subsidence	~4.25 metres (2004)

Strategic Importance of Indira Point

- Located close to the Six Degree Channel, one of the world's busiest sea lanes.
- Near the Malacca Strait maritime route, which carries a substantial share of global trade.

- Approximately 145 km from Indonesia's Rondo Island.
- Important for:
  - Maritime surveillance
  - Coastal security
  - Indo-Pacific strategy
  - Monitoring international shipping routes

Important Channels Around Andaman & Nicobar Islands

Channel	**Separates**
Ten Degree Channel	Andaman Islands and Nicobar Islands
Six Degree Channel	Great Nicobar and Sumatra (Indonesia)
Duncan Passage	South Andaman and Little Andaman
Preparis Channel	Andaman Islands and Myanmar

Great Nicobar Island:

Feature	**Details**
Largest Island of Nicobar Group	Yes
Biosphere Reserve	Great Nicobar Biosphere Reserve
Important Bay	Galathea Bay
National Parks	Campbell Bay National Park and Galathea National Park
Important Species	Leatherback Turtle, Nicobar Megapode, Saltwater Crocodile

IN NEWS:Convention centre, museum: Nod sought for work at Indira Point, India's southernmost tip

63.Battery Management System

Q.Consider the following statements regarding Battery Management Systems (BMS):

1. A Battery Management System continuously monitors parameters such as voltage, current, and temperature of battery cells.
2. The primary purpose of cell balancing in a BMS is to ensure uniform charging and discharging of all cells in a battery pack.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Explanation

Statement 1:A Battery Management System (BMS) acts as the electronic control unit of a battery pack. It continuously monitors critical parameters such as:

- Cell voltage
- Current flow
- Temperature
- State of Charge (SoC)
- State of Health (SoH)

This real-time monitoring helps ensure efficient operation and prevents battery damage due to overheating, overcharging, or deep discharge. **Hence, Statement 1 is correct.**

Statement 2: In a large lithium-ion battery pack, individual cells may charge and discharge at different rates. To address this, the BMS performs cell balancing, which:

- Equalizes charge levels among cells
- Prevents overcharging of stronger cells
- Prevents excessive discharge of weaker cells
- Improves battery life and performance .As a result, the entire battery pack functions as a single uniform power source. **Hence, Statement 2 is correct.**

Value Addition

Battery Management System (BMS)

Feature	**Details**
Purpose	Monitor, protect and optimize battery performance
Common Use	Electric Vehicles (EVs), Energy Storage Systems, Electronics
Battery Type	Mainly Lithium-ion Batteries
Function	Acts as the "Brain" of the Battery Pack
Key Parameters Monitored	Voltage, Current, Temperature, SoC, SoH

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Major Functions of BMS

1. Cell Monitoring

Tracks the condition of each battery cell individually.

2. Cell Balancing

Ensures uniform charging and discharging across all cells.

3. Thermal Management

Monitors temperature and prevents overheating.

4. Overcharge Protection

Disconnects the charging circuit if voltage exceeds safe limits.

5. Over-discharge Protection

Prevents battery damage due to excessive discharge.

6. Short-Circuit Protection

Cuts off power during electrical faults.

Term	Meaning
SoC (State of Charge)	Remaining battery charge (%)
SoH (State of Health)	Overall health and degradation level of battery
Cell Balancing	Equalizing charge among cells
Thermal Runaway	Uncontrolled rise in battery temperature leading to fire/explosion
MOSFET	Electronic switch used for protection and circuit control

Why BMS is Important for EVs?

- Enhances battery life
- Improves driving range
- Prevents battery fires
- Enables fast charging
- Provides real-time diagnostics
- Improves passenger safety

IN NEWS:E-rickshaw battery 'hacks' and Bluetooth BMS vulnerabilities

64. PROTAC Technology

Q.The term "PROTAC", recently seen in the news, refers to:

(a) A gene-editing technology that modifies defective DNA sequences using CRISPR-based enzymes.
(b) A targeted protein degradation technology that removes specific disease-causing proteins from cells by recruiting the cell's natural protein disposal machinery.
(c) A novel vaccine platform that uses engineered proteins to stimulate immune responses against cancer.
(d) A biotechnology technique used to artificially increase protein production in genetically modified organisms.

Answer: (b) A targeted protein degradation technology that removes specific disease-causing proteins from cells by recruiting the cell's natural protein disposal machinery.

Explanation

Option (a):PROTAC is not a gene-editing technology and does not alter DNA sequences like CRISPR-Cas systems. **Hence, Option (a) is incorrect.**

Option (b):PROTAC (Proteolysis-Targeting Chimera) is a targeted protein degradation technology designed to eliminate specific disease-causing proteins from cells.

A PROTAC molecule has two functional ends:

- One binds to the target protein.
- The other binds to an E3 ubiquitin ligase, an enzyme involved in the cell's protein degradation pathway.

This interaction marks the target protein for destruction by the cell's proteasome, thereby removing the protein entirely rather than merely blocking its activity.

The first FDA-approved PROTAC-based drug is Vepdegestrant (2025) for certain types of advanced breast cancer.**Hence, Option (b) is correct.**

Option (c):PROTAC technology is not a vaccine platform and is unrelated to immune-stimulation strategies used in vaccines. **Hence, Option (c) is incorrect.**

Option (d):PROTACs are designed to degrade proteins, not increase their production. **Hence, Option (d) is incorrect.**

Value Addition

PROTAC Technology

Feature	Details
Full Form	Proteolysis-Targeting Chimera
Developed	Early 2000s
Mechanism	Targeted Protein Degradation
Key Enzyme Recruited	E3 Ubiquitin Ligase

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Cellular Machinery Used	Ubiquitin-Proteasome System
First FDA-approved PROTAC Drug	Vepdegestrant (2025)
Major Application	Cancer Therapy

How PROTAC Works

Step 1: PROTAC binds to a disease-causing protein.

Step 2: It simultaneously recruits an E3 ligase.

Step 3: The target protein is tagged with ubiquitin molecules.

Step 4: The proteasome recognizes the tag and destroys the protein.

Step 5: The PROTAC molecule is released and can repeat the process.

Why is PROTAC Important?

- Removes harmful proteins instead of merely inhibiting them.
- Can target previously "undruggable" proteins.
- May work at lower doses because one molecule can degrade multiple protein copies.
- Has applications in:
  - Cancer
  - Neurodegenerative diseases
  - Inflammatory disorders
  - Muscle diseases

In NEWS:Why FDA nod for vepdegestrant is a major shift in cancer therapy

65.QUASARS

Q.Consider the following pairs:

Objects in Space	Description
1. Quasar	Active Galactic Nucleus powered by a supermassive black hole
2. Magnetar	Neutron star with an extremely strong magnetic field
3. Nebula	Collection of planets orbiting a star

How many of the above pairs are correctly matched?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two

Explanation

Pair 1: Quasar — Active Galactic Nucleus powered by a supermassive black hole

A Quasar (Quasi-Stellar Radio Source) is an extremely luminous type of Active Galactic Nucleus (AGN).

- At the center of a galaxy lies a supermassive black hole.
- When large amounts of gas and dust fall towards the black hole, they form an accretion disk.
- Friction and gravitational forces heat the disk to millions of degrees, producing enormous amounts of energy.
- As a result, quasars become among the brightest objects in the universe, often outshining their host galaxies. **Hence, Pair 1 is correctly matched.**

Pair 2: Magnetar — Neutron star with an extremely strong magnetic field

A Magnetar is a special type of neutron star possessing the strongest magnetic fields known in the universe.

- Formed after the collapse of a massive star.
- Magnetic field strength can be about 10^{14}–10^{15} gauss, trillions of times stronger than Earth's magnetic field.
- Magnetars emit intense X-rays and gamma rays.

Hence, Pair 2 is correctly matched.

Pair 3: Nebula — Collection of planets orbiting a star

A Nebula is a vast cloud of gas and dust in interstellar space.

- Nebulae are often regions where new stars are born.
- They are commonly called stellar nurseries.
- A collection of planets orbiting a star is known as a planetary system, not a nebula. **Hence, Pair 3 is incorrectly matched.**

Value Addition: Quasars

Quasar = Quasi-Stellar Radio Source

The name originated because early astronomers observed them as star-like objects that emitted powerful radio waves.

A quasar is actually:

An extremely luminous Active Galactic Nucleus (AGN) powered by a feeding supermassive black hole.

Relationship Between AGN and Quasars

- All Quasars are AGNs.
- All AGNs are not Quasars.

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Quasars are simply the most energetic and luminous class of AGNs.

Hierarchy

Galaxy → Active Galactic Nucleus (AGN) → Quasar (most luminous AGN)

Quasars Formation:

Step 1 : A galaxy contains a supermassive black hole at its centre.

Step 2 : Huge quantities of gas and dust fall towards the black hole.

Step 3 : The material forms a rapidly rotating accretion disk.

Step 4 : Friction and compression heat the disk to millions of degrees.

Step 5 : Enormous radiation is released in: Visible light, X-rays ,Ultraviolet rays , Radio waves

Step 6 : Some material is ejected as powerful relativistic jets.

Quasars Importance:

1. Study the Early Universe

Most quasars are located billions of light-years away.

2. Understand Galaxy Formation

They provide clues about:

- Early galaxies
- Supermassive black holes
- Cosmic evolution

3. Probe the Epoch of Reionisation

Ancient quasars help scientists study the period when:

- First stars formed
- First galaxies appeared
- Cosmic Dark Ages ended

Euclid Telescope and Recent Discovery

Recently, the Euclid Space Telescope discovered 31 ancient quasars, including the oldest known quasars.

Key facts:

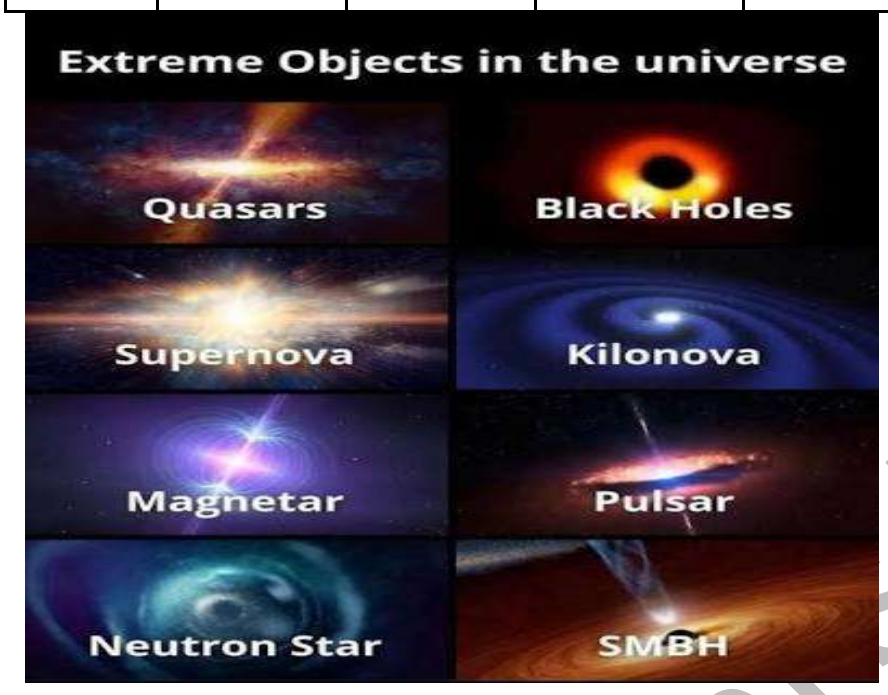
- Light originated when the universe was only about 670 million years old.
- This corresponds to roughly 5% of the present age of the universe.
- Helps scientists understand the Epoch of Reionisation.

Object	**Description**
Quasar	Brightest type of AGN
Pulsar	Rapidly rotating neutron star
Magnetar	Neutron star with extremely strong magnetic field
Nebula	Cloud of gas and dust
Black Hole	Region where gravity prevents escape of light
White Dwarf	Remnant of low-mass star
Red Giant	Expanded late-stage star
Protostar	Forming star inside a nebula

Object in Space	**meaning**	**Formation / Origin**	**Key Characteristics**	**Value Add**
Quasar	Extremely luminous Active Galactic Nucleus (AGN) powered by a supermassive black hole	Forms when huge amounts of gas and dust fall into a supermassive black hole, creating a hot accretion disk and energetic jets	Among the brightest objects in the universe; can outshine entire galaxies; emits radio waves, visible light, UV rays, X-rays, and gamma rays	All Quasars are AGNs, but all AGNs are not Quasars; Quasar = *Quasi-Stellar Radio Source*; used to study the early universe and Epoch of Reionisation
Magnetar	A special type of neutron star with an extremely powerful magnetic field	Formed after the collapse of a massive star during a supernova explosion	Magnetic field strength around 10^{14}–10^{15} gauss; emits intense X-rays and gamma rays; strongest known magnets in	Type of neutron star; examples include Soft Gamma Repeaters (SGRs) and Anomalous X-ray Pulsars (AXPs)

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			the universe	
Nebula	Vast cloud of gas and dust in interstellar space	Formed from remnants of dying stars or from interstellar gas clouds	Birthplace of stars (stellar nursery); contains hydrogen, helium, and dust particles	Example: Orion Nebula; can be Emission, Reflection, Dark, or Planetary Nebulae



In NEWS:Oldest quasars found deepen 'perplexing'space mystery

66.Coastal Regulation Zones(CRZ)

Q.Consider the following statements regarding the Coastal Regulation Zone (CRZ):

1. Under the CRZ Notification, 2019, CRZ-I is subdivided into CRZ-IA and CRZ-IB.
2. The Coastal Regulation Zone Notification, 2011 established Coastal Zone Management Authorities (CZMAs) at the national, state and district levels.
3. Under the CRZ Notification, 2019, the No Development Zone (NDZ) in densely populated rural areas was reduced to 50 metres from the High Tide Line.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation

Statement 1:

The CRZ Notification, 2019 divided CRZ-I into:

- CRZ-IA → Ecologically Sensitive Areas (ESAs) such as mangroves, coral reefs, sand dunes, turtle nesting grounds, etc.
- CRZ-IB → Intertidal zone between the High Tide Line (HTL) and Low Tide Line (LTL). **Hence, Statement 1 is correct.**

Statement 2:

The CRZ Notification, 2011 strengthened coastal governance by establishing:

- National Coastal Zone Management Authority (NCZMA)
- State Coastal Zone Management Authorities (SCZMAs)
- District-level coastal management mechanisms

These authorities oversee implementation and monitoring of CRZ provisions.**Hence, Statement 2 is correct.**

Statement 3:

Based on the recommendations of the Shailesh Nayak Committee, the CRZ Notification, 2019 reduced the No Development Zone (NDZ) in densely populated rural coastal areas (CRZ-IIIA) from 200 m to 50 m from the HTL.

- Population density ≥ 2161 persons/sq km → NDZ = 50 m
- Less populated areas (CRZ-IIIB) → NDZ = 200 m**Hence, Statement 3 is correct.**

Value Addition

CRZ Classification (2019)

Zone	**Area Covered**
CRZ-IA	Ecologically Sensitive Areas
CRZ-IB	Intertidal Zone (HTL–LTL)
CRZ-II	Developed Urban Coastal Areas
CRZ-IIIA	Densely Populated Rural Areas (NDZ = 50 m)
CRZ-IIIB	Less Populated Rural Areas (NDZ = 200 m)
CRZ-IVA	Water area up to 12 nautical miles
CRZ-IVB	Tidal influenced water bodies

High Tide Line (HTL) vs Low Tide Line (LTL)

Term	Meaning
HTL	Maximum reach of seawater during spring tide
LTL	Lowest reach of seawater during low tide
Intertidal Zone	Area between HTL and LTL

Shailesh Nayak Committee (2014)

Key Recommendations:

- Reduce NDZ in densely populated rural areas to 50 m.
- Promote tourism infrastructure.
- Decentralize coastal management.
- Identify and protect Ecologically Sensitive Areas.
- Allow certain public-interest infrastructure projects in coastal zones.

included in CRZ-IA:

- Mangroves
- Coral reefs
- Sand dunes
- Seagrass beds
- Turtle nesting grounds
- National Parks and Marine Protected Areas

IN NEWS: T.N. defends CRZ nod for Mamallan reservoir project

67.MAHENDRAGIRI

Q.Consider the following statements regarding INS Mahendragiri (F38):

1. INS Mahendragiri is the sixth indigenous stealth frigate under Project 17A and belongs to the Nilgiri-class.
2. It is designed by Mazagon Dock Shipbuilders Limited (MDL) and built by the Indian Navy's Warship Design Bureau.
3. It is powered by a Combined Diesel or Gas (CODOG) propulsion system.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (c) 1 and 3 only

Explanation

Statement 1 : INS Mahendragiri (F38) is the sixth Project 17A indigenous guided-missile stealth frigate of the Indian Navy.

- It belongs to the Nilgiri-class (Project 17A).
- It is designed as a multi-role stealth frigate capable of anti-air, anti-surface and anti-submarine warfare.
- It will strengthen India's maritime capabilities in the Indian Ocean Region (IOR) and Indo-Pacific.

Hence, Statement 1 is correct.

Statement 2 : Warship Design Bureau (WDB), Indian Navy - Designed the ship.

- Mazagon Dock Shipbuilders Limited (MDL), Mumbai - Built the ship.

Project 17A showcases India's growing expertise in indigenous warship design and construction under the Aatmanirbhar Bharat initiative. **Hence, Statement 2 is incorrect.**

Statement 3 : INS Mahendragiri is powered by a Combined Diesel or Gas (CODOG) propulsion system.

Under CODOG:

- Diesel engines are used for economical long-range cruising.
- Gas turbines are used for high-speed combat operations.
- Both systems do not operate simultaneously.

This provides a balance between endurance and speed. **Hence, Statement 3 is correct.**

Value Addition

Project 17A Frigates

Frigate	Pennant Number
INS Nilgiri	F34
INS Himgiri	F35
INS Udaygiri	F36
INS Dunagiri	F37
INS Mahendragiri	F38
INS Taragiri	F41
INS Vindhyagiri	F42

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Key Features of INS Mahendragiri

Feature	Details
Project	Project 17A
Class	Nilgiri-class
Type	Guided Missile Stealth Frigate
Indigenous Content	More than 75%
Builder	MDL, Mumbai
Designer	Warship Design Bureau (Indian Navy)
Propulsion	CODOG
Motto	Mighty – Majestic – Matchless

Major Weapon Systems

- BrahMos Supersonic Cruise Missile
- Barak-8 Long Range Surface-to-Air Missile
- Heavy-weight Torpedoes
- Anti-Submarine Rocket Launchers
- Advanced Main Gun
- Electronic Warfare Suite
- Indigenous Combat Management System (CMS)

Project 17A Importance:

- Enhances Blue Water Navy capabilities.
- Supports India's role as the Preferred Security Partner in the Indian Ocean Region.
- Strengthens Aatmanirbhar Bharat in defence manufacturing.
- Employs thousands of MSMEs in the defence supply chain.
- Incorporates advanced stealth technology and automation.

CODOG vs CODAG

Propulsion Type	Expansion	Feature
CODOG	Combined Diesel Or Gas	Either diesel or gas turbine operates
CODAG	Combined Diesel And Gas	Diesel and gas turbine can operate together

Previous Indian Navy Indigenous Warship Projects

Project	Class
Project 15A	Kolkata-class Destroyers
Project 15B	Visakhapatnam-class Destroyers
Project 17	Shivalik-class Frigates
Project 17A	Nilgiri-class Frigates
Project 75	Kalvari-class Submarines

IN NEWS:INDIAN NAVY SET TO COMMISSION MAHENDRAGIRI, THE SIXTH PROJECT 17A INDIGENOUS STEALTH FRIGATE

68.CANSCORA AGNI

Q.The term 'Canscora agni', sometimes seen in the news, refers to:

(a) A newly discovered species of dwarf herb adapted to fire-prone savanna ecosystems
(b) A rare medicinal tree species endemic to the Western Ghats rainforests
(c) A newly identified grass species found in the Thar Desert
(d) An invasive alien plant species threatening Indian savannas

Answer: (a) A newly discovered species of dwarf herb adapted to fire-prone savanna ecosystems

Explanation:

Canscora agni is a newly discovered species of flowering plant belonging to the genus *Canscora* of the family Gentianaceae. It was recently identified in the lateritic plateaus of the northern Western Ghats, particularly in fire-prone savanna-like ecosystems.

- Researchers discovered and described a new species named Canscora agni.
- The species name "agni" (meaning *fire* in Sanskrit) reflects its remarkable adaptation to habitats that experience periodic fires.
- The discovery highlights the ecological significance and biodiversity value of India's tropical savannas, which are often mistaken for degraded forests.

Key Features of *Canscora agni*

- It is a small annual dwarf herb.

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- Found in open savanna ecosystems on lateritic plateaus.
- Exhibits adaptations that allow it to survive and regenerate in areas affected by seasonal fires.
- Flowers during the post-monsoon season.
- Represents a specialized flora associated with India's threatened savanna landscapes.

Significance of the Discovery

- Demonstrates the rich and understudied biodiversity of Indian savannas.
- Reinforces the need to conserve savanna ecosystems, which support unique plant and animal communities.

Challenges the perception that all non-forested landscapes are degraded lands requiring afforestation.**Hence, Option (a) is correct.**

(b) A rare medicinal tree species endemic to the Western Ghats rainforests

Canscora agni is a dwarf herb, not a tree. It is associated with savanna habitats, not rainforest ecosystems.**Hence, Option (b) is incorrect.**

(c) A newly identified grass species found in the Thar Desert

It belongs to the genus Canscora and is not a grass species. It was discovered in the Western Ghats region, not the Thar Desert.**Hence, Option (c) is incorrect.**

(d) An invasive alien plant species threatening Indian savannas

It is a native species newly discovered in India. There is no evidence that it is invasive or alien.**Hence, Option (d) is incorrect.**

VALUE ADDITION:

Savanna Ecosystems

- Characterized by a mixture of grasses and scattered trees/shrubs.
- Maintained by seasonal fires and grazing.
- Support high levels of biodiversity and endemism.
- In India, savanna-like ecosystems occur in parts of the Western Ghats, Deccan Plateau, and central India.

Genus *Canscora*

- Belongs to the Gentianaceae family.
- Mostly consists of small herbaceous flowering plants.
- Several species are found in tropical Asia.

In NEWS:Meet Canscora agni, a new plant named to highlight the role of fire in the conservation of Indian Savannas

69.United Nations Security Council(UNSC)

Q.Consider the following statements regarding the United Nations Security Council (UNSC):

1. The UNSC is the only principal organ of the United Nations whose resolutions are legally binding on member states.
2. India has been elected as a non-permanent member of the UNSC eight times.
3. A retiring non-permanent member of the UNSC is eligible for immediate re-election.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two

Explanation:

Statement 1: The United Nations Security Council (UNSC) is one of the six principal organs of the United Nations and is entrusted with the primary responsibility for maintaining international peace and security. Under the UN Charter, the UNSC can adopt resolutions that are legally binding on all member states. Other principal organs such as the General Assembly generally issue recommendations that are not legally binding. Therefore, the UNSC is unique in possessing the authority to issue binding resolutions**. Hence, Statement 1 is correct.**

Statement 2: India is not a permanent member of the UNSC. However, it has been elected as a non-permanent member eight times. India's terms have been:

- 1950–51
- 1967–68
- 1972–73
- 1977–78
- 1984–85
- 1991–92
- 2011–12
- 2021–22

This reflects India's active engagement with the United Nations and its growing global role. **Hence, Statement 2 is correct.**

Statement 3: The ten non-permanent members of the UNSC are elected by the UN General Assembly for a term of two years. According to UNSC rules, a retiring non-permanent member cannot be immediately re-elected after the completion of its term. This provision ensures broader regional representation and rotation among member states.
Hence, Statement 3 is incorrect.

Correct Answer: (b) Only two

Value Addition:

Establishment

- Established in 1945 under the UN Charter.
- One of the six principal organs of the United Nations.

Headquarters

- Located in New York, USA.

Composition

- **Total Members: 15**
  - **5 Permanent Members (P5)**
    - China
    - France
    - Russia
    - United Kingdom
    - United States
  - **10 Non-Permanent Members**
    - Elected for a two-year term by the UN General Assembly.

Voting

- Each member has one vote.
- Procedural matters require 9 votes.
- Substantive matters require 9 votes including concurrence of P5 members.

Veto Power

- Available only to the P5 members.
- A negative vote by any permanent member blocks a substantive resolution irrespective of global support.

Why is UNSC Reform Demanded?

1. The UNSC reflects the geopolitical realities of 1945 rather than the contemporary world order.
2. Africa has 55 member states but no permanent representation in the UNSC.
3. China is currently the only P5 member opposing India's permanent membership bid.
4. Since 1945, UN membership has increased from 51 to 193 states, while UNSC membership expanded only once in 1965.

Important Groups Related to UNSC Reforms

G4 Group

- **Members:**
  - India
  - Brazil
  - Germany
  - Japan

The G4 countries support each other's candidature for permanent membership of the UNSC. Hence, the group advocates comprehensive Security Council reforms.

L.69 Group

It is a coalition of developing countries advocating:

- Expansion of permanent membership.
- Expansion of non-permanent membership.
- Better representation of developing countries, Africa, and Small Island Developing States (SIDS).

Hence, the L.69 Group is a pro-reform coalition.

Coffee Club (Uniting for Consensus)

It is an informal grouping of countries opposing the expansion of permanent membership.

Examples:

- Pakistan opposes India's candidature.
- Italy and Spain oppose Germany's candidature.

Hence, the Coffee Club remains a significant obstacle to UNSC reforms.

In NEWS:Jaishankar gearing up to lead India's campaign for non-permanent UNSC seat

70.Himalayan Pangolin

Q.Consider the following statements regarding the Himalayan Pangolin:

1. It is found in Nepal, South Tibet, and Northeast India.
2. It is a diurnal mammal.
3. Females usually give birth to a single offspring.
4. It is covered with overlapping keratin scales.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (c) Only three

Statement 1: The Himalayan Pangolin (*Manis aurita*) is distributed across the southern Himalayan foothills. Confirmed populations have been reported from Nepal, South Tibet, and Northeast India, including parts of Assam. Its evolutionary isolation was facilitated by geographical barriers such as the Brahmaputra River basin and the Arakan Mountains.
Hence, Statement 1 is correct.

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Statement 2: The Himalayan Pangolin is a strictly nocturnal mammal. It remains hidden in underground burrows during the day and emerges at night to forage for ants and termites. Like other pangolins, it relies on nocturnal activity to avoid predators and environmental stress. Therefore, it is not a diurnal animal. **Hence, Statement 2 is incorrect.**

Statement 3: Pangolins are characterized by a very low reproductive rate. Female Himalayan Pangolins generally give birth to a single offspring after a relatively long gestation period. The newborn remains dependent on its mother and is often carried on her tail until its scales harden. **Hence, Statement 3 is correct.**

Statement 4: One of the most distinctive features of the Himalayan Pangolin is the presence of overlapping keratin scales covering most of its body. These scales serve as protective armor against predators. When threatened, the animal curls into a tight ball, exposing only its scales. Keratin is the same protein found in human hair and nails. **Hence, Statement 4 is correct.**

Ecological Importance

- Controls populations of ants and termites.
- Maintains ecological balance in forest ecosystems.
- Acts as a natural pest regulator.

Threats

- Illegal wildlife trade for scales and meat.
- Habitat degradation.
- Low reproductive rate, making population recovery difficult.
- Considered among the most trafficked mammals in the world.

Himalayan Pangolin vs Chinese Pangolin

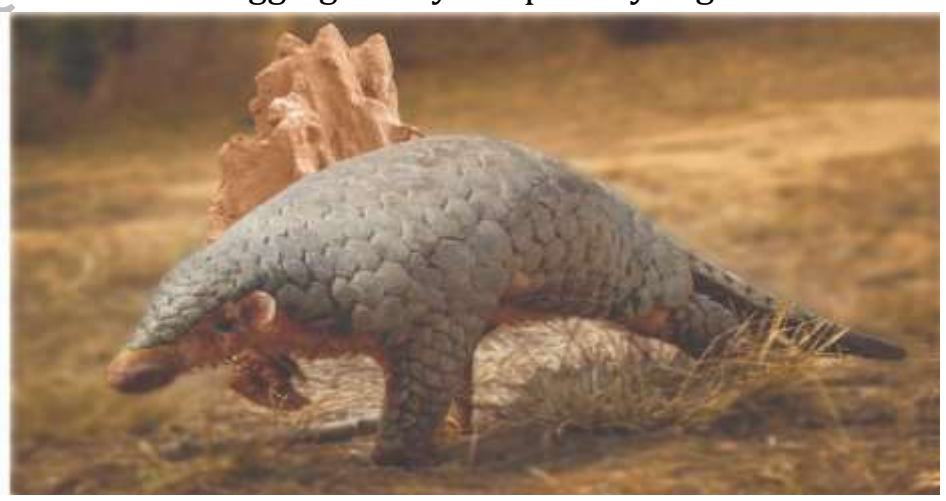
Feature	**Himalayan Pangolin (*Manis aurita*)**	**Chinese Pangolin (*Manis pentadactyla*)**
Taxonomic Status	Distinct species	Distinct species
Distribution	Nepal, South Tibet, Northeast India	China, Taiwan, parts of Southeast Asia
External Ears	Very small, highly reduced ear pinnae	Larger and more prominent ear pinnae
Nasal Structure	Shorter and broader nasal bones	Longer and narrower nasal bones
Evolutionary Refuge	Western Himalayan region	Southern Chinese forests
Body Size	Generally larger body and skull	Comparatively smaller
Inbreeding Pattern	Severe localized inbreeding spikes	More widespread baseline inbreeding
Habitat	Himalayan foothills	Subtropical and temperate forests of China

Natural Barriers Responsible for Speciation

- Brahmaputra River Basin
- Arakan Mountains These barriers prevented gene flow and contributed to the independent evolution of the Himalayan Pangolin.

Special Adaptations

- Enlarged olfactory bulb.
- Highly developed nasal turbinates.
- Long curved claws for digging termite mounds.
- Rolls into a defensive ball.
- Excellent digging ability but poor eyesight.



In NEWS:Himalayan pangolin confirmed as a distinct species, was earlier classified as subspecies of Chinese pangolin

71.Primary Agricultural Credit Societies(PACS)

Q.With reference to Primary Agricultural Credit Societies (PACS), consider the following statements:

1. PACS constitute the lowest tier in the short-term cooperative credit structure in India.
2. PACS provide short-term and medium-term agricultural loans to farmers.

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3. Membership is restricted only to large and medium farmers to ensure financial viability.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only
Explanation
Statement 1: PACS are village-level cooperative credit societies and form the last mile of the three-tier Short-Term Cooperative Credit Structure (STCCS):PACS directly interact with farmers and provide credit at the grassroots level.
State Cooperative Banks (SCBs) → District Central Cooperative Banks (DCCBs) → PACS.**Hence statement 1 is correct**
Statement 2: PACS provide:
- Short-term crop loans
- Medium-term loans for agricultural and allied activities
- Credit for seeds, fertilizers, irrigation, farm equipment, etc.

Their primary objective is to ensure timely and affordable institutional credit to farmers.**Hence statement 2 is correct**
Statement 3:PACS are cooperative institutions whose membership is open to eligible farmers within their area of operation.
Members include:
- Marginal farmers
- Small farmers
- Medium farmers
- Large farmers

Restricting membership to only large and medium farmers would violate the cooperative principles of inclusiveness and democratic participation.**Hence statement 3 is incorrect**
Value Addition
Three-Tier Cooperative Credit Structure

Level	**Institution**
State	State Cooperative Bank (SCB)
District	District Central Cooperative Bank (DCCB)
Village	Primary Agricultural Credit Society (PACS)

PACS - Importance :
- Last-mile delivery of agricultural credit
- Minimal paperwork and faster loan processing
- Accessible to small and marginal farmers
- Strengthen financial inclusion in rural areas
- Support implementation of government schemes

Recent Developments
- Ministry of Cooperation aims to establish PACS in every village by 2027.
- PACS are being computerized to improve transparency and efficiency.
- Government has approved 20+ new business activities for PACS, including:
  - Dairy
  - Fisheries
  - Storage and warehousing
  - Common Service Centres (CSCs)
  - LPG distribution
  - Custom Hiring Centres

Prelims Fact
- First PACS: 1904
- Registered PACS: Over 1 lakh
- Functional PACS: Around 63,000
- Member Base: More than 13 crore farmers

In NEWS:Grand ambitions -India's cooperative sector can serve as a worker-centric model of development

72.Indian Yak

Q.Consider the following statements regarding Yak:

1. The Indian yak is primarily found in the Trans-Himalayan and other high-altitude Himalayan regions, generally at elevations above 8,000 feet.
2. The wild yak (*Bos mutus*) is endemic to India and is primarily distributed in Arunachal Pradesh and Sikkim.
3. Scientists of the ICAR–National Research Centre on Yak have developed an IoT-based smart system for geo-fencing, health monitoring and stress prediction in yaks.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two

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Explanation

Statement 1:

The Indian yak, often called the "Ship of the Himalayas", is an important livestock species of the Trans-Himalayan and Himalayan regions.

It is generally found at elevations above 8,000 feet in:

- Ladakh
- Arunachal Pradesh
- Sikkim
- Himachal Pradesh
- Uttarakhand

Yaks are highly adapted to:

- Low oxygen conditions
- Extreme cold
- Rugged mountainous terrain **Hence, Statement 1 is correct.**

Statement 2:

The wild yak (*Bos mutus*) is not endemic to India.

Its primary natural distribution lies in:

- Tibet (China)
- Mongolia
- Russia

India mainly has domesticated yak populations (*Bos grunniens*) reared by high-altitude pastoral communities. **Hence, Statement 2 is incorrect.**

Statement 3:

Scientists from the ICAR–National Research Centre on Yak (NRC-Y), Dirang, Arunachal Pradesh, have developed an IoT-based smart collar system.

The system enables:

- Geo-fencing of yak herds
- Real-time movement tracking
- Health monitoring
- Stress prediction

The technology is especially useful in remote Himalayan grazing areas and near international border regions where physical monitoring is difficult.

Hence, Statement 3 is correct.

Value Addition

Yak

Feature	Details
Scientific Name (Domestic Yak)	*Bos grunniens*
Scientific Name (Wild Yak)	*Bos mutus*
Genus	*Bos*
Common Name	Ship of the Himalayas
Habitat	Trans-Himalayan and Himalayan high-altitude regions
Altitude	Generally above 8,000 feet
Major States in India	Ladakh, Arunachal Pradesh, Sikkim, Himachal Pradesh, Uttarakhand
Economic Importance	Milk, meat, fibre, dung fuel, transportation

Importance of Yak

Livelihood Support

- Source of milk and dairy products
- Provides meat and fibre
- Used as a pack animal in mountainous terrain
- Dung serves as fuel and manure

Ecological Importance

- Adapted to fragile mountain ecosystems
- Supports sustainable pastoralism
- Important component of high-altitude biodiversity

Scientific Importance

- Model species for studying:
  - Cold adaptation
  - High-altitude physiology
  - Hypoxia tolerance

ICAR–National Research Centre on Yak (NRC-Y)

Feature	Details
Established	1989
Location	Dirang, Arunachal Pradesh
Parent Organisation	ICAR
Focus Areas	Breeding, nutrition, health, genetic conservation

Recent Developments

IoT-based Smart Monitoring System

Features:

- Geo-fencing
- Health monitoring
- Stress prediction
- Real-time animal tracking

Benefits:

- Helps yak herders (Brokpas)
- Reduces livestock losses
- Enables early disease detection
- Useful in border and remote grazing areas

Yak Genome Research

Scientists used:

- Long-read sequencing technology
- Advanced bioinformatics tools

Applications:

- Disease resistance studies
- Fertility improvement
- Milk quality enhancement
- Climate adaptation research

Challenges Faced in Yak Rearing

- Harsh Terrain
- Climate Change
- Declining Yak Population
- Healthcare Constraints

IN NEWS:Scientists develop a health tracker for high-altitude yaks in Himalayas

73.Mission Drishti

Q.The term "Mission Drishti," recently seen in the news, is best described as:

(a) A deep-sea exploration mission initiated by the Indian Navy for maritime mapping.
(b) India's first indigenous space-based solar observatory to study coronal mass ejections.
(c) The world's first multi-sensor OptoSAR Earth observation satellite developed by a private start-up.
(d) A specialized drone-surveillance grid established along the international borders for anti-infiltration.

Answer: (c) The world's first multi-sensor OptoSAR Earth observation satellite developed by a private start-up.

Explanation

Option (a): Mission Drishti is not a deep-sea exploration mission of the Indian Navy. It is an Earth Observation (EO) satellite designed for remote sensing and geospatial applications. **Hence, Option (a) is incorrect.**

Option (b): Mission Drishti is not a space-based solar observatory. It is neither intended to study the Sun nor phenomena such as coronal mass ejections. **Hence, Option (b) is incorrect.**

Option (c): Mission Drishti is:

- The world's first multi-sensor Earth Observation satellite.
- The world's first OptoSAR satellite.
- Developed by Bengaluru-based space start-up GalaxEye.
- India's largest privately built Earth observation satellite.

It combines:

- Electro-Optical (EO) sensors for high-resolution imaging during daylight and clear-sky conditions.
- Synthetic Aperture Radar (SAR) sensors for all-weather, day-and-night imaging. **Hence, Option (c) is correct.**

Option (d):

Mission Drishti is not a drone-based surveillance grid. It is a satellite-based Earth observation platform designed for remote sensing and geospatial intelligence.**Hence, Option (d) is incorrect.**

Value Addition

Mission Drishti

Feature	Details
Developer	GalaxEye (Bengaluru)
Type	Earth Observation Satellite
Distinction	World's first OptoSAR satellite
Launch Vehicle	Falcon 9
Launch Date	May 2026
Weight	190 kg
Resolution	1.5 metre

What is OptoSAR?

OptoSAR combines:

- **Electro-Optical (EO) Sensors**
  - High-resolution imagery
  - Daylight and clear-sky observations
- **Synthetic Aperture Radar (SAR) Sensors**
  - All-weather imaging
  - Day and night operations
  - Cloud penetration capability

Applications

- Border surveillance
- Disaster management
- Defence
- Infrastructure monitoring
- Agriculture
- Financial and insurance assessment

IN NEWS:Contact lost with Drishti, world's first OptoSAR satellite

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74.Vaccine derived Polio Virus

Q.With reference to the Polio virus, vaccination types, and vaccine-derived infections, consider the following statements:

1. Inactivated Polio Vaccine (IPV) consists of weakened live strains of all three types of poliovirus and carries a rare risk of causing vaccine-derived infections.
2. The detection of a Vaccine-Derived Polio Virus (VDPV) in environmental sewage samples immediately strips a country of its official "polio-free" status.
3. Continuous circulation of the weakened oral vaccine virus among children in areas with low population immunity can directly lead to the development of VDPVs.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (b) 3 only

Explanation

Statement 1: The Inactivated Polio Vaccine (IPV), developed by Dr. Jonas Salk in 1955, consists of inactivated (killed) poliovirus strains of all three poliovirus types. Unlike the Oral Polio Vaccine (OPV), IPV does not contain weakened live virus and carries no risk of Vaccine-Derived Poliovirus (VDPV). **Hence, Statement 1 is incorrect.**

Statement 2: The detection of a Vaccine-Derived Poliovirus (VDPV) in environmental sewage samples does not affect India's polio-free status. India was declared free of wild poliovirus in 2014 after decades of mass immunisation. According to health authorities, only the detection of wild poliovirus can alter a country's polio-free status. **Hence, Statement 2 is incorrect.**

Statement 3:A Vaccine-Derived Poliovirus (VDPV) originates from the weakened live virus present in the Oral Polio Vaccine (OPV). According to WHO, in areas with low population immunity, the weakened vaccine virus may:

- Continue circulating from child to child,
- Regain its ability to transmit efficiently,
- Gradually mutate and acquire disease-causing potential.

This process can lead to the emergence of VDPVs. **Hence, Statement 3 is correct.**

Value Addition

Polio Virus

Feature	Details
Disease	Poliomyelitis (Polio)
Causative Agent	Poliovirus
Transmission	Primarily Faecal-Oral Route
Main Target	Young Children
Affected System	Nervous System
Prevention	Vaccination
Cure	No cure; preventable through immunisation

Symptoms

- Fever
- Fatigue
- Headache
- Vomiting
- Neck stiffness
- Limb pain

In severe cases:

- Paralysis (often permanent)

Types of Polio Vaccines

Feature	IPV	OPV
Full Form	Inactivated Polio Vaccine	Oral Polio Vaccine
Virus Type	Killed Virus	Weakened Live Virus
Route	Injection	Oral Drops
Risk of VDPV	No	Rarely Possible
Developer	Dr. Jonas Salk (1955)	Albert Sabin

Vaccine-Derived Poliovirus (VDPV)

A VDPV is a strain related to the weakened live poliovirus contained in OPV.

It may arise when:

- The vaccine virus circulates for prolonged periods in communities with low immunisation coverage.

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- The virus mutates and regains transmissibility.
- Immunocompromised children harbour the virus for long durations.

WHO Observation

VDPVs are extremely rare, but are more likely to occur in areas with:

- Low population immunity
- Incomplete vaccination coverage

India and Polio Eradication

Milestone	Year
Vaccination Started	1972
Universal Immunisation Programme	1985
Pulse Polio Programme	1995
Last Polio Case (Howrah, West Bengal)	2011
Declared Polio-Free	2014

Pulse Polio

- Launched in 1995
- Children aged 0–5 years are administered oral polio drops
- Famous slogan: "Do Boond Zindagi Ki"

Environmental Surveillance

India continues environmental surveillance even after becoming polio-free.

It involves:

- Regular testing of sewage samples
- Early detection of poliovirus circulation
- Monitoring under WHO and NCDC supervision

Recent Example

A VDPV was detected in a sewage sample in Ghaziabad, prompting intensified vaccination and surveillance measures. However, the detected strain remained contained and did not affect India's polio-free status.

In NEWS:After the detection of VDPV in a sewage sample in Ghaziabad, officials have intensified surveillance.

75.National Academic Depository

Q.With reference to the National Academic Depository (NAD), consider the following statements:

1. The National Academic Depository operates through the DigiLocker platform for digital storage, issuance, verification and authentication of academic records.
2. The Ministry of Education has designated the University Grants Commission (UGC) as the nodal agency for implementing NAD.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Explanation

Statement 1: Since 2020, the National Academic Depository (NAD) has been implemented through DigiLocker. NAD provides:

- Digital storage of academic records
- Issuance of academic awards
- Authentication of credentials
- Verification of documents

Students can access their academic credentials anytime through DigiLocker and share them through a consent-based mechanism. **Hence, Statement 1 is correct.**

Statement 2: The Ministry of Education is the parent ministry responsible for NAD. It has designated the University Grants Commission (UGC) as the nodal agency for implementing the National Academic Depository through DigiLocker. **Hence, Statement 2 is correct.**

Value Addition

National Academic Depository (NAD)

Feature	Details
Parent Ministry	Ministry of Education
Nodal Agency	UGC
Operational Platform	DigiLocker
Purpose	Digital storage, issuance, authentication and verification of academic records

Academic Records Covered

- Degrees
- Diplomas
- Certificates
- Mark Sheets

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- Evaluation Reports

Legal Framework

- Information Technology Act, 2000
- Digital Locker Rules, 2016
- National e-Authentication Framework (NeAF)

Stakeholders

- Students
- Universities
- School Boards
- Employers
- Government Agencies
- Banks
- Foreign Verification Agencies

NEP 2020 Linkages

NAD supports:

- Multiple Entry-Multiple Exit (MEME)
- National Credit Framework (NCrF)
- Academic Bank of Credits (ABC)
- APAAR (Automated Permanent Academic Account Registry)

In NEWS:National Academic Depository (NAD)

76.Non Banking Financial Companies(NBFC)

Q.With reference to the structural differences between Non-Banking Financial Companies (NBFCs) and Commercial Banks, consider the following statements:

1. Unlike banks, NBFCs are completely prohibited from accepting any form of customer deposits.
2. NBFCs do not form part of the payment and settlement system and cannot issue cheques drawn on themselves.
3. Depositors of NBFCs do not enjoy the deposit insurance facility provided by the Deposit Insurance and Credit Guarantee Corporation (DICGC).

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (b) 2 and 3 only

Explanation

Statement 1:NBFCs are not completely prohibited from accepting deposits.There are two types of NBFCs based on their liability structure:

- Deposit-taking NBFCs (NBFC-D) – can accept deposits from customers.
- Non-Deposit Taking NBFCs (NBFC-ND) – cannot accept customer deposits.

However, NBFCs cannot accept demand deposits, which is a key distinction from banks.**Hence, Statement 1 is incorrect.**

Statement 2:NBFCs do not form part of the payment and settlement system.Therefore, they:

- Cannot issue cheques drawn on themselves.
- Cannot provide payment and settlement services in the manner commercial banks do.**Hence, Statement 2 is correct.**

Statement 3: The Deposit Insurance and Credit Guarantee Corporation (DICGC) provides deposit insurance to bank depositors. This facility is not available to depositors of NBFCs, unlike commercial banks. **Hence, Statement 3 is correct.**

Value Addition

Non-Banking Financial Company (NBFC)

An NBFC is a company registered under Companies Act, 2013 and regulated by RBI under RBI Act, 1934. that provides:

- Loans
- Leasing
- Insurance-related services
- Investments and other financial services

Types of NBFCs

Type	**Features**
NBFC-D	Accepts customer deposits
NBFC-ND	Does not accept customer deposits

NBFC vs Bank

Parameter	**NBFC**	**Bank**
Demand Deposits	Not Allowed	Allowed
Payment & Settlement System	Not Part Of It	Part Of It
Cheque Issuance	Cannot Issue	Can Issue
DICGC Insurance	Not Available	Available
CRR Requirement	Not Required	Required
SLR Requirement	Not Required	Required

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Role of NBFCs

Financial Inclusion

- Serve areas where banks have limited presence.
- Extend credit to underserved sectors.

Customer-Centric Services

- Simplified procedures
- Faster loan processing
- Flexible products

Technology Adoption

- Use digital platforms, mobile applications and internet-based services.

Market Deepening

- Better understanding of local customer needs.
- Customized financial products.

Challenges Faced by NBFCs

Asset-Liability Mismatch

- Short-term borrowings funding long-term assets.

Liquidity Risks

- Absence of CRR and SLR requirements may increase vulnerability.

Governance Issues

- Connected lending.
- Rating agency and auditing concerns.

Example

- Infrastructure Leasing & Financial Services crisis highlighted systemic risks in the NBFC sector.

RBI Regulatory Measures

- Mandatory RBI registration for eligible NBFCs.
- Periodic reporting to RBI.
- Special Liquidity Scheme.
- Targeted Long-Term Repo Operations (TLTRO).
- Maintenance of liquid assets by certain deposit-taking NBFCs.

RBI Scale-Based Regulation Framework

Layer	Description
Base Layer	Non-systemically important NBFCs
Middle Layer	Systemically important NBFCs and NBFC-D
Upper Layer	Bank-like NBFCs with systemic risk
Top Layer	NBFCs posing extreme systemic risks

IN NEWS: NBFCs strike gold: Jewellery-backed loans jump 70%

77.SEED Scheme

Q.Consider the following statements regarding the Scheme for Economic Empowerment of DNTs (SEED):

1. It is a Central Sector Scheme launched by the Ministry of Tribal Affairs.
2. The scheme is implemented by the Development and Welfare Board for De-notified, Nomadic and Semi-Nomadic Communities (DWBDNC).
3. The scheme includes health insurance, livelihood assistance and coaching support.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two

Explanation

Statement 1: The SEED (Scheme for Economic Empowerment of DNTs) is a Central Sector Scheme, but it was launched by the Ministry of Social Justice & Empowerment, not the Ministry of Tribal Affairs. The scheme was launched in February 2022 to promote the socio-economic development of Denotified, Nomadic and Semi-Nomadic communities. **Hence, Statement 1 is incorrect.**

Statement 2: The scheme is implemented by the Development and Welfare Board for De-notified, Nomadic and Semi-Nomadic Communities (DWBDNC).

DWBDNC is responsible for:

- Registration of beneficiaries
- Monitoring implementation
- Delivery of benefits through an online portal

Hence, Statement 2 is correct.

Statement 3:SEED focuses on four major components:

- Free coaching for competitive examinations and professional courses
- Health insurance coverage through PM-JAY
- Livelihood support for income generation
- Housing assistance through schemes such as PMAY

Thus, health insurance, livelihood assistance and coaching support are important components of the scheme. **Hence, Statement 3 is correct.**

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Value Addition

SEED Scheme

Feature	Details
Full Form	Scheme for Economic Empowerment of DNTs
Launch Year	February 2022
Type	Central Sector Scheme
Ministry	Ministry of Social Justice & Empowerment
Implementing Agency	DWBDNC
Outlay	₹200 crore (Five Years)

Target Beneficiaries

Denotified Tribes DNTs

Communities that were earlier notified under the Criminal Tribes Act, 1871 during British rule and were de-notified after the Act was repealed in 1952.

Nomadic Tribes NTs

Communities that move from place to place and do not have permanent settlements.

Semi-Nomadic Tribes SNTs

Communities that migrate periodically but maintain some degree of permanent settlement.

Components of SEED

Component	Objective
Free Coaching	Competitive exams and professional courses
Health Insurance	Coverage through Ayushman Bharat–PMJAY
Livelihood Support	Income generation and entrepreneurship
Housing Support	Linkage with PMAY and other housing schemes

Importance:

Educational Empowerment

- Improves access to higher education.
- Enhances participation in competitive examinations.

Healthcare Access

- Health insurance through PM-JAY reduces out-of-pocket expenditure.

Economic Inclusion

- Encourages entrepreneurship and self-employment.
- Promotes sustainable livelihoods.

Social Justice

- Addresses historical exclusion and stigma faced by DNT, NT and SNT communities.

Communities in India

- Around 150 Denotified Tribes
- Around 500 Nomadic Communities
- Together account for nearly 10% of India's population

Challenges Faced by DNTs, NTs and SNTs

Various bodies such as:

- Kaka Kalelkar Commission
- Mandal Commission
- National Commission to Review the Working of the Constitution

have highlighted:

- Historical marginalisation
- Social stigma
- Poor educational outcomes
- Lack of access to healthcare
- Livelihood insecurity

IN NEWS:Over 1.10 lakh beneficiaries covered under SEED components with focus on education, livelihood and healthcare initiatives; Rs.53.06 Crores released for implementation during 2025-26

78.PM SETU

Q.With reference to the Pradhan Mantri Skilling and Employability through Upgraded ITIs (PM-SETU) Scheme, consider the following statements:

1. It is a flagship vocational training scheme implemented under the administrative control of the Ministry of Labour and Employment.
2. The scheme is executed via a hub-and-spoke model, where central "Hub" ITIs mentor a cluster of nearby "Spoke" ITIs by sharing advanced equipment and digital learning resources.
3. States like Odisha, Gujarat, and Telangana have finalized Strategic Investment Plans (SIPs) featuring specific corporate entities as Anchor Industry Partners.

Which of the statements given above are correct?

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(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (b) 2 and 3 only

Explanation

Statement 1:

The Pradhan Mantri Skilling and Employability through Upgraded ITIs (PM-SETU) Scheme is implemented by the Ministry of Skill Development and Entrepreneurship (MSDE). Recently, the Ministry constituted a National Steering Committee (NSC) as the apex body to guide the implementation of the scheme.Therefore, the scheme is not under the Ministry of Labour and Employment. **Hence, Statement 1 is incorrect.**

Statement 2:

PM-SETU adopts a Hub-and-Spoke Model for upgrading Industrial Training Institutes (ITIs).Under this model:

- 200 Hub ITIs will function as central institutions with advanced infrastructure.
- Each Hub ITI will support around four nearby Spoke ITIs.
- Hubs will share:
  - Modern equipment
  - Digital learning resources
  - Specialized training facilities

This approach aims to improve the quality and reach of vocational training across the country.**Hence, Statement 2 is correct.**

Statement 3:

The National Steering Committee recently approved **Strategic Investment Plans (SIPs)** in several states with designated **Anchor Industry Partners (AIPs)**.

Examples include:

- **Odisha** → Jindal Naveen Avsar Limited
- **Gujarat** → ArcelorMittal Nippon Steel India
- **Telangana** → Apollo MedSkills, Sri Siddharth Infratech & Services Pvt. Ltd. (SSISPL), and Neuland Foundation

These industry partners will play a major role in upgrading ITI clusters under the scheme.**Hence, Statement 3 is correct.**

Value Addition

PM-SETU Scheme

Feature	Details
Full Form	Pradhan Mantri Skilling and Employability through Upgraded ITIs
Ministry	Ministry of Skill Development and Entrepreneurship (MSDE)
Approval	Union Cabinet, 2025
Outlay	₹60,000 crore
Objective	Modernisation of ITIs and enhancement of employability

Key Features

Industry-Led Skilling Model

PM-SETU marks a transition from:

Government-driven training → Industry-led training

Industry partners participate in:

- Curriculum design
- Training methodology
- Infrastructure development
- Placement support

Hub-and-Spoke Model

Component	Role
Hub ITI	Advanced infrastructure and training facilities
Spoke ITI	Receives mentoring and support from Hub

Coverage

- 200 Hub ITIs
- Each Hub mentors about 4 Spoke ITIs

Special Purpose Vehicle (SPV)

Each upgraded ITI will be managed through an Industry-Government Partnership SPV.

Ownership Pattern

Stakeholder	Share
Industry Partner	51%
Government	49%

Funding Support

Industry partners can receive up to **83% government funding** for:

- Infrastructure upgrades

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- Training improvements

National Skill Training Institutes (NSTIs) to be Upgraded

The scheme will strengthen NSTIs at:

1. Bhubaneswar
2. Chennai
3. Hyderabad
4. Kanpur
5. Ludhiana

to transform them into **Global Centres of Excellence**.

New-Age Courses

The Directorate General of Training (DGT) has introduced **31 new-age courses** under the Craftsmen Training Scheme (CTS).

Focus Sectors

- Advanced Manufacturing
- Electronics
- Mobility
- Logistics

In NEWS:Jindal, ArcelorMittal, Apollo Med-Skills among partners who come forward to Anchor ₹ 1,237.58 Crore ITI Transformation as PM-SETU Goes Nationwide Across Odisha, Gujarat and Telangana

79.Designated Repositories

Q.With reference to the declaration of "Designated Repositories" in India, consider the following statements:

1. They are officially designated under the provisions of the Wildlife (Protection) Act, 1972.
2. The National Biodiversity Authority holds the absolute, independent statutory power to notify an institution as a Designated Repository.
3. The recently notified Natural History Museum (NHM) at Mizoram University is the first such repository established in the entire North-eastern region of India.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 3 only
(c) 1 and 3 only
(d) None of the above

Answer: (d) None of the above

Explanation

Statement 1: Designated Repositories are notified under the provisions of the **Biological Diversity Act, 2002**, and not under the **Wildlife (Protection) Act, 1972**. They serve as repositories for:

- Holotypes, isotypes and paratypes of newly discovered taxa in India.
- Biological resource samples accessed by foreign citizens or entities for research.
- Biological resources sent or carried abroad by Indian institutions or researchers for research purposes. **Hence, Statement 1 is incorrect.**

Statement 2: Under Section 39 of the Biological Diversity Act, 2002, the National Biodiversity Authority (NBA) can designate institutions as Designated Repositories in consultation with the Central Government.Therefore, the NBA does not possess an absolute and independent statutory power to notify institutions on its own. **Hence, Statement 2 is incorrect.**

Statement 3: The Natural History Museum (NHM), Mizoram University, Aizawl was recently designated as a Designated Repository.However, the given information only states that India now has 21 Designated Repositories and does not indicate that NHM, Mizoram University is the first repository in the North-Eastern region.**Hence, Statement 3 is incorrect.**

Value Addition

Designated Repository (DR)

Feature	**Details**
Legal Basis	Biological Diversity Act, 2002
Relevant Provision	Section 39
Designating Authority	National Biodiversity Authority (NBA) in consultation with Central Government
Purpose	Preservation and storage of biological resources and taxonomic specimens

Functions of Designated Repositories

Preservation of Taxonomic Specimens

Repositories maintain:

- Holotypes
- Isotypes
- Paratypes of newly discovered taxa from India.

Conservation of Biological Resources

They act as secure repositories for biological materials used in scientific research.

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Facilitation of Research

Support:

- Biodiversity studies
- Taxonomy
- Conservation biology
- Biological resource documentation

Recent Development

The Ministry of Environment, Forest and Climate Change has designated: Natural History Museum, Mizoram University as a Designated Repository under the Biological Diversity Act, 2002.

Institutional Framework

Institution	Role
Ministry of Environment, Forest and Climate Change	Policy and notification support
National Biodiversity Authority (NBA)	Designation of repositories under Section 39
Designated Repositories	Storage and maintenance of biological resources

In NEWS:Mizoram's Natural History Museum notified as India's 21st Designated Repository under Biological Diversity Act, 2002

80.Durand Cup

Q.Consider the following statements regarding the Durand Cup:

1. It holds the distinction of being Asia's oldest and the world's third-oldest football tournament, having its inaugural edition held in Shimla in 1888.
2. Uniquely, the winning team of the tournament is awarded three distinct trophies, including the original Durand Cup, the Shimla Trophy, and the President's Cup.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Explanation

Statement 1: The Durand Cup is Asia's oldest and the world's third-oldest football tournament.The inaugural edition was held in Shimla in 1888. Initially, it started as an Army Cup and was open only to troops of the British Indian Army stationed in India.The tournament is named after its founder Sir Henry Mortimer Durand.**Hence, Statement 1 is correct.**

Statement 2: The Durand Cup is unique among football tournaments because the winning team receives **three trophies**:

1. Durand Cup – The original trophy and a rolling trophy.
2. Shimla Trophy – Presented by the residents of Shimla in 1904 and also maintained as a rolling trophy.
3. President's Cup – Instituted in 1956 by Dr. Rajendra Prasad and awarded for permanent possession. **Hence, Statement 2 is correct.**

Value Addition

Durand Cup

Feature	Details
Established	1888
Venue of First Edition	Shimla
Founder	Sir Henry Mortimer Durand
Status	Asia's Oldest Football Tournament
Global Ranking	World's Third-Oldest Football Tournament

Historical Background

- Began in 1888 as an Army Cup.
- Initially open only to the British Indian Army.
- Started as a football competition among:
  - Military departments
  - Armed forces regiments
  - Princely state forces

After Independence, the tradition of military participation continued through teams representing various regiments of the Indian Armed Forces.

Three Trophies of the Durand Cup

Trophy	Significance
Durand Cup	Original tournament trophy and rolling trophy
Shimla Trophy	Donated by the residents of Shimla in 1904

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President's Cup	Instituted in 1956 by Dr. Rajendra Prasad and retained permanently by winners

Significance:

Sporting Importance

- One of India's most prestigious football tournaments.
- Features leading football clubs from across India.

Historical Significance

- Reflects the evolution of football in India from the colonial period to the present.
- Maintains one of the oldest continuous football traditions in the world.

Military Legacy

- Originated as a military football competition.
- Continues to retain links with the Indian Armed Forces through invited military teams.

In NEWS:PRESIDENT OF INDIA UNVEILS THE TROPHIES OF DURAND CUP TOURNAMENT

81.Chief Labour Commissioner

Q.With reference to the Chief Labour Commissioner (Central) [CLC(C)], consider the following statements:

1. The Chief Labour Commissioner (Central) Organisation functions under the Ministry of Labour & Employment.
2. It is primarily responsible for maintaining harmonious industrial relations in establishments falling under the Central Sphere.
3. It performs quasi-judicial functions under various labour laws.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (c) All three

EXPLANATION:

Statement 1: The Chief Labour Commissioner (Central) [CLC(C)] Organisation functions under the Ministry of Labour & Employment, Government of India. It is the apex industrial relations machinery of the Central Government and operates under the administrative control of the Ministry of Labour & Employment. **Hence, Statement 1 is correct.**

Statement 2: The organisation is responsible for maintaining harmonious industrial relations in establishments falling under the Central Sphere, such as:

- Railways
- Mines
- Major Ports
- Banking and Insurance establishments under Central jurisdiction
- Central Public Sector Enterprises

It undertakes conciliation and mediation to prevent and resolve industrial disputes. **Hence, Statement 2 is correct.**

Statement 3: The Chief Labour Commissioner (Central) Organisation performs various quasi-judicial functions under labour laws.

These include:

- Verification of Trade Union Membership
- Adjudicatory functions under labour legislations
- Enforcement-related proceedings under labour laws

Apart from conciliation and enforcement, the organisation also exercises statutory and quasi-judicial powers. **Hence, Statement 3 is correct.**

Value Addition

Chief Labour Commissioner (Central) [CLC(C)]

Feature	**Details**
Also Known As	Central Industrial Relations Machinery (CIRM)
Ministry	Ministry of Labour & Employment
Headquarters	New Delhi
Established	April 1945
Nature	Apex Industrial Relations Organisation
Jurisdiction	Central Sphere Establishments

Formation of CLC(C)

The organisation was established in April 1945 following the recommendations of the Royal Commission on Labour in India.

It was formed by combining:

- Conciliation Officer (Railways)
- Supervisor of Railway Labour
- Labour Welfare Advisor

Initially, the organisation was entrusted with:

- Prevention and settlement of industrial disputes

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- Enforcement of labour laws
- Promotion of labour welfare

Functions of CLC(C)

Industrial Relations Functions

- Prevention of industrial disputes
- Conciliation and mediation
- Promotion of industrial harmony

Labour Law Enforcement

- Enforcement of labour laws in Central Sphere establishments
- Monitoring compliance with labour standards

Quasi-Judicial Functions

- Verification of trade union membership
- Statutory adjudicatory functions under labour laws

Welfare Functions

- Protection of workers' interests
- Promotion of labour welfare measures

Importance of CLC(C)

Industrial Peace

- Facilitates amicable settlement of disputes.
- Prevents strikes and lockouts.

Labour Welfare

- Ensures implementation of labour welfare provisions.

Economic Stability

- Promotes uninterrupted industrial production and services.

Social Justice

- Protects workers' rights through enforcement of labour laws.

In NEWS:Chief Labour Commissioner (Central) Organisation Strengthens Labour Welfare and Compliance Across the Country Through Awareness, Proactive Interventions and Grievance Redressal Initiative

82.Index of Service Production

Q.With reference to the Index of Services Production (ISP), consider the following statements:

1. The Index of Services Production (ISP) is a short-term indicator designed to measure changes in the real volume of output produced by the services sector.
2. It is compiled and released by the Ministry of Commerce and Industry.
3. Public Administration and Defence are fully covered under the ISP framework.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (a) Only one

EXPLANATION:

Statement 1: The Index of Services Production (ISP) is a macro-economic short-term indicator that measures monthly changes in the real volume of output produced by the services sector. It has been developed as a direct counterpart to the Index of Industrial Production (IIP) for the services sector and helps capture high-frequency economic trends. **Hence, Statement 1 is correct.**

Statement 2: The ISP is compiled, managed and disseminated by the Ministry of Statistics and Programme Implementation (MoSPI). It is not compiled by the Ministry of Commerce and Industry. **Hence, Statement 2 is incorrect.**

Statement 3: The ISP explicitly excludes several government and non-market service activities, including:

- Public Administration and Defence
- Personal Services
- Gambling and Betting Activities
- Social Work Activities without Accommodation
- Central Bank Monetary Activities

Therefore, Public Administration and Defence are not covered under the ISP framework.**Hence, Statement 3 is incorrect.**

Value Addition

Index of Services Production (ISP)

Feature	Details
Nodal Ministry	Ministry of Statistics and Programme Implementation (MoSPI)
Nature	Short-term macroeconomic indicator
Measures	Real volume of services output
Counterpart	Index of Industrial Production (IIP)
Base Year	2024–25
Formula	Fixed-weight Laspeyres Volume Index

Frequency	Monthly
Release Schedule	29th day of every month
Time Lag	Around 60 days

Objectives of ISP

- Capture short-term movements in the services sector.
- Complement the Index of Industrial Production (IIP).
- Support economic forecasting and policy formulation.
- Monitor the performance of the services sector, which contributes more than 50% of India's Gross Value Added (GVA).

Major Data Sources

GST Database

- Uses Service Accounting Code (SAC) data from GSTR-1 filings.
- Captures service-sector turnover.

Administrative Records

- Air Transport
- Rail Transport
- Banking
- Insurance

ASISSE

- Annual Survey of Incorporated Services Sector Enterprises.
- Used for non-government Health and Education sectors.

Key Features

Formal Sector Focus

- Primarily captures the formal corporate services sector.
- Based largely on GST-registered entities.

Weight Assignment

- Weights are assigned according to sectoral contribution to Gross Value Added (GVA).

Strategic Exclusions

The ISP excludes:

- Public Administration and Defence
- Personal Services
- Gambling Activities
- Certain Social Work Activities
- Central Bank Monetary Activities

IN NEWS:Upcoming Index of Services Production is expected to rely heavily on GST figures

83.UDISE+

Q.With reference to the Unified District Information System for Education Plus (UDISE+), consider the following statements:

1. UDISE+ functions as the primary data repository for school education in India under the Department of School Education and Literacy.
2. According to the UDISE+ Report 2025–26, dropout rates have declined at both preparatory and secondary stages.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Statement 1: The Unified District Information System for Education Plus (UDISE+) functions as the primary and comprehensive data repository for school education in India. It operates under the Department of School Education and Literacy, Ministry of Education, and serves as the foundational database for:

- Educational planning
- Policy formulation
- Resource allocation
- Monitoring of school education indicators
- Tracking progress under the National Education Policy (NEP)

UDISE+ is one of the world's largest education management information systems, covering:

- More than 14.8 lakh schools
- More than 1.02 crore teachers
- More than 26 crore students **Hence, Statement 1 is correct.**

Statement 2: The UDISE+ Report 2025–26 highlights a significant improvement in student retention and a decline in dropout rates across major stages of schooling.

Key findings include:

- Preparatory-level dropout rate declined to 8%
- Secondary-level dropout rate declined to 7.0%

The decline in dropout rates indicates:

- Improved retention of students
- Better continuity in schooling
- Increased effectiveness of educational interventions

- Progress towards universal school education

Hence, Statement 2 is correct.

Value Addition

UDISE+

Feature	Details
Full Form	Unified District Information System for Education Plus
Ministry	Ministry of Education
Department	Department of School Education and Literacy
Nature	Education Management Information System (EMIS)
Coverage	Government and Private Schools
Schools Covered	14.8 lakh+
Teachers Covered	1.02 crore+
Students Covered	26 crore+

Major Findings of UDISE+ Report 2025–26

Improved Student Retention

- Retention improved at middle and secondary levels.
- Fewer students are dropping out before completing school education.

Reduced Dropout Rates

Stage	Trend
Preparatory Stage	Declined
Secondary Stage	Declined

Improved Transition Rates

Students are increasingly progressing from:

- Foundational → Preparatory
- Preparatory → Middle
- Middle → Secondary , reducing educational discontinuity.

Better Pupil-Teacher Ratio (PTR)

- PTR remains below the NEP benchmark of 30:1.
- Teacher availability has improved.

Expansion of Digital Infrastructure

- More schools now have:
  - Computers
  - Internet connectivity
  - Digital learning facilities

Challenges Highlighted in the Report

- Secondary-Level Retention Gap
- Children with Special Needs (CWSN)
- Single-Teacher Schools
- Sustainability Deficit

IN NEWS:The Union Ministry of Education Releases the Report on Unified District Information System for Education Plus (UDISE+) 2025-26 on School Education in India

84.Guwahati declaration

Q.The recently adopted Guwahati Declaration is primarily related to:

(a) Establishing a common BRICS currency for international trade
(b) Strengthening cooperation to combat illicit drug trafficking and transnational organized crime
(c) Promoting a unified BRICS framework for climate finance
(d) Creating a BRICS-wide digital payments settlement mechanism

Answer: (b) Strengthening cooperation to combat illicit drug trafficking and transnational organized crime

EXPLANATION:

Option (a): The Guwahati Declaration is not related to the establishment of a common BRICS currency or trade settlement arrangements. Issues relating to currency cooperation and financial architecture are discussed separately within BRICS forums. **Hence, Option (a) is incorrect.**

Option (b): The Guwahati Declaration, adopted by BRICS nations on 7 July 2026, focuses on strengthening cooperation to combat:

- Illicit drug trafficking
- Transnational organized crime
- Synthetic drug networks
- Cross-border narcotics smuggling

The declaration emphasizes:

- Timely intelligence sharing
- Exchange of best practices
- Data-driven law enforcement
- Use of innovative technologies
- International cooperation against organized crime **Hence, Option (b) is correct.**

Option (c): The declaration is not focused on climate finance, carbon markets, or climate-related cooperation. Its primary focus is counter-narcotics cooperation and law enforcement.**Hence, Option (c) is incorrect.**

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Option (d): The declaration does not deal with digital payment systems or the creation of a BRICS-wide payment settlement mechanism. Its focus remains drug control and combating organized crime.**Hence, Option (d) is incorrect.**

Value Addition

Guwahati Declaration (2026)

Feature	Details
Adopted By	BRICS Nations
Date	7 July 2026
Venue	Guwahati, India
Main Theme	Countering Illicit Drug Trafficking
Focus Area	Transnational Organized Crime
Nature	Counter-Narcotics Cooperation Framework

Major Objectives of the Declaration

Combat Illicit Drug Trafficking

- Disrupt international drug supply chains.
- Dismantle transnational criminal networks.

Strengthen Intelligence Sharing

- Real-time exchange of information.
- Enhanced cooperation among enforcement agencies.

Promote Technology-Based Enforcement

- Data analytics
- Digital investigation tools
- Monitoring misuse of virtual assets

Reduce Drug Demand

- Prevention of drug abuse
- Awareness campaigns
- Promotion of healthy lifestyles

Protect Vulnerable Groups

Special focus on:

- Children
- Adolescents
- Youth , through evidence-based prevention strategies.

India's Key Proposals

Zero-Tolerance Policy Against Drugs

India advocated a comprehensive approach against narcotics.

Three-Year Roadmap (2026–29)

Focus areas include:

- Dismantling drug trafficking networks
- Public awareness campaigns
- Strengthening rehabilitation systems
- Improving international cooperation

BRICS Virtual Workshop Group

India proposed:

- Capacity-building programmes
- Cross-border training initiatives
- Knowledge-sharing platforms

BRICS Members Associated with the Declaration

Original Members

- Brazil
- Russia
- India
- China
- South Africa

New Members

- Egypt
- Ethiopia
- Iran
- Saudi Arabia
- United Arab Emirates
- Indonesia

In NEWS:Significance of Guwahati Declaration adopted by BRICS nations

85.Performance Grading Index

Q.The Performance Grading Index (PGI) is primarily used to:

(a) Measure the performance of States/UTs in higher education institutions
(b) Measure the performance of States/UTs in school education
(c) Rank universities based on research output
(d) Evaluate vocational training institutes

Answer: (b) Measure the performance of States/UTs in school education

Option (a): The Performance Grading Index (PGI) is not designed to assess higher education institutions such as universities, colleges, or research institutes. Higher education institutions are assessed through other frameworks and ranking mechanisms.

Hence, Option (a) is incorrect.

Option (b): The Performance Grading Index (PGI) is an index developed by the Department of School Education & Literacy (DoSEL), Ministry of Education, to assess the performance of States and Union Territories in school education. It aims to:

- Evaluate educational performance on a uniform scale.

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- Encourage healthy competition among States and UTs.
- Identify strengths and gaps in school education.
- Support evidence-based policymaking.

The index evaluates States/UTs across multiple domains such as:

- Learning Outcomes and Quality
- Access
- Infrastructure & Facilities
- Equity
- Governance Processes
- Teacher Education & Training. **Hence, Option (b) is correct.**

Option (c): The PGI is not a university ranking framework and does not assess research output, patents, publications, or academic excellence in higher education institutions. **Hence, Option (c) is incorrect.**

Option (d): The PGI is focused on school education performance and not on vocational training institutes or skill-development centres. **Hence, Option (d) is incorrect.**

Value Addition

Performance Grading Index (PGI)

Feature	**Details**
Full Form	Performance Grading Index
Nodal Agency	Department of School Education & Literacy (DoSEL)
Ministry	Ministry of Education
Objective	Assess performance of States/UTs in school education
Nature	Performance Assessment Tool
Coverage	States and Union Territories

Objectives of PGI

Improve School Education

- Monitor educational performance.
- Encourage continuous improvement.

Promote Cooperative and Competitive Federalism

- Enables comparison among States and UTs.
- Encourages adoption of best practices.

Support Policy Formulation

- Identifies gaps in educational outcomes.
- Assists governments in targeted interventions.

Ensure Data-Driven Governance

- Uses measurable indicators for evaluation.
- Facilitates evidence-based decision-making.

PGI for States/UTs

Total Weightage

- 1000 Points

Indicators

- 70 Indicators

Categories

1. Outcome
2. Governance & Management

Domains

- Learning Outcomes and Quality
- Access
- Infrastructure & Facilities
- Equity
- Governance Processes
- Teacher Education & Training

Data Sources Used in PGI

The framework is aligned with data obtained from:

- UDISE+
- PARAKH Rashtriya Sarvekshan 2024
- PM POSHAN Portal
- PRABAND Portal
- Vidyanjali Portal

PGI Importance:

Accountability

Measures the effectiveness of educational policies implemented by States and UTs.

Better Learning Outcomes

Focuses attention on quality education rather than mere enrolment.

Resource Optimization

Helps governments allocate resources where deficiencies exist.

NEP Alignment

Supports monitoring of goals envisaged under the National Education Policy (NEP).

PGI vs PGI-D

Parameter	**PGI (States/UTs)**	**PGI-D (Districts)**
Coverage	States & UTs	Districts
Weightage	1000 Points	600 Points
Indicators	70	70
Purpose	State-level evaluation	District-level evaluation

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Nodal Agency	DoSEL	DoSEL

In NEWS:Union Ministry of Education Releases Report on Performance Grading Index 2.0 for States/UTs (PGI-S) and Performance Grading Index for Districts (PGI-D) for 2025-26

86.Prambanan Temple

Q.With reference to the Prambanan Temple, consider the following statements:

1. The temple complex is located on the island of Java in Indonesia.
2. Prambanan is the largest Hindu temple complex in Indonesia and is dedicated primarily to the Hindu Trimurti.
3. UNESCO inscribed the Prambanan Temple Compounds as a World Heritage Site in 1991.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (c) All three

Statement 1: The Prambanan Temple Compounds are located on the fertile volcanic plains of Java Island in Indonesia. The complex lies near Yogyakarta and straddles the administrative boundary between:

- Special Region of Yogyakarta
- Central Java Province

It is situated close to Mount Merapi, one of Indonesia's most active volcanoes. **Hence, Statement 1 is correct.**

Statement 2:

Prambanan is the largest Hindu temple complex in Indonesia and one of the largest Hindu temple complexes in Southeast Asia. The temple complex is primarily dedicated to the Hindu Trimurti, comprising:

- Shiva (Destroyer and Transformer)
- Vishnu (Preserver)
- Brahma (Creator)

Among these, the Shiva Temple is the tallest and most prominent structure, rising approximately 47 metres.The complex is regarded as the finest example of classical Shaivite Hindu architecture in Indonesia. **Hence, Statement 2 is correct.**

Statement 3:

After extensive restoration efforts, the Prambanan Temple Compounds were inscribed as a UNESCO World Heritage Site in 1991.UNESCO recognized the site for its:

- Outstanding Hindu architecture
- Cultural significance
- Historical value
- Exceptional stone craftsmanship

Later, in 1998, the Indonesian Government designated it as a National Cultural Property. **Hence, Statement 3 is correct.**

Value Addition

Prambanan Temple

Feature	Details
Country	Indonesia
Island	Java
Local Name	Roro Jonggrang (Loro Jonggrang)
Religion	Hinduism
Main Tradition	Shaivism
Dynasty	Sanjaya Dynasty
Period	c. 850 CE
UNESCO Status	World Heritage Site (1991)

Origin and History

Construction

- Built during the 8th–9th centuries AD.
- Constructed under the Sanjaya Dynasty.
- Symbolized the revival of Shaivite Hinduism in Java.

Decline

The complex was abandoned due to:

- Political shift of power to East Java
- Major earthquakes
- Volcanic eruptions from Mount Merapi

Rediscovery

- Rediscovered by British surveyors during the early nineteenth century.
- Systematic restoration began in 1918.

Architectural Features

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Concentric Square Layout

The temple follows a geometric layout consisting of three concentric zones:

Zone	Description
Outer Zone	Large courtyard and auxiliary structures
Middle Zone	224 Perwara (guardian) temples
Inner Zone	Sacred platform containing principal temples

Trimurti Temples

The Shiva Temple is the tallest and serves as the central shrine. The inner sacred zone contains three major temples dedicated to Three companion temples face the Trimurti shrines:

Temple	Deity	Vahana
Shiva Temple	Shiva	Nandi
Vishnu Temple	Vishnu	Garuda
Brahma Temple	Brahma	Hamsa

Cultural Significance

Ramayana Reliefs

The temple walls contain detailed stone carvings depicting:

- Ramayana
- Bhagavata Purana

These reliefs represent the Javanese adaptation of Indian epics.

Symbol of Indian Cultural Influence

Prambanan reflects the spread of:

- Hinduism
- Sanskritic traditions
- Temple architecture
- Indian cultural influence across Southeast Asia.

IN NEWS: India and Indonesia launch joint restoration project at iconic Prambanan Temple

87.Leucine

Q.The term "Leucine", recently seen in the news, is best described as:

(a) A neurotransmitter involved in signal transmission in the brain

(b) An essential amino acid belonging to the branched-chain amino acid (BCAA) family

(c) A hormone secreted by the pancreas that regulates blood sugar levels

(d) An enzyme involved in mitochondrial DNA replication

Answer: (b) An essential amino acid belonging to the branched-chain amino acid (BCAA) family

EXPLANATION:

Option (a):

Neurotransmitters are chemical messengers that transmit signals between nerve cells.Examples include:

- Dopamine
- Serotonin
- Acetylcholine Leucine is not a neurotransmitter.

Hence, Option (a) is incorrect.

Option (b):

Leucine is an essential amino acid and belongs to the family of Branched-Chain Amino Acids (BCAAs).The three Branched-Chain Amino Acids are:

- Leucine
- Isoleucine
- Valine

Since the human body cannot synthesize leucine, it must be obtained through diet.Recent research has shown that leucine acts as a protective shield by preventing the premature degradation of proteins present in the outer mitochondrial membrane.It does so by interacting with the protein SEL1L, which is involved in identifying and removing damaged or misfolded proteins. **Hence, Option (b) is correct.**

Option (c):

Insulin is the hormone produced by the pancreas that regulates blood glucose levels. Leucine is an amino acid and not a hormone. **Hence, Option (c) is incorrect.**

Option (d):

Leucine is neither an enzyme nor a component directly involved in mitochondrial DNA replication. It is an amino acid that contributes to protein synthesis and cellular functions.**Hence, Option (d) is incorrect.**

Value Addition

Leucine

Feature	Details
Type	Essential Amino Acid
Group	Branched-Chain Amino Acid (BCAA)
Related BCAAs	Isoleucine, Valine

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Synthesized in Body	No
Source	Diet
Major Role	Muscle growth, tissue repair, energy production

Branched-Chain Amino Acids (BCAAs):

BCAAs are a group of essential amino acids characterized by a branched molecular structure.The three BCAAs are:

BCAA	Function
Leucine	Muscle protein synthesis and mitochondrial protection
Isoleucine	Energy regulation and muscle metabolism
Valine	Tissue repair and energy production

These amino acids are especially important for:

- Muscles
- Nervous system
- Heart
- Brain

Since the body cannot produce them, they must be obtained from food.

Functions of Leucine

Muscle Growth

- Stimulates muscle protein synthesis.
- Supports muscle maintenance and recovery.

Tissue Repair

- Helps repair damaged tissues.
- Supports wound healing processes.

Energy Production

- Acts as an energy source during intense physical activity.

Mitochondrial Protection

Recent research has shown that leucine:

- Protects outer mitochondrial membrane proteins.
- Prevents premature protein degradation.
- Maintains mitochondrial integrity and function.

Interaction with SEL1L

Leucine interacts with:

- SEL1L protein

SEL1L normally identifies damaged or misfolded proteins for removal. Leucine helps regulate this process and protects healthy mitochondrial membrane proteins from unnecessary degradation.

Amino Acids:

Amino acids are organic molecules that combine to form proteins. They are known as the building blocks of life because proteins perform essential structural and functional roles in living organisms. Total Number of 20 standard amino acids.

Role

- Formation of proteins
- Enzyme production
- Hormone synthesis
- Tissue growth and repair

Essential vs Non-Essential Amino Acids

Type	Meaning
Essential Amino Acids	Cannot be synthesized by the body; must come from diet
Non-Essential Amino Acids	Can be synthesized by the body

Nine Essential Amino Acids

1. Histidine
2. Isoleucine
3. Leucine
4. Lysine
5. Methionine
6. Phenylalanine
7. Threonine
8. Tryptophan
9. Valine

IN NEWS:How the amino acid leucine helps our cells produce energy

88.Telecom Regulatory Authority of India(TRAI)

Q.With reference to the Telecom Regulatory Authority of India (TRAI), consider the following statements:

1. TRAI was established through an Act of Parliament in 1997.
2. TRAI is responsible for adjudicating disputes between telecom service providers.
3. The Central Government appoints the Chairperson and Members of TRAI.

Which of the statements given above is/are correct?

(a) 1 and 3 only
(b) 2 only
(c) 1 and 2 only
(d) 1, 2 and 3

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Answer: (a) 1 and 3 only

Statement 1:

The Telecom Regulatory Authority of India (TRAI) was established under the Telecom Regulatory Authority of India Act, 1997, enacted by Parliament.It functions as the statutory regulator of the telecom sector in India. **Hence, Statement 1 is correct.**

Statement 2:

TRAI is not responsible for adjudicating disputes between telecom service providers.This function is performed by the Telecom Disputes Settlement and Appellate Tribunal (TDSAT), which was established through an amendment to the TRAI Act in 2000. **Hence, Statement 2 is incorrect.**

Statement 3:

The Chairperson and Members of TRAI are appointed by the Central Government.The Authority consists of:

- One Chairperson
- Maximum two Full-Time Members
- Maximum two Part-Time Members **Hence, Statement 3 is correct.**

Value Addition

Telecom Regulatory Authority of India (TRAI)

Feature	Details
Established	1997
Statutory Basis	Telecom Regulatory Authority of India Act, 1997
Nature	Statutory Regulatory Body
Headquarters	New Delhi
Ministry	Ministry of Communications
Appointing Authority	Central Government

Composition of TRAI

Position	Strength
Chairperson	1
Full-Time Members	Maximum 2
Part-Time Members	Maximum 2

Tenure

- 3 years, or
- Up to 65 years of age, whichever is earlier.

Functions of TRAI

Regulation of Telecom Services

- Tariff fixation and revision
- Quality of Service (QoS)
- Consumer protection
- Interconnection regulations

Policy Recommendations

- Spectrum management
- Licensing framework
- Telecom sector reforms

Promotion of Competition

- Encourages fair competition
- Protects consumer interests

Telecom Disputes Settlement and Appellate Tribunal (TDSAT)

Feature	Details
Established	2000
Created Through	Amendment to TRAI Act
Nature	Quasi-Judicial Tribunal

Jurisdiction

Adjudicates disputes between:

- Licensor and Licensee
- Two or more Service Providers
- Service Provider and Group of Consumers

Appellate Function

Hears appeals against:

- Directions of TRAI
- Decisions of TRAI
- Orders of TRAI

TRAI vs TDSAT

Parameter	TRAI	TDSAT
Nature	Regulatory Body	Adjudicatory Tribunal
Established	1997	2000
Main Function	Telecom Regulation	Dispute Resolution
Tariff Regulation	Yes	No
Consumer Protection	Yes	No
Adjudication of Disputes	No	Yes

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In NEWS:TRAI issues Draft Telecom Consumer Protection Regulation 2026

89.Egyptian Vulture

Q.With reference to the Egyptian Vulture (Neophron percnopterus), consider the following statements:

1. It is one of the world's largest vultures.
2. It is endemic to the Indian subcontinent.
3. It is listed as Endangered by the International Union for Conservation of Nature (IUCN).

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (a) Only one

Statement 1: The Egyptian Vulture is not one of the world's largest vultures.In fact, it is recognized as one of the world's smallest vulture species.
Compared to species such as:

- Himalayan Griffon
- Eurasian Griffon
- Cinereous Vulture , the Egyptian Vulture is significantly smaller in size. **Hence, Statement 1 is incorrect.**

Statement 2: The Egyptian Vulture is not endemic to the Indian subcontinent.Its distribution extends across a vast geographical range including:

- Southern Europe
- Northern and Central Africa
- Middle East
- Central Asia
- Indian Subcontinent

An endemic species is one that is naturally restricted to a particular geographic region. Since the Egyptian Vulture occurs across multiple continents, it cannot be considered endemic to India or the Indian subcontinent.**Hence, Statement 2 is incorrect.**

Statement 3:

According to the International Union for Conservation of Nature (IUCN) Red List, the Egyptian Vulture is classified as Endangered.The species has experienced substantial population declines due to:

- Habitat loss
- Reduced food availability
- Poisoning
- Electrocution and collisions with power infrastructure
- Use of agricultural chemicals **Hence, Statement 3 is correct.**

Value Addition

Egyptian Vulture



Feature	Details
Common Name	Egyptian Vulture
Scientific Name	*Neophron percnopterus*
Family	Accipitridae
Type	Old World Vulture
IUCN Status	Endangered
Distribution	Europe, Africa, Middle East, Central Asia, Indian Subcontinent

Habitat

The species commonly inhabits:

- Open landscapes
- Semi-arid and arid regions
- Rocky cliffs
- Grazing lands
- Agricultural areas
- Human-dominated landscapes

Historically, it was frequently observed around:

- Villages
- Temple towns
- Cattle-grazing areas

Feeding Habits

The Egyptian Vulture is a scavenger with a highly diverse diet.
Food Sources:

- Carrion (animal carcasses)
- Insects
- Small reptiles
- Amphibians

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- Mammal faeces (They ingest animal feces to derive yellow carotenoid pigments for their faces.). This opportunistic feeding behaviour helps it survive in varied habitats.

Breeding Characteristics

Monogamous Species

- Forms long-term breeding pairs.
- Mates generally remain together across breeding seasons.

Nesting Behaviour

- Builds large nests on cliffs and rocky outcrops.
- Continuously adds nesting material during the breeding season.

Ecological Importance

Nature's Sanitation Worker

The Egyptian Vulture performs a vital ecological function by:

- Removing animal carcasses from the environment.
- Preventing the spread of diseases.
- Maintaining ecosystem health.

Ecosystem Services

- Accelerates decomposition processes.
- Supports nutrient recycling.

Major Threats

Habitat Loss

- Urban expansion
- Infrastructure development

Food Scarcity

- Decline in livestock carcasses available for scavenging.

Poisoning

- Agricultural chemicals
- Poison-laced carcasses

Power Infrastructure

- Electrocution
- Collision with transmission lines

In NEWS: Understanding the disappearance of Egyptian Vulture in Tamil Nadu

90.International Monetary Fund(IMF)

Q.With reference to the International Monetary Fund (IMF), consider the following statements:

1. The IMF provides long-term development loans for infrastructure projects.
2. India is a founding member of the IMF.
3. IMF quotas are denominated in Special Drawing Rights (SDRs).

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two

Statement 1:

The IMF does not provide long-term development loans for infrastructure projects.Its primary role is to:

- Maintain international monetary stability.
- Provide financial assistance to countries facing Balance of Payments (BoP) crises.
- Support macroeconomic stabilization.

Long-term development and infrastructure financing is mainly undertaken by institutions such as:

- World Bank
- Asian Development Bank
- New Development Bank **Hence, Statement 1 is incorrect.**

Statement 2:

India is a founding member of the IMF.India joined the IMF on 27 December 1945, shortly after the institution was established following the Bretton Woods Conference. **Hence, Statement 2 is correct.**

Statement 3:

Each member country's quota in the IMF is denominated in Special Drawing Rights (SDRs).Quotas determine:

- Financial contribution to the IMF
- Voting power
- Access to IMF financing
- Share in SDR allocations **Hence, Statement 3 is correct.**

Value Addition

International Monetary Fund (IMF)

Feature	Details
Established	1944
Operational Since	1945
Origin	Bretton Woods Conference
Headquarters	Washington, D.C., USA
Members	191 Countries
India	Founding Member
Main Objective	International Monetary Stability

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Objectives of IMF

Promote International Monetary Cooperation

- Facilitates cooperation among member countries.
- Provides a platform for consultation on monetary issues.

Ensure Exchange Rate Stability

- Prevents competitive currency devaluations.
- Promotes orderly exchange arrangements.

Facilitate International Trade

- Encourages balanced growth of world trade.

Provide Financial Assistance

- Assists countries facing external payment crises.

Reduce Global Financial Instability

- Helps countries restore macroeconomic stability.

SDR Currency Basket

The SDR value is based on:

1. US Dollar
2. Euro
3. Chinese Renminbi
4. Japanese Yen
5. British Pound Sterling

Uses of SDRs

- Supplement foreign exchange reserves.
- Enhance global liquidity.
- Facilitate transactions among IMF members.
- Strengthen financial stability.

Major IMF Reports

Report	**Purpose**
World Economic Outlook (WEO)	Global Economic Forecasts
Global Financial Stability Report (GFSR)	Financial Sector Risks
Fiscal Monitor	Public Finance & Fiscal Trends
External Sector Report	External Balances Analysis

IMF vs World Bank

Parameter	**IMF**	**World Bank**
Main Purpose	Monetary Stability	Development Financing
Lending Nature	Short- to Medium-Term	Long-Term
Focus	Balance of Payments Support	Infrastructure & Development
Established	1944	1944
Headquarters	Washington, D.C.	Washington, D.C.

In NEWS:IMF cuts global growth forecast assuming Strait of Hormuz opens mid-July, but Trump ends ceasefire

91.National Board for Wildlife (NBWL)

Q.With reference to the National Board for Wildlife (NBWL), consider the following statements:

1. NBWL is the apex advisory body for wildlife conservation in India.
2. NBWL controls poaching and illegal wildlife trade through direct enforcement powers.
3. The National Board for Wildlife is headed by the Minister for Environment, Forest and Climate Change.

Which of the statements given above is/are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 only

EXPLANATION

Statement 1:

The National Board for Wildlife (NBWL) is India's highest statutory advisory body on matters relating to wildlife conservation and protection. It was constituted by the Central Government in 2003 under Section 5A of the Wildlife (Protection) Act, 1972 through 2002 amendment.The Board advises the Central and State Governments on:

- Wildlife conservation policies
- Management of Protected Areas
- Wildlife protection measures
- Conservation planning
- Ecological sustainability

It is therefore regarded as the apex advisory body for wildlife conservation in India.**Hence, Statement 1 is correct.**

Statement 2:

The NBWL plays an advisory and policy-making role. It recommends measures for preventing poaching and illegal wildlife trade, but it does not possess

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direct enforcement powers.Actual enforcement is undertaken by agencies such as:

- State Forest Departments
- Wildlife Crime Control Bureau (WCCB)
- Police Authorities
- Customs Authorities
- Directorate of Revenue Intelligence (DRI)

NBWL can recommend stronger enforcement mechanisms but cannot directly investigate, arrest offenders, conduct raids or prosecute wildlife criminals. **Hence, Statement 2 is incorrect.**

Statement 3:

The National Board for Wildlife is not headed by the Minister for Environment, Forest and Climate Change.Under the Wildlife (Protection) Act, 1972:

- Prime Minister of India is the Chairperson of NBWL.
- Union Minister for Environment, Forest and Climate Change serves as the Vice-Chairperson.

Hence, Statement 3 is incorrect.

VALUE ADDITION

National Board for Wildlife (NBWL)

Feature	**Details**
Established	2003
Statutory Basis	Section 5A, Wildlife (Protection) Act, 1972
Nature	Statutory Body
Status	Apex Wildlife Advisory Body
Chairperson	Prime Minister of India
Vice-Chairperson	Union Minister for Environment, Forest and Climate Change
Members	The Board consists of the Prime Minister, Union Ministers, MPs, experts, NGOs and other nominated members
Member Secretary	Additional Director General of Forests (Wildlife) & Director, Wildlife Preservation

Composition of NBWL

Chairperson- Prime Minister of India

Vice-Chairperson- Minister for Environment, Forest and Climate Change

Other Members

- Chief of Army Staff
- Defence Secretary
- Expenditure Secretary
- Members of Parliament
- Wildlife Experts
- Conservationists
- Ecologists

Eminent Experts

- 10 members nominated by the Central Government from among:
  - Conservationists
  - Ecologists
  - Environmentalists

Functions of NBWL

1. Wildlife Conservation
2. Advisory Role
3. Protected Areas- Recommend creation and management of National Parks, Wildlife Sanctuaries , Conservation Reserves , Community Reserves
4. Environmental Appraisal
5. Monitoring
6. Wildlife Status Reports

Standing Committee of NBWL- The Standing Committee is the operational arm of the NBWL.

Feature	**Details**
Composition	Up to 10 Members
Chairperson	Environment Minister
Function	Examines project proposals affecting wildlife habitats
Focus	Land diversion and wildlife clearances

Wildlife Crime Control Bureau (WCCB)

Feature	**Details**
Established	2007
Ministry	MoEFCC
Nature	Statutory Enforcement Agency
Objective	Combat Wildlife Crime and Illegal Wildlife Trade

NBWL vs NTCA

Parameter	**NBWL**	**NTCA**
Full Form	National Board	National Tiger

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	for Wildlife	Conservation Authority
Scope	All Wildlife	Tigers and Tiger Reserves
Established	2003	2006
Chairperson	Prime Minister	Environment Minister
Nature	Apex Advisory Body	Statutory Conservation Authority
Act	Wildlife (Protection) Act, 1972	Wildlife (Protection) Act, 1972

In NEWS:PIL plea in Delhi HC against functioning of Wildlife Board

92.Lokayan 26

Q.Consider the following statements regarding Lokayan 26:

1. It is a transoceanic expedition undertaken by INS Sudarshini.
2. The expedition aims to strengthen maritime cooperation and advance the vision of MAHASAGAR.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

EXPLANATION

Statement 1: Lokayan 26 is a 10-month transoceanic expedition being undertaken by the Indian Naval Sail Training Ship (STS) INS Sudarshini.The expedition aims to project India's maritime heritage and strengthen international maritime engagement through port visits, cultural outreach and naval diplomacy.As part of the expedition, INS Sudarshini will sail more than 22,000 nautical miles and visit 18 foreign ports across 13 countries.**Hence, Statement 1 is correct.**

Statement 2: One of the major objectives of Lokayan 26 is to strengthen India's maritime cooperation with partner nations and promote goodwill across the oceans.The expedition also supports India's maritime vision of MAHASAGAR (Mutual and Holistic Advancement for Security and Growth Across Regions), which emphasizes:

- Maritime security
- Regional cooperation
- Capacity building
- Sustainable maritime development
- Collective prosperity across the Indo-Pacific and beyond

Through diplomatic engagements, sail training, cultural exchanges and international port calls, Lokayan 26 contributes directly to advancing the MAHASAGAR vision.**Hence, Statement 2 is correct.**

VALUE ADDITION

Lokayan 26

Feature	**Details**
Nature	Transoceanic Maritime Expedition
Vessel	INS Sudarshini
Duration	10 Months
Distance Covered	22,000+ Nautical Miles
Foreign Ports	18
Countries Visited	13
Participants	Indian Navy and Indian Coast Guard Trainees
Objective	Maritime Cooperation and Sail Training
Strategic Vision	MAHASAGAR

Objectives of Lokayan 26

Maritime Diplomacy

- Strengthen India's maritime partnerships.
- Enhance naval goodwill and cooperation.

Sail Training

- Provide practical training in:
  - Ocean navigation
  - Traditional seamanship
  - Survival at sea
  - Weather assessment

Cultural Outreach

- Showcase India's maritime heritage.

- Promote people-to-people ties.

Strategic Engagement

- Support India's role as a responsible maritime power.
- Expand cooperation in the Indo-Pacific and Atlantic regions.

INS Sudarshini

Feature	Details
Type	Sail Training Ship (STS)
Built By	Goa Shipyard Limited
Commissioned	27 January 2012
Home Base	Kochi
Naval Command	Southern Naval Command
Type of Vessel	Three-Masted Barque
Nature	Indigenously Built

MAHASAGAR

Mutual and Holistic Advancement for Security and Growth Across Regions

Objectives

- Maritime security
- Blue economy cooperation
- Disaster response
- Capacity building
- Sustainable maritime development
- Regional stability

IN NEWS:LOKAYAN 26 - INS SUDARSHINI CONCLUDES PORT CALL AT NEW YORK, USA

93.Semaglutide

Q.Which one of the following best describes Semaglutide, recently seen in the news?

(a) A monoclonal antibody used for the treatment of rheumatoid arthritis
(b) A GLP-1 receptor agonist used primarily for the treatment of Type 2 Diabetes Mellitus and obesity
(c) An antiviral drug used for the treatment of Hepatitis C
(d) A gene therapy used to treat inherited metabolic disorders

Answer: (b) A GLP-1 receptor agonist used primarily for the treatment of Type 2 Diabetes Mellitus and obesity

EXPLANATION

Option (a):

Monoclonal antibodies are laboratory-produced proteins designed to target specific molecules involved in diseases.Examples include:

- Adalimumab
- Infliximab
- Rituximab

These are commonly used for autoimmune disorders such as rheumatoid arthritis and certain cancers.Semaglutide is not a monoclonal antibody.**Hence, Option (a) is incorrect.**

Option (b):

Semaglutide is a GLP-1 (Glucagon-Like Peptide-1) Receptor Agonist primarily used for the treatment of:

- Type 2 Diabetes Mellitus (T2DM)
- Obesity and weight management

It mimics the action of the naturally occurring hormone GLP-1, which is released after meals.Semaglutide works by:

- Stimulating insulin secretion
- Reducing glucagon secretion
- Slowing gastric emptying
- Increasing satiety (feeling of fullness)

As a result, it helps lower blood glucose levels and promotes weight loss.It is the active ingredient in blockbuster drugs such as: Ozempic, Wegovy . **Hence, Option (b) is correct.**

Option (c):

Antiviral drugs are used to treat viral infections such as:

- Hepatitis C
- HIV/AIDS
- Influenza

Examples include: Sofosbuvir, Tenofovir, RemdesivirSemaglutide has no antiviral action. **Hence, Option (c) is incorrect.**

Option (d):

Gene therapies involve modification or replacement of defective genes to treat genetic disorders.Semaglutide is a pharmacological drug and not a gene therapy.Examples include:Luxturna, Zolgensma **Hence, Option (d) is incorrect.**

VALUE ADDITION

Semaglutide

Feature	Details
Drug Type	GLP-1 Receptor Agonist
Primary Use	Type 2 Diabetes Mellitus

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Additional Use	Obesity and Weight Management
Route	Injectable (primarily) and also oral formulations
Mimics	Glucagon-Like Peptide-1 (GLP-1)
Effect	Lowers Blood Sugar and Promotes Weight Loss

Glucagon-Like Peptide-1(GLP-1)

GLP-1 is a naturally occurring hormone secreted by intestinal cells after food intake.Semaglutide mimics these physiological effects.

Functions of GLP-1

- Stimulates insulin release
- Suppresses glucagon secretion
- Slows gastric emptying
- Promotes satiety
- Helps regulate blood glucose

Mechanism of Action of Semaglutide

Step 1: Increases Insulin Secretion

When blood glucose levels rise after a meal, Semaglutide stimulates the pancreas to release insulin.

Step 2: Decreases Glucagon Secretion

It reduces glucagon production, thereby limiting glucose release from the liver.

Step 3: Slows Gastric Emptying

Food remains in the stomach longer, leading to slower absorption of glucose.

Step 4: Reduces Appetite

It acts on appetite centres in the brain, increasing the feeling of fullness.

Clinical Benefits

Blood Sugar Control, Weight Reduction, Cardiovascular Benefits : Studies have shown reduction in:Risk of heart attack, Risk of stroke, Cardiovascular mortality in high-risk patients

Type 2 Diabetes Mellitus (T2DM)

Feature	**Type 2 Diabetes**
Cause	Insulin Resistance
Insulin Production	Present but inadequate
Common Age Group	Adults (increasingly seen in younger individuals)
Associated Factors	Obesity, Sedentary Lifestyle, Genetics
Blood Sugar	Elevated

Active Pharmaceutical Ingredient (API)

An Active Pharmaceutical Ingredient (API) is the biologically active component of a medicine that produces the intended therapeutic effect.Example:For Semaglutide injections, semaglutide itself is the API.

Generic Drugs

A generic medicine contains Same active ingredient, Same dosage form, Same safety profile, Same efficacy as the original patented drug.

Generic Drugs Importance:

- Lower cost
- Greater accessibility
- Increased competition
- Improved public health outcomes

IN NEWS:Dr. Reddy's halts semaglutide supply on quality, shares slide, Torrent recalls product

94.National Tiger Conservation Authority(NTCA)

Q.With reference to the National Tiger Conservation Authority (NTCA), consider the following statements:

1. The National Tiger Conservation Authority is a statutory body.
2. NTCA promotes coexistence measures for human-wildlife conflict in tiger landscapes
3. STRIDES 2026 is a report assessing infrastructure, ecology and social parameters of Tiger Reserves.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (c) All three

EXPLANATION

Statement 1:

The National Tiger Conservation Authority (NTCA) is a statutory body functioning under the Ministry of Environment, Forest and Climate Change (MoEFCC). It was established in 2006 through an amendment to the Wildlife (Protection) Act, 1972 to provide statutory backing to Project Tiger and strengthen tiger conservation in India. The Authority serves as the apex institution for tiger conservation and

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management of tiger reserves in the country.**Hence, Statement 1 is correct.**

Statement 2:

One of the important functions of NTCA under the Wildlife (Protection) Act, 1972 is to provide management focus and measures for addressing human-wildlife conflict and promoting coexistence in forest areas outside National Parks, Wildlife Sanctuaries and Tiger Reserves. The Authority emphasizes:

- Landscape-level conservation
- Protection of wildlife corridors
- Community participation
- Conflict mitigation strategies
- Sustainable coexistence between humans and wildlife

This reflects the shift from species-centric conservation towards ecosystem and landscape-based conservation.**Hence, Statement 2 is correct.**

Statement 3:

During the 29th Meeting of the NTCA held at CASFOS, Coimbatore (Tamil Nadu), the Union Minister for Environment, Forest and Climate Change released STRIDES 2026.STRIDES 2026 (Status of Tiger Reserves – Infrastructure, Development, Ecology & Social Parameters) is a comprehensive assessment report that evaluates:

- Conservation interventions
- Infrastructure status
- Ecological conditions
- Development indicators
- Social parameters across India's Tiger Reserves.

The report is intended to facilitate:

- Evidence-based planning
- Adaptive management
- Informed decision-making
- Improved conservation outcomes **Hence, Statement 3 is correct.**

VALUE ADDITION

National Tiger Conservation Authority (NTCA)

Feature	**Details**
Established	2006
Statutory Basis	Wildlife (Protection) Act, 1972 (Amended 2006)
Ministry	Ministry of Environment, Forest and Climate Change
Nature	Statutory Body
Chairperson	Union Minister for Environment, Forest and Climate Change
Objective	Tiger Conservation and Tiger Reserve Management
Headquarters	New Delhi

NTCA Creation:

The Tiger Task Force (2005) highlighted the need for a stronger institutional framework after concerns regarding declining tiger populations.

The NTCA was established to:

- Strengthen Project Tiger
- Provide statutory backing to conservation measures
- Improve accountability of States
- Ensure scientific management of Tiger Reserves
- Enhance parliamentary oversight

Composition of NTCA

Position	**Member**
Chairperson	Union Environment Minister
Vice-Chairperson	Minister of State (MoEFCC)
Members	Members of Parliament
Members	Wildlife Experts
Members	Senior Government Officials
Member Secretary	Senior MoEFCC Official

Powers and Functions of NTCA

Conservation Planning

- Approves Tiger Conservation Plans prepared by States.

Habitat Protection

- Evaluates ecological sustainability.
- Can oppose ecologically destructive projects.

Research and Monitoring

- Coordinates research on Tigers, Co-predators , Prey species, Habitats

Human-Wildlife Conflict Management

- Promotes coexistence measures.
- Supports conflict mitigation programmes.

Capacity Building

- Training of forest officers and frontline staff.

Tourism Regulation

- Prescribes tourism guidelines in Tiger Reserves.

Tiger Corridors

- Ensures connectivity between protected areas.

Project Tiger

Feature	Details
Launched	1973
Type	Centrally Sponsored Scheme
Ministry	MoEFCC
Objective	In-situ conservation of tigers
Implementing Agency	NTCA

Objectives

- Protect tiger populations.
- Conserve habitats.
- Maintain ecological balance.
- Reduce poaching.

STRIDES 2026: Status of Tiger Reserves – Infrastructure, Development, Ecology & Social Parameters

Purpose

To provide a holistic assessment of Tiger Reserves by evaluating:

- Infrastructure
- Development interventions
- Ecological health
- Social dimensions
- Conservation effectiveness

Importance

- Evidence-based conservation planning
- Better allocation of resources
- Identification of management gaps
- Strengthening adaptive management

Other Important Current Affairs from the 29th NTCA Meeting

Roadmap to Rescue

A strategic framework for:

- Rescue
- Rehabilitation
- Release of wild animals within tiger landscapes.

Significance

- Strengthens preparedness for wildlife emergencies.
- Addresses rising human-wildlife interactions.
- Standardizes rescue and rehabilitation protocols.

Management Effectiveness Evaluation (MEE)

The Authority deliberated on the 6th Cycle of Management Effectiveness Evaluation (MEE).

Purpose

To assess:

- Quality of management
- Protection effectiveness
- Habitat conservation
- Governance standards , across Tiger Reserves.

STRIPES: Symposium on Tiger Research, Innovation, Policy, Ecology & Sustainability

Objective

A national platform for:

- Sharing research findings
- Disseminating best practices
- Knowledge exchange among managers and researchers
- Promoting evidence-based tiger conservation

Tiger Conservation Foundations (TCFs)

The Authority discussed using Tiger Conservation Foundations for mobilizing resources through:

- Corporate partnerships
- CSR funding
- Philanthropic organisations
- Public participation ,This supports long-term conservation financing.

In NEWS:Shri Bhupender Yadav chairs the 29th Meeting of the National Tiger Conservation Authority at Coimbatore

95.Astra Missiles

Q.With reference to the Astra Missile Programme, consider the following statements:

1. Astra Mk 1 is an indigenous Beyond Visual Range Air-to-Air Missile (BVRAAM) developed by the Defence Research and Development Organisation (DRDO).
2. Astra Mk 1 is set to become India's first exported air-to-air missile system.
3. Astra Mk 2 is powered by a dual-pulse solid rocket motor.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (c) All three

EXPLANATION

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Statement 1: Astra Mk 1 is an indigenous Beyond Visual Range Air-to-Air Missile (BVRAAM) developed by the Defence Research and Development Organisation (DRDO).It is designed to engage and destroy enemy aircraft at ranges beyond visual contact and significantly enhances India's air combat capability.The missile has been developed under the Atmanirbhar Bharat initiative to reduce dependence on imported air-to-air missiles.It is being integrated with:

- Sukhoi Su-30 MKI
- Tejas Mk 1A
- Rafale (planned)
- Future naval fighter platforms **Hence, Statement 1 is correct.**

Statement 2: India and Indonesia recently signed an agreement for the supply of Astra Mk 1 missiles for Indonesia's Su-30 fighter fleet.This marks the first export of an Indian air-to-air missile system to a foreign country.The deal represents a major milestone in India's growing defence exports and demonstrates international confidence in indigenous defence technologies developed by DRDO.The export is also aligned with India's objective of becoming a major global defence manufacturing hub. **Hence, Statement 2 is correct.**

Statement 3: Astra Mk 2 is a more advanced version of Astra Mk 1.Unlike Astra Mk 1, which uses a single-pulse smokeless solid-fuel motor, Astra Mk 2 is powered by a dual-pulse solid rocket motor.Advantages of a dual-pulse motor include:

- Better energy management during flight
- Extended operational range
- Re-ignition capability during terminal phase
- Improved ability to engage manoeuvring targets
- Reduced chances of target escape

Due to these enhancements, Astra Mk 2's range has increased to approximately 200 km. **Hence, Statement 3 is correct.**

VALUE ADDITION

Astra Missile Family

Feature	**Details**
Type	Beyond Visual Range Air-to-Air Missile (BVRAAM)
Developer	DRDO
User	Indian Air Force and Indian Navy
Guidance	Inertial Navigation + Mid-course Update + Active Radar Homing
Role	Air-to-Air Combat
Status	Indigenous Missile Programme

BVRAAM:

Beyond Visual Range Air-to-Air Missile These missiles can engage targets beyond the pilot's visual range.Generally capable of striking targets beyond:

- 20 nautical miles
- Approximately 37 km

They are among the most important weapons in modern aerial warfare.

Astra Variants

Variant	**Status**	**Range**
Astra Mk 1	Inducted	80–110 km
Astra Mk 2	Advanced Testing	~200 km
Astra Mk 3 (Gandiva)	Under Development	350+ km

Astra Mk 1

Feature	**Details**
Range	80–110 km
Speed	Mach 4.5
Altitude	Up to 20 km
Motor	Single-pulse smokeless solid-fuel motor
Guidance	Active Radar Homing

Key Features

- Indigenous RF seeker
- High kill probability
- All-weather capability
- Fire-and-forget functionality
- Integrated with Su-30 MKI

Astra Mk 2

Feature	**Details**
Range	Around 200 km
Propulsion	Dual-pulse Solid Rocket Motor
Status	Under testing

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Objective	Counter long-range enemy missiles

Advantages of Dual-Pulse Motor

- Conserves energy during cruise phase
- Re-accelerates during terminal phase
- Increases lethality
- Better against highly manoeuvrable aircraft

Astra Mk 3 (Gandiva)

Feature	Details
Official Name	Gandiva
Status	Under Development
Expected Range	350+ km
Engine	Solid Fuel Ducted Ramjet (SFDR)

Importance:

Unlike conventional rocket motors, SFDR technology provides:

- Sustained thrust throughout flight
- Higher speed retention
- Longer engagement envelope
- Better interception capability

Guidance System of Astra

1. Inertial Guidance

Guides missile during the initial phase using onboard navigation systems.

2. Mid-Course Update

Receives target updates from launch aircraft.

3. Terminal Active Radar Homing

Missile's own radar locks onto and destroys the target during the final phase.

Strategic Importance of Astra

Self-Reliance

Reduces dependence on imported missiles such as:

- Russian R-77
- European Meteor

Force Multiplier

Enhances combat capability of:

- Su-30 MKI
- Tejas
- Rafale
- Future naval aircraft

Export Potential

Successful export to Indonesia opens opportunities in:

- Southeast Asia
- Africa
- Latin America
- Friendly Indian Ocean Region countries

Atmanirbhar Bharat

Demonstrates India's growing capability in:

- Missile technology
- Radar seekers
- Propulsion systems
- Defence manufacturing

India–Indonesia Defence Cooperation

Recent agreements include:

- Astra Mk 1 Missile Export
- BrahMos Missile Cooperation
- Maritime Security Cooperation
- Defence Technology Transfer
- Critical Minerals Cooperation

The Astra deal marks the first foreign export of India's indigenous air-to-air missile system, strengthening defence ties between India and Indonesia.

In NEWS:The significance of India's Astra missiles, which Indonesia is buying

96.Tribex

Q.With reference to the TribeX Platform, consider the following statements:

1. It is a digital learning platform dedicated to tribal arts, culture, traditional knowledge and skill development.
2. It includes a Repository Management System for knowledge preservation and progress tracking.
3. It is implemented by the Ministry of Culture.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two

EXPLANATION

Statement 1: The TribeX Platform is a first-of-its-kind digital learning platform dedicated to:

- Tribal arts
- Tribal culture
- Traditional knowledge systems
- Skill development

It has been conceived as a comprehensive digital ecosystem to promote, preserve and disseminate India's rich tribal heritage.The platform provides structured learning opportunities through:

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- Free certificate courses
- UGC-recognized Post Graduate Diploma programmes
- Tribal language learning modules
- Traditional art and craft courses

It serves as a one-stop digital platform for learning, documentation and preservation of tribal knowledge.

Hence, Statement 1 is correct.

Statement 2: The TribeX Platform integrates advanced digital infrastructure, including:

Learning Management System (LMS)

Used for : Online learning , Course delivery , Learner assessment , Progress monitoring

Repository Management System (RMS)

Used for: Knowledge preservation , Digital archiving , Resource management , Documentation of tribal traditions. Together, these systems provide a secure and seamless digital ecosystem for:

- Learning
- Knowledge management
- Progress tracking **Hence, Statement 2 is correct.**

Statement 3: The TribeX Platform has been launched by the Ministry of Tribal Affairs, Government of India.Its objective is to preserve and promote tribal languages, art forms, crafts and indigenous knowledge systems.The Ministry of Culture is not the implementing ministry for the platform.**Hence, Statement 3 is incorrect.**

VALUE ADDITION

TribeX Platform

Feature	Details
Launching Ministry	Ministry of Tribal Affairs
Nature	Digital Learning and Knowledge Preservation Platform
Focus	Tribal Arts, Culture and Traditional Knowledge
Coverage	Tribal Languages, Crafts, Art Forms and Indigenous Knowledge
Type	First-of-its-kind Tribal Digital Learning Ecosystem

Key Objectives

Preservation

- Tribal languages
- Oral traditions
- Folk knowledge
- Indigenous practices

Promotion

- Tribal art forms
- Handicrafts
- Handloom traditions
- Cultural heritage

Capacity Building

- Skill development
- Academic learning
- Certification programmes

Courses Offered

The platform currently hosts 20 free certificate courses covering:Tribal paintings, Handicrafts ,Handloom traditions ,Artefacts ,Traditional musical instruments

It also offers:

- UGC-recognized Post Graduate Diploma programmes
- Tribal language courses
- Traditional knowledge programmes

Significance of TribeX

1. Cultural Preservation
2. Knowledge Transmission
3. Digital Inclusion
4. Academic Recognition
5. Global Accessibility

IN NEWS:Union Tribal Affairs Minister Shri Jual Oram Launches TribeX – A Digital Learning Platform for Tribal Arts, Culture and Traditional Knowledge

97.Quality council of India

Q.With reference to the Quality Council of India (QCI), consider the following statements:

1. QCI was established through an Act of Parliament.
2. The National Accreditation Board for Certification Bodies (NABCB) functions under QCI.
3. QCI functions under the Ministry of Consumer Affairs, Food and Public Distribution.

How many of the statements given above are *incorrect*?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two

EXPLANATION:

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Statement 1: The Quality Council of India (QCI) was not established through an Act of Parliament.It is a non-profit autonomous organisation registered under the Societies Registration Act XXI of 1860.QCI was established in 1997 through a collaborative initiative between:

- Government of India
- Associated Chambers of Commerce and Industry of India (ASSOCHAM)
- Confederation of Indian Industry (CII)
- Federation of Indian Chambers of Commerce and Industry (FICCI)

Therefore, QCI is a registered society and not a statutory body created by legislation. **Hence, Statement 1 is incorrect.**

Statement 2: The National Accreditation Board for Certification Bodies (NABCB) is one of the constituent boards functioning under the Quality Council of India.NABCB provides accreditation to:

- Certification Bodies
- Inspection Bodies
- Validation and Verification Bodies

It promotes internationally accepted standards relating to:

- Quality Management Systems
- Environmental Management Systems
- Food Safety Management Systems
- Product Certification **Hence, Statement 2 is correct.**

Statement 3:QCI functions under the administrative control of the Department for Promotion of Industry and Internal Trade (DPIIT) under the Ministry of Commerce and Industry.It does not function under the Ministry of Consumer Affairs, Food and Public Distribution.DPIIT acts as the nodal department for policy coordination and administrative oversight of QCI. **Hence, Statement 3 is incorrect.**

Value Addition

Quality Council of India (QCI)

Feature	Details
Established	1997
Nature	Non-profit Autonomous Organisation
Registration	Societies Registration Act XXI of 1860
Nodal Department	DPIIT
Ministry	Ministry of Commerce & Industry
Members	38
Role	National Accreditation Body

Objectives of QCI

Independent Third-Party Assessment : Products, Services, Processes

Promote Quality Ecosystem : Standardisation, Accreditation, Certification, Quality Assurance

National Quality Movement

- Leads the National Quality Campaign
- Encourages adoption of global quality standards

Boards under QCI

Board	Function
NABL	Testing & Calibration Laboratories
NABH	Hospitals & Healthcare Providers
NABET	Education & Training
NABCB	Certification Bodies
NBQP	Quality Promotion

NABCB (National Accreditation Board for Certification Bodies)

Functions

Provides accreditation to:Certification Bodies, Inspection Bodies, Validation & Verification Bodies

Promotes Standards

Standard	Purpose
ISO 9001	Quality Management System
ISO 14001	Environmental Management System
ISO 22000	Food Safety Management System

Composition of QCI

Particular	Details
Total Members	38
Government Representation	Equal
Industry Representation	Equal
Other Stakeholders	Equal

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IN NEWS:Quality Council of India and National Small Industries Corporation Sign MoU to Strengthen Quality, Competitiveness and Market Access for MSMEs

98.World Intellectual Property Organisation (WIPO)

Q.With reference to the World Intellectual Property Organization (WIPO), consider the following statements:

1. WIPO is a specialized agency of the United Nations headquartered in Geneva, Switzerland.
2. WIPO administers international treaties relating to patents, trademarks and copyrights.
3. The *World Intangible Investment Report* is published by WIPO in collaboration with Luiss Business School.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (c) All three
EXPLANATION:
Statement 1: The World Intellectual Property Organization (WIPO) is a specialized agency of the United Nations responsible for promoting the protection of intellectual property (IP) worldwide through cooperation among nations.

- Headquarters: Geneva, Switzerland
- Established under the WIPO Convention
- Became a specialized UN agency in 1974
- India became a member in 1975 **Hence, Statement 1 is correct.**

Statement 2:WIPO administers numerous international treaties dealing with various forms of intellectual property, including:

- Patents – Patent Cooperation Treaty (PCT)
- Trademarks – Madrid Protocol
- Copyrights – Berne Convention, WIPO Copyright Treaty
- Industrial Designs – Locarno Agreement
- Geographical Indications and Industrial Property – Paris Convention

It provides global registration services, dispute resolution mechanisms, and policy forums for IP governance.**Hence, Statement 2 is correct.**

Statement 3: The World Intangible Investment Report 2026 was launched jointly by:

- World Intellectual Property Organization (WIPO)
- Luiss Business School (Italy)

The report assesses investments in intangible assets such as:

- Research & Development (R&D)
- Software and Databases
- Intellectual Property
- Organizational Capital
- Brand Value
- Innovation Capabilities

The report highlighted that India recorded the fastest growth in intangible investment among the world's 15 largest economies, reaching approximately USD 78.2 billion in 2023.**Hence, Statement 3 is correct.**

Value Addition

World Intellectual Property Organization (WIPO)

Feature	**Details**
Established	1967 (WIPO Convention)
Became UN Specialized Agency	1974
Headquarters	Geneva, Switzerland
Members	193 Member States
India Joined	1975
Parent System	United Nations
Objective	Promotion and protection of Intellectual Property worldwide

Major Functions of WIPO

IP Protection

- Promotes global protection of intellectual property rights.

Treaty Administration

- Administers international IP treaties.

Global Registration Services

- Patents (PCT)
- Trademarks (Madrid System)
- Industrial Designs (Hague System)

Dispute Resolution

- Resolves cross-border IP disputes through arbitration and mediation.

Capacity Building

- Assists developing countries in strengthening IP ecosystems.

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Important WIPO-Administered Treaties

Treaty	Subject
Paris Convention (1883)	Industrial Property
Berne Convention (1886)	Copyright Protection
Patent Cooperation Treaty (PCT)	International Patent Filing
Madrid Protocol	International Trademark Registration
Marrakesh Treaty	Access to Books for Visually Impaired
WIPO Copyright Treaty	Copyright in Digital Environment

World Intangible Investment Report (WIIR)

Feature	Details
Published By	WIPO & Luiss Business School
Nature	Global assessment of intangible investments
Focus Areas	R&D, Software, IP, Innovation, Branding, Organizational Capital
Latest Report	World Intangible Investment Report 2026

Intangible Assets Include:

- Research & Development (R&D)
- Software and Databases
- Patents
- Copyrights
- Trademarks
- Brand Value
- Human Capital
- Organizational Know-how

Reports Published by WIPO

Report	Purpose
Global Innovation Index (GII)	Measures innovation performance of countries
World Intellectual Property Indicators (WIPI)	Global IP statistics
World Intellectual Property Report (WIPR)	Trends in innovation and IP
World Intangible Investment Report (WIIR)	Tracks intangible investments globally

India and WIPO: World Intangible Investment Report 2026

- India recorded the fastest growth in intangible investment among the world's 15 largest economies.
- Intangible investment reached USD 78.2 billion in 2023.
- Growth rate: 7.9%.
- Key drivers:
  - Startup ecosystem
  - Digital transformation
  - R&D spending
  - Software development
  - Intellectual property creation
  - Innovation-led growth

In NEWS:WIPO report reaffirms India's emergence as a global innovation powerhouse

99.Pachpadra Refinery-Cum-Petrochemical Complex

Q.With reference to the recently inaugurated PachpadraRefinery – Cum - Petrochemical Complex, consider the following statements:

1. It has been developed jointly by HPCL and the Government of Rajasthan.
2. It is India's first greenfield integrated refinery-cum-petrochemical complex.
3. The project has a Nelson Complexity Index (NCI) of 17.0.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (c) All three

EXPLANATION:

Statement 1:

The Pachpadra Refinery-Cum-Petrochemical Complex has been developed jointly by:

- Hindustan Petroleum Corporation Limited (HPCL)

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- Government of Rajasthan

The project represents one of India's largest energy-sector investments and is expected to transform Rajasthan into a major petrochemical hub.**Hence, Statement 1 is correct.**

Statement 2:

The facility is recognized as India's first greenfield integrated refinery-cum-petrochemical complex.A greenfield project refers to a facility built from scratch on previously undeveloped land.The complex integrates:

- Crude oil refining
- Petrochemical manufacturing

within a single site, resulting in greater operational efficiency, higher value addition and improved profitability.**Hence, Statement 2 is correct.**

Statement 3:

The Pachpadra Complex has a Nelson Complexity Index (NCI) of 17.0, making it one of the most sophisticated refinery complexes globally.A higher NCI indicates:

- Advanced refining capabilities
- Ability to process a wider variety of crude oils
- Greater production of high-value petroleum products and petrochemicals

The NCI of 17.0 places the refinery among the most technologically advanced refining facilities in the world.**Hence, Statement 3 is correct.**

Value Addition

Pachpadra Refinery-Cum-Petrochemical Complex

Feature	**Details**
Location	Pachpadra, Balotra District, Rajasthan
Type	First Greenfield Integrated Refinery-Cum-Petrochemical Complex
Developed By	HPCL & Government of Rajasthan
Status	India's First Greenfield Integrated Refinery-Cum-Petrochemical Complex
Crude Processing Capacity	9 MMTPA
Petrochemical Production Capacity	2.4 MMTPA
Nelson Complexity Index	17.0
Petrochemical Yield	More than 26%

Greenfield Project:

A Greenfield Project is a project developed:

- On previously undeveloped land
- Without using existing infrastructure
- Built entirely from scratch

Advantages

- Modern technology adoption
- Higher efficiency
- Better environmental compliance
- Optimized infrastructure planning

Nelson Complexity Index (NCI)

The Nelson Complexity Index is a globally accepted indicator used to measure the sophistication and technological advancement of an oil refinery.

Significance

Higher NCI	**Lower NCI**
Advanced refinery	Basic refinery
More value-added products	Limited product range
Processes heavy crude	Processes simpler crude
Higher profitability	Lower profitability

Pachpadra Refinery

- NCI: 17.0
- Among the highest globally.

Significance of the Project

- Energy Security
- Industrial Growth
- Economic Diversification
- Employment Generation

In NEWS:Pachpadra refinery-cum-petrochemical complex: A new era for India's energy and manufacturing sectors

100. Understanding Lifelong Learning for All in Society (ULLAS)

Q.With reference to the ULLAS (Understanding Lifelong Learning for All in Society) Initiative, consider the following statements:

1. It is a Centrally Sponsored Scheme implemented under the National Education Policy (NEP) 2020.

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2. Its key objective is to provide Foundational Literacy and Numeracy (FLN) to non-literate persons aged 15 years and above.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2
EXPLANATION:
Statement 1:
ULLAS (Understanding Lifelong Learning for All in Society), also known as the New India Literacy Programme (NILP), is a Centrally Sponsored Scheme (CSS).It aligns with the vision of the National Education Policy (NEP) 2020, which emphasizes lifelong learning, adult education, and universal literacy.The scheme is being implemented during the period 2022–2027 by the Department of School Education and Literacy (DoSEL) under the Ministry of Education.**Hence, Statement 1 is correct.**
Statement 2:
One of the primary objectives of ULLAS is to provide Foundational Literacy and Numeracy (FLN) to non-literate persons aged 15 years and above who could not access formal education.The programme aims to transform literacy from merely signing one's name to acquiring:

- Reading skills
- Writing skills
- Numeracy skills
- Digital literacy
- Financial literacy
- Essential life skills

The scheme targets 5 crore learners over a five-year period through a hybrid mode of learning.**Hence, Statement 2 is correct.**
Value Addition
ULLAS (Understanding Lifelong Learning for All in Society)

Feature	**Details**
Full Form	Understanding Lifelong Learning for All in Society
Other Name	New India Literacy Programme (NILP)
Type	Centrally Sponsored Scheme (CSS)
Implementation Period	2022–2027
Nodal Ministry	Ministry of Education
Nodal Department	Department of School Education and Literacy (DoSEL)
Target Group	Non-literate persons aged 15 years and above
Target	5 crore learners
Vision	Jan Jan Saakshar
Assessment Tool	FLNAT
Certification	NIOS & DoSEL

ULLAS:
ULLAS is a nationwide adult literacy initiative aimed at creating a literate and empowered society through community participation and volunteerism.The scheme is based on the spirit of:
Kartavya Bodh (Sense of Duty)- where volunteers contribute towards making Bharat fully literate.
Five Components of ULLAS
1. Foundational Literacy and Numeracy (FLN)- Basic reading, writing and arithmetic skills.
2. Critical Life Skills- Financial literacy, digital literacy and legal awareness.
3. Basic Education- Equivalent learning opportunities for school dropouts.
4. Vocational Skills- Skill development and employability enhancement.
5. Continuing Education- Lifelong learning opportunities for adults.
Significance of ULLAS

- Social Inclusion
- Human Capital Development
- Digital Empowerment
- Gender Empowerment
- Inclusive Development

FLNAT (Foundational Literacy and Numeracy Assessment Test)
FLNAT is the assessment framework used under ULLAS to evaluate literacy and numeracy outcomes.
Features

- Conducted periodically.

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- Measures reading, writing and arithmetic abilities.
- Successful learners receive literacy certification.

Digital Features of ULLAS

ULLAS Mobile App & Portal

Used for:

- Learner registration
- Progress tracking
- Learning resources
- Assessment support

DIKSHA Platform

Provides digital learning content in multiple regional languages.

In NEWS:Uttarakhand becomes country's sixth fully literate State

101. Hummus Trial

Q.The term "Hummus Trail", recently seen in the news, best describes:

(a) A proposed India–Middle East trade corridor connecting ports through West Asia

(b) An informal travel circuit across India popular among Israeli tourists, particularly young veterans after compulsory military service

(c) A pilgrimage route linking important Jewish heritage sites in India

(d) A bilateral tourism initiative jointly administered by the Governments of India and Israel

Answer: (b) An informal travel circuit across India popular among Israeli tourists, particularly young veterans after compulsory military service

EXPLANATION

Option (a):

The Hummus Trail is not a trade or connectivity corridor. It has no relation to initiatives such as the India–Middle East–Europe Economic Corridor (IMEC). **Hence, Option (a) is incorrect.**

Option (b):

The Hummus Trail refers to an informal network of destinations across India frequented by Israeli tourists, especially those undertaking the **Tiul Gadol** (post-military "Big Trip") after completing mandatory military service.

Popular destinations include:

- Kasol
- Goa
- Hampi
- Gokarna
- Dharamkot
- Pushkar
- Rishikesh
- Varanasi
- Kodaikanal **Hence, Option (b) is correct.**

Option (c):

The Hummus Trail is not a religious pilgrimage route and is not specifically associated with Jewish heritage sites. **Hence, Option (c) is incorrect.**

Option (d):

The Hummus Trail is a colloquial term that emerged from travel patterns of Israeli backpackers. It is not an officially recognized tourism programme or bilateral initiative. **Hence, Option (d) is incorrect.**

VALUE ADDITION:

The Hummus Trail has recently drawn legal and security scrutiny following human rights petitions alleging violations of international humanitarian law by certain foreign nationals visiting or stationed in India.The HRF alleged that Gilboa's actions violated the Fourth Geneva Convention, to which India is a signatory.

Geneva Conventions (1949)

Convention	Protected Persons	Main Objective
First Geneva Convention	Wounded and sick members of armed forces on land	Protection and medical care of wounded and sick soldiers during land warfare
Second Geneva Convention	Wounded, sick and shipwrecked members of armed forces at sea	Protection and treatment of military personnel affected during naval warfare
Third Geneva Convention	Prisoners of War (POWs)	Humane treatment, protection of rights and welfare of prisoners of war
Fourth Geneva Convention	Civilians in conflict zones and occupied territories	Protection of civilians during armed conflicts and military occupation

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Geneva Conventions Act, 1960

Key Provision

India can prosecute any person, irrespective of nationality, for committing a grave breach of the Geneva Conventions, even if the offence occurred outside India.

Grave Breaches Include:

- Wilful killing of civilians.
- Torture or inhuman treatment.
- Extensive destruction of civilian property.
- Unlawful deportation or confinement.

Universal Jurisdiction

A principle of international law under which a State may prosecute certain serious crimes regardless of:

- Where the crime occurred.
- Nationality of the accused.
- Nationality of the victims.

Commonly Applied To:

- Genocide
- War crimes
- Crimes against humanity
- Piracy

India and Geneva Conventions

Feature	**Details**
Geneva Conventions Signed	1949
Implementing Law	Geneva Conventions Act, 1960
Scope	Grave breaches of humanitarian law
Jurisdiction	Applies even to foreigners for specified offences

In NEWS: What is the 'Hummus trail' and why is it under scrutiny?

102. Partnership on Cyber, Critical Technologies and Supply Chains(PACTS)

Q.With reference to the Partnership on Cyber, Critical Technologies and Supply Chains (PACTS), consider the following statements:

1. It seeks to strengthen cooperation on undersea cable security through the Quad Partnership for Cable Connectivity and Resilience.
2. The National Security Council Secretariat (NSCS) is the Indian lead agency for both the Supply Chain Resilience and Critical Technology pillars.
3. The Critical Technology pillar includes collaboration in Artificial Intelligence (AI), biotechnology, telecommunications, space technologies and advanced materials.

Which of the statements given above are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

Statement 1:

Under the Supply Chain Resilience and Diversification Pillar, India and Australia have agreed to strengthen cooperation for developing secure, resilient and trusted technology supply chains.The framework also seeks to enhance cooperation on undersea cable security through the Quad Partnership for Cable Connectivity and Resilience, recognising the strategic importance of submarine communication infrastructure for global connectivity and digital security.**Hence, Statement 1 is correct.**

Statement 2:

The National Security Council Secretariat (NSCS) is the designated Indian lead agency for:

- Supply Chain Resilience and Diversification Pillar
- Critical Technology Pillar

The NSCS will coordinate policy implementation and strategic cooperation with Australian counterparts in these domains.**Hence, Statement 2 is correct.**

Statement 3:

The Critical Technology Pillar aims to deepen bilateral cooperation in several emerging and strategic technologies, including:

- Artificial Intelligence (AI)
- Biotechnology
- Telecommunications
- Space Technologies
- Advanced Materials

The objective is to promote innovation, technological resilience and secure technological ecosystems.**Hence, Statement 3 is correct.**

Value Addition

PACTS at a Glance

Feature	Details
Full Form	Partnership on Cyber, Critical Technologies and Supply Chains
Countries Involved	India and Australia
Replaces	2020 Framework Arrangement on Cyber and Cyber Enabled Critical Technology Cooperation
Nature	Bilateral Strategic Partnership
Oversight (India)	Deputy National Security Advisor
Oversight (Australia)	Deputy Secretary, International and Security Group

Pillars of PACTS

Pillar	Indian Lead Agency
Supply Chain Resilience & Diversification	National Security Council Secretariat (NSCS)
Critical Technology	National Security Council Secretariat (NSCS)
Cyber Security	Cyber Diplomacy Division, Ministry of External Affairs
Digital Resilience	Oceania Division, Ministry of External Affairs
Defence Research Collaboration	Ministry of Defence

Critical Technology Areas under PACTS

Sector	Focus
Artificial Intelligence	Emerging technologies and innovation
Space Technologies	Strategic and civilian applications
Telecommunications	Secure communication networks
Biotechnology	Research and technological cooperation
Advanced Materials	High-end manufacturing and defence applications

Supply Chain Resilience Pillar

Focus Area	Objective
Trusted Technology Supply Chains	Reduce vulnerabilities and dependencies
Supply Chain Diversification	Enhance resilience against disruptions
Undersea Cable Security	Protect critical communication infrastructure
Quad Partnership for Cable Connectivity and Resilience	Strengthen regional digital connectivity

IN NEWS:Australia-India Partnership on Cyber, Critical Technologies and Supply Chains (PACTS)

103. PaBV-4

Q.With reference to Parrot Bornavirus 4 (PaBV-4), consider the following statements:

1. It is a single-stranded RNA virus.
2. It is the primary causative agent of Proventricular Dilatation Disease (PDD).
3. The virus can spread through contaminated food, water, saliva and feces.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

Statement 1:

IN NEWS:Deadly bird virus PaBV-4 identified for first time in India

Hence, Statement 1 is correct.

Statement 2:

PaBV-4 is the principal etiological agent responsible for Proventricular Dilatation Disease (PDD), a progressive neurological and gastrointestinal disease affecting psittacine birds such as:

- Parrots
- Macaws
- Cockatiels
- Lovebirds

- Parakeets

The disease was earlier known as Macaw Wasting Disease. **Hence, Statement 2 is correct.**

Statement 3:

The virus spreads through multiple transmission routes, including:

- Contaminated food
- Contaminated water
- Saliva
- Feces
- Vertical transmission from infected parents to eggs

A major concern is that infected birds may remain asymptomatic while continuing to shed the virus. **Hence, Statement 3 is correct.**

Value Addition

PaBV-4 at a Glance

Feature	Details
Type	Single-stranded RNA Virus
Species	*Orthobornavirus alphapsittaciforme*
Disease Caused	Proventricular Dilatation Disease (PDD)
Earlier Name of Disease	Macaw Wasting Disease
Main Hosts	Psittacine birds (Parrots, Macaws, Cockatiels, Lovebirds, Parakeets)
First Link with PDD Established	2008
First Confirmation in India	2026

Key Examination Facts

Feature	Details
Nature of Infection	Systemic non-cytolytic infection
Transmission	Food, water, saliva, feces, eggs
Diagnostic Method	RT-PCR
Major Threat	Silent transmission by asymptomatic birds
Impact	Digestive dysfunction, weight loss, neurological damage, mortality

IN NEWS:Deadly bird virus PaBV-4 identified for first time in India

104. Cyananthus Hookeri

Q.With reference to Cyananthus hookeri, recently seen in the news, consider the following statements:

1. It is a rare alpine flowering herb.
2. It is endemic only to the Western Himalayas.
3. The species has been recommended for classification as Critically Endangered in India under IUCN criteria.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (a) Only one

EXPLANATION

Statement 1: Cyananthus hookeri is a rare flowering alpine herb belonging to the Campanulaceae (bellflower) family. It is adapted to high-altitude habitats of the Eastern Himalayas and was recently rediscovered in Arunachal Pradesh after 158 years. **Hence, Statement 1 is correct.**

Statement 2: The species is not endemic to the Western Himalayas. It was historically recorded from Sikkim and has now been rediscovered in Arunachal Pradesh, both of which form part of the Eastern Himalayas, a global biodiversity hotspot. **Hence, Statement 2 is incorrect.**

Statement 3: Researchers recommended the species for classification as Endangered in India under IUCN criteria due to its extremely small population size. It has not been recommended as Critically Endangered. **Hence, Statement 3 is incorrect.**

Value Addition

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Cyananthus hookeri

Feature	Details
Scientific Name	*Cyananthus hookeri*
Family	Campanulaceae (Bellflower Family)
Plant Type	Alpine flowering herb
Flower Colour	Purple-blue
Habitat	High-altitude alpine regions
Rediscovered At	Chuna Valley, Tawang District
State	Arunachal Pradesh
Altitude	Around 3,600 metres
Rediscovered By	Botanical Survey of India (BSI)
Last Indian Record	1867
Recorded By	Sir Joseph Dalton Hooker
Gap Since Last Record	158 years
Mature Plants Observed	Fewer than 50

Botanical Survey of India (BSI)

Feature	Details
Established	1890
Ministry	Ministry of Environment, Forest and Climate Change
Headquarters	Kolkata
Function	Plant exploration, taxonomy and documentation
Role	Survey and conservation of India's floral diversity

Eastern Himalayas: Biodiversity Importance

- Recognized as a global biodiversity hotspot.
- Home to numerous endemic plant and animal species.
- Characterized by alpine meadows, temperate forests and high-altitude ecosystems.
- Arunachal Pradesh contains some of India's richest floral diversity.

Campanulaceae (Bellflower Family)

Feature	Details
Common Name	Bellflower Family
Plant Type	Mostly herbaceous flowering plants
Distribution	Temperate and alpine regions worldwide
Distinctive Feature	Bell-shaped flowers
Example	*Cyananthus hookeri*

In NEWS: Purple Himalayan flower makes a comeback after 158 years — for the first time in Arunachal

105. Samriddh Gram

Q.The term "Samriddh Gram", recently seen in the news, best describes:

(a) A centrally sponsored scheme for promoting organic farming in aspirational districts
(b) An integrated rural digital transformation initiative built on the BharatNet network for delivering citizen-centric services through village-level centres
(c) A financial inclusion programme aimed at providing interest-free loans to Self-Help Groups in rural areas
(d) A digital platform for monitoring the implementation of MGNREGS works across Gram Panchayats

Answer: (b) An integrated rural digital transformation initiative built on the BharatNet network for delivering citizen-centric services through village-level centres
EXPLANATION

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Option (a): Samriddh Gram is not an organic farming scheme. Although it provides agriculture-related services such as IoT-enabled soil testing and drone-assisted spraying, its scope extends far beyond agriculture.**Hence, Option (a) is incorrect.**

Option (b): Samriddh Gram is the Department of Telecommunications' flagship rural digital transformation initiative built on the BharatNet network.

At its core are Samriddhi Kendras, which function as village-level one-stop centres delivering:

- Healthcare services
- Education and skilling
- Agriculture support
- Financial inclusion
- E-governance
- E-commerce
- Digital connectivity

The initiative recently won the WSIS Prize 2026 under Action Line C6 (Enabling Environment).**Hence, Option (b) is correct.**

Option (c):

The initiative is not a credit or loan programme for Self-Help Groups.**Hence, Option (c) is incorrect.**

Option (d):

Samriddh Gram is not an MGNREGS monitoring platform.**Hence, Option (d) is incorrect.**

Value Addition

Samriddh Gram

Feature	Details
Ministry	Department of Telecommunications (DoT)
Nature	Integrated Rural Digital Transformation Initiative
Backbone	BharatNet Network
Village Centres	Samriddhi Kendras
Objective	Convert digital connectivity into socio-economic outcomes
Approach	Telecom infrastructure + Local facilitation + Community participation

Services Provided through Samriddhi Kendras

Sector	Services
Healthcare	Teleconsultation, Health ATMs, PM Jan Aushadhi
Education	Smart classrooms, AR/VR learning
Agriculture	IoT soil testing, drone spraying, smart irrigation
Financial Inclusion	CSC services, Banking Correspondents
E-Commerce	Marketing support for local products
Connectivity	FTTH, PM-WANI Wi-Fi
Governance	Digital citizen services
Security	CCTV-based village surveillance

WSIS (World Summit on the Information Society)

Feature	Details
Organised By	International Telecommunication Union (ITU) and UN agencies
Forum 2026 Venue	Geneva, Switzerland
Purpose	Promote ICTs for sustainable development
WSIS Prizes	Recognise innovative ICT-based development projects
Samriddh Gram Award	Winner under Action Line C6 (Enabling Environment)

BharatNet

Feature	Details
Nature	Rural Broadband Connectivity Project
Implemented By	Department of Telecommunications
Objective	Connect Gram Panchayats through high-speed broadband
Significance	World's largest rural telecom project

In NEWS:DoT's Samriddh Gram Wins WSIS Prize 2026 for Driving Rural Digital Transformation

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106. Mugger Crocodiles

Q.With reference to the Mugger Crocodile (*Crocodylus palustris*), recently seen in the news, consider the following statements:

1. It is a freshwater crocodile species found in rivers, lakes and marshes of the Indian subcontinent.
2. It is primarily a herbivorous reptile feeding on aquatic vegetation.
3. It is listed as Vulnerable in the IUCN Red List and included in Schedule I of the Wildlife (Protection) Act, 1972.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two

EXPLANATION

Statement 1:

The Mugger Crocodile (*Crocodylus palustris*) is primarily a freshwater crocodile found in:

- Rivers
- Lakes
- Marshes
- Reservoirs

Its distribution extends across India, Pakistan, Nepal, Sri Lanka and parts of Iran. **Hence, Statement 1 is correct.**

Statement 2:

The Mugger Crocodile is not herbivorous. It is a carnivorous predator that feeds on:

- Fish
- Frogs
- Crustaceans
- Insects
- Birds
- Small mammals

As an opportunistic predator, it plays an important role in maintaining the ecological balance of freshwater ecosystems. **Hence, Statement 2 is incorrect.**

Statement 3:

The species enjoys the highest level of legal protection in India and is recognized internationally for conservation.

- IUCN Red List: Vulnerable
- CITES: Appendix I
- Wildlife (Protection) Act, 1972: Schedule I **Hence, Statement 3 is correct.**

Value Addition

Mugger Crocodile (*Crocodylus palustris*)

Feature	**Details**
Common Name	Mugger Crocodile / Marsh Crocodile
Scientific Name	*Crocodylus palustris*
Family	Crocodylidae
Habitat	Rivers, lakes, marshes, reservoirs
Distribution	India, Pakistan, Nepal, Sri Lanka, Iran
Diet	Carnivorous
Nesting Type	Hole-nesting
Breeding Season	Dry season
Distinctive Feature	Broad snout among living crocodiles

Conservation Status

Convention/Act	**Status**
IUCN Red List	Vulnerable
CITES	Appendix I
Wildlife (Protection) Act, 1972	Schedule I

Threats to the Mugger Crocodile

- Habitat destruction and degradation
- Wetland loss
- Illegal hunting for skin
- Poaching for traditional medicine markets
- Water pollution and pesticide contamination

Aldrin

Feature	**Details**
Type	Organochlorine pesticide
Status in India	Banned
Reason for Ban	Persistent environmental pollutant
Impact	Contaminates soil and water; bioaccumulates in food chains

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Health Effects	Harmful to the nervous system of humans and wildlife

Crocodile Species in India

Feature	**Gharial**	**Mugger Crocodile (Marsh Crocodile)**	**Saltwater Crocodile**
Scientific Name	*Gavialis gangeticus*	*Crocodylus palustris*	*Crocodylus porosus*
Habitat	Freshwater rivers	Freshwater rivers, lakes, marshes	Saltwater, brackish water, estuaries, mangroves
Distribution in India	Chambal, Son, Gandak, Ghagra, Hooghly river systems	Throughout India	East Coast, Bhitarkanika, Sundarbans, Andaman & Nicobar Islands
Distinctive Feature	Long, narrow snout	Broad U-shaped snout	Large size, V-shaped snout
IUCN Status	Critically Endangered (CR)	Vulnerable (VU)	Least Concern (LC)
CITES Status	Appendix I	Appendix I	Appendix I(Indian)
CMS Status	Appendix I	—	Appendix II
Wildlife Protection Act, 1972	Schedule I	Schedule I	Schedule I
Major Threats	Dams, sand mining, pollution	Habitat destruction, illegal hunting	Habitat loss, hunting
Nesting Type	Nests on river sandbanks	Hole-nesting species	Mound-nesting species

IN NEWS:Kota mugger crocodile deaths: Banned pesticide Aldrin found in Chandraloi river, forest department denies responsibility

107. Sea of Azov

Q.Which one of the following water-body sequences correctly links the Sea of Azov to the Atlantic Ocean?

(a) Sea of Azov - Black Sea - Sea of Marmara - Aegean Sea - Mediterranean Sea - Atlantic Ocean
(b) Sea of Azov - Caspian Sea - Persian Gulf - Arabian Sea - Atlantic Ocean
(c) Sea of Azov - Red Sea - Mediterranean Sea - Atlantic Ocean
(d) Sea of Azov - Baltic Sea - North Sea - Atlantic Ocean

Answer: (a) Sea of Azov - Black Sea - Sea of Marmara - Aegean Sea - Mediterranean Sea - Atlantic Ocean

EXPLANATION

Option (a):

The Sea of Azov is connected to the Black Sea through the Strait of Kerch. From the Black Sea, ships move through:

- Bosporus Strait → Sea of Marmara
- Dardanelles Strait → Aegean Sea
- Mediterranean Sea
- Atlantic Ocean through the Strait of Gibraltar

Sea of Azov → Kerch Strait → Black Sea → Bosporus → Sea of Marmara → Dardanelles → Aegean Sea → Mediterranean Sea → Strait of Gibraltar → Atlantic Ocean **Hence, Option (a) is correct.**



Option (b):

The Caspian Sea is a landlocked water body and has no natural connection with the Sea of Azov. Therefore, this sequence is geographically incorrect. **Hence, Option (b) is incorrect.**

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Option (c):

The Sea of Azov has no direct connection with the Red Sea. The Red Sea is located between Africa and the Arabian Peninsula and is connected to the Mediterranean through the Suez Canal.**Hence, Option (c) is incorrect.**



Option (d):

The Baltic Sea and North Sea are located in Northern Europe and have no direct maritime connection with the Sea of Azov. **Hence, Option (d) is incorrect.**



Value Addition

Sea of Azov



Feature	Details
Type	Inland Sea / Marginal Sea
Bordered By	Russia and Ukraine
Connected To	Black Sea
Connecting Strait	Strait of Kerch
Major Rivers	Don River, Kuban River
Important Ports	Taganrog, Mariupol, Berdyansk
Salinity	Relatively low due to large freshwater inflow
Climate	Continental climate with seasonal ice formation

Important Straits in the Region

Strait	Connects
Kerch Strait	Sea of Azov ↔ Black Sea
Bosporus Strait	Black Sea ↔ Sea of Marmara
Dardanelles Strait	Sea of Marmara ↔ Aegean Sea
Strait of Gibraltar	Mediterranean Sea ↔ Atlantic Ocean

IN NEWS: Ukrainian drones hit southern Russian refineries and Azov port

Source:https://epaper.thehindu.com/ccidist-ws/th/th_international/issues/193611/OPS/G30G6SBG4.1+GCRG819NT.1.html

108. Sub-Mission on Agriculture(SMAM)

Q.With reference to the Sub-Mission on Agricultural Mechanization (SMAM), consider the following statements:

1. It is a Centrally Sponsored Scheme under Rashtriya Krishi Vikas Yojana (RKVY).
2. It supports the establishment of Custom Hiring Centres (CHCs).
3. SMAM provides subsidy support for agricultural machinery through the Direct Benefit Transfer (DBT) system.
4. Drones under SMAM are supported only for crop surveillance purposes.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (c) Only three

EXPLANATION:

Statement 1: The Sub-Mission on Agricultural Mechanization (SMAM) was launched in 2014–15 as a Centrally Sponsored Scheme under the Rashtriya Krishi Vikas Yojana (RKVY).Its objective is to promote farm mechanization among small and marginal farmers, women farmers, SC/ST farmers, FPOs, SHGs and rural entrepreneurs.**Hence, Statement 1 is correct.**

Statement 2:SMAM supports the establishment of Custom Hiring Centres (CHCs), which provide agricultural machinery and equipment on a rental basis.These centres help farmers, particularly small and marginal farmers, access costly machinery without having to purchase it.**Hence, Statement 2 is correct.**

Statement 3: The scheme provides financial assistance for the procurement of agricultural machinery through the Direct Benefit Transfer (DBT) mechanism.Subsidy support is provided directly to eligible beneficiaries, ensuring transparency and efficient fund transfer.**Hence, Statement 3 is correct.**

Statement 4:SMAM promotes the use of Kisan Drones for multiple agricultural activities such as:

- Application of fertilizers
- Spraying of pesticides and agro-chemicals
- Nutrient management
- Precision agriculture demonstrations

Drones are not supported only for crop surveillance.

Hence, Statement 4 is incorrect.

Value Addition

Sub-Mission on Agricultural Mechanization (SMAM)

Feature	**Details**
Launch Year	2014–15
Type	Centrally Sponsored Scheme
Umbrella Scheme	Rashtriya Krishi Vikas Yojana (RKVY)
Ministry	Ministry of Agriculture & Farmers Welfare
Objective	Promote farm mechanization and improve productivity
Key Beneficiaries	Small & Marginal Farmers, Women, SC/ST, FPOs, SHGs
DBT Support	Yes
Custom Hiring Centres	Supported
Farm Machinery Banks	Supported
Drone Promotion	Yes
Special Focus	North-Eastern States and underserved regions

Strategic Pillars of SMAM

Pillar	**Objective**
Promotion of Farm Mechanization	Training, testing, demonstrations and mechanized farming
Financial Assistance	Subsidy for purchase of agricultural machinery
Custom Hiring Centres (CHCs)	Rental access to machinery for small farmers
Farm Machinery Banks (FMBs)	Shared machinery pool for community use

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Hi-Tech Hubs	Access to advanced and high-capacity farm equipment
Drone Promotion	Precision agriculture and input application
NER Focus	Enhanced mechanization support in North-Eastern States

In NEWS:Sub-Mission on Agricultural Mechanization

109. Zero FIR

Q.With reference to Zero FIR, consider the following statements:

1. It can be filed at any police station irrespective of territorial jurisdiction.
2. It was introduced on the recommendation of the Justice Verma Committee.
3. It can be registered only for non-cognizable offences.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only
EXPLANATION:
Statement 1:
Zero FIR is a special type of FIR that can be lodged at any police station, regardless of whether the offence occurred within that police station's territorial jurisdiction.The objective is to ensure that victims do not face delays in reporting serious crimes because of jurisdictional issues. Once registered, the Zero FIR is transferred to the police station having the appropriate jurisdiction for investigation.Under **Section 173(1) of the BNSS, 2023**, if a Zero FIR is recorded through electronic communication (e-FIR), it is not legally operational until the informant signs it **within three days**. **Hence, Statement 1 is correct.**
Statement 2:
The concept of Zero FIR was introduced in 2013 following the recommendations of the Justice J.S. Verma Committee, which was constituted after the 2012 Delhi gangrape case.The committee recommended measures to make the criminal justice system more responsive, especially in cases involving sexual offences and other grave crimes.**Hence, Statement 2 is correct.**
Statement 3:
Zero FIR applies to cognizable offences, i.e., offences in which the police can:

- Register a case without prior approval of a Magistrate.
- Arrest an accused without a warrant.

Examples include:

- Murder
- Rape
- Kidnapping
- Robbery

It is not limited to non-cognizable offences.**Hence, Statement 3 is incorrect.**
Value Addition
Zero FIR at a Glance

Feature	**Details**
Introduced	2013
Recommended By	Justice Verma Committee
Purpose	Prevent delay in registration of serious crimes
Jurisdiction	Can be filed at any police station
Initial Number	"0" (Zero)
Investigation	Conducted by police station having jurisdiction
Applicable For	Cognizable offences
Legal Basis	Section 173, Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023 for FIR registration

FIR vs Zero FIR

Parameter	**FIR**	**Zero FIR**
Jurisdiction	Only concerned police station	Any police station
Registration	Regular FIR number	Initially numbered "0"

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Transfer	No	Yes
Purpose	Normal registration	Immediate registration in urgent cases

Cognizable vs Non-Cognizable Offence

Cognizable Offence	**Non-Cognizable Offence**
Police can arrest without warrant	Police cannot arrest without warrant
Investigation can begin immediately	Magistrate's permission required
Murder, Rape, Kidnapping	Defamation, Public Nuisance, Simple Hurt

In NEWS:Crackdown on cybercrime: Rs 25,000 crore saved, Rs 323 crore refunded, e-Zero FIR rolled out

110. International Bank for Reconstruction and Development(IBRD)

Q.With reference to the World Bank Group and the Livable Planet Fund, consider the following statements:

1. The International Bank for Reconstruction and Development (IBRD) is one of the two institutions that together constitute what is commonly referred to as the "World Bank."
2. The IBRD primarily provides concessional grants and interest-free loans to the world's poorest countries.
3. The Livable Planet Fund is an initiative of the World Bank aimed at mobilising additional resources for global challenges such as climate change, biodiversity loss and pandemic preparedness.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two
EXPLANATION:

Statement 1:
The term World Bank generally refers to two institutions of the World Bank Group:
- International Bank for Reconstruction and Development (IBRD)
- International Development Association (IDA)

IBRD is therefore one of the two institutions constituting the World Bank.**Hence, Statement 1 is correct.**

Statement 2:
IBRD does not primarily provide concessional grants and interest-free loans.
Its main function is to provide:
- Loans
- Guarantees
- Development financing to middle-income and creditworthy low-income countries.

Concessional loans and grants to the poorest countries are provided by the International Development Association (IDA).**Hence, Statement 2 is incorrect.**

Statement 3:
The Livable Planet Fund (LPF) is a World Bank initiative intended to mobilize additional resources for global public goods such as climate action, biodiversity and pandemic preparedness.The fund seeks to complement traditional development financing by attracting additional donor and private-sector resources.**Hence, Statement 3 is correct.**

Value Addition
World Bank Group Structure

Institution	**Full Form**	**Primary Function**
IBRD	**International Bank for Reconstruction and Development**	Provides loans and development assistance to middle-income and creditworthy low-income countries
IDA	**International Development Association**	Provides concessional loans (credits) and grants to the world's poorest countries
IFC	**International Finance Corporation**	Promotes private sector development through investments, loans and advisory services

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MIGA	**Multilateral Investment Guarantee Agency**	Provides political risk insurance and credit enhancement to encourage foreign investment
ICSID	**International Centre for Settlement of Investment Disputes**	Facilitates arbitration and conciliation of investment disputes between governments and foreign investors

IBRD vs IDA

Feature	**IBRD**	**IDA**
Established	1944	1960
Target Countries	Middle-income & creditworthy low-income countries	Poorest countries
Funding Source	Global capital markets	Donor contributions
Assistance	Loans and guarantees	Grants and concessional credits

Livable Planet Fund (LPF)

Feature	**Details**
Institution	World Bank
Objective	Address global challenges beyond national borders
Focus Areas	Climate change, biodiversity, pandemics, resilience
Purpose	Mobilise additional development finance and global public goods funding

In NEWS:World Bank commits$890 mn for solar rooftops in India

111. INS Mahendragiri

Q. With reference to INS Mahendragiri, recently commissioned by the Indian Navy, consider the following statements:

1. It is the sixth ship under Project 17A.
2. It was designed by the Warship Design Bureau and built by Mazagon Dock Shipbuilders Limited.
3. It is equipped with a Combined Diesel or Gas (CODOG) propulsion system.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
Explanation:
In News: Indian Navy commissions INS Mahendragiri
1. Project 17A
Project 17A is a follow-on programme to the Shivalik-class (Project 17) frigates and represents a major advancement in India's indigenous warship-building capability. These stealth-guided missile frigates are designed as versatile multi-mission platforms capable of operating across the full spectrum of maritime warfare. INS Mahendragiri is the sixth ship constructed under Project 17A. **Hence, Statement 1 is correct.**
2. Design and Construction
INS Mahendragiri was designed by the Indian Navy's Warship Design Bureau (WDB), which is responsible for the indigenous design of naval warships. The vessel was constructed by Mazagon Dock Shipbuilders Limited (MDSL), Mumbai. The project showcases India's growing self-reliance in defence manufacturing and complex warship construction. **Hence, Statement 2 is correct.**
3. Propulsion and Operational Capabilities
The ship uses a Combined Diesel or Gas (CODOG) propulsion system, where diesel engines provide fuel-efficient cruising and gas turbines deliver high-speed performance when required. It is equipped with advanced weapons and sensors for anti-air, anti-surface and anti-submarine warfare. The vessel can also perform maritime security operations, Search and Rescue (SAR), Humanitarian Assistance and Disaster Relief (HADR), and sustained deployment missions. **Hence, Statement 3 is correct.**

112. Vande Mataram

Q. With reference to *Vande Mataram*, consider the following statements:

1. It was originally included in Bankim Chandra Chattopadhyay's novel *Anandamath*.
2. It was adopted as the National Song of India by the Constituent Assembly on 24 January 1950.
3. Respect for *Vande Mataram* can be linked to the Fundamental Duty of respecting the ideals and institutions of the freedom struggle.
4. The Constitution of India explicitly defines the status and protocol of the National Song.

Which of the statements given above is/are correct?

(a) 1, 2 and 3 only
(b) 1 and 4 only
(c) 2, 3 and 4 only
(d) 1, 2, 3 and 4

Answer: (a) 1, 2 and 3 only

Explanation:

In News: Strictly comply with Vande Mataram directive, says govt.

1. Historical Origin

Vande Mataram was composed by **Bankim Chandra Chattopadhyay** in the late nineteenth century and later incorporated into his famous novel **Anandamath (1882).** The song personifies India as the motherland and became one of the most influential literary works associated with Indian nationalism. Its stirring message of devotion to the motherland transformed it from a literary composition into a symbol of patriotic awakening. **Hence, Statement 1 is correct.**

Role in the Freedom Struggle

The song emerged as a rallying cry during the Swadeshi Movement (1905) against the Partition of Bengal. The slogan "Vande Mataram" was widely used in protests, public meetings and nationalist campaigns. British authorities attempted to restrict its public use due to its growing influence on anti-colonial mobilisation. Freedom fighters, students and political leaders invoked it as a symbol of resistance and national unity. Owing to this historic contribution, the song occupies a unique place in India's freedom movement.

2. Status as the National Song

On 24 January 1950, the Constituent Assembly resolved the issue of national symbols. Dr. Rajendra Prasad announced that while Jana Gana Mana would be the National Anthem, *Vande Mataram*, because of its exceptional role in the struggle for independence, would be honoured equally and accorded the status of the National Song. This recognition was based on the decision of the Constituent Assembly rather than a constitutional provision. **Hence, Statement 2 is correct.**

3. Link with Fundamental Duties

Although *Vande Mataram* is not specifically mentioned in Article 51A, respect for it can be associated with the spirit of the Fundamental Duties. In particular:

- Article 51A(a) requires every citizen to abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem.
- Article 51A(b) requires citizens to cherish and follow the noble ideals that inspired the national struggle for freedom.

Since *Vande Mataram* played a historic role in mobilising the freedom movement and symbolises the ideals of nationalism and sacrifice, respecting the National Song is closely connected with the spirit of Article 51A(b). **Hence, Statement 3 is correct.**

4. Constitutional Position

The Constitution of India does not explicitly mention the National Song, nor does it prescribe its status, lyrics, duration or protocol. Unlike the National Anthem, there is no specific constitutional provision dealing with *Vande Mataram*. Its recognition flows from the Constituent Assembly's decision and subsequent governmental practice. **Hence, Statement 4 is incorrect.**

Why in News?

- The Ministry of Home Affairs (MHA) has directed all States and Union Ministries to strictly comply with the protocol that Vande Mataram (National Song) must be sung or played before Jana Gana Mana (National Anthem) whenever both are performed at official events.
- The MHA also emphasized the use of the correct script, text and pronunciation while rendering both the National Song and the National Anthem.
- Earlier, the Ministry had advised that all six stanzas of Vande Mataram be sung or played at official functions.
- The Union Cabinet has approved a proposal to amend the Prevention of Insults to National Honour

Act, 1971, to make insult to or obstruction of the singing of Vande Mataram punishable, subject to enactment by Parliament.

Additional Facts

- **Composer: Bankim Chandra Chattopadhyay**
- **Novel: Anandamath (1882)**
- **First sung at an Indian National Congress session: 1896**
- **Sung by: Rabindranath Tagore**
- **Became a major symbol of the Swadeshi Movement (1905)**
- **National Song status accorded on: 24 January 1950**
- **Not expressly mentioned in the Constitution**
- **Linked to Article 51A(b) through its role in the freedom struggle**

113. India - New Zealand Ties

Q. With reference to the recent developments in India–New Zealand relations, consider the following statements:

1. India and New Zealand have elevated their bilateral relationship to a Strategic Partnership.
2. The two countries agreed to establish a Maritime Security Dialogue and a Joint Working Group on Counter-Terrorism.
3. Both countries reiterated their support for freedom of navigation and overflight in accordance with the United Nations Convention on the Law of the Sea (UNCLOS), 1982.
4. New Zealand is a member of the Quad grouping along with India, Japan and Australia.

Which of the statements given above is/are correct?

(a) 1, 2 and 3 only
(b) 1 and 4 only
(c) 2, 3 and 4 only
(d) 1, 2, 3 and 4

Answer: (a) 1, 2 and 3 only

Explanation:

In News: India and New Zealand set trade goal, elevate ties

1. Strategic Partnership

During Prime Minister Narendra Modi's visit to New Zealand in 2026, the two countries elevated their ties to a Strategic Partnership. A four-year roadmap was adopted to deepen cooperation across trade, security, maritime affairs and people-to-people exchanges. Hence, Statement 1 is correct.

2. Security Cooperation

As part of the enhanced partnership, India and New Zealand agreed to establish a Maritime Security Dialogue and a Joint Working Group on Counter-Terrorism. The two countries also reiterated a zero-tolerance approach towards terrorism and emphasized disrupting terror financing networks and safe havens. Hence, Statement 2 is correct.

3. Indo-Pacific and UNCLOS

Both countries reaffirmed their commitment to a free, open, inclusive and rules-based Indo-Pacific. They supported freedom of navigation, overflight and other lawful uses of the seas in accordance with international law, particularly the 1982 United Nations Convention on the Law of the Sea (UNCLOS). Hence, Statement 3 is correct.

4. Quad Membership

The Quad consists of India, the United States, Japan and Australia. New Zealand is not a member of the Quad, though it cooperates with Quad countries on several regional issues. Hence, Statement 4 is incorrect.

Additional Facts



- Target bilateral trade: ₹35,000 crore by 2030
- New Zealand committed to invest USD 20 billion in India over the next 15 years
- Agreements included hydrographic data sharing, mutual naval logistics, and enhanced maritime engagement
- Both countries expressed support for a rules-based international order
- Concern expressed over disruptions to maritime commerce through the Strait of Hormuz

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114. Māori Tribe

Q. With reference to the Māori tribe, recently seen in the news, consider the following statements:

1. Prime Minister Narendra Modi was accorded a traditional Powhiri Ceremony featuring the Māori Haka performance.
2. The Māori are the indigenous people of New Zealand and belong to the Polynesian cultural group.
3. Haka is performed exclusively as a war dance during armed conflicts.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation:

In News: Māori Haka



1. Traditional Māori Welcome

During Prime Minister Narendra Modi's visit to New Zealand, he was accorded a traditional **Powhiri** ceremonial welcome by the Māori people. The ceremony included a **Haka** performance, a distinctive Māori cultural expression used to honour guests and mark significant occasions. **Hence, Statement 1 is correct.**

2. Indigenous Polynesian Community

The **Māori** are the indigenous people of New Zealand and have inhabited the islands for centuries. They are culturally **Polynesian**, with their closest links to the peoples of Eastern Polynesia. Māori settlements were historically concentrated in the northern parts of New Zealand's North Island, while the South Island was comparatively sparsely populated. **Hence, Statement 2 is correct.**

Language and Cultural Identity

The Māori language (*Te Reo Māori*) belongs to the **Polynesian Group of the Eastern Oceanic Branch of the Austronesian language family**. It remains an important symbol of Māori identity and heritage. Traditional Māori attire, customs, and rituals reflect their strong connection to ancestral lands, nature, and spiritual traditions.

3. Haka Performance

Although Haka originated as a warrior tradition and is often described as a war dance, it is **not performed exclusively during warfare**. Today, Haka is performed during welcoming ceremonies, state visits, cultural gatherings, sporting events, celebrations, and funerals. It is characterized by vigorous movements, rhythmic chanting, and expressive facial gestures. **Hence, Statement 3 is incorrect.**

Ta Moko: Traditional Māori Tattoos

One of the most recognizable features of Māori culture is **Ta Moko**, the traditional art of tattooing. These tattoos are highly symbolic and unique to each individual. The designs often represent a person's ancestry, genealogy, social status, achievements, and role within the community. Ta Moko is regarded as a sacred cultural practice and an important marker of Māori identity.

Additional Facts

- Indigenous people of **New Zealand**
- Cultural group: **Polynesian**
- Language family: **Austronesian**
- Traditional welcome ceremony: **Powhiri**
- Traditional ceremonial performance: **Haka**
- Traditional tattoo art: **Ta Moko**
- Concentrated historically in the **North Island** of New Zealand
- Māori constitute a significant indigenous community within New Zealand's population.

115. Raindrop Evaporation

Q. With reference to the recent IITM-Pune study on monsoon rainfall, consider the following statements:

1. A significant fraction of monsoon rainfall can evaporate before reaching the ground.
2. Evaporation of falling raindrops cools the surrounding air and can influence atmospheric convection.
3. Raindrop evaporation can affect the accuracy of monsoon and climate predictions.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
Explanation:
In News: One-fourth of India's monsoon rain evaporates mid-air, says new study
1. Raindrop Evaporation
Raindrop evaporation is a sub-cloud process where falling raindrops partially or completely vaporize before hitting the ground. A recent study by the Indian Institute of Tropical Meteorology (IITM), Pune, found that a significant portion of monsoon rainfall evaporates before reaching the ground. Over the northwestern Ghats, researchers estimated that nearly 25% of the rain mass evaporates during its descent, though the rate varies depending on atmospheric conditions. Hence, Statement 1 is correct.
2. Impact on Atmospheric Convection
When raindrops evaporate as they fall, they absorb heat from the surrounding air, thereby cooling the sub-cloud layer. This cooling can generate downdrafts and cold pools near the surface, which in turn influence atmospheric convection and affect the formation and intensity of subsequent rainfall events. Hence, Statement 2 is correct.
3. Importance for Weather and Climate Models
Raindrop evaporation plays a crucial role in determining rainfall distribution, atmospheric cooling and storm dynamics. Climate and monsoon models often struggle to accurately represent this process. Improved estimates of raindrop evaporation can therefore enhance the accuracy of weather forecasts, monsoon predictions and climate simulations. Hence, Statement 3 is correct.

116. Interpol Red Corner Notice

Q. With reference to the Interpol Red Corner Notice (Red Notice), consider the following statements:

1. It is issued by Interpol at the request of a member country or an international tribunal.
2. A Red Notice is an international arrest warrant that legally binds all member countries to arrest the individual concerned.
3. It seeks the location and provisional arrest of a wanted person pending extradition, surrender or similar legal action.

Which of the statements given above is/are correct?

(a) 1 and 3 only
(b) 2 and 3 only
(c) 1 and 2 only
(d) 1, 2 and 3

Answer: (a) 1 and 3 only
Explanation:
In News: Interpol Red Corner Notice issued against Hizbul terrorist
1. Issuance of Red Notice
A **Red Notice** is issued by **Interpol** at the request of a member country seeking assistance in locating a fugitive. Requests may also originate from international criminal courts or tribunals. Interpol publishes the notice and circulates it among its member countries. **Hence, Statement 1 is correct.**



2. Legal Status
A Red Notice is **not an international arrest warrant**. Interpol does not possess powers to compel member countries to arrest an individual. Each country decides, according to its domestic laws, what legal value it attaches to a Red Notice and whether any action should be taken. **Hence, Statement 2 is incorrect.**
3. Purpose of Red Notice
The primary purpose of a Red Notice is to **locate and provisionally arrest a wanted person** pending extradition, surrender or other legal proceedings. It contains identifying information such as name,

nationality, photographs, biometric details and information regarding the alleged offences. Hence, **Statement 3 is correct**.

Key Facts about Interpol

- Full Form: **International Criminal Police Organization (ICPO)**
- Common Name: **Interpol**
- Headquarters: **Lyon, France**
- Members: **195 countries**
- India's National Central Bureau (NCB): **Central Bureau of Investigation (CBI)**
- Global police communication network: **I-24/7**
- Facilitates international police cooperation, information sharing and fugitive tracking.

117. PaBV-4

Q. With reference to the recently reported (PaBV-4) in India, consider the following statements:

1. The term psittacine birds refers to birds belonging to the parrot family, including parrots, macaws, cockatiels and parakeets.
2. PaBV-4 is associated with Proventricular Dilatation Disease (PDD) in psittacine birds.
3. Infected birds may remain asymptomatic while continuing to spread the virus.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation:

In News: Deadly bird virus PaBV-4 identified for first time in India

1. Psittacine Birds



The term **psittacine** refers to birds belonging to the order **Psittaciformes**, commonly known as the parrot family. This group includes parrots, macaws, parakeets, cockatoos, cockatiels, budgerigars and lovebirds. The recent Indian study tested several such species maintained in captivity. **Hence, Statement 1 is correct.**

2. Proventricular Dilatation Disease (PDD)

Parrot Bornavirus 4 (PaBV-4) is a highly contagious **single-stranded RNA virus** belonging to the species *Orthobornavirus alphapsittaciforme*. It is the most veterinary-significant bornavirus affecting parrots and is closely associated with **Proventricular Dilatation Disease (PDD)**, a progressive disease affecting the digestive and nervous systems. Infected birds may experience enlargement of the proventriculus (part of the stomach), impaired digestion, weight loss, neurological symptoms and, in severe cases, death. While bornaviruses have been detected in some other bird groups, PaBV-4 primarily affects **psittacine birds**. **Hence, Statement 2 is correct.**

3. Asymptomatic Transmission

One of the major concerns highlighted by the study is that infected birds do not always show visible signs of illness. Apparently healthy birds can continue to shed the virus through saliva, food, water, and droppings, facilitating its spread within aviaries and captive breeding centres. The study found infection even among clinically healthy cage mates, demonstrating the potential for silent transmission. **Hence, Statement 3 is correct.**

Why in News?

For the first time in India, scientists led by the **Assam Veterinary and Fishery University** identified and genetically characterised **PaBV-4** in captive psittacine birds. The virus was detected in birds from Assam, Karnataka and West Bengal, raising concerns for aviculture, exotic pet management and conservation of threatened parrot species.

Additional Facts

- Virus: **Parrot Bornavirus 4 (PaBV-4)**
- Type: **Single-stranded RNA virus**
- Disease: **Proventricular Dilatation Disease (PDD)**
- First linked to PDD globally: **2008**
- Common hosts: Parrots, macaws, parakeets, cockatoos, cockatiels and lovebirds
- Transmission: Contaminated food, water, saliva, droppings and possibly from parent birds to eggs
- First confirmed report from India: **2026**

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118. FSSAI

Q. With reference to the Food Safety and Standards Authority of India (FSSAI), recently seen in the news, consider the following statements:

1. FSSAI is a statutory body established under the Food Safety and Standards Act, 2006.
2. It functions under the Ministry of Health and Family Welfare.
3. FSSAI is responsible for regulating and monitoring the manufacture, storage, distribution, sale and import of food in India.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
Explanation:
In News: FSSAI pulls up Instamart for supply of 'rotten food items'
Why in News?
The Food Safety and Standards Authority of India (FSSAI) issued notices to a quick-commerce platform following consumer complaints alleging the supply of expired, spoiled, rotten and contaminated food products. The regulator sought an explanation and compliance report under the provisions of the Food Safety and Standards Act, 2006.
1. Statutory Body
FSSAI is a statutory body established under the Food Safety and Standards Act, 2006. The Act consolidated various food-related laws into a single regulatory framework to ensure the availability of safe and wholesome food for human consumption. **Hence, Statement 1 is correct.**
2. Administrative Ministry
FSSAI functions under the Ministry of Health and Family Welfare. It serves as India's apex food regulator and is responsible for framing science-based standards for food products. **Hence, Statement 2 is correct.**
3. Regulatory Functions
Under Section 16 of the Food Safety and Standards Act, 2006, FSSAI regulates and monitors the manufacture, processing, storage, distribution, sale and import of food to ensure food safety and quality. It also issues licenses to Food Business Operators (FBOs), sets standards for food products and oversees food safety compliance. **Hence, Statement 3 is correct.**
Additional Information
- **Established:** under the Food Safety and Standards Act, 2006 and became operational in 2008.
- **Nature:** Autonomous statutory body under the **Ministry of Health and Family Welfare**.
- **Headquarters:** New Delhi.
- **Composition:** Chairperson and **22 Members**, of whom **one-third shall be women** .
- **Food Business Operators (FBOs)** are issued a **14-digit FSSAI licence number**.
- **India's apex food regulator** responsible for setting science-based food standards and regulating manufacture, storage, distribution, sale and import of food.
- Operates through the **Food Safety and Standards Rules, 2011** and **Food Safety and Standards Regulations, 2011**.
- Accredits food testing laboratories and certification bodies.
- Promotes consumer awareness through campaigns such as **Eat Right India**.

Important FSSAI Initiatives
- **Eat Right India (2018)** – Promotes safe, healthy and sustainable diets.
- **RUCO (Repurpose Used Cooking Oil)** – Converts used cooking oil into biodiesel.
- **FoSTaC (Food Safety Training and Certification)** – Training programme for food business operators.
- **DART (Do-It-Yourself Adulteration Testing)** – Enables consumers to detect food adulteration at home.
- **Jaivik Bharat** – Promotes certified organic food products.
- **Clean Street Food Hub Initiative** – Improves hygiene standards among street food vendors.
- **Aaj Se Thoda Kam Campaign** – Encourages reduced consumption of salt, sugar and fat.
- **Eat Right Campus & School Programme** – Promotes healthy eating habits in educational institutions.

Emerging Challenges
- Regulation of **e-commerce and online food delivery platforms**.
- Monitoring **chemical contaminants, toxins and pesticide residues** in food.

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- Shortage of accredited testing laboratories and enforcement personnel.
- Need for greater consumer awareness regarding food safety standards.

119. Green Shipping & Green Ports

Q. With reference to Green Shipping and Green Port initiatives, consider the following statements:

1. Green ports promote the adoption of renewable energy sources such as solar and wind power in port operations.
2. Green hydrogen, green ammonia and green methanol are being explored as alternative fuels for decarbonising the maritime sector.
3. The V.O. Chidambaranar Port Authority recently signed an MoU with Belgian institutions to cooperate in capacity building, green port initiatives and digital transformation of port operations.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
Explanation:
In News: V.O.C. Port signs MoU with Port of Antwerp-Bruges
1. Renewable Energy Integration
Green ports seek to reduce the environmental footprint of maritime activities through the adoption of renewable energy sources such as **solar and wind power**, electrification of equipment and energy-efficient infrastructure. These measures help lower carbon emissions and promote sustainable port operations. **Hence, Statement 1 is correct.**
2. Alternative Green Fuels
India's green shipping transition emphasizes the use of **green hydrogen, green ammonia and green methanol** as alternative fuels for the maritime sector. These fuels are expected to play a key role in decarbonising shipping and achieving India's climate commitments. **Hence, Statement 2 is correct.**
3. India–Europe Maritime Cooperation
Recently, the **V.O. Chidambaranar Port Authority** signed an MoU with **Port of Antwerp-Bruges International (Belgium)** and **APEC Antwerp-Flanders Port Training Center (Belgium)** to strengthen cooperation in **capacity building, green port initiatives and digital transformation of port operations**. The agreement also focuses on training, workshops, and sharing of best practices in sustainable port management. **Hence, Statement 3 is correct.**
Additional Information
- **Maritime India Vision 2030** aims to modernise Indian ports through sustainability, efficiency and technology-driven initiatives.
- **Maritime Amrit Kaal Vision 2047** seeks to position India as a global maritime leader through green shipping and green ports.
- **Harit Sagar Green Port Guidelines** target a reduction in carbon emissions per tonne of cargo and increased use of renewable energy.
- **National Green Hydrogen Mission** envisages the development of hydrogen hubs at select ports.
- VOC Port is pursuing projects related to **solar energy, wind energy, green hydrogen, green ammonia and green methanol production and bunkering**.

120. PMFME – Pradhan Mantri Formalisation of Micro Food Processing Enterprises Scheme

Q. With reference to the Pradhan Mantri Formalisation of Micro Food Processing Enterprises (PMFME) Scheme, consider the following statements:

1. It is a Centrally Sponsored Scheme implemented by the Ministry of Food Processing Industries.
2. The scheme follows the One District One Product (ODOP) approach to promote value addition and local food processing clusters.
3. New units under the scheme are eligible for support irrespective of whether they are linked to the One District One Product (ODOP) approach.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only
Explanation:

In News: PMFME scheme sanctions loans to over two lakh micro food processing enterprises, says Union Minister Chirag Paswan

1. Implementing Ministry

The Pradhan Mantri Formalisation of Micro Food Processing Enterprises (PMFME) Scheme was launched on 29 June 2020 as a Centrally Sponsored Scheme by the Ministry of Food Processing Industries (MoFPI). It aims to enhance the competitiveness of micro food processing enterprises by providing financial, technical and institutional support. **Hence, Statement 1 is correct.**

2. One District One Product (ODOP) Approach

A key feature of the PMFME Scheme is the One District One Product (ODOP) approach. Under this framework, each district identifies a product with growth potential and focuses resources on value addition, processing, branding and marketing of that product. This helps create local food-processing clusters and strengthens the supply chain. **Hence, Statement 2 is correct.**

3. Eligibility of New Units

The scheme does not support new units irrespective of product selection. New units are eligible for assistance only if they are linked to the identified ODOP product of the district. This ensures focused development and better utilisation of resources. **Hence, Statement 3 is incorrect.**

Additional Information

- **Ministry:** Ministry of Food Processing Industries (MoFPI)
- **Launch Date:** 29 June 2020
- **Outlay:** ₹10,000 crore
- **Coverage:** 2 lakh micro food processing enterprises
- **Funding Pattern:**
  - 60:40 (Centre–State)
  - 90:10 (North-Eastern and Himalayan States)
  - 60:40 (UTs with Legislature)
  - 100% Central funding (Other UTs)
- Supports **FPOs, SHGs, Cooperatives and individual entrepreneurs**.
- Provides support for **branding, marketing, skill development, common infrastructure and credit access**.
- SHGs are eligible for **seed capital assistance of ₹4 lakh per SHG**.

121. One-horned Rhino

Q.With reference to the Greater One-Horned Rhinoceros (*Rhinoceros unicornis*), consider the following statements:

1. It is primarily found in the grasslands and riverine ecosystems of the Indo-Gangetic and Brahmaputra plains.
2. It is classified as Endangered under the IUCN Red List.
3. The species is listed in Schedule I of the Wildlife (Protection) Act, 1972.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (b) 1 and 3 only

EXPLANATION:

In NEWS: Nod to profile rhinos through DNA indexing

Statement 1:

The Greater One-Horned Rhinoceros is mainly distributed in the alluvial floodplains, grasslands, riverine forests, marshes and savanna ecosystems of the Indo-Gangetic and Brahmaputra plains.Major populations occur in:

- Kaziranga National Park (Assam)
- Pobitora Wildlife Sanctuary (Assam)
- Orang National Park (Assam)
- Jaldapara National Park (West Bengal)
- Chitwan National Park (Nepal) **Hence, Statement 1 is correct.**

Statement 2:

The Greater One-Horned Rhinoceros is currently classified as Vulnerable (VU) under the IUCN Red List.It was once reduced to fewer than 200 individuals due to hunting and habitat loss, but intensive conservation efforts have significantly increased its population.**Hence, Statement 2 is incorrect.**

Statement 3:

The species receives the highest legal protection in India under the Wildlife (Protection) Act, 1972, being listed in Schedule I.This provides stringent penalties for poaching, hunting and illegal trade.**Hence, Statement 3 is correct.**

VALUE ADDITION



Greater One-Horned Rhinoceros: Quick Facts

Feature	**Details**
Scientific Name	*Rhinoceros unicornis*
Common Name	Indian Rhinoceros / Greater One-Horned Rhinoceros
Family	Rhinocerotidae
Distribution	India and Nepal
Habitat	Riverine grasslands, marshes, floodplains
Diet	Herbivorous grazer
Horn	Single horn made of keratin
Largest Population	Kaziranga National Park (Assam)

Conservation Status

Convention/Act	**Status**
IUCN Red List	Vulnerable (VU)
CITES	Appendix I
Wildlife (Protection) Act, 1972	Schedule I

Rhino DNA Indexing System

Feature	**Details**
Approved By	Standing Committee of National Board for Wildlife (SC-NBWL)
Purpose	Create a genetic database of rhinos
Uses	Individual identification, age/sex determination, population monitoring
Conservation Benefit	Supports scientific breeding and genetic diversity assessment
Anti-Poaching Role	Provides forensic evidence in wildlife crime investigations

National Board for Wildlife (NBWL)

Feature	**Details**
Statutory Basis	Wildlife (Protection) Act, 1972
Chairperson	Prime Minister of India
Standing Committee Chairperson	Union Minister for Environment, Forest and Climate Change
Function	Advises on wildlife conservation and protected areas

122. World Economic Outlook

Q.With reference to the World Economic Outlook (WEO), consider the following statements:

1. It is released twice a year by the International Monetary Fund (IMF).
2. Countries are ranked solely on the basis of GDP at Purchasing Power Parity (PPP) in the WEO

Which of the statements given above are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 and 2

Answer: (a) 1 only

EXPLANATION

Statement 1:

The World Economic Outlook (WEO) is a flagship publication of the International Monetary Fund (IMF) and is released twice every year.It provides:

- Global economic forecasts
- Growth projections
- Inflation outlook
- Trade and financial market assessments

for advanced, emerging and developing economies.

Hence, Statement 1 is correct.

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Statement 2:

The IMF's World Economic Outlook does **not rank countries solely on the basis of GDP at Purchasing Power Parity (PPP).**

The WEO publishes data using **both:**

- **Nominal GDP (current US dollars)** and
- **GDP based on Purchasing Power Parity (PPP).**

In international discussions regarding the world's largest economies (e.g., USA, China, Germany, India, Japan), **Nominal GDP rankings are generally used.**

PPP-based GDP is primarily used for:

- Comparing standards of living
- Measuring real purchasing power
- Comparing economic size after adjusting for price differences

Therefore, it is incorrect to state that countries are ranked **solely** on the basis of PPP in the WEO.**Hence, Statement 2 is incorrect.**

VALUE ADDITION

World Economic Outlook (WEO)

Feature	Details
Released By	International Monetary Fund (IMF)
Frequency	Twice a year
Purpose	Global economic analysis and projections
Coverage	Advanced, Emerging & Developing Economies
Key Indicators	GDP Growth, Inflation, Trade, Fiscal Outlook

Nominal GDP vs PPP GDP

Feature	Nominal GDP	PPP GDP
Basis	Market Exchange Rate	Purchasing Power Adjustment
Used For	Global Economic Rankings	Living Standards Comparison
Sensitive to Currency Movements	Yes	No
IMF ranking method	yes	no

IMF

Feature	Details
Established	1944 (Bretton Woods Conference)
Headquarters	Washington D.C., USA
Members	191 Countries
Objective	Global Monetary Cooperation and Financial Stability
Publications	World Economic Outlook (WEO), Global Financial Stability Report (GFSR), Fiscal Monitor

IN NEWS:India Among Fastest-Growing Major Economies with 6.4% GDP Growth Projected for FY27: IMF

123. Cyanobacteria

Q.With reference to Cyanobacteria (Blue-Green Algae), consider the following statements:

1. They are among the oldest known fossil organisms on Earth.
2. cyanobacteria are autotrophic and perform photosynthesis.
3. Cyanobacteria secrete exopolysaccharides (EPS), a sticky sugar-rich substance.
4. EPS produced by cyanobacteria can bind and help remove lead from contaminated water.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (d) All four

EXPLANATION

Statement 1:

Cyanobacteria are regarded as some of the oldest known life forms on Earth, with fossil evidence dating back more than 3.5 billion years.They played a crucial role in the Great Oxygenation Event, releasing oxygen into the atmosphere through photosynthesis and transforming Earth's environment.**Hence, Statement 1 is correct.**

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Statement 2:

Cyanobacteria are prokaryotic, autotrophic organisms capable of carrying out oxygenic photosynthesis.Like plants, they contain chlorophyll and use sunlight, water and carbon dioxide to produce food.They are often referred to as blue-green algae, although scientifically they are bacteria and not true algae.**Hence, Statement 2 is correct.**

Statement 3:

Cyanobacteria naturally secrete Exopolysaccharides (EPS), a sticky extracellular material composed mainly of complex sugars.

Functions of EPS include:

- Protection against environmental stress
- Cell adhesion
- Biofilm formation
- Retention of moisture and nutrients
- Binding of heavy metals **Hence, Statement 3 is correct.**

Statement 4:

Researchers at IIT Guwahati demonstrated that cyanobacterial EPS can effectively absorb lead (Pb) from contaminated water.The negatively charged functional groups present in EPS bind heavy metal ions and immobilize them.The study found that EPS could remove about 66.2% of lead contamination, providing a low-cost and eco-friendly alternative to conventional treatment methods.Also Can remove Cadmium (Cd) , Chromium (Cr) **Hence, Statement 4 is correct.**

VALUE ADDITION

Cyanobacteria :

Feature	**Details**
Common Name	Blue-Green Algae
Kingdom	Monera (Bacteria)
Cell Type	Prokaryotic
Nutrition	Autotrophic
Photosynthesis	Oxygenic Photosynthesis
Age	More than 3.5 Billion Years
Habitat	Freshwater, Marine, Brackish Water, Moist Soil
Reproduction	Asexual (Binary Fission, Fragmentation)
Nitrogen Fixation	Present in several species
Cell Wall	Peptidoglycan
Ecological Role	Primary Producers

Cyanobacteria Importantance:

Role	**Significance**
Oxygen Production	Responsible for oxygenation of Earth's atmosphere
Nitrogen Fixation	Converts atmospheric nitrogen into usable forms
Biofertilizer	Enhances soil fertility in paddy fields
Carbon Sequestration	Absorbs atmospheric CO_2
Wastewater Treatment	Removes pollutants and heavy metals
Symbiotic Associations	Forms lichens and plant symbioses

Cyanobacteria (Blue-Green Algae)

◆ Prokaryotic organisms (not true algae)

🔻 Many species fix atmospheric nitrogen through heterocysts : Thick-walled cells, Site of nitrogen fixation, Protect nitrogenase enzyme from oxygen

🔻 Important biofertilizers:

- Anabaena
- Nostoc
- Aulosira

🔻 Symbiotic association:

- Azolla–Anabaena
- Lichens (Cyanobacteria + Fungus)

🔻 Can form harmful algal blooms in nutrient-rich waters

🔻 Produce toxins such as Microcystin and Anatoxin

IN NEWS:IIT Guwahati team finds blue-green algae can remove lead from water

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124. Helium

Q.With reference to Helium, consider the following statements:

1. Helium is formed in the Earth's crust through the radioactive decay of uranium and thorium.
2. Helium is the lightest element in the periodic table.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (a) 1 only
EXPLANATION
Statement 1:
Helium is a naturally occurring non-renewable noble gas generated deep within the Earth's crust.

- Radioactive decay of uranium and thorium releases alpha particles.
- These alpha particles capture electrons and become helium atoms.
- Over millions of years, helium accumulates in underground reservoirs. **Hence, Statement 1 is correct.**

Statement 2:
Helium is not the lightest element in the periodic table.

- Hydrogen (Atomic Number 1) is the lightest element.
- Helium (Atomic Number 2) is the second lightest element. **Hence, Statement 2 is incorrect.**

VALUE ADDITION

Feature	Details
Symbol	He
Atomic Number	2
Group	Group 18 (Noble Gases)
Nature	Chemically inert
Lightness	Second lightest element after Hydrogen
Boiling Point	−268.9°C
Flammability	Non-flammable
Toxicity	Non-toxic but can cause suffocation by displacing oxygen
Renewability	Non-renewable resource

Major Uses of Helium

Sector	Application
Medical	MRI magnet cooling
Semiconductor Industry	Cooling silicon wafers
Quantum Computing	Cryogenic cooling
Aerospace	Pressurising rocket fuel tanks
Optical Fibre	Cooling molten glass
Industry	Leak detection

In NEWS:China's temporary export ban on helium

125. Jodhpuri Mojari

Q.Consider the following statements regarding Jodhpuri Mojari:

1. It has recently been granted a Geographical Indication (GI) tag.
2. It is traditionally crafted by the Jinagar community of Rajasthan.
3. It is primarily made from bamboo fibres and cotton.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only
EXPLANATION
Statement 1:
Jodhpuri Mojari has recently been granted a **Geographical Indication (GI) tag**.

- It is a traditional handcrafted footwear of Rajasthan.

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- The GI certificate was awarded by the GI Registry under the Geographical Indications of Goods (Registration and Protection) Act, 1999.
- The recognition helps protect the craft from imitation and promotes its market value.**Hence, Statement 1 is correct.**

Statement 2:

Jodhpuri Mojari is traditionally crafted by the **Jinagar (Jingar) community** of Rajasthan.

- The Jinagars were historically known for making **jeen (saddles)** and leather accessories for royal armies.
- Over time, they developed expertise in shoemaking and became associated with the Mojari craft.
- Even today, thousands of artisan households in Jodhpur are engaged in this traditional occupation.**Hence, Statement 2 is correct.**

Statement 3:

Jodhpuri Mojari is **not made from bamboo fibres and cotton**.

- It is a traditional **leather footwear** crafted using leather and decorative embroidery.
- Its design evolved to suit Rajasthan's hot and arid climate.
- Bamboo fibres and cotton are associated with certain handicrafts and textiles, but not with Mojari production.**Hence, Statement 3 is incorrect.**

VALUE ADDITION



Aspect	**Jodhpuri Mojari**
State	Rajasthan
GI Tag	Granted in 2026
Traditional Community	Jinagar (Jingar)
Material	Leather
Type	Slip-on footwear
Specialty	Hand embroidery and leather craftsmanship
Significance	Protection of traditional artisan heritage

Geographical Indication (GI) –

Aspect	**Details**
Legal Framework	GI Act, 1999
Implemented	2003
Registry Location	Chennai
Nodal Department	DPIIT
WTO Link	TRIPS Agreement
Validity	10 years (renewable)

Recent GI Tags (2026)

Product	**State**
Jodhpuri Mojari	Rajasthan
Ponduru Khadi	Andhra Pradesh
Khurasani Imli	Madhya Pradesh
Jolbhora Sandesh	West Bengal
Monohara	West Bengal
Munda Jewellery	Jharkhand
Bhagaiya Silk	Jharkhand

In NEWS:200-year old Jodhpuri Mojari gets GI tag

126. Molluscs

Q.With reference to Molluscs, consider the following statements:

1. Molluscs occur in marine, freshwater and terrestrial habitats.
2. Molluscs constitute nearly one-fourth of all living marine species.
3. Molluscs are among the most diverse groups of invertebrate animals.

Which of the statements given above are correct?

(a) 1 and 2 only

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(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
EXPLANATION
IN NEWS:Desert rain frog, over half of molluscs pushed towards extinction: IUCN Red List
Statement 1:
Molluscs are highly adaptable invertebrate animals found in a wide range of habitats.They occur in:

- Marine ecosystems (oceans and seas)
- Freshwater ecosystems (rivers, lakes and ponds)
- Terrestrial ecosystems (moist land environments)

Examples:

- Marine: Octopus, squid, oysters
- Freshwater: Freshwater mussels
- Terrestrial: Land snails and slugs **Hence, Statement 1 is correct.**

Statement 2:
Molluscs account for approximately 25% of all living marine species, making them one of the most important animal groups in marine ecosystems.They play significant ecological roles as:

- Grazers
- Filter feeders
- Predators
- Prey species

They are also an important source of food for humans. **Hence, Statement 2 is correct.**
Statement 3:
Mollusca is one of the largest and most diverse animal phyla.The group includes:

- Snails
- Slugs
- Clams
- Oysters
- Mussels
- Squid
- Octopus
- Cuttlefish
- Chitons

The phylum exhibits enormous diversity in:

- Body forms
- Habitats
- Feeding mechanisms
- Behavioural adaptations **Hence, Statement 3 is correct.**

VALUE ADDITION
Mollusca:

Feature	**Details**
Phylum	Mollusca
Type	Invertebrate Animals
Habitat	Marine, Freshwater and Terrestrial
Importance	Food source, ecological balance, biodiversity
Marine Diversity	Nearly 25% of all living marine species
Body Structure	Soft body, often protected by shell

Major Groups of Mollusca

Class	**Characteristics**	**Examples**
Polyplacophora	Eight overlapping shell plates	Chitons
Gastropoda	Largest molluscan class; true snails and slugs	Snails, Slugs
Bivalvia	Two-part hinged shell	Clams, Oysters, Mussels, Scallops
Cephalopoda	Large head, well-developed eyes and tentacles	Octopus, Squid, Cuttlefish, Nautilus

Ecological Importance of Molluscs

Role	**Significance**
Filter Feeders	Improve water quality
Grazers	Control algal growth
Predators	Maintain ecosystem balance
Prey Species	Support marine food webs
Shell Formation	Contribute to calcium cycling

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Threats to Molluscs

Threat	Impact
Deep-Sea Mining	Habitat destruction
Ocean Acidification	Weakens shell formation
Marine Pollution	Toxic contamination
Climate Change	Habitat alteration
Overexploitation	Population decline

Deep-Sea Mining

Feature	Details
Purpose	Extraction of critical minerals from seabed
Key Minerals	Nickel, Cobalt, Copper, Manganese
Concern	Damage to deep-sea biodiversity
Affected Groups	Molluscs, Corals, Sponges and other benthic fauna
Concern Raised By	IUCN

127. Phu Quoc island

Q.With reference to Phu Quoc Island, consider the following statements:

1. Phu Quoc Island is located in the Gulf of Thailand.
2. The island has a tropical climate with distinct wet and dry seasons.
3. The island is recognized as a UNESCO Biosphere Reserve.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION

Statement 1:

Phu Quoc Island is located in the Gulf of Thailand.

- Phu Quoc, also known as the Pearl Island, is a Vietnamese island situated in the Gulf of Thailand.
- It is Vietnam's largest island and forms part of an archipelago of 22 islands.
- Geographically, it lies closer to Cambodia than mainland Vietnam.**Hence, Statement 1 is correct.**

Statement 2:

The island has a tropical climate with distinct wet and dry seasons.

- Phu Quoc experiences a tropical monsoon climate.
- It has two distinct seasons:
  - Dry season: November to April
  - Rainy season: May to October
- The island receives abundant rainfall during the monsoon months.**Hence, Statement 2 is correct.**

Statement 3:

The island is recognized as a UNESCO Biosphere Reserve.

- A large part of the island is protected under Phu Quoc National Park.
- The island has been recognized as a UNESCO Biosphere Reserve for its rich biodiversity and ecological significance.
- The reserve includes forests, wetlands, marine ecosystems and coastal habitats.**Hence, Statement 3 is correct.**



IN NEWS:15 Indians die in Vietnam boat capsize, were part of group on holiday organised by cellphone firm

128. South China sea

Q. Consider the following countries:

1. China
2. Vietnam
3. Malaysia
4. Cambodia

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5. Philippines
6. Brunei

How many of the above-mentioned countries border the South China Sea?

(a) Only three
(b) Only four
(c) Only five
(d) All six

Answer: (c) Only five
EXPLANATION



China:
China has a coastline along the **South China Sea**.
- It lies on the northern boundary of the South China Sea.
- China is a major claimant in the South China Sea disputes.**Hence, China borders the South China Sea.**

Vietnam:
Vietnam has a long eastern coastline along the **South China Sea**.
- It is one of the major claimant countries in the region.
- The Paracel and Spratly Islands are disputed between Vietnam and China.**Hence, Vietnam borders the South China Sea.**

Malaysia:
Malaysia borders the **South China Sea**.
- Both Peninsular Malaysia and East Malaysia (Sabah and Sarawak) have coastlines facing the South China Sea.
- Malaysia is also a claimant in the Spratly Islands.**Hence, Malaysia borders the South China Sea.**

Cambodia:
Cambodia does **not** border the South China Sea.
- Its coastline lies along the **Gulf of Thailand**.
- It has no direct coastline on the South China Sea.**Hence, Cambodia does not border the South China Sea.**

Philippines:
The Philippines borders the **South China Sea**.
- The western coast of the Philippines faces the South China Sea.
- The country refers to part of it as the **West Philippine Sea**. **Hence, the Philippines borders the South China Sea.**

Brunei:
Brunei has a coastline along the **South China Sea**.
- It is located on the northern coast of Borneo Island.
- Brunei is also among the claimants in the Spratly Islands region. **Hence, Brunei borders the South China Sea.**

VALUE ADDITION:
- Largest marginal sea of the Western Pacific Ocean.
- Major island groups: Paracel Islands and Spratly Islands.
- Connects the Pacific Ocean with the Indian Ocean through strategic straits.
- Nearly one-third of global maritime trade passes through it.
- Rich in fisheries, oil and natural gas resources.

In NEWS:EU and 14 countries reaffirm ruling on South China Sea

129. Bio-Ethanol

Q.With reference to Bioethanol, consider the following statements:

1. Bioethanol is produced through fermentation of biomass using microorganisms.
2. Ethanol blending with petrol helps reduce carbon monoxide emissions from vehicles.
3. Flexible Fuel Vehicles (FFVs) can operate on high ethanol blends such as E85.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only

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(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION

Statement 1:

Bioethanol is ethyl alcohol (C_2H_5OH) produced through the fermentation of biomass feedstocks such as:

- Sugarcane juice
- Molasses
- Corn (maize)
- Damaged food grains
- Agricultural residues (for 2G ethanol)

The fermentation process is carried out by microorganisms, primarily Saccharomyces cerevisiae (yeast). **Hence, Statement 1 is correct.**

Statement 2:

Ethanol is an oxygenated fuel that promotes more complete combustion of petrol.

Benefits of ethanol blending include:

- Reduction in carbon monoxide (CO) emissions
- Lower hydrocarbon emissions
- Reduced dependence on crude oil imports
- Improved octane rating of fuel **Hence, Statement 2 is correct.**

Statement 3:

Flexible Fuel Vehicles (FFVs) are specially designed vehicles capable of operating on varying ethanol-petrol blends.

Examples:

- E20 → 20% ethanol + 80% petrol
- E85 → 85% ethanol + 15% petrol
- E100 → Pure ethanol

FFVs can automatically adjust engine operation according to the ethanol concentration in the fuel.**Hence, Statement 3 is correct.**

VALUE ADDITION

Bioethanol at a Glance

Feature	**Details**
Type	Renewable Liquid Biofuel
Chemical Formula	C_2H_5OH
Production Method	Fermentation of biomass
Major Feedstocks	Sugarcane, Molasses, Corn, Damaged Grains, Crop Residues
Major Microorganism	Saccharomyces cerevisiae (Yeast)
Main Use	Blending with Petrol
Renewable Source	Yes

Generations of Ethanol

Generation	**Feedstock**
1G Ethanol	Sugarcane, Molasses, Food Grains
2G Ethanol	Rice Straw, Crop Residues, Bamboo, Bagasse
3G Ethanol	Algae-Based Feedstocks
4G Ethanol	Genetically Engineered Algae/Cyanobacteria

Ethanol Blending Programme (EBP)

Feature	**Details**
Nodal Ministry	Ministry of Petroleum & Natural Gas
Policy Support	National Policy on Biofuels, 2018
Target	20% Ethanol Blending (E20)
Target Year	2025–26
Benefits	Energy Security, Lower Emissions, Farmer Income Support

Advantages of Bioethanol

- Renewable and biodegradable fuel.
- Reduces carbon monoxide and particulate emissions.
- Improves fuel octane rating.
- Lowers crude oil import dependence.
- Supports sugarcane farmers and rural economy

In NEWS:Insular incentive- Ethanol-blended fuel should not be forced on consumers irrationally

130. PM SVANidhi

Q.Consider the following statements regarding the Prime Minister Street Vendor's AtmaNirbhar Nidhi (PM SVANidhi) scheme:

1. It is a Central Sector Scheme jointly implemented by the Ministry of Housing and Urban Affairs (MoHUA) and the Ministry of Rural Development to cover both urban and rural street vendors.

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2. Under the recently introduced 'Street Food Hub' plan of the scheme, selected tourist and transit towns receive financial assistance to establish organized, hygienic culinary vending spaces.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (b) 2 only
EXPLANATION
Statement 1:
PM SVANidhi is a Central Sector Scheme launched in June 2020 by the Ministry of Housing and Urban Affairs (MoHUA) in collaboration with the Department of Financial Services (DFS).

- It was launched to provide collateral-free working capital loans to urban street vendors affected by the Covid-19 pandemic.
- The scheme is specifically designed for urban street vendors.
- It is not jointly implemented with the Ministry of Rural Development.
- Rural vendors are not covered under the scheme.

Hence, Statement 1 is incorrect.

Statement 2:
Under the recently introduced Street Food Hub initiative of PM SVANidhi, selected tourist, pilgrimage and transit towns are provided financial assistance to create organized and hygienic vending spaces.

- The initiative aims to develop world-class food streets showcasing regional cuisine.
- Preference is given to towns with tourism potential, heritage significance and unique food traditions.
- Each approved Street Food Hub receives financial assistance of up to ₹4 crore.
- Lakhanpur (Jammu & Kashmir) was among the first towns approved under this initiative.**Hence, Statement 2 is correct.**

VALUE ADDITION
PM SVANidhi – Quick Facts

Feature	**Details**
Launch Year	2020
Ministry	Ministry of Housing & Urban Affairs (MoHUA)
Type	Central Sector Scheme
Target Group	Urban Street Vendors
Implementing Partner	SIDBI
Nature of Loan	Collateral-free Working Capital Loan
Extended Till	March 2030

Loan Structure under PM SVANidhi

Loan Tranche	**Amount**
First	₹15,000
Second	₹25,000
Third	₹50,000

Incentives under PM SVANidhi

Incentive	**Benefit**
Interest Subsidy	7% per annum
Digital Transactions	Cashback incentives
Credit Expansion	UPI-linked RuPay Credit Card up to ₹30,000

SVANidhi se Samriddhi
Objective
To link PM SVANidhi beneficiaries and their families with major social welfare schemes.
Important Schemes Covered

- PM Jan Dhan Yojana
- PM Jeevan Jyoti Bima Yojana (PMJJBY)
- PM Suraksha Bima Yojana (PMSBY)
- One Nation One Ration Card (ONORC)
- PM Shram Yogi Maandhan
- PM Matru Vandana Yojana (PMMVY)
- Janani Suraksha Yojana (JSY)

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Street Food Hub Initiative

Feature	Details
Ministry	MoHUA
Under	PM SVANidhi
Number Proposed	Up to 50 hubs
Assistance	₹4 crore per project
Focus Areas	Tourist, Heritage, Pilgrimage & Transit Towns
Convergence	Swadesh Darshan, PRASHAD, UNESCO Sites
Partners	Ministry of Tourism, FSSAI

In NEWS:Lakhanpur among first towns selected under PM SVANidhi 'Street Food Hub' Plan; Dr. Jitendra Singh thanks PM Modi for giving high priority to Jammu & Kashmir

131. Paizhen Fault

Q.With reference to the Paizhen Fault, consider the following statements:

1. It is part of a network of tectonic faults in the eastern Himalayan region.
2. It originated due to the collision of the Indian and Eurasian tectonic plates.
3. It has remained active since the Early Pleistocene and continues to show activity during the Holocene epoch.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

In NEWS: Chinese geologists question safety of Brahmaputra Mega-Dam in Tibet being built over active seismic fault line

Statement 1:

It is part of a network of tectonic faults in the eastern Himalayan region.The Paizhen Fault is located in the eastern Himalayan tectonic zone and forms part of a complex fault network associated with active crustal deformation in the Himalayas. **Hence, Statement 1 is correct.**

Statement 2:

It originated due to the collision of the Indian and Eurasian tectonic plates.The Himalayan mountain system and associated fault networks, including the Paizhen Fault, were formed due to the ongoing convergence and collision of the Indian Plate with the Eurasian Plate.**Hence, Statement 2 is correct.**

Statement 3:

It has remained active since the Early Pleistocene and continues to show activity during the Holocene epoch.Geological studies indicate that the fault has been active since the Early Pleistocene (part of the Quaternary Period) and continues to exhibit tectonic activity during the present Holocene epoch.**Hence, Statement 3 is correct.**

VALUE ADDITION

Fault

A fracture or zone of fractures in the Earth's crust along which rocks have moved relative to each other.

Tectonic Plates

Large rigid slabs of the Earth's lithosphere that move over the asthenosphere.

Seismic Belt

A region characterized by frequent earthquakes due to active tectonic movements.

Geological Time Scale

Epoch	Duration
Pleistocene	~2.58 million years ago to 11,700 years ago
Holocene	11,700 years ago to Present

Himalayan Seismic Belt

- One of the most earthquake-prone regions in the world.
- Formed due to active convergence of Indian and Eurasian Plates.
- Includes major fault systems:
  - Main Central Thrust (MCT)
  - Main Boundary Thrust (MBT)
  - Himalayan Frontal Thrust (HFT)
  - Paizhen Fault

Brahmaputra River: Course

Country	Name
Tibet (China)	Yarlung Tsangpo
India	Brahmaputra
Bangladesh	Jamuna



Key Facts

- Origin: Chemayungdung Glacier, Tibet.
- Enters India near Gelling (Arunachal Pradesh).
- Forms the world's deepest gorge: Yarlung Tsangpo Grand Canyon.
- Flows through India and Bangladesh before joining the Ganga-Meghna system.

132. NIDAR 2.0

Q.With reference to the National Innovation Challenge for Drone Application and Research (NIDAR 2.0), consider the following statements:

1. It has been launched by the Ministry of Electronics and Information Technology (MeitY) in collaboration with the Drone Federation India (DFI).
2. It is being implemented under the SwaYaan initiative.
3. NIDAR 2.0 includes a component innovation track focused on developing indigenous flight controllers based on the VEGA processor.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION

IN NEWS:Ministry of Electronics and Information Technology (MeitY) and Drone Federation India (DFI) launch NIDAR 2.0

Statement 1:

NIDAR 2.0 (National Innovation Challenge for Drone Application and Research) has been launched jointly by the Ministry of Electronics and Information Technology (MeitY) and the Drone Federation India (DFI). **Hence, Statement 1 is correct.**

Statement 2:

NIDAR 2.0 is being conducted under the SwaYaan Initiative, which aims to strengthen India's drone ecosystem through capacity building, research and innovation.**Hence, Statement 2 is correct.**

Statement 3:

Under the Component Innovation Track, teams are required to develop an indigenous flight controller and autopilot system using the VEGA processor and indigenous electronic components.**Hence, Statement 3 is correct.**

VALUE ADDITION

NIDAR 2.0:

Feature	Details
Full Form	National Innovation Challenge for Drone Application and Research
Launched By	MeitY + Drone Federation India
Under	SwaYaan Initiative
Focus	Drone Innovation & Component Innovation
Prize Pool	More than ₹65 lakh
Target Group	Students and Researchers
Objective	Indigenous Drone Technologies

VEGA Processor

Feature	Details
Developed By	C-DAC
Ministry	MeitY
Architecture	RISC-V
Type	Indigenous Microprocessor
Objective	Semiconductor Self-Reliance
Programme	Digital India RISC-V (DIR-V)

SwaYaan Initiative:

Feature	Details
Full Form	Capacity Building for Human Resource Development in unmanned aircraft system(UAS)
Ministry	MeitY
Approved	July 2022
Outlay	₹89.87 crore
Duration	5 Years
Institutions	IITs, IISc, IIITs, NITs, C-DAC, NIELIT
Model	Hub-and-Spoke
Goal	Make India a Global Drone Hub by 2030

133. NAMASTE SCHEME

Q.Which of the following correctly describes the NAMASTE Scheme?

(a) A programme for rural sanitation under the Ministry of Jal Shakti
(b) A scheme to promote mechanised sanitation and improve the safety, dignity and livelihood of sanitation workers
(c) A programme for rehabilitation of bonded labourers under the Ministry of Labour and Employment
(d) A scheme for urban housing development under PMAY-U

Answer: (b) A scheme to promote mechanised sanitation and improve the safety, dignity and livelihood of sanitation workers

EXPLANATION :

IN NEWS: Ministry of Social Justice & Empowerment to Mark 3rd NAMASTE Day on July 14

Option (a):

A programme for rural sanitation under the Ministry of Jal Shakti

The NAMASTE Scheme is not a rural sanitation programme under the Ministry of Jal Shakti. It is jointly implemented by the Ministry of Social Justice and Empowerment (MoSJE) and the Ministry of Housing and Urban Affairs (MoHUA). Its focus is on Sewer and Septic Tank Sanitation Workers (SSWs) and mechanised sanitation systems.**Hence, Option (a) is incorrect.**

Option (b):

A scheme to promote mechanised sanitation and improve the safety, dignity and livelihood of sanitation workers

The National Action for Mechanised Sanitation Ecosystem (NAMASTE) Scheme aims to:

- Eliminate hazardous manual cleaning of sewers and septic tanks.
- Promote mechanised sanitation practices.
- Ensure occupational safety and health of sanitation workers.
- Eliminate direct contact with human faecal matter.
- Improve dignity, livelihood and social security of sanitation workers.
- Promote entrepreneurship and Self-Help Groups (SHGs).**Hence, Option (b) is correct.**

Option (c):

A programme for rehabilitation of bonded labourers under the Ministry of Labour and Employment

NAMASTE has no connection with bonded labour rehabilitation.It specifically targets sanitation workers engaged in sewer and septic tank cleaning.**Hence, Option (c) is incorrect.**

Option (d):

A scheme for urban housing development under PMAY-U

PMAY-U (Pradhan Mantri Awas Yojana-Urban) deals with affordable housing for urban residents.NAMASTE is focused on sanitation worker welfare and mechanised sanitation ecosystems.**Hence, Option (d) is incorrect.**

VALUE ADDITION

NAMASTE Scheme:

Feature	Details
Full Form	National Action for Mechanised Sanitation Ecosystem
Launch	2023
Ministries	MoSJE and MoHUA
Implementing Agency	National Safai Karamcharis Finance and Development Corporation (NSKFDC)
Target Group	Sewer and Septic Tank Sanitation Workers (SSWs)
Duration	FY 2023-24 to FY 2025-26

Key Objectives

- Zero fatalities in sanitation work.
- Elimination of manual entry into sewers and septic tanks.
- 100% mechanised cleaning.
- Occupational safety and health protection.
- Skill development and capacity building.
- Promotion of SHGs and entrepreneurship.
- Strengthening Emergency Response Sanitation Units (ERSUs).

Institutions in News

Institution	Function
MoSJE	Nodal Ministry for social justice and worker welfare
MoHUA	Urban sanitation and implementation support
NSKFDC	Implementation of NAMASTE Scheme
National Commission for Safai Karamcharis (NCSK)	Safeguards welfare of sanitation workers

134. JNANPITH AWARD

Q.With reference to the Jnanpith Award, consider the following statements:

1. It is the highest literary honour in India awarded for outstanding contribution to literature.
2. It is open to creative writers in any language listed in the Eighth Schedule of the Constitution as well as English.
3. A language that receives the award in a particular year is not eligible for consideration for the next two years.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION

In NEWS: Tamil poet and lyricist Vairamuthu conferred Jnanpith Award

Statement 1:

The Jnanpith Award is regarded as the highest literary award in India.It is conferred upon an author for outstanding contribution to Indian literature through creative writing.The award recognizes a writer's overall literary excellence rather than a single work.**Hence, Statement 1 is correct.**

Statement 2:

The award is given to writers producing literary works in:

- Any language listed in the Eighth Schedule of the Constitution
- English

Only Indian citizens who are alive can be considered for the award.**Hence, Statement 2 is correct.**

Statement 3:

As per Jnanpith Award rules:

- Once a language receives the award in a particular year,
- That language becomes ineligible for consideration for the next two years.

This provision ensures broader linguistic representation among Indian languages.**Hence, Statement 3 is correct.**

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VALUE ADDITION

Jnanpith Award:

Feature	Details
Instituted	1961
First Award Given	1965
Sponsor	Bharatiya Jnanpith
Nature	Highest Literary Honour in India
Award Components	cash + Citation + Bronze Statue of Vagdevi
Eligibility	Living Indian citizens
Languages Covered	Eighth Schedule Languages + English
Frequency	Annual

First Jnanpith Award Winner

Feature	Details
First Recipient	G. Sankara Kurup
Language	Malayalam
Awarded For	*Odakkuzhal*
Year Awarded	1965

Tamil Recipients of Jnanpith Award

Writer	Year
Akilan	1975
Jayakanthan	2002
R. Vairamuthu	2025

Eighth Schedule of the Constitution

Feature	Details
Constitutional Provision	Articles 344 & 351
Number of Languages	22
Latest Addition	Bodo, Dogri, Maithili, Santali (92nd Constitutional Amendment Act, 2003)

135. UMANG PORTAL

Q.With reference to the Unified Mobile Application for New-age Governance (UMANG), consider the following statements:

1. It was launched in 2017 under the Digital India Programme.
2. It was developed jointly by the Ministry of Electronics and Information Technology (MeitY) and the National e-Governance Division (NeGD).
3. It acts as a unified platform providing access to services offered by Central and State Governments.
4. UMANG provides access to services such as EPFO, DigiLocker, e-Shram, Bharat Gas and Indian Railways.

How many of the above statements are correct?

(a) Only One
(b) Only Two
(c) Only Three
(d) All Four

Answer: (d) All Four

EXPLANATION

In NEWS:User data exposed due to flaws in UMANG portal: researchers

Statement 1:

UMANG was launched in November 2017 as part of the Government of India's Digital India Initiative.Its objective is to provide a single platform for accessing multiple government services through mobile and web applications.**Hence, Statement 1 is correct.**

Statement 2:

UMANG was developed by:

- Ministry of Electronics and Information Technology (MeitY)
- National e-Governance Division (NeGD)

NeGD functions under MeitY and plays a key role in implementing various Digital India initiatives.**Hence, Statement 2 is correct.**

Statement 3:

UMANG follows a single-window service delivery model.It integrates services provided by:

- Central Government Ministries
- State Governments
- Local Bodies
- Autonomous Organizations

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- Statutory Bodies

This eliminates the need for citizens to download multiple applications.**Hence, Statement 3 is correct.**

Statement 4:

UMANG offers access to more than 2,400 government services, including:

- EPFO services
- DigiLocker
- e-Shram
- Bharat Gas
- Indian Railways
- Mera Ration
- Atal Pension Yojana
- PM Jan Dhan Yojana **Hence, Statement 4 is correct.**

VALUE ADDITION

UMANG: Quick Facts

Feature	**Details**
Full Form	Unified Mobile Application for New-age Governance
Launch Year	2017
Initiative	Digital India
Developed By	MeitY & NeGD
Platform	Mobile App & Web Portal
Objective	Single-point access to government services
Services Available	2400+ Services
Nature	Citizen-Centric Digital Governance Platform

Major Services Available on UMANG

Sector	**Services**
Social Security	EPFO, APY
Labour	e-Shram
Documents	DigiLocker
Banking	PM Jan Dhan Yojana
Utilities	Bharat Gas
Food Security	Mera Ration
Transport	Indian Railways
Health	Ayushman Bharat-related services

UMANG vs DigiLocker

Feature	**UMANG**	**DigiLocker**
Purpose	Access Government Services	Store & Access Digital Documents
Platform Type	Service Delivery Platform	Digital Document Wallet
Developed By	MeitY & NeGD	MeitY
Includes Documents	Yes (through DigiLocker integration)	Primary Function

136. DARK PATTERN

Q.With reference to Dark Patterns, consider the following statements:

1. Dark patterns exploit users' cognitive biases to influence decision-making.
2. Drip Pricing refers to revealing additional charges only after the consumer has begun the purchase process.
3. Disguised Advertisements are designed to appear as independent content or user-generated content.
4. The Central Consumer Protection Authority (CCPA) issued the Guidelines for Prevention and Regulation of Dark Patterns in 2023.

How many of the above statements are correct?

(a) Only One
(b) Only Two
(c) Only Three
(d) All Four

Answer: (d) All Four

EXPLANATION

IN NEWS: What are dark patterns and why are they under scrutiny

Statement 1:

Dark patterns exploit users' cognitive biases to influence decision-making.Dark Patterns are deceptive user-interface designs that manipulate users into taking actions they may not otherwise choose.These designs exploit psychological tendencies such as:

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- Fear of Missing Out (FOMO)
- Loss aversion
- Default bias
- Scarcity bias
- Urgency bias

Their objective is often to increase sales, collect personal data, or retain subscriptions**. Hence, Statement 1 is Correct.**

Statement 2:

Drip Pricing refers to revealing additional charges only after the consumer has begun the purchase process.Drip Pricing is one of the 13 dark patterns identified by CCPA.

Under this practice:

- The initial price appears low.
- Additional charges are gradually added during checkout.
- Taxes, convenience fees, platform fees, handling charges etc. appear later. . This prevents consumers from making informed price comparisons. **Hence, Statement 2 is Correct.**

Statement 3:

Disguised Advertisements are designed to appear as independent content or user-generated content.Disguised Ads are advertisements presented in a manner that makes them appear as:

- News articles
- User reviews
- Editorial content
- Social media posts . Consumers may mistake advertisements for unbiased information.**Hence, Statement 3 is Correct.**

Statement 4:

The Central Consumer Protection Authority (CCPA) issued Guidelines for Prevention and Regulation of Dark Patterns in 2023.The CCPA issued these guidelines on 1 December 2023 under the Consumer Protection Act, 2019.The guidelines identify and regulate 13 specific dark patterns used by digital platforms and online businesses.

Objective:

- Protect consumer rights
- Ensure informed consent
- Prevent unfair trade practices **Hence, Statement 4 is Correct.**

VALUE ADDITION

Dark Patterns: Quick Facts

Feature	**Details**
Term Coined By	Harry Brignull
Year	2010
Nature	Deceptive User Interface Design
Objective	Manipulate consumer behaviour
Regulator in India	CCPA
Guidelines Issued	2023
Legal Basis	Consumer Protection Act, 2019

CCPA's 13 Dark Patterns

Dark Pattern	**Meaning**
False Urgency	Fake scarcity or deadlines
Basket Sneaking	Adding products without consent
Confirm Shaming	Guilt-based persuasion
Forced Action	Mandatory unwanted action
Nagging	Repeated interruptions
Subscription Trap	Difficult cancellation
Bait and Switch	Different product delivered
Rogue Malware	Fake security threats
Disguised Ads	Ads appearing as content
Interface Interference	Manipulative design layout
Drip Pricing	Hidden charges revealed later
Trick Question	Confusing wording
SaaS Billing	Hidden recurring billing

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CCPA (Central Consumer Protection Authority)

Feature	Details
Established Under	Consumer Protection Act, 2019
Operational Since	July 2020
Parent Ministry	Ministry of Consumer Affairs
Objective	Protect consumer rights
Powers	Investigation, Recall Orders, Penalties, Regulation of Misleading Advertisements

137. SMALL MODULAR REACTORS

Q.With reference to Small Modular Reactors (SMRs), consider the following statements:

1. SMRs are nuclear fission reactors with a power generation capacity of up to 300 MW per module.
2. SMRs can complement renewable energy sources in hybrid energy systems.
3. SMRs are characterized by modular construction and factory-based assembly.

Which of the statements given above is/are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION

IN NEWS:The right path for India's nuclear power development

Statement 1:

According to the International Atomic Energy Agency (IAEA), Small Modular Reactors are nuclear reactors with an electricity generation capacity of up to 300 MW(e) per module, roughly one-third of conventional large nuclear reactors.They use the same basic principle of controlled nuclear fission to generate heat and electricity.**Hence, Statement 1 is correct.**

Statement 2:

SMRs provide stable baseload power and can be integrated with:

- Solar power
- Wind energy
- Hydrogen production systems
- Energy storage systems

This helps address the intermittency associated with renewable energy sources.**Hence, Statement 2 is correct.**

Statement 3:

The term "Modular" refers to:

- Factory fabrication of components
- Transportation of modules to the site
- Faster installation and commissioning

Benefits include:

- Lower construction time
- Reduced capital cost
- Standardized manufacturing
- Better quality control **Hence, Statement 3 is correct.**

VALUE ADDITION

Small Modular Reactors (SMRs): Quick Facts

Feature	Details
Definition	Nuclear reactors producing up to 300 MW(e) per module
Technology	Nuclear Fission
Regulator Reference	IAEA
Fuel	Generally Low-Enriched Uranium (LEU)
Construction	Factory-built and modular
Refuelling Cycle	3–7 years (some designs longer)
Main Objective	Safe, flexible and scalable nuclear power generation

Meaning of SMR

Term	Meaning
Small	Smaller physical size than conventional reactors
Modular	Factory-manufactured modules assembled at site
Reactor	Uses controlled nuclear fission to generate electricity

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Types of SMRs

Category	Example
Land-based Water-cooled SMRs	RITM-200N (Russia)
Marine/Floating SMRs	Akademik Lomonosov (Russia)
High Temperature Gas-cooled Reactors	HTR-PM (China)
Liquid Metal Fast Reactors	BREST (Russia)
Molten Salt Reactors	Under Development
Microreactors	Up to 10 MW

Global Operational SMRs

Country	Reactor
Russia	Akademik Lomonosov (Floating SMR)
China	HTR-PM

Only two SMR projects are currently operational globally.

Opportunities for India

- Achieving Net Zero by 2070
- Power supply to AI Data Centres
- Deployment in remote and Northeastern regions
- Replacement of decommissioned thermal power plants
- Support for nuclear-powered submarines
- Expansion of clean baseload power

India's Nuclear Programme:

Stage	Technology
Stage I	Pressurised Heavy Water Reactors (PHWRs) using Natural Uranium
Stage II	Fast Breeder Reactors (FBRs) using Plutonium
Stage III	Thorium-based Reactors (U-233 Cycle)

India recently commissioned its 500 MW Prototype Fast Breeder Reactor (PFBR), a major milestone in the three-stage nuclear programme.

138. HORN OF AFRICA

Q. Consider the following countries:

1. Djibouti
2. Eritrea
3. Ethiopia
4. Somalia
5. Sudan

How many of the above are generally considered part of the Horn of Africa?

(a) Only Two
(b) Only Three
(c) Only Four
(d) All Five

Answer: (c) Only Four

EXPLANATION:

In NEWS: Red Sea: a hinge between West Asia and Horn of Africa

Country 1: Djibouti

Djibouti is one of the four core countries of the Horn of Africa. It occupies a strategic location near the Bab-el-Mandeb Strait, connecting the Red Sea and the Gulf of Aden.It hosts several foreign military bases due to its strategic maritime position.**Hence, Djibouti is part of the Horn of Africa.**

Country 2: Eritrea

Eritrea is a core constituent of the Horn of Africa.It has a long coastline along the Red Sea and occupies the northeastern part of the region.**Hence, Eritrea is part of the Horn of Africa.**

Country 3: Ethiopia

Ethiopia is the largest country in the Horn of Africa by population.The Ethiopian Plateau (Abyssinian Highlands) is one of the most prominent geographical features of the region.**Hence, Ethiopia is part of the Horn of Africa.**

Country 4: Somalia

Somalia forms the easternmost projection of Africa into the Indian Ocean.The region is often referred to as the Somali Peninsula because Somalia occupies a major portion of it.**Hence, Somalia is part of the Horn of Africa.**

Country 5: Sudan

Sudan is generally not included in the core definition of the Horn of Africa.Although Sudan borders the Red Sea and is sometimes included in broader geopolitical discussions involving the Horn of Africa, it is not one of the four principal countries

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constituting the region.**Hence, Sudan is not part of the core Horn of Africa.**

VALUE ADDITION



Core Countries of Horn of Africa

Country	**Capital**
Djibouti	Djibouti City
Eritrea	Asmara
Ethiopia	Addis Ababa
Somalia	Mogadishu

Important Water Bodies

Water Body	**Significance**
Red Sea	Connects to Suez Canal
Gulf of Aden	Connects Red Sea to Arabian Sea
Indian Ocean	Eastern maritime boundary
Bab-el-Mandeb Strait	Separates Africa from Arabia

Bab-el-Mandeb Strait

Feature	**Details**
Location	Between Djibouti/Eritrea and Yemen
Connects	Red Sea and Gulf of Aden
Importance	Global oil and container shipping route
Strategic Relevance	Gateway to Suez Canal

139. CONSUMER PRICE INDEX

Q.With reference to the Consumer Price Index (CPI) in India, consider the following statements:

1. The CPI is the primary measure of retail inflation in India.
2. The CPI is compiled and released by the Reserve Bank of India (RBI).
3. The weight assigned to food items has been reduced in the new CPI series with base year 2024 compared to the previous series.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (b) 1 and 3 only

EXPLANATION

In NEWS:Retail inflation climbs past RBI target to 4.4%

Statement 1:

The CPI is the primary measure of retail inflation in India.The Consumer Price Index (CPI) measures changes in the prices of goods and services consumed by households and is the principal indicator of retail inflation in India.The Reserve Bank of India (RBI) uses CPI (Combined) as the nominal anchor for its inflation-targeting monetary policy framework under the Monetary Policy Committee (MPC).**Hence, Statement 1 is correct.**

Statement 2:

The CPI is compiled and released by the Reserve Bank of India (RBI).The Consumer Price Index is compiled and released by the National Statistical Office (NSO) under the Ministry of Statistics and Programme Implementation (MoSPI).The RBI uses CPI data for monetary policy decisions but does not compile or publish the index.**Hence, Statement 2 is incorrect.**

Statement 3:

The weight assigned to food items has been reduced in the new CPI series with base year 2024 compared to the previous series.The new CPI series (Base Year: 2024) reflects changing consumption patterns arising from rising incomes, urbanisation and greater spending on services.As a result, the relative weight of food items in the CPI basket has been reduced compared to the previous CPI series.A lower food weight is expected to reduce volatility in headline

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inflation since food prices are highly sensitive to monsoon conditions and supply-side shocks.**Hence, Statement 3 is correct.**

VALUE ADDITION

Consumer Price Index (CPI): Quick Facts

Feature	**Details**
Full Form	Consumer Price Index
Measures	Retail Inflation
Compiled By	National Statistical Office (NSO)
Ministry	Ministry of Statistics and Programme Implementation (MoSPI)
Current Base Year	2024
Frequency	Monthly
Coverage	Rural, Urban and Combined
Used By	RBI for Monetary Policy

New CPI Series (Base Year 2024)

Aspect	**Change**
Base Year	Revised from 2012 to 2024
Consumption Basket	Updated to reflect current spending patterns
New Items Added	Headphones, Earphones, Bluetooth Devices
Obsolete Items Removed	CDs, DVDs and similar products
Food Weight	Reduced
Objective	Better representation of contemporary household consumption

CPI vs WPI

Feature	**CPI**	**WPI**
Measures	Retail Prices	Wholesale Prices
Released By	NSO, MoSPI	Office of Economic Adviser, DPIIT
Includes Services	Yes	No
RBI Inflation Targeting	Yes	No

140. Foreigners Tribunal

Q.With reference to Foreigners Tribunals (FTs), consider the following statements:

1. Foreigners Tribunals are quasi-judicial bodies constituted under the Foreigners Act framework to determine whether a person is a foreigner or not.
2. As of now, Foreigners Tribunals currently function only in Assam.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Explanation

In NEWS: Citizenship rulings must be 'fair, lawful': SC

Statement 1:

Foreigners Tribunals (FTs) are quasi-judicial bodies established under:

- Foreigners Act, 1946
- Foreigners (Tribunals) Order, 1964

Their role is to determine:

- Determine whether a person is a foreigner under the Foreigners Act, 1946.
- Citizenship disputes arising from suspected illegal migration.
- Cases involving D-Voters and NRC-related references.

The recent Supreme Court judgment itself arose from orders passed by Foreigners Tribunals declaring individuals to be foreigners.**Hence, Statement 1 is correct.**

Statement 2:

Although the Foreigners (Tribunals) Order, 1964 permits establishment of tribunals elsewhere, at present:

- Foreigners Tribunals are operational only in Assam.
- Their functioning is closely linked with:
  - Illegal migration issues
  - NRC process

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○ Assam Accord, 1985 **Hence, Statement 2 is correct.**

VALUE ADDITION

- Legal Basis: Foreigners (Tribunals) Order, 1964
- Parent Act: Foreigners Act, 1946
- Function:Determine whether a person is a foreigner under the Foreigners Act.
- Composition: Headed by judges, advocates, or experienced civil servants with judicial background
- Recent Development: Under the Immigration and Foreigners Order, 2025, FTs have been given powers of a Judicial Magistrate First Class in specified matters.

Foreigners Tribunals (FTs):

Feature	Details
Nature	Quasi-Judicial Body
Legal Basis	Foreigners Act, 1946
Operational Framework	Foreigners (Tribunals) Order, 1964
Purpose	Determine whether a person is a foreigner
Present Location	Assam
Referring Authorities	Border Police, Election Authorities, Government Agencies

Foreigner:

Section 2(a) of the Foreigners Act, 1946

A foreigner means a person who is not a citizen of India.

Burden of Proof:

Aspect	Provision
Applicable Law	Foreigners Act, 1946
Relevant Section	Section 9
Burden of Proof	Lies on the person concerned

Unlike ordinary criminal law, where the prosecution proves guilt, under the Foreigners Act: The individual must prove that he/she is an Indian citizen.

Foreigners Tribunal vs NRC

Feature	Foreigners Tribunal	NRC
Nature	Quasi-Judicial Body	Citizenship Register
Purpose	Decide citizenship disputes	Prepare list of genuine citizens
Legal Basis	Foreigners Act & FT Order	Citizenship Act, 1955
Operational Area	Assam	Assam (currently)

Assam Accord, 1985

Feature	Details
Signed Between	Government of India & AASU
Year	1985
Cut-off Date	24 March 1971
Objective	Detection and deportation of illegal migrants

The functioning of FTs is closely linked to the implementation of the Assam Accord.

141. Index of Service Production

Q.With reference to the Index of Services Production (ISP), consider the following statements:

1. The Index of Services Production (ISP) is a short-term indicator designed to measure changes in the real volume of output produced by the services sector.
2. It is compiled and released by the Ministry of Commerce and Industry.
3. Public Administration and Defence are fully covered under the ISP framework.
4. It currently covers only the formal services sector.

How many of the statements given above are correct?

(a) Only one
(b) Only two

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(c) Only three
(d) All four

Answer: (b) Only two
EXPLANATION:
In NEWS: Government launches first trial Index of Services Production for formal services sector
Statement 1:
The Index of Services Production (ISP) is a short-term indicator designed to measure changes in the real volume of output produced by the services sector.

- ISP is a monthly/high-frequency indicator.
- It measures real output of the services sector.
- It is conceptually similar to the Index of Industrial Production (IIP). **Hence, Statement 1 is correct.**

Statement 2:

- ISP is compiled by the National Statistical Office (NSO).
- NSO functions under the Ministry of Statistics and Programme Implementation (MoSPI).
- Ministry of Commerce and Industry is not responsible for ISP. **Hence, Statement 2 is incorrect.**

Statement 3:

- ISP primarily covers formal market-based services.
- Public Administration and Defence are core government/non-market activities.
- These are not fully covered under the present ISP framework. **Hence, Statement 3 is incorrect.**

Statement 4:

- ISP relies mainly on GST data, administrative data and ASISSE.
- These sources largely capture formal sector activity.
- Large parts of the informal services sector remain outside coverage. **Hence, Statement 4 is correct.**

VALUE ADDITION : India's Major High-Frequency Indicators

Indicator	Measures	Released By
CPI	Retail Inflation	NSO (MoSPI)
IIP	Industrial Output	NSO (MoSPI)
WPI	Wholesale Prices	DPIIT
PMI	Business Sentiment	S&P Global
ISP	Services Output	NSO (MoSPI)

IIP vs ISP

Feature	IIP	ISP
Sector	Industry	Services
Coverage	Mining, Manufacturing, Electricity	Trade, Transport, Banking, Telecom etc.
Nature	Physical Production	Service Output
Base Year	2022-23 (new series proposed)	2024-25
Frequency	Monthly	Monthly

ISP = Services Equivalent of IIP

- IIP measures output of Mining, Manufacturing and Electricity.
- ISP measures output of the Services Sector.
- Both are monthly high-frequency indicators.
- ISP is often called the "Services-sector counterpart of IIP".

Base Year

- The base year of ISP is 2024-25.
- First trial release was issued in July 2026.

Developed By

- ISP is compiled by the National Statistical Office (NSO).
- NSO functions under the Ministry of Statistics and Programme Implementation (MoSPI).

Technical Advisory Committee (TAC-ISP)

- Constituted in May 2025.
- Comprised 24 experts.
- Provided conceptual and methodological guidance for ISP.

Major Data Sources
ISP uses three major sources:

- GST Network (GSTN) Data
- Administrative Data
- ASISSE Data (Annual Survey of Incorporated Services Sector Enterprises)

Coverage

- Current ISP covers only the formal services sector.
- Trial release covers 19 sub-sectors.
- These account for about 60% of the services sector.

ISP need

- Services contribute ~55% of India's GVA.
- Unlike industry (IIP), no high-frequency output indicator existed for services.
- Helps RBI, policymakers and investors assess economic activity in near real time.

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142. codex alimentarius commission

Q.With reference to the Codex Alimentarius Commission (CAC), consider the following statements:

1. It was established jointly by the Food and Agriculture Organization (FAO) and the World Health Organization (WHO).
2. Its primary objectives include protecting consumer health and ensuring fair practices in food trade.
3. Membership of the Commission is restricted only to countries that are members of both FAO and WHO.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two
EXPLANATION
IN NEWS:India Strengthens Global Leadership in Food Standards at 49th Session of the Codex Alimentarius Commission (CAC49)
Statement 1:
It was established jointly by the Food and Agriculture Organization (FAO) and the World Health Organization (WHO).

- The Codex Alimentarius Commission (CAC) was established in 1963.
- Codex standards are recognized under:WTO Agreement on Sanitary and Phytosanitary Measures (SPS Agreement)
- It was jointly created by FAO and WHO.
- It serves as the international food standards-setting body. **Hence, Statement 1 is correct.**

Statement 2:
Its primary objectives include protecting consumer health and ensuring fair practices in food trade.

- Consumer protection is one of the core objectives of CAC.
- It develops science-based international food standards.
- It also aims to promote fair practices in international food trade. **Hence, Statement 2 is correct.**

Statement 3:
Membership of the Commission is restricted only to countries that are members of both FAO and WHO.

- Membership is open to Member Nations and Associate Members of FAO or WHO interested in food standards.
- It is not mandatory to be a member of both organizations.
- CAC currently has 189 members. **Hence, Statement 3 is incorrect.**

143. Caspian Sea

Q.With reference to the Caspian Sea, consider the following statements:

1. It is the world's largest enclosed inland water body.
2. It is an endorheic basin with no natural outflow to the oceans.
3. It is bordered by Russia, Kazakhstan, Turkmenistan, Armenia and Iran

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two
EXPLANATION :
IN NEWS: How the shrinking Caspian Sea intensifies ecological stress
Statement 1:

- World's largest enclosed inland water body.
- Despite its name, it is often regarded as the **world's largest lake** because it has no natural connection with the oceans.
- It covers an area of about **371,000 sq km**. **Hence, Statement 1 is correct.**

Statement 2:

- The Caspian Sea is an **endorheic basin**, meaning water flows into it but does not flow out to any sea or ocean.
- Water entering the basin is lost mainly through **evaporation**.
- Major rivers draining into it include:
  - **Volga**
  - **Ural**
  - **Terek** **Hence, Statement 2 is correct.**

Statement 3:

It is bordered by Russia, Kazakhstan, Turkmenistan, Iran and Armenia.The Caspian Sea has five littoral (coastal) countries:

- Russia
- Kazakhstan
- Turkmenistan
- Iran
- Azerbaijan

Armenia does not have a coastline on the Caspian Sea. Although Armenia is located in the South Caucasus region and is geographically close to the Caspian basin, it is a landlocked country and does not directly border the Caspian Sea.**Hence, Statement 3 is incorrect.**

Caspian Sea



Feature	Fact
Largest enclosed water body	Caspian Sea
Type	Endorheic Basin
Continents	Europe & Asia
Bordering Countries	Russia, Kazakhstan, Turkmenistan, Iran, Azerbaijan
Largest Inflow River	Volga River
Major City	Baku (Azerbaijan)
Resources	Oil & Natural Gas
Salinity	Nearly fresh in north; saltier in south

144. Agriphotovoltaics

Q.With reference to Agri-Photovoltaics (AgriPV), consider the following statements:

1. Agri-Photovoltaics increases land-use efficiency by generating food and electricity from the same land.
2. AgriPV systems are suitable only for arid and semi-arid regions.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (a) 1 only

EXPLANATION

In NEWS: Road to energy security passes through our farms

Statement 1:

- Agri-Photovoltaics (AgriPV) enables the **simultaneous use of agricultural land for crop cultivation and solar power generation**.
- It helps address the **"Food vs Fuel" dilemma** by avoiding competition between agriculture and renewable energy projects.
- Farmers can earn income from both:
  - Agricultural production
  - Sale of solar electricity **Hence, Statement 1 is correct.**

Statement 2:

- AgriPV is **not limited to arid and semi-arid regions**.
- Different AgriPV designs can be adapted to various agro-climatic conditions.
- Suitable crops and system designs vary across states:
  - **Madhya Pradesh:** Tomato, onion, turmeric, tulsi
  - **Karnataka & Maharashtra:** Ragi, jowar, grapes, banana, brinjal
- The choice depends on:
  - Crop type
  - Solar radiation
  - Irrigation availability
  - Local climate **Hence, Statement 2 is incorrect.**

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AgriPV – Key Benefits

Benefit	Significance
Dual Land Use	Food + Electricity from same land
Income Diversification	Additional revenue from power generation
Water Conservation	Reduced evapotranspiration
Climate Resilience	Protection from heatwaves and hailstorms
Renewable Energy	Supports India's 300 GW solar target

Types of AgriPV Systems

Type	Feature
Elevated System	Panels mounted above crops
Row-Based System	Panels installed between crop rows
Vertical System	Bifacial panels mounted vertically
Greenhouse-Integrated	Panels incorporated into greenhouse structures

145. India-UK Comprehensive Economic and Trade Agreement(CETA)

Q.With reference to the India–UK Comprehensive Economic and Trade Agreement (CETA), consider the following statements:

1. The agreement provides duty-free access to all Indian exports entering the UK.
2. Indian professionals such as chefs, yoga instructors and IT specialists are among those expected to benefit from improved mobility provisions.
3. The Double Contribution Convention (DCC) is aimed at preventing dual social security contributions by workers on temporary assignments.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two

EXPLANATION

In NEWS:India-UK Comprehensive Economic and Trade Agreement Comes into Force today

Statement 1:

The agreement provides duty-free access to all Indian exports entering the UK.

- The UK will eliminate tariffs on about 99% of tariff lines covering almost the entire value of Indian exports.
- The agreement does **not** provide duty-free access to **all** Indian exports. **Hence, Statement 1 is incorrect.**

Statement 2:

Indian professionals such as chefs, yoga instructors and IT specialists are among those expected to benefit from improved mobility provisions.

- CETA provides assured temporary access for several categories of Indian professionals.
- Beneficiaries include **chefs, yoga instructors, IT professionals and other skilled workers**.
- This is aimed at enhancing professional mobility between the two countries.**Hence, Statement 2 is correct.**

Statement 3:

The Double Contribution Convention (DCC) is aimed at preventing dual social security contributions by workers on temporary assignments.

- DCC is the social security agreement accompanying CETA.
- It exempts eligible Indian workers and employers from making social security contributions in both countries simultaneously.
- The exemption period has been extended from **3 years to 5 years**. **Hence, Statement 3 is correct.**

India-UK CETA: Key Gains

India:

- Textiles
- Gems & Jewellery
- Marine Products
- Leather
- Engineering Goods

UK:

- Whisky
- Automobiles
- Financial Services

146. Nemawashi

Q.What is the Nemawashi Method?

(a) A Japanese silvicultural practice used for regeneration of degraded forest ecosystems
(b) A traditional Japanese technique for preparing and relocating mature trees with minimal damage to their root system

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(c) A method of afforestation using seed balls in urban areas
(d) A technique of bonsai cultivation for dwarf trees

Answer: (b) A traditional Japanese technique for preparing and relocating mature trees with minimal damage to their root system

EXPLANATION

IN NEWS:Japan does not cut down centuries-old trees for development. Instead, experts spend months preparing their roots and relocating them

Option (a):A Japanese silvicultural practice used for regeneration of degraded forest ecosystems

Silviculture refers to the cultivation and management of forests for regeneration, productivity and ecological sustainability.Nemawashi is not a forest regeneration technique and is not primarily concerned with restoring degraded forest lands. **Hence, Option (a) is incorrect.**

Option (b):A traditional Japanese technique for preparing and relocating mature trees with minimal damage to their root system

Nemawashi has **two meanings**:

1. Horticulture/tree transplantation
2. Japanese management practice of informal consensus-building.

Nemawashi is a traditional Japanese horticultural practice involving:

- Gradual root pruning
- Formation of a compact root ball
- Wrapping roots using straw mats and ropes
- Acclimatisation before transplantation
- Relocation using specialised equipment

The technique maximises tree survival after transplantation and is commonly used for culturally, historically and ecologically significant trees. **Hence, Option (b) is correct.**

Option (c):A method of afforestation using seed balls in urban areas

Seed-ball technology involves dispersing seeds enclosed in clay and organic matter to promote afforestation.Nemawashi does not involve seed dispersal or afforestation. **Hence, Option (c) is incorrect.**

Option (d):A technique of bonsai cultivation for dwarf trees

Bonsai is the Japanese art of growing miniature trees through pruning and shaping.Nemawashi is related to transplantation of mature trees and not bonsai cultivation. **Hence, Option (d) is incorrect.**

VALUE ADDITION

Nemawashi: Quick Facts

Feature	**Details**
Country	Japan
Literal Meaning	Going around the roots / Laying the groundwork
Nature	Tree-transplantation technique
Purpose	Relocation of mature trees
Duration	Several months to over a year
Significance	Conservation-oriented infrastructure development

Stages of Nemawashi

Stage	**Activity**
Root Pruning	Gradual trimming of peripheral roots
Root Ball Formation	Encourages compact root growth
Wrapping	Straw mats and natural ropes
Acclimatisation	6–12 months
Relocation	Cranes, rollers and transport systems



147. Indian Grey hornbills

Q.With reference to the Indian Grey Hornbill, consider the following statements:

1. It is endemic to the Western Ghats of India.
2. It plays an important role in seed dispersal.
3. It is primarily found in deciduous forests, open woodlands and urban parks.and largely arboreal
4. It is classified as Endangered by the IUCN.

Which of the statements given above are correct?

(a) 2 and 3 only
(b) 1, 2 and 4 only
(c) 2, 3 and 4 only
(d) 1, 2, 3 and 4

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Answer: (a) 2 and 3 only

EXPLANATION:

IN NEWS:Grey hornbills thrive again in Gir as four years of successful breeding revive a lost species

Statement 1:

The Indian Grey Hornbill is distributed across much of the Indian subcontinent, including:

- India
- Nepal
- Pakistan
- Bangladesh . Hence, it is not endemic to the Western Ghats. **Hence, Statement 1 is incorrect.**

Statement 2:

Indian Grey Hornbills are important long-distance seed dispersers.Since they feed mainly on fruits, especially figs, they help:

- Forest regeneration
- Plant diversity maintenance
- Ecosystem functioning **Hence, Statement 2 is correct.**

Statement 3:largely arboreal.

The species inhabits:

- Deciduous forests
- Thorn forests
- Open woodlands
- Agricultural landscapes
- Urban gardens and parks **Hence, Statement 3 is correct.**

Statement 4:

The IUCN Red List categorizes the Indian Grey Hornbill as Least Concern (LC). **Hence, Statement 4 is incorrect.**

VALUE ADDITION



Indian Grey Hornbill: Quick Facts

Feature	Details
Scientific Group	Hornbill
Distribution	Indian Subcontinent
Habitat	Deciduous forests, woodlands, urban parks
Diet	Fruits (especially figs)
Ecological Role	Seed Disperser
IUCN Status	Least Concern

Hornbill Species in India

Species	Distribution	IUCN Status
Indian Grey Hornbill	Peninsular India	Least Concern
Malabar Grey Hornbill	Western Ghats (Endemic)	Vulnerable
Great Hornbill	Western Ghats & NE India	Vulnerable
Narcondam Hornbill	Narcondam Island (A&N)	Endangered
Oriental Pied Hornbill	Northeast India	Least Concern
Rufous-necked Hornbill	Northeast India	Vulnerable

Reintroduction Programme

Reintroduction

Release of a species into an area where it has become locally extinct.

Stages

1. Captive breeding / rescue
2. Habitat assessment
3. Release
4. Monitoring through tagging
5. Population establishment

Examples in India

Species	Location
Cheetah	Kuno NP
Indian Grey Hornbill	Gir
Gharial	Chambal
Great Indian Bustard	Conservation Breeding Programme

Satellite Telemetry

The Gir project used **satellite transmitters**.

Purpose

- Track movement

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- Study habitat use
- Understand migration/dispersal
- Monitor breeding success

Difference

Term	Meaning
Relocation	Moving individuals from one place to another
Reintroduction	Releasing species into an area where it became extinct locally
Translocation	Deliberate movement from one habitat to another

148. Biosensors

Q.With reference to Protein Biosensors, consider the following statements:

1. They specifically detect disease-related proteins.
2. They can be used for rapid diagnosis of conditions such as sepsis and cancer.
3. They primarily rely on aptamers or antibodies for target recognition.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

In NEWS: How protein biosensors can shrink the timeline of medical emergencies

Statement 1:

Protein biosensors are a specialized class of biosensors designed to identify and measure specific proteins (biomarkers) present in biological samples such as blood, saliva, or tissue.detect and quantify specific proteins, including disease biomarkers.

Proteins are important diagnostic targets because they:

- Regulate biological functions.
- Change concentration during disease conditions.
- Often provide early indications of illness.

Examples:

- Troponin → Heart attack
- Procalcitonin → Bacterial infection and sepsis
- Cancer biomarkers → Early cancer detection

Hence, Statement 1 is correct.

Statement 2:

One of the major advantages of protein biosensors is their ability to provide rapid and real-time detection.

Applications include:

- Early detection of sepsis.
- Identification of cancer biomarkers.
- Monitoring inflammatory diseases.
- Detection of infectious diseases such as malaria and COVID-19.

Unlike conventional laboratory tests that may take hours or days, protein biosensors can produce results within minutes. **Hence, Statement 2 is correct.**

Statement 3:

The most important component of a protein biosensor is the bioreceptor, which specifically recognizes the target protein.aptamers, antibodies, enzymes or receptors may act as bioreceptors.

Common bioreceptors include:

Antibodies

- Naturally produced by the immune system.
- Bind specific proteins with high specificity.
- Widely used in traditional biosensors.

Aptamers

- Short synthetic DNA or RNA strands.
- Bind selectively to target proteins.
- More stable and cheaper than antibodies.
- Easier to manufacture. . Modern protein biosensors increasingly use aptamers because of their high sensitivity and stability. **Hence, Statement 3 is correct.**

VALUE ADDITION

Protein Biosensor: Basic Components

Component	Function
Bioreceptor	Recognizes target protein
Transducer	Converts biological interaction into signal
Signal Processor	Displays and interprets result

Common Protein Biomarkers

Biomarker	Disease/Condition
Troponin	Heart Attack
Procalcitonin	Sepsis/Bacterial Infection
PSA	Prostate Cancer
C-Reactive Protein (CRP)	Inflammation
Cytokines	Immune Response Monitoring

Aptamers vs Antibodies

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Feature	Aptamers	Antibodies
Nature	Synthetic DNA/RNA	Proteins
Stability	High	Moderate
Cost	Lower	Higher
Production	Laboratory synthesis	Biological systems
Storage	Easier	Requires careful storage

Types of Biosensors

Type	Example
Electrochemical	Glucose meter
Optical	Fluorescence-based
Piezoelectric	Mass detection
Protein Biosensor	Disease biomarkers

149. Archaeological survey of India

Q.With reference to the Archaeological Survey of India (ASI), consider the following statements:

1. It was established in 1861 by Sir Alexander Cunningham.
2. It regulates archaeological activities under the Ancient Monuments and Archaeological Sites and Remains Act, 1958.
3. It protects monuments declared by both the Central and State Governments as monuments of national importance.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

EXPLANATION

In NEWS:SC suggests temporary prayer space for Muslims near Bhojshala complex

Statement 1:

The Archaeological Survey of India (ASI) was established in 1861 by the British Government under the initiative of Sir Alexander Cunningham, who is regarded as the Father of Indian Archaeology.Later, it was revived as a separate department in 1871, with Cunningham serving as its first Director-General.**Hence, Statement 1 is correct.**

Statement 2:

ASI functions under the provisions of the Ancient Monuments and Archaeological Sites and Remains (AMASR) Act, 1958.Under this Act, ASI:

- Protects monuments of national importance.
- Regulates excavation activities.
- Controls construction activities in protected and regulated areas around monuments.
- Conducts archaeological surveys and excavations.**Hence, Statement 2 is correct.**

Statement 3:

Only the Central Government can declare a monument as a Monument of National Importance under the AMASR Act, 1958.

- ASI protects and maintains monuments declared by the Central Government.
- State-protected monuments are managed by the respective State Archaeology Departments.

Therefore, ASI does not protect monuments declared by State Governments as monuments of national importance.**Hence, Statement 3 is incorrect.**

VALUE ADDITION

ASI Protected Areas

- Centrally Protected Monuments (CPMs)
- Centrally Protected Sites and Remains

Prohibited Area

100 metres around a protected monument.

Regulated Area

Next 200 metres.

Archaeological Survey of India (ASI):

Feature	Details
Established	1861(ASI was suspended in 1866 and revived in 1871.)
Founder	Sir Alexander Cunningham
Headquarters	New Delhi
Ministry	Ministry of Culture
Parent Act	AMASR Act, 1958
Protected Monuments	3,600+
Logo Inspired By	Sanchi Stupa

Major Functions of ASI

Function	Description
Conservation	Maintenance of protected monuments
Excavation	Scientific archaeological excavations
Exploration	Discovery of archaeological sites

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Epigraphy	Study of inscriptions
Underwater Archaeology	Exploration of submerged heritage
Museum Management	Maintenance of site museums

Important Acts Associated with ASI

Act	**Purpose**
AMASR Act, 1958	Protection of monuments and archaeological sites
Antiquities and Art Treasures Act, 1972	Regulation of antiquities and prevention of smuggling

150. Chandipura virus

Q.With reference to Chandipura Virus (CHPV), consider the following statements:

1. It primarily causes Acute Encephalitis Syndrome (AES).
2. It is transmitted mainly through infected mosquitoes.
3. A vaccine against CHPV is currently available in India.

Which of the statements given above is/are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 only

Explanation

In NEWS: Three children die of Chandipura virus in Gujarat in latest outbreak; 4 under treatment

Statement 1:

Chandipura Virus is a neurotropic virus that mainly affects the central nervous system and is known to cause Acute Encephalitis Syndrome (AES), characterized by inflammation of the brain. AES can lead to:

- High fever
- Convulsions
- Altered consciousness
- Neurological complications **Hence, Statement 1 is correct.**

Statement 2:

CHPV is primarily transmitted through the bite of infected sand flies (Phlebotomus species).Some studies have suggested a possible role of ticks in transmission. However, mosquitoes are not considered the primary vector of Chandipura Virus. **Hence, Statement 2 is incorrect.**

Statement 3:

At present:

- No approved vaccine is available against Chandipura Virus.
- No specific antiviral treatment exists.
- Management is based on supportive care and symptomatic treatment. **Hence, Statement 3 is incorrect.**

VALUE ADDITION

Chandipura Virus (CHPV)

Feature	**Details**
Family	Rhabdoviridae
First Identified	1965
Place of Discovery	Chandipura Village, Maharashtra
Disease Caused	Acute Encephalitis Syndrome (AES)
Major Vector	Sand Fly
Other Suspected Vector	Ticks
Human-to-Human Transmission	Not reported
Vaccine	Not Available
Treatment	Supportive Care

Acute Encephalitis Syndrome (AES)

Feature	**Details**
Meaning	Acute inflammation of the brain
Symptoms	Fever, headache, seizures, confusion
Affected Group	Mainly children
Causes	Viruses, bacteria, fungi, parasites
Public Health Concern	High mortality and neurological sequelae

Family Comparison

Virus	**Family**
Chandipura	Rhabdoviridae
Rabies	Rhabdoviridae
Dengue	Flaviviridae
Zika	Flaviviridae
Chikungunya	Togaviridae

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151. Import Cover

Q. Consider the following statements:

1. Import cover refers to the number of months of imports that can be financed by a country's foreign exchange reserves.
2. A rise in crude oil prices, with foreign exchange reserves remaining unchanged, can reduce India's import cover.
3. India's foreign exchange reserves provide an import cover of more than one year, reflecting a strong external sector position.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation:

In News: Iran threatens 'to halt all regional oil exports'

1. Meaning of Import Cover

Import cover is an indicator of a country's external sector strength. It measures the number of months for which imports can be financed using existing foreign exchange reserves. A higher import cover enhances a country's ability to withstand external shocks such as capital outflows, exchange rate volatility, and disruptions in international trade. **Hence, Statement 1 is correct.**

2. Impact of Rising Crude Oil Prices

India imports more than 85% of its crude oil requirements. An increase in global crude oil prices raises the country's import bill and demand for foreign exchange. If foreign exchange reserves remain unchanged, the same stock of reserves can finance imports for a shorter duration, leading to a decline in import cover. **Hence, Statement 2 is correct.**

3. India's Current Import Cover Position

As of March 2026, India's foreign exchange reserves stood at about $710 billion, sufficient to cover more than 12 months of imports. This provides a significant buffer against external sector shocks such as geopolitical conflicts, oil price spikes, and foreign capital outflows. **Hence, Statement 3 is correct.**

Value Addition

During the 1991 Balance of Payments Crisis, India's foreign exchange reserves had fallen to levels sufficient for only 2–3 weeks of imports, forcing the government to pledge gold and seek IMF assistance. The crisis underscored the importance of maintaining adequate forex reserves and import cover for macroeconomic stability.

152. Great Green Wall

Q. With reference to global initiatives aimed at combating desertification and land degradation, consider the following statements:

1. China's Three-North Protective Forest Program (Green Great Wall) was launched before the African Great Green Wall Initiative.
2. The African Great Green Wall Initiative is an African Union-led initiative supported by international organisations including UNCCD.
3. Both China's Green Great Wall and Africa's Great Green Wall primarily seek to restore degraded land and reduce the spread of desertification.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation:

In News: China's 'Great Green Wall' tames desert growth, but scientists say fight is not over

1. Timeline of the Two Initiatives

China's Three-North Protective Forest Program, popularly known as the 'Great Green Wall', was launched in 1978 to reduce desertification and sandstorms across northern China. The programme involves large-scale afforestation, shelterbelt plantations, and dune stabilization measures.

The Great Green Wall Initiative is an African-led initiative launched in 2007 by the African Union to combat desertification and land degradation across the Sahel region, stretching from Senegal in the west to Djibouti in the east. Hence, Statement 1 is correct.

2. Role of UNCCD in the African Great Green Wall

The Great Green Wall Initiative aims to restore degraded landscapes, enhance food security, improve livelihoods, and strengthen climate resilience across participating African countries. It is supported by

several international partners, including the United Nations Convention to Combat Desertification (UNCCD). UNCCD supports the initiative by facilitating coordination, knowledge sharing and resource mobilisation. Hence, Statement 2 is correct.

3. Common Objective of Both Initiatives

Despite differences in geography and implementation, both initiatives seek to address desertification and land degradation through ecological restoration. China's programme focuses primarily on afforestation and sandstorm control, while Africa's Great Green Wall combines land restoration with socio-economic objectives such as employment generation and sustainable development. Hence, Statement 3 is correct.

Value Addition

Feature	China's Green Great Wall	Africa's Great Green Wall
Launch Year	1978	2007
Region	Northern China	Sahel Region of Africa
Lead Institution	Government of China	African Union
UNCCD Role	Domestic Chinese programme	International supporting partner
Main Objective	Control desertification and sandstorms	Restore degraded land, improve livelihoods, and build climate resilience

Only the African Great Green Wall has a formal UNCCD support and coordination mechanism. China's Great Green Wall is a domestic Chinese programme.

Straw Checkerboards Technique (China)

- Straw checkerboards are a traditional sand-control technique widely used in China's Green Great Wall programme.
- Workers place straw or reed stalks in a criss-cross grid pattern on shifting dunes.
- The grids reduce wind speed at the ground surface, stabilize loose sand, and prevent dune movement.
- Saplings are then planted within the checkerboard squares, increasing the chances of vegetation establishment.
- The technique has been extensively used in the Gobi Desert and Kubuqi Desert to combat desertification.

153. Nuclear Non-Proliferation Treaty (NPT) & NSG

Q. Consider the following statements:

1. India is not a signatory to the Nuclear Non-Proliferation Treaty (NPT).
2. India follows a policy of No-First Use of nuclear weapons.
3. India is a member of the Nuclear Suppliers Group (NSG).

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation:

In News: "The Treaty on the Non-Proliferation of Nuclear Weapons institutionalised a hierarchy rather than dismantling it"

1. India's Status in the NPT

India is not a signatory to the Nuclear Non-Proliferation Treaty (NPT). India has consistently maintained that the treaty is discriminatory because it recognizes only five countries as Nuclear Weapon States and creates a distinction between nuclear "haves" and "have-nots". Hence, Statement 1 is correct.

2. No-First Use Policy

India's Nuclear Doctrine is based on the principle of No-First Use (NFU). Under this policy, India will not be the first to use nuclear weapons and will resort to them only in retaliation against a nuclear attack. India follows No-First Use policy, but reserves the right to retaliate massively against nuclear, chemical or biological attacks. Hence, Statement 2 is correct.

3. Membership of the NSG

India is not a member of the Nuclear Suppliers Group (NSG). Although India obtained an NSG waiver in 2008 that allows civilian nuclear commerce with NSG members, it is yet to secure full membership of the group. Hence, Statement 3 is incorrect.

NPT membership is not an explicit legal requirement for NSG membership, but several members consider it an important criterion. This

remains one of the major issues in India's NSG membership bid.

Additional Facts

Nuclear Non-Proliferation Treaty (NPT)

- Entered into force in **1970**.
- Based on three pillars:
  - **Non-Proliferation**
  - **Disarmament**
  - **Peaceful Use of Nuclear Energy**
- Recognizes only five Nuclear Weapon States:
  - United States
  - Russia
  - United Kingdom
  - France
  - China

Nuclear Suppliers Group (NSG)

- Established in **1974** following India's **Pokhran-I (Smiling Buddha)** nuclear test.
- Aims to prevent the proliferation of nuclear weapons through export controls on nuclear materials and technology.
- Functions on the principle of **consensus**.
- Currently consists of **48 member countries**.

Key Features of India's Nuclear Doctrine (2003)

- **No-First Use (NFU)**
- **Credible Minimum Deterrence**
- **Massive Retaliation** in response to a nuclear attack
- **Non-use against non-nuclear weapon states**
- Commitment to **universal, non-discriminatory nuclear disarmament**

154. TribeX Platform

Q. With reference to the TribeX Platform, recently seen in the news, consider the following statements:

1. It is a digital learning platform dedicated to tribal arts, culture, traditional knowledge and skill development.
2. It serves as a digital repository for preserving tribal languages, oral traditions and indigenous knowledge systems.
3. It hosts only UGC-recognized Post Graduate diploma programmes and does not offer certificate courses.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation:

In News: Union Tribal Affairs Minister Shri Jual Oram Launches TribeX

1. Digital Learning Platform

TribeX is a first-of-its-kind digital learning platform launched by the Ministry of Tribal Affairs to promote tribal arts, culture, traditional knowledge and skill development. It provides structured learning opportunities through digital courses. **Hence, Statement 1 is correct.**

2. Digital Repository for Tribal Heritage

Apart from being a learning platform, TribeX functions as a digital repository for documenting, preserving and promoting tribal languages, oral traditions, performing arts, cultural practices and indigenous knowledge systems. **Hence, Statement 2 is correct.**

3. Courses Offered

TribeX offers both **free certificate courses** and **UGC-recognized Post Graduate Diploma programmes**. It currently offers 20 certificate courses in areas such as painting, handicrafts, handloom, artefacts, and traditional musical instruments. **Hence, Statement 3 is incorrect.**

Additional Information

Key Features of TribeX

- Launched by: **Ministry of Tribal Affairs**
- Learning ecosystem for tribal knowledge and skills
- Offers:
  - Free Certificate Courses
  - UGC-recognized PG Diploma Programmes
- Integrates:
  - **Learning Management System (LMS)**
  - **Repository Management System**

Heritage Archive

- Contains **more than 5,000 multimedia resources**
- Includes: Audio recordings, Videos, Literature, Documentation of tribal festivals, Oral traditions and socio-cultural practices
- Planned expansion to **10,000 resources**

Tribal Research Institutes (TRIs)

- Nodal institutions for research, documentation and preservation of tribal culture and traditions.
- Support evidence-based tribal policymaking and the implementation of tribal welfare programmes.

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● TribeX was launched during the **National Workshop on Strengthening Tribal Research Institutes (TRIs)**.

155. Alzheimer's Disease & Dementia

Q. With reference to Dementia, recently seen in the news, consider the following statements:

1. Alzheimer's disease is the most common cause of dementia.
2. Dementia is a normal consequence of ageing and affects all elderly persons.
3. According to the World Health Organization, a significant proportion of dementia risk can be prevented or delayed by addressing modifiable risk factors.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (c) 1 and 3 only
Explanation:
In News: Dementia risk that could be prevented or delayed
1. Alzheimer's Disease and Dementia
Alzheimer's disease is the leading cause of dementia worldwide, accounting for a majority of dementia cases. **Hence, Statement 1 is correct.**
2. Dementia and Ageing
Dementia is **not a normal part of ageing**. It is a clinical syndrome caused by diseases or injuries affecting the brain and results in deterioration of memory, thinking and daily functioning. **Hence, Statement 2 is incorrect.**
3. Preventable Risk Factors
The World Health Organization has stated that up to **45% of dementia risk could be prevented or delayed** by addressing modifiable risk factors such as tobacco use, air pollution, physical inactivity and unhealthy lifestyles. **Hence, Statement 3 is correct.**
Value Addition:
Dementia: Global Burden
● Dementia is a clinical syndrome characterized by deterioration in memory, thinking, behaviour and the ability to perform everyday activities.
● It is the seventh leading cause of death globally, according to the World Health Organization (WHO).
● Alzheimer's disease is the most common cause of dementia, accounting for about 60–80% of cases.
Preventable Risk Factors
According to the WHO (2026):
● Up to 45% of dementia risk could be prevented or delayed by addressing modifiable risk factors.
● Important risk factors include:
○ Tobacco use
○ Air pollution
○ Physical inactivity
○ Unhealthy diet
○ Excessive alcohol consumption
○ Hypertension and other cardiovascular diseases
○ Social isolation
Importance of Prevention
WHO emphasizes that even delaying the onset of dementia can substantially reduce its overall burden on individuals, families, and healthcare systems. Consequently, public health measures focusing on healthy lifestyles and environmental risk reduction play a crucial role in combating dementia.
Source:https://epaper.thehindu.com/ccidist-ws/th/th_international/issues/194466/OPS/GQUG7JGS7.1+G6LG8MTOQ.1.html

156. India's First Hydrogen Train

Q. With reference to the Hydrogen Fuel Cell-Powered Train, recently seen in the news, consider the following statements:

1. Hydrogen fuel cells generate electricity by combining hydrogen with oxygen.
2. Water is the only direct emission/by-product during operation.
3. Unlike conventional electric trains, hydrogen trains do not require continuous overhead electrification infrastructure.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
Explanation:
In News: PM to flag off India's first Hydrogen train between Jind and Sonipat
1. Generation of Electricity through Fuel Cells
Hydrogen trains operate using hydrogen fuel cell technology. In the fuel cell, hydrogen stored onboard

reacts electrochemically with oxygen from the atmosphere to produce electricity. This electricity powers the train's traction motors. Unlike diesel locomotives, no combustion takes place during this process. **Hence, Statement 1 is correct.**



2. Zero-Emission Technology

The electrochemical reaction between hydrogen and oxygen produces electricity, heat(not an emission), and water vapour. Since no fossil fuel is burned, there are no emissions of carbon dioxide, sulphur oxides, or particulate matter during operation. Therefore, hydrogen trains are considered a clean transportation technology. **Hence, Statement 2 is correct.**

3. No Need for Continuous Electrification

Conventional electric trains require electricity to be supplied continuously through overhead wires. In contrast, hydrogen trains generate electricity onboard through fuel cells and can therefore operate without continuous overhead electrification infrastructure. This makes them particularly suitable for routes where electrification is expensive or difficult. **Hence, Statement 3 is correct.**

Additional Information

India's First Hydrogen Train

- Marks India's entry into the group of countries operating hydrogen-powered trains.
- Developed using indigenous engineering and integration capabilities.
- Features a 10-coach configuration (8 passenger coaches and 2 power cars).
- Among the world's longest hydrogen-powered passenger trainsets.
- Equipped with a 3,200 HP (2400 kW) propulsion system.

How Does the Train Work?

Hydrogen Storage

- Hydrogen is stored onboard under high pressure.
- India's hydrogen train carries about 440 kg of hydrogen.

Fuel Cell Operation

- Hydrogen from storage tanks combines with oxygen from the air.
- The reaction generates electricity to power the train.
- Excess electricity is stored in lithium ferro-phosphate batteries.

Hybrid Power System

- Fuel cells provide continuous power.
- Batteries supply additional power during acceleration and peak demand.
- Batteries recharge when power demand falls.

Challenges

- Hydrogen Storage: Hydrogen is highly flammable and must be stored under very high pressure, making storage and transportation technologically challenging.
- Production Constraint: Large-scale production of green hydrogen remains costly and energy-intensive.
- Infrastructure Requirement: Dedicated hydrogen production, storage, transportation, and refuelling facilities are required.

Hydrogen Fuel Cell vs Conventional Electric Train

Feature	**Hydrogen Train**	**Conventional Electric Train**
Source of Power	Hydrogen fuel cells	Overhead electric lines
Emissions	Water vapour only	Depends on electricity source
Electrification Needed	No	Yes
Suitable for Remote Routes	High	Limited by electrification
Dependence on Fossil Fuel Imports	Lower	Depends on electricity mix

National Green Hydrogen Mission

- Launched in 2023.
- Nodal Ministry: Ministry of New and Renewable Energy (MNRE).
- Objective: Make India a global hub for the production, utilization and export of Green Hydrogen.
- Target: Production of 5 million metric tonnes of Green Hydrogen annually by 2030.

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157. Semaglutide

Q. With reference to Semaglutide, consider the following statements:

1. It belongs to the class of GLP-1 receptor agonists.
2. It is used in the treatment of Type 2 diabetes mellitus.
3. It promotes weight loss by slowing gastric emptying and reducing appetite.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

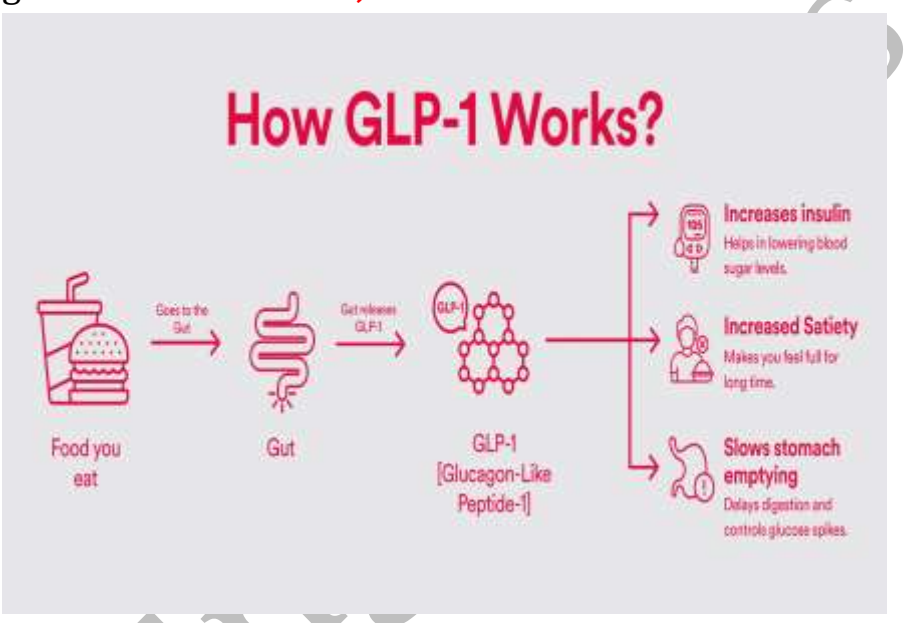
Answer: (d) 1, 2 and 3

Explanation:

In News: Semaglutide: inflection point for Indian generics?

1. GLP-1 Receptor Agonist

Semaglutide belongs to a class of drugs known as **Glucagon-Like Peptide-1 (GLP-1) receptor agonists**. GLP-1 is a naturally occurring hormone released by the intestine after food intake. These drugs mimic its action and help regulate blood glucose levels. **Hence, Statement 1 is correct.**



2. Treatment of Type 2 Diabetes Mellitus

Semaglutide is primarily prescribed for the treatment of **Type 2 Diabetes Mellitus**. It increases insulin secretion when blood glucose levels are elevated and reduces glucose production by the liver, thereby improving glycaemic control. **Hence, Statement 2 is correct.**

3. Role in Weight Management

Semaglutide slows the emptying of food from the stomach (gastric emptying), prolonging the feeling of fullness and reducing appetite. This helps lower calorie intake and supports weight loss. **Hence, Statement 3 is correct.**

What is GLP-1?

GLP-1 (Glucagon-Like Peptide-1) is a hormone released from the intestine after meals. It:

- Stimulates insulin secretion.
- Suppresses glucagon release.
- Slows gastric emptying.
- Promotes satiety (feeling of fullness).
- Helps regulate blood sugar levels.

Semaglutide

- **Active ingredient in Ozempic and Wegovy.**
- Used for:
  - Type 2 diabetes management.
  - Weight loss and obesity treatment.
  - Reduction of cardiovascular risks in certain patients.

An **API (Active Pharmaceutical Ingredient)** is the biologically active component of a medicine that produces the intended therapeutic effect.

Example: In Ozempic and Wegovy, **Semaglutide itself is the API.**

Value Addition

Examples of GLP-1 Receptor Agonists:

- Semaglutide
- Liraglutide
- Dulaglutide

News Context: Recently, Dr. Reddy's Laboratories launched a generic version of Semaglutide in Canada after patent expiries opened opportunities for generic manufacturers.

158. SEMICON 2.0

Q. Consider the following statements regarding the India Semiconductor Mission (ISM) 2.0 / Semicon 2.0 Programme:

1. It aims to strengthen India's semiconductor design and manufacturing ecosystem.
2. The programme focuses only on semiconductor fabrication units (fabs) and excludes semiconductor design and research-related activities.
3. It seeks to strengthen the ATMP-OSAT industry, which is involved in semiconductor assembly, packaging and testing.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (b) 1 and 3 only
Explanation:
In News: Govt approves Semicon 2.0 to strengthen semiconductor ecosystem
1. India Semiconductor Mission (ISM) 2.0 / Semicon 2.0
The **India Semiconductor Mission (ISM) 2.0**, also known as **Semicon 2.0**, aims to strengthen India's semiconductor ecosystem by promoting **semiconductor design, manufacturing, packaging, testing, equipment and research & development (R&D)**.
The programme seeks to position India as a reliable and competitive player in the global semiconductor value chain and reduce dependence on imports for critical technologies. **Hence, Statement 1 is correct.**
2. Focus Areas of Semicon 2.0
The programme does **not focus only on semiconductor fabrication units (fabs)**. Semiconductor manufacturing involves multiple stages, including:
- **Design:** Developing chip architecture and semiconductor intellectual property (IP).
- **Fabrication (Fabs):** Manufacturing integrated circuits (ICs) on silicon wafers.
- **Machines and Materials:** Developing semiconductor equipment, chemicals and gases required for chip production.
- **ATMP-OSAT:** Assembly, packaging and testing of semiconductor chips.
- **Research and Development:** Promoting innovation and advanced semiconductor technologies.

Since Semicon 2.0 follows a holistic approach covering the entire semiconductor ecosystem, it does not exclude design or R&D activities. **Hence, Statement 2 is incorrect.**
3. ATMP-OSAT Industry
ATMP-OSAT (Assembly, Testing, Marking and Packaging – Outsourced Semiconductor Assembly and Test) represents the backend segment of semiconductor manufacturing.
These facilities are involved in:
- **Assembly** of semiconductor dies into packages,
- **Packaging** to protect chips and enable their use in electronic devices,
- **Testing** to ensure chip functionality and quality.

Strengthening the ATMP-OSAT industry is an important component of Semicon 2.0 as India aims to build a complete semiconductor ecosystem.
Hence, Statement 3 is correct.
Additional Information
Semiconductor Value Chain
Design → Fabrication (Fab) → Assembly, Packaging & Testing → Electronic Devices
Key Focus Areas of Semicon 2.0
- Semiconductor design and indigenous chip development
- Semiconductor fabrication units (fabs)
- Semiconductor equipment and materials
- ATMP-OSAT facilities
- Research and development
- Skill development in semiconductor technology

Importance of Semiconductor Self-Reliance
- Reduces dependence on global supply chains
- Enhances technological security
- Supports critical sectors such as defence, telecom, automobiles and electronics
- Promotes India's role in the global semiconductor market

159. Kudankulam Nuclear Power Plant

Q. Consider the following statements regarding nuclear power in India:

1. Kudankulam Nuclear Power Plant is located in Tamil Nadu and uses Russian-designed VVER reactors for electricity generation.
2. India has set a long-term target of achieving 100 GW of nuclear power capacity by 2047 under the Nuclear Energy Mission.
3. The Prototype Fast Breeder Reactor (PFBR) at Kalpakkam is developed by the Nuclear Power Corporation of India Limited (NPCIL).

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only
Explanation:
In News: Leak of Kudankulam nuclear plant data sparks 'commotion'

1. Kudankulam Nuclear Power Plant

The Kudankulam Nuclear Power Plant (KKNPP) is located in Tirunelveli district of Tamil Nadu. It was developed through India-Russia cooperation and uses Russian-designed VVER-1000 reactors for electricity generation.

VVER (Water-Water Energetic Reactor) is a type of Pressurised Water Reactor (PWR) developed by Russia. **Hence, Statement 1 is correct.**

2. Nuclear Energy Mission

The Government of India has set a target of achieving 100 GW of nuclear power capacity by 2047 under the Nuclear Energy Mission.

The mission focuses on:

- Expansion of nuclear power capacity
- Development of indigenous nuclear technologies
- Promotion of Small Modular Reactors (SMRs)
- Strengthening India's energy security

Nuclear energy is considered important because it provides reliable baseload electricity with very low carbon emissions. **Hence, Statement 2 is correct.**

3. Prototype Fast Breeder Reactor (PFBR)

The 500 MWe Prototype Fast Breeder Reactor (PFBR) at Kalpakkam, Tamil Nadu, has been developed by Bharatiya Nabhikiya Vidyut Nigam Limited (BHAVINI), not NPCIL.

BHAVINI was established to develop and operate Fast Breeder Reactors (FBRs) as part of India's three-stage nuclear power programme. **Hence, Statement 3 is incorrect.**

Important Organisations:

- **NPCIL** → Operates nuclear power plants in India
- **CERT-In** → National agency for cybersecurity incident response
- **DAE (Department of Atomic Energy)** → Overall nuclear energy administration in India

Additional Information

Atomic Energy vs Nuclear Energy

The terms **atomic energy** and **nuclear energy** are often used interchangeably, but there is a slight difference.

Atomic Energy

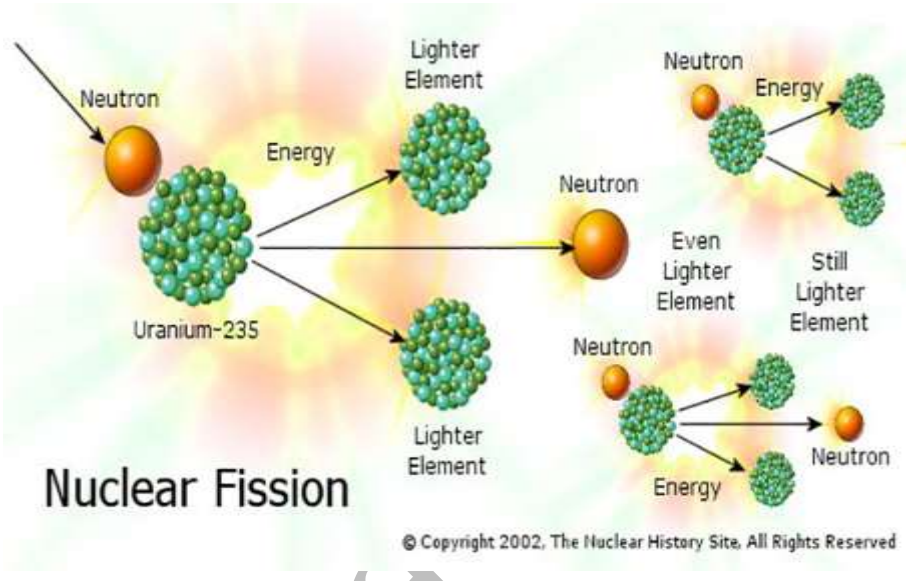
- Refers broadly to energy associated with **atoms**.
- It includes energy released from changes in the atom, especially the nucleus.

Nuclear Energy

- A specific type of atomic energy obtained from the **nucleus of atoms**.
- It is produced through:
  - **Nuclear fission** → Splitting of heavy atoms (Uranium-235, Plutonium-239) to release energy.
  - **Nuclear fusion** → Combining light nuclei (future energy technology).

India's nuclear power plants mainly use nuclear fission.

Example: Uranium nucleus → Fission → Heat → Steam → Turbine → Electricity



Nuclear Power in India

- Nuclear power is the **fifth largest source of electricity generation in India** after thermal, renewable, hydro and other sources.
- India has around **24 operational nuclear reactors** with an installed capacity of about **8.78 GW**.
- Nuclear power contributes around **3% of India's electricity generation**.
- It provides continuous electricity supply as a **baseload source**.

India's Three-Stage Nuclear Power Programme

Designed by **Dr. Homi J. Bhabha** to utilise India's limited uranium resources and abundant thorium reserves.

Stage 1: Pressurised Heavy Water Reactors (PHWRs)

- Fuel: Natural uranium
- Produces plutonium as a by-product

Stage 2: Fast Breeder Reactors (FBRs)

- Uses plutonium fuel
- Produces more fissile material than it consumes

Example: **PFBR, Kalpakkam**

Stage 3: Thorium-based Reactors

- Utilises India's large thorium reserves
- Aims for long-term energy security

160. National Investment Policy for Urea

Q. Consider the following statements regarding the National Investment Policy for Urea-2026 for Atmanirbhar Bharat (NIPU-2026):

1. It aims to promote new investments in gas-based urea manufacturing units in India.

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2. It seeks to reduce India's dependence on imported urea by increasing domestic production capacity.
3. It is a subsidy programme aimed at reducing the retail price of urea for farmers through changes in the fertilizer subsidy mechanism.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only
Explanation:

In News: Cabinet approves National Investment Policy for Urea-2026 for Atmanirbhar Bharat (NIPU-2026)

1. Promotion of Gas-Based Urea Manufacturing Units

The **National Investment Policy for Urea-2026 (NIPU-2026)** aims to encourage fresh investments in the urea sector by setting up **new gas-based urea manufacturing units**. The policy focuses on expanding domestic production capacity and strengthening fertilizer security. **Hence, Statement 1 is correct.**

2. Reducing Import Dependence

India has historically depended on imports to bridge the gap between domestic urea production and demand. NIPU-2026 aims to increase indigenous production and move towards **self-sufficiency in urea production**.

Increasing domestic manufacturing will help reduce reliance on imported urea and improve fertilizer availability. **Hence, Statement 2 is correct.**

3. Nature of NIPU-2026

NIPU-2026 is **not a subsidy programme**. It is an **investment policy** aimed at encouraging the establishment of new urea manufacturing capacity.

It does not:

- Change the **Maximum Retail Price (MRP)** of urea.
- Modify the **Direct Benefit Transfer (DBT)** mechanism.
- Alter the existing fertilizer subsidy framework.

Hence, Statement 3 is incorrect.

Additional Information

Urea Fertilizer

- Urea is a **nitrogenous fertilizer**.
- It contains around **46% nitrogen**.
- It promotes vegetative growth in crops.
- It is the most widely used fertilizer in India.

Progress of India's Urea Industry

- Domestic urea production increased from around **22 million tonnes (2011-12)** to around **31.4 million tonnes (2023-24)**.
- Six new urea plants have contributed significantly to increasing production.
- New plants are more energy-efficient compared to older units.

Challenges

- Gap between domestic production and demand.
- Dependence on imported urea.
- Excessive use of urea due to low prices leading to nutrient imbalance.

161. CAMPA

Q.With reference to the Compensatory Afforestation Fund Management and Planning Authority (CAMPA), consider the following statements:

1. The National Compensatory Afforestation Fund is maintained under the Public Account of India.
2. CAMPA was established by the Supreme Court to monitor and evaluate compensatory afforestation activities.
3. Under the Compensatory Afforestation Fund Act, 2016, the entire fund collected is retained by the Central Government.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two
EXPLANATION:
In NEWS: While compensatory afforestation is suitable for a developing economy to maintain sustainability, one of its main drawbacks is that plantations are inherently different from natural ecosystems - HINDU
Statement 1:
The Compensatory Afforestation Fund (CAF) Act, 2016 created:

- National Compensatory Afforestation Fund (NCAF) under the Public Account of India.

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- State Compensatory Afforestation Funds (SCAF) under the Public Account of the respective States.

The money collected from user agencies for diversion of forest land is deposited in these funds and utilized for afforestation and forest conservation activities.**Hence, statement 1 is correct.**

Statement 2:

The Supreme Court established CAMPA in 2002 as an ad hoc body to:

- Monitor compensatory afforestation activities.
- Provide technical assistance.
- Evaluate afforestation projects.
- Administer the funds collected for compensatory afforestation.

CAMPA was created because large amounts of money collected for forest diversion were lying unutilized.**Hence, statement 2 is correct.**

Statement 3:

The CAF Act, 2016 does not allow the Central Government to retain the entire fund.Instead:

- 90% of the funds are transferred to the respective State Compensatory Afforestation Funds.
- 10% remains in the National Compensatory Afforestation Fund.**Hence, statement 3 is incorrect.**

Value Addition

- Established: CAMPA (2002) by Supreme Court
- CAF Act: 2016, Rules Notified: 2018
- Fund Distribution: States – 90% ,Centre – 10%
- CAF funds are non-lapsable.
- Accounts: National CAF → Public Account of India, State CAF → Public Account of respective States
- Compensatory Afforestation:When forest land is diverted for non-forest purposes (mining, industries, roads, dams etc.), the user agency must: Provide equivalent non-forest land for afforestation, or Provide twice the area of degraded forest land if non-forest land is unavailable.
- Funds can be used for:Afforestation and reforestation,Wildlife protection and management, Catchment area treatment,Forest management,Human-wildlife conflict mitigation,Relocation of villages from protected areas

162. Cyclosporiasis

Q. Cyclosporiasis is a disease caused by:

(a) A virus transmitted by mosquitoes
(b) A bacterium contaminating dairy products
(c) A microscopic parasite spread through contaminated food and water
(d) A fungus affecting the respiratory tract

Answer: (c) A microscopic parasite spread through contaminated food and water

EXPLANATION

IN NEWS: What is cyclosporiasis? | Explained - HINDU

Option (a): A virus transmitted by mosquitoes

Cyclosporiasis is not caused by a virus. It is also not a vector-borne disease transmitted by mosquitoes.Diseases such as Dengue, Zika and Chikungunya are mosquito-borne viral diseases. **Hence, Option (a) is incorrect.**

Option (b): A bacterium contaminating dairy products

Cyclosporiasis is not caused by bacteria. The causative organism is a protozoan parasite.Food-borne bacterial diseases include:

- Cholera (*Vibrio cholerae*)
- Typhoid (*Salmonella typhi*) **Hence, Option (b) is incorrect.**

Option (c): A microscopic parasite spread through contaminated food and water

Cyclosporiasis is caused by Cyclospora cayetanensis, a microscopic protozoan parasite. Transmission occurs through the faeco-oral route, mainly by consuming contaminated:

- Water
- Fruits
- Vegetables
- Leafy greens

The disease commonly causes:

- Explosive watery diarrhoea
- Abdominal cramps
- Bloating
- Weight loss
- Fatigue **Hence, Option (c) is correct.**

Option (d): A fungus affecting the respiratory tract

Cyclosporiasis is neither fungal nor a respiratory disease.It primarily affects the gastrointestinal tract. **Hence, Option (d) is incorrect.**

VALUE ADDITION

Cyclosporiasis: Quick Facts

Feature	**Details**
Disease	Cyclosporiasis
Causative Agent	*Cyclospora cayetanensis*
Type of Organism	Protozoan Parasite
Transmission	Faeco-oral route
Main Symptoms	Watery diarrhoea, cramps, fatigue
Diagnosis	Stool Microscopy, PCR
Treatment	Trimethoprim-Sulfamethoxazole (TMP-SMX)
Human-to-Human Spread	Rare/Not common

Cyclospora vs Cholera

Feature	**Cyclospora**	**Cholera**
Organism	Protozoan Parasite	Bacterium
Disease	Cyclosporiasis	Cholera
Transmission	Contaminated food/water	Contaminated food/water
Main Symptom	Prolonged watery diarrhoea	Rice-water stools
Treatment	TMP-SMX	ORS + Antibiotics

163. REWARD Programme

Q.With reference to the Rejuvenating Watersheds for Agricultural Resilience through Innovative Development (REWARD) Programme, consider the following statements:

1. The programme primarily aims to strengthen watershed management for climate resilience and livelihood enhancement.
2. It is assisted by the World Bank and operates on the Programme-for-Results (PforR) financing instrument.
3. It is implemented by the Ministry of Agriculture and Farmers Welfare.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer:(a) 1 and 2 only

EXPLANATION :

IN NEWS: National Rainfed Area Authority Organizes Brainstorming Workshop on Satellite and Drone-based Survey for Watershed Planning and Monitoring under REWARD Programme - PIB

Statement 1:

The objective of REWARD is to strengthen the capacity of national and state institutions for improved watershed management, enhance farmers' resilience to climate change, and support livelihood generation through value chains.The programme focuses on:

- Climate-resilient watershed development
- Sustainable livelihoods
- Agricultural resilience **Hence, Statement 1 is correct.**

Statement 2:

REWARD is a **World Bank-assisted programme**.

It operates through the **Programme-for-Results (PforR)** financing mechanism, under which World Bank disbursements are linked to the achievement of pre-defined **Disbursement Linked Indicators (DLIs). Hence, Statement 2 is correct.**

Statement 3:

The programme is implemented by the **Department of Land Resources (DoLR)** under the **Ministry of Rural Development**.It is **not** implemented by the Ministry of Agriculture and Farmers Welfare. **Hence, Statement 3 is incorrect.**

VALUE ADDITION

REWARD Programme: Quick Facts

Feature	**Details**
Full Form	Rejuvenating Watersheds for Agricultural Resilience through Innovative Development
Duration	2021–2026
Assistance	World Bank
Implementing Agency	Department of Land Resources (DoLR)

Ministry	Ministry of Rural Development
States Covered	Karnataka and Odisha
Knowledge Partner	National Rainfed Area Authority (NRAA)
Financing Model	Programme-for-Results (PforR)

PforR (Programme-for-Results)

Feature	Details
Institution	World Bank
Basis of Funding	Achievement of results
Tool Used	Disbursement Linked Indicators (DLIs)
Focus	Outcomes rather than expenditure

National Rainfed Area Authority (NRAA)

Feature	Details
Full Name	National Rainfed Area Authority (NRAA)
Established	2006
Ministry	Ministry of Agriculture & Farmers Welfare
Nature	Advisory and Policy Support Body
Headquarters	New Delhi
Chairman	Appointed by Government of India
Primary Focus	Sustainable development of rainfed and drought-prone areas
Key Areas	Watershed Development, Natural Resource Management, Climate Resilience
Current Role in REWARD	Knowledge Partner to Department of Land Resources (DoLR)
Major Objective	Improve productivity and livelihoods in rainfed regions through integrated watershed management

164. Mobile Phone Manufacturing Scheme (MPMS)

Q.With reference to the Mobile Phone Manufacturing Scheme (MPMS) and India's mobile phone manufacturing sector, consider the following statements:

1. India is the world's second-largest mobile phone manufacturer by volume.
2. The Mobile Phone Manufacturing Scheme (MPMS) provides incentives linked to eligible sales of mobile phones manufactured in India.
3. Under MPMS, incentives are available only for exports and not for domestic sales.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

EXPLANATION

In NEWS: Cabinet approves Mobile Phone Manufacturing Scheme (MPMS) - PIB

Statement 1:

India has emerged as the world's second-largest mobile phone manufacturer by volume.Currently, about 99.2% of mobile phones used in India are manufactured domestically, reflecting the success of initiatives such as Make in India and PLI-LSEM.**Hence, Statement 1 is correct.**

Statement 2:

MPMS provides incentive support on eligible sales of mobile phones manufactured in India.The incentive rates range from 2.25% to 5%, with additional incentives for domestic sourcing and R&D activities.**Hence, Statement 2 is correct.**

Statement 3:

MPMS incentives are not restricted to exports alone.The scheme provides incentives based on eligible sales, which include production and sales from manufacturing activities in India. The objective is to promote both domestic manufacturing and exports.**Hence, Statement 3 is incorrect.**

VALUE ADDITION

Mobile Phone Manufacturing Scheme (MPMS): Quick Facts

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Feature	Details
Launch Year	2026
Outlay	₹62,500 crore
Duration	FY 2026-27 to FY 2030-31
Objective	Scale up production and domestic value addition
Incentive Rate	2.25%–5% on eligible sales
Additional Incentive	Up to 1.5% for domestic sourcing
R&D Incentive	Additional 3% for Indian brands

Additional Incentives under MPMS

Component	Incentive
Eligible Sales	2.25%–5%
Domestic Sourcing of Components	Up to 1.5%
Design & R&D for Indian Brands	3%

India's Mobile Manufacturing Sector

Indicator	Status
Global Rank	2nd Largest by Volume
Domestic Manufacturing Share	99.2% of phones used in India
Largest Export Category (2025)	Smartphones
Previous Scheme	The Production Linked Incentive Scheme for Large Scale Electronics Manufacturing (PLI-LSEM)
PLI-LSEM Tenure Ended	31 March 2026

165. Gandak River

Q.With reference to the Gandak River, consider the following statements:

1. It is a right-bank tributary of the Ganga.
2. It enters India at Triveni near the Indo–Nepal border.
3. It forms one of the deepest river gorges in the world while flowing through Nepal.

Which of the statements given above are correct?

(a) 2 and 3 only
(b) 1 and 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 2 and 3 only

EXPLANATION

In NEWS: Gandak River crosses danger mark in Bihar amid rising water levels-AIR

Statement 1:

The Gandak is one of the major rivers of North India and an important left-bank tributary of the Ganga River.It joins the Ganga near Hajipur in Bihar.**Hence, Statement 1 is incorrect.**

Statement 2:

The river enters India near Valmikinagar in Bihar.Its entry point at the Indo–Nepal border is known as Triveni.After entering India, it flows mainly through Bihar.**Hence, Statement 2 is correct.**

Statement 3:

While flowing through the Nepal Himalayas, the river forms the Kali Gandaki Gorge, which is regarded as one of the deepest river gorges in the world.The gorge lies between the Dhaulagiri and Annapurna mountain ranges.**Hence, Statement 3 is correct.**

VALUE ADDITION

Gandak River

- One of the major left-bank tributaries of the Ganga.
- Mentioned in the Mahabharata.
- Originates near Dhaulagiri Mountain in the Himalayas.
- Formed by the confluence of Kali River and Trisuli River in Nepal.
- Known as Kali Gandaki and Narayani in Nepal.
- Enters India at Triveni near Valmikinagar (Bihar).
- Joins the Ganga near Hajipur, opposite Patna.
- Total length: ~630 km.
- The basin lies between Kosi Basin (East) and Karnali/Ghaghara Basin (West).

Kali Gandaki Gorge

- One of the deepest river gorges in the world.
- Located in Nepal.
- Lies between Dhaulagiri and Annapurna Massif.
- Formed by the Kali Gandaki River.

Important Tributaries

- Budhi Gandaki

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- Marsyandi
- Seti
- Madi
- Daraudi

Basin Features

- Contains about 1,710 glaciers.
- Has 300+ lakes in the upper catchment.
- Includes three peaks above 8,000 m:
  - Dhaulagiri
  - Manaslu
  - Annapurna

Protected Areas

- Chitwan National Park (Nepal)
- Valmiki Tiger Reserve (India)



166. Windfall Tax

Q.With reference to Windfall Tax, consider the following statements:

1. It is imposed on extraordinary profits arising from unforeseen external events.
2. It is generally levied on revenue earned by a company rather than its profits.
3. Oil and gas companies are among the sectors commonly subjected to windfall taxes.

Which of the statements given above are correct?

(a) 1 and 3 only
(b) 2 and 3 only
(c) 1 and 2 only
(d) 1, 2 and 3

Answer: (a) 1 and 3 only

EXPLANATION

IN NEWS: Windfall tax on diesel, ATF up, levy on petrol exports cut- HINDU

Statement 1:

Windfall tax is imposed on extraordinary or unexpected profits earned due to external factors beyond the control of a company.Such profits may arise because of:

- Wars
- Geopolitical tensions
- Commodity shortages
- Pandemics
- Sudden policy changes

These profits are not the result of additional effort, investment or innovation by the company.**Hence, Statement 1 is correct.**

Statement 2:

Windfall tax is generally imposed on extraordinary profits or gains, not merely on total revenue earned by a company.Revenue represents total receipts, whereas windfall tax seeks to capture a portion of the abnormal profits generated due to favourable external circumstances.**Hence, Statement 2 is incorrect.**

Statement 3:

Oil and gas companies are among the most common targets of windfall taxation.Sharp increases in global crude oil and natural gas prices often result in unusually high profits for these companies, prompting governments to levy windfall taxes.**Hence, Statement 3 is correct.**

VALUE ADDITION

Windfall Tax

- A tax imposed on unexpected or extraordinary profits.
- Levied when gains arise from external events rather than business decisions.
- Commonly associated with:
  - Oil sector
  - Gas sector
  - Mining sector
- Used by governments as an additional revenue source.
- Based on the principle of sharing extraordinary gains with society.

Conditions Leading to Windfall Profits

- Wars and geopolitical conflicts
- Supply chain disruptions
- Commodity price spikes
- Pandemics
- Government policy changes

Objectives

- Mobilize additional government revenue.
- Redistribute a portion of extraordinary gains.
- Fund welfare and developmental expenditure.
- Prevent excessive profiteering during crises.

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Countries that have imposed Windfall Taxes

- India
- United Kingdom
- Italy
- Spain
- Hungary

167. Jagannath Temple

Q.With reference to the architecture of the Jagannath Temple in Puri, consider the following statements:

1. It is designed in the traditional Dravidian style of temple architecture, characterized by a stepped pyramid-shaped vimana.
2. The temple complex is historically referred to as the "White Pagoda", while the Sun Temple of Konark is known as the "Black Pagoda".
3. The main temple layout consists of four distinct sequential structures: the Vimana, the Jagamohana, the Nata Mandira, and the Bhoga Mandapa.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (b) 2 and 3 only

EXPLANATION:

IN NEWS:Puri resonating with unparalleled spiritual fervour as Rath Yatra commences - AIR

Statement 1:

The Jagannath Temple is **not built in the Dravidian style**.It is constructed in the **Kalinga style of architecture**, a regional variant of the Nagara style that flourished in present-day Odisha.

Key features of Kalinga architecture include:

- Curvilinear Shikhara (Rekha Deula)
- Rich sculptural ornamentation
- Elaborate carvings
- Distinct temple components arranged along a common axis

The stepped pyramidal Vimana is a characteristic feature of **Dravidian architecture**, seen prominently in South Indian temples.**Hence, Statement 1 is incorrect.**

Statement 2:

The Jagannath Temple has historically been referred to as the **"White Pagoda"** by European sailors.Similarly, the **Konark Sun Temple** was known as the **"Black Pagoda"** because of its dark appearance and its use as a navigational landmark along the Odisha coast.**Hence, Statement 2 is correct.**

Statement 3:

The main temple complex consists of four principal structures aligned in sequence:

- **Vimana (Deula)** – Sanctum sanctorum housing the deities
- **Jagamohana** – Assembly/prayer hall
- **Nata Mandira** – Dance and festival hall
- **Bhoga Mandapa** – Hall for offerings and rituals

This arrangement is a defining feature of mature Kalinga temple architecture.**Hence, Statement 3 is correct.**

VALUE ADDITION

Key Facts about Jagannath Rath Yatra

- Held annually at **Puri, Odisha**.
- Begins on **Ashadha Shukla Dwitiya**.
- Celebrates the journey of **Lord Jagannath, Lord Balabhadra and Goddess Subhadra** from Jagannath Temple to Gundicha Temple.
- One of the **oldest and largest chariot festivals** in the world.
- Considered an important **Vaishnavite tradition**.
- Chariots are **constructed afresh every year** using traditional methods.

Important Rituals (Chronology)

Snana Purnima → Anavasara → Nava Jaubana Darshan → Pahandi Bije → Chhera Pahanra → Rath Yatra → Hera Panchami → Bahuda Yatra → Suna Besha → Niladri Bijay

Three Chariots

Deity	Chariot
Lord Jagannath	Nandighosha
Lord Balabhadra	Taladhwaja
Goddess Subhadra	Darpadalana (Devadalana)

Cultural Significance

- Symbol of **equality and inclusiveness**.
- Lord Jagannath comes out of the temple to bless all devotees irrespective of caste or social status.
- Reflects the synthesis of **tribal traditions, Vaishnavism, Shaivism and Shaktism**.

Associated Temple Facts

- Jagannath Temple is built in **Kalinga Architecture**.
- Known as the **White Pagoda**.

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- One of the **Char Dham** pilgrimage sites.
- Built by **Anantavarman Chodaganga Deva**.
- **Nilachakra** atop the temple is made of **Ashtadhatu**.





168. BIMSTEC

Q.With reference to BIMSTEC, consider the following statements:

1. All member states of BIMSTEC share a direct maritime coastline with the Bay of Bengal.
2. The first-ever BIMSTEC Traditional Music Festival, titled 'Sapt-Sur', was inaugurated in 2025 at Bharat Mandapam in New Delhi.
3. The Permanent Secretariat of the organization is located in Bangkok, Thailand, where the founding declaration was signed.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (a) Only one

EXPLANATION:

In NEWS: India calls for stronger BIMSTEC cooperation on security and economy- AIR

Statement 1:

BIMSTEC comprises seven countries:

- Bangladesh
- Bhutan
- India
- Myanmar
- Nepal
- Sri Lanka
- Thailand

Not all member states have a coastline along the Bay of Bengal.Bhutan and Nepal are landlocked countries and do not possess any maritime coastline.**Hence, Statement 1 is incorrect.**

Statement 2:

The first-ever BIMSTEC Traditional Music Festival, known as 'Sapt-Sur', was inaugurated at Bharat Mandapam, New Delhi.The event aimed to showcase the rich musical traditions of BIMSTEC countries and strengthen people-to-people ties.**Hence, Statement 2 is correct.**

Statement 3:

Although BIMSTEC was established through the Bangkok Declaration (1997), its Permanent Secretariat is located in Dhaka, Bangladesh, not Bangkok.The Secretariat became operational in 2014.**Hence, Statement 3 is incorrect.**

Value Addition

BIMSTEC at a Glance

Feature	**Details**
Full Form	Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation
Established	1997
Founding Declaration	Bangkok Declaration
Secretariat	Dhaka
Member Countries	7
Population	About 1.7 billion people,accounts for nearly **22% of the world's population**.

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Combined GDP	About USD 5.2 trillion

BIMSTEC Members

South Asia	**Southeast Asia**
Bangladesh	Myanmar
Bhutan	Thailand
India	
Nepal	
Sri Lanka	

Key Highlights of the 6th BIMSTEC Summit (2025)

- Adoption of Bangkok Vision 2030
- Proposal for BODHI Programme
- BIMSTEC Centres of Excellence in India
- Promotion of Digital Public Infrastructure (DPI)
- Proposal to link UPI with BIMSTEC payment systems
- BIMSTEC Athletics Meet (2025)
- First BIMSTEC Games (2027)

BIMSTEC - India

- Supports Act East Policy
- Complements Neighbourhood First Policy
- Serves as an alternative platform to the stalled South Asian Association for Regional Cooperation
- Strengthens India's role in the Bay of Bengal and Indo-Pacific region
- Enhances connectivity, trade, energy, and security cooperation

India's Lead Sectors in BIMSTEC

- Transport & Communication
- Tourism
- Environment & Disaster Management
- Counter Terrorism & Transnational Crime

169. Bio Economy Roadmap

Q.With reference to the NITI Aayog's roadmap for building India as a leading bioeconomy powerhouse by 2035, consider the following statements:

1. The proposed BioEconomy Growth Fund is intended to support commercialization of biotechnology innovations.
2. It proposes the establishment of a National BioData Council to govern biological and health data.
3. It recommends introducing a dedicated Production-Linked Incentive (PLI) scheme for biomanufacturing to reduce import dependence on critical bio-inputs.
4. The roadmap calls for treating biotechnology as national infrastructure, analogous to Digital Public Infrastructure (DPI).

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer:(d) All four

EXPLANATION

In NEWS:NITI Aayog suggests ₹50,000-crore fund for biotech growth-HINDU

Statement 1:

The roadmap proposes a **₹50,000-crore BioEconomy Growth Fund** for the period 2026–2035.The fund aims to bridge the biotechnology sector's **"Valley of Death"**, i.e., the gap between laboratory research and commercial-scale production.

It seeks to support:

- Biomanufacturing
- Synthetic biology
- Diagnostics
- Advanced therapeutics
- Fermentation technologies

Thus, one of its primary objectives is the **commercialization of biotechnology innovations**.**Hence, Statement 1 is correct.**

Statement 2:

The roadmap proposes the establishment of a **National BioData Council** to govern and facilitate the use of biological, genomic and health-related data.The objective is to:

- Improve data governance.
- Enable secure data sharing.
- Support AI-driven biotechnology research.
- Create interoperable biological and health data ecosystems.
- Accelerate innovation in precision medicine, drug discovery and genomics.**Hence, Statement 2 is correct.**

Statement 3:

The roadmap specifically recommends a **dedicated Production-Linked Incentive (PLI) Scheme for biomanufacturing**.The objective is to:

- Scale domestic manufacturing capacity
- Improve supply-chain resilience
- Promote indigenous biotechnology capabilities
- Reduce dependence on imported bio-inputs and technologies **Hence, Statement 3 is correct.**

Statement 4:

A key recommendation of the roadmap is to treat biotechnology as a strategic national capability.It advocates building biotechnology infrastructure in a manner similar to India's successful **Digital Public Infrastructure (DPI)** model.The report argues that biotechnology should be viewed as a foundational growth infrastructure for:

- Health security
- Food security
- Climate resilience
- Industrial innovation **Hence, Statement 4 is correct.**

VALUE ADDITION

BioEconomy Growth Fund

- Proposed Corpus: **₹50,000 crore**
- Period: **2026–2035**
- Objective: Bridge the **Valley of Death** in biotechnology innovation.
- Supports:
  - Biomanufacturing
  - Synthetic Biology
  - Diagnostics
  - Advanced Therapeutics
  - Fermentation Technologies

"Valley of Death"

- Gap between:
  - Proof-of-concept research
  - Commercial-scale deployment
- Major challenge for startups and deep-tech ventures.
- Due to:
  - High capital requirements
  - Long gestation periods
  - Regulatory hurdles

Six Proposed National BioMissions

Mission	Focus Area
GeneIndia	Affordable gene & cell therapies
AgriBio 2.0	Climate-resilient gene-edited crops
BioX Foundry	Synthetic biology commercialization
One Health Grid	Disease & AMR surveillance
Marine Biotechnology	Seaweed and marine bio-products
BioPharmaNext	Biologics, biosimilars & AI-driven drug discovery

Bioeconomy Targets

Year	Target Size
2025	$195.3 Billion
2035	$691 Billion
2047	$2.6 Trillion

Expected Outcomes

- Over **30 million high-value jobs**
- Stronger biotechnology startups
- Enhanced global competitiveness
- Reduced import dependence
- Expansion of biomanufacturing ecosystem

Related Policy

BioE3 Policy

BioE3 = Biotechnology for Economy, Environment and Employment

Focus Areas:

- Economic growth
- Sustainable development
- Employment generation
- Biomanufacturing expansion

170. PARIVARTAN Scheme

Q.With reference to the PARIVARTAN Scheme, consider the following statements:

1. It aims to replace older polluting trucks and buses in the National Capital Region (NCR) with BS-VI compliant or electric vehicles.
2. The scheme provides incentives exclusively for replacement with electric vehicles.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

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Answer:(a) 1 only

Explanation

In NEWS: MoHUA Approves Guidelines for PARIVARTAN Scheme to Accelerate Replacement of Old Trucks and Buses in NCR-PIB

Statement 1:

PARIVARTAN stands for **Programme for Accelerated Renewal and Incentivization of Vehicle Assets for Reducing Transport Air Pollution and Network Emissions**.

The scheme seeks to facilitate the replacement of:

- Old trucks
- Old buses
- Highly polluting commercial vehicles

with:

- **BS-VI compliant vehicles**
- **Electric Vehicles (EVs)**

The objective is to improve air quality and reduce transport-related emissions in the NCR region. **Hence, Statement 1 is correct.**

Statement 2:

The scheme does **not provide incentives exclusively for electric vehicles**.

Benefits are available for replacement with:

- BS-VI compliant diesel/CNG vehicles
- Electric vehicles

The scheme offers:

- Motor Vehicle Tax concessions
- Registration fee waiver
- 5% interest subvention on vehicle loans
- Minimum 8% OEM discount
- Fuel voucher support for eligible diesel/CNG replacement vehicles
- One-time financial assistance for EV replacement vehicles

Therefore, incentives are available for both cleaner conventional vehicles and EVs. **Hence, Statement 2 is incorrect.**

VALUE ADDITION

PARIVARTAN Scheme

- Full Form: Programme for Accelerated Renewal and Incentivization of Vehicle Assets for Reducing Transport Air Pollution and Network Emissions
- Approved by: Ministry of Housing & Urban Affairs (MoHUA)
- Implemented by: Ministry of Road Transport and Highways (MoRTH)
- Funded by: National Capital Region Planning Board (NCRPB)

Objective

- Reduce vehicular pollution in NCR.
- Promote fleet modernization.
- Encourage adoption of cleaner transport technologies.
- Improve regional air quality.

Coverage

- Delhi (NCT)
- Haryana
- Uttar Pradesh
- Rajasthan

Incentives

- Motor Vehicle Tax concessions
- Registration fee waiver
- 5% interest subvention
- Minimum 8% OEM discount
- Fuel voucher support
- EV replacement assistance
- Certificate of Deposit (CoD) trading

Digital Ecosystem

The scheme will integrate with:

- VAHAN
- V-Scrap
- DigiELV
- PFMS

171. Operation Southern Readiness 26-2

Q.With reference to Operation Southern Readiness 26-2, consider the following statements:

1. It is a multinational maritime training exercise hosted by the Indian Navy at Kochi.
2. Training under the exercise covers Maritime Domain Awareness, Counter-Narcotics operations, and maritime Uncrewed Systems.
3. The exercise is conducted in partnership with the Combined Maritime Forces (CMF).
4. The Combined Maritime Forces (CMF) is a maritime partnership comprising only Indian Ocean littoral countries.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (c) Only three

EXPLANATION:

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In NEWS: Indian Navy to organise Operation Southern Readiness 26-2 in Kochi

Statement 1:

Operation Southern Readiness 26-2 is a multinational maritime training exercise hosted by the Indian Navy at the Southern Naval Command, Kochi.The exercise brings together personnel from CMF partner nations for maritime security training and capacity building.**Hence, Statement 1 is correct.**

Statement 2:

The training programme includes:

- Maritime Domain Awareness (MDA)
- Counter-Narcotics Operations
- Maritime Law
- Information Sharing
- Force Protection
- Asymmetric Threats
- Maritime Uncrewed Systems

It also includes practical training in firefighting, boarding operations and survival at sea.**Hence, Statement 2 is correct.**

Statement 3:

The four-day training engagement is conducted in partnership with the Combined Maritime Forces (CMF) under the Indian Navy-led Combined Task Force 154 (CTF 154), CMF's dedicated Training Task Force.**Hence, Statement 3 is correct.**

Statement 4:

.CMF is a multinational maritime partnership comprising more than 40 nations from different regions of the world.Its membership is not limited to Indian Ocean littoral countries.**Hence, Statement 4 is incorrect.**

VALUE ADDITION

Operation Southern Readiness 26-2

Feature	**Details**
Nature	Multinational Maritime Training Exercise
Host	Indian Navy
Venue	Southern Naval Command, Kochi
Partner Organisation	Combined Maritime Forces (CMF)
Training Task Force	Combined Task Force 154 (CTF 154)
Objective	Capacity Building and Maritime Security Cooperation

Combined Maritime Forces (CMF)

Feature	**Details**
Nature	Multinational Maritime Partnership
Membership	More than 40 Nations
Headquarters	Bahrain
Focus Areas	Maritime Security, Counter-Terrorism, Counter-Piracy, Capacity Building
Indian Role	Active participant; currently leads CTF 154

Major Training Areas

- Maritime Law
- Maritime Domain Awareness (MDA)
- Information Sharing
- Counter-Narcotics Operations
- Force Protection
- Asymmetric Threats
- Maritime Uncrewed Systems
- Damage Control and Firefighting
- Survival at Sea
- Boarding Procedures

172. MALVAN

Q. With reference to the Anti-Submarine Warfare Shallow Water Craft (ASW-SWC) 'Malvan', consider the following statements:

1. Malvan is the second indigenous Mahe-class Anti-Submarine Warfare Shallow Water Craft commissioned by the Indian Navy.
2. It has been designed primarily for anti-submarine operations in coastal and shallow waters.
3. The vessel was constructed by Mazagon Dock Shipbuilders Limited (MDL), Mumbai.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (b) Only two

EXPLANATION

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IN NEWS:Anti-marine Warfare Malvan set to join Indian Navy

Statement 1:

Malvan is the second vessel of the Mahe-class ASW-SWC being inducted into the Indian Navy.The class has been specifically developed to enhance anti-submarine warfare capabilities in shallow coastal waters. **Hence, Statement 1 is correct.**

Statement 2:

The vessel is optimized for:

- Detection and tracking of submarines
- Countering underwater drones
- Coastal and littoral defence operations

It is specifically designed for shallow-water operations, where larger warships face operational limitations. **Hence, Statement 2 is correct.**

Statement 3:

Malvan was built by Cochin Shipyard Limited (CSL), Kochi, Kerala.It was not constructed by MDL.**Hence, Statement 3 is incorrect.**

VALUE ADDITION

Malvan (ASW-SWC): Quick Facts

Feature	**Details**
Class	Mahe-class ASW-SWC
Type	Anti-Submarine Warfare Shallow Water Craft
Builder	Cochin Shipyard Limited (CSL), Kochi
Indigenous Content	Over 80%
Role	Coastal Anti-Submarine Warfare
Operating Area	Shallow and Littoral Waters

Key Features

- Hull-mounted Sonar
- Towed Array Sonar
- Lightweight Torpedoes
- Anti-Submarine Rocket Launchers
- Mine-Laying Capability
- High Speed and Maneuverability

Strategic Importance

- Enhances coastal security.
- Protects ports and offshore assets.
- Counters hostile submarines and underwater drones.
- Supports the Aatmanirbhar Bharat initiative in defence manufacturing

173. Island in News

Q.Consider the following pairs:

Island	**Country**
1. Greater Tunb island	Iran
2. Falkland Islands (Islas Malvinas)	United Kingdom
3. Phu Quoc Island	Vietnam

How many of the above pairs are correctly matched?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (c) All three

EXPLANATION :

IN NEWS: What is Greater Tunb Island, the strategic point in the Strait of Hormuz targeted by U.S. airstrikes? l Falkland Islands: Why Argentina's celebration after beating England has sparked controversy l 15 Indian tourists killed as speedboat capsizes near Vietnam's Phu Quoc Island

Pair 1:

Greater Tunb – Iran

Greater Tunb is a small island located in the Persian Gulf near the Strait of Hormuz.It has been under Iranian control since 1971, although the United Arab Emirates continues to claim sovereignty over it. **Hence, Pair 1 is correctly matched.**

Pair 2:

Falkland Islands (Islas Malvinas) – United Kingdom

The Falkland Islands are a self-governing British Overseas Territory located in the South Atlantic Ocean.Argentina also claims the islands and refers to them as Islas Malvinas, but the islands remain under British administration.**Hence, Pair 2 is correctly matched.**

Pair 3:

Phu Quoc Island – Vietnam

Phu Quoc is the largest island of Vietnam, located in the Gulf of Thailand off the southwestern coast of the country.It is an important tourism hub and holds strategic significance in the region.**Hence, Pair 3 is correctly matched.**

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VALUE ADDITION

Important Islands in News

Island	Country/ Administration	Water Body
Greater Tunb	Iran (disputed by UAE)	Persian Gulf
Falkland Islands (Islas Malvinas)	United Kingdom (claimed by Argentina)	South Atlantic Ocean
Phu Quoc Island	Vietnam	Gulf of Thailand

Greater Tunb



- Located near the Strait of Hormuz.
- Along with Lesser Tunb and Abu Musa, it overlooks a major global oil shipping route.
- Strategic military importance for Iran.

Falkland Islands



- British Overseas Territory.
- Capital: Stanley.
- Site of the 1982 Falklands War between the UK and Argentina.

Phu Quoc Island



- Largest island of Vietnam.
- Located in the Gulf of Thailand.
- Known for tourism, fisheries, and maritime significance.

174. INTRON

Q.Which one of the following best describes an "Intron"?

(a) A coding segment of DNA that is translated into protein
(b) A non-coding sequence of a gene that is removed from pre-mRNA during splicing
(c) A three-nucleotide sequence that codes for an amino acid
(d) A cellular organelle responsible for protein synthesis

Answer: (b) A non-coding sequence of a gene that is removed from pre-mRNA during splicing

EXPLANATION

In NEWS: Why scientists watched a gene hop from a predator into its prey

Option (a): A coding segment of DNA that is translated into protein

This describes an Exon, which remains in mature mRNA and is translated into protein..Exons contain genetic information that ultimately determines the amino acid sequence of a protein. During RNA processing, exons are retained and joined together to form mature mRNA. **Hence, Option (a) is incorrect.**

Option (b): A non-coding sequence of a gene that is removed from pre-mRNA during splicing

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Introns are non-coding regions present in genes. They are transcribed into pre-mRNA but are removed during RNA splicing before translation occurs.After introns are removed, the remaining exons join together to form mature mRNA, which is then translated into a protein.**Hence, Option (b) is correct.**

Option (c): A three-nucleotide sequence that codes for an amino acid

This refers to a Codon.Each codon on mRNA specifies either an amino acid or a stop signal during translation. For example, AUG codes for Methionine and also acts as the start codon.

Hence, Option (c) is incorrect.

Option (d): A cellular organelle responsible for protein synthesis

This describes a Ribosome.Ribosomes read the codons present on mRNA and assemble amino acids into proteins. They are cellular organelles and not genetic sequences.

Hence, Option (d) is incorrect.

VALUE ADDITION

Intron: Quick Facts

Feature	**Details**
Nature	Non-coding DNA/RNA sequence
Present In	Genes of eukaryotes
Removed By	RNA Splicing
Present in Mature mRNA	No
Also Known As	Intervening Sequences

Introns vs Exons

Feature	**Introns**	**Exons**
Function	Non-coding	Coding
Present in pre-mRNA	Yes	Yes
Present in mature mRNA	No	Yes
Removed during splicing	Yes	No
Translated into protein	No	Yes

Splicing

Splicing is the process by which non-coding **introns** are removed from a newly formed **pre-mRNA**, and the coding **exons** are joined together.

DNA → Transcription → pre-mRNA (Introns + Exons) → Splicing → Mature mRNA (Only Exons) → Protein

Without splicing, non-coding sequences would remain in mRNA and interfere with proper protein formation.

175. AI SQUATTING

Q.Which one of the following best describes "AI Squatting"?

(a) A cyberattack that exploits software vulnerabilities in Artificial Intelligence systems to gain unauthorized access

(b) A cyberattack technique that exploits AI-generated hallucinations by creating fake domains, websites, or software packages suggested by AI models

(c) A method of training AI models using malicious datasets to alter their outputs permanently

(d) A process of encrypting AI-generated content to prevent unauthorized access

Answer: (b) A cyberattack technique that exploits AI-generated hallucinations by creating fake domains, websites, or software packages suggested by AI models

EXPLANATION

IN NEWS: What is AI squatting? An emerging cyber threat targeting AI hallucinations

Option (a): A cyberattack that exploits software vulnerabilities in Artificial Intelligence systems to gain unauthorized access

This describes conventional cyberattacks that exploit software bugs, vulnerabilities, or misconfigurations. AI Squatting does not require exploiting any software vulnerability. **Hence, Option (a) is incorrect.**

Option (b): A cyberattack technique that exploits AI-generated hallucinations by creating fake domains, websites, or software packages suggested by AI models

AI Squatting takes advantage of instances where AI systems generate convincing but non-existent websites, domains, or software packages. Attackers register these fake assets and use them for:

- Phishing attacks
- Credential theft
- Malware distribution

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- Supply-chain attacks. **Hence, Option (b) is correct.**

Option (c): A method of training AI models using malicious datasets to alter their outputs permanently

This is closer to concepts such as **data poisoning** or **model poisoning**, where training data is manipulated to influence model behavior.It is different from AI Squatting.**Hence, Option (c) is incorrect.**

Option (d): A process of encrypting AI-generated content to prevent unauthorized access

Encryption is a cybersecurity mechanism used to protect information and communications.It has no relation to AI Squatting.**Hence, Option (d) is incorrect.**

VALUE ADDITION

Types of AI Squatting

Type	Target
Phantom Squatting	Fake AI-generated website domains
HalluSquatting	Fake AI-generated software packages / repositories

How it Works

1. AI hallucinates a non-existent website or software package.
2. Attackers identify these recurring hallucinations.
3. They register the domain or upload a malicious package.
4. Users or AI agents access the fake resource.
5. Malware, phishing, or credential theft occurs.

Why It Is Dangerous

- Exploits trust in AI-generated outputs.
- Does not require software vulnerabilities.
- Can affect software supply chains.
- Creates risks for autonomous AI agents.
- Difficult to eliminate completely due to the probabilistic nature of LLMs.

Related Terms

Term	Meaning
AI Hallucination	AI generates false but plausible information
Prompt Injection	Manipulating AI through crafted inputs
Data Poisoning	Corrupting training data
AI Squatting	Exploiting hallucinated domains or software packages

176. Congo Colobus

Q.Which one of the following primates does not possess thumbs as an adaptation for arboreal movement?

(a) Rhesus Macaque
(b) Bonnet Macaque
(c) Congo Colobus (*Colobus congoensis*)
(d) Lion-tailed Macaque

Answer: (c) Congo Colobus (*Colobus congoensis*)

EXPLANATION

IN NEWS: Scientists Just Discovered A New Monkey Species. Here's What Makes It So Special

Option (a): Rhesus Macaque

The Rhesus Macaque (*Macaca mulatta*) is a widespread Old World monkey found across South and Southeast Asia.It possesses well-developed opposable thumbs that assist in climbing, grasping objects, and foraging. **Hence, Option (a) is incorrect.**



Option (b): Bonnet Macaque

The Bonnet Macaque (*Macaca radiata*) is endemic to southern India.Like other macaques, it possesses opposable thumbs that aid in arboreal and terrestrial movement.**Hence, Option (b) is incorrect.**



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Option (c): Congo Colobus (*Colobus congoensis*)

The newly identified Congo Colobus belongs to the Colobus genus, a group of African monkeys characterized by the absence or extreme reduction of thumbs. This adaptation allows the hand to function like a hook, facilitating rapid swinging, grasping, and movement through dense forest canopies. **Hence, Option (c) is correct.**



Option (d): Lion-tailed Macaque

The Lion-tailed Macaque (*Macaca silenus*) is endemic to the Western Ghats of India.It possesses opposable thumbs which help in climbing and handling food while living in forest canopies.**Hence, Option (d) is incorrect.**



VALUE ADDITION

Congo Colobus (*Colobus congoensis*)

Feature	**Details**
Scientific Name	*Colobus congoensis*
Local Name	Likweli
Location	Democratic Republic of Congo
Habitat	Tropical Rainforest Canopy
Lifestyle	Arboreal
Distinctive Feature	Absence of Thumbs
Diet	Leaves and Fruits

Colobus Monkeys Lack Thumbs

- Their thumbs are highly reduced or absent.
- Hands function as curved hooks for gripping branches.
- Enhances agility while leaping and moving through dense forest canopies.
- Considered an evolutionary adaptation to an arboreal lifestyle.

Other Well-Known Colobus Monkeys

- Black-and-White Colobus (*Colobus guereza*)
- Angolan Colobus (*Colobus angolensis*)
- King Colobus (*Colobus polykomos*)
- Congo Colobus (*Colobus congoensis*)

177. KOALA

Q.Consider the following characteristics:

1. It is a marsupial mammal endemic to Australia.
2. It is primarily dependent on eucalyptus trees for food and shelter.
3. It is classified as Vulnerable under the IUCN Red List.
4. The female possesses a pouch for the development of offspring.

The above characteristics best describe:

(a) Wombat
(b) Tasmanian Devil
(c) Koala
(d) Kangaroo

Answer: (c) Koala

EXPLANATION:

IN NEWS: Scientists are deep-freezing koala eggs and sperm to protect the species from extinction

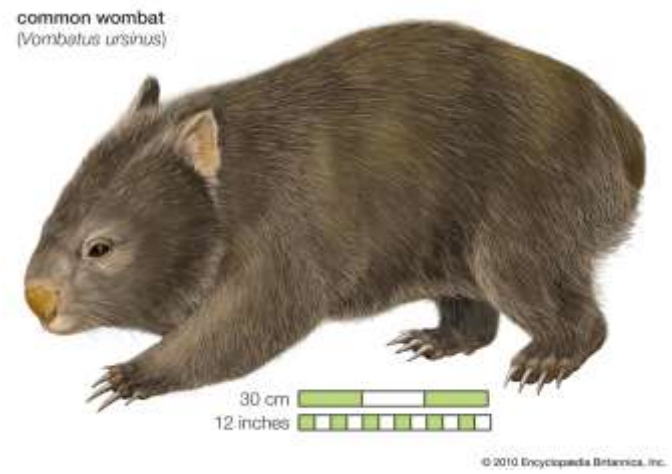
Option (a): Wombat

Wombats are marsupials native to Australia and are among the closest living relatives of koalas.

However:

- They are primarily terrestrial, not arboreal.
- They do not depend on eucalyptus trees for food and shelter.
- The characteristics given in the question do not specifically match wombats. **Hence, Option (a) is incorrect.**

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Option (b): Tasmanian Devil

The Tasmanian Devil is a carnivorous marsupial found mainly in Tasmania.

However:

- It is not dependent on eucalyptus trees.
- It is a scavenger and predator.
- The species described in the question does not match the Tasmanian Devil. **Hence, Option (b) is incorrect.**



Option (c): Koala

The given characteristics correctly describe the **Koala (*Phascolarctos cinereus*)**.

Key features:

- Endemic to Australia.
- Arboreal marsupial mammal.
- Feeds mainly on eucalyptus leaves.
- Females possess a pouch for carrying and nurturing young.
- Classified as **Vulnerable** under the IUCN Red List.

Hence, Option (c) is correct.



Option (d): Kangaroo

Kangaroos are marsupials native to Australia.

However:

- They are predominantly terrestrial.
- They do not rely on eucalyptus trees for food and shelter.
- The description given is specific to koalas rather than kangaroos. **Hence, Option (d) is incorrect.**



VALUE ADDITION

Koala: Quick Facts

Feature	**Details**
Scientific Name	*Phascolarctos cinereus*
Common Name	Koala
Type	Marsupial Mammal
Family	Phascolarctidae
Closest Living Relative	Wombat
Distribution	Eastern Australia
Habitat	Eucalyptus Forests and Woodlands
Conservation Status	Vulnerable (IUCN)

Marsupials

Marsupials are mammals whose young complete development in a pouch (marsupium).

Examples:

- Koala
- Kangaroo
- Wombat
- Tasmanian Devil

Koalas Threatened Reason:

- Habitat loss and fragmentation

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- Bushfires
- Climate change
- Disease (Chlamydia infections)
- Road accidents and dog attacks

Special Adaptation

- Sleep up to **18–20 hours per day**.
- Digest toxic eucalyptus leaves through a specialized digestive system.
- Spend most of their lives in trees.

178. WHITE RABBIT TECHNOLOGY

Q.With reference to the White Rabbit Technology-based Indian Standard Time (IST) Distribution Network, consider the following statements:

1. It enables highly precise synchronization of Indian Standard Time through Ethernet and
2. fibre-optic networks.
3. The system relies primarily on foreign GPS satellites for generating India's official time reference.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (a) 1 only

EXPLANATION:

In NEWS:'One Nation, One Time': White Rabbit Technology-based network demonstration inaugurated

Statement 1:

The White Rabbit Technology Demonstration Network is India's first high-precision Ethernet-based time distribution facility.

It uses:

- Fibre-optic and Ethernet networks
- Precision Time Protocol (PTP) based on IEEE 1588 standards
- Automatic clock calibration mechanisms

The technology can achieve sub-nanosecond (<1 ns) synchronization across connected devices over long distances. **Hence, Statement 1 is correct.**

Statement 2:

One of the primary objectives of the project is to eliminate India's dependence on foreign timing sources such as GPS.Indian Standard Time (IST) is generated through sovereign national infrastructure developed by:

- National Physical Laboratory (NPL)
- ISRO

The indigenous system enhances cybersecurity and protects critical infrastructure from risks such as GPS spoofing and timing attacks.**Hence, Statement 2 is incorrect.**

VALUE ADDITION

White Rabbit Technology Demonstration Network

Feature	Details
Initiative	White Rabbit Technology-based IST Distribution Network
Vision	One Nation, One Time
Nodal Agency	Legal Metrology Division, Ministry of Consumer Affairs
Key Partners	NPL, ISRO, BSNL, SEBI
Time Accuracy	Sub-nanosecond (<1 ns)
Communication Standard	IEEE 1588 Precision Time Protocol (PTP)
Technology Origin	CERN Open-Source White Rabbit Technology

Key Features of White Rabbit Technology

- Ultra-high precision synchronization.
- Deterministic communication with fixed latency.
- Supports thousands of devices across networks extending beyond 1,000 km.
- Reduces dependence on foreign GPS-based timing systems.
- Enhances cyber resilience of critical infrastructure.

Strategic Applications

Financial Sector

- Stock exchanges
- High-frequency trading
- Digital banking and payment systems

Telecommunications

- 5G and future 6G networks

Critical Infrastructure

- National power grids
- Defence communication systems
- Nuclear facilities

Emerging Technologies

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- Quantum communication
- Space operations

One Nation, One Time

The initiative seeks to create a uniform, secure and sovereign national time infrastructure by distributing Indian Standard Time across the country through indigenous networks while remaining compliant with global Coordinated Universal Time (UTC) standards.

179. Index of Core Industries

Q.. Consider the following industries:

1. Coal
2. Steel
3. Cement
4. Textiles
5. Fertilizers

Which of the above are included in the Index of Eight Core Industries?

(a) 1, 2, 3 and 5 only
(b) 1, 2 and 4 only
(c) 2, 3, 4 and 5 only
(d) 1, 2, 3, 4 and 5

Answer: (a) 1, 2, 3 and 5 only

EXPLANATION

In NEWS:Core Industries Index set to get an upgrade

Option 1: Coal

Coal is one of the **Eight Core Industries** and plays a critical role in India's energy and industrial sectors.

Hence, Coal is correctly included.

Option 2: Steel

Steel is one of the Eight Core Industries and serves as a key input for infrastructure, construction, automobile, and manufacturing sectors.**Hence, Steel is correctly included.**

Option 3: Cement

Cement is included among the Eight Core Industries due to its importance in infrastructure and construction activities.**Hence, Cement is correctly included.**

Option 4: Textiles

Textiles are an important manufacturing sector but are **not included** in the Index of Eight Core Industries.**Hence, Textiles is incorrectly included.**

Option 5: Fertilizers

Fertilizers form one of the Eight Core Industries because of their importance to agricultural productivity and food security.**Hence, Fertilizers is correctly included.**

VALUE ADDITION

Eight Core Industries

1. Coal
2. Crude Oil
3. Natural Gas
4. Refinery Products
5. Fertilizers
6. Steel
7. Cement
8. Electricity

Importance of Core Industries

- Account for **40.27%** of the weight in the Index of Industrial Production (IIP).
- Act as a leading indicator of industrial growth.
- Influence GDP growth, employment, and infrastructure development.
- Provide critical inputs to various sectors of the economy.

Index of Eight Core Industries (ICI)

Feature	**Details**
Released By	Office of Economic Adviser (OEA)
Department	DPIIT
Ministry	Ministry of Commerce & Industry
Frequency	Monthly
Current Base Year	2011-12
Revised Base Year	2022-23

180. National Commission for Minorities

Q.With reference to the National Commission for Minorities (NCM), consider the following statements:

1. It is a statutory body.
2. While investigating complaints, it enjoys powers of a civil court.
3. It consists of a Chairperson, a Vice-Chairperson and five Members, and they are appointed by the President of India.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation

In NEWS:Harjit Singh Grewal appointed chief of Minorities Commission

Statement 1:

The National Commission for Minorities is a **statutory body** established under the **National Commission for Minorities Act, 1992**.It is not a constitutional body.

Hence, Statement 1 is correct.

Statement 2:

Under the NCM Act, while investigating complaints and matters relating to minorities, the Commission enjoys powers of a **Civil Court**.These include:

- Summoning and enforcing attendance of persons.
- Receiving evidence on affidavits.
- Requisitioning public records.
- Examining witnesses and documents.

Hence, Statement 2 is correct.

Statement 3:

As per Section 3 of the NCM Act, 1992, the Commission consists of:

- Chairperson
- Vice-Chairperson
- Five Members

However, they are **nominated by the Central Government**, not appointed by the President of India.

Hence, Statement 3 is incorrect.

VALUE ADDITION

Composition of NCM

Component	**Number**
Chairperson	1
Vice-Chairperson	1
Members	5
Total Strength	7

Key Features

- Established: 1992
- Act: National Commission for Minorities Act, 1992
- Nature: Statutory Body
- Tenure: 3 Years
- Nodal Ministry: Ministry of Minority Affairs

Minority Communities Notified by Central Government

1. Muslims
2. Christians
3. Sikhs
4. Buddhists
5. Parsis (Zoroastrians)
6. Jains

Powers of Civil Court

The Commission can:

- Summon persons and examine them on oath.
- Require production of documents.
- Receive evidence on affidavits.
- Requisition public records.
- Issue commissions for examination of witnesses.

181. Vikram- 1 Rocket

Q.With reference to the Vikram-1 Rocket, consider the following statements:

1. Vikram-1 is India's first privately developed orbital-class rocket developed by Skyroot Aerospace.
2. All four stages of Vikram-1 use solid-fuel propulsion systems.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (a) 1 only

EXPLANATION:

In NEWS: India's first privately made orbital rocket Vikram-1 places payloads into orbit

Statement 1:

Vikram-1 is a privately developed launch vehicle designed by Skyroot Aerospace, a Hyderabad-based space startup. It is intended to place small satellites into Low Earth Orbit (LEO) and represents a significant step in India's private space sector following space sector reforms.**Hence, Statement 1 is correct.**

Statement 2:

Vikram-1 is a four-stage launch vehicle, but not all stages use solid propulsion.

- Stage 1: Kalam-100 solid motor
- Stage 2: Kalam-250 solid motor
- Stage 3: Kalam-100 solid motor
- Stage 4: Raman engine-based liquid propulsion system

The fourth stage uses a liquid-fuel propulsion system, enabling precise orbital insertion and maneuverability.**Hence, Statement 2 is incorrect.**

VALUE ADDITION

Vikram-1: Quick Facts

Feature	Details
Developer	Skyroot Aerospace
Type	Orbital-Class Launch Vehicle
Nature	India's First Privately Developed Orbital Rocket
Mission	Launch Small Satellites into Orbit
Named After	Dr. Vikram Sarabhai
Launch Site	Satish Dhawan Space Centre (SDSC), Sriharikota

Stages of Vikram-1

Stage	Propulsion
Stage 1	Kalam-100 (Solid)
Stage 2	Kalam-250 (Solid)
Stage 3	Kalam-100 (Solid)
Stage 4	Raman Engine (Liquid)

Key Features

- Four-stage launch vehicle.
- Carbon-composite structure.
- Designed for small satellite launches.
- Supports commercial space missions.
- Developed under India's expanding private space ecosystem.

Skyroot Aerospace

Feature	Details
Headquarters	Hyderabad
Founded	2018
Sector	Private Space Technology
Notable Achievement	Launched Vikram-S, India's first private rocket (2022)

Vikram Series

Rocket	Significance
Vikram-S	India's first private suborbital rocket
Vikram-1	India's first private orbital launch vehicle

182. UNESCO World Heritage Sites

Q.With reference to UNESCO World Heritage Sites, consider the following statements:

1. India signed the World Heritage Convention in 1977.
2. A country must cede a portion of its sovereignty over a site after its inscription on the World Heritage List.
3. The International Council on Monuments and Sites (ICOMOS) evaluates both natural and cultural site nominations.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (a) Only one

EXPLANATION

IN NEWS:Indian Embassy in Azerbaijan inaugurates permanent exhibition on UNESCO World Heritage Sites

Statement 1:

India formally signed the World Heritage Convention on 14 November 1977 and became a State Party to the Convention.**Hence, Statement 1 is correct.**

Statement 2:

World Heritage Sites remain under the sovereignty and ownership of the respective State even after inscription.UNESCO recognition does not transfer ownership or require any country to cede sovereignty over the designated site. However, the State assumes responsibility for conserving and protecting the site for future generations.**Hence, Statement 2 is incorrect.**

Statement 3:

The evaluation of World Heritage nominations is carried out by UNESCO's advisory bodies:

- ICOMOS (International Council on Monuments and Sites) – evaluates Cultural Sites
- IUCN (International Union for Conservation of Nature) – evaluates Natural Sites

Therefore, ICOMOS does not evaluate both natural and cultural site nominations.**Hence, Statement 3 is incorrect.**

VALUE ADDITION

UNESCO World Heritage Convention

Feature	**Details**
Adopted	1972
Entered into Force	1975
India Signed	14 November 1977
Objective	Protection of Cultural and Natural Heritage of Outstanding Universal Value
Governing Body	World Heritage Committee

Advisory Bodies of UNESCO

Advisory Body	**Function**
ICOMOS	Evaluates Cultural Heritage Sites
IUCN	Evaluates Natural Heritage Sites
ICCROM	Provides Conservation and Restoration Expertise

UNESCO World Heritage Sites in India (2026)

Category	**Number**
Cultural Sites	36
Natural Sites	7
Mixed Sites	1
Total	44

Only Mixed Site: Khangchendzonga National Park (Sikkim)

183. VISHWAS scheme

Q.With reference to the VISHWAS 2026 Scheme, consider the following statements:

1. It is a one-time dispute resolution scheme launched by the Employees' Provident Fund Organisation (EPFO) under the Ministry of Labour & Employment.
2. The scheme covers disputes relating to damages and penalties levied under the Employees' Provident Funds and Miscellaneous Provisions Act, 1952 and the Code on Social Security, 2020.
3. Cases involving fraud, misappropriation, or deliberate falsification of records are eligible for settlement under the scheme.
4. Employers opting for settlement under the scheme must agree not to pursue further appeals relating to the dispute being settled.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (c) Only three

EXPLANATION :

IN NEWS: EPFO launches scheme VISHWAS to amicable settlement of damages & penalty-related disputes

Statement 1:

VISHWAS 2026 is a one-time dispute resolution initiative launched by the Employees' Provident Fund Organisation (EPFO) under the Ministry of Labour & Employment. **Hence, Statement 1 is correct.**

Statement 2:

The scheme applies to disputes relating to damages and penalties levied under:

- Employees' Provident Funds and Miscellaneous Provisions Act, 1952
- Code on Social Security, 2020. **Hence, Statement 2 is correct.**

Statement 3:

The scheme specifically excludes:

- Cases involving fraud
- Misappropriation
- Deliberate falsification of records

Therefore, such cases are not eligible for settlement. **Hence, Statement 3 is incorrect.**

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Statement 4:

Employers seeking settlement must furnish an undertaking/declaration agreeing to:

- Withdraw pending appeals, and
- Not pursue further litigation relating to the dispute settled under the scheme. **Hence, Statement 4 is correct.**

VALUE ADDITION

VISHWAS 2026: Quick Facts

Feature	Details
Implementing Agency	EPFO
Ministry	Ministry of Labour & Employment
Nature	One-time dispute resolution scheme
Launch Year	2026
Duration	Six months from notification
Objective	Promote voluntary compliance and reduce litigation

Eligible Cases

- Cases pending before judicial forums/tribunals/courts.
- Final penalty/damages orders where recovery is pending or partial.
- Cases where show-cause notices have been issued but final orders are pending.
- Cases where contribution delays occurred but notices have not yet been issued.

Excluded Cases

- Damages/penalties already fully recovered.
- Fraud and misappropriation cases.
- Deliberate falsification of records.
- Cases where statutory interest has not been fully deposited.

Concessional Damages Structure

Period of Default	Damages Rate
Up to 2 months	0.25% per month
2 to < 4 months	0.50% per month
More than 4 months	1.00% per month

Cut-off Date

Concessional rates apply only to defaults occurring on or before 14 June 2024.

184. E-Commerce Council of India

Q.With reference to the E-Commerce Council of India (ECCI), consider the following statements:

1. It is a national platform launched to promote collaborative development of India's digital commerce ecosystem.
2. It is a statutory body under the Ministry of Commerce and Industry.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (a) 1 only

EXPLANATION

In NEWS: IAMAI launches E-Commerce Council of India to unify $120 bn digital commerce ecosystem

Statement 1:

The E-Commerce Council of India (ECCI) is a first-of-its-kind national platform launched to advance India's digital commerce ecosystem through collaborative engagement among industry, government, and other stakeholders.The Council aims to promote policy advocacy, research, and industry engagement across various sectors of digital commerce.**Hence, Statement 1 is correct.**

Statement 2:

The ECCI is not a statutory body and has not been established under any Act of Parliament.It was launched by the Internet and Mobile Association of India (IAMAI), a not-for-profit industry body, and functions as an industry-led platform rather than a government regulatory institution.**Hence, Statement 2 is incorrect.**

VALUE ADDITION

E-Commerce Council of India (ECCI)

Feature	Details
Launched By	Internet and Mobile Association of India (IAMAI)
Nature	Industry-led National Platform
Objective	Strengthen India's Digital Commerce Ecosystem

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Focus Areas	E-commerce, Logistics, MSMEs, AI, Digital Payments, Cross-border Trade
Functions	Policy Advocacy, Research, Industry Engagement

Internet and Mobile Association of India (IAMAI)

Feature	**Details**
Established	2004
Nature	Not-for-Profit Industry Body
Registration	Societies Registration Act, 1860
Members	500+ Companies and Startups
Objective	Promote Internet and Digital Economy Growth

185. PM - E Drive

Q.With reference to the PM E-DRIVE Scheme, consider the following statements:

1. It aims to promote the adoption of electric vehicles and develop charging infrastructure in India.
2. The scheme provides incentives for electric two-wheelers, three-wheelers, e-ambulances and e-trucks.
3. The scheme is implemented by the Ministry of New and Renewable Energy.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

EXPLANATION

IN NEWS: Goa to establish 70 new electric vehicle charging stations under PM E-Drive Scheme

Statement 1:

PM E-DRIVE (Electric Drive Revolution in Innovative Vehicle Enhancement) was launched in October 2024 to accelerate EV adoption, establish charging infrastructure, and strengthen the EV manufacturing ecosystem in India.**Hence statement 1 is correct**

Statement 2:

The scheme provides demand incentives for:

- Electric two-wheelers (e-2W)
- Electric three-wheelers (e-3W)
- E-ambulances
- E-trucks
- Other emerging EV categories.**Hence statement 2 is correct**

Statement 3:

The scheme is implemented by the Ministry of Heavy Industries (MHI), not the Ministry of New and Renewable Energy. **Hence statement 3 is incorrect**

Value Addition

PM E-DRIVE Scheme: Quick Facts

Feature	**Details**
Full Form	PM Electric Drive Revolution in Innovative Vehicle Enhancement (PM E-DRIVE) Scheme
Ministry	Ministry of Heavy Industries
Launch	2024
Outlay	₹10,900 crore
Duration	Operational till 31 March 2026 (as originally approved).
Objective	Promote electric mobility and strengthen the EV ecosystem
Replaces	EMPS-2024

Eligible Vehicle Categories

- e-Two Wheelers (e-2W)
- e-Three Wheelers (e-3W)
- e-Rickshaws
- e-Carts
- e-Buses
- e-Trucks
- e-Ambulances

Unique Feature

E-Voucher System

- Aadhaar-authenticated e-voucher.
- Generated at the time of EV purchase.
- Enables direct transfer of incentives.

Objectives

- Accelerate EV adoption.
- Reduce fossil fuel dependence.
- Lower vehicular emissions.
- Develop charging infrastructure.
- Support green mobility transition.

Source:

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186. Mansbal lake

Q.With reference to Manasbal Lake, consider the following statements:

1. It is a monomictic lake.
2. It is considered the deepest freshwater lake in India.
3. Its waters ultimately drain into the Chenab River.
4. Recently, the return of migratory birds has been viewed as a sign of its ecological recovery.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (c) Only three

EXPLANATION

IN NEWS: J&K's Mansbal lake starts hosting migratory birds again

Statement 1:

Manasbal Lake is a **monomictic lake**, meaning its water column mixes completely once a year. It exhibits thermal stratification for much of the year.**Hence, Statement 1 is correct.**

Statement 2:

Manasbal Lake, located in the Ganderbal district of Jammu & Kashmir, is regarded as the **deepest freshwater lake in India**, with a depth of about 13 metres.**Hence, Statement 2 is correct.**

Statement 3:

The lake's outflow is regulated into the **Jhelum River**, not the Chenab River.**Hence, Statement 3 is incorrect.**

Statement 4:

Recently, the **return of migratory birds** to Manasbal Lake has been considered an indicator of its improving ecological health and recovery.**Hence, Statement 4 is correct.**

VALUE ADDITION

Manasbal Lake: Quick Facts

Feature	**Details**
Location	Ganderbal District, Jammu & Kashmir
Type	Freshwater Lake
Special Feature	Deepest Freshwater Lake in India
Altitude	1,585–1,600 m
Water Source	Precipitation and 1,200+ Springs
Outflow	Jhelum River
Lake Type	Monomictic
Recent News	Return of Migratory Birds

187. BOSS SCAM

Q.Which one of the following best describes a "Boss Scam"?

(a) A cyberattack that encrypts an organization's data and demands payment for restoring access.
(b) A cyber fraud in which attackers impersonate senior executives using techniques such as deepfakes, voice cloning, or compromised communication accounts to trick employees into transferring funds.
(c) A financial fraud involving the creation of fake online investment platforms promising unusually high returns.
(d) A cyberattack that disrupts critical infrastructure through Distributed Denial-of-Service (DDoS) attacks.

Answer: (b) A cyber fraud in which attackers impersonate senior executives using techniques such as deepfakes, voice cloning, or compromised communication accounts to trick employees into transferring funds.

EXPLANATION:

IN NEWS:SEBI cautions regulated entities against 'Boss Scam'

Option (a): A cyberattack that encrypts an organization's data and demands payment for restoring access.

This describes Ransomware, where attackers lock or encrypt systems and demand a ransom for decryption.**Hence, Option (a) is incorrect.**

Option (b): A cyber fraud in which attackers impersonate senior executives using techniques such as deepfakes, voice cloning, or compromised communication accounts to trick employees into transferring funds.

Boss Scam involves:

- AI-generated voice cloning
- Deepfake video calls
- Fake social media or messaging groups

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- Hijacked WhatsApp accounts
- Impersonation of CEOs, MDs, or senior executives

The objective is to persuade finance personnel to transfer money to fraudulent accounts, often under the guise of urgent or confidential transactions.**Hence, Option (b) is correct.**

Option (c): A financial fraud involving the creation of fake online investment platforms promising unusually high returns.

This refers to investment fraud or Ponzi-type scams, not Boss Scam.**Hence, Option (c) is incorrect.**

Option (d): A cyberattack that disrupts critical infrastructure through Distributed Denial-of-Service (DDoS) attacks.

A DDoS attack floods systems with traffic to make services unavailable and is unrelated to executive impersonation.**Hence, Option (d) is incorrect.**

VALUE ADDITION

Boss Scam: Quick Facts

Feature	Details
Nature	Executive Impersonation Cyber Fraud
Target	Finance and Accounts Personnel
Regulator Warning	SEBI
Objective	Unauthorized Transfer of Funds
Key Technologies Used	Deepfakes, Voice Cloning, Malware

Common Methods Used

Method	Description
Deepfake Calls	AI-generated voice/video of executives
Voice Cloning	Mimicking CEO/MD voice
Fake Groups	Fraudulent WhatsApp/Telegram groups
Malware ZIP Files	Compromise devices and messaging accounts
WhatsApp Hijacking	Gain access to active WhatsApp Web sessions

Red Flags

- Urgent requests for fund transfers.
- Instructions to maintain secrecy.
- References to Unpublished Price Sensitive Information (UPSI).
- Requests through unfamiliar numbers or communication channels.

Related Cybersecurity Terms

Term	Meaning
Deepfake	AI-generated fake audio/video content
Voice Cloning	AI replication of a person's voice
Phishing	Fraudulent attempt to obtain sensitive information
Ransomware	Malware that encrypts data and demands payment
Business Email Compromise (BEC)	Fraud using compromised or spoofed business communications

188. EXERCISE PITCH BLACK

Q.Which one of the following best describes Exercise Pitch Black?

(a) A biennial multinational naval exercise hosted by Australia to enhance maritime security in the Indo-Pacific region.

(b) A multinational air combat exercise hosted by the Royal Australian Air Force to enhance interoperability and operational coordination among participating air forces.

(c) A trilateral military exercise conducted by India, Australia and Japan focusing on amphibious warfare operations.

(d) A joint special forces exercise between India and Australia aimed at counter-terrorism operations.

Answer: (b) A multinational air combat exercise hosted by the Royal Australian Air Force to enhance interoperability and operational coordination among participating air forces.

EXPLANATION:

In NEWS:Exercise Pitch Black 2026: India's Rafale fighter aircraft & C-17 arrive in Australia for air combat exercise

Option (a): A biennial multinational naval exercise hosted by Australia to enhance maritime security in the Indo-Pacific region.

This description is incorrect because Exercise Pitch Black is an air combat exercise, not a naval exercise.

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It is conducted by the Royal Australian Air Force (RAAF) and focuses on air warfare training and interoperability among participating air forces.**Hence, Option (a) is incorrect.**

Option (b): A multinational air combat exercise hosted by the Royal Australian Air Force to enhance interoperability and operational coordination among participating air forces.

Exercise Pitch Black is the RAAF's premier multinational air combat exercise. It involves participation from numerous countries and focuses on:

- Operational interoperability
- Air combat training
- Mission planning
- Military cooperation
- High-intensity warfare scenarios

The exercise is conducted in northern Australia and is regarded as one of the Indo-Pacific's most important air warfare exercises.**Hence, Option (b) is correct.**

Option (c): A trilateral military exercise conducted by India, Australia and Japan focusing on amphibious warfare operations.

This description is incorrect because Exercise Pitch Black is:

- Not a trilateral exercise
- Not limited to India, Australia and Japan
- Not focused on amphibious warfare

It is a multinational air combat exercise involving more than 19 countries. **Hence, Option (c) is incorrect.**

Option (d): A joint special forces exercise between India and Australia aimed at counter-terrorism operations.

This description does not match Exercise Pitch Black. The exercise is conducted by air forces and focuses on air combat operations rather than special forces or counter-terrorism training.**Hence, Option (d) is incorrect.**

VALUE ADDITION

Exercise Pitch Black: Quick Facts

Feature	**Details**
Nature	Multinational Air Combat Exercise
Host	Royal Australian Air Force (RAAF)
Country	Australia
Domain	Air Warfare
Frequency	Biennial
Edition	Pitch Black 2026
Duration	20 July – 7 August 2026
Main Locations	Darwin, Tindal and Amberley

Objectives

- Enhance operational interoperability
- Improve combat readiness
- Strengthen military cooperation
- Conduct complex air warfare training
- Build long-term defence partnerships

Indian Participation in Pitch Black 2026

Component	**Details**
Fighter Aircraft	4 Rafale
Transport Aircraft	2 C-17 Globemaster III
Personnel	120+ Air Warriors

Australia-India Defence Exercises

Exercise	**Service**
AUSTRA HIND	Army
AUSINDEX	Navy
Pitch Black	Air Force
MALABAR	Multilateral Naval Exercise

189. CAFE III Norms

Q.With reference to the Draft Corporate Average Fuel Efficiency (CAFE) Norms, consider the following statements:

1. CAFE norms prescribe fuel efficiency and emission targets based on the average performance of a manufacturer's vehicle fleet.
2. Compliance under CAFE norms is monitored using the Modified Indian Driving Cycle (MIDC).
3. Manufacturers may offset compliance deficits through approved carbon-credit trading mechanisms administered under the regulatory framework.

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How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (c) All three
EXPLANATION
In NEWS:Draft CAFE-3 Norms: Govt eases penalties, focuses on carbon credit trading for auto sector
Statement 1:
CAFE norms are government-mandated standards that require automobile manufacturers to meet prescribed fuel efficiency and CO_2 emission targets across their **entire fleet of vehicles**, rather than for individual models.**Hence, Statement 1 is correct.**
Statement 2:
Compliance with CAFE norms is assessed through standardized testing procedures, including the **Modified Indian Driving Cycle (MIDC)**, which measures fuel consumption and emissions under prescribed conditions.**Hence, Statement 2 is correct.**
Statement 3:
Under the proposed **CAFE-3 norms**, Manufacturers may offset compliance deficits through approved carbon-credit trading mechanisms administered under the regulatory framework. **Hence, Statement 3 is correct.**
VALUE ADDITION
Corporate Average Fuel Efficiency (CAFE) Norms

Feature	**Details**
Introduced in India	2017 (CAFE-1)
Objective	Improve fuel efficiency and reduce CO_2 emissions
Applicability	Fleet-wide average emissions
Monitoring Standard	Modified Indian Driving Cycle (MIDC)
Next Phase	CAFE-3 (From April 2027)

Key Features of Draft CAFE-3

Feature	**Details**
Carbon Credit Trading	Introduced
Credit Purchase	Through BEE
Compliance Approach	Flexible and Incentive-based
Emission Target	113 gCO_2/km (FY27) → 78.9 gCO_2/km (FY32)
Promoted Technologies	EVs, Hybrids, Flex-fuel Vehicles

190. Investment Friendliness Index

Q.With reference to the Investment Friendliness Index (IFI), consider the following statements:

1. The Investment Friendliness Index is developed and published by NITI Aayog.
2. The Index evaluates States and Union Territories across eight pillars, including infrastructure, regulatory ease, and environmental resilience.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2
Explanation
In NEWS: Gujarat tops NITI first Investment Friendliness Index 2026
Statement 1:
The **Investment Friendliness Index (IFI)** is a comprehensive assessment framework developed and published by **NITI Aayog** to evaluate the investment attractiveness of States and Union Territories. **Hence, Statement 1 is correct.**
Statement 2:
The Index assesses investment friendliness across **eight pillars**, namely:

- Infrastructure
- Business Climate
- Resources
- Government Policy
- Regulatory Ease

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- Institutional Environment
- Financial Health
- Environmental Resilience. **Hence, Statement 2 is correct.**

VALUE ADDITION

Investment Friendliness Index (IFI)

Feature	**Details**
Released By	NITI Aayog
Objective	Promote Competitive & Cooperative Federalism
Coverage	28 States and 8 UTs
Assessment Basis	84 Indicators under 8 Pillars
Investor Input	1,850 Investor Responses
Vision	Viksit Bharat @2047

Four-Tier Performance Banding

Category	**Score Range**
Top Performers	Above 50
Frontrunners	45–50
Emerging Performers	40–45
Aspiring States	Below 40

191. Spider Species

Q.The organisms Simonia lawbah and Hamataliwa papikonda, recently seen in the news, are:

(a) Amphibians
(b) Spiders
(c) Beetles
(d) Molluscs

Answer: (b) Spiders

EXPLANATION:

In NEWS: ZSI Researchers discover 2 New Spider Species in Subterranean Caves of Meghalaya and Canopies of Andhra Pradesh

Option (a): Amphibians

Amphibians are vertebrates such as frogs, toads, salamanders, and caecilians that generally depend on water for reproduction.*Simonia lawbah* and *Hamataliwa papikonda* belong to the class Arachnida, not Amphibia.**Hence, option (a) is incorrect.**

Option (b): Spiders

Both newly discovered organisms are spider species.

Simonia lawbah

- Belongs to the family Theridiosomatidae (Ray Spiders).
- Discovered from Krem Lawbah Cave, Meghalaya.
- First record of the genus Simonia in India.
- Constructs specialized cone-shaped "ray webs" that function like slingshots to trap prey.
- A troglophilic (cave-loving) species.

Hamataliwa papikonda

- Belongs to the family Oxyopidae (Lynx Spiders).
- Discovered from Papikonda National Park, Andhra Pradesh.
- First record of the genus Hamataliwa in Andhra Pradesh.
- Does not build webs; instead, actively hunts prey using agility and spiny legs.
- Characterized by a unique hexagonal arrangement of eight eyes. **Hence, option (b) is correct.**

Option (c): Beetles

Beetles belong to the order Coleoptera, characterized by hardened forewings (elytra).The newly discovered species are arachnids and not insects.**Hence, option (c) is incorrect.**

Option (d): Molluscs

Molluscs include snails, slugs, oysters, octopuses, and squids.The organisms mentioned are spiders and have no taxonomic relation to molluscs.**Hence, option (d) is incorrect.**

VALUE ADDITION

Newly Discovered Spider Species

Feature	**Simonia lawbah**	**Hamataliwa papikonda**
Family	Theridiosomatidae (Ray Spiders)	Oxyopidae (Lynx Spiders)
Location	Krem Lawbah Cave, Meghalaya	Papikonda National Park, Andhra Pradesh
Habitat	Cave ecosystem (Troglophilic)	Forest canopy
Hunting Strategy	Cone-shaped slingshot web	Active hunting without webs

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Significance	First record of genus *Simonia* in India	First record of genus *Hamataliwa* in Andhra Pradesh

Spider Families in News

Family	Common Name	Distinctive Feature
Theridiosomatidae	Ray Spiders	Cone-shaped ray webs
Oxyopidae	Lynx Spiders	Excellent vision and active hunting
Araneidae	Orb-weaver Spiders	Circular orb webs
Salticidae	Jumping Spiders	Powerful jumping ability and keen eyesight

Zoological Survey of India (ZSI)

Feature	Details
Established	1916
Headquarters	Kolkata
Ministry	Ministry of Environment, Forest and Climate Change
Function	Faunal exploration, taxonomy and biodiversity documentation
Motto	Survey, research and conservation of animal diversity

192. Japan Open 2026

Q.With reference to the Badminton World Federation (BWF), consider the following statements:

1. It organizes the BWF World Championships.
2. The BWF World Championships are conducted annually except in Olympic years.
3. P. V. Sindhu became the first Indian to win the Japan Open title in 2026, and the tournament is conducted under the BWF World Tour Super 750 category.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION

IN NEWS: Prime Minister congratulates PV Sindhu on winning Japan Open 2026 title

Statement 1:

The Badminton World Federation (BWF) is the international governing body for badminton. It organizes major international tournaments, including:

- BWF World Championships
- Thomas Cup
- Uber Cup
- Sudirman Cup
- BWF World Tour events **Hence, Statement 1 is correct.**

Statement 2:

Since 2005, the BWF World Championships have generally been held every year. However, the championship is not conducted during Summer Olympic years. **Hence, Statement 2 is correct.**

Statement 3:

Sindhu won the Japan Open 2026 by defeating Japan's Akane Yamaguchi. With this victory, she became the first Indian to win the Japan Open title. The Japan Open is part of the BWF World Tour Super 750 circuit. **Hence, Statement 3 is correct.**

VALUE ADDITION

Badminton World Federation (BWF)

Feature	Details
Established	1934
Former Name	International Badminton Federation (IBF)
Headquarters	Kuala Lumpur, Malaysia
Governing Body For	Badminton
Member Associations	200+

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Major BWF Championships

Tournament	Category
BWF World Championships	Individual World Championship
Thomas Cup	Men's Team Championship
Uber Cup	Women's Team Championship
Sudirman Cup	Mixed Team Championship
World Tour Finals	Season-ending Championship

BWF World Tour Structure

Category	Level
Super 1000	Highest
Super 750	Second Highest
Super 500	Third Highest
Super 300	Fourth Highest
Super 100	Entry Level

P. V. Sindhu – Key Achievements

Achievement	Year
Olympic Silver Medal	2016
Olympic Bronze Medal	2021
BWF World Champion	2019
Japan Open Champion	2026

193. Marine cloud Brightening

Q.With reference to Marine Cloud Brightening (MCB), consider the following statements:

1. It involves spraying tiny seawater droplets into the atmosphere to increase the reflectivity of marine clouds.
2. MCB is primarily aimed at increasing the absorption of solar radiation by the oceans.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (a) 1 only

EXPLANATION :

IN NEWS: Seeding clouds with seawater could change Earth's weather and prevent a super El Nino: Study

Statement 1:

Marine Cloud Brightening (MCB) is a proposed **solar geoengineering technique** that involves spraying microscopic seawater droplets into the lower atmosphere over oceans.As these droplets evaporate, they leave behind **sea salt particles**, which act as cloud condensation nuclei. This increases the number of cloud droplets, making clouds **brighter and more reflective**.As a result, more sunlight is reflected back into space, cooling the ocean surface below. **Hence, Statement 1 is correct.**

Statement 2:

MCB aims to achieve the exact opposite effect. Its objective is to **increase cloud albedo (reflectivity)** so that more incoming solar radiation is reflected back into space and **less sunlight is absorbed by the oceans**.This cooling effect may help mitigate global warming and weaken climate phenomena such as El Niño. **Hence, Statement 2 is incorrect.**

VALUE ADDITION

Marine Cloud Brightening (MCB): Quick Facts

Feature	Details
Type	Solar Geoengineering Technique
Method	Spraying seawater droplets into atmosphere
Main Aerosol Used	Sea Salt
Objective	Increase cloud reflectivity (Albedo)
Effect	Reflects sunlight back into space
Result	Cooling of ocean surface

MCB Working:

Step	Process
1	Seawater droplets are sprayed into the atmosphere
2	Water evaporates leaving sea salt particles

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3	Sea salt acts as cloud condensation nuclei
4	More cloud droplets are formed
5	Clouds become brighter and whiter
6	More sunlight reflected back to space
7	Ocean surface cools

Benefits of MCB

- May reduce global warming.
- Can weaken extreme El Niño events.
- May lower risks of marine heatwaves.
- Could help reduce coral bleaching.

Concerns Associated with MCB

- Alteration of rainfall patterns.
- Changes in regional weather systems.
- Ecological uncertainties.
- Potential unintended climatic consequences.

194. UN HABITAT

Q.With reference to the United Nations Human Settlements Programme (UN-Habitat), consider the following statements:

1. It was established following the Habitat I Conference held in Vancouver in 1976.
2. Its headquarters is located in Nairobi, Kenya.
3. It is the lead United Nations agency for sustainable urban development.
4. It supports the implementation of Sustainable Development Goal (SDG) 11.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (d) All four

EXPLANATION

In NEWS:Andhra Pradesh signs MoU with UN-Habitat for Inclusive urban development

Statement 1:

UN-Habitat traces its origin to the United Nations Conference on Human Settlements (Habitat I) held in Vancouver, Canada, in 1976.The conference highlighted growing concerns regarding urbanisation, housing shortages, and human settlements, leading to the establishment of the UN Centre for Human Settlements, which later evolved into UN-Habitat. **Hence, Statement 1 is correct.**

Statement 2:

UN-Habitat is headquartered in Nairobi, Kenya.It is one of the few major UN agencies headquartered in the Global South. **Hence, Statement 2 is correct.**

Statement 3:

UN-Habitat is the UN system's focal agency for:

- Sustainable urbanisation
- Human settlements
- Housing policies
- Urban governance
- Inclusive and resilient cities **Hence, Statement 3 is correct.**

Statement 4:

UN-Habitat is the custodian agency for several indicators under SDG 11 (Sustainable Cities and Communities) and assists countries in achieving sustainable urban development targets.**Hence, Statement 4 is correct.**

VALUE ADDITION

UN-Habitat: Quick Facts

Feature	**Details**
Full Name	United Nations Human Settlements Programme
Origin	Habitat I, 1976)
Headquarters	Nairobi, Kenya
Parent Organisation	United Nations
Focus Area	Sustainable Urban Development
Key Mandate	Adequate shelter and sustainable cities
Major Global Agenda	New Urban Agenda
Related SDG	SDG 11

Habitat Conferences

Conference	**Year**	**Location**	**Outcome**
Habitat I	1976	Vancouver, Canada	Foundation of global urban agenda
Habitat II	1996	Istanbul, Türkiye	Habitat Agenda adopted

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Habitat III	2016	Quito, Ecuador	New Urban Agenda adopted

Sustainable Development Goals (SDGs)

Feature	**Details**
Adopted By	United Nations General Assembly
Year	2015
Agenda	2030 Agenda for Sustainable Development
Total Goals	17
Total Targets	169
Target Year	2030

SDGs

SDG	**Goal**
SDG 1	No Poverty
SDG 2	Zero Hunger
SDG 3	Good Health and Well-being
SDG 4	Quality Education
SDG 5	Gender Equality
SDG 6	Clean Water and Sanitation
SDG 7	Affordable and Clean Energy
SDG 11	Sustainable Cities and Communities
SDG 13	Climate Action
SDG 14	Life Below Water
SDG 15	Life on Land
SDG 16	Peace, Justice and Strong Institutions
SDG 17	Partnerships for the Goals

195. Lestes Paloti

Q.With reference to *Lestes paloti*, recently discovered in the Western Ghats, consider the following statements:

1. It inhabits scrub vegetation around homesteads and laterite hillocks.
2. It is commonly found only near large freshwater lakes.
3. It represents the first addition to the Western Ghats spread wing damselfly fauna in nearly a century.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (c) 1 and 3 only

EXPLANATION

IN NEWS: New damselfly species discovered in Kerala's Western Ghats

Statement 1:

The newly discovered species has been recorded from several districts of Kerala and is known to inhabit **scrub vegetation around homesteads and laterite hillocks**, often away from open water bodies.**Hence, Statement 1 is correct.**

Statement 2:

The species does **not** show a preference for large freshwater lakes. Researchers observed that it is often found in **scrub habitats and laterite hillocks**, frequently away from open water sources.**Hence, Statement 2 is incorrect.**

Statement 3:

Lestes paloti is the **first new species of the genus Lestes discovered from the Western Ghats since *Lestes malabaricus* was described in 1929**, making it the first addition to the region's spreadwing fauna in almost 100 years.**Hence, Statement 3 is correct.**

VALUE ADDITION

Lestes paloti: Quick Facts

Feature	**Details**
Scientific Name	*Lestes paloti*
Group	Spreadwing Damselfly
Genus	*Lestes*

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Location	Western Ghats, Kerala
Published In	International Journal of Odonatology
Named After	Muhamed Jafer Palot (ZSI Scientist)
Significance	First new *Lestes* species from Western Ghats since 1929

Habitat Characteristics

Feature	**Details**
Habitat	Scrub vegetation, homesteads, laterite hillocks
Association with Water	Often away from open water bodies
Seasonal Colour Change	Brown in dry season, bluish during monsoon
Distribution	Kannur, Kozhikode, Wayanad, Malappuram, Palakkad and Ernakulam

Damselflies vs Dragonflies

Feature	**Damselflies**	**Dragonflies**
Body	Slender	Robust
Wings at Rest	Folded together over body	Held open horizontally
Flight	Weaker, fluttering	Strong and fast
Example	*Lestes paloti*	*Orthetrum* spp.

196. Specified Non-Financial Assets(SNFA)

Q.With reference to the recently introduced concept of Specified Non-Financial Assets (SNFAs) by the RBI, consider the following statements:

1. SNFAs refer to immovable properties acquired by banks from defaulting borrowers.
2. Banks can acquire an SNFA only after the loan is classified as a Non-Performing Asset (NPA).
3. SNFAs are included in the calculation of Gross NPAs and Net NPAs.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

EXPLANATION

IN NEWS: What is the RBI's new Specified Non-Financial Assets (SNFAs) and SARFAESI Act?

Statement 1:

SNFAs are **immovable properties** acquired by banks in settlement of outstanding debt when borrowers default on loan repayments.These may include:

- Residential buildings
- Commercial properties
- Industrial land
- Other immovable real estate assets **Hence, Statement 1 is correct.**

Statement 2:

According to RBI's revised directions, banks may acquire an SNFA **only after the borrower's loan account has been officially classified as a Non-Performing Asset (NPA)**.The acquisition must involve:

- Full settlement, or
- Partial settlement of the bank's outstanding exposure. **Hence, Statement 2 is correct.**

Statement 3:

The RBI has specifically clarified that SNFAs will **not be treated as part of Gross NPAs, Net NPAs, or stressed assets**.Instead, they will be reported separately in bank balance sheets under:**"Non-banking assets acquired in satisfaction of claims." Hence, Statement 3 is incorrect.**

VALUE ADDITION

Specified Non-Financial Assets (SNFAs): Quick Facts

Feature	**Details**
Introduced By	Reserve Bank of India (RBI)
Framework	Third Amendment Directions, 2026
Nature	Immovable assets acquired from defaulting borrowers
Examples	Residential buildings, commercial properties, industrial land
Acquisition Condition	Loan must first become NPA

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Disposal Method	Primarily through public auction

Key Provisions of RBI's SNFA Framework

Provision	**Details**
NPA Requirement	Asset can be acquired only after loan becomes NPA
Internal Policy	Mandatory for every commercial bank
Public Auction	Preferred mode of disposal
Sale to Borrower	Prohibited
Sale to Related Parties	Prohibited
Balance Sheet Treatment	Reported separately as non-banking assets

SNFAs vs NPAs

Feature	**SNFA**	**NPA**
Nature	Immovable property acquired by bank	Defaulted loan account
Asset Type	Non-financial asset	Financial asset
Included in Gross NPA	No	Yes
Appears in Balance Sheet As	Non-banking asset acquired in satisfaction of claims	Gross/Net NPA

Term	**Meaning**	**Key Fact**
SNFA (Specified Non-Financial Asset)	Immovable property acquired by banks from defaulting borrowers	Acquired only after loan becomes NPA
NPA (Non-Performing Asset)	Loan overdue for more than 90 days	RBI norm: >90 days overdue
SMA-0(Special Mention Account)	Early stress account	Not overdue beyond 30 days, but signs of stress
SMA-1	Stressed account	Overdue 31–60 days
SMA-2	Highly stressed account	Overdue 61–90 days
Sub-standard Asset	NPA for up to 12 months	First stage after becoming NPA
Doubtful Asset	NPA for more than 12 months	Higher recovery risk
Loss Asset	Uncollectible asset of little value	Not necessarily written off
Gross NPA (GNPA)	Total NPAs before provisioning	Indicator of asset quality
Net NPA (NNPA)	GNPA minus provisions	Actual burden on bank
Stressed Assets	NPAs, restructured assets and written-off assets.	Broader measure of financial stress
Security Interest	Legal right over collateral pledged for a loan	Core concept under SARFAESI
Securitisation	Converting loans into marketable securities	Basis of SARFAESI Act
ARC (Asset Reconstruction Company)	Company that buys bad loans and recovers them	Regulated by RBI
SARFAESI Act, 2002	Law enabling recovery of bad loans without court intervention	Notice period: 60 days

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When Does a Loan Become an NPA?

Category	Overdue Period
SMA-0	Signs of stress; not overdue beyond 30 days
SMA-1	31–60 days overdue
SMA-2	61–90 days overdue
NPA	More than 90 days overdue

197. Central Consumer Protection Authority (CCPA)

Q.. With reference to the Central Consumer Protection Authority (CCPA), consider the following statements:

1. It is a statutory body established under the Consumer Protection Act, 2019.
2. It can order recall of goods and reimbursement of the prices paid by consumers.
3. It can investigate violations of consumer rights through an Investigation Wing headed by a Director-General.
4. It functions under the Ministry of Law and Justice.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (c) Only three

EXPLANATION:

IN NEWS:CCPA initiates suo motu action against 41 restaurants over service charge

Statement 1:

The CCPA was established under Section 10 of the Consumer Protection Act, 2019 to protect, promote, and enforce consumer rights.Since it is created by an Act of Parliament, it is a statutory body. **Hence, Statement 1 is correct.**

Statement 2:

The Consumer Protection Act, 2019 empowers the CCPA to:

- Recall unsafe goods and services.
- Order reimbursement/refund of the prices paid by consumers.
- Discontinue unfair trade practices.
- Take action against misleading advertisements.

Hence, Statement 2 is correct.

Statement 3:

The CCPA has an Investigation Wing for conducting inquiries and investigations into violations of consumer rights.This wing is headed by a Director-General. **Hence, Statement 3 is correct.**

Statement 4:

The CCPA functions under the Ministry of Consumer Affairs, Food and Public Distribution, not under the Ministry of Law and Justice. **Hence, Statement 4 is incorrect.**

VALUE ADDITION

Central Consumer Protection Authority (CCPA): Quick Facts

Feature	Details
Established Under	Consumer Protection Act, 2019
Section	Section 10
Nature	Statutory Body
Nodal Ministry	Ministry of Consumer Affairs, Food & Public Distribution
Operational Since	20 July 2020
Objective	Protection, promotion and enforcement of consumer rights

Composition of CCPA - 3 members

Component	Details
Head	Chief Commissioner
Commissioners	Two
Areas Handled	Goods-related issues and Service-related issues

Powers of CCPA

Power	Description
Recall of Goods	Can order recall of unsafe goods
Refund/Reimbursement	Can direct refund of prices paid
Investigation	Through Investigation Wing

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Misleading Advertisements	Can impose penalties and order discontinuation
Unfair Trade Practices	Can issue directions to stop such practices
Class Action	Can initiate action on behalf of consumers as a class

198. Gyan Bharatam Mission

Q.With reference to the Gyan Bharatam Mission, consider the following statements:

1. It is a flagship initiative of the Ministry of Culture aimed at conserving and digitising India's manuscript heritage.
2. One of its core objectives is to establish a National Digital Repository (NDR) for manuscripts.
3. It aligns with UNESCO's Memory of the World Programme.

Which of the statements given above are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

In NEWS: Over 8 crore manuscripts digitised under Gyan Bharatam Mission: Union Minister Dr Mansukh Mandaviya

Statement 1:

The Gyan Bharatam Mission is a flagship initiative of the Ministry of Culture launched to identify, document, conserve, digitise and promote India's rich manuscript heritage.**Hence, Statement 1 is correct.**

Statement 2:

A key objective of the mission is the creation of a National Digital Repository (NDR), which will function as a centralized digital platform hosting manuscripts from universities, libraries, monasteries and research institutions.**Hence, Statement 2 is correct.**

Statement 3:

The mission supports the preservation of India's intellectual and cultural heritage and aligns with UNESCO's Memory of the World Programme, which seeks to preserve and provide access to documentary heritage worldwide.**Hence, Statement 3 is correct.**

VALUE ADDITION

Gyan Bharatam Mission: Quick Facts

Feature	**Details**
Ministry	Ministry of Culture
Objective	Conservation, Digitisation and Promotion of Manuscripts
Digital Platform	National Digital Repository (NDR)
Coverage	Manuscripts in Sanskrit, Pali, Prakrit, Tamil, Persian, Arabic and other languages
Supporting Initiative	Digital India
Global Linkage	UNESCO Memory of the World Programme

National Digital Repository (NDR)

Feature	**Details**
Nature	Centralised Digital Manuscript Archive
Purpose	Preservation and Public Access
Sources	Libraries, Universities, Monasteries, Research Institutes
Significance	India's Largest Knowledge Archive

UNESCO Memory of the World Programme

Feature	**Details**
Launched	1992
Organisation	UNESCO
Objective	Preservation of Documentary Heritage
Coverage	Manuscripts, Archives, Rare Documents, Audiovisual Records
Aim	Universal Access to Documentary Heritage

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Six Thematic Verticals of Gyan Bharatam Mission

Vertical	Focus Area
Survey & Cataloguing	Identification of Manuscripts
Conservation & Capacity Building	Restoration and Training
Technology & Digitisation	Digital Preservation
Linguistics & Translation	Translation of Rare Texts
Research & Publication	Academic Research
Outreach & Public Awareness	Public Participation

199. UNFCCC

Q.With reference to the United Nations Framework Convention on Climate Change (UNFCCC), consider the following statements:

1. It was adopted at the Rio Earth Summit held in 1992.
2. It is the principal global treaty for coordinating international responses to climate change.
3. The Kyoto Protocol and Paris Agreement were adopted under the framework of the UNFCCC.
4. The UNFCCC has binding emission reduction targets for all Parties.

Which of the statements given above are correct?

(a) 1, 2 and 3 only
(b) 1 and 4 only
(c) 2, 3 and 4 only
(d) 1, 2, 3 and 4

Answer: (a) 1, 2 and 3 only

EXPLANATION

In NEWS:Union Minister Bhupender Yadav meets UNFCCC Executive Secretary Simon Stiell in New Delhi

Statement 1:

The **UNFCCC was adopted in 1992** at the **Rio Earth Summit (Earth Summit)** held in Rio de Janeiro, Brazil.**Hence, Statement 1 is correct.**

Statement 2:

The UNFCCC serves as the **principal global treaty** for coordinating international efforts to address climate change and its impacts. **Hence, Statement 2 is correct.**

Statement 3:

The **Kyoto Protocol (1997)** and the **Paris Agreement (2015)** were negotiated and adopted under the framework of the UNFCCC.**Hence, Statement 3 is correct.**

Statement 4:

The UNFCCC itself does **not prescribe binding emission reduction targets for all Parties**. Binding quantitative emission reduction targets were introduced later through the **Kyoto Protocol**, and only for developed countries.**Hence, Statement 4 is incorrect.**

VALUE ADDITION

UNFCCC: Quick Facts

Feature	Details
Adopted	1992
Venue	Rio Earth Summit, Brazil
Nature	Global Climate Treaty
Parties	198 (197 States + European Union)
Objective	Stabilisation of Greenhouse Gas Concentrations
Annual Meeting	Conference of Parties (COP)
Scientific Basis	IPCC Assessments

Three Rio Conventions

Convention	Focus Area
UNFCCC	Climate Change
CBD	Biodiversity Conservation
UNCCD	Desertification & Land Degradation

Kyoto Protocol

Feature	Details
Adopted	1997
Nature	Legally Binding
Applicable To	Developed Countries

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Target	~5% reduction below 1990 levels (2008–2012)
Market Mechanism	Clean Development Mechanism (CDM)

Paris Agreement

Feature	Details
Adopted	COP21, Paris (2015)
Temperature Goal	Well below 2°C; pursue 1.5°C
Applicable To	All Countries
Key Instrument	Nationally Determined Contributions (NDCs)
Principle	Common but Differentiated Responsibilities (CBDR-RC)

IPCC

Feature	Details
Full Form	Intergovernmental Panel on Climate Change
Established	1988
Established By	UNEP & WMO
Role	Scientific Assessment of Climate Change

200. Electronic Gold Receipts(EGR)

Q.With reference to Electronic Gold Receipts (EGRs), consider the following statements:

1. EGRs are exchange-traded securities representing ownership of physical gold stored in SEBI-regulated vaults.
2. EGRs can be held in a demat account and converted into physical gold.
3. EGRs follow a T+1 settlement cycle.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (c) All three

Explanation

In NEWS: Electronic Gold Receipts: A new way to own gold

Statement 1: Electronic Gold Receipts (EGRs) are securities that represent ownership of a specified quantity and purity of physical gold stored in **SEBI-regulated vaults**. They provide investors with a regulated and transparent mechanism to own gold. **Hence, Statement 1 is correct.**

Statement 2:EGRs are held electronically in a **demat account** and can be bought and sold on stock exchanges. Investors may also convert their EGR holdings into physical gold through the prescribed redemption process.**Hence, Statement 2 is correct.**

Statement 3: Like equity shares, EGR transactions follow a **T+1 settlement cycle**, under which receipts are credited to the buyer's demat account on the next trading day after the transaction.**Hence, Statement 3 is correct.**

VALUE ADDITION

Electronic Gold Receipts (EGRs): Quick Facts

Feature	Details
Nature	Exchange-traded Security
Backed By	Physical Gold
Storage	SEBI-Regulated Vaults
Holding Mode	Demat Account
Settlement Cycle	T+1
Regulator	SEBI
Conversion	Can be converted into Physical Gold
Recognised as Security	Securities Contracts (Regulation) Act, 1956
Vault Manager	Regulated by SEBI under Vault Managers Regulations, 2021

Available Purity Standards

Purity	Gold Content
999	99.9% Pure Gold
995	99.5% Pure Gold

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Available Denominations

- 10 mg
- 100 mg
- 1 g
- 10 g
- 100 g
- 1 kg

Advantages of EGRs

- Transparent price discovery
- Assured purity standards
- No personal storage concerns
- High liquidity through exchange trading
- Easy conversion into physical gold
- Portfolio diversification

201. High-speed Maglev Trains

Q.Consider the following statements:

Statement-I: High-speed maglev trains are designed with a front portion resembling a "bird's beak" to enhance their operational efficiency.
Statement-II: Because maglev trains float above the track, they eliminate both rolling friction and aerodynamic resistance at speeds exceeding 300 km/hr.

Which one of the following is correct in respect of the above statements?

(a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I.
(b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I.
(c) Statement-I is correct but Statement-II is incorrect.
(d) Statement-I is incorrect but Statement-II is correct.

Answer: (c) Statement-I is correct but Statement-II is incorrect.
Explanation:

- The extended "bird's beak" nose design (famously inspired by the kingfisher's beak, as seen in Japan's Shinkansen and high-speed maglev prototypes) is an aerodynamic innovation. When a high-speed train enters a tunnel, it pushes air forward, creating atmospheric pressure waves that produce a loud "tunnel boom" (sonic boom) at the exit and cause severe drag. The long, tapered beak shape allows air to flow smoothly over the train, suppressing pressure waves, lowering noise pollution, and significantly improving operational energy efficiency.**Hence Statement-I is correct.**
- While it is true that maglev trains eliminate **rolling friction** because they float, they **do not eliminate aerodynamic resistance**. In fact, the sources state that a train moving over 300 km/hr experiences significant **aerodynamic resistance** due to air flow, which is the exact reason why the "bird's beak" design is necessary.**Hence Statement-II is incorrect.**

In NEWS: How magnetic levitation lifts and propels high-speed maglev trains

202. Right to Protest

Q. With reference to the "Right to Protest" in the context of the CJP stir, consider the following statements:

1. The right to assemble and march to Parliament is an absolute right that cannot be restricted by local police authorities if the protest is declared "peaceful" by its leaders.
2. Legal actions such as registering an FIR for "rioting" and "assault on public servants" act as "reasonable restrictions" under the framework of maintaining public order.
3. An internet shutdown in a specific district during a protest is a procedural tool used by the state to prevent the escalation of unruly or violent behavior.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (b)2 and 3 only
Explanation:
Under **Article 19(1)(b)** of the Indian Constitution, citizens have the right to assemble peaceably and without arms. However, this right is **not absolute**. Under **Article 19(3)**, the State can impose **reasonable restrictions** in the interests of the sovereignty and integrity of India, or public order. Local police and law enforcement authorities have the power to regulate, alter, or deny permissions for protest marches/routes (such as marching to Parliament or sensitive administrative zones) if there

is a perceived threat to public safety, traffic, or order—even if organizers declare the protest "peaceful."**Hence Statement 1 is INCORRECT.**

Registering an FIR for offenses such as rioting or assault on public servants fall under statutory provisions designed to enforce law and order. The framework of **"public order"** under Article 19(3) permits legal actions against actions that disrupt peace or involve confrontation with state machinery, acting as legitimate, legal mechanisms of restriction.**Hence Statement 2 is CORRECT.**

Imposing internet shutdowns (under the *Temporary Suspension of Telecom Services Rules, 2017* framing the Telegraph Act) is a preventive and procedural mechanism used by administration and local police to check the spread of rumors, prevent mob mobilization, and maintain public order during volatile protests (subject to judicial review guidelines laid down in cases like *Anuradha Bhasin v. Union of India*).**Hence Statement 3 is CORRECT.**

In NEWS: Several hurt in crackdown on CJP stir

203. Internet Suspension

Q. With reference to Internet suspension in India, consider the following statements:

1. Suspension of telecom services can be ordered only under the Temporary Suspension of Telecom Services Rules.
2. Every order suspending Internet services must necessarily receive judicial approval before it comes into force.
3. Such suspension cannot be indefinite in duration.

Which of the statements given above are correct?

(a) 1 and 3 only
(b) 2 only
(c) 3 only
(d) 1, 2 and 3

Answer:(c)3 only

Explanation:

While the *Temporary Suspension of Telecom Services (Public Emergency or Public Safety) Rules* (notified in 2017 under the Indian Telegraph Act, 1885 and later superseded by the 2024 Rules under the Telecommunications Act, 2023) are the primary procedural guidelines, they are **not the only mechanism**. State governments and local law enforcement authorities have historically ordered (and frequently continue to order) internet suspensions under **Section 144 of the Code of Criminal Procedure (CrPC)**. Therefore, stating that telecom services can be ordered *only* under these specific rules is factually incorrect.**Hence Statement 1 is INCORRECT.**

There is **no requirement for prior judicial approval** before an internet suspension order comes into force. Orders are issued by executive authorities (such as the Union Home Secretary or State Home Secretary). Post-issuance oversight is conducted by an executive **Review Committee** (headed by the Cabinet Secretary or Chief Secretary) within 5 working days. While orders are subject to *judicial review* (courts can hear challenges later), they do not need prior judicial sanction.**Hence Statement 2 is INCORRECT.**

In ***Anuradha Bhasin v. Union of India (2020)***, the Supreme Court unequivocally held that **an indefinite suspension of internet services is illegal and impermissible** under Indian law. Under current statutory rules, a single suspension order cannot exceed **15 days** at a time, and any continuation requires periodic review.**Hence Statement 3 is CORRECT.**

In NEWS: Several hurt in crackdown on CJP stir

204. FCRA

Q. With reference to the Foreign Contribution (Regulation) Act (FCRA), consider the following statements:

1. The Act regulates the acceptance and utilisation of foreign contributions by certain individuals and organisations.
2. The objective of the Act is to ensure that foreign contributions do not adversely affect the sovereignty and integrity of India.
3. Every individual receiving any amount of foreign contribution is exempt from registration under the Act.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

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Answer: (a)1 and 2 only

Explanation:

The Foreign Contribution (Regulation) Act (FCRA), 2010 regulates the acceptance and utilization of foreign contributions (donations, securities, or foreign currency) and foreign hospitality by specified individuals, associations, non-governmental organizations (NGOs), and companies.**Hence Statement 1 is CORRECT.**

As outlined in the Preamble and Section 11 of the Act, the primary objective of FCRA is to prohibit and regulate the acceptance and utilization of foreign funds for any activities that are **detrimental to the national interest**, including safeguarding the sovereignty and integrity of India, public interest, and the security of the State.**Hence Statement 2 is CORRECT.**

There is **no blanket exemption** for every individual receiving foreign contributions. Any individual or entity seeking to accept a foreign contribution for a cultural, economic, educational, religious, or social program **must register under FCRA** or obtain prior permission from the Ministry of Home Affairs (MHA).**Hence Statement 3 is INCORRECT.**

- *Exceptions exist only for specific non-contribution transactions* (e.g., normal business transactions/fees, salary, or personal gifts up to a specified monetary limit like ₹1 lakh), but this does not mean *every* individual receiving foreign contributions is exempt from registration

In NEWS: Kerala Catholic Church reiterates concerns overFCRA amendment Bill

205. Question Hour

Q. With reference to the Question Hour in Parliament, consider the following statements:

1. Question Hour is the first hour of a sitting of the Lok Sabha, unless otherwise decided.
2. It is meant to enable Members of Parliament to seek information and hold the Executive accountable.
3. Questions can be addressed only to the Prime Minister.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation:

As per the *Rules of Procedure and Conduct of Business in Lok Sabha*, the **first hour** of every parliamentary sitting is generally devoted to asking and answering questions, known as the **Question Hour** (starts at 11:00 AM), unless the Speaker directs otherwise.**Hence Statement 1 is CORRECT.**

Question Hour is one of the most critical parliamentary devices used to enforce **executive accountability** under the principle of collective responsibility (**Article 75(3)**). It allows MPs to demand details on government policies, administration, and performance.**Hence Statement 2 is CORRECT.**

Questions are addressed to **individual Ministers** heading respective ministries (e.g., Finance, Home Affairs, Defense) and not solely to the Prime Minister. Furthermore, under Rule 40 of the Lok Sabha rules, a question may also be addressed to a **private member** (an MP who is not a Minister) if the subject matter relates to a bill, resolution, or matter connected with the business of the House for which that member is responsible.**Hence Statement 3 is INCORRECT.**

In NEWS: Monsoon Session off to a stormy start amid Opposition protests

206. Judges of the Supreme Court of India

Q. With reference to the Judges of the Supreme Court of India, consider the following statements:

1. The Constitution fixes the maximum number of judges of the Supreme Court.
2. Parliament may by law increase or decrease the number of judges of the Supreme Court.
3. The Chief Justice of India is included in the sanctioned strength prescribed under the Supreme Court (Number of Judges) Act.

Which of the statements given above are correct?

(a) 2 and 3 only
(b) 1 and 2 only
(c) 3 only
(d) 1, 2 and 3

Answer: (a)2 and 3 only

Explanation:

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Article 124(1) of the Indian Constitution originally provided for a Chief Justice and 7 other judges, but explicitly stated: *"until Parliament by law prescribes a larger number."* Thus, the **Constitution itself does not fix a permanent maximum ceiling**. It leaves the authority to determine and increase the maximum sanctioned strength entirely to the Parliament through legislation.**Hence Statement 1 is INCORRECT.**

Under **Article 124(1)**, Parliament has the exclusive legislative authority to alter (increase or decrease) the sanctioned strength of Supreme Court judges by enacting or amending a law. This is done via the *Supreme Court (Number of Judges) Act, 1956* and its periodic amendments.**Hence Statement 2 is CORRECT.**

The statutory sanctioned strength under the *Supreme Court (Number of Judges) Act* **includes the Chief Justice of India (CJI)**. For instance, when the sanctioned strength was fixed at 34 judges via the 2019 Amendment, it comprised **1 Chief Justice of India + 33 other judges**.**Hence Statement 3 is CORRECT.**

In NEWS: Govt. introduces Bill in Lok Sabha to raise strength of judges in SC to 38

207. Contempt of Courts Act 1971

Q. With reference to the Contempt of Courts Act, 1971, consider the following statements:

1. Civil contempt relates to wilful disobedience of a court's judgment, decree, direction, order, writ or other process.
2. Criminal contempt includes publication that tends to obstruct the administration of justice.
3. Every criticism of a judicial decision amounts to contempt of court.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 only
(c) 1 only
(d) 1, 2 and 3

Answer: (a)1 and 2 only
Explanation:
Under **Section 2(b)**, *Civil Contempt* is defined as wilful disobedience to any judgment, decree, direction, order, writ, or other process of a court, or wilful breach of an undertaking given to a court.**Hence Statement 1 is correct.**
Under **Section 2(c)**, *Criminal Contempt* includes the publication (whether by words, spoken or written, or by signs, or by visible representation) of any matter or the doing of any other act whatsoever which lowers the authority of any court, scandalizes a court, or prejudices/interferes with the administration of justice.**Hence Statement 2 is correct.**
Under **Section 5** (*Fair criticism of judicial act not contempt*), fair, reasonable, and bona fide criticism of any judicial decision on its merits does not constitute contempt of court.**Hence Statement 3 is incorrect.**

In NEWS: Don't level baseless graft claims against judiciary: SC

208. Dengue Infection

Q. Consider the following statements regarding dengue infection:

1. Infection with one dengue serotype provides lifelong immunity against all four serotypes.
2. A subsequent infection with a different serotype may increase the risk of severe dengue.
3. Severe dengue may involve plasma leakage, bleeding manifestations, and organ impairment.

Which of the statements given above are correct?

(a) 2 and 3 only
(b) 1 only
(c) 1 and 2 only
(d) 1, 2 and 3

Answer: (a)2 and 3 only
Explanation:
Infection with one dengue serotype provides **lifelong immunity only against that specific serotype** (homologous immunity). It provides only temporary, short-term cross-immunity (heterologous immunity) to the other three serotypes, which fades after a few months. **Hence, Statement 1 is incorrect.**
A secondary infection with a different serotype increases the risk of severe dengue (such as Dengue Hemorrhagic Fever or Dengue Shock Syndrome) due to a phenomenon called **Antibody-Dependent Enhancement (ADE)**, where non-neutralizing antibodies from the first infection help the new

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serotype infect host cells more efficiently. **Hence Statement 2 is correct.**
Severe dengue is characterized by potential complications such as **plasma leakage** (leading to shock or fluid accumulation), **severe bleeding**, and **severe organ impairment** (such as liver or heart involvement). **Hence Statement 3 is correct.**

In NEWS: India approves first dengue vaccine for persons aged 4-60

209. Landslides

Q. Consider the following factors:

1. Slope angle
2. Geological structure
3. Intense rainfall
4. Road cutting on hill slopes

Which of the above can increase the susceptibility of an area to landslides?

(a) 1 and 2 only
(b) 1, 2 and 3 only
(c) 2, 3 and 4 only
(d) 1, 2, 3 and 4

Answer: (d)1,2,3 and 4
Explanation:
1. Slope angle (Topographic factor): Steeper angles increase the gravitational force pulling slope materials downward. Steeper slopes naturally carry a much higher risk of mass wasting.
2. Geological structure (Endogenic/Geological factor): Fault lines, heavily fractured or weathered rocks, loose uncompacted soil, and dip slopes (where rock layers slope in the same direction as the surface hill) create inherent lines of weakness.
3. Intense rainfall (Climatic trigger): Water adds significant weight to the soil mass and creates pore-water pressure, which lubricates fault planes and reduces the shear strength/friction holding the slope together.
4. Road cutting on hill slopes (Anthropogenic factor): Excavating and cutting into the toe (base) of a slope removes its natural structural support, leaving an unstable overhang that easily collapses under gravity or rain. **Hence option(d) is correct.**

In NEWS: 7 killed in landslide in Jammu and Kashmir as rain toll hits 21

210. Inflation

Q. With reference to inflation in India, consider the following statements:

1. Cost-push inflation arises due to an increase in the cost of production.
2. Demand-pull inflation occurs when aggregate demand exceeds aggregate supply.
3. A rise in international crude oil prices can contribute to cost-push inflation in India.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d)1,2 and 3
EXPLANATION:
Cost-push inflation occurs when overall prices increase due to a rise in the costs of wages, raw materials, or intermediate inputs. As production costs go up, producers decrease total supply or raise prices to maintain profit margins. **Hence Statement 1 is correct.**
Demand-pull inflation occurs when the aggregate demand for goods and services in an economy outpaces the aggregate supply (often summarized as *"too much money chasing too few goods"*). **Hence Statement 2 is correct.**
India imports over 80% of its crude oil requirements. A spike in international crude oil prices increases transportation, freight, and energy costs across almost all economic sectors, acting as a classic **imported cost-push factor**. **Hence Statement 3 is correct.**

211. Q Denga

Q. With reference to dengue vaccine QDENGA (TAK-003), consider the following statements:

1. It is India's first approved dengue vaccine approved for market authorisation.
2. It has been approved by the Drug Controller General of India.
3. It is authorised for use in persons aged 4–60 years.
4. It prevents malaria and dengue infections.

How many of the above statements are correct?

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(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (c) Only three
EXPLANATION:
In NEWS: CDSCO grants marketing authorisation for India's first Dengue vaccine-AIR
Statement 1:
QDENGA (TAK-003) has become the first dengue vaccine to receive market authorisation in India. The approval marks a major milestone in India's efforts to reduce the burden of dengue, a major mosquito-borne viral disease. **Hence, Statement 1 is correct.**
Statement 2:
The Drug Controller General of India (DCGI), India's national drug regulatory authority, granted market authorisation for QDENGA after reviewing safety and efficacy data. **Hence, Statement 2 is correct.**
Statement 3:
The vaccine has been approved for administration to individuals between 4 and 60 years of age, providing protection across a broad population group vulnerable to dengue infection. **Hence, Statement 3 is correct.**
Statement 4:
QDENGA is specifically designed to protect against dengue virus infection. It does not provide protection against malaria, which is caused by *Plasmodium* parasites and transmitted by *Anopheles* mosquitoes.**Hence, Statement 4 is incorrect.**
VALUE ADDITION
QDENGA (TAK-003): Quick Facts

Feature	**Details**
Vaccine Name	QDENGA (TAK-003)
Developer	Takeda Pharmaceuticals
Approval Authority	DCGI
Approval Status	First Dengue Vaccine Approved in India
Eligible Age Group	4–60 years
Disease Targeted	Dengue
Disease Type	Mosquito-borne Viral Disease
Public Health Significance	Reduces dengue infections, hospitalisations and severe cases

Dengue: Quick Facts

Feature	**Details**
Causative Agent	Dengue Virus (DENV)
Vector	*Aedes aegypti* mosquito
Peak Season	Monsoon and Post-Monsoon
Symptoms	High fever, headache, muscle pain, joint pain, rash
Severe Forms	Dengue Haemorrhagic Fever (DHF), Dengue Shock Syndrome (DSS)
Prevention	Mosquito control and protection from mosquito bites

Difference Between Dengue and Malaria

Feature	**Dengue**	**Malaria**
Cause	Dengue Virus	*Plasmodium* Parasite
Vector	*Aedes* Mosquito	*Anopheles* Mosquito
Disease Type	Viral	Parasitic
Vaccine	QDENGA (TAK-003)	RTS,S and R21 (limited use globally)

212. SMILE SCHEME

Q.With reference to the SMILE (Support for Marginalized Individuals for Livelihood and Enterprise) Scheme, consider the following statements:

1. It is a Central Sector Scheme implemented by the Ministry of Social Justice and Empowerment.
2. The scheme includes a component for the comprehensive rehabilitation of persons engaged in the act of begging.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2

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(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

EXPLANATION

IN NEWS:SMILE Scheme Rehabilitates Over 10,200 Beggars Across 216 Cities

Statement 1:

SMILE is a Central Sector Scheme launched and implemented by the Ministry of Social Justice and Empowerment. The scheme aims to ensure social and economic rehabilitation of marginalized individuals through a dignity-based and livelihood-oriented approach.**Hence, Statement 1 is correct.**

Statement 2:

The SMILE Scheme comprises two sub-schemes:

- Central Sector Scheme for Comprehensive Rehabilitation for Welfare of Transgender Persons.
- Central Sector Scheme for Comprehensive Rehabilitation of Persons Engaged in the Act of Begging.

The latter focuses on identification, rescue, shelter, healthcare, counselling, skill development, education and sustainable livelihood opportunities for persons engaged in begging.**Hence, Statement 2 is correct.**

VALUE ADDITION

SMILE Scheme: Quick Facts

Feature	**Details**
Full Form	Support for Marginalized Individuals for Livelihood and Enterprise (SMILE)
Ministry	Ministry of Social Justice and Empowerment
Type	Central Sector Scheme
Vision	"Bhiksha Vritti Mukt Bharat" (Begging-Free India)
Target Group	Persons engaged in begging and Transgender persons
Approach	Dignity-based rehabilitation and livelihood support
Implementation	Convergence-based approach with States, UTs and local bodies

Two Components of SMILE Scheme

Sub-Scheme	**Objective**
Comprehensive Rehabilitation for Welfare of Transgender Persons	Social inclusion, welfare and livelihood support
Comprehensive Rehabilitation of Persons Engaged in the Act of Begging	Rescue, rehabilitation, skill development and reintegration

Key Features of SMILE Beggary Scheme

Feature	**Details**
Cities Covered	216 cities (2026)
Target Coverage	295 cities by FY 2030-31
Rehabilitation Target	35,000 persons
Persons Rehabilitated	Over 10,200
Digital Platform	SMILE-Beggary National Portal
Convergence Schemes	DAY-NULM, NRLM, Mission Vatsalya
Focus Areas	Metropolitan, religious, tourist and historical cities

Best Practice

Indore (Madhya Pradesh) became India's first "Beggar-Free City" through a sustained rehabilitation and social reintegration campaign, often cited as a model for other cities

213. COAL GASIFICATION

Q.With reference to Coal Gasification, consider the following statements:

1. Coal gasification converts coal into syngas consisting mainly of carbon monoxide, hydrogen, carbon dioxide, and methane.
2. Syngas produced through coal gasification can be used for the production of fertilizers such as urea.
3. Nitroglycerine can be produced directly through coal gasification processes.
4. Coal gasification is considered cleaner than direct coal combustion because pollutants can be removed before power generation.

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Which of the statements given above are correct?

(a) 1, 2 and 4 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2, 3 and 4

Ans: (a) 1, 2 and 4 only
EXPLANATION :
IN NEWS:Government Targets 100 MT Coal Gasification Capacity by 2030
Statement 1:
Coal gasification is a process in which coal is converted into **syngas (synthesis gas)** by reacting it with a controlled amount of oxygen and steam at very high temperatures, generally above 1000°C. The resulting syngas mainly contains **carbon monoxide (CO), hydrogen (H_2), carbon dioxide (CO_2), methane (CH_4), and water vapour**.**Hence, Statement 1 is correct.**
Statement 2:
Syngas can be further processed to produce **ammonia** through the Haber-Bosch process. Ammonia serves as a key input for the manufacture of **urea and other nitrogenous fertilizers**.**Hence, Statement 2 is correct.**
Statement 3:
Nitroglycerine is not produced through coal gasification. It is manufactured by the **nitration of glycerol using concentrated nitric acid and sulfuric acid**. This process is unrelated to syngas production or coal gasification.**Hence, Statement 3 is incorrect.**
Statement 4:
Coal gasification is regarded as a relatively cleaner technology because contaminants such as **sulfur compounds, tar, and particulate matter** can be removed from the syngas before it is used for electricity generation or chemical production.**Hence, Statement 4 is correct.**
VALUE ADDITION
Coal Gasification: Quick Facts

Feature	**Details**
Process	Conversion of coal into syngas
Main Inputs	Coal, Oxygen, Steam
Temperature	Generally above 1000°C
Main Product	Syngas (Synthesis Gas)
Major Components of Syngas	CO, H_2, CO_2, CH_4
End Uses	Power generation, fertilizers, chemicals, fuels
Advantage	Cleaner than direct coal combustion

Uses of Syngas

Sector	**Application**
Fertilizers	Ammonia and Urea Production
Chemicals	Methanol, Synthetic Fuels
Energy	Power Generation
Industry	Feedstock for Chemical Manufacturing

Why is Coal Gasification Important for India?

Aspect	**Significance**
Energy Security	Reduces dependence on imported natural gas
Chemical Industry	Supports coal-to-chemical production
Resource Utilisation	Utilises abundant domestic coal reserves
Cleaner Technology	Lower emissions compared to direct coal burning
National Target	100 Million Tonnes of Coal Gasification by 2030

India and Coal Reserves

Rank	**Country**
1	United States
2	Russia
3	Australia
4	India

India possesses the **fourth-largest coal reserves in the world**, making coal gasification an important component of its energy and industrial strategy.

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214. MISSION GOLDEN SPICE

Q.Consider the following statements:

1. Mission Golden Spice aims to develop an integrated value chain ecosystem for Lakadong Turmeric in Meghalaya.
2. Lakadong Turmeric is mainly known for its low curcumin content compared to other turmeric varieties.
3. Lakadong Turmeric received the Geographical Indication (GI) tag in 2024.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (b) 1 and 3 only

EXPLANATION

In NEWS: Union Minister Jyotiraditya M. Scindia launches Mission "Golden Spice" for Meghalaya's Lakadong Turmeric

Statement 1: Mission Golden Spice is an Integrated Lakadong Turmeric Value Chain Enhancement Project designed to establish a comprehensive ecosystem covering cultivation, processing, value addition, branding, market linkages and exports in Meghalaya.**Hence, Statement 1 is correct.**

Statement 2: Lakadong Turmeric is renowned for its exceptionally high curcumin content, generally ranging from 7% to 10%, which is significantly higher than that of most turmeric varieties worldwide. Curcumin is the principal bioactive compound responsible for turmeric's medicinal properties.**Hence, Statement 2 is incorrect.**

Statement 3: Lakadong Turmeric, cultivated primarily in the Jaintia Hills of Meghalaya, was granted the Geographical Indication (GI) tag in 2024, recognizing its unique characteristics and regional identity.**Hence, Statement 3 is correct.**

VALUE ADDITION

About the Mission

- Launched by Union Ministry of Development of North Eastern Region (MDoNER).
- Anchored by MDoNER and North Eastern Council (NEC).
- Project outlay: ₹175.45 crore.
- Duration: 2025–2030.
- Implemented in two phases:
  - Phase I – Value Chain Strengthening
  - Phase II – Expansion & Scale-up
- Part of PM Modi's "Ashtalakshmi Vision" for North-East development.
- Aims to make Lakadong Turmeric a global premium/luxury spice brand.
- Focuses on GI monetisation, branding and exports.
- Target to increase cultivation area from ~2,500 ha to ~7,500 ha.
- Aims to increase farmer remuneration from ₹30–40/kg to nearly ₹80/kg by 2030.

Lakadong Turmeric

Location

- Grown mainly in Lakadong region of West Jaintia Hills and East Jaintia Hills, Meghalaya.

Unique Characteristics

- Curcumin content: 7–10%
- World average turmeric curcumin content: around 2–3%
- Therefore Lakadong turmeric contains nearly 3–4 times higher curcumin than ordinary turmeric.
- Darker colour and stronger aroma.
- Mostly cultivated organically.

GI Tag

- Received Geographical Indication (GI) Tag in 2024.

Medicinal Importance

Curcumin possesses:

- Anti-inflammatory properties.
- Antioxidant properties.
- Neuroprotective properties.
- Nutraceutical and pharmaceutical applications.

Curcumin

- Principal bioactive compound in turmeric.
- Responsible for yellow colour.
- Gives medicinal value to turmeric.

GI Tag

- Granted under Geographical Indications of Goods (Registration and Protection) Act, 1999.
- Valid for 10 years and renewable.
- Indicates product's quality/reputation linked to a geographical area.

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215. OPERATION VAJRA

Q.Operation Vajra, recently in the news, is associated with:

(a) Rehabilitation of surrendered militants in Northeast India.
(b) Prevention of transnational trafficking and diversion of synthetic opioid drugs.
(c) Border infrastructure development in high-altitude regions.
(d) Humanitarian assistance during international disasters.

Answer: (b) Prevention of transnational trafficking and diversion of synthetic opioid drugs.

EXPLANATION

IN NEWS: Central Bureau of Narcotics (CBN) India thwarts illegal trafficking of 12 crore Tramadol Hydrochloride, a.k.a. "fighter drug" tablets under "Operation Vajra"

Option (a): Rehabilitation of surrendered militants in Northeast India.

This is not related to Operation Vajra. Such initiatives are generally associated with rehabilitation and development programmes in insurgency-affected regions. **Hence, Option (a) is incorrect.**

Option (b): Prevention of transnational trafficking and diversion of synthetic opioid drugs.

Operation Vajra was an intelligence-led anti-drug trafficking operation conducted by the **Central Bureau of Narcotics (CBN)** to prevent the diversion of a large consignment of **Tramadol Hydrochloride**, a synthetic opioid often referred to as a "fighter drug", into illicit international trafficking networks. **Hence, Option (b) is correct.**

Option (c):Border infrastructure development in high-altitude regions.

Operation Vajra has no connection with border roads, strategic infrastructure or high-altitude development projects.**Hence, Option (c) is incorrect.**

Option (d): Humanitarian assistance during international disasters.

Operation Vajra is not a disaster relief or humanitarian assistance operation.**Hence, Option (d) is incorrect.**

VALUE ADDITION

Operation Vajra: Quick Facts

Feature	Details
Operation Name	Operation Vajra
Conducted By	Central Bureau of Narcotics (CBN)
Ministry	Ministry of Finance
Department	Department of Revenue
Objective	Prevent transnational drug trafficking
Drug Involved	Tramadol Hydrochloride
Quantity Seized	12 Crore Tablets
Weight	Approximately 30 Metric Tonnes
Law Invoked	NDPS Act, 1985

Term	Meaning
Tramadol	A synthetic opioid analgesic used for pain relief but prone to misuse and illegal trafficking.
Synthetic Opioid	Laboratory-produced substances that act on opioid receptors and can be highly addictive.
Drug Diversion	Redirection of legally manufactured drugs into illegal markets.
INCB	International Narcotics Control Board, an independent body monitoring implementation of UN drug control conventions.

Agencies Involved

- Central Bureau of Narcotics (CBN)
- Directorate of Revenue Intelligence (DRI)
- International Narcotics Control Board (INCB)
- Other national enforcement agencies

Central Bureau of Narcotics (CBN)

- **Nodal Ministry:** Ministry of Finance (**Department of Revenue**)
- **Headquarters: Gwalior**, Madhya Pradesh
- **Head: Narcotics Commissioner**
- **Legal Framework: NDPS Act, 1985**

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- **International Framework:** Functions under **UN Drug Control Conventions**
- **Key Role:** Regulates and monitors **Narcotic Drugs, Psychotropic Substances and Precursor Chemicals**
- **Supervises** legal (**licit**) cultivation of **opium poppy** in India
- **Investigates** offences under the NDPS Act
- Issues:
  - Licences for **opium poppy cultivation**
  - Licences for **synthetic narcotic drugs manufacture**
  - **Export/Import Authorisations** for narcotic drugs and psychotropic substances
  - **NOCs** for selected precursor chemicals
- **Major Opium-Growing States:** Rajasthan, Madhya Pradesh, Uttar Pradesh

NDPS Act, 1985

Feature	Details
Full Form	Narcotic Drugs and Psychotropic Substances Act, 1985
Objective	Control and regulate narcotic drugs and psychotropic substances
Nodal Ministry	Ministry of Finance (Department of Revenue)
Scope	Production, possession, transport, sale and trafficking of narcotic substances

216. WTO Agreement on Fisheries

Q.Consider the following statements regarding the WTO Agreement on Fisheries Subsidies (AFS):

1. India recently deposited its Instrument of Acceptance of the WTO Agreement on Fisheries Subsidies.
2. It prohibits subsidies linked to Illegal, Unreported and Unregulated (IUU) fishing.
3. It covers marine wild capture fishing but excludes aquaculture and inland fisheries.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION

IN NEWS: India joins WTO's agreement on fisheries subsidies

Statement 1:

India has formally deposited its Instrument of Acceptance of the WTO Agreement on Fisheries Subsidies, thereby becoming a Party to the Agreement. India is the **123rd WTO Member** to complete this process.**Hence, Statement 1 is correct.**

Statement 2:

One of the key provisions of the Agreement is the prohibition of subsidies that contribute to **Illegal, Unreported and Unregulated (IUU) fishing**, which is a major cause of depletion of global fish stocks and marine biodiversity loss.**Hence, Statement 2 is correct.**

Statement 3:

The Agreement focuses on subsidies related to **marine wild capture fishing and fishing-related activities at sea**. However, **aquaculture and inland fisheries are explicitly excluded** from its scope.**Hence, Statement 3 is correct.**

VALUE ADDITION

WTO Agreement on Fisheries Subsidies: Quick Facts

Feature	Details
Adopted At	12th WTO Ministerial Conference (MC12), Geneva
Year of Adoption	2022
Nature	Multilateral WTO Agreement
Objective	Environmental Sustainability of Fisheries
Entry into Force	15 September 2025
India's Status	123rd Member to deposit Instrument of Acceptance
First of its Kind	First WTO Agreement with an environmental sustainability objective

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Key Provisions

Provision	Purpose
Ban on IUU Fishing Subsidies	Prevent illegal fishing activities
Ban on Subsidies for Overfished Stocks	Promote recovery of fish populations
Discipline on Harmful Subsidies	Prevent overexploitation of marine resources
Protection of Small Fishers	Supports sustainable livelihoods
Equitable Fisheries Regime	Addresses distortions caused by heavily subsidised industrial fleets

Term	Meaning
IUU Fishing	Illegal, Unreported and Unregulated Fishing carried out in violation of national or international fisheries laws.
Marine Wild Capture Fishing	Harvesting naturally occurring fish and marine organisms from oceans and seas.
Aquaculture	Farming of fish, shellfish, aquatic plants and other aquatic organisms under controlled conditions.
Instrument of Acceptance	A formal legal document through which a country agrees to be bound by an international agreement.

Scope of the Agreement

Covered	Not Covered
Marine Wild Capture Fisheries	Aquaculture
Fishing-related Activities at Sea	Inland Fisheries
Ocean-based Fishing Subsidies	Freshwater Fish Farming

217. White-bellied Heron

Q.Consider the following statements about the White-bellied Heron:

1. It is an indicator species of a pristine riverine ecosystem.
2. It requires clear, shallow and fast-flowing streams for feeding.
3. It is endemic to India.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

EXPLANATION:

IN NEWS: India's rarest heron faces a silent end as new dam ignores its home

Statement 1:

The White-bellied Heron is considered an indicator species of pristine habitats, particularly undisturbed riverine and wetland ecosystems. Its presence signifies healthy freshwater ecosystems with minimal human disturbance. **Hence, Statement 1 is correct.**

Statement 2:

The species prefers clear, shallow, and fast-flowing river streams for fishing. It primarily feeds on fish and depends on clean freshwater habitats for survival. **Hence, Statement 2 is correct.**

Statement 3:

The White-bellied Heron is not endemic to India. At present, it is found in parts of Arunachal Pradesh (India), Bhutan and Myanmar. Historically, its range extended across Nepal, Sikkim, Bangladesh and Myanmar. **Hence, Statement 3 is incorrect.**

VALUE ADDITION



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White-bellied Heron: Quick Facts

Feature	Details
Scientific Name	*Ardea insignis*
Common Names	White-bellied Heron, Imperial Heron
Family	Ardeidae
Size	Second-largest heron species
Habitat	Fast-flowing rivers and forested wetlands
Distribution	India (Arunachal Pradesh), Bhutan, Myanmar
Diet	Primarily fish and other aquatic prey
Significance	Indicator species of pristine habitats

Habitat Requirements

Requirement	Importance
Clear Water	Enables visual hunting of fish
Shallow Streams	Facilitates feeding
Fast-flowing Rivers	Preferred riverine habitat
Forest Cover	Provides nesting and roosting sites
Minimal Disturbance	Essential for breeding success

Threats to Survival

- Habitat loss and fragmentation.
- Wetland conversion for agriculture.
- Expansion of human settlements.
- Disturbance from resource extraction.
- Poaching and human interference.
- River ecosystem degradation.

Conservation Status

Convention/Act	Status
IUCN Red List	Critically Endangered (CR)
Wildlife (Protection) Act, 1972	Schedule I

Terminology Gist

Term	Meaning
Pristine Habitat	An ecosystem that remains largely undisturbed by human activities and retains its natural ecological characteristics.
Indicator Species	A species whose presence, absence, or abundance reflects the health of a particular ecosystem.
Endemic Species	A species naturally restricted to a specific geographic region and found nowhere else in the wild.
Riverine Ecosystem	An ecosystem associated with rivers, streams and their surrounding habitats.

Other Important Herons

Species	Key Feature
White-bellied Heron (*Ardea insignis*)	Second-largest heron; Critically Endangered
Grey Heron (*Ardea cinerea*)	Widely distributed across Europe, Asia and Africa
Purple Heron (*Ardea purpurea*)	Found in marshes and reed beds; occurs in India
Great Egret (*Ardea alba*)	Large white heron-like bird commonly seen in wetlands
Indian Pond Heron (*Ardeola grayii*)	Common wetland bird found across India

218. Movie Angen

Q.Consider the following statements with regard to the film 'Angen':

1. *Angen* became the first Santhali film to win a National Film Award.
2. It won the 72nd National Film Award in the category of Best Debut Film of a Director (Non-Feature).
3. The film was directed by Ravi Raj Murmu.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only

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(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
EXPLANATION:

In NEWS:'Angen' wins National Award, bringing Santhali cinema into the national spotlight
Statement 1:
Angen marks a historic milestone for Santhali cinema as it is the first film in the Santhali language to receive a National Film Award, earning national acclaim for Jharkhand's rich tribal cultural heritage.**Hence, Statement 1 is correct.**
Statement 2:
The film's director, Ravi Raj Murmu, received the 72nd National Film Award for Best Debut Film of a Director in the Non-Feature category for *Angen*.**Hence, Statement 2 is correct.**
Statement 3:
Angen was directed by Ravi Raj Murmu, whose work highlights themes of cultural identity, migration, memory, and the importance of staying connected to one's roots.**Hence, Statement 3 is correct.**
VALUE ADDITION
Angen: Quick Facts

Feature	**Details**
Film Name	Angen
Language	Santhali
Category	Short Film (Non-Feature)
Duration	12 minutes 10 seconds
Director	Ravi Raj Murmu
Award	72nd National Film Award
Award Category	Best Debut Film of a Director (Non-Feature)
Significance	First Santhali Film to Win a National Film Award

About Santhali Language

Feature	**Details**
Language Family	Austroasiatic (Munda Group)
Major States	Jharkhand, Odisha, West Bengal, Bihar
Constitutional Status	Included in the Eighth Schedule (92nd Constitutional Amendment Act, 2003)
Script	Ol Chiki (developed by Pandit Raghunath Murmu)

219. GEO TAGGING

Q.With reference to Geotagging of River Origin Points, consider the following statements:

1. Geotagging is the process of adding geographic location data such as latitude and longitude to a photo's metadata.
2. Geotagging helps identify and digitally map river source points for monitoring and restoration planning.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2
EXPLANATION
In NEWS:After HC rap over river pollution, how Chhattisgarh is course-correcting
Statement 1:
Geotagging refers to the process of attaching **geographical coordinates** such as **latitude, longitude, and sometimes altitude** to digital photographs, videos, maps, or other data. This enables the precise identification of a location on the earth's surface.**Hence, Statement 1 is correct.**
Statement 2:
Geotagging enables authorities to accurately identify river origin points using GPS-based coordinates and digitally map them on platforms such as **PM Gati Shakti**. It supports scientific river rejuvenation by facilitating monitoring, catchment treatment, afforestation, desiltation, pollution control, and protection against encroachments and illegal activities.**Hence, Statement 2 is correct.**

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VALUE ADDITION

Geotagging: Quick Facts

Feature	**Details**
Meaning	Adding geographical location information to digital data
Location Data	Latitude, Longitude, Altitude
Technology Used	GPS, GIS, Remote Sensing
Output	Digitally mapped and location-specific information
Major Applications	Mapping, Planning, Monitoring, Asset Tracking

Uses of Geotagging in River Rejuvenation

Application	**Significance**
Identification of River Sources	Accurate mapping of origin points
Monitoring	Tracks encroachments and pollution threats
Restoration Planning	Supports afforestation, desiltation and catchment treatment
Digital Governance	Integration with PM Gati Shakti platform
Resource Management	Facilitates scientific water resource planning

PM Gati Shakti

Feature	**Details**
Launch Year	2021
Nature	National Master Plan for Multi-modal Connectivity
Nodal Ministry	Ministry of Commerce and Industry
Platform	GIS-based Digital Planning Platform
Objective	Integrated infrastructure planning and monitoring

River Rejuvenation Measures Adopted by Chhattisgarh

- Geo-tagging of river origin points.
- District-wise River Revival Plans.
- Scientific monitoring of river discharge.
- Groundwater level monitoring.
- Water quality assessment.
- Catchment area treatment.
- Desiltation and riverbank protection.
- Collaboration with institutions such as IIT Roorkee, NIT Raipur and CSVTU.

220. Batukeshwar Dutt

Q.Consider the following statements regarding Batukeshwar Dutt:

1. He was a member of the Hindustan Republican Association (HRA).
2. He was sentenced to life imprisonment for his role in the Central Legislative Assembly Bombing.
3. He participated in the Quit India Movement.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation

In NEWS: HM Amit Shah & Defence Minister Rajnath Singh pay tribute to freedom fighter Batukeshwar Dutt on his death anniversary

Statement 1:

Batukeshwar Dutt joined the **Hindustan Republican Association (HRA)** and became actively involved in revolutionary activities. During this period, he also learned bomb-making techniques and worked closely with revolutionaries such as Bhagat Singh and Chandrashekhar Azad.**Hence, Statement 1 is correct.**

Statement 2:

On **8 April 1929**, Batukeshwar Dutt and Bhagat Singh threw non-lethal bombs in the Central Legislative Assembly in New Delhi to protest against the **Public Safety Bill** and **Trade Disputes Bill**. After voluntarily surrendering, both were tried and sentenced to **life imprisonment**.**Hence, Statement 2 is correct.**

Statement 3:

After his release from prison, Batukeshwar Dutt continued his involvement in the freedom struggle and actively participated in the **Quit India Movement (1942)**. He was subsequently

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imprisoned again for several years by the British authorities.**Hence, Statement 3 is correct.**

VALUE ADDITION

Batukeshwar Dutt: Quick Facts

Feature	**Details**
Full Name	Batukeshwar Dutt (B.K. Dutt)
Birth	18 November 1910
Birth Place	Burdwan, Bengal Presidency
Revolutionary Organisation	Hindustan Republican Association (HRA)
Close Associates	Bhagat Singh, Chandrashekhar Azad
Famous Event	Central Legislative Assembly Bombing (1929)
Death	20 July 1965
Cremation Site	Hussainiwala National Martyrs Memorial, Punjab

Central Legislative Assembly Bombing (1929)

Feature	**Details**
Date	8 April 1929
Participants	Bhagat Singh and Batukeshwar Dutt
Objective	Protest against Public Safety Bill and Trade Disputes Bill
Slogan Raised	**"Inquilab Zindabad"**
Famous Quote	"If the deaf are to hear, the sound has to be very loud."
Outcome	Both revolutionaries courted arrest

221. Dakar Port, Senegal

Q. With reference to the recent concerns regarding illegal shark fin trade through Dakar Port, consider the following statements:

1. Dakar Port is located in Senegal on the Atlantic coast of West Africa.
2. Shark finning refers to the practice of removing a shark's fins and often discarding the remaining body at sea.
3. All species of sharks are listed as Critically Endangered in the IUCN Red List.
4. Sharks play an important role as apex predators in maintaining marine ecosystem balance.

Which of the statements given above are correct?

(a) 1, 2 and 4 only
(b) 1 and 3 only
(c) 2, 3 and 4 only
(d) 1, 2, 3 and 4

Answer:(a) 1, 2, 4 only

EXPLANATION:

In NEWS:'Cruel, wasteful': Dakar port a hotspot for illegal shark fins

Statement 1:

Dakar is the capital of Senegal and hosts one of Africa's busiest fishing and commercial ports. It is situated on the Atlantic coast of West Africa and serves as an important maritime hub. **Hence, Statement 1 is correct.**

Statement 2:

Shark finning involves cutting off a shark's fins and throwing the body back into the sea, often while the shark is still alive. The shark subsequently dies from suffocation, blood loss, or predation.The fins are highly valued for:

- Shark fin soup
- Traditional medicine markets **Hence, Statement 2 is correct.**

Statement 3:

Different shark species are placed under different IUCN categories:

- Least Concern (LC)
- Near Threatened (NT)
- Vulnerable (VU)
- Endangered (EN)
- Critically Endangered (CR)

The IUCN estimates that nearly 70% of pelagic shark species are threatened with extinction, but not all are Critically Endangered.**Hence, Statement 3 is incorrect.**

Statement 4:

Sharks occupy the top of marine food chains and regulate populations of prey species. Their presence helps maintain:

- Biodiversity
- Food-web stability
- Healthy coral reef and marine ecosystems

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The loss of sharks can trigger trophic cascades, disrupting entire marine ecosystems.**Hence, Statement 4 is correct.**

VALUE ADDITION

Sharks Ecologically Important

Key Functions

- Apex predators of marine ecosystems.
- Control prey populations.
- Prevent ecological imbalance.
- Maintain species diversity.
- Support healthy coral reefs and seagrass ecosystems.
- Indicators of ocean health.
- Removal of apex predators often causes Trophic Cascade — a chain reaction affecting multiple levels of the food web.

Shark Finning:

- Considered cruel and wasteful.
- Often linked with:
  - IUU Fishing
  - Bycatch
  - Human rights abuses aboard fishing vessels.
- Increases profitability because fins occupy less storage space than whole shark bodies.
- Compared by conservationists to killing a rhinoceros solely for its horn.

Dakar Port:

- Major Atlantic fishing hub.
- Important landing port for international fishing fleets.
- Strategic location connecting Africa, Europe and the Americas.

Feature	Details
Country	Senegal
Region	West Africa
Coast	Atlantic Ocean
Importance	One of Africa's busiest fishing ports

Global Status of Sharks

- Global shark and ray populations have declined by 71% since 1970.
- Approximately 80–101 million sharks are killed annually due to fishing activities.
- Industrial fishing remains the primary threat.
- Longline tuna fisheries contribute significantly to shark bycatch.

ICCAT (Important Organisation)

International Commission for the Conservation of Atlantic Tunas

- Regulates tuna and tuna-like fisheries in the Atlantic Ocean.
- Senegal is a party to ICCAT.
- Prohibits retention of certain shark species.
- Imposes regulations to curb shark finning.

Feature	Details
Full Form	International Commission for the Conservation of Atlantic Tunas
Established	1966
Headquarters	Madrid, Spain
Focus Area	Atlantic Tuna & Associated Species Conservation

Important Terms

Term	Meaning
Apex Predator	Predator at the top of the food chain
Bycatch	Unintentional capture of non-target species
IUU Fishing	Illegal, Unreported and Unregulated Fishing
Pelagic Species	Species living in the open ocean
Trophic Cascade	Ecological chain reaction caused by removal of key species

222. PM- YASASVI

Q.With reference to the PM-YASASVI (Prime Minister Young Achievers Scholarship Award Scheme for Vibrant India), consider the following statements:

1. PM-YASASVI is an umbrella scheme aimed at supporting students belonging to OBCs, EBCs and Denotified, Nomadic and Semi-Nomadic Tribes (DNTs).
2. The scheme is implemented by the Department of Social Justice and Empowerment under the Ministry of Social Justice and Empowerment.

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Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2
EXPLANATION
IN NEWS:Government Strengthens Educational Opportunities for OBC, EBC and DNT Students through PM-YASASVI
Statement 1:
PM-YASASVI is a comprehensive umbrella scholarship scheme designed to provide financial assistance to students belonging to:

- Other Backward Classes (OBCs)
- Economically Backward Classes (EBCs)
- Denotified, Nomadic and Semi-Nomadic Tribes (DNTs)

Its objective is to reduce financial barriers and improve access to quality education. **Hence, Statement 1 is correct.**
Statement 2:
The scheme is implemented by the Department of Social Justice and Empowerment, functioning under the Ministry of Social Justice and Empowerment. **Hence, Statement 2 is correct.**
VALUE ADDITION
PM-YASASVI: Key Highlights

- Full Form: Prime Minister Young Achievers Scholarship Award Scheme for Vibrant India.
- Launched as an umbrella scheme from 2021-22.
- Focuses on educational empowerment of socially and economically disadvantaged groups.
- Seeks to improve access, retention and completion of education.
- Several earlier scholarship schemes have been subsumed under PM-YASASVI.

Scholarship Components

Component	**Coverage**
Pre-Matric Scholarship	Classes IX and X
Post-Matric Scholarship	Post-secondary education
Top Class Education in Schools	Quality school education
Top Class Education in Colleges	Higher education in premier institutions

Eligibility for Pre-Matric Scholarship

- Applicable to students of Classes IX and X.
- Students should belong to eligible OBC/EBC/DNT categories.
- Annual family income must be below ₹2.5 lakh.
- Applicable to students studying in government schools.

223. National Investment Policy for Urea

Q.With reference to the National Investment Policy for Urea-2026 (NIPU-2026), consider the following statements:

1. Urea is the only major fertilizer whose Maximum Retail Price (MRP) is statutorily fixed by the Government of India.
2. NIPU-2026 aims to promote investment in new gas-based urea manufacturing units.
3. The policy was approved by the Cabinet Committee on Economic Affairs (CCEA).

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (c) All three
EXPLANATION :
IN NEWS:National Investment Policy for Urea-2026: What's new and why now?
Statement 1:
Urea is the only major fertilizer in India whose Maximum Retail Price (MRP) is statutorily fixed by the Government.Unlike urea, Phosphatic and Potassic (P&K) fertilizers are covered under the Nutrient Based Subsidy (NBS) Scheme, where prices are not directly fixed by the Government. **Hence, Statement 1 is correct.**
Statement 2:
NIPU-2026 has been launched to encourage investments in new gas-based urea manufacturing units across the country.The policy aims to:

- Increase domestic production of urea.
- Reduce dependence on imports.
- Improve fertilizer self-sufficiency.
- Support the vision of Atmanirbhar Bharat. **Hence, Statement 2 is correct.**

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Statement 3:

The Cabinet Committee on Economic Affairs (CCEA) approved the National Investment Policy for Urea-2026.The policy replaces the investment framework under New Investment Policy (NIP)-2012, whose investment window ended in 2019. **Hence, Statement 3 is correct.**

VALUE ADDITION

National Investment Policy for Urea-2026 (NIPU-2026)

Feature	Details
Approved By	Cabinet Committee on Economic Affairs (CCEA)
Objective	Increase domestic urea production
Focus	New gas-based urea plants
Capacity Addition	About 10 Million Tonnes
Proposed Plants	8–9 new plants
Expected Output per Plant	Around 12.7 LMT
Vision	Atmanirbhar Bharat
Nodal Ministry	Department of Fertilizers, Ministry of Chemicals & Fertilizers

Key Reforms under NIPU-2026

• Cost Separation

- Fixed and variable costs separated for greater transparency.

• Assured Returns

- Return on Equity (RoE) band:
  - Minimum: **12%**
  - Maximum: **16%**

• Forex Risk Mitigation

- Fixed costs converted into INR after four years based on prevailing exchange rates.

• Equal Incentives

- Applicable to: Private Sector, Public Sector, Cooperative Sector

NIPU-2026 Need

• Demand-Supply Gap

- India's annual urea demand: ~40 Million Tonnes.
- Domestic production: ~30 Million Tonnes.
- Remaining requirements met through imports.

• Expiry of NIP-2012

• Energy & Fertilizer Security

- Reduce vulnerability to global LNG and fertilizer market disruptions.

• Promote Self-Reliance

- Support Atmanirbhar Bharat in the fertilizer sector.

Urea: Quick Facts

Feature	Details
Chemical Formula	NH_2CONH_2
Type	Nitrogenous Fertilizer
Nitrogen Content	~46%
Appearance	White crystalline solid
Major Crops	Paddy, Wheat, Sugarcane
Subsidy Status	Highly Subsidized
MRP	Fixed by Government

Agricultural Urea vs Industrial Urea

Feature	Agricultural Urea	Industrial Urea
Purpose	Fertilizer	Industrial Raw Material
Subsidy	Subsidized	No Subsidy
Price	Government Controlled	Market Determined
Neem Coating	Mandatory	Not Required
Regulation	Fertiliser Control Order	Industrial Standards
Uses	Crop Nutrition	Resins, Adhesives, Textiles, Melamine

Related Fertilizer Initiatives

Neem Coated Urea (NCU)

- Mandatory 100% neem coating since 2015.
- Improves Nitrogen Use Efficiency (NUE).
- Prevents diversion of subsidized urea.

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- Incentivizes States to reduce excessive chemical fertilizer use.

Soil Health Card Scheme

- Provides soil nutrient status to farmers.
- Encourages balanced fertilization.

Nano Urea

- Developed by IFFCO.
- Enhances nutrient-use efficiency.

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- Reduces requirement of conventional urea.

One Nation One Fertilizer

- All subsidized fertilizers marketed under the **"Bharat"** brand

224. Cauvery River

Q.With reference to the Cauvery River and its significance, consider the following statements:

1. The Cauvery River is known as the “Dakshina Ganga” of India.
2. The Western Ghats portion of the Cauvery Basin is a biodiversity hotspot.
3. The Cauvery Water Management Authority (CWMA) was established following Supreme Court directions.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION

In NEWS:Maintain *status quo* on water release, says Cauvery body

Explanation

Statement 1:

The Cauvery is often referred to as the "Dakshina Ganga" (Ganga of the South) because of its immense cultural, religious, and economic significance in southern India. It is one of the most important peninsular rivers and supports millions of people through irrigation, drinking water supply, and agriculture.**Hence, Statement 1 is correct.**

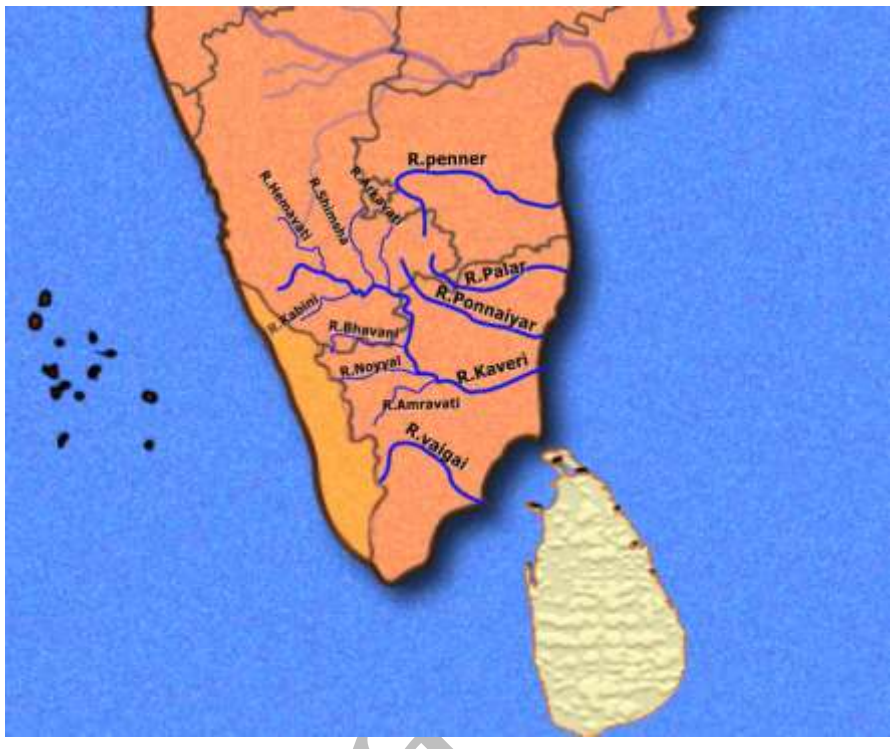
Statement 2:

The Cauvery originates in the Brahmagiri Hills of the Western Ghats, which form part of one of the world's eight hottest biodiversity hotspots. The basin supports rich flora and fauna and contains important protected areas such as Nagarhole National Park, Bandipur Tiger Reserve, Mudumalai Tiger Reserve, and Brahmagiri Wildlife Sanctuary.**Hence, Statement 2 is correct.**

Statement 3:

The Cauvery Water Management Authority (CWMA) was constituted by the Government of India in 2018 after the Supreme Court's judgment on the Cauvery water dispute. The Authority oversees the implementation of water-sharing arrangements among Karnataka, Tamil Nadu, Kerala, and Puducherry.**Hence, Statement 3 is correct.**

Value Addition



Cauvery River: At a Glance

Feature	**Details**
Source	Talakaveri, Brahmagiri Hills, Kodagu (Karnataka)
Length	~800 km
Drainage	Bay of Bengal
Basin Area	~81,155 sq km
States/UT Covered	Karnataka, Tamil Nadu, Kerala & Puducherry
Popular Name	Dakshina Ganga (Ganga of South India)
Major Distributary	Kollidam (Coleroon)
Delta Region	Thanjavur Delta (Rice Bowl of Tamil Nadu)

Major Tributaries of Cauvery

Bank	**Major Tributaries**
Left Bank	Harangi, Hemavati, Shimsha, Arkavati
Right Bank	Lakshmana Tirtha, Kabini, Suvarnavati, Bhavani, Noyyal, Amaravati

Biodiversity Significance of Cauvery Basin

- Originates in the Western Ghats, one of the world's 8 hottest biodiversity hotspots.

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- Supports important tiger and elephant habitats.
- Provides critical ecosystem services such as:
  - Groundwater recharge
  - Soil conservation
  - Watershed protection
  - Climate regulation
- Home to several endemic species of flora and fauna.

Important Protected Areas in the Cauvery Landscape

Protected Area	State
Nagarhole National Park	Karnataka
Bandipur Tiger Reserve	Karnataka
Brahmagiri Wildlife Sanctuary	Karnataka
Mudumalai Tiger Reserve	Tamil Nadu
Sathyamangalam Tiger Reserve	Tamil Nadu

Many of these protected areas form part of the Nilgiri Biosphere Reserve, India's first Biosphere Reserve.

Important Waterfalls Associated with Cauvery

Waterfall	Location
Shivanasamudra Falls	Karnataka
Hogenakkal Falls	Tamil Nadu
Chunchanakatte Falls	Karnataka

Shivanasamudra Hydroelectric Station (1902) is among the earliest hydroelectric power projects in Asia.

Important Dams on the Cauvery System

Dam	River	State
Krishnaraja Sagar (KRS)	Cauvery	Karnataka
Mettur Dam (Stanley Reservoir)	Cauvery	Tamil Nadu
Kabini Dam	Kabini	Karnataka
Hemavathi Reservoir	Hemavati	Karnataka
Bhavanisagar Dam	Bhavani	Tamil Nadu

Cauvery Water Management Authority (CWMA)

Feature	Details
Established	2018
Basis	Supreme Court Judgment (2018)
Ministry	Ministry of Jal Shakti
Purpose	Implement Cauvery water-sharing arrangements
Assisted By	Cauvery Water Regulation Committee (CWRC)

Cauvery Water Dispute: Timeline

Year	Event
1892	Agreement between Madras Presidency & Mysore Kingdom
1924	Revised Agreement
1990	Cauvery Water Disputes Tribunal constituted
2007	Tribunal Final Award
2018	Supreme Court Verdict
2018	CWMA & CWRC established

225. Aravalli Range

Q.With reference to the Aravalli Range, consider the following statements:

1. The Aravalli Range is one of the oldest fold mountain systems in the world.
2. Guru Shikhar, the highest peak of the Aravalli Range, is located in the Mount Abu region of Rajasthan.
3. The Aravalli Range acts as a natural barrier preventing the eastward expansion of the Thar Desert.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

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EXPLANATION

IN NEWS: Supreme Court-appointed Aravali panel's consultation process excludes rural people, say environmentalists

Statement 1:

The Aravalli Range is one of the oldest fold mountain systems in the world, formed during the Proterozoic Era approximately 1.8–2.5 billion years ago.Over millions of years, the range has undergone extensive denudation and erosion, resulting in its present form as discontinuous ridges and residual hills.**Hence, Statement 1 is correct.**

Statement 2:

Guru Shikhar is the highest peak of the Aravalli Range.It is located in the Mount Abu region of Rajasthan and rises to an elevation of about 1,722 metres (5,650 feet) above sea level.**Hence, Statement 2 is correct.**

Statement 3:

The Aravalli Range acts as a natural ecological barrier against the eastward spread of the Thar Desert.It helps:

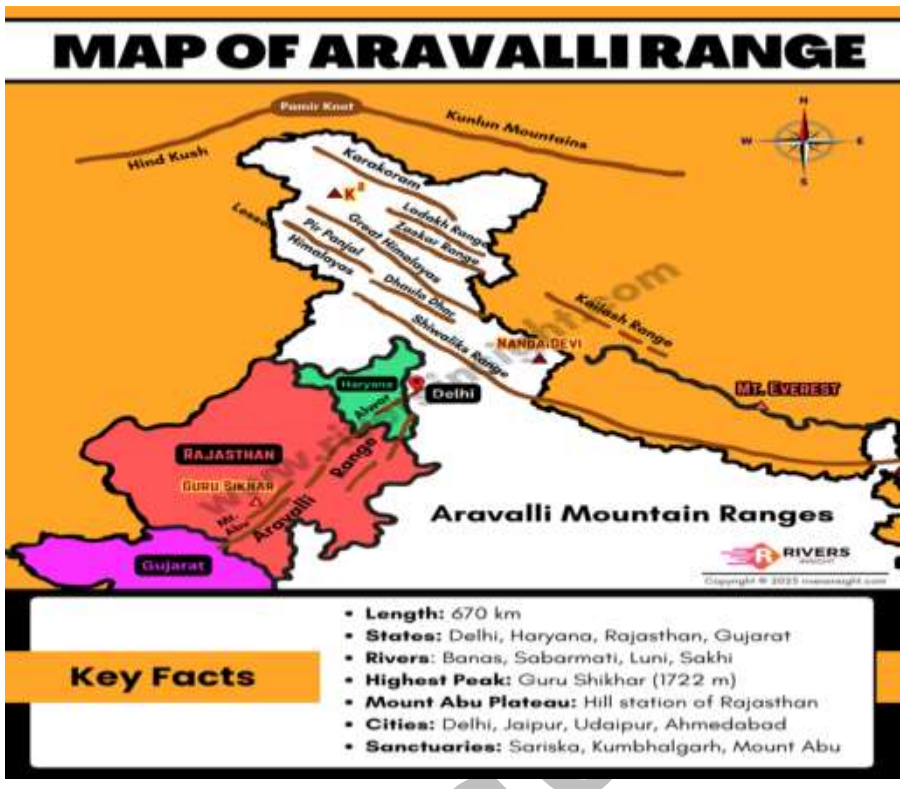
- Reduce desertification.
- Check sand movement and dry winds.
- Support groundwater recharge.
- Protect biodiversity and forest ecosystems.
- Conserve soil moisture in adjoining regions.

Hence, Statement 3 is correct.

VALUE ADDITION

Aravalli Range:

Feature	Details
Type	Residual Fold Mountain System
Geological Age	Proterozoic Era
Length	Approximately 670 km
Orientation	South-West to North-East
Extent	Gujarat – Rajasthan – Haryana – Delhi
Highest Peak	Guru Shikhar (1,722 m)
State with Maximum Extent	Rajasthan
Importance	Ecological Barrier against Desertification



Physiographical Significance of Aravallis

• Natural Barrier

- Prevents eastward expansion of the Thar Desert.
- Reduces wind erosion and sand encroachment.

• Watershed Function

- Forms an important drainage divide in north-western India.
- Rivers drain towards both the Arabian Sea and Yamuna basin.

• Climatic Influence

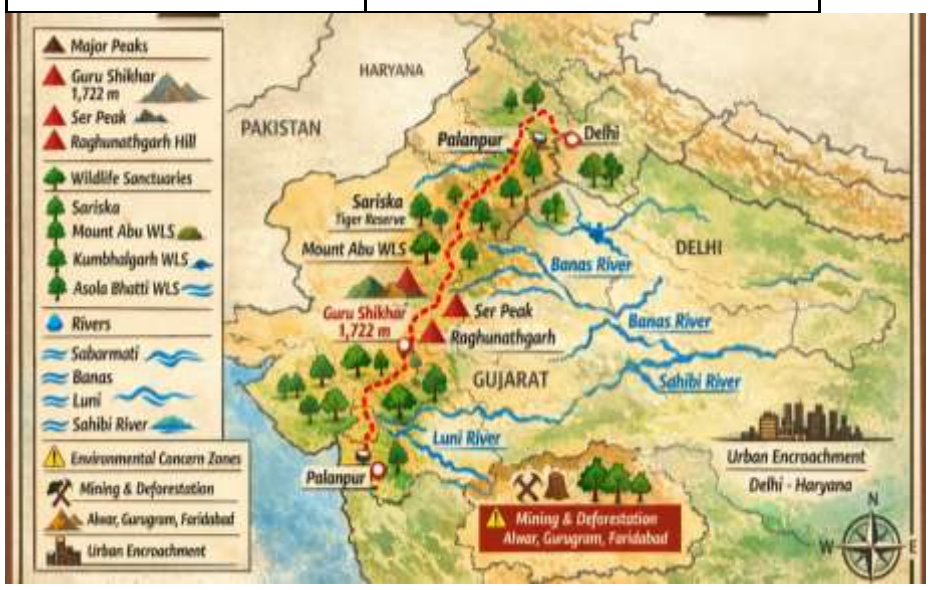
- Influences local rainfall patterns.
- Moderates desert conditions in adjoining regions.

• Groundwater Recharge

- Supports aquifers through forest cover and hill ecosystems.

Important Peaks of the Aravalli Range

Peak	Location
Guru Shikhar	Mount Abu, Rajasthan
Ser Peak	Rajasthan
Achalgarh Peak	Mount Abu Region



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Rivers Associated with the Aravalli Range

River	Drainage
Luni	Arabian Sea Basin
Sabarmati	Arabian Sea Basin
Banas	Tributary of Chambal
Berach	Chambal Basin
Sahibi	Yamuna Basin

Important Note:

- Godavari originates near Trimbakeshwar (Western Ghats).
- Krishna originates near Mahabaleshwar (Western Ghats).
- Therefore, neither river originates from the Aravalli Range.

Ecological Importance

• Biodiversity Hotspot

- Supports dry deciduous forests.
- Habitat for leopard, hyena, sloth bear, and numerous bird species.

• Mineral Resources

- Rich deposits of:
  - Copper
  - Zinc
  - Lead
  - Marble
  - Granite

• Soil Conservation

- Prevents land degradation and desertification.

• Carbon Sink

- Forests contribute to carbon sequestration and climate regulation.

Threats to the Aravallis

• Illegal Mining
• Urbanisation
• Deforestation
• Groundwater Depletion

226. e VTOL

Q.With reference to Electric Vertical Take-off and Landing (eVTOL) aircraft, consider the following statements:

1. An eVTOL aircraft can take off and land vertically without requiring a conventional runway.
2. eVTOL aircraft are generally powered by multiple electric motors and propellers using distributed electric propulsion technology.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

EXPLANATION

IN NEWS:India unveils its first air taxi prototype: Here's how eVTOLs work

Statement 1:

An **eVTOL (Electric Vertical Take-off and Landing) aircraft** is a battery-powered aircraft capable of taking off and landing vertically like a helicopter and then transitioning to forward flight like a conventional airplane. Since it can operate vertically, it does not require long runways. **Hence, Statement 1 is correct.**

Statement 2:

Most eVTOL aircraft use **Distributed Electric Propulsion (DEP)** technology, where multiple electric motors and propellers work together to provide lift, propulsion, efficiency and redundancy. This improves safety because the failure of one motor can be compensated by others. **Hence, Statement 2 is correct.**

VALUE ADDITION

eVTOL Aircraft: Quick Facts

Feature	Details
Full Form	Electric Vertical Take-off and Landing Aircraft
Power Source	Battery-powered electric motors
Take-off & Landing	Vertical
Runway Requirement	Not required
Technology	Distributed Electric Propulsion (DEP)
Emissions	Zero tailpipe emissions
Primary Use	Urban Air Mobility

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Major Applications of eVTOLs

- Air taxi services
- Emergency Medical Services (EMS)
- Cargo and logistics transport
- Airport transfers
- Disaster response and rescue operations
- Recreation and tourism

India's First eVTOL – e200X

Feature	Details
Developer	The ePlane Company (IIT Madras incubated)
Aircraft Type	eVTOL Air Taxi
Payload Capacity	1 pilot + 200 kg payload
Passenger Capacity	Up to 2 passengers
Range	Approximately 110 km
Structure	Carbon-fiber composite
Key Advantage	Can operate from existing helipads and hospital rooftops

227. Allographa

Q.The term "Allographa", recently in the news, is best described as:

(a) A genus of lichen-forming fungi found in tropical regions.
(b) A medicinal plant endemic to the Western Ghats.
(c) A species of cyanobacteria used in agriculture.
(d) A fungal disease affecting forest trees.

Answer: (a) A genus of lichen-forming fungi found in tropical regions.

EXPLANATION:

IN NEWS:Five new species of Allographa discovered in the Western Ghats

Option (a): A genus of lichen-forming fungi found in tropical regions.

Allographa is a genus of script lichens or lichen-forming fungi belonging to the family Graphidaceae. It is characterized by elongated fruiting bodies and comprises more than 200 species globally. Several species occur in tropical and subtropical regions, including the Western Ghats of India. These lichens are highly sensitive to environmental changes and serve as important indicators of ecosystem health. **Hence, Option (a) is correct.**

Option (b): A medicinal plant endemic to the Western Ghats.

Allographa is neither a medicinal plant nor endemic to the Western Ghats. It is a fungal genus associated with lichens and is distributed across different tropical regions of the world. **Hence, Option (b) is incorrect.**

Option (c): A species of cyanobacteria used in agriculture.

Allographa is not a cyanobacterium. Although some lichens may contain cyanobacteria as their photosynthetic partner, Allographa itself refers to a genus of lichen-forming fungi. **Hence, Option (c) is incorrect.**

Option (d): A fungal disease affecting forest trees.

Allographa is not a disease-causing organism. It is a lichen-forming fungal genus that grows on tree bark, rocks and other substrates. **Hence, Option (d) is incorrect.**

VALUE ADDITION

Allographa: Quick Facts

Feature	Details
Type	Genus of lichen-forming fungi(a fungus that teams up with algae or cyanobacteria to create a lichen).
Family	Graphidaceae
Common Group	Script lichens
Global Diversity	More than 200 species
Indian Distribution	Primarily Western Ghats and other humid forest regions
Ecological Importance	Bioindicator of ecosystem health

Newly Discovered Allographa Species from Western Ghats

- Allographa keralensis
- Allographa paraconsanguinea
- Allographa persistinspersa
- Allographa semicarbonisata
- Allographa Nayakae

Lichens

- Lichens are a symbiotic association between:
  - A fungus (mycobiont) providing structure and protection.

○ An alga or cyanobacterium (photobiont) performing photosynthesis.

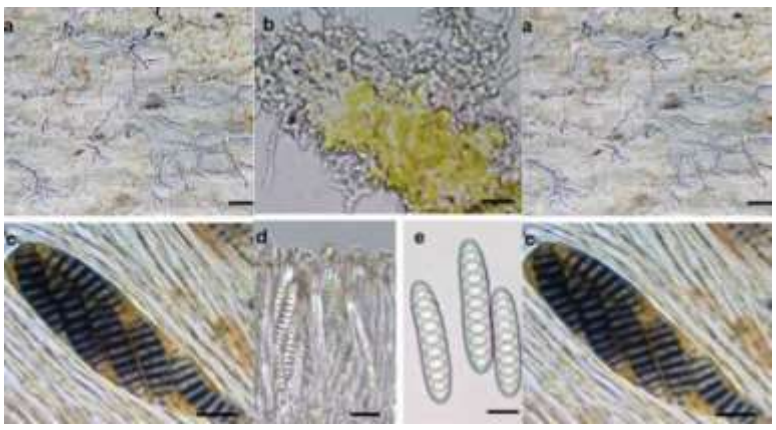
- They are not a single organism but a mutually beneficial partnership.

Ecological Significance of Lichens

- Act as bioindicators of air quality and environmental health.
- Sensitive to pollution, climate change and habitat disturbance.
- Help in soil formation by weathering rocks.
- Contribute to nutrient cycling in ecosystems.
- Provide food and habitat for various organisms.

Lichens as Good Bioindicators

- Absorb water and nutrients directly from the atmosphere.
- Highly sensitive to sulfur dioxide and other pollutants.
- Changes in lichen diversity often indicate environmental degradation.



228. Bal Gangadhar

Q.Consider the following statements:

1. Bal Gangadhar Tilak was one of the prominent leaders of the extremist wing of the Indian National Congress.
2. Tilak founded the Indian Home Rule League, while Annie Besant founded a separate Home Rule League
3. British writer Valentine Chirol described Tilak as the “Father of Indian Unrest”.
4. The Lokmanya Tilak National Award is conferred annually by the Tilak Smarak Mandir Trust on individuals who have made remarkable contributions to nation-building.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (d) All four

EXPLANATION:

In NEWS:NSA Ajit Doval Selected for Lokmanya Tilak National Award l Prime Minister pays tributes to Lokmanya Tilak on his birth anniversary

Statement 1:

Bal Gangadhar Tilak was among the leading figures of the extremist (assertive nationalist) wing of the Indian National Congress. Along with Lala Lajpat Rai and Bipin Chandra Pal, he formed the famous Lal-Bal-Pal trio, which advocated Swaraj, Swadeshi, Boycott and National Education.**Hence, Statement 1 is correct.**

Statement 2:

In 1916, Tilak launched the Home Rule Movement and established the Indian Home Rule League in Maharashtra, Karnataka, Berar and the Central Provinces. Annie Besant simultaneously led the Home Rule League in the rest of India. Together, they popularized the demand for self-government.**Hence, Statement 2 is correct.**

Statement 3:

British journalist and author Sir Valentine Chirol referred to Tilak as the "Father of Indian Unrest" because of his strong opposition to British rule and his role in mobilizing nationalist sentiment.**Hence, Statement 3 is correct.**

Statement 4:

The Lokmanya Tilak National Award was instituted in 1983 by the Tilak Smarak Mandir Trust. It is conferred annually on 1 August, the death anniversary of Lokmanya Tilak, on individuals who have made outstanding contributions towards the progress and development of the nation. **Hence, Statement 4 is correct.**

VALUE ADDITION



Lokmanya Bal Gangadhar Tilak: Quick Facts

Feature	Details
Original Name	Keshav Gangadhar Tilak

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Popular Title	Lokmanya ("Accepted by the people as their leader")
Birth	23 July 1856, Ratnagiri, Maharashtra
Political Group	Extremist Wing of INC
Famous Trio	Lal-Bal-Pal
Famous Slogan	"Swaraj is my birthright and I shall have it"
British Epithet	Father of Indian Unrest
Death	1 August 1920

Major Contributions of Tilak

- Popularised the demand for Swaraj (Self-rule).
- Advocated the four-point programme:
  - Swadeshi
  - Boycott
  - National Education
  - Swaraj
- Converted Ganesh Utsav into a public festival to promote nationalism.
- Started Shivaji Jayanti celebrations.
- Founded:
  - New English School (1880)
  - Deccan Education Society (1884)
- Played a crucial role in the Home Rule Movement (1916).

Literary Works

Newspaper	Language	Founded
Kesari	Marathi	1881
Mahratta	English	1881

Work	Theme
Gita Rahasya	Interpretation of Bhagavad Gita
The Arctic Home in the Vedas	Theory regarding Aryan homeland
The Orion	Study of Vedic chronology

Lokmanya Tilak National Award

Feature	Details
Instituted	1983
Instituting Body	Tilak Smarak Mandir Trust
Occasion	1 August (Death Anniversary of Tilak)
Objective	Honour contributions to nation-building

229. Tenzing Norgay National Adventure Award

Q.With reference to the Tenzing Norgay National Adventure Award, consider the following statements:

1. One of the objectives of the award is to encourage young people to develop endurance, risk-taking ability, teamwork and participation in adventure activities.
2. The award enjoys a status equivalent to the Arjuna Award and is presented along with the National Sports Awards.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

EXPLANATION

In NEWS:Online Nominations Open for Tenzing Norgay National Adventure Award 2025

Statement 1:

The award aims to recognize excellence in adventure activities and promote the spirit of adventure among youth. Its objectives include encouraging endurance, risk-taking, cooperative teamwork, quick decision-making and participation in adventure activities, thereby inspiring young people to engage in challenging pursuits. **Hence, Statement 1 is correct.**

Statement 2:

The Tenzing Norgay National Adventure Award enjoys a status equivalent to the Arjuna Award and is conferred by the Government of India along with the National Sports Awards. **Hence, Statement 2 is correct.**

VALUE ADDITION

Tenzing Norgay National Adventure Award: Quick Facts

Feature	Details
Highest Recognition	India's highest national award for adventure sports
Named After	Tenzing Norgay, one of the first climbers to summit Mount Everest
Nodal Ministry	Ministry of Youth Affairs and Sports
Status	Equivalent to Arjuna Award
Presented Along With	National Sports Awards
Prize Money	₹15 lakh
Other Components	Bronze statuette, certificate, blazer with silken tie/saree

Categories of the Award

- Land Adventure
- Water (Sea) Adventure
- Air Adventure
- Lifetime Achievement in Adventure

Objectives of the Award

- Recognize outstanding achievements in adventure activities.
- Encourage the spirit of adventure among youth.
- Promote endurance and risk-taking abilities.
- Foster teamwork and leadership qualities.
- Motivate participation in adventure sports and expeditions.

230. Mission Vatsalaya

Q.With reference to the Mission Vatsalya Portal, consider the following statements:

1. The Mission Vatsalya Scheme is a Central Sector Scheme fully funded and implemented by the Union Government.
2. The Mission Vatsalya Portal has interoperability with the Crime and Criminal Tracking Network & Systems (CCTNS).
3. Child Welfare Committees (CWCs) and Juvenile Justice Boards (JJBs) are stakeholders in the implementation of the portal.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (b) 2 and 3 only

Explanation

In NEWS:Ministry of Women and Child Development has developed integrated unified Mission Vatsalya Portal

Statement 1:

Mission Vatsalya is a **Centrally Sponsored Scheme (CSS)** implemented through States and Union Territories.The scheme supports:

- Children in Need of Care and Protection (CNCP)
- Children in Conflict with Law (CCL)

While the Ministry of Women and Child Development is the nodal ministry, implementation is carried out by State Governments and UT Administrations. **Hence, Statement 1 is incorrect.**

Statement 2:

The Mission Vatsalya Portal is integrated with **CCTNS**, a platform under the Ministry of Home Affairs.This interoperability enables:

- Matching of FIRs related to missing children.
- Cross-verification with the Mission Vatsalya database.
- Faster tracing and restoration of missing children.

Hence, Statement 2 is correct.

Statement 3:

The TrackChild module under Mission Vatsalya is implemented with the involvement of multiple stakeholders including:

- Child Welfare Committees (CWCs)
- Juvenile Justice Boards (JJBs)
- Ministry of Home Affairs
- Ministry of Railways
- National Legal Services Authority (NALSA)
- State Governments and UT Administrations

These bodies play an important role in child protection, rehabilitation, and restoration. **Hence, Statement 3 is correct.**

VALUE ADDITION

Mission Vatsalya: Key Highlights

- Centrally Sponsored Scheme (since 2021-22).
- Nodal Ministry: Ministry of Women & Child Development.
- Supports:
  - Children in Need of Care and Protection (CNCP).
  - Children in Conflict with Law (CCL).

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- Based on:
  - Juvenile Justice (Care and Protection of Children) Act, 2015.
  - POCSO Act, 2012.
- Motto: **"Leave No Child Behind"**.
- Aligned with Sustainable Development Goals (SDGs).

Integrated Digital Platforms

Portal	Purpose
TrackChild	Tracking missing and found children
Khoya-Paya	Citizen reporting of missing/sighted children
GHAR (Go Home and Re-Unite)	Restoration and repatriation of children

CCTNS:

Feature	Details
Full Form	Crime and Criminal Tracking Network & Systems
Ministry	Ministry of Home Affairs
Objective	Digital crime and criminal tracking
Relevance	Matching FIRs of missing children with Mission Vatsalya database

231. ASEAN

Q. Consider the following statements regarding the Association of Southeast Asian Nations (ASEAN):

1. India is a member of ASEAN.
2. ASEAN is India's fourth-largest trading partner.
3. ASEAN is a military alliance similar to NATO.
4. A ASEAN member is also a member of the Regional Comprehensive Economic Partnership (RCEP).

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (b) Only two

EXPLANATION:

In NEWS:India strongly condemns Pakistan's comments at ASEAN as baseless and unwarranted

Statement 1:

India is not a member of ASEAN. ASEAN currently consists of 11 Southeast Asian countries. India is a Dialogue Partner, Summit-Level Partner, and an important strategic partner of ASEAN, but it is not a member of the organization. **Hence, Statement 1 is incorrect.**

Statement 2:

ASEAN is among India's most important economic partners and is currently India's fourth-largest trading partner. Trade between India and ASEAN has expanded significantly following the implementation of the ASEAN-India Free Trade Agreement (AIFTA). **Hence, Statement 2 is correct.**

Statement 3:

ASEAN is primarily a political, economic, and socio-cultural organization aimed at promoting regional cooperation and stability. Unlike NATO, ASEAN does not have a collective defence clause or a military command structure.

ASEAN follows principles such as:

- Consensus-based decision-making
- Non-interference in internal affairs
- Peaceful settlement of disputes **Hence, Statement 3 is incorrect.**

Statement 4:

The Regional Comprehensive Economic Partnership (RCEP) includes all ten original ASEAN members along with ASEAN's FTA partners such as China, Japan, South Korea, Australia, and New Zealand. Timor-Leste's ASEAN membership is recent, but ASEAN as a grouping forms the core of RCEP. **Hence, Statement 4 is correct.**

VALUE ADDITION

ASEAN: Quick Facts

Feature	Details
Full Form	Association of Southeast Asian Nations
Established	8 August 1967
Founding Document	Bangkok Declaration
Headquarters	Jakarta, Indonesia
Members	11

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Latest Member	Timor-Leste
Motto	One Vision, One Identity, One Community

ASEAN vs NATO

Feature	**ASEAN**	**NATO**
Nature	Regional Cooperation Organisation	Military Alliance
Established	1967	1949
Headquarters	Jakarta	Brussels
Collective Defence Clause	No	Yes (Article 5)
Decision Making	Consensus	Political-Military Coordination
Primary Focus	Economic, Political & Security Cooperation	Collective Defence & Security

ASEAN and India

Major Areas of Cooperation

- Trade and Investment
- Maritime Security
- Digital Connectivity
- Defence Cooperation
- Indo-Pacific Cooperation
- Cultural and Educational Exchanges

RCEP: Quick Facts

Feature	**Details**
Full Form	Regional Comprehensive Economic Partnership
Entered into Force	2022
Members	15
Core Group	ASEAN-led FTA
World's Largest Trade Bloc	Yes (by population)
India	Not a member

RCEP Members

ASEAN Members	**Non-ASEAN Members**
Brunei	China
Cambodia	Japan
Indonesia	South Korea
Laos	Australia
Malaysia	New Zealand
Myanmar	
Philippines	
Singapore	
Thailand	
Vietnam	
Timor-Leste*	

*Timor-Leste joined ASEAN recently and its RCEP participation remains a separate process.

232. Pickaxe mountain

Q.Pickaxe Mountain, recently seen in the news, is located in which country?

(a) Iraq
(b) Iran
(c) Syria
(d) Saudi Arabia

Answer: (b) Iran

EXPLANATION

IN NEWS:Trump's bunker buster gamble on Pickaxe Mountain

Option (a): Iraq

Pickaxe Mountain is not located in Iraq. The facility that has recently attracted international attention is situated within Iran and is linked to concerns regarding Iran's nuclear infrastructure.**Hence, Option (a) is incorrect.**

Option (b): Iran

Pickaxe Mountain (Kūh-e Kolang Gaz Lā) is a heavily fortified underground facility located in Isfahan Province, Iran, approximately 220 km south of Tehran and close to the Natanz nuclear enrichment

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complex.The facility is excavated deep inside the Zagros Mountains and is believed to be buried more than 100 metres underground, making it highly resistant to conventional military strikes.**Hence, Option (b) is correct.**

Option (c): Syria

Pickaxe Mountain is not located in Syria. Although Syria has been associated with regional geopolitical and security issues, the facility in question is situated in Iran.**Hence, Option (c) is incorrect.**

Option (d): Saudi Arabia

Pickaxe Mountain is not located in Saudi Arabia. The facility lies within Iranian territory and is associated with Iran's nuclear programme.**Hence, Option (d) is incorrect.**

VALUE ADDITION



Pickaxe Mountain: Quick Facts

Feature	Details
Also Known As	Kūh-e Kolang Gaz Lā
Country	Iran
Province	Isfahan
Mountain Range	Zagros Mountains
Distance from Tehran	~220 km south
Nearby Nuclear Facility	Natanz Nuclear Enrichment Complex
Elevation	~1,600 m above sea level
Depth of Facility	More than 100 m underground
Geological Formation	Granite Bedrock

Why is Pickaxe Mountain Important?

- Deeply buried and heavily fortified underground facility.
- Located near the strategic Natanz nuclear complex.
- Suspected by Western intelligence agencies to have a role in Iran's nuclear programme.
- Iran maintains that it is intended for centrifuge assembly and production.
- Limited access for the International Atomic Energy Agency (IAEA) has raised concerns regarding transparency and nuclear non-proliferation.

Natanz Nuclear Complex

Feature	Details
Location	Isfahan Province, Iran
Function	Uranium Enrichment Facility
Significance	Core component of Iran's nuclear programme
IAEA Monitoring	Subject to international scrutiny

International Atomic Energy Agency (IAEA)

Feature	Details
Established	1957
Headquarters	Vienna, Austria
Related To	United Nations System
Objective	Promote peaceful use of nuclear energy and prevent military use of nuclear technology
Motto	Atoms for Peace and Development

233. Great scheme

Q.With reference to the Grant for Research & Entrepreneurship across Aspiring Innovators in Technical Textiles (GREAT) Scheme, consider the following statements:

1. It was launched under the National Technical Textiles Mission (NTTM).
2. It provides financial assistance only to government research institutions working in technical textiles.
3. The scheme supports start-ups in translating innovative ideas into commercial technologies and products.

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Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (b) 1 and 3 only
IN NEWS:GREAT SCHEME
Statement 1:
The GREAT Scheme was launched in **August 2023** under the **Research, Development and Innovation (R&D&I) component** of the National Technical Textiles Mission (NTTM).**Hence, Statement 1 is Correct.**
Statement 2:
The scheme is designed to support **start-ups, entrepreneurs, innovators, scientists, and technologists** for developing prototypes and commercializing technologies in Technical Textiles. It is not limited to government research institutions.**Hence, Statement 2 is Incorrect.**
Statement 3:
One of the primary objectives of the GREAT Scheme is to help innovators convert research ideas into **commercially viable technologies and products**, thereby strengthening India's Technical Textiles ecosystem.**Hence, Statement 3 is Correct.**
Value Addition
GREAT Scheme – Key Facts

Particulars	**Details**
Launch Year	2023
Under	National Technical Textiles Mission (NTTM)
Ministry	Ministry of Textiles
Objective	Support innovation and commercialization in Technical Textiles
Beneficiaries	Start-ups, entrepreneurs, scientists, technologists
Financial Assistance	Up to ₹50 lakh
Duration of Support	Up to 18 months
Coverage	Pan-India

234. 4.Picrorhiza kurroa

Q.With reference to Picrorhiza kurroa (locally known as Kutki), consider the following statements:

1. It is a high-value medicinal herb native to the alpine and sub-alpine regions of the Himalayas.
2. The roots of the plant are the primary source of its medicinal properties and are harvested for trade.
3. It is classified as 'Endangered' under the IUCN Red List and is listed under Appendix II of CITES.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (c) 1 and 3 only
EXPLANATION
In NEWS:Himachal farmers pioneer sustainable cultivation of endangered herb
Statement 1:
Picrorhiza kurroa, commonly known as **Kutki**, is a medicinal herb naturally found in the **alpine and sub-alpine zones of the Himalayas**, generally at elevations ranging from **2,700–4,500 metres**.It is widely used in traditional medicinal systems such as **Ayurveda**, particularly for liver-related disorders and immune support.**Hence, Statement 1 is correct.**
Statement 2:
For many years, there was a misconception that the **roots** constituted the medicinal part of the plant. However, researchers have clarified that the commercially valuable and medicinally important part is the **stolon**—a creeping stem-like side branch possessing nodes.The roots are fibrous and do not contain significant medicinal compounds. Sustainable harvesting now focuses on collecting only the stolons while leaving the mother plant intact.**Hence, Statement 2 is incorrect.**
Statement 3:
Picrorhiza kurroa is categorized as **Endangered** on the **IUCN Red List** due to overexploitation and destructive harvesting practices.To regulate international trade and prevent further depletion of wild populations, it has been included in **Appendix II of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES)**.**Hence, Statement 3 is correct.**
VALUE ADDITION

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Picrorhiza kurroa (Kutki): Quick Facts

Feature	Details
Common Name	Kutki
Scientific Name	*Picrorhiza kurroa*
Type	Medicinal Herb
Habitat	Alpine and Sub-alpine Himalayas
Altitude Range	2,700–4,500 m
Major States	Himachal Pradesh, Uttarakhand, Jammu & Kashmir, Sikkim
Conservation Status	Endangered (IUCN)
CITES Status	Appendix II

Sustainable Harvesting Model

Traditional Method

- The entire plant was uprooted from the wild.
- Led to population decline and habitat degradation.
- Threatened long-term survival of the species.

Scientific "Stolon Harvesting" Method

- Only side stolons are harvested.
- The mother plant remains intact.
- Allows regeneration and perennial cultivation.
- Reduces pressure on wild populations.
- Enhances farmer income while supporting conservation.

IUCN Red List Categories

Category	Meaning
CR	Critically Endangered
EN	Endangered
VU	Vulnerable
NT	Near Threatened
LC	Least Concern

Kutki is currently listed as Endangered (EN).

CITES Appendix Classification

Appendix	Significance
Appendix I	Trade generally prohibited except under exceptional circumstances
Appendix II	Regulated trade through permits to prevent overexploitation
Appendix III	Species protected in at least one country seeking international cooperation

Kutki is included in Appendix II.

Importance of Medicinal Plants in Himalayan Economy

- Source of livelihood for mountain communities.
- Supports traditional healthcare systems.
- Promotes biodiversity conservation.
- Encourages sustainable agro-forestry practices.
- Contributes to the AYUSH sector and herbal pharmaceutical industry.

235. East Asia Summit(EAS)

Q.With reference to the East Asia Summit (EAS), consider the following countries:

1. Mongolia
2. Canada
3. India
4. Russia
5. United States
6. Australia

How many of the above are members of the East Asia Summit (EAS)?

(a) Only two
(b) Only three
(c) Only four
(d) Only five

Answer: (c) Only four

EXPLANATION

IN NEWS: Global waterways should be safe, unimpeded: Jaishankar

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Statement 1: Mongolia

Mongolia is not a member of the East Asia Summit. Although it maintains close relations with ASEAN and participates in various regional forums, it has not been admitted as an EAS member. **Hence, Statement 1 is incorrect.**

Statement 2: Canada

Canada is not a member of the East Asia Summit. While Canada is a Dialogue Partner of ASEAN, it is not part of the EAS framework. **Hence, Statement 2 is incorrect.**

Statement 3: India

India is one of the founding participating countries of the East Asia Summit and has been a member since the inaugural summit in 2005. **Hence, Statement 3 is correct.**

Statement 4: Russia

Russia joined the East Asia Summit in 2011 along with the United States, expanding the forum's strategic significance. **Hence, Statement 4 is correct.**

Statement 5: United States

The United States became a member of the East Asia Summit in 2011 and actively participates in discussions relating to regional security, connectivity and economic cooperation. **Hence, Statement 5 is correct.**

Statement 6: Australia

Australia is a founding member of the East Asia Summit and has been participating since its inception in 2005. **Hence, Statement 6 is correct.**

Therefore, India, Russia, United States and Australia are members of the East Asia Summit.Hence, only four of the above countries are members of the EAS.

VALUE ADDITION

East Asia Summit (EAS): Quick Facts

Feature	**Details**
Established	2005
First Summit	Kuala Lumpur, Malaysia
Nature	Leaders-led forum for strategic, political and economic dialogue
Total Members	18
ASEAN Members	10
Non-ASEAN Members	India, China, Japan, South Korea, Australia, New Zealand, United States, Russia
Secretariat Support	ASEAN Secretariat, Jakarta

Membership Criteria for EAS

A country must:

- Be a Dialogue Partner of ASEAN
- Accede to the Treaty of Amity and Cooperation (TAC)
- Have substantial relations with ASEAN

Priority Areas of Cooperation

Area	**Focus**
Environment & Energy	Sustainable development
Education	Human resource development
Finance	Regional economic cooperation
Global Health	Pandemic preparedness
Disaster Management	Humanitarian cooperation
ASEAN Connectivity	Physical and digital connectivity

Members

ASEAN Members	**Non-ASEAN Members**
Brunei	India
Cambodia	China
Indonesia	Japan
Laos	South Korea
Malaysia	Australia
Myanmar	New Zealand
Philippines	United States
Singapore	Russia
Thailand	

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236. PM Viswakarma

Q.With reference to the PM Vishwakarma Scheme, consider the following statements:

1. It is a Central Sector Scheme implemented by the Ministry of Micro, Small and Medium Enterprises.
2. It aims to support traditional artisans and craftspeople engaged in family-based occupations.
3. Beneficiaries are eligible for collateral-free enterprise development loans.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION

IN NEWS:The Ministry of MSME is implementing it's PM Vishwakarma Scheme in West Bengal in collaboration with the State Government

Explanation

Statement 1:

PM Vishwakarma is a **Central Sector Scheme** launched by the **Ministry of Micro, Small and Medium Enterprises (MSME)** to support traditional artisans and craftspeople across India.**Hence, Statement 1 is Correct.**

Statement 2:

The scheme seeks to strengthen and nurture the **Guru-Shishya Parampara** (family-based practice of traditional skills) by supporting artisans and craftspeople working with their hands and tools.**Hence, Statement 2 is Correct.**

Statement 3:

Under the credit support component, beneficiaries can avail **collateral-free enterprise development loans up to ₹3 lakh** in two tranches (₹1 lakh and ₹2 lakh) at a concessional interest rate of 5%.**Hence, Statement 3 is Correct.**

Value Addition

Feature	**PM Vishwakarma Scheme**
Type	Central Sector Scheme
Ministry	Ministry of MSME
Launch Period	FY 2023-24 to FY 2027-28
Target Group	Traditional artisans and craftspeople
Coverage	Rural and Urban Areas
Training Stipend	₹500 per day
Toolkit Incentive	Up to ₹15,000
Credit Support	Up to ₹3 lakh (Collateral-free)
Interest Rate	5%

237. Zoopharmacognosy and Geophagy

Q.Consider the following statements:

1. Zoopharmacognosy refers to the practice of animals consuming soil, clay, or earth materials for therapeutic purposes.
2. Geophagy refers to the behaviour of animals self-medicating using plants, roots, fruits, or other natural substances.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (d) Neither 1 nor 2

EXPLANATION:

IN NEWS:Zoopharmacognosy — when animals medicate themselves

Statement 1:

Zoopharmacognosy is the study of animal self-medication, where animals use plants, roots, fruits, insects, or other natural substances to prevent or treat diseases.

Examples:

- Orangutans applying medicinal plant extracts on wounds.
- Chimpanzees consume specific plants to eliminate parasites. **Hence, Statement 1 is incorrect.**

Statement 2:

Geophagy refers to the deliberate consumption of soil, clay, or earth materials by animals (and sometimes humans).

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Examples:

- Elephants, monkeys, and other animals consuming clay to obtain minerals or relieve digestive disorders.It does not refer to animal self-medication using plants. **Hence, Statement 2 is incorrect.**

VALUE ADDITION

Important Terms

Term	Meaning
Zoopharmacognosy	Animal self-medication using plants, roots, fruits, insects, etc.
Geophagy	Consumption of soil, clay, or earth materials
Self-medication	Behaviour through which animals reduce illness or parasite load using natural substances

Examples of Zoopharmacognosy

Animal	Behaviour
Orangutan	Applies medicinal plant paste on wounds
Chimpanzee	Consumes medicinal plants to expel parasites
Elephant	Eats oregano leaves for respiratory relief
Dogs & Cats	Eat grass to aid digestion and remove parasites

Examples of Geophagy

Animal	Purpose
Elephants	Mineral supplementation
Monkeys	Detoxification and digestive aid
Parrots	Neutralize plant toxins
Sheep	Nutrient supplementation

238. 8.Geothermal Energy

Q.With reference to Geothermal Energy in India, consider the following statements:

1. Geothermal energy is considered renewable because the Earth's internal heat is continuously replenished.
2. India commissioned its first and deepest geothermal wells at Puga Valley in Ladakh.
3. The project site is located within the Himalayan Geothermal Province along a tectonic plate collision zone.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

In NEWS:Puga Valley: What does it mean to be India's first geothermal power project?

Statement 1:

Geothermal energy is derived from the Earth's interior. The heat is continuously replenished through residual heat from Earth's formation and the natural decay of radioactive elements, making it a renewable energy source.**Hence, Statement 1 is Correct.**

Statement 2:

India recently commissioned its first geothermal wells at Puga Valley, drilled to a depth of 1,000 metres, making them the deepest geothermal wells in the country.**Hence, Statement 2 is Correct.**

Statement 3:

Puga Valley falls within the Himalayan Geothermal Province, one of India's recognised geothermal provinces. The geothermal activity in the region is associated with the Indian and Eurasian tectonic plate collision zone, which creates favourable conditions for geothermal resources.**Hence, Statement 3 is Correct.**

Value Addition

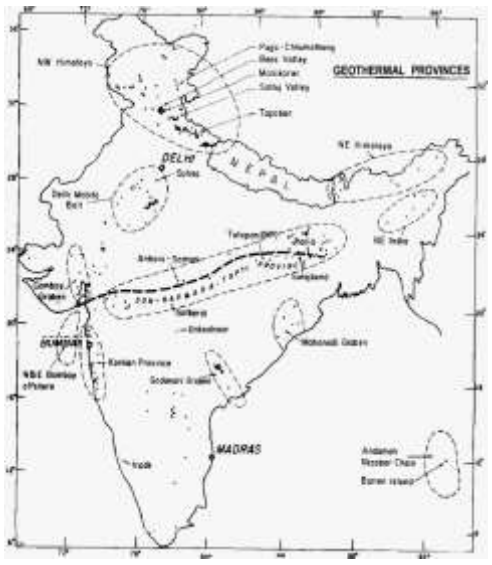
Puga Geothermal Project – Key Facts

Particulars	Details
Location	Puga Valley, Ladakh
Region	Changthang Plateau

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Well Depth	1,000 metres
Altitude	More than 14,000 feet
Proposed Capacity	1 MW Pilot Plant
Implementing Agency	ONGC Energy Centre
Renewable Source	Geothermal Energy

India's Geothermal Provinces



- Himalayan Province
- Naga-Lusai Province
- Andaman & Nicobar Province
- Son–Narmada–Tapi Province
- Cambay Graben Province
- West Coast Province
- Aravalli Province
- Mahanadi Province
- Godavari Province
- South Indian Cratonic Province

Renewable Energy Comparison

Parameter	Geothermal	Solar	Wind
Availability	24×7 Baseload Power	Daytime only	Depends on wind
Weather Dependence	No	High	High
Land Requirement	Low	High	Moderate
Capacity Factor	Very High (70–90%)	Moderate (15–25%)	Moderate (25–40%)
Emissions	Very Low	Very Low	Very Low

Types of Geothermal Resources

Type	Description
Hydrothermal Resources	Naturally occurring hot water/steam reservoirs
Geopressured Resources	Hot brine reservoirs under high pressure
Hot Dry Rock (HDR)	Hot rocks with little or no water
Enhanced Geothermal Systems (EGS)	Artificially fractured rocks to extract heat

Why Puga Valley is Important

- Among the most promising geothermal sites in India.
- Located in the Changthang region of Ladakh.
- Presence of:
  - Hot springs
  - Fumaroles (steam vents)
  - Mud pools
  - Sulphur deposits
- Estimated reservoir temperatures suitable for commercial power generation.

Geothermal Manifestations (UPSC Fact)

Term	Meaning
Hot Spring	Naturally emerging heated groundwater
Geyser	Intermittent eruption of hot water and steam
Fumarole	Opening emitting steam and gases
Mud Pool	Pool of bubbling mud heated by geothermal activity

Advantages of Geothermal Energy

- 24×7 power generation.
- Low carbon footprint.
- Small land requirement.
- High reliability and capacity factor.
- Suitable for remote regions such as Ladakh.
- Can provide both electricity and direct heating.

Challenges in India

- High initial exploration cost.
- Technological limitations.
- Remote location of resources.
- Lack of private investment.
- Geological uncertainties.

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National Policy on Geothermal Energy, 2025

Objectives:

- Promote exploration and development of geothermal resources.
- Encourage public-private partnerships.
- Facilitate pilot projects.
- Support India's net-zero and clean energy goals.

Nodal Ministry: Ministry of New and Renewable Energy (MNRE)

Global Leaders in Geothermal Power (2024)

Country	**Significance**
United States	Largest geothermal power producer
Indonesia	Second-largest producer
Philippines	Major geothermal producer
Turkey	Rapidly growing sector
Kenya	Leading producer in Africa

239. Borophene

Q.Recently, borophene was in the news for its application as:

(a) A catalyst for uranium enrichment
(b) A lubricant additive to improve the performance of castor oil
(c) A substitute for lithium in batteries
(d) A coating material for thermal power plants

Answer: (b) A lubricant additive to improve the performance of castor oil

EXPLANATION

In NEWS:Borophene-based green lubricant can help save energy

Option (a):

A catalyst is a substance that speeds up a chemical reaction without being consumed in the process. The recent research on borophene was not related to nuclear technology or uranium enrichment. Instead, it focused on reducing friction and wear in mechanical systems.**Hence, Option (a) is Incorrect.**

Option (b):

Scientists recently demonstrated the use of **borophene as a lubricant additive in castor oil**, a renewable and biodegradable lubricant. Borophene dispersed uniformly in the oil and formed a protective layer called a **tribofilm** on contact surfaces.The study found that adding only **0.1% borophene by weight** reduced friction by about **42%**, while also improving wear resistance and load-bearing capacity. This breakthrough highlights the potential of borophene in green lubrication technologies, renewable energy systems, marine applications, and sustainable manufacturing.**Hence, Option (b) is Correct.**

Option (c):

Although borophene is being explored for various advanced applications such as energy storage and electronics, the recent news was not related to replacing lithium in batteries.**Hence, Option (c) is Incorrect.**

Option (d):

The recent research did not involve thermal power plants or coating technologies. The focus was specifically on lubrication and friction reduction.**Hence, Option (d) is Incorrect.**

Value Addition: Borophene

- **Borophene** is a **single-atom-thick, two-dimensional (2D) allotrope of boron**.
- Borophene is a metallic 2D material unlike graphene, which is a semimetal.
- It was **theoretically predicted in the mid-1990s** and **experimentally confirmed in 2015**.
- Unlike graphene, borophene is **highly polymorphic**, meaning it can exist in multiple structural forms.
- Key properties:
  - Ultra-lightweight
  - Highly flexible and strong
  - Excellent thermal and electrical conductivity
  - Metallic nature
  - Highly reactive
- Applications:
  - Electronics and optoelectronics
  - Biosensors and chemical sensors
  - Batteries and supercapacitors
  - Energy storage and catalysis
  - Advanced lubricants
- **Recent Breakthrough:** Indian scientists used borophene as an additive in **castor oil**, reducing friction by about **42%** and improving wear resistance.
- During operation, borophene forms a protective **tribofilm** that minimizes friction and surface damage.

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2D Materials at a Glance

Material	Element
Graphene	Carbon
Borophene	Boron
Silicene	Silicon
Germanene	Germanium
Phosphorene	Phosphorus

240. Methane

Q.With reference to Methane, consider the following statements:

1. Methane is the primary component of natural gas and is a colourless, odourless gas.
2. Methane remains in the atmosphere for a longer period than carbon dioxide.
3. Agriculture is one of the major anthropogenic sources of methane emissions.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (b) 1 and 3 only

Explanation

In NEWS:India's rice fields now rank among world's top methane hotspots

Statement 1:

Methane is a hydrocarbon and the primary component of natural gas. It is a **colourless, odourless, and highly flammable gas**, often referred to as **marsh gas**.**Hence, Statement 1 is Correct.**

Statement 2:

Methane has a relatively short atmospheric lifetime of about **12 years**, whereas carbon dioxide can remain in the atmosphere for **hundreds to thousands of years**.Although methane is much more effective at trapping heat in the short term, it does **not** remain in the atmosphere longer than carbon dioxide.**Hence, Statement 2 is Incorrect.**

Statement 3:

The major anthropogenic (human-induced) sources of methane emissions are:

- **Agriculture** (paddy cultivation, livestock)
- **Energy sector** (oil, gas, coal extraction)
- **Waste sector** (landfills, wastewater)

Therefore, agriculture is one of the major sources of methane emissions.**Hence, Statement 3 is Correct.**

VALUE ADDITION

Methane (CH_4)

Methane (CH_4) is the simplest hydrocarbon, the primary component of **natural gas**, and is also known as **marsh gas**.

- It is a **colourless, odourless, highly flammable greenhouse gas** responsible for nearly **30% of global warming since the Industrial Revolution**.
- Though it remains in the atmosphere for only about **12 years**, it is a much stronger greenhouse gas than CO_2, trapping **86 times more heat over 20 years** and **28 times more heat over 100 years**.
- **Major natural sources:** wetlands, termites, oceans, and permafrost.
- **Major anthropogenic sources:** agriculture (especially paddy cultivation and livestock), energy sector (oil, gas, coal), and waste sector (landfills and wastewater).
- **Paddy fields** emit methane because flooded conditions create an **anaerobic (low-oxygen) environment** where methanogenic microbes thrive.
- India is among the major methane emitters from agriculture, and **eastern Indian rice fields** are identified as important methane hotspots.
- Important mitigation measures include **Alternate Wetting and Drying (AWD)**, **System of Rice Intensification (SRI)**, **Direct Seeded Rice (DSR)**, improved manure management, and biogas production.
- Globally, methane reduction efforts are promoted through the **Global Methane Pledge (2021)**, which aims to reduce methane emissions by **30% by 2030**.

241. Jantar Mantar

Q. With reference to the Jantar Mantar observatories in India, consider the following statements:

1. Maharaja Sawai Jai Singh II established observatories at Delhi, Jaipur, Ujjain, Varanasi and Mathura.
2. The Jantar Mantar at Jaipur is a UNESCO World Heritage Site and houses the world's largest stone sundial.
3. The Misra Yantra, found at the Delhi Jantar Mantar, is used to determine the shortest and longest

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days of the year.
4. All observations at Jantar Mantar were originally carried out using telescopes.

Which of the statements given above are correct?

(a) 1, 2 and 3 only
(b) 1 and 4 only
(c) 2 and 3 only
(d) 1, 2, 3 and 4

Answer: (a) 1, 2 and 3 only
EXPLANATION:
In NEWS: What Does Jantar Mantar Mean?
1. Five Observatories Established by Jai Singh II
Maharaja Sawai Jai Singh II established five observatories at Delhi, Jaipur, Ujjain, Varanasi and Mathura during the early eighteenth century to improve the accuracy of astronomical observations. The Mathura observatory was later destroyed before the Revolt of 1857. **Hence, Statement 1 is correct.**
2. UNESCO World Heritage Site and Largest Stone Sundial
The Jaipur Jantar Mantar was inscribed as a UNESCO World Heritage Site in 2010. It houses the Samrat Yantra, which is recognised as the world's largest stone sundial and can measure time with remarkable precision. **Hence, Statement 2 is correct.**
3. Misra Yantra at Delhi
The Misra Yantra is a unique instrument located at the Delhi Jantar Mantar. It is used to determine the shortest and longest days of the year and perform other astronomical calculations. **Hence, Statement 3 is correct.**
4. Naked-Eye Observations
The Jantar Mantar observatories were designed for naked-eye astronomical observations. Their large masonry instruments enabled astronomers to measure celestial positions without the use of telescopes. **Hence, Statement 4 is incorrect.**

242. National Testing Agency

Q. With reference to the National Testing Agency (NTA), consider the following statements:

1. The National Testing Agency was established by the Ministry of Education as an autonomous organisation to conduct entrance examinations for higher educational institutions.
2. The National Testing Agency conducts the National Eligibility-cum-Entrance Test (NEET-UG), Joint Entrance Examination (JEE Main), and Common University Entrance Test (CUET).
3. The Union Public Service Commission (UPSC) functions under the administrative control of the National Testing Agency for conducting the Civil Services Examination.
4. The National Testing Agency was established to promote transparent, efficient and technology-driven examination systems.

Which of the statements given above are correct?

(a) 1, 2 and 4 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2, 3 and 4

Answer: (a) 1, 2 and 4 only
EXPLANATION:

IN NEWS: Centre sacks 47 NTA officials, tightens exam security law
1. Establishment of the National Testing Agency
The National Testing Agency (NTA) was established in **2018** by the **Ministry of Education** (then Ministry of Human Resource Development) as an **autonomous and specialised testing organisation**. It was created to conduct high-quality entrance examinations for admissions into higher educational institutions across India.
The objective was to establish a professional testing body capable of conducting examinations in a standardised, transparent and efficient manner while reducing the burden on universities and other institutions. **Hence, Statement 1 is correct.**
2.Conducts Major National Entrance Examinations
NTA is responsible for conducting several major examinations, including:
- **NEET-UG (National Eligibility-cum-Entrance Test)** for medical admissions.
- **JEE Main (Joint Entrance Examination)** for engineering admissions.
- **CUET (Common University Entrance Test)** for undergraduate and postgraduate admissions in central and participating universities.
- **UGC-NET** for determining eligibility for Assistant Professorship and Junior Research Fellowship.
- **CSIR-UGC NET** and other specialised examinations.

Over the years, NTA has become one of the world's largest examination-conducting agencies, handling millions of candidates annually. **Hence, Statement 2 is correct.**

3. UPSC is an Independent Constitutional Body

The **Union Public Service Commission (UPSC)** is a **constitutional body** established under **Article 315 of the Constitution of India**.

It conducts examinations such as:

- Civil Services Examination (CSE)
- Indian Forest Service Examination
- Engineering Services Examination
- NDA and CDS Examinations

UPSC functions independently and is not under the administrative control of the National Testing Agency. The NTA has no role in conducting UPSC examinations. **Hence, Statement 3 is incorrect.**

4. Objectives Behind Creation of NTA

The NTA was established to:

- Conduct **transparent and merit-based examinations**.
- Improve efficiency through **technology-driven testing systems**.
- Standardise examination processes across the country.
- Reduce administrative burdens on educational institutions.
- Strengthen fairness and credibility in admissions.

The agency also expanded computer-based testing and introduced modern examination-management practices. **Hence, Statement 4 is correct.**

Additional Points for UPSC

Parliamentary Standing Committee Observations (371st Report, 2025)

The Parliamentary Standing Committee on Education expressed concerns regarding NTA's functioning and noted that:

- Out of 14 examinations conducted in 2024, at least 5 faced significant issues.
- UGC-NET, CSIR-NET and NEET-PG witnessed postponements.
- NEET-UG faced allegations of paper leaks.
- CUET experienced delays in result declaration.
- JEE Main 2025 saw withdrawal of multiple questions from the final answer key due to errors.

Challenges Faced by NTA

- Paper leaks and examination security concerns.
- Cybersecurity threats and dark-web-based dissemination of questions.
- Logistical management across thousands of examination centres.
- Technical glitches in computer-based tests.
- Maintaining public trust in high-stakes examinations.

243. Pachycheles ravichandrani

Q. With reference to *Pachycheles ravichandrani*, recently seen in the news, consider the following statements:

1. It is a newly discovered species of porcelain crab found in the shallow coral reefs of Lakshadweep.
2. The species was discovered near Agatti Island and was found associated with coralline algae-covered rock boulders.
3. The species has been named in honour of Dr. M. Ravichandran for his contributions to India's Deep Ocean Mission and marine exploration.

Which of the statements given above are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION :

IN NEWS: New species of 'porcelain crab' with a striking red shell discovered in Lakshadweep's coral reefs

1. Newly Discovered Porcelain Crab from Lakshadweep

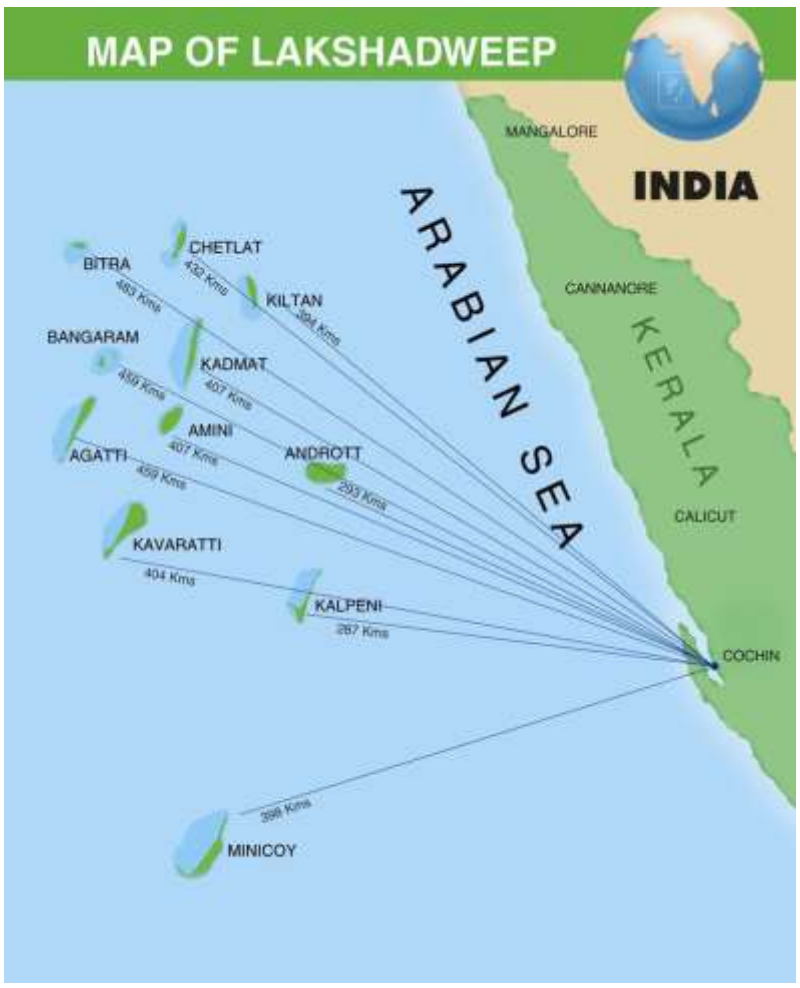


Pachycheles ravichandrani is a newly identified species of **porcelain crab** discovered from the **Lakshadweep Islands** in the Arabian Sea. The discovery adds to the known marine biodiversity of India and highlights the ecological significance of India's coral reef ecosystems.

Porcelain crabs belong to the family **Porcellanidae** and are known for their delicate body structure, flattened appearance and ability to shed limbs as a

defence mechanism against predators. **Hence, Statement 1 is correct.**

2. Discovery Near Agatti Island



The species was discovered during a marine biodiversity survey conducted along the rocky shores of **Agatti Island** in Lakshadweep.

Researchers found the crab hiding among **rock boulders covered with coralline algae** at depths of less than one metre. Such habitats provide shelter and food resources for a variety of marine organisms and are important components of shallow reef ecosystems.

The discovery demonstrates that even relatively accessible coastal habitats may contain previously unknown species. **Hence, Statement 2 is correct.**

3. Named After Dr. M. Ravichandran

The species has been named ***Pachycheles ravichandrani*** in honour of **Dr. M. Ravichandran**, Secretary, Ministry of Earth Sciences.

The naming recognises his leadership in advancing India's marine scientific programmes, particularly the **Deep Ocean Mission**, which aims to strengthen deep-sea exploration, marine resource assessment and ocean research capabilities.

Naming newly discovered species after eminent scientists is a common scientific practice to acknowledge their contributions to a particular field. **Hence, Statement 3 is correct.**

Unique Features of the Species

- Exhibits a distinctive **deep carmine-red colouration**.
- Possesses a **white blotch on the dorsal shell** and white spots on the claws.
- Walking legs display characteristic **horizontal carmine stripes**.
- Has a unique **sub-circular carapace (shell)**.
- Features specialised hair-like structures called **setae**.
- Displays a distinct gap between the fingers of its smaller claw, helping distinguish it from related species.

244. NSE Ahimsa Index

Q. With reference to the NSE Ahimsa Index, recently seen in the news, consider the following statements:

1. The Nifty 500 Ahimsa Index tracks companies whose business practices align with the principle of Ahimsa (non-violence) and animal welfare.
2. The index was developed jointly by NSE Indices and the Ahimsagain Foundation.
3. Only companies classified under the Green band of the Ahimsa Investment Movement (AIM) framework are included in the index.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

In NEWS: Signals from NSE's new Ahimsa index: When love for animals meets investments

1. Ethical Investing Based on Ahimsa

The **Nifty 500 Ahimsa Index** was launched by the National Stock Exchange (NSE) on **10 July 2026** as India's first thematic stock market index focused exclusively on **animal cruelty-free companies**.

The index seeks to promote **ethical investing** by tracking companies whose business practices are consistent with the principle of **Ahimsa (non-violence)**, a concept deeply rooted in Indian philosophical traditions. It offers investors an opportunity to align investment decisions with concerns regarding animal welfare. **Hence, Statement 1 is correct.**

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2. Developed by NSE Indices and Ahimsagain Foundation

The index was jointly developed by **NSE Indices** and the **Ahimsagain Foundation**, a non-profit organisation established in 2024 to promote ethical finance and animal welfare-conscious investing.

The collaboration aims to create a transparent framework through which investors can identify companies that avoid business activities involving cruelty to animals. **Hence, Statement 2 is correct.**

3. Green Band Companies Only

The index uses the **Ahimsa Investment Movement (AIM) Framework**, which categorises companies into:

- **Green Band** – Fully compliant with animal-cruelty-free criteria.
- **Orange Band** – Partially compliant.
- **Red Band** – Non-compliant.

Only companies placed in the **Green Band** are eligible for inclusion in the Nifty 500 Ahimsa Index. **Hence, Statement 3 is correct.**

Value Addition:

What Does the Index Exclude?

The index excludes companies significantly involved in:

- Animal testing for commercial purposes.
- Dairy, meat and poultry businesses.
- Leather-based products.
- Wool-based products.
- Certain cosmetics and fashion products linked to animal-derived materials.

Significance of the Index

- India's first dedicated **animal welfare-focused investment index**.
- Expands the scope of ethical investing beyond conventional ESG metrics.
- May serve as a benchmark for future **ETFs and index funds**.
- Reflects the growing influence of values-based investing among younger investors.

NSE at a Glance:

Feature	**Details**
Established	1992
Operations Began	1994
Headquarters	Mumbai
Regulator	SEBI
Recognition	Securities Contracts (Regulation) Act, 1956
Clearing Agency	NSE Clearing Limited
Depository Partner	NSDL

245. Samudra Pradikshana

Q. Samudra Pradakshina, recently seen in the news, is best described as:

(a) India's first indigenous deep-sea exploration mission under the Deep Ocean Mission
(b) The first-ever Tri-service all-women circumnavigation sailing expedition
(c) A joint India-France naval exercise in the Indian Ocean Region
(d) An Indian Navy mission to map the seabed of the Southern Ocean

Answer: (b) The first-ever Tri-service all-women circumnavigation sailing expedition

EXPLANATION:

IN NEWS: Raksha Mantri flags in first-ever Tri-service all-women circumnavigation sailing expedition

Samudra Pradakshina was the **first-ever Tri-service all-women circumnavigation sailing expedition**, bringing together women officers from the **Indian Army, Indian Navy and Indian Air Force** on a single mission.

The expedition was conducted aboard the **Indian Army Sailing Vessel (IASV) Triveni** and completed a **314-day voyage covering nearly 25,500 nautical miles** across four oceans. The crew fulfilled all internationally recognised criteria for circumnavigation, including crossing all meridians of longitude, crossing the **Equator twice, crossing the International Date Line and rounding the Three Great Capes**. **Hence, option (b) is correct.**

246. Deepfakes

Q. With reference to Deepfakes, recently seen in the news, consider the following statements:

1. Deepfakes are synthetic media generated using artificial intelligence techniques that can create realistic but fabricated images, videos or audio.
2. Deepfake technology primarily relies on machine learning and neural networks to manipulate or generate content.

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3. Deepfakes can be used for disinformation campaigns, identity theft and financial fraud.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
EXPLANATION:
IN NEWS: Union Minister Piyush Goyal files police complaint over AI deepfake video
1. AI-Generated Synthetic Media
Deepfakes are synthetic media created using artificial intelligence (AI) that can depict events that never occurred or make individuals appear to say or do things they never said or did. They may take the form of images, videos, or audio recordings. **Hence, Statement 1 is correct.**
2. Based on Machine Learning and Neural Networks
Deepfakes are created using machine learning models, particularly deep learning techniques and neural networks. These systems are trained on large datasets of images, videos or voice samples to generate highly realistic content. **Hence, Statement 2 is correct.**
3. Potential for Misuse
Deepfake technology has increasingly been used for malicious purposes such as misinformation campaigns, election manipulation, identity theft, financial fraud, scams, social engineering attacks, and non-consensual content creation. **Hence, Statement 3 is correct.**

247. Viksit Vibrant Village Programme

Q. With reference to the Viksit Vibrant Village Program (VVVP) 2026, consider the following statements:

1. It is a youth-led initiative implemented by Mera Yuva Bharat (MY Bharat) to strengthen grassroots engagement and national integration in border villages.
2. The programme is being implemented in collaboration with the Ministry of Home Affairs and the Indo-Tibetan Border Police (ITBP).
3. Under the programme, volunteers are deployed in border villages of Himachal Pradesh, Uttarakhand and Ladakh for immersive community engagement.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
EXPLANATION:
IN NEWS: PM Modi to interact with participants of Viksit Vibrant Village Programme 2026 tomorrow
1. Youth-led Initiative through MY Bharat
The **Viksit Vibrant Village Program (VVVP) 2026** is a youth-led initiative launched by the **Department of Youth Affairs** through **Mera Yuva Bharat (MY Bharat)**. It aims to strengthen grassroots engagement, promote national integration and support sustainable development in India's border villages. **Hence, Statement 1 is correct.**
2. Collaboration with MHA and ITBP
The programme is being implemented in collaboration with the **Ministry of Home Affairs (MHA)** and the **Indo-Tibetan Border Police (ITBP)**. The partnership helps facilitate youth engagement in strategically important border regions while ensuring coordination with local authorities and security agencies. **Hence, Statement 2 is correct.**
3. Focus on Border Villages of Himachal Pradesh, Uttarakhand and Ladakh
Under the programme, **500 MY Bharat volunteers** selected through a nationwide online quiz competition are being deployed in phases across border villages of **Himachal Pradesh, Uttarakhand and Ladakh**. The volunteers participate in a seven-day residential immersion programme involving community interaction, village surveys, Gram Sabha participation and developmental activities. **Hence, Statement 3 is correct.**
Value Addition:
- **Nodal Ministry:** Ministry of Youth Affairs & Sports.
- **Implementing Platform:** Mera Yuva Bharat (MY Bharat).
- **Phase I:** 250 volunteers across 43 villages.
- **Phase II:** 250 volunteers across 50 villages.
- **Nation First Challenge Themes:**
  - Adoption of Swadeshi products
  - Healthy cooking practices
  - Public transport usage and fuel conservation
  - Promotion of natural farming
  - Support for local tourism

248. Bab el-Mandeb Strait

Q. The Bab el-Mandeb Strait, recently seen in the news, connects which of the following water bodies?

(a) Mediterranean Sea and Black Sea
(b) Red Sea and Gulf of Aden
(c) Arabian Sea and Persian Gulf
(d) Gulf of Oman and Arabian Sea

Answer: (b) Red Sea and Gulf of Aden
EXPLANATION:

In NEWS: What is the Bab el-Mandeb Strait, and why does the Houthi 'blockade' threat against Saudi Arabia matter?



The **Bab el-Mandeb Strait** is one of the world's most important maritime chokepoints. Its name means **"Gate of Tears"** in Arabic, a reference to the numerous shipwrecks historically associated with the region.

The strait is located at the **southern entrance of the Red Sea**, lying between **Yemen** on the Arabian Peninsula and the **Horn of Africa** (primarily Djibouti and Eritrea). It serves as a narrow waterway connecting the **Red Sea** to the **Gulf of Aden**, which in turn opens into the **Arabian Sea and the Indian Ocean**.

Hence, option (b) is correct.

Geographical Significance

- It separates **Asia (Arabian Peninsula)** from **Africa (Horn of Africa)**.
- It forms a vital maritime link between the **Indian Ocean** and the **Mediterranean Sea** through the **Red Sea and the Suez Canal**.
- The strait is approximately **30 km wide** at its narrowest point.
- It is divided into two channels by **Perim Island (Yemen)**.

Strategic Importance



The Bab el-Mandeb Strait is a crucial component of global trade and energy transportation because:

- Ships travelling between **Europe and Asia** via the **Suez Canal** must pass through it.
- A significant portion of the world's **oil, LNG and container traffic** transits this route.
- Any disruption can force ships to take the much longer route around the **Cape of Good Hope** in South Africa.
- It is therefore considered one of the world's most important maritime chokepoints, along with the **Strait of Hormuz**, **Strait of Malacca** and **Suez Canal**.

249. BHAVYA Rasayan Scheme

Q. With reference to the Bharat Audyogik Vikas Yojana Rasayan (BHAVYA Rasayan) Scheme, recently approved by the Union Cabinet, consider the following statements:

1. The scheme aims to establish three dedicated Chemical Parks in the country.
2. The Chemical Parks will be developed by State Governments through a Challenge Route.
3. The parks will provide common infrastructure facilities such as Common Effluent Treatment Plants (CETPs), hazardous waste management facilities and logistics infrastructure.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
EXPLANATION

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In NEWS: CCEA approves three dedicated Chemical Parks with Centre-State contributions

1. Establishment of Three Dedicated Chemical Parks

The **Bharat Audyogik Vikas Yojana Rasayan (BHAVYA Rasayan) Scheme** was approved by the Union Cabinet for establishing **three dedicated Chemical Parks** in India. The scheme aims to strengthen the chemical manufacturing ecosystem by creating integrated industrial clusters with shared infrastructure facilities. **Hence, Statement 1 is correct.**

2. Development by State Governments through Challenge Route

The Chemical Parks will be developed by **State Governments through a Challenge Route**. Under this approach, States will compete by submitting proposals based on predefined criteria such as availability of suitable land, infrastructure readiness, investment potential and commitment towards project implementation.

Each park will require a minimum of **8 sq. km. (2,000 acres) of contiguous encumbrance-free land**. The Challenge Route promotes **competitive federalism** by selecting States that are better prepared to implement the scheme. **Hence, Statement 2 is correct.**

3. Provision of Common Infrastructure Facilities

The Chemical Parks will provide **plug-and-play infrastructure** and common facilities required for chemical industries, reducing the burden on individual companies. These include:

- **Common Effluent Treatment Plants (CETPs)**
- **Treatment, Storage and Disposal Facilities (TSDFs)**
- Hazardous waste management infrastructure
- Water supply and distribution systems
- Solvent recovery and distillation facilities
- Steam generation and distribution networks
- Interconnected pipeline networks
- Logistics and warehousing facilities

These shared facilities will improve efficiency, reduce costs, and ensure better compliance with environmental regulations. **Hence, Statement 3 is correct.**

Cabinet Committee on Economic Affairs (CCEA)

- The **Cabinet Committee on Economic Affairs (CCEA)** is one of the **Cabinet Committees of the Union Government**.
- It is headed by the **Prime Minister of India**.
- It deals with major economic matters that require approval at the highest level of the government.
- It takes decisions on:
  - Industrial policy and development.
  - Public sector enterprises.
  - Infrastructure projects.
  - Government schemes involving significant financial expenditure.
  - Pricing policies of important commodities.
  - Allocation of resources for major economic programmes.
- The CCEA approves major government schemes that have significant financial implications, such as the **BHAVYA Rasayan Scheme**.

250. Fast Track Courts

Q. With reference to Fast Track Courts and Fast Track Special Courts (FTSCs) in India, consider the following statements:

1. Fast Track Special Courts (FTSCs) established under the Centrally Sponsored Scheme are specifically meant for speedy disposal of rape cases and offences under the Protection of Children from Sexual Offences (POCSO) Act.
2. Fast Track Courts announced for examination paper leak cases are part of the existing Fast Track Special Courts Scheme funded through the Nirbhaya Fund.
3. Fast Track Courts are created with the objective of reducing delays in judicial proceedings and ensuring speedy disposal of cases.

Which of the statements given above are correct?

(a) 1 and 3 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 3 only

Explanation

In NEWS: What new fast-track courts can address: Lapses in exam leak probes, weak legal provisions

1. Fast Track Special Courts (FTSCs) for Rape and POCSO Cases

Fast Track Special Courts (FTSCs) are dedicated courts established under the **Centrally Sponsored Scheme of Fast Track Special Courts** for the **speedy disposal of pending rape cases and**

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offences under the Protection of Children from Sexual Offences (POCSO) Act, 2012.

The scheme was introduced in **2019** following:

- The **Criminal Law (Amendment) Act, 2018**, which strengthened laws relating to sexual offences.
- Directions of the Supreme Court in **Suo Motu Writ (Criminal) proceedings, 2019** regarding speedy trials.

These courts aim to:

- Reduce pendency of sensitive criminal cases.
- Ensure timely justice for survivors.
- Improve efficiency of the criminal justice system.

Hence, Statement 1 is correct.

2. Fast Track Courts for Examination Paper Leak Cases

Recently announced fast-track courts for dealing with **examination paper leak cases** are **different from the existing FTSC Scheme**.

The existing FTSCs are limited to **rape and POCSO cases** and are supported through the **Nirbhaya Fund**.

Paper leak cases generally involve offences such as:

- Cheating,
- Criminal conspiracy,
- Forgery,
- Fraud and other related offences.

These cases are normally investigated by police or specialised agencies and tried before competent criminal courts. If the government creates dedicated fast-track courts for paper leak cases, they would require a **separate legal or administrative framework**. Therefore, fast-track courts for paper leaks **are not part of the existing FTSC Scheme**.

Hence, Statement 2 is incorrect.

3. Objective of Fast Track Courts

The broad purpose of **Fast Track Courts** is to ensure **speedy disposal of cases** by prioritising specific categories of cases and reducing judicial delays.

They are created to:

- Address case pendency.
- Provide quicker justice.
- Improve efficiency of courts.
- Ensure timely conclusion of trials.

The category of cases handled depends on the purpose for which the fast-track court is established.

Hence, Statement 3 is correct.

Difference Between Fast Track Courts and FTSCs

Fast Track Courts	**Fast Track Special Courts (FTSCs)**
General term for courts created for speedy disposal of cases	Specific category of fast-track courts
Can be created for different categories of cases	Specifically created for rape and POCSO cases
Scope depends on government decision	Operates under the FTSC Centrally Sponsored Scheme
Example: Proposed fast-track courts for paper leak cases	Existing FTSCs for sexual offences

Fast Track Special Courts (FTSCs):

- **Nodal Ministry:** Ministry of Law and Justice (Department of Justice)
- **Type:** Centrally Sponsored Scheme
- **Funding:** Nirbhaya Fund
- **Implemented with:** State Governments and High Courts

Important Concept:

A fast-track court is a broader concept, whereas Fast Track Special Court (FTSC) is a specific scheme-based court created for rape and POCSO cases.

Fast Track Courts are created through administrative and financial measures and are not separate courts established under the Constitution.

251. Sarnath

Q. With reference to Sarnath, consider the following statements:

1. Sarnath is the place where Lord Buddha delivered his first sermon after attaining enlightenment.
2. The site includes monuments such as the Dhamekh Stupa, Mulagandkuti Vihara and the Ashokan Pillar.
3. Sarnath is India's 45th UNESCO World Heritage Site.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (c) All three

EXPLANATION:

In NEWS: Sarnath inscribed on UNESCO World Heritage List; PM Modi calls it a proud moment

Statement 1:

Sarnath is one of the holiest pilgrimage sites in Buddhism. It is the place where Lord Buddha delivered his first sermon after attaining enlightenment at Bodh Gaya, thereby setting in motion the Dharmachakra Pravartan (Turning of the Wheel of Dharma). This event marked the beginning of the spread of Buddhist teachings and the formation of the Buddhist Sangha. **Hence, Statement 1 is correct.**

Statement 2:

The UNESCO-inscribed property of Sarnath includes several significant Buddhist monuments and archaeological remains. Prominent among them are the Dhamekh Stupa, built at the site of Buddha's first sermon;,Mulagandhakuti (Mulagandkuti) Vihara is traditionally believed to mark the place where Buddha spent his first rainy season after delivering his first sermon; and the Ashokan Pillar, erected by Emperor Ashoka to commemorate the sacred site. These monuments reflect the site's continuous religious and historical importance. **Hence, Statement 2 is correct.**

Statement 3:

Recently, the Ancient Buddhist Site of Sarnath was inscribed on the UNESCO World Heritage List. With this inscription, Sarnath became India's 45th UNESCO World Heritage Site and Uttar Pradesh's fourth World Heritage Site, recognizing its outstanding universal value as a centre of Buddhist heritage and archaeology. **Hence, Statement 3 is correct.**

Value Addition:

- Location: Varanasi, Uttar Pradesh.
- Ancient Names: Rishipattana, Isipatana, Mrigadava.
- UNESCO Inscription: 2026.
- Key Components: Chaukhandi Stupa, Dhamekh Stupa, Dharmarajika Stupa, Mulagandkuti Vihara, Ashokan Pillar.
- Historical Span: Archaeological remains dating from the 5th–4th century BCE to the 12th century CE.
- Significance: Site of the First Sermon (Dharmachakra Pravartan) and establishment of the Buddhist Sangha.
- Ashokan Lion Capital from Sarnath is the **National Emblem of India**

252. Project Kusha

Q. With reference to Project Kusha, consider the following statements:

1. Project Kusha is also known as the Extended Range Air Defence System (ERADS).
2. It is an indigenous long-range Surface-to-Air Missile (SAM) defence system being developed by DRDO.
3. The system comprises three interceptor variants with engagement ranges extending up to 350–400 km.
4. It is exclusively designed to intercept ballistic missiles.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (c) Only three

EXPLANATION:

IN NEWS: DRDO conducts successful maiden flight-test of 'Kusha' Long-Range Surface-to-Air Missile

Statement 1:

Project Kusha is also known as the Extended Range Air Defence System (ERADS) and is being developed as part of India's future air defence network. **Hence, Statement 1 is correct.**

Statement 2:

It is an indigenous long-range Surface-to-Air Missile (SAM) defence system being developed by the Defence Research and Development Organisation (DRDO). **Hence, Statement 2 is correct.**

Statement 3:

Project Kusha is planned as a three-tier interceptor system with engagement ranges of approximately 150 km, 250 km, and up to 350–400 km. **Hence, Statement 3 is correct.**

Statement 4:

The system is capable of engaging a wide variety of aerial threats, including fighter aircraft, missiles, UAVs, and large enemy aircraft.Project Kusha is designed as a long-range air defence system capable of engaging multiple aerial threats such as aircraft, UAVs and missiles. Therefore, it is not limited to ballistic missile interception alone. **Hence, Statement 4 is not correct.**

Value Addition:

- Also Known As: Extended Range Air Defence System (ERADS)
- Developer: DRDO with industry partners
- Also called India's indigenous answer to S-400
- Air Defence Umbrella: Part of the proposed Sudarshan Chakra
- Integration: Integrated Air Command and Control System (IACCS)
- Comparable Systems: S-400 (Russia), THAAD (USA), Patriot PAC-3 (USA)
- Recent Development: Maiden flight-test conducted from APJ Abdul Kalam Island, Odisha (July 2026).

253. Plurilateral Pacts

Q. Plurilateral Trade Agreements (PTAs) are best described as:

(a) Trade agreements that are negotiated and binding on all members of the World Trade Organization (WTO).
(b) Trade agreements negotiated among a subset of WTO members and binding only on participating countries.
(c) Bilateral trade agreements signed exclusively between two countries outside the WTO framework.
(d) Non-binding arrangements for economic cooperation without legal obligations.

Answer: (b) Trade agreements negotiated among a subset of WTO members and binding only on participating countries

EXPLANATION :

IN NEWS:India & most WTO members remain at loggerheads on plurilateral pacts

Option (a)

This describes **multilateral agreements**, where all WTO members participate and the outcomes are binding on the entire membership. **Hence, Option (a) is not correct.**

Option (b)

Plurilateral Trade Agreements (PTAs) are trade agreements negotiated by a **subset of WTO members** and are legally binding only on the countries that choose to participate. **Hence, Option (b) is correct.**

Option (c)

This describes **bilateral trade agreements (BTAs)** between two countries. PTAs involve more than two countries but not necessarily all WTO members. **Hence, Option (c) is not correct.**

Option (d)

PTAs are **formal trade agreements with legal commitments** for participating members and are not merely voluntary cooperation arrangements. **Hence, Option (d) is not correct.**

Value Addition:

- **WTO Basis:** Permitted under **Annex 4 of the Marrakesh Agreement**.
- **Examples within WTO:** Agreement on Government Procurement (GPA), Agreement on Trade in Civil Aircraft.
- **Modern Forms:** Joint Statement Initiatives (JSIs) on E-commerce, Investment Facilitation, MSMEs.
- **Key Advantage:** Faster negotiations among like-minded countries.
- **India's Concern:** May weaken the WTO's consensus-based multilateral trading system and reduce policy space for developing countries.
- Information Technology Agreement (ITA) — a plurilateral agreement under WTO framework.

254. Biofouling

Q. Biofouling is best described as:

(a) The accumulation of aquatic organisms on surfaces immersed in or exposed to water environments.
(b) The excessive growth of phytoplankton caused by nutrient pollution in oceans.
(c) The release of toxic chemicals from ships into marine ecosystems.
(d) The process of natural weathering of rocks in aquatic environments.

Answer: (a) The accumulation of aquatic organisms on surfaces immersed in or exposed to water environments.

EXPLANATION:

In NEWS: Thousands of stranded vessels in the Persian Gulf may unintentionally spread invasive marine life worldwide: Study

Option (a)

Biofouling is the **accumulation of aquatic organisms** such as microorganisms, algae, barnacles, mussels, and other marine organisms on surfaces immersed in or exposed to water. **Hence, Option (a) is correct.**

Option (b)

This describes **algal bloom/eutrophication**, a phenomenon where excess nutrients (mainly nitrogen and phosphorus) lead to rapid growth of phytoplankton or algae in water bodies. It is different

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from biofouling, which involves the attachment of organisms to submerged surfaces. **Hence, Option (b) is not correct.**

Option (c)

This refers to **marine pollution caused by discharge of toxic substances** such as oil, chemicals, or waste from ships. Biofouling involves biological growth on surfaces and not the release of pollutants. **Hence, Option (c) is not correct.**

Option (d)

This describes **weathering or erosion processes**, where rocks break down due to physical, chemical, or biological factors in aquatic environments. It is unrelated to the accumulation of living organisms on submerged structures. **Hence, Option (d) is not correct.**

Value Addition

- **Examples of Biofouling Organisms:** Algae, bacteria, barnacles, mussels, sea squirts.
- **Major Impact:** Spread of invasive marine species through international shipping.
- **Economic Impact:** Increased drag on ships leading to higher fuel consumption.
- **Biosecurity Concern:** Organisms attached to ship hulls can establish themselves in new ecosystems when vessels move between ports.
- **Major Concern:** Facilitates the spread of invasive marine species.
- **Common Surfaces Affected:** Ship hulls, offshore platforms, pipelines, aquaculture equipment.
- **Recent Context:** Prolonged anchorage of vessels in the Persian Gulf has raised concerns about a potential marine invasive species "super-spreader" event due to increased biofouling.

IMO Convention

- International Maritime Organization (IMO) has issued **Biofouling Guidelines (2011, revised later)** to reduce transfer of invasive aquatic species.

255. Aerosols

Q.Consider the following statements regarding aerosols:

1. Aerosols are tiny particles suspended in the atmosphere that can either reflect or absorb solar radiation.
2. Aerosols influence cloud formation and rainfall patterns.
3. Aerosols can alter Earth's energy balance.
4. Aerosols may originate from both natural and anthropogenic sources.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (d) All four

EXPLANATION:

IN NEWS: Unraveling global aerosol diversity provides the key to decoding Climate Change

Statement 1: Aerosols are microscopic solid or liquid particles suspended in the atmosphere. They can either reflect sunlight back into space or absorb solar radiation, thereby influencing climate. **Hence, Statement 1 is correct.**

Statement 2: Aerosols act as cloud condensation nuclei and influence cloud formation, precipitation processes, and rainfall patterns. **Hence, Statement 2 is correct.**

Statement 3: By scattering and absorbing solar radiation, aerosols modify the Earth's radiative balance (energy budget) and affect surface and atmospheric temperatures. **Hence, Statement 3 is correct.**

Statement 4: Aerosols originate from both natural sources (dust storms, volcanic eruptions, sea spray, forest fires) and anthropogenic sources (vehicle emissions, industrial pollution, biomass burning). **Hence, Statement 4 is correct.**

VALUE ADDITION

Types

- Primary Aerosols → emitted directly
- Secondary Aerosols → formed through atmospheric chemical reactions

Study Finding

- Strongly Absorbing Aerosols (SA) produce highest atmospheric forcing.
- South Asia is largely dominated by Dust-Dominated Mixtures (DDM).

Feature	Details
Definition	Tiny particles suspended in the atmosphere
Natural Sources	Desert dust, sea spray, volcanic ash, forest fires
Anthropogenic Sources	Vehicle emissions, industrial pollution, biomass burning

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Climate Impact	Influence radiation balance, clouds and rainfall
Types	Primary aerosols and Secondary aerosols

256. Ginger Lily

Q. With reference to *Hedychium zubeengargianum*, recently seen in the news, consider the following statements:

1. It is a newly identified species of ginger lily
2. The species flowers primarily during April–May.
3. It is characterized by creamy-white fragrant flowers.

Select the correct answer using the code below:

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
EXPLANATION:
IN NEWS: Scientists name new ginger lily 'Hedychium zubeengargianum' after Zubeen Garg
Statement 1: *Hedychium zubeengargianum* is a newly identified species of ginger lily belonging to the **Zingiberaceae (Ginger) family**. It was identified by botanists from the **Botanical Survey of India (BSI), Shillong**, after detailed taxonomic studies. **Hence, Statement 1 is correct.**
Statement 2: The species flowers primarily during **April–May**, which is an important characteristic used to distinguish it from closely related species. Its earlier flowering season is one of its key identifying features. **Hence, Statement 2 is correct.**
Statement 3: *Hedychium zubeengargianum* is characterized by **creamy-white fragrant flowers**. It also possesses **yellow anthers** and a **densely hairy flowering stalk**, which differentiate it from its nearest relatives. **Hence, Statement 3 is correct.**
Value Addition



- **Scientific Name:** *Hedychium zubeengargianum*
- **Family:** Zingiberaceae

Botanical Classification

- Genus: Hedychium
- Second plant named after Zubeen Garg in 2026.
- Earlier: *Osbeckia zubeengargiana*.
- **Named After:** Assamese singer and environmental advocate **Zubeen Garg**
- **Discovery Location:** Identified by scientists of BSI, Shillong, from specimens collected in Assam.
- **India's Diversity:** Around **41 taxa of Hedychium** occur mainly in **Northeast India** and the **Western Ghats**.

257. Portfolio Management Services

Q. With reference to Portfolio Management Service (PMS), consider the following statements:

1. PMS is regulated by the Securities and Exchange Board of India (SEBI).
2. PMS is primarily designed for retail investors with small investment amounts.
3. Only entities registered with SEBI as Portfolio Managers can offer PMS .

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (b) 1 and 3 only
EXPLANATION:
IN NEWS: Overhaul of portfolio management services: What SEBI has proposed and what it means for investors
Statement 1: Portfolio Management Service (PMS) is regulated by the Securities and Exchange Board of India (SEBI) under the SEBI (Portfolio Managers) Regulations, 2020. These regulations govern registration, compliance requirements, disclosures, and investor protection measures for portfolio managers. **Hence, Statement 1 is correct.**
Statement 2: PMS is not designed for retail investors with small investment amounts. SEBI mandates a minimum investment requirement of ₹50 lakh per client, making PMS a product primarily meant for High Net Worth Individuals (HNIs) and sophisticated investors. **Hence, Statement 2 is incorrect.**
Statement 3: Portfolio Management Services can be offered only by entities registered with SEBI under the SEBI (Portfolio Managers) Regulations, 2020.

This ensures regulatory oversight and accountability in managing client portfolios. **Hence, Statement 3 is correct.**

Value Addition: Portfolio Management Service (PMS)

- Regulator: Securities and Exchange Board of India (SEBI).
- Legal Framework: SEBI (Portfolio Managers) Regulations, 2020.
- **Minimum Net Worth Requirement-** ₹5 crore (existing PMS regulations)
- Minimum Investment: ₹50 lakh per client.
- Target Investors: High Net Worth Individuals (HNIs) and sophisticated investors.
- Offer PMS: SEBI-registered corporate entities, companies, and LLPs.
- Types of PMS:
  - Discretionary PMS: Portfolio manager takes investment decisions.
  - Non-Discretionary PMS: Investor takes final investment decisions.
  - Advisory PMS: Portfolio manager provides only investment advice.
- Custodian Requirement: Independent custodian holds client assets to avoid conflict of interest.
- Recent SEBI Proposal (2026):
  - Introduction of a Mutual Fund-only PMS (MF-PMS) category.
  - Proposal to allow investments in overseas securities and to-be-listed securities.
  - Proposed minimum investment of ₹25 lakh for MF-PMS.

258. Chenab River

Q.With reference to the Chenab River, consider the following statements:

1. It is formed by the confluence of the Chandra and Bhaga rivers in Himachal Pradesh.
2. The Chandra and Bhaga rivers, which form the Chenab, originate in the Lahaul-Spiti region.
3. It is one of the western rivers covered under the Indus Waters Treaty.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

In NEWS:Baseless allegations, says India on Pakistan's claims of creating a flood-like situation by increasing flow of Chenab

Statement 1: The Chenab River is formed by the confluence of the Chandra and Bhaga rivers at Tandi in the Lahaul region of Himachal Pradesh. In its upper reaches, it is also known as Chandrabhaga. **Hence, Statement 1 is correct.**

Statement 2: The Chenab is formed at Tandi by the confluence of Chandra and Bhaga rivers, which originate from glaciers in the Lahaul-Spiti region. **Hence, Statement 2 is correct.**

Statement 3: The Chenab is one of the Western Rivers (Indus, Jhelum and Chenab) covered under the Indus Waters Treaty, 1960 between India and Pakistan. The waters of these rivers are primarily allocated to Pakistan, subject to specified rights of India. **Hence, Statement 3 is correct.**

Value Addition

- River System: Indus River System.
- Upper Course Name: Chandrabhaga.
- Major Dams: Salal, Baglihar, Dul and Ratle Hydroelectric Projects.
- States/UTs Traversed: Himachal Pradesh and Jammu & Kashmir.
- Indus Waters Treaty (1960): Brokered by the World Bank between India and Pakistan.

Indus Western Rivers

- Indus
- Jhelum
- Chenab

Eastern Rivers

- Ravi
- Beas
- Sutlej



259. Amrit Bharat Station Scheme

Q.With reference to the Amrit Bharat Station Scheme (ABSS), consider the following statements:

1. The scheme emphasizes preparation of station-specific Master Plans with a long-term vision for development.
2. One of the objectives of the scheme is to promote multimodal connectivity and improve accessibility for passengers.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

In NEWS:Prime Minister, Shri Narendra Modi Dedicates to the Nation 75 Amrit Bharat Stations Across India

Statement 1: The Amrit Bharat Station Scheme (ABSS), launched in 2022, envisages redevelopment of railway stations based on station-specific Master Plans with a long-term vision. The development is undertaken according to the needs and future requirements of each station. **Hence, Statement 1 is correct.**

Statement 2: The scheme focuses on enhancing passenger experience through multimodal connectivity, better accessibility, improved amenities, and integrated station infrastructure. **Hence, Statement 2 is correct.**

VALUE ADDITION

Feature	Details
Launch Year	2022
Ministry	Ministry of Railways
Approach	Station-specific Master Planning
Key Objective	Modern, passenger-centric railway stations
Focus Areas	Multimodal connectivity, accessibility, amenities, heritage-based design

Recent Progress

- 103 stations dedicated in May 2025.
- 75 stations dedicated in July 2026.

Heritage Angle

- Station architecture reflects local culture and regional heritage

260. Startup Village Entrepreneurship Programme

Q. With reference to the Start-up Village Entrepreneurship Programme (SVEP), consider the following statements:

1. It is a sub-scheme of the Deendayal Antyodaya Yojana – National Rural Livelihoods Mission (DAY-NRLM).
2. The programme focuses on promoting non-farm rural enterprises through support to Self Help Groups (SHGs) and their family members.
3. SVEP is implemented directly by NITI Aayog through District Industries Centres.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation

In NEWS: Start-up Village Entrepreneurship Programme

Statement 1: The Start-up Village Entrepreneurship Programme (SVEP) was launched in 2016 as a sub-scheme of the **Deendayal Antyodaya Yojana – National Rural Livelihoods Mission (DAY-NRLM)**. **Hence, Statement 1 is correct.**

Statement 2:SVEP promotes **non-farm rural enterprises** by supporting **Self Help Groups (SHGs)** and their family members through financial assistance, training, business development services, and institutional support. **Hence, Statement 2 is correct.**

Statement 3:SVEP is implemented through **State Rural Livelihoods Missions (SRLMs)**. At the block level, **Block Resource Centres–Enterprise Promotion (BRC-EPs)** and **Community Resource Persons–Enterprise Promotion (CRP-EPs)** facilitate implementation. It is **not implemented directly by NITI Aayog**. **Hence, Statement 3 is not correct.**

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Value Addition:

- **Launch:** 2016
- **Parent Scheme:** DAY-NRLM
- **Objective:** Promote rural entrepreneurship and generate local employment through non-farm enterprises.
- **Implementation:** SRLMs → BRC-EPs → CRP-EPs
- **Beneficiaries:** SHGs, women entrepreneurs, marginalized communities, rural youth.
- **Achievement:** Supported **4.32 lakh rural enterprises** across India (June 2026).

Community Institutions

- BRC-EP → Block Resource Centre–Enterprise Promotion
- CRP-EP → Community Resource Person–Enterprise Promotion

Special Feature

- Community-based funding decisions.

261. Praxis Mission

Q.Recently in the news, PRAXIS is associated with:

(a) Planetary ring exploration and in-situ sampling mission
(b) Mars sample return mission
(c) Lunar habitat construction programme
(d) Exoplanet detection mission

Answer: (a) Planetary ring exploration and in-situ sampling mission

Explanation

In NEWS: PRAXIS: a NASA mission to graze planetary rings

Option (a): Planetary ring exploration and in-situ sampling mission

PRAXIS is a proposed NASA mission concept designed to directly explore and study planetary ring systems. It aims to collect and analyze ring particles through in-situ sampling, helping scientists understand the composition, structure, evolution, and origins of rings around Saturn, Uranus, Neptune, and ringed Centaurs such as Chariklo and Chiron. **Hence, Option (a) is correct.**

Option (b): Mars sample return mission

PRAXIS is not related to Mars exploration or the retrieval of Martian rock and soil samples. Its focus is on planetary ring systems rather than the surface of Mars. **Hence, Option (b) is incorrect.**

Option (c): Lunar habitat construction programme

The mission has no objective related to establishing human settlements or habitats on the Moon. It is a robotic scientific exploration concept focused on ring particles.**Hence, Option (c) is incorrect.**

Option (d): Exoplanet detection mission

PRAXIS is not designed to discover or study exoplanets. Its mission objective is confined to the direct investigation of ring systems within our Solar System.**Hence, Option (d) is incorrect.**

Value Addition

PRAXIS: Quick Facts

- Full Form: Planetary Rings Autonomous EXploration with In-situ Sampling
- Agency: NASA
- Programme: NIAC (NASA Innovative Advanced Concepts)
- Mission Type: AI-enabled robotic ring explorer
- Key Objective: Direct sampling of planetary ring particles
- Sampling Method: Touch-and-Go sampling using a deployable boom

Potential Targets

- Saturn's Rings
- Uranus' Rings
- Neptune's Rings
- Chariklo (Ringed Centaur)
- Chiron (Ringed Centaur)

Key Technologies

- Artificial Intelligence (AI)-based autonomous navigation
- Collision avoidance system
- Precision particle sampling
- Real-time onboard scientific analysis
- Sport-casting-inspired particle capture mechanism

Scientific Significance

- First direct observations of mm–cm scale ring particles
- Study of particle size, porosity, and composition
- Understanding the origin and evolution of planetary rings
- Addresses a priority identified in the Planetary Science Decadal Survey

NIAC Programme

- Full Form: NASA Innovative Advanced Concepts
- Supports early-stage futuristic space concepts
- Phase-I funding is for feasibility studies

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262. INSAT - 3DR

Q. With reference to INSAT-3DR, consider the following statements:

1. INSAT-3DR is an advanced meteorological satellite developed by ISRO.
2. It is positioned in a geostationary orbit.
3. It is intended to improve weather forecasting and cyclone monitoring.

Which of the statements given above are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

IN NEWS:Isro satellite images show clouds over nearly two-thirds of India: What to expect

Statement 1:INSAT-3DR (Indian National Satellite-3DR) is an advanced meteorological satellite developed by the Indian Space Research Organisation (ISRO). It was launched on 8 September 2016 aboard the GSLV-F05 launch vehicle and serves as a successor to INSAT-3D, enhancing India's weather observation capabilities.**Hence, Statement 1 is correct.**

Statement 2:INSAT-3DR operates from a geostationary orbit, enabling continuous observation of the same geographical region. This allows the satellite to provide regular weather updates and half-hourly multispectral images over the Indian subcontinent and adjoining oceans.**Hence, Statement 2 is correct.**

Statement 3: One of the primary objectives of INSAT-3DR is to improve weather forecasting, cyclone tracking, disaster warning, and environmental monitoring. Its advanced imaging and atmospheric sounding capabilities help meteorological agencies monitor cloud patterns, storms, cyclones, fog, and sea surface temperatures with greater accuracy.**Hence, Statement 3 is correct.**

Value Addition

INSAT-3DR: Quick Facts

Feature	Details
Launch Date	8 September 2016
Launch Vehicle	GSLV-F05
Developed By	ISRO
Orbit	Geostationary Orbit
Type	Advanced Meteorological Satellite
Predecessor	INSAT-3D-2013

Key Applications

- Cyclone tracking
- Fog detection
- Monsoon monitoring
- Sea Surface Temperature estimation
- Search & Rescue services

Key Payloads

- Imaging System for cloud and weather monitoring.
- Atmospheric Sounder for vertical profiling of temperature and humidity.
- Data Relay Transponder (DRT) for data collection from remote stations.
- Search and Rescue Transponder (SART) for distress alert services.

Major Improvements over INSAT-3D

- Imaging in the Middle Infrared Band for nighttime fog and low-cloud detection.
- Two Thermal Infrared Bands for more accurate Sea Surface Temperature (SST) estimation.
- Higher spatial resolution in Visible and Thermal Infrared channels.
- Enhanced cyclone monitoring and disaster warning capabilities.

Geostationary Orbit

- Altitude: ~35,786 km above the Equator.
- Satellite appears stationary relative to Earth.
- Ideal for continuous weather observation and communication services.

Source:https://www.indiatoday.in/science/story/isro-satellite-images-india-cloud-cover-insat-monsoon-imd-2026-rain-science-news-2951126-2026-07-19

263. Undavalli Caves

Q. The Undavalli Caves are situated overlooking which of the following rivers?

(a) Godavari
(b) Penna
(c) Krishna
(d) Tungabhadra

Answer: (c) Krishna

Explanation

In NEWS: Telugu Mahotsavam begins with a celebration of literature and heritage

Option (a): Godavari

The Godavari is India's second-longest river and flows through Maharashtra, Telangana and Andhra Pradesh. However, the Undavalli Caves are not located along the Godavari River. **Hence, Option (a) is incorrect.**

Option (b): Penna

The Penna (Pennar) River flows through Karnataka and Andhra Pradesh before draining into the Bay of Bengal. The Undavalli Caves have no geographical association with the Penna River**. Hence, Option (b) is incorrect.**

Option (c): Krishna

The Undavalli Caves are located in **Undavalli Village, Guntur district, Andhra Pradesh**, on a hill overlooking the **Krishna River**. The caves are among the finest examples of ancient rock-cut architecture in India and are famous for the gigantic reclining statue of Lord Vishnu. **Hence, Option (c) is correct.**

Option (d): Tungabhadra

The Tungabhadra is a major tributary of the Krishna River and is closely associated with Hampi and the Vijayanagara Empire. However, the Undavalli Caves are not situated on its banks. **Hence, Option (d) is incorrect.**

Value Addition

Undavalli Caves: Quick Facts

- Located in **Guntur district, Andhra Pradesh**.
- Situated on a hill overlooking the **Krishna River**.
- Built between the **4th–5th centuries CE**.
- Associated with the **Vishnukundina dynasty (420–620 CE)**.
- Carved out of **solid sandstone**.
- One of the finest examples of **rock-cut architecture** in South India.
- Features a **four-storeyed cave complex**.

Architectural Significance

- Contains a massive **reclining Vishnu (Anantasayana)** sculpture.
- Vishnu idol carved from a **single block of granite**.
- Shrines dedicated to the **Trimurti—Brahma, Vishnu and Shiva**.
- The third floor is adorned with sculptures of **lions, elephants and other motifs**.

Religious Evolution

- Initially developed as a **Buddhist monastery (Vihara)**.
- Later associated with **Jain traditions**.
- Eventually converted into **Hindu temples**.
- Represents the transition from **Buddhist to Hindu religious traditions**.

Cave Site	State	Associated River
Undavalli Caves	Andhra Pradesh	Krishna
Ajanta Caves	Maharashtra	Waghora
Ellora Caves	Maharashtra	Near Elaganga
Elephanta Caves	Maharashtra	Mumbai Harbour
Badami Caves	Karnataka	Agastya Lake

264. Black Sea

Q. Consider the following countries:

1. Türkiye
2. Moldova
3. Ukraine
4. Romania
5. Russia
6. Georgia

How many of the above-mentioned countries border the Black Sea?

(a) Only three
(b) Only four
(c) Only five
(d) All six

Answer: (c) Only five

Explanation

In NEWS: Govt. warns seafarers about jobs in 'volatile' Black Sea

Explanation:

The Black Sea is a large inland sea located at the southeastern extremity of Europe and is considered one of the marginal seas of the Atlantic Ocean. Six countries border it:

- Türkiye
- Bulgaria

- Romania
- Ukraine
- Russia
- Georgia



Among the countries mentioned in the question, all except **Moldova** border the Black Sea. Moldova is a landlocked country, though it lies close to the Black Sea region. **Hence, the correct answer is: (c) Only five**

Key Facts about the Black Sea

- Surface area: Approximately 436,000 sq. km.
- Connected to the Mediterranean Sea through the Bosporus Strait, the Sea of Marmara, and the Dardanelles Strait.
- Linked to the Sea of Azov through the Kerch Strait.
- Major rivers draining into it include the Danube, Dnieper, Dniester, Southern Bug, and Rioni.
- The Black Sea is known for its low salinity and extensive anoxic deep-water zones.

Important Straits

Strait	Connects
Bosporus	Black Sea–Sea of Marmara
Dardanelles	Sea of Marmara–Aegean Sea
Kerch Strait	Black Sea–Sea of Azov



Strategic Importance

- An important route for grain and energy exports, especially for Russia and Ukraine.
- The region has high geopolitical significance due to the NATO-Russia naval presence.
- Russia's Black Sea Fleet is based at Sevastopol in Crimea.

265. UNCITRAL

Q.With reference to the United Nations Commission on International Trade Law (UNCITRAL), consider the following statements:

1. The Commission currently has 70 member states elected for six-year terms.
2. All member states of the United Nations are automatically members of UNCITRAL.
3. India is not a founding member of UNCITRAL.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) 3 only
(d) None

Answer: (a) 1 only

EXPLANATION:

IN NEWS: EAM Jaishankar calls for inclusive global trade laws as India hosts UNCITRAL's 60th anniversary conference

Statement 1: UNCITRAL currently consists of **70 member states**, which are elected by the **UN General Assembly**. Members serve **six-year terms**, and every three years the terms of half of the members expire to ensure continuity in the Commission's work. **Hence, Statement 1 is correct.**

Statement 2: UNCITRAL does not include all UN member states automatically. Membership is limited to **70 elected states**, chosen to ensure equitable geographical representation among different regions of the world. **Hence, Statement 2 is incorrect.**

Statement 3: India is a **founding member** of UNCITRAL and has actively contributed to the development of international trade law since the Commission's establishment in 1966. **Hence, Statement 3 is incorrect.**

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Value Addition

UNCITRAL: Quick Facts

Feature	**Details**
Established	1966
Parent Body	UN General Assembly
Members	70 Member States
Term of Members	6 Years
Retirement Cycle	Half the members retire every 3 years
Headquarters	New York
Annual Sessions	Alternately in New York and Vienna
Mandate	Harmonization and unification of international trade law
India's Status	Founding Member

Five Regional Groups in UNCITRAL

- African States
- Asian States
- Eastern European States
- Latin American & Caribbean States
- Western European & Other States

UNCITRAL Landmark Laws

- UNCITRAL Model Law on Arbitration (1985)
- Model Law on Electronic Commerce (1996)
- Model Law on Cross-Border Insolvency (1997)

India

- Arbitration and Conciliation Act, 1996 is based substantially on UNCITRAL Model Law.

266. Brahmaputra

Q.Which of the following rivers originate outside India and later become part of the Brahmaputra River System?

1. Lohit
2. Subansiri
3. Teesta
4. Kameng

Select the correct answer using the code given below:

(a) 1 and 2 only
(b) 1, 2 and 4 only
(c) 2 and 3 only
(d) 1, 2, 3 and 4

Answer: (a) 1 and 2 only

EXPLANATION:

IN NEWS: Eroding identities of Brahmaputra's children-Across Assam, decades of govt.-issued records are failing the citizenship test before Foreigners Tribunals and the Gauhati High Court as spelling variations, age mismatches, and shifting residences come under scrutiny

Statement 1: Lohit

The Lohit River originates in eastern Tibet before entering Arunachal Pradesh and eventually joining the Siang and Dibang rivers to form the Brahmaputra. Due to its origin outside India, it is considered a transboundary tributary of the Brahmaputra River System. **Hence, Statement 1 is correct.**

Statement 2: Subansiri

The Subansiri River originates in Tibet and flows through Arunachal Pradesh and Assam before joining the Brahmaputra. It is the largest tributary of the Brahmaputra and is also a transboundary river because its source lies outside India. **Hence, Statement 2 is correct.**

Statement 3: Teesta

The Teesta River originates from the Pauhunri Glacier in the Eastern Himalayas in Sikkim, India. Although it later flows into Bangladesh and is therefore a transboundary river, its origin is within India. The question specifically asks for rivers that originate outside India. **Hence, Statement 3 is incorrect.**

Statement 4: Kameng

The Kameng River originates near the Tawang region of Arunachal Pradesh in India and flows through Arunachal Pradesh and Assam before joining the Brahmaputra. Since its source lies within India, it does not satisfy the condition given in the question. **Hence, Statement 4 is incorrect.**

Brahmaputra

- Only major river in India with a **masculine name** ("Son of Brahma").
- One of the **largest braided rivers** in the world.
- Carries one of the **highest sediment loads** among global rivers.
- Creates extensive floodplains and riverine islands.

● Forms the **world's largest inhabited river island — Majuli**.
● Known for frequent channel shifting and bank erosion.

Brahmaputra River Tributaries

Right-Bank Tributaries (North Bank / Himalayan Side)	**Left-Bank Tributaries (South Bank / Naga & Assam Hills)**
Subansiri	Dibang
Kameng (Jia Bharali)	Lohit
Manas	Dhansiri (South)
Sankosh	Burhi Dihing
Teesta	Dikhow
Raidak	Kopili
Jaldhaka	Kulsi
Beki / Pagladiya / Puthimari	Digaru / Krishnai / Noa Dihing

When Brahmaputra flows east → west through Assam:

● North bank tributaries come from Himalayas.
● South bank tributaries come from Meghalaya, Nagaland, Assam Hills.

Formation of Brahmaputra in India

The Brahmaputra is formed after the confluence of:

● Siang (Dihang)
● Dibang
● Lohit

near **Sadiya (Assam).**

Different Names of Brahmaputra

Region	**Name**
Tibet	Yarlung Tsangpo
Arunachal Pradesh	Siang / Dihang
Assam	Brahmaputra
Bangladesh	Jamuna
After joining Ganga	Padma
Before entering Bay of Bengal	Meghna

Important National Parks in Brahmaputra Floodplains

● Kaziranga National Park
● Manas National Park
● Orang National Park
● Dibru-Saikhowa National Park
● Nameri National Park

Species Associated with Brahmaputra Basin

● One-Horned Rhinoceros
● Gangetic River Dolphin(national aquatic animal)
● Wild Water Buffalo
● Bengal Florican
● Hoolock Gibbon
● Asian Elephant
● Golden Langur

Ramsar Sites in Brahmaputra Landscape

● Deepor Beel (Assam)
● Loktak Lake linkage through Northeast river systems (indirect basin relevance)

Deepor Beel

● Freshwater lake near Guwahati.
● Important bird habitat.
● Ramsar Site.



Foreigners Tribunal:

● Legal Basis: Foreigners (Tribunals) Order, 1964
● Parent Act: Foreigners Act, 1946
● Function:Determine whether a person is a foreigner under the Foreigners Act.
● Composition: Headed by judges, advocates, or experienced civil servants with judicial background
● Recent Development: Under the Immigration and Foreigners Order, 2025, FTs have been given powers of a Judicial Magistrate First Class in specified matters.
● Foreigners Tribunals have predominantly functioned in Assam, particularly in relation to NRC and illegal migration issues

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267. ASEAN Regional Forum

Q.With reference to the ASEAN Regional Forum (ARF), consider the following statements:

1. The Forum operates on consensus-based decision-making.
2. The Chair of the ARF is the ASEAN Chair for that year.
3. India is a member of the ARF.

Which of the statements given above are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

IN NEWS: EAM Jaishankar meets Chinese Counterpart Wang Yi at ASEAN Regional Forum Meetings in Philippines

Statement 1: The ASEAN Regional Forum (ARF) functions on the principle of consensus-based decision-making, which is a hallmark of ASEAN-led mechanisms. Decisions are taken only when all members agree, thereby promoting dialogue, confidence-building, and cooperation among participating countries. **Hence, Statement 1 is correct.**

Statement 2: The Chair of the ASEAN Regional Forum is the ASEAN Chair of that particular year. Since ASEAN centrality is the guiding principle of the Forum, the country holding the annual ASEAN Chairmanship automatically presides over ARF meetings and activities. **Hence, Statement 2 is correct.**

Statement 3: India is a member of the ASEAN Regional Forum. It participates actively in discussions related to regional security, maritime cooperation, counter-terrorism, cybersecurity, disaster management, and Indo-Pacific affairs. **Hence, Statement 3 is correct.**

VALUE ADDITION

ASEAN Regional Forum (ARF): Quick Facts

- Established: 1994
- Nature: Security dialogue platform in the Indo-Pacific
- Members: 27
- Chair: ASEAN Chair of the year
- Decision-Making: Consensus-based
- Secretariat: Jakarta, Indonesia (through ASEAN Secretariat)

Objectives of ARF

- Foster dialogue on political and security issues
- Promote confidence-building measures
- Advance preventive diplomacy
- Enhance regional peace and stability

ARF Membership Composition

Category	Number
ASEAN Members	10
ASEAN Dialogue Partners	10
Other Participants	7
Total	27

India and ARF

- Member since 1996
- Supports ASEAN Centrality
- Promotes Indo-Pacific cooperation
- Participates in maritime security and counter-terrorism initiatives

ASEAN-Led Mechanisms

- ASEAN Regional Forum (ARF)
- East Asia Summit (EAS)
- ASEAN Defence Ministers' Meeting Plus (ADMM-Plus)
- ASEAN Plus Three (APT)

268. Corporate Mitra scheme

Q.With reference to the Corporate Mitra Scheme, consider the following statements:

1. It aims to support Micro, Small and Medium Enterprises (MSMEs) through trained paraprofessionals known as Corporate Mitras.
2. The scheme primarily focuses on Tier-II and Tier-III cities to improve outreach beyond metropolitan areas.
3. The scheme is implemented by the Ministry of MSME

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

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Answer: (a) 1 and 2 only

Explanation

In NEWS: Ministry of Corporate Affairs' 'Corporate Mitra Scheme' Awareness Webinar Successfully Organized by IICA Shillong for the youth and enterprises of Arunachal Pradesh

Statement 1:

The Corporate Mitra Scheme, announced in the Union Budget 2026-27, aims to empower MSMEs through trained and certified paraprofessionals called Corporate Mitras.They provide affordable support in areas such as:

- Regulatory compliance
- GST matters
- Accounting
- Financial guidance
- Cost accounting
- Secretarial services **Hence, Statement 1 is correct.**

Statement 2:

To ensure wider outreach beyond metropolitan centres, Corporate Mitras will primarily operate in Tier-II and Tier-III cities, helping MSMEs in smaller towns access professional services.**Hence, Statement 2 is correct.**

Statement 3:

The scheme is implemented by the Ministry of Corporate Affairs (MCA) in collaboration with:

- Institute of Chartered Accountants of India
- Institute of Company Secretaries of India
- Institute of Cost Accountants of India

It is not implemented by the Ministry of MSME **Hence, Statement 3 is incorrect.**

Value Addition

Corporate Mitra Scheme

Feature	Details
Announced In	Union Budget 2026-27
Nodal Ministry	Ministry of Corporate Affairs
Target Group	MSMEs
Key Objective	Provide affordable compliance and business support
Focus Areas	GST, accounting, finance, secretarial services
Geographical Focus	Tier-II and Tier-III cities
Training Structure	6 months academic training + 6 months on-the-job training
Platform	SWAYAM Plus
Age Limit	Up to 30 years

Implementing Institutions

- ICAI
- ICSI
- ICMAI
- IICA

Training Structure

- 6 months classroom
- 6 months internship

269. Indus Water Treaty

Q.Consider the following statements regarding the Indus Waters Treaty:

1. The treaty was brokered by the World Bank.
2. India has exclusive rights over the eastern rivers Ravi, Beas and Sutlej.
3. Pakistan has unrestricted rights over all western rivers without any Indian usage**.**

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation

In NEWS: A year on, India's case for freezing it and what it plans next

Statement 1: The **Indus Waters Treaty (IWT)** was signed on **19 September 1960** at Karachi between **Indian Prime Minister Jawaharlal Nehru** and **Pakistani President Ayub Khan**. The treaty was negotiated with the assistance of the **World Bank (International Bank for Reconstruction and Development - IBRD)**, which acted as a facilitator and signatory to the agreement.**Hence, Statement 1 is correct.**

Statement 2: Under the Indus Waters Treaty, the waters of the **Eastern Rivers—Ravi, Beas, and Sutlej**—were allocated for the **exclusive use of India**. India enjoys unrestricted rights over these rivers for irrigation, storage, and other

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developmental purposes.**Hence, Statement 2 is correct.**

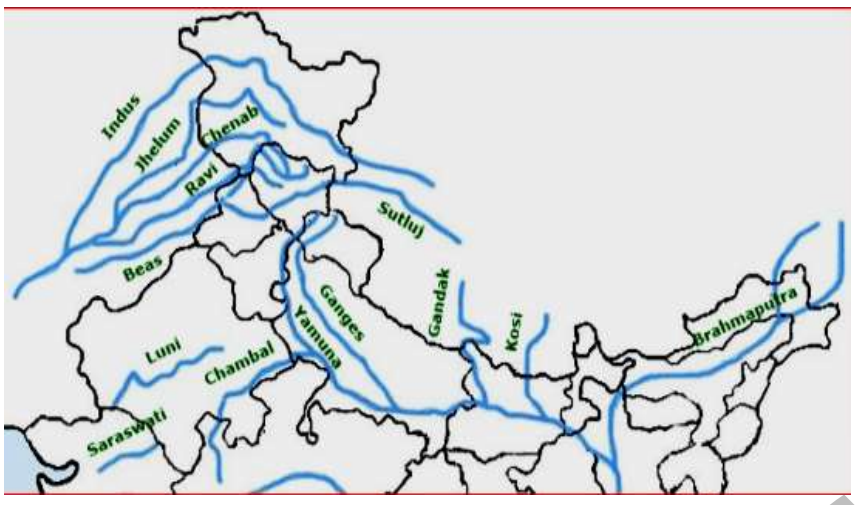
Statement 3:

The **Western Rivers—Indus, Jhelum, and Chenab**—were allocated primarily to Pakistan. However, India's rights over these rivers are not completely barred. The treaty permits India to undertake certain uses such as:

- Domestic use
- Navigation
- Fish culture
- Run-of-the-River (RoR) hydroelectric projects
- Limited irrigation under specified conditions

Therefore, it is incorrect to state that Pakistan has completely unrestricted rights with no Indian usage permitted.**Hence, Statement 3 is incorrect.**

VALUE ADDITION



River Allocation

India (Eastern Rivers)	Pakistan (Western Rivers)
Ravi	Indus
Beas	Jhelum
Sutlej	Chenab

Mnemonic: RBS → India | IJC → Pakistan

- **Water Allocation Ratio:** In terms of total water volume, the treaty allocates roughly 80% of the Indus system's water to Pakistan and about 20% to India.
- **Dispute Hierarchy:** The treaty outlines a three-step dispute resolution process:

1. Questions are handled by the Permanent Indus Commission.
2. Differences are referred to a World Bank-appointed Neutral Expert.
3. Disputes are referred to a seven-member Court of Arbitration.

Dispute Resolution Mechanism

1. Permanent Indus Commission (PIC)
2. Neutral Expert
3. Court of Arbitration

Permanent Indus Commission (PIC)

- One Commissioner from each country.
- Exchanges river data and resolves technical issues.

Projects

- Kishanganga Project → Jhelum Basin
- Baglihar Project → Chenab Basin
- Ratle Project → Chenab Basin

270. Caspian Sea

Q.Consider the following statements:

1. The Caspian Sea is an endorheic basin with no natural outflow to the oceans.
2. The Volga River, Europe's longest river, drains into the Caspian Sea.
3. The Caspian Sea is shared by exactly five littoral states.
4. The Caspian Sea has uniformly high salinity comparable to that of the open oceans.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (c) Only three

Explanation

In NEWS: Iran slams Ukrainian attack on its vessel in Caspian Sea

Explanation

Statement 1: The Caspian Sea is an endorheic basin, meaning it is a closed drainage basin with no natural outflow to any ocean or sea. Water entering the basin is lost primarily through evaporation. This characteristic makes it the world's largest enclosed inland water body.**Hence, Statement 1 is correct.**

Statement 2: The Volga River, the longest river in Europe, is the largest river flowing into the Caspian Sea and contributes a major share of its freshwater inflow. Other important rivers draining into the Caspian include the Ural and Terek rivers.**Hence, Statement 2 is correct.**

Statement 3: The Caspian Sea is bordered by five littoral states:

- Russia
- Kazakhstan
- Turkmenistan
- Iran

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- Azerbaijan

These countries share maritime, energy, and environmental interests in the region.**Hence, Statement 3 is correct.**

Statement 4: The salinity of the Caspian Sea is not uniform. The northern part receives substantial freshwater inflow from rivers such as the Volga and is relatively less saline, whereas the southern part is considerably more saline. Therefore, it does not possess uniformly high salinity comparable to open oceans.**Hence, Statement 4 is incorrect.**

VALUE ADDITION

Facts

- Largest enclosed water body in the world.
- No natural outlet to any ocean.
- Salinity increases from north to south.
- Rich in offshore hydrocarbon reserves.
- Often described as both a sea and a lake due to its unique characteristics.

Caspian Sea: Quick Facts

Feature	**Details**
Type	World's largest enclosed inland water body
Basin Type	Endorheic Basin
Location	Between Europe and Asia
Littoral States	Russia, Kazakhstan, Turkmenistan, Iran, Azerbaijan
Largest Inflowing River	Volga River
Major Resources	Oil and Natural Gas
Largest City	Baku (Azerbaijan)

Important Rivers Draining into Caspian Sea

- Volga
- Ural
- Terek

271. Places in news

Q.Consider the following:

Row	**Country**	**Feature / Detail**
I.	Moldova	Landlocked country with capital at Chișinău
II.	Romania	Bounded by the Black Sea with capital at Bucharest
III.	North Macedonia	Landlocked country with capital at Skopje

In how many of the above rows is the given information correctly matched?

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: (c) All three

Explanation

In NEWS:President Murmu winds up three-nation European tour; departs for Delhi

Row I: Moldova – Landlocked country with capital at Chișinău

Moldova is a landlocked country in Eastern Europe, situated between the Prut and Dniester Rivers. It is bordered by Romania to the west and Ukraine to the north, east, and south. The capital and largest city of Moldova is Chișinău.**Hence, Row I is correctly matched.**

Row II: Romania – Bounded by the Black Sea with capital at Bucharest

Romania is located in Southeastern Europe and has a coastline along the Black Sea. Its capital city is Bucharest. Romania is also a member of NATO (2004) and the European Union (2007).**Hence, Row II is correctly matched.**

Row III: North Macedonia – Landlocked country with capital at Skopje

North Macedonia is a landlocked Balkan country in Southeast Europe. Its capital city is Skopje. The country adopted the name "North Macedonia" under the Prespa Agreement (2018) with Greece, enabling its accession to NATO.**Hence, Row III is correctly matched.**

Value Addition: Moldova – Romania – North Macedonia

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Feature	Moldova	Romania	North Macedonia
Capital	Chișinău	Bucharest	Skopje
Location	Eastern Europe	Southeastern Europe	Balkan Peninsula (Southeast Europe)
Landlocked	Yes	No	Yes
Sea Access	None	Black Sea Coast	None
Major River	Dniester, Prut	Danube, Prut	Vardar
Highest Peak	Bălănești Hill	Mount Moldoveanu	Mount Korab
Official Language	Romanian	Romanian	Macedonian
Currency	Moldovan Leu	Romanian Leu	Macedonian Denar
NATO Member	No	Yes (2004)	Yes (2020)
EU Member	No (Candidate Country)	Yes (2007)	No (Candidate Country)
Former Soviet Republic	Yes	No	No (Former Yugoslavia)

Country	Best Known For
Moldova	Former Soviet republic; located between Romania and Ukraine
Romania	Black Sea coastline; Carpathian Mountains; EU & NATO member
North Macedonia	Prespa Agreement (2018); name changed from Macedonia; NATO member

Country	Borders
Moldova	Romania, Ukraine
Romania	Ukraine, Moldova, Hungary, Serbia, Bulgaria, Black Sea
North Macedonia	Kosovo, Serbia, Bulgaria, Greece, Albania

272. Hepatitis

Q.Consider the following statements regarding the modes of transmission of Hepatitis viruses:

1. Hepatitis A and Hepatitis E are primarily transmitted via the faecal-oral route through contaminated food and water.
2. Hepatitis B, C, and D are mainly transmitted through infected blood, body fluids, and parenteral/perinatal routes.
3. A person can contract a Hepatitis D infection independently without a co-existing Hepatitis B infection.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2, and 3

Answer: (a) 1 and 2 only

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EXPLANATION:

IN NEWS:India's Fight to Combat Hepatitis

Statement 1: Hepatitis A (HAV) and Hepatitis E (HEV) are primarily transmitted through the faecal-oral route, usually via contaminated food and water. These infections are strongly associated with poor sanitation, unsafe drinking water, and inadequate hygiene practices. Both HAV and HEV generally cause acute infections and are often outbreak-prone in areas with poor sanitation.**Hence, Statement 1 is correct.**

Statement 2: Hepatitis B (HBV), Hepatitis C (HCV), and Hepatitis D (HDV) are mainly transmitted through contact with infected blood and body fluids. Common routes include:

- Mother-to-child transmission (especially HBV)
- Unsafe injections and medical procedures
- Sharing contaminated needles and syringes
- Blood transfusions with infected blood
- Other parenteral exposures

HBV and HCV are transmitted through infected blood and body fluids, while HDV occurs only in individuals infected with HBV. Therefore, these viruses are primarily blood-borne rather than food- or water-borne.**Hence, Statement 2 is correct.**

Statement 3: Hepatitis D (HDV) is a defective virus that cannot complete its life cycle independently. It can infect only those individuals who are already infected with Hepatitis B virus (HBV) because HDV requires the Hepatitis B surface antigen (HBsAg) for replication.Therefore, a person cannot contract Hepatitis D independently without a co-existing Hepatitis B infection.**Hence, Statement 3 is incorrect.**

Value Addition

Transmission Routes of Hepatitis Viruses

Virus	Major Transmission Route
Hepatitis A(HAV)	Contaminated food and water (Faecal-Oral)
Hepatitis B(HBV)	Blood, body fluids, mother-to-child
Hepatitis C(HCV)	Infected blood, unsafe injections
Hepatitis D(HDV)	Along with Hepatitis B infection
Hepatitis E(HEV)	Contaminated food and water (Faecal-Oral)

Hepatitis: Vaccine Availability

Hepatitis Virus	Vaccine Available
Hepatitis A	Yes
Hepatitis B	Yes
Hepatitis C	No
Hepatitis D	Prevented through HBV vaccination
Hepatitis E	No (not widely available)

Hepatitis: Acute vs Chronic

Usually Acute	Can Become Chronic
Hepatitis A	Hepatitis B
Hepatitis E	Hepatitis C
	Hepatitis D

- HBV is the most common cause of chronic viral hepatitis globally.
- HCV is curable using Direct-Acting Antiviral (DAA) drugs.
- HDV is a defective virus and requires HBV for replication.
- HAV and HEV outbreaks are linked to poor sanitation and contaminated water.
- Chronic HBV and HCV can lead to Cirrhosis and Hepatocellular Carcinoma (Liver Cancer).

National Viral Hepatitis Control Programme (NVHCP)

- Launched: 2018
- Ministry: Ministry of Health & Family Welfare
- Target: Elimination of Viral Hepatitis as a public health threat by 2030
- Free Services: Diagnosis and treatment for Hepatitis B & C
- Digital Platform: NVHCP-MIS Portal

273. World Junior Squash Championship

Q. Consider the following statements:

I. Anahat Singh became the first Indian squash player to win the World Junior Squash Championship title.
II. Squash is included in the Olympic Games programme for Los Angeles 2028.

Which of the statements given above is/are correct?

(a) I only
(b) II only
(c) Both I and II
(d) Neither I nor II

Answer: (c) Both I and II
Explanation
In NEWS: Prime Minister praises historic feat at the 2026 World Junior Squash Championship
Statement I:
Anahat Singh won the **Gold Medal at the 2026 World Junior Squash Championship** and became the **first Indian squash player ever to win the World Junior Championship title**.**Hence, Statement I is correct.**
Statement II:
The **International Olympic Committee (IOC)** approved the inclusion of **Squash** in the **Los Angeles 2028 Olympic Games**, marking the sport's Olympic debut.**Hence, Statement II is correct.**
Value Addition
Squash: Quick Facts
- **Olympic Debut:** Los Angeles 2028
- **Governing Body:** World Squash Federation (WSF)
- **Court Type:** Four-walled indoor court
- **Ball Used:** Hollow rubber ball
- **Format:** Singles and Doubles competitions

Indian Squash Personalities
- **Anahat Singh** – World Junior Champion 2026
- **Saurav Ghosal** – First Indian to break into World Top 10
- **Dipika Pallikal** – First Indian woman in World Top 10
- **Joshna Chinappa** – Multiple-time national champion

Sports in LA 2028 Added by IOC
- Squash
- Cricket (T20)
- Baseball/Softball
- Flag Football
- Lacrosse

274. Flue-cured Virginia Tobacco (FCV)

Q.Consider the following statements regarding Flue-Cured Virginia (FCV) Tobacco:

1. It contains high sugar content and medium to high levels of nicotine.
2. It is predominantly used for cigarette production rather than cigars.
3. In India, its cultivation is majorly concentrated in the states of Punjab and Haryana.

Which of the statements given above is/are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2, and 3

Answer: (b) 1 and 2 only
Explanation
In NEWS: Department of Commerce Reviews Flue-Cured Virginia (FCV) Tobacco Market Situation in Andhra Pradesh
Statement 1: Flue-Cured Virginia (FCV) Tobacco is characterized by its **high sugar content** and **medium to high levels of nicotine**. These properties contribute to its distinct flavour, aroma, and widespread commercial use. **Hence, Statement 1 is correct.**
Statement 2:FCV Tobacco is primarily used in the manufacture of **cigarettes**. Its mild and slightly sweet flavour, resulting from the flue-curing process and natural tannins, makes it particularly suitable for cigarette production rather than cigars. **Hence, Statement 2 is correct.**
Statement 3: In India, FCV Tobacco cultivation is concentrated mainly in **Andhra Pradesh and Karnataka**, which account for the bulk of the country's FCV tobacco production. Punjab and Haryana are not major producers of FCV tobacco. **Hence, Statement 3 is incorrect.**
Value Addition
FCV Tobacco: Quick Facts
- **Full Form:** Flue-Cured Virginia Tobacco
- **Major Use:** Cigarette manufacturing
- **Sugar Content:** High
- **Nicotine Level:** Medium to High
- **Flavour:** Mild and slightly sweet
- **Major Producing States:** Andhra Pradesh and Karnataka
- **Regulator:** Tobacco Board

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- **Ministry:** Ministry of Commerce & Industry

Tobacco Curing Methods

Method	Key Feature
Air-Cured	Natural air drying
Fire-Cured	Smoke-assisted curing
Flue-Cured	Indirect heat curing through flues

Why is Tobacco Cured?

- Removes excess moisture from harvested leaves.
- Reduces chlorophyll content.
- Enhances flavour and aroma.
- Makes tobacco suitable for smoking and storage.

FCV Tobacco: Key Characteristics

- Produces primarily cigarette tobacco.
- Rich in natural tannins.
- Distinct mild and sweet aroma.
- Standardized curing process ensures uniform quality

275. Caridina Sharmai

Q.With reference to Caridina sharmai, recently discovered in India, consider the following statements:

1. It is a freshwater shrimp species.
2. It possesses brush-like hairs on its front legs.
3. The species is solitary in nature and avoids group living.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

EXPLANATION:

IN NEWS: New species of freshwater shrimp discovered camouflaging in Uttar Pradesh's Gomti River

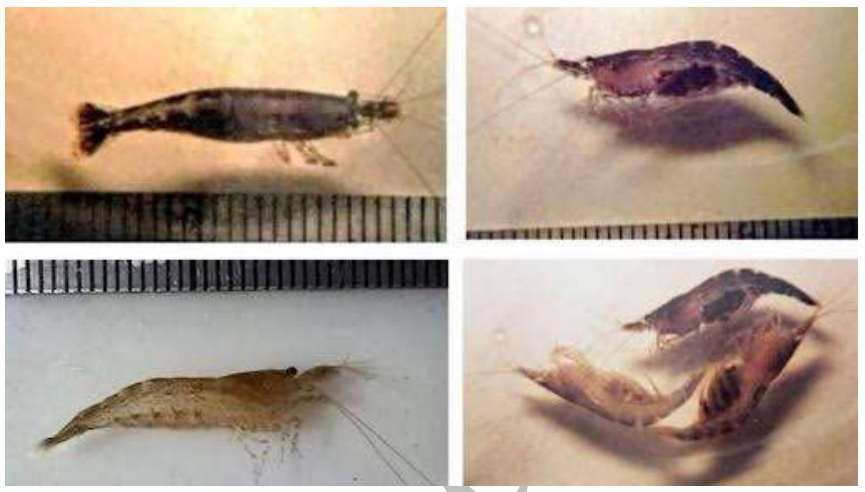
Statement 1: Caridina sharmai is a newly identified species of freshwater shrimp belonging to the genus *Caridina*.**Hence, Statement 1 is correct.**

Statement 2: The shrimp possess **long brush-like hairs on its front legs**, which help it scrape algae from aquatic plants and filter food particles from the water.**Hence, Statement 2 is correct.**

Statement 3: Researchers observed that *Caridina sharmai* is a **calm and social species that prefers to live in groups**, often clinging to underwater vegetation. Therefore, it is not solitary in nature.**Hence, Statement 3 is incorrect.**

Value Addition

Caridina sharmai: Quick Facts



- **Type:** Freshwater shrimp
- **Discovered in:** Gomti River, Ganga basin, Uttar Pradesh
- **River System:** Ganga Basin
- **Length:** 15–24 mm
- **Behaviour:** Social and group-living
- **Feeding Adaptation:** Brush-like hairs on front legs
- **Colour:** Dark brown/black with yellow-grey stripes
- **Special Trait:** Becomes nearly transparent in aquariums

Distinctive Features

- Short **knife-shaped rostrum**
- Lacks median spine on **telson**
- Algae scraper and filter feeder
- Prefers submerged aquatic vegetation

276. Asiatic Wild Dog

Q.With reference to the Asiatic Wild Dog (Dhole), consider the following statements:

1. Dholes are hypercarnivores, with more than 70% of their diet consisting of meat.
2. The Asiatic Wild Dog is listed as Endangered on the IUCN Red List.
3. The Asiatic Wild Dog is endemic to India.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

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EXPLANATION:

IN NEWS:What do Asiatic wild dogs or dholes in India eat? Mostly chital and sambar and hardly any livestock, says study

Statement 1:

Dholes are classified as hypercarnivores, meaning that more than 70% of their diet consists of animal flesh. Recent studies in Central India show that their prey base is dominated by wild herbivores such as sambar and chital rather than livestock. **Hence, Statement 1 is correct.**

Statement 2:

The Asiatic Wild Dog (Cuon alpinus) is listed as Endangered on the IUCN Red List due to habitat fragmentation, declining prey populations, disease transmission from domestic dogs, and human-wildlife conflict. **Hence, Statement 2 is correct.**

Statement 3:

The Dhole is not endemic to India. It is distributed across parts of South Asia, Southeast Asia, and East Asia, though India currently supports one of the largest remaining populations. **Hence, Statement 3 is incorrect.**

Value Addition



Dhole: Quick Facts

- Scientific Name: Cuon alpinus
- Family: Canidae
- IUCN Status: Endangered
- WPA, 1972: Schedule I
- CITES: Appendix II
- Other Names:
  - Indian Wild Dog
  - Red Dog
  - Red Wolf
  - Whistling Dog

Distribution in India

- Western Ghats
- Eastern Ghats
- Central Indian Landscape
- Northeast India

Unique Features

- Uses a distinctive whistling call for communication.
- Highly social; lives in packs of 2–25 individuals.
- Excellent runner, swimmer, and jumper.
- One of India's major pack-hunting predators.

Major Prey Species

- Sambar
- Chital
- Gaur calves
- Wild pig

Threats

- Habitat fragmentation
- Decline of prey base
- Diseases from domestic dogs
- Human-wildlife conflict

277. Exomoon

Q.The recently reported CD-35 2722 (or CD-35 2772) system has gained attention in astronomy due to the detection of:

(a) A terrestrial exoplanet in the habitable zone
(b) A candidate giant gaseous exomoon (exosatellite) orbiting a brown dwarf
(c) The nearest known black hole to Earth
(d) An exoplanet with signatures of atmospheric water vapor

Answer: (b) A candidate giant gaseous exomoon (exosatellite) orbiting a brown dwarf

EXPLANATION:

IN NEWS:Astronomers Discover the First Known Moon Outside our Solar System—and What a Moon it Is

Option (a): A terrestrial exoplanet in the habitable zone

The discovery is not related to an Earth-like rocky planet located within the habitable zone of a star. The reported object is associated with a brown dwarf system and is considered a possible exomoon.**Hence, Option (a) is incorrect.**

Option (b): A candidate giant gaseous exomoon (exosatellite) orbiting a brown dwarf

Researchers detected a giant gaseous object with a mass comparable to Jupiter orbiting a brown dwarf in the CD-35 2722 system. The discovery was made using the radial velocity method, which measures the wobble of a celestial object caused by an orbiting companion.

Since the object appears to orbit a brown dwarf that itself revolves around a central star, astronomers

classify it as a candidate exomoon (exosatellite). Its exact nature—whether a giant planet or a moon—is still under investigation.**Hence, Option (b) is correct.**

Option (c): The nearest known black hole to Earth

The CD-35 2722 system is not associated with the discovery of a black hole. The recent findings relate to a possible moon-like object in a brown dwarf system.**Hence, Option (c) is incorrect.**

Option (d): An exoplanet with signatures of atmospheric water vapor

No evidence of atmospheric water vapor or biosignatures has been reported from this system. The discovery concerns a candidate exomoon.**Hence, Option (d) is incorrect.**

Value Addition

Exomoon vs Exoplanet

Term	Meaning
Exoplanet	Planet outside the Solar System
Exomoon (Exosatellite)	Natural satellite orbiting an exoplanet or similar body
Candidate Exomoon	Possible exomoon awaiting confirmation

Brown Dwarf

Feature	Description
Nature	"Failed Star"
Mass Range	Between planets and stars
Hydrogen Fusion	Cannot sustain stable hydrogen fusion
Example	CD-35 2722 B

Radial Velocity Method

- Based on the Doppler Effect.
- Detects the wobble of a star or brown dwarf.
- Measures shifts in spectral lines.
- Helps estimate the mass of the orbiting object.

ESO Very Large Telescope (VLT)

- Operated by the European Southern Observatory (ESO).
- Located in Chile's Atacama Desert.
- Among the world's most powerful optical telescopes.

Quick Facts

- Distance from Earth: ~73 light-years
- Candidate object mass: ~90% of Jupiter's mass
- Orbital Period: ~170 days
- Status: Not yet conclusively confirmed

278. Anti defection Law

Q.With reference to the Anti-Defection Law under the Tenth Schedule of the Constitution, consider the following statements:

1. Under the Tenth Schedule, a merger of a legislative party is valid only if at least two-thirds of its members agree to it.
2. The earlier provision allowing a one-third split of a legislative party as an exemption from disqualification was removed by the 91st Constitutional Amendment Act, 2003.
3. Under the Anti-Defection Law, the decision on the disqualification of members is taken by the Election Commission of India.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (a) 1 and 2 only

Explanation

In NEWS: Court seeks Centre's reply on plea over anti-defection law

Rajya Sabha member Kapil Sibal has challenged the interpretation of the 'merger' exception under Paragraph 4 of Tenth Schedule, contends issue is unlikely to be resolved through legislative process

Statement 1:

Under Paragraph 4 of the Tenth Schedule, a merger of a legislative party is considered valid only when **at least two-thirds of the members of the legislature party agree to such a merger**. Members supporting a merger under this provision are protected from disqualification. **Hence, Statement 1 is correct.**

Statement 2:

The Anti-Defection Law originally contained a provision allowing **one-third of the members of a legislative party to split** without attracting disqualification. However, this provision was widely misused. To curb defections, the **91st Constitutional Amendment Act, 2003** deleted the split provision and retained only the merger provision requiring the support of two-thirds of the members. **Hence, Statement 2 is correct.**

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Statement 3:

The power to decide questions relating to disqualification under the Tenth Schedule rests with the **Speaker of the Legislative Assembly/Lok Sabha or the Chairman of the Legislative Council/Rajya Sabha**, depending on the House concerned. The **Election Commission of India has no role in deciding disqualification under the Anti-Defection Law. Hence, Statement 3 is incorrect.**

Value Addition

Anti-Defection Law: Quick Facts

- **Constitutional Basis:** Tenth Schedule
- **Inserted By:** 52nd Constitutional Amendment Act, 1985
- **Popular Name:** Anti-Defection Law
- **Objective:** Curb political defections and ensure stability of governments
- **Disqualification Decided By:** Speaker/Chairman of the concerned House
- **Split Provision (1/3):** Removed by 91st Constitutional Amendment Act, 2003
- **Merger Requirement:** Support of at least **2/3rd members** of the legislative party

Grounds for Disqualification

- Voluntarily giving up membership of a political party
- Voting or abstaining contrary to the party whip without prior permission
- Independent member joining a political party after election
- Nominated member joining a political party after six months of taking seat

Constitutional Amendments

Amendment	Significance
52nd Amendment Act, 1985	Added Tenth Schedule
91st Amendment Act, 2003	Removed split provision; strengthened anti-defection law

Important Cases

- **Kihoto Hollohan v. Zachillhu (1992)** – Upheld constitutional validity of the Anti-Defection Law; Speaker's decision subject to judicial review.
- **Keisham Meghachandra Singh v. Speaker, Manipur Assembly (2020)** – Supreme Court recommended timely disposal of disqualification petitions.

279. PM Vidyalakshmi scheme

Q.With reference to the PM Vidyalaxmi Scheme, consider the following statements:

1. It is a Central Sector Scheme aimed at supporting meritorious students pursuing higher education.
2. There is no prescribed upper limit on the amount of education loan that can be sanctioned under the scheme.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Explanation

In NEWS: PM-Vidyalaxmi Portal for Higher Education Loans

Statement 1:

PM Vidyalaxmi is a **Central Sector Scheme** launched by the Government of India to provide financial assistance to **meritorious students** pursuing higher education in Quality Higher Education Institutions (QHEIs). The scheme aims to ensure that financial constraints do not prevent deserving students from accessing quality education. **Hence, Statement 1 is correct.**

Statement 2:

Under the PM Vidyalaxmi Scheme, there is **no fixed upper limit** on the amount of education loan that can be sanctioned. The loan amount depends on the actual requirements of the student, including tuition fees, hostel charges, mess fees, laptop expenses, living expenses, and other educational costs charged by the institution. **Hence, Statement 2 is correct.**

Value Addition

PM Vidyalaxmi: Quick Facts

- **Scheme Type:** Central Sector Scheme
- **Target Group:** Meritorious students
- **Eligible Institutions:** Top **860** Quality Higher Educational Institutions (**QHEIs)** (identified through NIRF Applicable to both Government and Private Higher Educational Institutions included in the approved list.)
- **Loan Limit:** No upper ceiling
- **Application Platform:** PM-Vidyalaxmi Portal
- **Interest Subvention:** 3% on loans up to ₹10 lakh

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- **Income Eligibility for Interest Subvention:** Family income up to ₹8 lakh per annum
- **Credit Guarantee:** Available for loans up to ₹7.5 lakh
- **Government Guarantee Coverage:** 75% of outstanding default

280. Gobind Sagar Lake

Q. With reference to Gobind Sagar Lake, consider the following statements:

1. It is one of the deepest man-made lakes in the world.
2. It is the third-largest reservoir in India in terms of water storage.
3. It was formed due to the construction of the Bhakra Dam on the Sutlej River.
4. It was recognized as a waterfowl refuge in 1962.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (d) All four

Explanation

In NEWS: Himachal's first island tourism project at Gobind Sagar Lake to be ready by Sept-end

Statement 1:

Gobind Sagar Lake is a man-made reservoir located in Bilaspur district of Himachal Pradesh. It is regarded as one of the deepest man-made lakes in the world and is an important freshwater resource in northern India.**Hence, Statement 1 is correct.**

Statement 2:

Gobind Sagar is the **third-largest reservoir in India in terms of water storage capacity**. It plays a major role in irrigation, hydropower generation, and water supply for several north Indian states.**Hence, Statement 2 is correct.**

Statement 3:

The lake was created following the construction of the **Bhakra Dam** across the **Sutlej River** during the 1950s and 1960s. The dam is among the world's highest gravity dams.**Hence, Statement 3 is correct.**

Statement 4:

In **1962**, Gobind Sagar Lake was officially recognized as a **Waterfowl Refuge** because of its importance as a habitat for migratory and resident aquatic birds.**Hence, Statement 4 is correct.**

Value Addition

Gobind Sagar Lake



- Named after Guru Gobind Singh.
- Located in Bilaspur, Himachal Pradesh.
- Formed by Bhakra Dam on Sutlej River.
- Third-largest reservoir in India (water storage).
- Declared Waterfowl Refuge in 1962.
- Important for irrigation, fisheries and hydropower.

Flora & Fauna

Notable Fish Species

- Tor Putitora (Golden Mahseer)
- Labeo Dero
- Mystus Seenghala
- Mirror Carp

Important Wildlife

- Sambar
- Nilgai
- Chinkara
- Sloth Bear
- Panther
- Wild Boar

Bhakra Dam

- Built on Sutlej River.
- One of the world's highest gravity dams.
- Key component of the Bhakra-Nangal Project.

Waterfowl Refuge

- Protected habitat for migratory and resident aquatic birds.
- Supports nesting, breeding and feeding.
- Important for wetland conservation and biodiversity.

Beas–Sutlej Link

- Completed in 1976.
- Diverts Beas waters to Gobind Sagar.
- Helps maintain reservoir levels.

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Reservoir	River
Gobind Sagar	Sutlej
Hirakud	Mahanadi
Nagarjuna Sagar	Krishna
Sardar sarovar	Narmada
Tehri	Bhagirathi

281. Bharat Taxi

Q.Consider the following statements regarding Bharat Taxi:

1. It is India's first cooperative-led, driver-owned ride-hailing platform.
2. It has been developed by Sahkar Taxi Cooperative Limited.
3. It aims to transfer the full fare amount directly to drivers without commission deductions.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
Explanation
In NEWS: Availability of Bharat Taxi in Purvanchal Areas of UP
Statement 1:
Bharat Taxi is India's first **cooperative-led, driver-owned ride-hailing platform**. It has been designed on the principle of **platform cooperativism**, where drivers are not merely service providers but also stakeholders in the cooperative structure.**Hence, Statement 1 is correct.**

Statement 2:
The platform is operated and developed by **Sahakar Taxi Cooperative Limited (STCL)**, a Multi-State Cooperative Society registered under the **Multi-State Cooperative Societies (MSCS) Act, 2002**.STCL undertakes driver onboarding, customer registration, platform management, expansion, and operational activities.**Hence, Statement 2 is correct.**

Statement 3:
Bharat Taxi follows a zero-commission model under which no commission is deducted from the fare earned by a Sarathi. The fare is transferred directly to the driver. However, a separate nominal subscription/platform usage fee may be charged to support technology maintenance and platform expansion.**Hence, Statement 3 is correct.**

Bharat Taxi:

Feature	Details
Nature	Cooperative-led, driver-owned ride-hailing platform
Operated By	Sahakar Taxi Cooperative Limited (STCL)
Legal Status	Multi-State Cooperative Society
Registered Under	Multi-State Cooperative Societies (MSCS) Act, 2002
Ministry Support	Ministry of Cooperation
Technical Support	National e-Governance Division (NeGD)
Ownership Model	Drivers become cooperative members and shareholders
Drivers Called	**Sarathis**
Platform Components	Sarathi App, Passenger App, Backend Management System
Core Objective	Transparent, driver-centric mobility service
Commission Model	Zero commission on fare; separate platform usage/subscription fee may apply
Fare Policy	No Surge Pricing; transparent fares
Funding Source	STCL's own capital; no Government expenditure
Governance Model	Cooperative-based platform governance
Expansion Model	Phased rollout based on operational readiness and statutory approvals

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Current Operations	Delhi-NCR, Gujarat, Maharashtra
Digital Integrations	DigiLocker, UMANG, API Setu
Data Security	Complies with Government data protection and cybersecurity norms
Driver Benefits	Loans, insurance support, business expansion opportunities
Review Mechanism	Periodic review of customer experience, grievances, fares, and platform performance
Public Awareness	Social media outreach, stakeholder engagement, awareness campaigns
State-Level Coordination	Transport Departments, Urban Local Bodies, Cooperation Departments
Launch Condition	Operational preparedness + statutory approvals + administrative clearances
Cooperative Vision	"Sahkar se Samriddhi" (Prosperity through Cooperation)
Unique Feature	India's first cooperative-led ride-hailing platform

Institutional Support

STCL is backed by major cooperative institutions:

- NCDC
- NDDB
- NAFED
- IFFCO
- KRIBHCO
- NABARD
- NCEL
- GCMMF (Amul)

Bharat Taxi vs Private Aggregators

Feature	Bharat Taxi	Conventional Aggregators
Ownership	Driver-Owned Cooperative	Corporate-Owned
Driver Status	Shareholder &	Service
	Member	Provider
Commission	Zero	Platform Commission Charged
Pricing	No Surge Pricing	Dynamic/Surge Pricing
Decision Making	Cooperative Governance	Corporate Management
Welfare Support	Loans, Insurance, Cooperative Benefits	Limited Platform Benefits
Objective	Driver Empowerment	Profit Maximization

282. World largest grain storage

Q.Consider the following statements regarding the World's Largest Grain Storage Plan in the Cooperative Sector:

1. The plan aims to create decentralized agricultural infrastructure at the level of Primary Agricultural Credit Societies (PACS).
2. PACS form the village-level unit and the last link in the three-tier cooperative credit structure.
3. The plan has been integrated with the storage requirements of the Food Corporation of India (FCI).

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

IN NEWS:World's Largest Grain Storage Plan in the Cooperative Sector

Statement 1:

The World's Largest Grain Storage Plan aims to establish decentralized agricultural infrastructure at the level of Primary Agricultural Credit Societies (PACS). The plan includes the creation of godowns, custom hiring centres, processing units, and Fair Price Shops to strengthen rural agricultural value chains and reduce post-harvest losses.**Hence, Statement 1 is correct.**

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Statement 2:

Primary Agricultural Credit Societies (PACS) are village-level cooperative credit institutions and constitute the last mile of the three-tier cooperative credit structure.

The structure consists of:

- State Cooperative Banks (SCBs) – State Level
- District Central Cooperative Banks (DCCBs) – District Level
- PACS – Village Level**Hence, Statement 2 is correct.**

Statement 3:

FCI has mapped an additional storage requirement of approximately 26.03 lakh metric tonnes (LMT) across 216 locations in 18 States/UTs. To encourage warehouse creation, FCI provides a 9-year hiring assurance for eligible warehouses.**Hence, Statement 3 is correct.**

World's Largest Grain Storage Plan:

- Nodal Ministry: Ministry of Cooperation
- Objective: Creation of decentralized agricultural infrastructure at the village level.
- Implementation: Through PACS and other cooperative societies.
- Infrastructure Components: Godowns, Processing Units, Custom Hiring Centres (CHCs), and Fair Price Shops (FPSs).
- PACS Identified: 1,015 across the country.
- Godowns Completed: 313 PACS.
- Storage Capacity Created: 1.80 Lakh Metric Tonnes (LMT).
- FCI Integration: Aligned with Food Corporation of India's storage requirements.
- FCI Storage Gap Identified: 26.03 LMT.
- Potential Storage Locations: 216 locations mapped by FCI.
- Geographical Coverage: 18 States/Union Territories.
- Implementation Strategy: Convergence of existing schemes such as AIF, AMI, SMAM, and PMFME.
- Key Goal: Reduce post-harvest losses and strengthen decentralized food grain storage.
- Cooperative Focus: Transform PACS into multi-service rural economic hubs.
- FCI Support: 9-year hiring assurance for eligible warehouses.
- Vision: Supports the Ministry of Cooperation's "Sahkar Se Samriddhi" initiative.

PACS in Cooperative Credit Structure

Tier	Institution	Level
Apex Tier	State Cooperative Banks (SCBs)	State Level
Intermediate Tier	District Central Cooperative Banks (DCCBs)	District Level
Grassroots Tier	Primary Agricultural Credit Societies (PACS)	Village Level

Schemes Converged Under the Plan

Scheme	Purpose
Agriculture Infrastructure Fund (AIF)	Credit support for infrastructure
Agricultural Marketing Infrastructure (AMI)	Storage and marketing facilities
SMAM	Farm mechanization support
PMFME	Food processing infrastructure

283. Blue straggler star

Q.With reference to Blue Straggler Stars (BSS), recently seen in the news, consider the following statements:

1. They are main-sequence stars found in star clusters that appear hotter, bluer, and more massive than expected for the cluster's age.
2. Their existence challenges the standard theory of single-star evolution as they appear younger than neighbouring stars formed at the same time.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Explanation

In NEWS: Astronomers discover an actively forming luminous blue straggler star

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Statement 1:

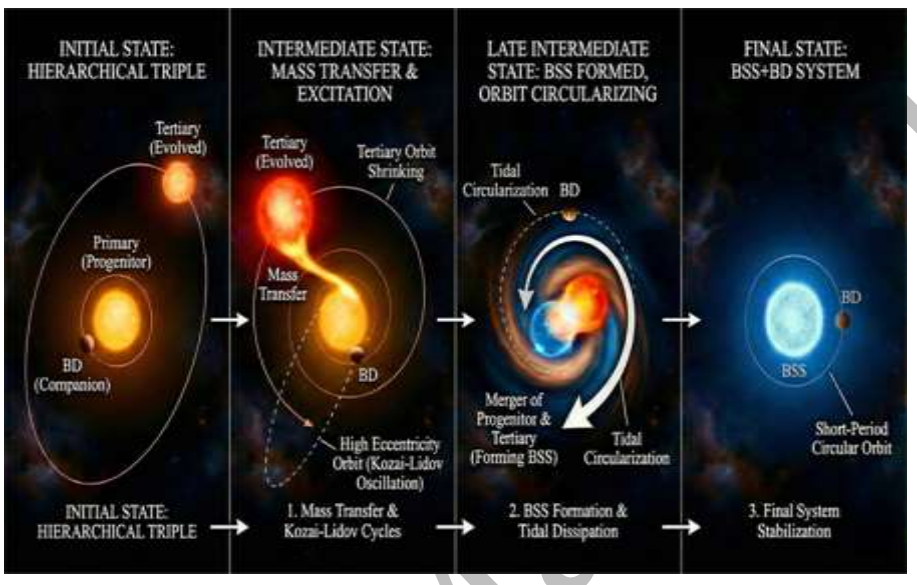
Blue Straggler Stars are a peculiar class of **main-sequence stars** found in star clusters. They appear **hotter, bluer, brighter, and more massive** than other stars that formed at the same time within the cluster.

Normally, stars of similar age in a cluster evolve together. However, Blue Stragglers appear unusually young and continue to occupy an extended region of the main sequence.**Hence, Statement 1 is correct.**

Statement 2:

According to standard **single-star evolution theory**, stars born together in a cluster should evolve at comparable rates.Blue Straggler Stars appear younger than their peer stars despite belonging to the same stellar population. This apparent rejuvenation cannot be explained by normal stellar evolution alone and is believed to result from **mass transfer in binary systems or stellar mergers**.Thus, they challenge the predictions of standard single-star evolutionary models.**Hence, Statement 2 is correct.**

Value Addition



Why are Blue Stragglers Special?

- Appear younger than the cluster they belong to.
- Remain on the **main sequence** longer than expected.
- Are hotter and bluer than surrounding stars.
- Often called **"rejuvenated stars."**
- Provide insights into **binary star evolution**.

Blue Straggler vs Normal Cluster Star

Feature	**Blue Straggler Star**	**Normal Cluster Star**
Age Appearance	Appears younger	Consistent with cluster age
Colour	Blue	Redder/Cooler
Temperature	Higher	Lower
Mass	Higher than expected	Normal for age
Evolution	Rejuvenated	Standard stellar evolution

Formation Mechanisms

Mechanism	**Description**
Mass Transfer	Companion star transfers matter to the BSS
Stellar Merger	Two stars collide and merge
Triple-Star Interaction	Third star facilitates mass transfer

BSS commonly found

- Globular Clusters
- Open Clusters
- Dwarf Galaxies
- Galactic Field

284. Bhargavastra

Q.With reference to Bhargavastra, recently seen in the news, consider the following statements:

1. It is India's first indigenous micro-missile-based counter-drone system designed to neutralise drone swarms.
2. It employs both kinetic and non-kinetic methods for countering hostile drones.
3. It can detect larger UAVs up to 10 km and smaller drones up to 6 km.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation

In NEWS: Army gets demo of indigenous counter-drone system 'Bhargavas

Statement 1:

Bhargavastra is India's first indigenous micro-missile-based counter-drone system, developed by Economic Explosives Ltd. in collaboration with Solar

Defence and Aerospace Ltd. (SDAL). It has been specifically designed to neutralise both individual drones and coordinated drone swarms. **Hence, Statement 1 is correct.**

Statement 2:

Bhargavastra employs a combination of kinetic (hard-kill) and non-kinetic (soft-kill) technologies.

- Kinetic methods: Unguided micro-rockets and precision-guided micro-missiles destroy hostile drones physically.
- Non-kinetic methods: Electronic jamming disrupts communication, navigation, and control systems of drones.

This layered defence mechanism enhances operational effectiveness against diverse UAV threats.**Hence, Statement 2 is correct.**

Statement 3:

The system incorporates advanced surveillance and detection technologies.

- Detection range for larger UAVs: up to 10 km
- Detection range for smaller drones: up to 6 km

It uses Radar, Electro-Optical/Infrared (EO/IR) sensors, and Passive Radio Frequency (RF) sensors integrated through a C4I architecture for target acquisition and tracking.**Hence, Statement 3 is correct.**

Value Addition

Bhargavastra: Quick Facts

Feature	**Details**
Type	Counter-Swarm Drone System
Developed By	Economic Explosives Ltd. & SDAL
Significance	India's first indigenous micro-missile counter-drone system
Target	Individual drones and drone swarms
Detection Range	10 km (large UAVs), 6 km (small drones)
Interception Range	Up to 2.5 km
Kill Mechanism	Dual-layered (Hard Kill + Soft Kill)
Mobility	Mobile platform-based
Terrain Capability	Deserts, plains, high-altitude regions (>5,000 m)

Detection Technologies Used

Technology	**Function**
Radar	Long-range target detection
EO/IR Sensors	Day-night visual and thermal tracking
Passive RF Sensors	Detect drone communication signals
C4I System	Integrated command and control architecture

Hard Kill vs Soft Kill

Method	**Description**
Hard Kill	Physical destruction using rockets and micro-missiles
Soft Kill	Electronic jamming and disruption of drone systems

Counter-Drone Methods

Soft Kill	**Hard Kill**
Jamming	Missile
Spoofing	Rocket
RF Disruption	Laser/Kinetic Intercept

285. Atal Bemmit vyakthi kalyan yojana

Q.With reference to the Atal Beemit Vyakti Kalyan Yojana (ABVKY), consider the following statements:

1. It is an unemployment relief scheme implemented by the Employees' State Insurance Corporation (ESIC).
2. It provides temporary financial assistance to insured employees who lose their jobs.
3. The benefit under the scheme can be availed for a maximum period of 90 days of unemployment.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

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IN NEWS: ATAL BEEMIT VYAKTI KALYAN YOJANA

Statement 1: The **Atal Beemit Vyakti Kalyan Yojana (ABVKY)** is an unemployment relief scheme for workers covered under the **Employees' State Insurance (ESI) Act, 1948**. It is implemented by the **Employees' State Insurance Corporation (ESIC)**.The scheme was introduced on **1 July 2018** to provide income support to insured workers who become unemployed.**Hence, Statement 1 is correct.**

Statement 2: The scheme provides **temporary financial assistance** to insured employees who lose their jobs due to involuntary unemployment. The objective is to help them meet essential expenses while searching for new employment opportunities.**Hence, Statement 2 is correct.**

Statement 3: Under the scheme, eligible beneficiaries receive unemployment relief equal to **50% of their average daily earnings** during the previous contribution periods.The benefit is payable for a **maximum period of 90 days of unemployment** and can be availed only once during the insured person's lifetime.**Hence, Statement 3 is correct.**

Value Addition

ABVKY: Quick Facts

Feature	**Details**
Full Form	Atal Beemit Vyakti Kalyan Yojana
Launched	1 July 2018
Implementing Agency	Employees' State Insurance Corporation (ESIC)
Scheme Type	Unemployment Relief Scheme
Coverage	Insured Persons under ESI Act, 1948
Relief Amount	50% of average daily earnings
Maximum Benefit Period	90 days
Benefit Frequency	Once in lifetime
Payment Mode	Direct Benefit Transfer (DBT)
Current Extension	1 July 2026 – 30 June 2027

Eligibility Conditions

- Covered under the **ESI Act, 1948**.
- Must be unemployed during the claim period.
- Minimum prescribed contribution under the ESI scheme.
- Aadhaar and bank account should be linked.
- Claim to be submitted within **one year of unemployment**.

Relief Not Available In

- Voluntary retirement.
- Voluntary abandonment of employment.
- Lockout.
- Illegal strike.
- Superannuation.
- Dismissal due to disciplinary action.
- Re-employment during the relief period.

286. Eco Sensitive zones

Q.With reference to Eco-Sensitive Zones (ESZs), consider the following statements:

1. Eco-Sensitive Zones are notified around National Parks and Wildlife Sanctuaries to act as shock absorbers for Protected Areas.
2. Eco-Sensitive Zones are notified under the provisions of the Environment (Protection) Act, 1986.
3. The National Wildlife Action Plan (2002–2016) recommended declaration of land within 10 km of Protected Areas as Eco-Sensitive Zones wherever feasible.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

IN NEWS:Centre reissues draft on Western Ghats eco-zone

Statement 1: Eco-Sensitive Zones (ESZs) are areas notified around Protected Areas such as National Parks and Wildlife Sanctuaries to create a transition zone between regions of high protection and areas involving human activities. They act as "shock absorbers" by minimizing the negative impacts of developmental activities on wildlife habitats and ecosystems.**Hence, Statement 1 is correct.**

Statement 2: Eco-Sensitive Zones are notified by the Ministry of Environment, Forest and Climate Change (MoEFCC) under the provisions of the **Environment**

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(Protection) Act, 1986. They are not notified under the Wildlife (Protection) Act, 1972.**Hence, Statement 2 is correct.**

Statement 3: The **National Wildlife Action Plan (2002–2016)** recommended that lands within **10 km of the boundaries of National Parks and Wildlife Sanctuaries** should be declared as Eco-Sensitive Zones wherever feasible to provide additional protection to fragile ecosystems.**Hence, Statement 3 is correct.**



Eco-Sensitive Zones (ESZs): Quick Facts

Feature	Details
Legal Basis	Environment (Protection) Act, 1986
Notified By	Ministry of Environment, Forest and Climate Change (MoEFCC)
Purpose	Protect fragile ecosystems around Protected Areas
Coverage	National Parks and Wildlife Sanctuaries
Function	Buffer/Transition Zone (Shock Absorber)
Recommendation	Up to 10 km around Protected Areas wherever feasible (NWAP 2002–16)
Nature	Site-specific; no uniform width

Activities in ESZs

Prohibited Activities	Regulated Activities	Permitted Activities
Mining & Quarrying	Tourism Activities	Organic Farming
Polluting Industries	Road Expansion	Rainwater Harvesting
Hazardous Waste Handling	Construction Activities	Renewable Energy Projects
Commercial Use of Wood	Hydropower Projects	Local Community Activities

ESA vs ESZ

Feature	Ecologically Sensitive Area (ESA)	Eco-Sensitive Zone (ESZ)
Scale	Large landscape/ecosystem	Area around Protected Areas
Example	Western Ghats ESA	ESZ around Bandipur NP
Objective	Landscape-level conservation	Buffer protection
Legal Basis	Environment (Protection) Act, 1986	Environment (Protection) Act, 1986
Coverage	Ecologically important regions	National Parks & Sanctuaries

Western Ghats ESA is **not an ESZ** around a National Park; it is a large landscape-level conservation area.

Activities Proposed in Western Ghats ESA

Prohibited

- Mining
- Quarrying
- Sand Mining
- New Thermal Power Plants
- Red Category/Polluting Industries
- Large Construction Projects

Regulated

- Hydropower Projects
- Infrastructure Development
- Tourism Activities

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Gadgil Committee vs Kasturirangan Committee

Feature	Gadgil Committee (WGEEP)	Kasturirangan Committee (HLWG)
Year	2011	2013
Chairman	Madhav Gadgil	K. Kasturirangan
Area Proposed	Entire Western Ghats (~1,29,000 sq km)	~59,940 sq km
Coverage	100% Western Ghats	~37% Western Ghats
Approach	Conservation-Centric	Conservation + Development Balance
Status	Not accepted	Basis for ESA notification

National Wildlife Action Plan (2002–2016)

- Recommended declaration of land within **10 km** of Protected Areas as ESZs wherever feasible.
- Emphasized landscape-level conservation and wildlife corridors.
- Promoted integration of conservation with local livelihoods.

National Board for Wildlife (NBWL)

Feature	Details
Constitutional/Statutory Status	Statutory Body
Constituted Under	Wildlife (Protection) Act, 1972
Chairperson	Prime Minister of India
Vice-Chairperson	Union Minister for Environment, Forest and Climate Change
Role	Apex advisory body on wildlife conservation

Western Ghats:

- UNESCO World Heritage Site (2012).
- One of the world's **8 "Hottest Hotspots" of Biodiversity**.
- Older than the Himalayas.
- Runs parallel to the western coast of India.
- Covers approximately **1,60,000 sq km**.
- Intercepts southwest monsoon winds and influences India's climate.
- Source of major peninsular rivers:
  - Godavari
  - Krishna
  - Kaveri
  - Tungabhadra
  - Periyar
- Contains high levels of endemism in:
  - Amphibians
  - Reptiles
  - Flowering plants



287. Atmanirbhar Panchayat Program

Q.With reference to the Atmanirbhar Panchayat Programme, consider the following statements:

1. The programme aims to help Panchayats develop local assets into revenue-generating projects to strengthen Own Source Revenue (OSR).
2. The programme is implemented exclusively through grants provided by the Union Government.
3. Financing under the programme can be facilitated through Public-Private Partnerships (PPPs), CSR contributions, bank finance, and convergence with existing schemes.

Which of the statements given above are correct?

(a) 1 and 3 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

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Answer: (a) 1 and 3 only

EXPLANATION:

IN NEWS:Union Minister Shri Rajiv Ranjan Singh Launches Atmanirbhar Panchayat Program, SAMARTH Portal & Releases Model OSR Rules to Strengthen Panchayati Raj Institutions in India

Statement 1: The Atmanirbhar Panchayat Programme enables Gram Panchayats and Block Panchayats to identify local assets, resources, and opportunities and convert them into bankable revenue-generating projects. The objective is to build sustainable Own Source Revenue (OSR) and reduce dependence on external funding.**Hence, Statement 1 is correct.**

Statement 2: The programme is not implemented exclusively through Union Government grants. Instead, financing is mobilized through multiple channels including Public-Private Partnerships (PPP), Corporate Social Responsibility (CSR) contributions, bank finance, and convergence with existing government schemes.**Hence, Statement 2 is incorrect.**

Statement 3: The programme specifically promotes diversified financing mechanisms such as:

- Public-Private Partnerships (PPP)
- CSR funding
- Bank finance
- Convergence with other government schemes

This approach aims to improve the financial viability and sustainability of Panchayat projects.**Hence, Statement 3 is correct.**

Value Addition

Atmanirbhar Panchayat Programme: Key Facts

Feature	**Details**
Nodal Ministry	Ministry of Panchayati Raj
Objective	Strengthen Panchayat Own Source Revenue (OSR)
Target	350 projects over 4 years
Eligible Institutions	Gram Panchayats & Block Panchayats
Institutional Partners	NABARD, HUDCO
Financing Sources	PPP, CSR, Bank Finance, Scheme Convergence
Focus	Asset Monetization & Revenue Generation

Important Pointers

- Own Source Revenue (OSR) refers to revenue raised by Panchayats through taxes, fees, user charges, and local assets.
- The programme seeks to create financially self-reliant Panchayats aligned with Viksit Bharat @2047.
- 50 projects are planned in the first year, followed by 100 projects annually for the next three years.
- Gram Panchayats require a minimum OSR of ₹50 lakh for eligibility.
- Block Panchayats require a minimum OSR of ₹1 crore.
- NABARD and HUDCO provide institutional and technical support.
- The programme promotes asset-based local economic development rather than grant-dependent governance.
- **Article 243H** of the Constitution, which empowers State Legislatures to authorize Panchayats to levy, collect, and appropriate local taxes/fees (the constitutional bedrock for OSR).

OSR vs Grants

Own Source Revenue (OSR)	**Grants**
Generated by Panchayats themselves	Provided by Centre/States
Sustainable revenue source	External financial support
Includes taxes, fees, user charges	Finance Commission & scheme-based transfers
Enhances financial autonomy	Creates dependence on transfers

SAMARTH Panchayat Portal

- Launched by: Ministry of Panchayati Raj (2026)
- Purpose: Strengthen Own Source Revenue (OSR) of Gram Panchayats through digital technology.
- Nature: First-of-its-kind national digital platform for Panchayat revenue management.
- Key Function: Manages the complete revenue lifecycle:
  - Taxpayer Registration
  - Demand Generation
  - Online Payment Collection
  - Receipt Generation
  - Real-Time Monitoring
- Revenue Covered:

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- Property Tax
- User Charges
- Fees
- Other Non-Tax Revenues

● Significance:

- Enhances transparency and accountability.
- Improves efficiency in revenue collection.
- Reduces manual record-keeping and leakages.
- Strengthens financial sustainability of Panchayats.

● Aligned With: Recommendations of the 16th Finance Commission on increasing Own Source Revenue (OSR) of Local Bodies.

288. Fire clouds

Q.With reference to Pyrocumulonimbus (PyroCb) clouds, recently seen in the news, consider the following statements:

1. They are giant thunderstorm clouds generated by intense heat and smoke plumes from severe wildfires.
2. They can reach heights of 10–15 km and penetrate the Earth's stratosphere.
3. They are capable of producing their own weather systems.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation

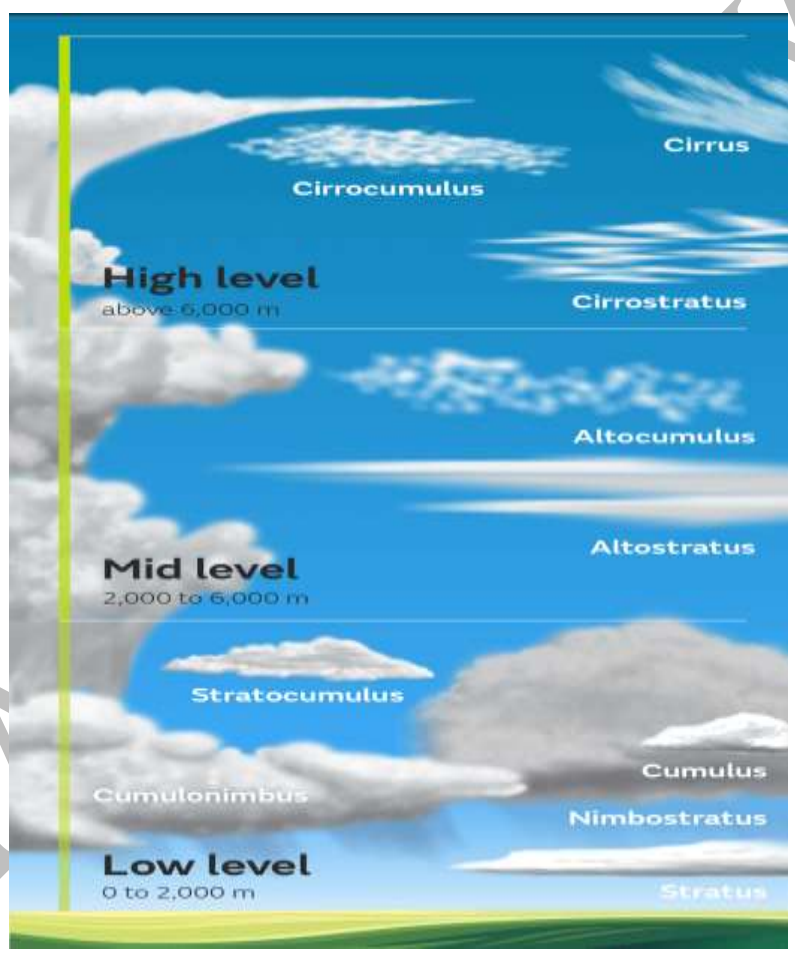
In NEWS: What is a 'fire cloud', and how have they worsened wildfires in Europe?

Statement 1: Pyrocumulonimbus (PyroCb) clouds are giant thunderstorm clouds generated by extremely intense wildfires. Massive fires release enormous amounts of heat, smoke, ash, and water vapour into the atmosphere. The rapidly rising hot air forms a towering plume that can eventually develop into a thunderstorm-like cloud.**Hence, Statement 1 is correct.**

Statement 2: When wildfire intensity is sufficiently high, the rising smoke plume can form a PyroCb cloud that reaches altitudes of approximately **10–15 km**. Such clouds can penetrate the **stratosphere**, allowing smoke and aerosols to be transported over long distances.**Hence, Statement 2 is correct.**

Statement 3: Pyrocumulonimbus clouds can generate their own weather systems. They may produce strong winds, turbulence, lightning, and thunderstorm-like conditions. These weather phenomena can accelerate wildfire spread and even ignite new fires through lightning strikes.**Hence, Statement 3 is correct.**

Value Addition



Pyrocumulonimbus (PyroCb): Quick Facts

Feature	**Details**
Common Name	Fire Cloud
Full Form	Pyrocumulonimbus
Trigger	Intense Wildfires
Nature	Fire-generated Thunderstorm
Height	Up to 10–15 km
Atmospheric Layer Reached	Stratosphere
NASA Description	"Fire-Breathing Dragon of Clouds"

How Fire Clouds Form?

Intense Wildfire → Massive Heat Release → Rising Smoke & Ash Plume → Cooling & Condensation → Cloud Formation → Pyrocumulonimbus (Fire Cloud)

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Why are PyroCb Clouds Dangerous?

- Generate strong and unpredictable winds.
- Accelerate wildfire spread.
- Produce lightning strikes.
- Can ignite secondary fires far from the original fire.
- Create dangerous conditions for firefighters.
- Inject smoke and aerosols into the stratosphere.

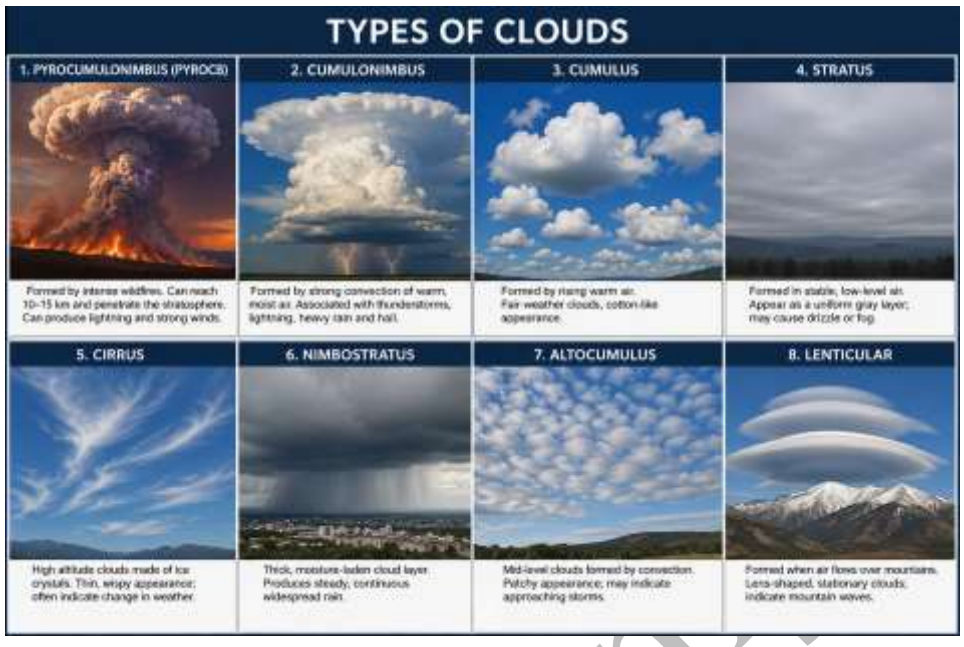
Climate Change Link

PyroCb events are becoming more frequent due to:

- Rising global temperatures
- More intense heatwaves
- Prolonged droughts
- Longer wildfire seasons
- Increased dry vegetation acting as fuel

PyroCb vs Other Important Clouds

Cloud Type	Formation	Key Feature
Pyrocumulonimbus (PyroCb)	Intense wildfires	Fire-generated thunderstorm; may reach stratosphere
Cumulonimbus	Strong convection of warm moist air	Thunderstorms, lightning, heavy rain
Cumulus	Rising warm air	Fair-weather "cotton-like" clouds
Stratus	Low-level stable air	Uniform grey cloud layer; drizzle/fog
Cirrus	Ice crystals at high altitude	Thin, wispy clouds; indicate weather change
Nimbus (Nimbostratus)	Thick moisture-laden cloud	Continuous widespread rainfall
Altocumulus	Mid-level convection	Patchy clouds; may indicate approaching storms
Lenticular	Air flowing over mountains	Lens-shaped stationary cloud



289. Mahanadhi River

Q.With reference to the Mahanadi River, consider the following statements:

1. The Mahanadi River originates in the Sihawa Hills of Chhattisgarh.
2. The Tel and Ong rivers are right-bank tributaries of the Mahanadi.
3. The Hirakud Dam, built across the Mahanadi, is among the world's longest earthen dams.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer (d) 1, 2 and 3

Explanation

In NEWS:ONGC spuds first deepwater exploratory well in Mahanadi Basin

Statement 1: The Mahanadi is one of India's major east-flowing peninsular rivers. It originates from the **Sihawa Hills in Dhamtari district of Chhattisgarh** and flows through Chhattisgarh and Odisha before draining into the Bay of Bengal.**Hence, Statement 1 is correct.**

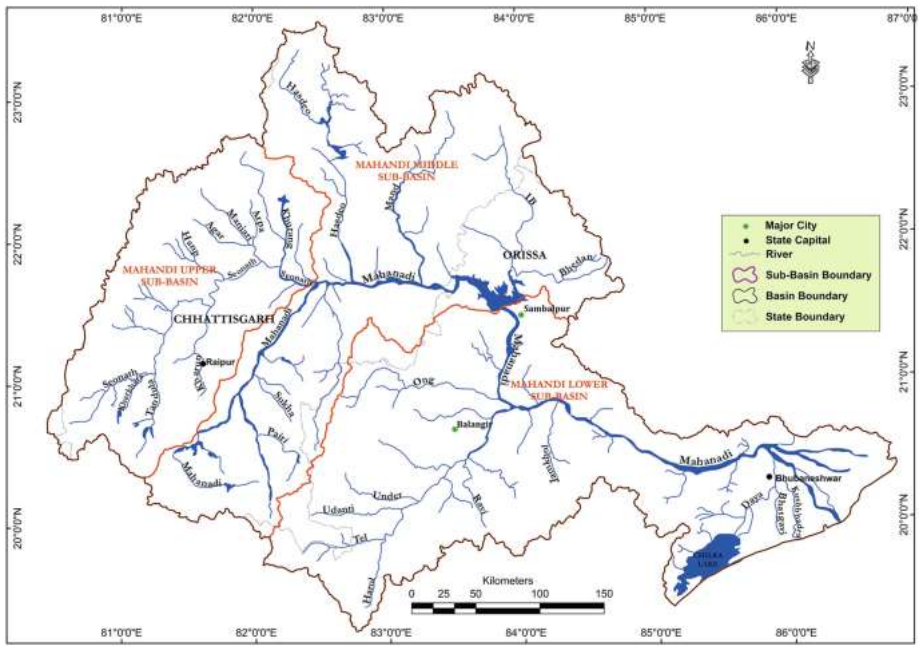
Statement 2: The Mahanadi's major **right-bank tributaries** are **Jonk, Ong and Tel**, while the major **left-bank tributaries** include **Seonath (Shivnath), Hasdeo, Mand and Ib**.**Hence, Statement 2 is correct.**

Statement 3: The **Hirakud Dam** near Sambalpur in Odisha is constructed across the Mahanadi River. It is renowned as one of the **world's longest earthen dams** and plays a vital role in irrigation, flood control and hydropower generation.**Hence, Statement 3 is correct.**

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Value Addition

Mahanadi River: Quick Facts



Feature	Details
Origin	Sihawa Hills, Dhamtari (Chhattisgarh)
Length	~851 km
Drainage Basin	~1.4 lakh sq. km
Outfall	Bay of Bengal near Paradip (False Point)
Major States	Chhattisgarh, Odisha
Basin Extends Into	Maharashtra, Madhya Pradesh, Jharkhand

Tributaries at a Glance

Left Bank	Right Bank
Seonath (Shivnath)	Jonk
Hasdeo	Ong
Mand	Tel
Ib	—

Hirakud Dam

- Located near **Sambalpur, Odisha**
- Built across the **Mahanadi River**
- One of the **world's longest earthen dams**
- Major functions:
  - Flood Control
  - Irrigation
  - Hydroelectric Power Generation
- Creates the **Hirakud Reservoir**

Mahanadi Basin:

- Extends across:
  - Chhattisgarh
  - Odisha
  - Jharkhand
  - Maharashtra
  - Madhya Pradesh
- Receives most rainfall from the **Southwest Monsoon**
- Divided into:
  - Upper Mahanadi Basin
  - Middle Mahanadi Basin
  - Lower Mahanadi Basin

Mahanadi Offshore Basin

- Hydrocarbon-rich sedimentary basin in the Bay of Bengal.
- Formed due to large-scale sediment deposition by the Mahanadi River.
- Important area for:
  - Deepwater oil exploration
  - Natural gas exploration
  - Energy security

Samudra Manthan Mission

- Also known as the **National Deep Water Exploration Mission**.
- Aims to explore India's offshore hydrocarbon reserves.
- Supports:
  - Energy self-reliance
  - Reduced crude oil imports
  - Enhanced domestic production

290. Andromeda

Q. Andromeda, recently seen in the news, is:

(a) A dwarf galaxy orbiting the Milky Way
(b) A galaxy outside the Local Group
(c) The nearest major galaxy to the Milky Way
(d) A satellite galaxy of the Milky Way

Answer : (c) The nearest major galaxy to the Milky Way

Explanation

In NEWS: Andromeda Galaxy entering 'middle age'

Option (a): A dwarf galaxy orbiting the Milky Way

Andromeda (M31) is not a dwarf galaxy. It is a large spiral galaxy and is actually bigger than the Milky Way in terms of diameter. It is an independent galaxy and does not orbit the Milky Way. **Hence, Option (a) is incorrect.**

Option (b): A galaxy outside the Local Group

Andromeda is a member of the **Local Group**, a collection of more than 80 galaxies that includes the

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Milky Way, Andromeda, the Triangulum Galaxy, and numerous dwarf galaxies. **Hence, Option (b) is incorrect.**

Option (c): The nearest major galaxy to the Milky Way

Andromeda is the closest major galaxy to the Milky Way, located about 2.5 million light-years away. It is expected to merge with the Milky Way in the distant future due to their mutual gravitational attraction. **Hence, Option (c) is correct.**

Option (d): A satellite galaxy of the Milky Way

Satellite galaxies are smaller galaxies gravitationally bound to larger galaxies. Andromeda itself hosts satellite galaxies such as M32 and M110 and is not a satellite of the Milky Way. **Hence, Option (d) is incorrect.**

Value Addition

Related Terms

Term	Meaning
Galaxy	Massive system of stars, gas, dust, and dark matter
Exoplanet	Planet outside the Solar System
Exomoon	Moon orbiting an exoplanet
Nebula	Cloud of gas and dust where stars may form
Local Group	Galaxy group containing Milky Way, Andromeda, M33, and dwarf galaxies

Andromeda Galaxy: Quick Facts

Feature	Details
Other Name	M31 (Messier 31)
Type	Spiral Galaxy
Distance from Earth	~2.5 million light-years
Constellation	Andromeda
Local Group Member	Yes
Nearest Major Galaxy to Milky Way	Yes
Diameter	~220,000 light-years
Satellite Galaxies	M32, M110
Future Interaction	Likely merger with Milky Way

Local Group: Major Members

Galaxy	Type
Milky Way	Spiral Galaxy
Andromeda (M31)	Spiral Galaxy
Triangulum (M33)	Spiral Galaxy
Large Magellanic Cloud	Dwarf Galaxy
Small Magellanic Cloud	Dwarf Galaxy

Value Addition:

- **Andromeda = M31 (Messier 31)**, the largest galaxy in the Local Group.
- Located approximately **2.5 million light-years** from Earth.
- Light seen from Andromeda today began its journey when early human ancestors were evolving on Earth.
- One of the few galaxies visible to the **naked eye** under dark-sky conditions.
- Contains an estimated **1 trillion stars**, compared to about **100–400 billion stars** in the Milky Way.
- Classified as a **barred spiral galaxy (SA(s)b/Sb type)**.
- Has at least **30+ known satellite galaxies**, including **M32** and **M110**.
- Approaching the Milky Way at about **110 km/s** due to gravitational attraction.
- Expected to merge with the Milky Way in roughly **4–5 billion years**, forming a giant elliptical or lenticular galaxy sometimes nicknamed **"Milkomeda."**
- Houses a **supermassive black hole** at its center, estimated at around **100 million solar masses**.
- Along with the Milky Way and Triangulum (M33), it forms the core of the **Local Group**.
- The **Andromeda Paradox** is a famous concept in relativity associated with the constellation's direction.
- First catalogued by **Charles Messier** in 1764 as **M31**.
- The first galaxy proven to exist outside the Milky Way by **Edwin Hubble (1920s)**, establishing the concept of external galaxies.
- Often called the **"Great Andromeda Galaxy."**

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291. Tigers outside tiger reserves

Q.With reference to the Tigers Outside Tiger Reserves (TOTR) initiative, consider the following statements:

1. It promotes coexistence-based conservation across shared landscapes.
2. It treats local communities as partners in conservation.
3. It promotes the use of Artificial Intelligence (AI), drones and camera traps for monitoring.
4. It is being implemented simultaneously across all tiger-bearing forest divisions in India.

Which of the statements given above are correct?

(a) 1, 2 and 3 only
(b) 1 and 4 only
(c) 2, 3 and 4 only
(d) 1, 2, 3 and 4

Answer: (a) 1, 2 and 3 only

Explanation

In NEWS: Co-existing with tigers outside reserves

Statement 1:TOTR aims to facilitate **coexistence-based conservation** by managing tigers in shared landscapes comprising forests, wildlife corridors, agricultural areas and village fringes. It shifts the focus from reserve-centric conservation to landscape-level conservation. **Hence, Statement 1 is correct.**

Statement 2: The initiative recognizes that long-term tiger conservation outside protected areas is not possible without community support. Therefore, local communities are treated as **conservation partners** through awareness programmes, community engagement and participatory approaches. **Hence, Statement 2 is correct.**

Statement 3:TOTR promotes the use of modern technologies such as **Artificial Intelligence (AI), drones, camera traps, GPS-enabled monitoring systems and early-warning systems** to strengthen surveillance, monitoring and conflict mitigation. **Hence, Statement 3 is correct.**

Statement 4: The initiative is **not being implemented across all tiger-bearing forest divisions**. It is being launched as a pilot project in **40 forest divisions across 9 States**, selected based on the All India Tiger Estimation (2022). **Hence, Statement 4 is incorrect.**

Value Addition

Tigers Outside Tiger Reserves (TOTR): Quick Facts

Feature	Details
Launched By	Ministry of Environment, Forest and Climate Change
Objective	Conservation of tigers outside Tiger Reserves
Approach	Landscape-based conservation
Focus Areas	Reserve forests, corridors, agricultural lands, village fringes
Key Technologies	AI, Drones, Camera Traps, GPS Monitoring
Conflict Mitigation	Rapid Response Teams, Rescue Infrastructure, Early Warning Systems
Community Role	Conservation Partners
Pilot Coverage	40 Forest Divisions
States Covered	9 States
Key Challenge Addressed	Human-Tiger Conflict

Why TOTR?

- India's tiger population increased from **1,411 (2006)** to **3,682 (2022)**.
- India hosts **over 70% of the world's wild tigers**.
- Nearly **35–40% of India's tigers now live outside notified Tiger Reserves**.
- Represents a shift from **Protected Area Conservation → Landscape Conservation → Coexistence Conservation**.

States Selected in Phase-I

- Uttarakhand
- Uttar Pradesh
- Rajasthan
- Madhya Pradesh
- Maharashtra
- Karnataka
- Tamil Nadu
- Telangana
- Assam

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292. Semiconductor Mission 2.0

Q.Consider the following statements regarding India Semiconductor Mission (ISM) 2.0:

1. Unlike ISM 1.0, which focused primarily on semiconductor fabrication and packaging facilities, ISM 2.0 extends support to upstream supply-chain industries such as semiconductor materials, specialty gases, chemicals, and equipment manufacturing.
2. ISM 2.0 retains the uniform 50% capital subsidy for all semiconductor manufacturing segments.
3. Under ISM 2.0, the Government may acquire equity stakes in companies and startups engaged in semiconductor design and other strategic segments of the semiconductor value chain.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (b) 1 and 3 only
EXPLANATION:

IN NEWS:India Semiconductor Mission 2.0: How is India betting big on semiconductors now?
Statement 1:
ISM 1.0 (2021) largely focused on establishing semiconductor fabrication plants (fabs), assembly, testing, marking and packaging (ATMP/OSAT) facilities, and display manufacturing units. ISM 2.0 marks a significant shift by expanding incentives to the broader semiconductor value chain, including:

- Semiconductor equipment manufacturing
- Specialty gases
- Commodity and specialty chemicals
- Semiconductor materials
- Design tools
- Research & Development (R&D)

The objective is to build a resilient domestic semiconductor ecosystem rather than focusing only on chip fabrication. **Hence, Statement 1 is correct.**
Statement 2:
ISM 1.0 provided a uniform 50% capital subsidy for approved semiconductor projects.However, ISM 2.0 has rationalised incentives:

- Silicon Fabs – 40%
- Other Fabs – 35%
- Advanced Packaging – 35%
- Conventional Packaging – 25%

Therefore, the uniform 50% subsidy structure has not been retained. **Hence, Statement 2 is incorrect.**
Statement 3:
Under ISM 2.0, the Government may provide grants and also acquire equity stakes in strategically important semiconductor companies and startups.
This support is intended to encourage:

- Semiconductor design
- Indigenous intellectual property (IP)
- Advanced technologies
- Supply-chain resilience

The extent of support depends on factors such as private capital mobilisation and strategic importance. **Hence, Statement 3 is correct.**
Value Addition
ISM 1.0 vs ISM 2.0

Feature	**ISM 1.0**	**ISM 2.0**
Launch	2021	2026
Focus	Fabs & Packaging	Entire Semiconductor Value Chain
Approach	Infrastructure-centric	Ecosystem-centric
Incentive Structure	Uniform 50% subsidy	Differential subsidy
Design Support	Limited	Expanded DLI + Equity Support
R&D Support	Limited	Dedicated incentives
Talent Development	Limited	Dedicated incentives
Upstream Supply Chain	Not a major focus	Major focus

Semiconductor Value Chain
Design → Fabrication → Assembly/Testing/Packaging → Electronics Manufacturing
Key Inputs:

- Silicon Wafers
- Specialty Gases
- Specialty Chemicals
- Lithography Equipment
- Electronic Design Automation (EDA) Tools

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Why Semiconductors Matter?

Semiconductors are the foundation of:

- Artificial Intelligence (AI)
- Quantum Computing
- Smartphones
- Electric Vehicles (EVs)
- Telecommunications (5G/6G)
- Defence Systems
- Satellites & Space Technology
- Medical Devices

Semiconductor Types

Type	Uses
Logic Chips	CPUs, AI
Memory Chips	RAM, Storage
Analog Chips	Sensors
Power Semiconductors	EVs, Renewable Energy

Global Semiconductor Ecosystem

Segment	Major Countries/Companies
Advanced Manufacturing	Taiwan, South Korea
Chip Design	USA
Lithography Machines	Netherlands (ASML)
Specialty Chemicals	Japan
Semiconductor Materials	Japan, Germany, USA
Electronics Assembly	China, Vietnam

293. SHet A2 Therapy

Q.SHetA2, recently seen in the news, is best described as:

(a) A prophylactic vaccine against Human Papillomavirus (HPV) infection

(b)A small-molecule oral therapy targeting HPV-induced cervical cancer progression

(c) A gene-editing technology for correcting cervical cancer mutations

(d) A monoclonal antibody used against breast cancer

Answer: (b) A small-molecule oral therapy targeting HPV-induced cervical cancer progression

Explanation

In NEWS:New cervical cancer pill shows promise by stopping HPV from turning cancerous

Option (a): A prophylactic vaccine against Human Papillomavirus (HPV) infection

SHetA2 is **not a vaccine**. Vaccines such as HPV vaccines are administered to prevent HPV infection before exposure. SHetA2 is a therapeutic drug intended for individuals who already have HPV-induced precancerous or cancerous lesions. **Hence, Option (a) is incorrect.**

Option (b): A small-molecule oral therapy targeting HPV-induced cervical cancer progression

SHetA2 is a **small-molecule therapy**, meaning it is a chemically manufactured, low-molecular-weight compound that can be easily absorbed by the body. It is currently being evaluated in oral capsule form.

The therapy targets host proteins such as:

- Mortalin
- Heat Shock Cognate 70 (Hsc70)
- Glucose-Regulated Protein 78 (Grp78)

These proteins help the HPV-E7 oncoprotein survive and promote cancer development. By disrupting this pathway, SHetA2 allows the body's natural defence mechanisms to eliminate the harmful viral protein. **Hence, Option (b) is correct.**

Option (c): A gene-editing technology for correcting cervical cancer mutations

Gene-editing technologies such as **CRISPR-Cas9** modify DNA sequences. SHetA2 does not alter or edit genes. Instead, it works by interfering with protein interactions involved in HPV-induced carcinogenesis.**Hence, Option (c) is incorrect.**

Option (d): A monoclonal antibody used against breast cancer

Monoclonal antibodies are laboratory-produced proteins that target specific antigens. SHetA2 is neither a monoclonal antibody nor a breast cancer therapy. It is a small-molecule drug being developed primarily for HPV-associated cervical lesions and cervical cancer. **Hence, Option (d) is incorrect.**

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Value Addition: SHetA2 at a Glance

Feature	Details
Type	Small-Molecule Therapy
Form	Oral Capsule
Target Disease	HPV-induced Cervical Dysplasia & Cervical Cancer
Key Viral Target	HPV-E7 Oncoprotein
Host Proteins Targeted	Mortalin, Hsc70, Grp78
Purpose	Prevent progression of HPV infection into cancer
Developer	University of Oklahoma & ICMR-NICPR Collaboration
Current Status	Clinical Development Stage
Not a	Vaccine / Gene Therapy / Monoclonal Antibody

294. Malvi camel

Q.With reference to the Malvi Camel, consider the following statements:

1. It is considered the smallest among India's recognised camel breeds.
2. It is well adapted to semi-arid conditions and is used for transportation and agricultural work.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Explanation

In NEWS: Rare camel breed faces extinction threat; conservation measures initiated

Statement 1:

The Malvi camel is one of the **nine recognised camel breeds of India**. It is known for its distinctive appearance and is regarded as the **smallest among the recognised camel breeds**. The breed is primarily distributed across the **Malwa region of Madhya Pradesh** and adjoining parts of Rajasthan.**Hence, Statement 1 is correct.**

Statement 2:

Unlike desert-adapted camel breeds found in the Thar region, the Malvi camel is adapted to **semi-arid and relatively humid conditions** prevailing in the Malwa plateau. Owing to its endurance and strength, it is traditionally used for **transportation, carrying loads, and agricultural operations**. It is also valued for its **milk production**, with females yielding around 4–5 kg of milk per cycle. **Hence, Statement 2 is correct.**

Value Addition: Malvi Camel at a Glance

Feature	Details
Breed Status	One of the 9 recognised camel breeds of India
Distribution	Madhya Pradesh (Mandsaur, Neemuch, Ratlam, Gwalior) and Rajasthan (Kota, Bundi, Jhalawar, Baran)
Distinctive Feature	Smallest recognised camel breed
Adaptation	Semi-arid and moderately humid conditions
Utility	Transportation, draught power, agricultural work
Milk Yield	Around 4–5 kg per lactation cycle
Conservation Agency	National Research Centre on Camel (NRCC), Bikaner
Conservation Concern	Rapid decline in population; threat of extinction

Camel Breeds of India

India has **9 recognised camel breeds**:

- Bikaneri
- Jaisalmeri
- Kachchhi
- Mewari
- Jalori
- Sindhi
- Marwari
- Kharai *(swimming camel of Gujarat)*
- **Malvi**

Special Breed to Remember:

Kharai Camel

- Found in Gujarat.

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- Known as the **"Swimming Camel"**.
- Can swim through seawater creeks to graze on mangroves.



295. Pithora Painting

Q.With reference to Pithora Painting, consider the following statements:

1. It is a ritualistic tribal art form practiced mainly in Gujarat and is executed by ritual painters called Lakharas.
2. The paintings depict deities, animals, and scenes from everyday tribal life.
3. Traditionally, natural pigments derived from minerals and plants are used in the paintings.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
EXPLANATION:
IN NEWS: GI Tag to Gujarat's Pithora Painting boosts tribal art and livelihood
Statement 1: Pithora Painting is a **ritualistic tribal wall art** practiced mainly in the tribal regions of Gujarat, particularly by the **Rathwa, Bhil and Bhilala communities**. The paintings are traditionally created by specialist ritual painters known as **Lakharas**, often under the guidance of a priest called **Badwa**. **Hence, Statement 1 is correct.**
Statement 2: Pithora paintings portray **tribal and Hindu deities, horses, animals, birds, and scenes from daily life**, including farming, hunting, dancing, and festivals. The artwork serves as a visual record of tribal beliefs, customs, and traditions. **Hence, Statement 2 is correct.**
Statement 3: Traditionally, artists use **natural colours obtained from minerals, plants, and other organic materials**. Common colours include red, yellow, green, blue, and white. These pigments are applied using bamboo sticks, cotton, and wooden stencils. **Hence, Statement 3 is correct.**

Value Addition: Pithora Painting

Feature	**Details**
State	Gujarat
Associated Tribes	Rathwa, Bhil, Bhilala
Tribal God	Pithora De
Ritual Painters	Lakharas
Priest	Badwa
Traditional Surface	Walls coated with cow dung and clay
Themes	Deities, horses, animals, farming, hunting, dancing
Colours	Natural pigments from minerals and plants
GI Tag	2026

296. Sacred Groves

Q.With reference to Sacred Groves in India, consider the following statements:

1. They are patches of forest protected due to religious and cultural beliefs.
2. They often harbor rare, endemic, and medicinal plant species.
3. They are legally protected under the Wildlife (Protection) Act, 1972 through the provision of Community Reserves.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3
EXPLANATION:
IN NEWS:Unprotected protectors: Bundelkhand's sacred groves are thriving without any law to save them

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Statement 1:

Sacred Groves are forest fragments traditionally protected by local communities because they are associated with:

- Village deities
- Ancestral spirits
- Religious traditions
- Cultural taboos

Activities such as:

- Tree felling
- Hunting
- Grazing
- Fuelwood collection

are often restricted or prohibited.Thus, conservation is achieved through **community stewardship and faith-based protection** rather than formal state control.**Hence, Statement 1 is correct.**

Statement 2:

Due to minimal human disturbance, Sacred Groves act as:

- Biodiversity hotspots
- Gene pools of native species
- In-situ conservation sites

Recent research from Bundelkhand (2026) found:

- Sacred groves contained **70–72 plant species**
- Nearby degraded forests had only **40–51 species**

Rare medicinal plants such as:

- **Curcuma caesia (Black Turmeric)**
- **Chlorophytum borivilianum (Safed Musli)**

were found within sacred groves.These groves often preserve species that have disappeared from surrounding landscapes due to deforestation and habitat degradation.**Hence, Statement 2 is correct.**

Statement 3:

Traditionally, Sacred Groves were protected through customary practices.However, the **Wildlife (Protection) Amendment Act, 2002** introduced:

- **Community Reserves**
- **Conservation Reserves**

Sacred Groves can be brought under this legal framework if notified by the State Government. However, many Sacred Groves continue to be protected primarily through customary community practices **Hence, Statement 3 is correct.**

Value Addition: Sacred Groves (UPSC Quick Facts)

Feature	**Details**
Conservation Type	In-situ biodiversity conservation
Protection	Religious and cultural beliefs
Basis	
Estimated Number in India	~1–1.5 lakh sacred groves
Biodiversity Role	Refuges for rare, endemic and medicinal species
Legal Protection	Can be notified as **Community Reserves** under WPA, 1972; SC (2024) advocated protection through FRA & forest classification
Famous Names	Orans (Rajasthan), Devrai (Maharashtra), Kavu (Kerala), Devarakadu (Karnataka), Law Kyntang (Meghalaya)
Ecosystem Services	Soil conservation, groundwater recharge, carbon sequestration, pollinator habitat
Community Role	Managed largely by local communities and Gram Sabhas
Women's Role	Tribal and Dalit women often act as key custodians of sacred groves

Bundelkhand Study (2026)

Indicator	**Sacred Groves**	**Degraded Forests**
Plant Species	70–72	40–51
Medicinal Plants	High	Low
Invasive Species	Low	High (Lantana camara)

Significance

- Sacred Groves function as **"living gene banks"** and **micro-climax ecosystems**, conserving biodiversity even in ecologically degraded regions.
- Sacred Groves represent the integration of Traditional Ecological Knowledge (TEK) with biodiversity conservation.

Community Conservation Models in India

Model	**Region**
Sacred Groves	Across India
Orans	Rajasthan

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Devarakadu	Karnataka
Kavu	Kerala
Van Panchayats	Uttarakhand
Community Reserves	Various States

Supreme Court (2024)

- Directed mapping and inventory of Sacred Groves across India.
- Recommended legal recognition.
- Suggested involvement of Gram Sabhas under Forest Rights Act.
- Rajasthan Orans case became a landmark judgment.

297. Marine Micro Plastic Pollution

Q.Consider the following statements regarding marine microplastic pollution:

1. Synthetic filaments are among the most common forms of microplastics found along the Indian coast.
2. Abandoned, Lost or Discarded Fishing Gear (ALDFG) is a major source of marine microplastics.
3. Riverine discharge contributes to the transport of microplastics into marine ecosystems.

Which of the statements given above are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

IN NEWS:PARLIAMENT QUESTION: NCCR Studies on Microplastic Pollution

Statement 1:NCCR monitoring studies have reported that synthetic filaments are the predominant type of microplastic present in India's coastal and marine environments. These filaments mainly originate from synthetic textiles, ropes, fishing nets and other plastic-based materials that release microscopic fibres into water bodies.**Hence, Statement 1 is correct.**

Statement 2:ALDFG, often referred to as "ghost gear", includes discarded fishing nets, lines and traps. Over time, these plastic-based materials degrade into smaller fragments and microplastics, contributing significantly to marine pollution and posing threats to marine organisms through entanglement and ingestion.**Hence, Statement 2 is correct.**

Statement 3: Rivers act as major pathways carrying plastic waste from inland areas to seas and oceans. Plastic debris entering rivers undergoes fragmentation and eventually reaches marine ecosystems as microplastics. NCCR has identified riverine discharge as one of the major contributors to coastal microplastic pollution.**Hence, Statement 3 is correct.**

Value Addition: Microplastics

Feature	**Details**
Definition	Plastic particles smaller than 5 mm
Categories	Primary and Secondary microplastics
Primary Microplastics	Manufactured small particles (microbeads, textile fibres)
Secondary Microplastics	Formed from breakdown of larger plastic items
Major Coastal Sources in India	Riverine discharge, ALDFG
Predominant Type Along Indian Coast	Synthetic filaments
Monitoring Agency	National Centre for Coastal Research (NCCR)
Coastal Cleanliness Tool	Clean Coast Index (CCI)
Major Impacts	Bioaccumulation, biomagnification, ecosystem damage
Key Policy Measures	SUP Ban (2022), EPR Guidelines under PWM Rules, 2022

Size Classification

Type	**Size**
Macroplastic	>25 mm
Mesoplastic	5–25 mm
Microplastic	<5 mm
Nanoplastic	<1000 nm

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NCCR (National Centre for Coastal Research)

- An autonomous institute under the **Ministry of Earth Sciences (MoES)**.
- Headquarters: **Chennai, Tamil Nadu**.
- Established in **2018** (successor to Integrated Coastal and Marine Area Management Project Directorate).
- NCCR monitors microplastics, mesoplastics, macroplastics and marine litter along India's coastline.
- Conducts research on:
  - Coastal erosion
  - Marine pollution
  - Microplastics and marine litter
  - Shoreline changes
  - Coral reefs and coastal ecosystems
  - Sea-level rise and climate impacts

CCI (Clean Coast Index)

- Developed by **NCCR**.
- Used to assess the cleanliness of coastal areas.
- Based on the abundance of:
  - **Microplastics**
  - **Macroplastics (visible litter)**

Classification

The index categorizes beaches/coastal stretches into cleanliness levels such as:

- Very Clean
- Clean
- Moderately Clean
- Dirty
- Extremely Dirty

Gulf of Mannar and Thoothukudi coast are classified as "Moderately Clean" under CCI

298. Polypores

Q.With reference to Polypores, consider the following statements:

1. They are also known as bracket fungi.
2. Their spores are produced inside pores present on the underside of the fruiting body.
3. They play a role in the decomposition of dead wood.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer (d) 1, 2 and 3

Explanation

In NEWS: New Fungus Species Discovered, Named after Kolhapur

Statement 1: Polypores are commonly called **bracket fungi** or **shelf fungi** because their fruiting bodies often project from tree trunks or logs in a shelf-like or bracket-like form.**Hence, Statement 1 is correct.**

Statement 2: Polypores derive their name from the presence of numerous **pores on the underside of the fruiting body**. These pores contain spore-producing structures and help increase the surface area available for spore dispersal.**Hence, Statement 2 is correct.**

Statement 3: Many polypores are important **decomposer fungi**. They break down dead wood and lignin-rich plant material, thereby recycling nutrients and maintaining forest ecosystem health.**Hence, Statement 3 is correct.**

Value Addition: Polypores (Bracket Fungi)

Feature	**Details**
Other Names	Bracket fungi, Shelf fungi
Kingdom	Fungi
Reproduction	Through spores produced in pores
Habitat	Dead logs, rotting wood, living trees
Ecological Role	Wood decomposition and nutrient recycling
Fruiting Body	Shelf-like or bracket-shaped
Importance	Maintains forest nutrient cycles

Quick Facts

- Polypores are among the most important **wood-decaying fungi**.
- They may occur on both **living and dead trees**.
- The visible bracket structure is only the **fruiting body**; the fungal mycelium remains embedded within the wood.
- Some species are indicators of **old-growth forests and ecosystem health**.

Microporellus kolhapurensis

- **Type:** Newly discovered species of **polypore (bracket) fungus**.

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- **Discovery Location: Bhudargad taluka, Kolhapur district, Maharashtra**.
- **Habitat:** Found growing on **rotting wood**.
- **Fruiting Season:** Mainly **July–September**.
- **Growth Pattern:** Usually grows **singly**.
- **Distinct Features:**
  - Umbrella-shaped fruiting body
  - Lateral stalk
  - Large pores
  - Teardrop-shaped spores
  - Thick-walled cystidia
- **Ecological Role:** Acts as a **decomposer**, helping break down dead wood and recycle nutrients in forest ecosystems.
- **Significance:** Adds to India's documented **fungal biodiversity** and highlights the rich yet understudied diversity of Western Ghats fungi.

299. Lead Pollution

Q.Consider the following statements regarding Lead:

1. Lead is a cumulative toxicant that can remain stored in the human body for long periods.
2. No level of lead exposure is considered safe for humans.
3. Informal recycling of lead-acid batteries is a significant source of lead pollution.
4. Elevated blood pressure can occur even in the absence of obvious symptoms of lead poisoning

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2, 3 and 4 only
(c) 1, 3 and 4 only
(d) 1, 2, 3 and 4

Answer: (d) 1, 2, 3 and 4

Explanation

In NEWS:Urban Indian private schools at risk of proximity to toxic lead sites: report

Statement 1:

Lead is a toxic heavy metal that accumulates in the body over time rather than being easily eliminated.Once absorbed, lead can:

- Enter the bloodstream
- Get deposited in bones and teeth
- Remain stored for years or even decades
- Re-enter the bloodstream during pregnancy, ageing, or illness

Because of this tendency to accumulate gradually, lead is classified as a **cumulative toxicant**.

"cumulative toxicant" –

A cumulative toxicant is a substance whose harmful effects increase over time because it accumulates faster than the body can remove it.**Hence, Statement 1 is correct.**

Statement 2:

According to the World Health Organization (WHO), there is **no known safe blood lead concentration**.Even low levels of exposure can cause:

- Reduced intelligence quotient (IQ)
- Learning disabilities
- Behavioral problems
- Neurological impairment

Children are particularly vulnerable because their brains and nervous systems are still developing.Therefore, unlike many pollutants that have permissible thresholds, lead has **no safe exposure limit**.**Hence, Statement 2 is correct.**

Statement 3:

Lead-acid batteries account for nearly **86% of global lead consumption**.In many developing countries, including India, used batteries are often recycled in the informal sector through:

- Manual dismantling
- Open burning
- Crude smelting techniques

These practices release lead into:

- Air
- Soil
- Water

As a result, battery recycling has emerged as one of the largest anthropogenic sources of lead contamination globally.**Hence, Statement 3 is correct.**

Statement 4:

Lead poisoning is often called a **"silent health threat."**Many individuals may not show acute symptoms initially.However, long-term exposure can lead to:

- Hypertension (high blood pressure)
- Cardiovascular diseases
- Kidney damage
- Increased risk of heart attacks

Thus, even apparently healthy individuals may suffer adverse health impacts from chronic lead exposure.**Hence, Statement 4 is correct.**

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Value Addition

Lead: Quick Facts

Feature	Details
Nature	Toxic Heavy Metal
Status	No Safe Exposure Level
WHO Position	Lead poisoning is entirely preventable
Type	Cumulative Toxicant
Major Storage Site	Bones and Teeth
Major Exposure Routes	Air, Water, Soil, Food
Most Vulnerable Group	Children and Pregnant Women

Major Sources of Lead Pollution

Source	Significance
Lead-acid battery recycling	Largest source globally
E-waste recycling	Growing source
Metal smelting	Industrial emissions
Contaminated soil	Legacy pollution
Industrial waste	Heavy metal contamination
Lead-based paints	Important household source

Health Impacts of Lead Exposure

Children

- Reduced IQ
- Learning disabilities
- Developmental delays
- Attention disorders
- Behavioural problems
- Reduced academic performance

Adults

- High blood pressure
- Kidney damage
- Cardiovascular diseases
- Reproductive disorders
- Nervous system damage

300. Smooth-Coated Otter

Q. With reference to the Smooth-coated Otter (*Lutrogale perspicillata*), consider the following statements:

1. It is the largest otter species found in Southeast Asia.
2. It is listed as Endangered on the IUCN Red List.
3. It is known for group hunting and often swims in a V-formation while fishing.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer (b) 1 and 3 only

Explanation

In NEWS: Mayiladuthurai college study identifies major otter habitat in Mettur region

Statement 1:

The Smooth-coated Otter is recognized as the **largest otter species in Southeast Asia**. It is distinguished by its smooth, short fur, flattened tail, webbed feet, and strong swimming ability. **Hence, Statement 1 is correct.**

Statement 2:

The Smooth-coated Otter is currently classified as **Vulnerable (VU)** on the IUCN Red List, not Endangered. Its population is declining due to habitat loss, wetland degradation, pollution, and poaching. **Hence, Statement 2 is incorrect.**

Statement 3

Smooth-coated Otters are highly social animals and frequently hunt in groups. During fishing, they often move in a **V-formation**, which helps them coordinate movements and herd fish efficiently. **Hence, Statement 3 is correct.**

Value Addition: Smooth-coated Otter

Feature	Details
Scientific Name	*Lutrogale perspicillata*
Genus	*Lutrogale* (only extant species)
IUCN Status	Vulnerable (VU)
Distribution	India, South & Southeast Asia, China, Iraq

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Habitat	Rivers, wetlands, mangroves, lakes, rice paddies
Distinctive Feature	Smooth, short fur and flattened tail
Behaviour	Social, group hunting, V-formation swimming
Conservation Threats	Habitat loss, pollution, poaching

Quick Facts

- One of the **three otter species found in India**.
- Equally comfortable on **land and water**.
- Strong swimmer with **webbed feet and dexterous paws**.
- Indicator species for the health of **freshwater ecosystems**.
- Often found in **mangroves, wetlands, large rivers, and reservoirs**.



301. Daanveer Initiative

Q. DAANVEER Initiative, recently launched by the Government of India, is primarily related to:

(a) Promotion of digital payments in rural areas
(b) Voluntary contribution of computers to strengthen digital infrastructure in Gram Panchayats
(c) Financial assistance for rural entrepreneurs through SHGs
(d) Digital literacy training for Panchayat representatives

Answer: (b) Voluntary contribution of computers to strengthen digital infrastructure in Gram Panchayats

Explanation

In NEWS: DAANVEER Initiative Launched; To Boost Digital Infrastructure Across India's Gram Panchayats

Option (a): Promotion of digital payments in rural areas

The DAANVEER Initiative is not aimed at promoting digital payments or financial transactions in rural areas. Its primary objective is to strengthen the digital infrastructure of Gram Panchayats by facilitating donations of computer systems and related equipment. Hence, Option (a) is incorrect.

Option (b): Voluntary contribution of computers to strengthen digital infrastructure in Gram Panchayats

DAANVEER is a citizen participation initiative that provides an end-to-end digital platform through which individuals and organisations can voluntarily donate computers to Gram Panchayats. The initiative operates through the **Meri Panchayat App**, integrated with the **DigiHaat marketplace**, enabling transparent procurement, delivery tracking, and installation monitoring. Hence, Option (b) is correct.

Option (c): Financial assistance for rural entrepreneurs through SHGs

The initiative does not provide financial support, credit assistance, or entrepreneurship funding through Self-Help Groups (SHGs). Its focus is exclusively on enhancing Panchayat-level digital infrastructure. Hence, Option (c) is incorrect.

Option (d): Digital literacy training for Panchayat representatives

Although improved digital infrastructure may indirectly support digital governance and capacity building, DAANVEER is not a digital literacy or training programme for Panchayat representatives. Hence, Option (d) is incorrect.

Value Addition:

Feature	**Details**
Ministry	Ministry of Panchayati Raj
Developed With	National Informatics Centre (NIC) and DigiHaat
Platform	Meri Panchayat Mobile Application
Objective	Strengthen digital infrastructure in Gram Panchayats
Mode	Voluntary donation of computers by citizens and organisations

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Marketplace Integration	DigiHaat
Transparency Mechanism	End-to-end digital and verifiable tracking
Donor Benefit	Digital Certificate of Appreciation
Vision	Digitally Empowered and Self-Reliant Rural India

Quick Facts

- **DAANVEER** promotes **Jan Bhagidari (People's Participation)** in rural governance.
- Supports the objectives of **Digital India** and **e-Panchayat** initiatives.
- Enables donors to select **pre-approved computer bundles** based on state-specific technical requirements.
- Delivery and installation can be tracked digitally from dispatch to deployment.

302. Udham Singh

Q.The revolutionary freedom fighter who avenged the Jallianwala Bagh Massacre by assassinating Michael O'Dwyer at Caxton Hall, London, was:

(a) Madan Lal Dhingra
(b) Bhagat Singh
(c) Udham Singh
(d) Ras Behari Bose

Answer: (c) Udham Singh.

EXPLANATION:

IN NEWS: Nation pays homage to freedom fighter Shaheed Udham Singh on his 87th martyrdom day

Option (a): Madan Lal Dhingra

Madan Lal Dhingra was an Indian revolutionary associated with India House in London. In 1909, he assassinated **Sir William Hutt Curzon Wyllie**, a British official. However, he was not associated with avenging the Jallianwala Bagh Massacre. **Hence, Option (a) is incorrect.**

Option (b): Bhagat Singh

Bhagat Singh was one of the most prominent revolutionaries of the Indian freedom struggle and was associated with the **Hindustan Socialist Republican Association (HSRA)**. Although he inspired many revolutionaries, including Udham Singh, he did not assassinate Michael O'Dwyer. **Hence, Option (b) is incorrect.**

Option (c): Udham Singh

Shaheed Udham Singh witnessed the aftermath of the **Jallianwala Bagh Massacre**, which profoundly shaped his revolutionary commitment. To avenge the massacre, he assassinated **Michael O'Dwyer**, the then Lieutenant Governor of Punjab during the massacre, at **Caxton Hall, London, on 13 March 1940**. During his trial, he adopted the name **"Ram Muhammad Singh Azad"** to symbolize India's religious unity. **Hence, Option (c) is correct.**

Option (d): Ras Behari Bose

Ras Behari Bose was a prominent revolutionary who played a key role in the Ghadar Movement and later contributed to the formation of the **Indian National Army (INA)** in Southeast Asia. He was not involved in the assassination of Michael O'Dwyer. **Hence, Option (d) is incorrect.**

Value Addition: Shaheed Udham Singh

Feature	**Details**
Birth Name	Sher Singh
Born	26 December 1899, Sunam (Punjab)
Associated Organisations	Ghadar Party, HSRA
Inspiration	Bhagat Singh
Historic Act	Assassination of Michael O'Dwyer
Place of Assassination	Caxton Hall, London
Date	13 March 1940
Trial Name	Ram Muhammad Singh Azad
Execution	Pentonville Prison, London
Martyrdom	31 July 1940
Ashes Returned to India	1974

Quick Facts

- Witnessed the impact of the **Jallianwala Bagh Massacre (1919)**.
- Joined the **Ghadar Party** in 1924.
- Worked in **Mesopotamia**, **Africa**, and **America** before actively participating in revolutionary activities.

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● Revered as **"Shaheed-i-Azam"** for his sacrifice against British imperialism.

303. PM CARES

Q.With reference to the PM CARES Fund, consider the following statements:

1. It has been registered as a Public Charitable Trust.
2. The Prime Minister is the ex-officio Chairperson of the Trust, while the Ministers of Defence, Home Affairs and Finance are ex-officio Trustees.
3. Donations to the Fund are exempt from the provisions of the Foreign Contribution (Regulation) Act (FCRA), enabling it to receive foreign contributions.
4. Donations made to the Fund qualify for 100% deduction under Section 80G of the Income Tax Act, 1961.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (d) All four

Explanation

In NEWS:Activists raise alarm over PM CARES denial of access to audit statements

Statement 1:PM CARES Fund was established in March 2020 and has been registered as a **Public Charitable Trust**. The Trust Deed was registered under the **Registration Act, 1908** at New Delhi on 27 March 2020. **Hence, Statement 1 is correct.**

Statement 2: The **Prime Minister of India** serves as the **ex-officio Chairperson** of the PM CARES Fund. The **Union Ministers of Defence, Home Affairs and Finance** serve as **ex-officio Trustees** of the Trust. **Hence, Statement 2 is correct.**

Statement 3:PM CARES Fund has been granted exemption from the operation of the **Foreign Contribution (Regulation) Act (FCRA), 2010**, enabling it to receive donations from individuals and organisations located outside India through designated foreign contribution accounts. **Hence, Statement 3 is correct.**

Statement 4: Contributions made to the PM CARES Fund are eligible for **100% deduction under Section 80G of the Income Tax Act, 1961**, subject to applicable provisions. It is also one of the few funds whose contributions qualify as **CSR expenditure** under the Companies Act, 2013. **Hence, Statement 4 is correct.**

Value Addition: PM CARES Fund Quick Facts

Feature	**Details**
Full Form	Prime Minister's Citizen Assistance and Relief in Emergency Situations Fund
Established	March 2020
Legal Status	Public Charitable Trust
Registration	Registration Act, 1908
Chairperson	Prime Minister of India (Ex-officio)
Ex-officio Trustees	Defence, Home Affairs and Finance Ministers
Tax Benefit	100% deduction under Section 80G
CSR Eligibility	Yes
Foreign Donations	Permitted through FCRA exemption
Administrative Support	Prime Minister's Office (PMO)
Main Objective	Relief during public health emergencies, disasters and distress situations
Audit	Independent Chartered Accountant
CAG Audit	Not mandated

PM CARES vs PMNRF

Feature	**PM CARES**	**PMNRF**
Established	2020	1948
CSR Eligibility	Yes	No
Created For	Emergencies & disasters	Relief and rehabilitation
Foreign Donations	Allowed	Allowed

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Tax & Financial Benefits

Provision	Benefit
Section 80G	100% Tax Deduction
Section 10(23C)	Income Tax Exemption
CSR Eligibility	Yes
Foreign Donations	Permitted
FCRA Status	Exempted from provisions of FCRA

304. Alanine Amino Transferase

Q.Alanine Aminotransferase (ALT) recently seen in the news in the context of liver health, is related to:

(a) A hormone responsible for regulating blood sugar levels
(b) An enzyme that serves as an early marker of liver injury
(c) A protein involved in blood clotting
(d) A diagnostic imaging technique used to detect liver fibrosis

Answer: (b) An enzyme that serves as an early marker of liver injury

Explanation

In NEWS: What Alanine aminotransferase level can reveal about liver health

Option (a): A hormone responsible for regulating blood sugar levels

Hormones are chemical messengers secreted by endocrine glands that regulate various physiological functions in the body. For example, insulin, produced by the pancreas, lowers blood glucose levels by facilitating the uptake of glucose into cells, while glucagon increases blood glucose levels by stimulating the release of stored glucose from the liver. ALT (Alanine Aminotransferase) is neither secreted by an endocrine gland nor involved in glucose regulation. Instead, it is an intracellular enzyme primarily found in liver cells and is used as a biochemical marker of liver injury. **Hence, Option (a) is incorrect.**

Option (b): An enzyme that serves as an early marker of liver injury

Alanine Aminotransferase (ALT), also known as Serum Glutamic-Pyruvic Transaminase (SGPT), is an enzyme predominantly present inside hepatocytes (liver cells). It plays an important role in amino acid metabolism by facilitating the transfer of amino groups during biochemical reactions. Under normal conditions, ALT remains largely confined within liver cells. However, when the liver is damaged due to conditions such as viral hepatitis, fatty liver disease (MASLD), alcohol-related liver disease, drug-induced liver injury, or autoimmune hepatitis, ALT leaks into the bloodstream. Therefore, elevated ALT levels often serve as one of the earliest laboratory indicators of liver cell damage or inflammation. **Hence, Option (b) is correct.**

Option (c): A protein involved in blood clotting

Blood clotting is mediated by a complex cascade of proteins known as clotting factors, most of which are synthesized in the liver. Important clotting proteins include fibrinogen, prothrombin, and Factors VII, VIII, IX, and X, which help prevent excessive bleeding after injury. Although severe liver disease can affect the production of these clotting factors, ALT itself is not a clotting protein and has no direct role in the coagulation process. It merely indicates liver cell injury rather than the liver's clotting function. **Hence, Option (c) is incorrect.**

Option (d): A diagnostic imaging technique used to detect liver fibrosis

ALT is a blood-based biochemical marker measured through laboratory testing and not an imaging technique. Diagnostic imaging methods used to evaluate liver fibrosis, cirrhosis, or structural abnormalities include FibroScan (Transient Elastography), Ultrasound, Computed Tomography (CT), and Magnetic Resonance Imaging (MRI). These techniques provide information about liver stiffness, fat accumulation, and tissue architecture. In contrast, ALT only reflects ongoing liver cell injury and cannot independently determine the degree of fibrosis or cirrhosis. **Hence, Option (d) is incorrect.**

Value Addition: ALT (Alanine Aminotransferase)

Feature	Details
Full Form	Alanine Aminotransferase
Alternate Name	SGPT (Serum Glutamic-Pyruvic Transaminase)
Type	Enzyme
Major Location	Liver Cells (Hepatocytes)

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Function	Amino Acid Metabolism
Clinical Significance	Early marker of liver injury
Test Type	Blood Test
Elevated In	Hepatitis, MASLD, Alcoholic Liver Disease, Autoimmune Hepatitis

Quick Facts

- ALT is one of the key components of the **Liver Function Test (LFT)**.
- Elevated ALT indicates **liver cell damage**, not necessarily liver failure.
- Normal ALT levels do **not always rule out liver disease**.
- People with **MASLD**, cirrhosis, or chronic hepatitis may have normal or mildly elevated ALT levels despite significant liver damage.
- ALT is present in smaller amounts in the **heart and skeletal muscles**, so elevation may occasionally occur due to non-liver causes.

Related Liver Diseases

Disease	**Key Feature**
Hepatitis B	Viral liver infection
Hepatitis C	Chronic viral liver disease
MASLD	Fat accumulation in liver associated with metabolic dysfunction
Alcoholic Liver Disease	Caused by excessive alcohol consumption
Autoimmune Hepatitis	Immune-mediated liver inflammation

305. AI Palletising

Q.AI Palletising, recently seen in the news, is primarily associated with:

(a) Warehouse automation using AI-driven robotic stacking and unstacking systems
(b) AI-powered cargo ships
(c) Drone-based delivery systems
(d) Smart container manufacturing

Answer: (a) Warehouse automation using AI-driven robotic stacking and unstacking systems.

EXPLANATION:

IN NEWS: How AI palletising technology is reshaping warehouse automation

Option (a): Warehouse automation using AI-driven robotic stacking and unstacking systems

AI palletising and de-palletising refers to the use of Artificial Intelligence (AI), computer vision, machine learning, and robotic arms to automate the process of stacking goods onto pallets and unloading them. Unlike traditional robotic systems that rely on fixed programming, AI-powered systems can identify cartons of different shapes and sizes, detect damaged packages, calculate optimal gripping points, and determine stable stacking patterns without manual reprogramming. These systems are increasingly being used in warehouses, e-commerce fulfilment centres, manufacturing units, and logistics hubs to improve efficiency and reduce dependence on manual labour. **Hence, Option (a) is correct.**

Option (b): AI-powered cargo ships

AI-powered cargo ships use artificial intelligence for navigation, route optimisation, fuel efficiency, and autonomous maritime operations. While they are part of the broader logistics ecosystem, they are unrelated to palletising and de-palletising operations carried out inside warehouses and distribution centres. **Hence, Option (b) is incorrect.**

Option (c): Drone-based delivery systems

Drone delivery systems use unmanned aerial vehicles (UAVs) to transport goods over short distances, particularly for last-mile delivery. AI palletising, on the other hand, focuses on warehouse handling operations such as stacking, sorting, and unloading packages before they enter the delivery chain. **Hence, Option (c) is incorrect.**

Option (d): Smart container manufacturing

Smart container manufacturing involves the production of containers equipped with technologies such as sensors, GPS tracking, and IoT devices for monitoring cargo movement and environmental conditions. AI palletising does not deal with container manufacturing but with the automated handling of packages within warehouses and logistics facilities. **Hence, Option (d) is incorrect.**

Value Addition: AI Palletising

Feature	**Details**
Technology Used	AI, Computer Vision, Machine Learning, Robotics
Main Function	Automated stacking and

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	unstacking of goods on pallets
Key Advantage	Handles mixed-size and irregular packages
Real-Time Capability	Detects damaged cartons and optimises gripping points
Learning Ability	Improves performance through Machine Learning
Major Users	E-commerce, Logistics, FMCG, Manufacturing, Pharmaceuticals

How It Works

1. **Computer Vision** scans and identifies package dimensions, orientation, and condition.
2. **AI Algorithms** calculate stable stacking patterns and optimal gripping points.
3. **Machine Learning** improves handling decisions using operational data.
4. **Robotic Arms** perform lifting, stacking, and unloading operations.

Major Applications

- Mixed-case palletising
- Mixed-case de-palletising
- Layer picking
- Container unloading
- Bag and sack handling
- E-commerce fulfilment centres

Advantages

- Reduces manual labour dependency
- Improves warehouse productivity
- Enhances workplace safety
- Handles thousands of different SKUs
- Operates continuously with minimal downtime

Challenges

- High initial investment cost
- Integration with legacy warehouse systems
- Compatibility with Warehouse Management Systems (WMS)
- Need for skilled workforce and maintenance support

Quick Facts

- China is projected to record the fastest growth in AI palletising adoption.
- India is expected to witness significant growth due to expanding logistics infrastructure and warehouse modernisation.
- AI palletising is a key component of **Industry 4.0** and **smart warehouse automation**.

306. Houbara bustard

Q.With reference to the Asian Houbara Bustard (*Chlamydotis macqueenii*), consider the following statements:

1. It is a ground-dwelling migratory bird adapted to arid and semi-arid habitats.
2. It is listed as Vulnerable on the IUCN Red List.
3. India serves as one of its wintering grounds, particularly the Thar Desert and Rann of Kutch.

Which of the statements given above are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

EXPLANATION:

IN NEWS:Vulnerable bustard species being captive-bred and released to restore Aral Sea, destroyed by Soviet folly

Statement 1: The **Asian Houbara Bustard (*Chlamydotis macqueenii*)** is a medium-to-large **ground-dwelling migratory bird** belonging to the bustard family (**Otididae**). It is specially adapted to survive in **arid and semi-arid ecosystems** such as sandy deserts, gravel plains, dry steppes, and open shrublands. Its cryptic coloration provides excellent camouflage in desert landscapes. **Hence, Statement 1 is correct.**

Statement 2: According to the **IUCN Red List**, the Asian Houbara Bustard is classified as **Vulnerable** due to threats such as habitat degradation, illegal hunting, poaching, and loss of migratory stopover sites. The species is also protected under **CITES Appendix I** and **CMS Appendix I**. **Hence, Statement 2 is correct.**

Statement 3: Northern populations of the Asian Houbara Bustard undertake long-distance seasonal migrations from Central Asia to warmer regions during winter. **India serves as an important wintering ground**, particularly in the **Thar Desert of Rajasthan** and the **Rann of Kutch in Gujarat**, which provide suitable desert habitats for the species. **Hence, Statement 3 is correct.**

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Value Addition: Asian Houbara Bustard

Feature	Details
Scientific Name	*Chlamydotis macqueenii*
Family	Otididae
IUCN Status	Vulnerable
CITES	Appendix I
CMS	Appendix I
Habitat	Deserts, dry steppes, shrublands
Indian Wintering Areas	Thar Desert, Rann of Kutch
Type	Ground-dwelling Migratory Bird

Quick Facts

- Known for its **elaborate courtship display** involving black-and-white neck feathers.
- Primarily feeds on **seeds, shoots, insects, beetles, locusts, and small reptiles**.
- Acts as a **flagship species for desert ecosystem conservation**.
- Conservation of the species helps protect broader **arid and semi-arid biodiversity**.



307. World Gold Council

Q.Consider the following statements regarding the World Gold Council (WGC):

1. India is a member country of the World Gold Council.
2. The World Gold Council played a pioneering role in developing and launching the world's first gold-backed ETF.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (b) 2 only

EXPLANATION:

IN NEWS:Jewellery loses sheen; gold ETFs glitter, says WGC

Statement 1: India is a member country of the World Gold Council.

The World Gold Council is not an intergovernmental organisation with country membership. It is a non-profit association of leading gold mining companies. Therefore, countries, including India, are not members. **Hence, Statement 1 is incorrect.**

Statement 2: The World Gold Council pioneered and promoted the world's first Gold Exchange-Traded Fund (Gold ETF), which enabled investors to gain exposure to gold through stock exchanges without physically holding the metal. **Hence, Statement 2 is correct.**

Value Addition:

- Established: 1987
- Headquarters: London, United Kingdom
- Nature: Non-profit market development organisation for the global gold industry.
- Membership: Comprises leading gold mining companies; countries are not members.
- Coverage: Represents markets accounting for about three-fourths of global annual gold consumption.
- Key Functions:
  - Promotes demand for gold.
  - Develops gold industry standards.
  - Supports responsible and sustainable gold mining.
  - Provides market intelligence and research on gold.
- Important Publications:
  - Central Bank Gold Reserves (CBGR) Survey
  - Gold Demand Trends Report (Quarterly)
  - Gold Market Commentary
- Global Authority on Gold: Widely regarded as the leading source of data and analysis on gold markets, investment, technology, and central bank reserves.

About Exchange-Traded Funds (ETFs)

Key Features of ETFs

- Traded on stock exchanges like shares
- Usually passively managed and track an index

- Lower expense ratio compared to many actively managed mutual funds
- Offer liquidity and transparency
- Provide diversification through a basket of securities

Advantages of ETFs

- Cost-effective investment option
- Real-time trading flexibility
- Diversification benefits
- Suitable for long-term passive investment strategies

Limitations of ETFs

- Subject to market volatility
- Tracking error may occur
- Liquidity may vary across ETFs
- Sector-specific ETFs may carry concentrated risk

Difference between ETFs and Mutual Funds

ETF	Mutual Fund
Traded on stock exchanges like shares	Bought/redeemed at end-of-day NAV(Net Asset Value)
Can be traded throughout the day	Transactions occur after market closing
Usually lower expense ratio	Generally higher management cost
Mostly passively managed	Often actively managed

308. Cloud burst

Q.With reference to Cloudbursts, consider the following statements:

1. According to the India Meteorological Department (IMD), a cloudburst occurs when an area of about 20–30 sq. km receives 10 cm or more rainfall in one hour.
2. Orographic lifting plays an important role in the formation of cloudbursts in mountainous regions.
3. Cloudbursts generally occur over relatively small geographical areas.
4. Cloudbursts are more difficult to forecast than cyclones.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1, 2 and 3 only
(c) 2, 3 and 4 only
(d) 1, 2, 3 and 4

Answer: (d) 1, 2, 3 and 4

Explanation

In NEWS: How common are cloudbursts in India? When is heavy rainfall a cloudburst, and when is it not?

Statement 1:

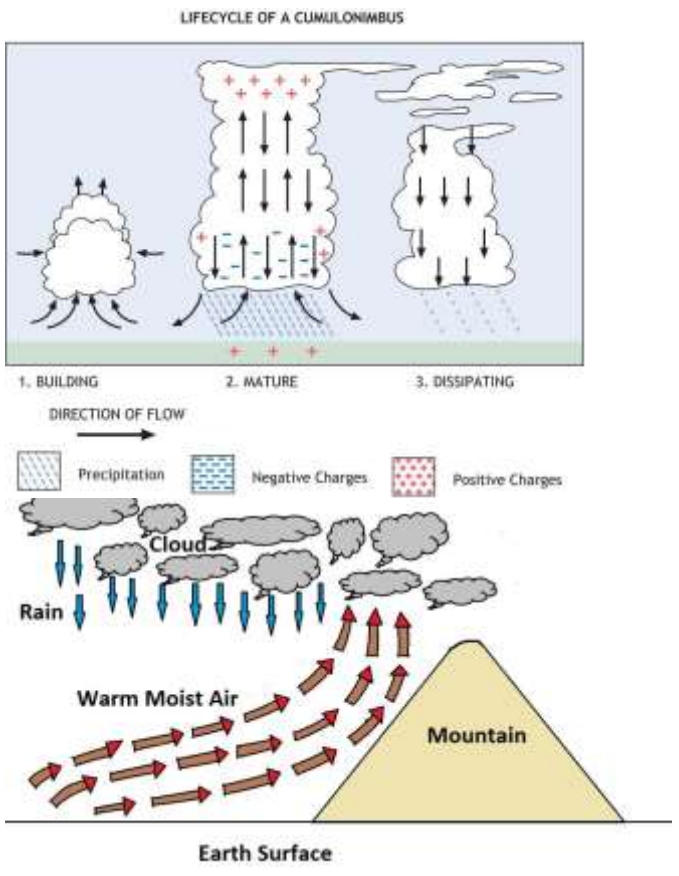
This is an extremely intense rainfall event concentrated over a very limited geographical area. Such rainfall far exceeds the infiltration capacity of the soil, causing sudden runoff and flash floods.For perspective, a single cloudburst can dump nearly **10% of Indore's average annual rainfall** in just one hour.**Hence, Statement 1 is correct.**

Statement 2:

When moisture-laden monsoon winds encounter mountain barriers such as the Himalayas, they are forced to rise upward.As the air rises:

- It cools rapidly.
- Water vapour condenses.
- Massive **cumulonimbus clouds** develop.

Strong updrafts inside these clouds keep water droplets suspended for longer periods. When the updrafts weaken or can no longer support the accumulated water, intense rainfall occurs over a short duration, resulting in a cloudburst.**Hence, Statement 2 is correct.**



Statement 3:

Unlike widespread monsoon rainfall, cloudbursts generally affect a very small area, often limited to **20–30 sq. km**.This localized nature makes them particularly dangerous because:

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- Huge quantities of water are concentrated in a small region.
- Rivers and streams swell rapidly.
- Flash floods and landslides occur within minutes.

The localized nature also explains why a cloudburst may cause severe destruction in one valley while nearby areas receive normal rainfall.**Hence, Statement 3 is correct.**

Statement 4:

Cyclones:

- Develop over several days.
- Cover large geographical areas.
- Can be tracked using satellites and ocean observations.

Cloudbursts:

- Develop rapidly within a few hours.
- Occur over very small areas.
- Often occur in rugged mountainous terrain.
- Require ultra-high-resolution weather models and dense observation networks.

Additionally, mountains create **blind spots for Doppler Weather Radars**, making real-time monitoring more challenging.**Hence, Statement 4 is correct.**

Value Addition: Why Cloudbursts Are Difficult to Predict

Factor	**Explanation**
Small Spatial Scale	Occur over areas smaller than many weather-model grid cells
Rapid Development	Form and dissipate within a few hours
Mountain Terrain	Creates radar blind spots
Sparse Weather Stations	Limited observations in remote regions
High Computing Requirement	Hyperlocal forecasting needs high-resolution models

Value Addition: IMD Definition

Parameter	**Cloudburst**
Rainfall Threshold	≥ 10 cm in 1 hour
Area Covered	Approximately 20–30 sq. km
Nature	Highly localized
Major Impact	Flash floods, landslides, erosion
Common Regions	Himalayas (Uttarakhand, Himachal Pradesh, J&K) and occasionally Western Ghats.

Government Initiatives

Mission Mausam

- Expansion of Doppler Weather Radar network.
- Use of Artificial Intelligence (AI) for hyperlocal forecasting.
- Improved monitoring of extreme weather events.

Nowcasting

- Very short-term weather forecasts (next few hours).
- Provides rapid warnings for thunderstorms, heavy rainfall, and cloudburst-prone conditions.

309. Himalayan Chandra Telescope

Q.With reference to the Himalayan Chandra Telescope (HCT), consider the following statements:

1. It is a 2-metre optical-infrared telescope.
2. It is located at the Indian Astronomical Observatory (IAO), Hanle in Ladakh.
3. It is currently among the highest optical telescopes in the world.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: (d) 1, 2 and 3

Explanation

In NEWS:Celebrating 25 years of the Himalayan Chandra Telescope in Hanle

Statement 1: The **Himalayan Chandra Telescope (HCT)** is a **2-metre optical-infrared telescope** named after Nobel Laureate **Subrahmanyan Chandrasekhar**.The telescope is used for optical and near-infrared astronomical observations and has contributed significantly to studies of **supernovae, exoplanets, gamma-ray bursts, comets, stars, and galaxies**.**Hence, Statement 1 is correct.**

Statement 2: The telescope is situated at the **Indian Astronomical Observatory (IAO), Hanle**, in the Union Territory of **Ladakh**.Hanle was selected after extensive site surveys because of its:

- High altitude desert environment
- More than **250 clear nights annually**
- Extremely low atmospheric water vapour
- Highly transparent skies
- Minimal light pollution

These conditions make it one of the finest locations in India for optical and infrared astronomy.**Hence, Statement 2 is correct.**

Statement 3: Located at an altitude of approximately **4,517 metres above sea level**, the Himalayan Chandra Telescope is currently among the **highest optical telescopes in the world**.Its high-altitude location reduces atmospheric disturbances and improves observation quality, especially for infrared astronomy.**Hence, Statement 3 is correct.**

Value Addition: Himalayan Chandra Telescope (HCT)

Feature	**Details**
Telescope Name	Himalayan Chandra Telescope (HCT)
Named After	Nobel Laureate Subrahmanyan Chandrasekhar
Type	2-metre Optical-Infrared Telescope
Location	Indian Astronomical Observatory (IAO), Hanle, Ladakh
Altitude	4,517 metres
Operator	Indian Institute of Astrophysics (IIA)
Administrative Ministry	Department of Science & Technology (DST)
First Light	26 September 2000
Dedicated to Nation	August 2001
Remote Operation	Since June 2001 from Hosakote, Bengaluru
Satellite Link	INSAT-3B (ISRO)
Clear Nights	More than 250 per year

Why is Hanle Ideal for Astronomy?

Factor	**Significance**
High Altitude	Reduced atmospheric interference
Low Water Vapour	Excellent infrared observations
Transparent Skies	Better visibility of celestial objects
Minimal Light Pollution	Improved observation quality
Dry Cold Desert Climate	Stable atmospheric conditions

Scientific Contributions of HCT

- First scientific result on a **Gamma-Ray Burst afterglow** (2001).
- Participation in discovery of **TRAPPIST-1b exoplanet**.
- Spectroscopy of **comets**.
- Research on **supernovae and novae**.
- Studies of **nebulae, variable stars, binary stars, quasars and galaxies**.
- Reverberation mapping of **Active Galactic Nuclei (AGN)**.

Current Instruments on HCT

Instrument	**Function**
HFOSC	Optical Camera & Spectrograph
uTIRSPEC	Near-Infrared Spectrometer & Imager
HESP	High-Resolution Echelle Spectrograph

Future Projects at Hanle

Union Budget Announcements

- **3.7-metre Upgraded Himalayan Chandra Telescope (UHCT)**
- **13.7-metre National Large Optical-Infrared Telescope (NLOT)**

These projects will be led by the **Indian Institute of Astrophysics (IIA)** and are expected to significantly enhance India's astronomical research capabilities.

Hanle: India's Astronomy Hub

Major facilities located at Hanle:

1. Himalayan Chandra Telescope (HCT)
2. GROWTH-India Telescope
3. MACE (Major Atmospheric Cherenkov Experiment)
4. HAGAR (High Altitude Gamma Ray Telescope)

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5. Hanle Dark Sky Reserve (2022) — India's first Dark Sky Reserve.

310. Cabinet Committee on Security

Q.Consider the following statements regarding the Cabinet Committee on Security (CCS):

1. It is the highest decision-making body on national security matters in the Government of India.
2. It is mentioned in the Constitution of India.
3. The National Security Advisor is a member of the CCS.
4. The Prime Minister chairs the committee.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Answer: (b) Only two

Explanation

In NEWS: PM Modi stresses on safeguarding citizens' interests including seafarers during CCS meeting on West Asia

Statement 1: The CCS is the highest decision-making committee of the Union Government on issues concerning **national security, defence policy, defence expenditure, intelligence matters, foreign affairs, and strategic issues**. It plays a central role in addressing urgent security concerns and approving major national security decisions. **Hence, Statement 1 is correct.**

Statement 2: The CCS is **not mentioned in the Constitution of India**. It is one of the Cabinet Committees constituted by the Government to facilitate decision-making on specific subjects. Therefore, it is an **extra-constitutional body** and derives its authority from executive rules and governmental practice rather than constitutional provisions. **Hence, Statement 2 is incorrect.**

Statement 3: The **National Security Advisor (NSA)** acts as a **secretary-level coordinator** for matters relating to the CCS and assists in coordination and implementation. However, the NSA is **not a formal member** of the committee. The core membership consists of the Prime Minister, Home Minister, Defence Minister, Finance Minister, and External Affairs Minister. **Hence, Statement 3 is incorrect.**

Statement 4: The **Prime Minister of India** serves as the **Chairperson of the Cabinet Committee on Security**. The committee functions under the leadership of the Prime Minister and is responsible for taking critical decisions related to national security and strategic affairs. **Hence, Statement 4 is correct.**

Value Addition: Cabinet Committee on Security (CCS)

Feature	**Details**
Status	Extra-Constitutional Body
Chairperson	Prime Minister
Members	Home Minister, Defence Minister, Finance Minister, External Affairs Minister
Coordinator	National Security Advisor (NSA)
Records Maintained By	Cabinet Secretariat
Deals With	Defence, Foreign Affairs, Intelligence, Nuclear & Space Issues
Key Role	Apex National Security Decision-Making Body

Quick Facts

- A committee similar to the CCS existed in **1947** under Prime Minister **Jawaharlal Nehru**.
- The CCS evolved into its present formal structure after the **Kargil War (1999)**.
- It is the apex body for major **national security appointments**.
- It discusses issues relating to **internal security, law and order, defence procurement, atomic energy, and strategic affairs**.
- Senior military officials and bureaucrats may be invited depending on the issue under consideration.

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